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CLINICAL PRACTICE GUIDELINES

ផ្នែក

FOR

វេជ្ជសាស្ត្រទូទៅ
MEDICINE

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នាយកដ្ឋានសេវាសុខភាព

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Kingdom of Cambodia
Nation Religion King



**CLINICAL PRACTICE
GUIDELINES**

FOR

MEDICINE

Part 2

Department of Health Services

December 2025



Ministry of Health

CLINICAL PRACTICE GUIDELINES FOR MEDICINE

Part
2

December 2025

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CHAPTER III RESPIRATORY SYSTEM

1. Acute rhinopharyngitis
2. Acute Sinusitis
3. Acute and Chronic dyspnea in adult
4. Acute and Chronic cough in adult
5. Asthma on adult
6. Hemoptysis
7. Chronic obstructive pulmonary disease
8. Pulmonary Fibrosis
9. Community acquired pneumonia
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11. Obstructive sleep apnea
12. Pulmonary tuberculosis in adult
13. Pulmonary Aspergillosis
14. Pleural effusion
15. Pneumothorax
16. Bronchiectasis
17. Lung abscess
18. Lung cancer

ACUTE RHINOPHARYNGITIS

By: Asso. Prof NY Chanty, Prof CHAN Sarin, Asso. Prof SAINT Saly, Dr CHHAR Bunpaul, Dr TANN Bunly, Dr CHHORN Sotheary, Prof VANN Mich

I. DEFINITION

The rhinitis or rhino pharyngitis is the common cold, a minor viral infection that frequently affects young children but also adults. It can be recognized by its inflammation of the nose (blocked nose, runny nose) and throat (pain and irritation), sometimes headaches and/or a little fever.

II. EPIDEMIOLOGY

2.1- Viruses (40%)

- Mainly rhinovirus, adenovirus, Influenza A and B
- Adenovirus and herpes virus: lower incidence, but more severe episodes.

2.2- Bacteria (30%)

- Mainly group A beta-hemolytic streptococcus (GABHS, also called *S. pyogenes*)
- More rarely: *Streptococcus viridans*, *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, *Hemophilus influenza*.

2.3- Unknown origin (30%): no detectable pathogen

III. ETIOLOGY

The etiology of rhinopharyngitis is in three factors:

- pathogenic organisms,
- environmental irritants, and
- structural factors.

IV. PATHOPHYSIOLOGY (SUMMARY)

Infection of the adenoids by a virus causes an inflammatory reaction with a tingle or even burning sensation in the cavum, accompanied by mucoid, then purulent, secretions.

Its direct (airborne flow) or indirect (hands) contagiousness explains its rapid spread.

V. DIAGNOSIS

1. Clinic

A- VIRAL INFECTION: Rhinopharyngitis presents the inflammatory syndrome of the upper level of the pharynx (cavum) with which nasal damage is variably associated:

- Rhinorrhea, sneezing, nasal obstruction, fever, cough.
- These signs can be associated with extra-respiratory signs; aches, myalgia, headache.
- Clinical examination finds more or less significant inflammation of the pharynx, rhinorrhea which can be seromucous, purulent or mucopurulent.
- It is a benign pathology, with spontaneously favorable evolution in 7 to 10 days in the vast majority of cases.
- The purulent of rhinorrhea and the existence of a fever (within the normal time course for rhinopharyngitis) are not synonymous with bacterial infection or superinfection.

B- BACTERIAL SURINFECTION: As soon as the bacterial super infection occurs if there is the symptoms persist beyond a week or the character of purulent mucopurulent rhinorrhea.

They are dominated by acute otitis media (AOM) and sinusitis
The common causative germs are *Haemophilus influenzae*, *Moraxella catarrhalis*.
Treatment is essentially local and oral antibiotic therapy is indicated.

In practice, clinical signs may suggest bacterial superinfection as follows:

- respiratory difficulty
- a fever that persists more than 3 days, or appears later after this period,
- the persistence, without any tendency towards improvement, of other symptoms (cough, rhinorrhea, nasal obstruction) more than 10 days,
- irritability, nighttime awakenings, ear pain, otorrhea,
- digestive disorders (anorexia, vomiting, diarrhea) a rash

2. Investigation

1- Biological examination:

Complete blood count is not necessary because of acute viral pharyngitis. The prescription can be given when there is suspicion of acute purulent pharyngitis or presence of complications of orbital infection or osteomyelitis or bacterial meningitis or dental origin.

2- Radiological examination:

Standard radiographic examination is not recommended routinely.

It can also be justified in the event of failure of a first antibiotic therapy and applied in the ENT specialty.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnoses of rhinopharyngitis:

- allergic rhinitis,
- vasomotor rhinitis,
- intranasal foreign body, and
- sinusitis.

VII. MANAGEMENT

1. Objectives

All patients must benefit from symptomatic and etiological treatment if the cause is bacterial.

- Patient comfort
- Symptomatic treatment
- Etiological treatment
- Avoid complication

2. Critical Sign

As soon as the bacterial infection occurs if there is the symptoms persist beyond a week or the character of purulent mucopulent rhinorrhea.

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- respiratory difficulty
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- the persistence, without any tendency towards improvement, of other symptoms (cough, rhinorrhea, nasal obstruction) beyond 10 days,
- irritability, nighttime awakenings, ear pain, otorrhea,

- digestive problems (anorexia, vomiting, diarrhea) a skin rash

3. Mortality risk factor

Not mortality risk

4. Outpatient Management

SYMPTOMATIC TREATMENT

In cases of uncomplicated rhinopharyngitis, treatment is essentially symptomatic. A minimal therapeutic attitude: antipyretics (Acetaminophen (Paracetamol*) 1g 3 times a day if the temperature $> 38^{\circ}\text{C}$), washing of the nasal passages (with physiological serum 0.9% 3 – 4 times/day), sometimes treatments premises, is most often sufficient.

Use of nonsteroidal anti-inflammatory drugs (NSAIDs) and systemic corticosteroids in acute rhinopharyngitis is not recommended.

ANTIBIOTIC TREATMENT

Which antibiotic to choose when antibiotic therapy is indicated?

A. Penicillins

Oral penicillin (ORACILLIN*) remains the treatment of first choice, due to its proven effectiveness, its narrow spectrum, its low toxicity and its low cost. Plus, and this is a major advantage. A treatment lasting 10 days is necessary in order to achieve a maximum rate of eradication of streptococci in the pharynx.

B. Alternatives

- Macrolides: Macrolides (preferably erythromycin) can be used in cases of penicillin allergy. The association between emergence of resistance and increasing consumption of macrolides has clearly been demonstrated.

- First- and second-generation cephalosporin: clinical and microbiological responses are equal (or even superior) to those obtained with penicillin. They can be used in patients who are allergic to penicillin but do not have an immediate hypersensitivity reaction.

5. First-line treatment

Table 1: Antibiotic recommended and dosages

Place	Antibiotic	Adult dose
First choice	Phenoxymethylpenicillin	1 Million IU x 3/day in 7-10 days
	Clometocilline	500mg x 2/day in 7-10 days
Second choice	Cefadroxil	500mg x 2/day in 7-10 days
	Amoxillin	500mg x 2 /day or 1g x 2/day in 7-10 days
Third choice	Erythromycin or neo-macrolide	500mg x 2 /day in 7-10 days

6. Treatment adjustment No significant

7. Algorithm

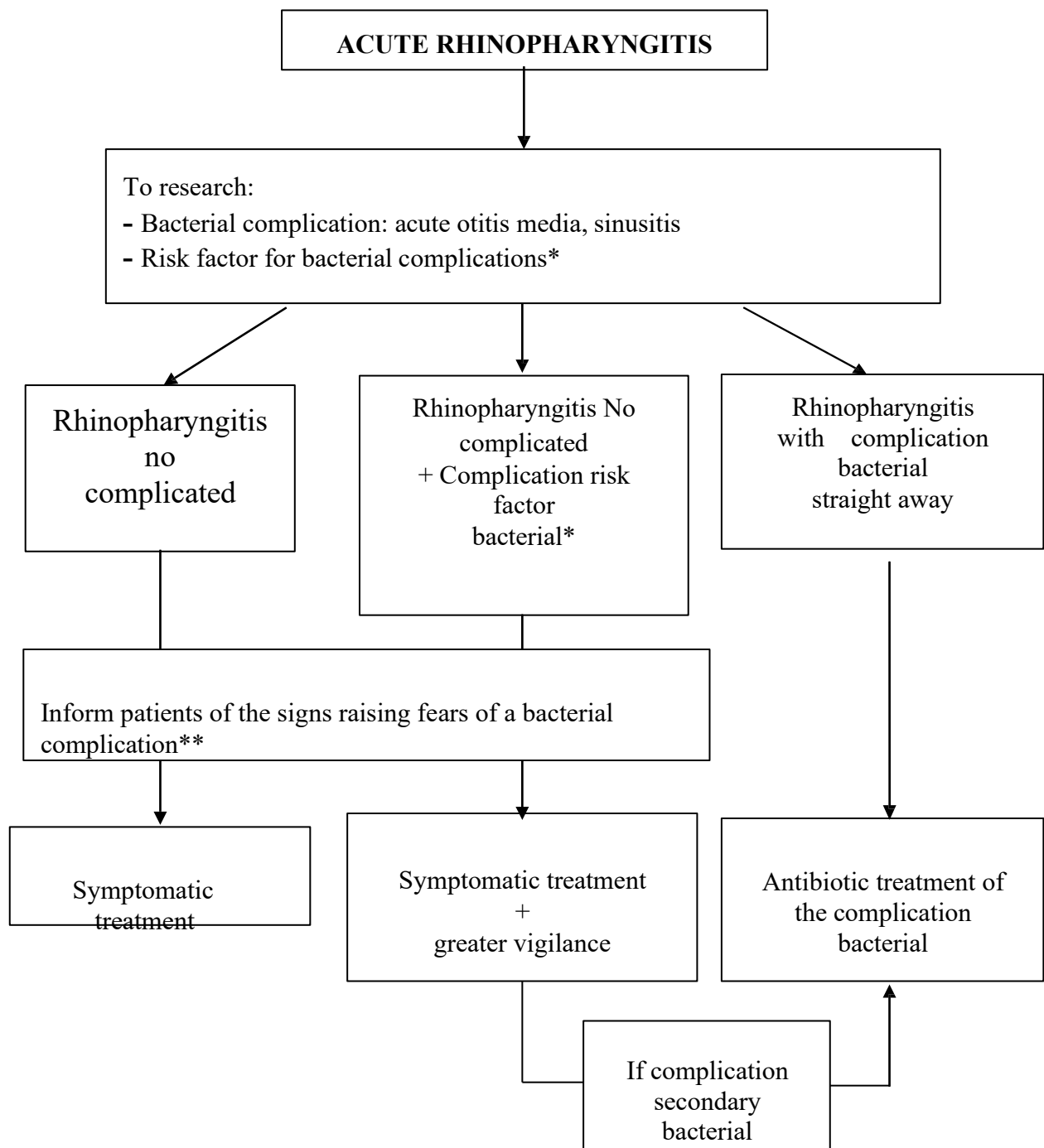


Figure 1: Acute nasopharyngitis: attitude practical

VIII. SPECIAL CASES

The differentiation of respiratory distress syndrome from influenza is caused by the virus of AH1N1 and H5N1 and COVID-19 sometimes difficult, but if apparition of fever, cough and dyspnea, it is necessary to transfer to the emergency department.

IX. EVOLUTION AND PROGNOSIS

Normaly, rhinopharyngitis is not severe and it can heal spontaneously when the patient stays in bed for a few days, except bacterial infection on mortality subject.

X. COMPLICATIONS

The complication of rhinopharyngitis can progress to severity in:

- immunodepression.
- history of recurrent AOM (more than 3 times in 6 months) or the presence of serous otitis increase the risk of occurrence of AOM.

XI. PREVENTION AND RECOMMENDATION

Patient Advice

In the type of mode of contamination by air flow and hands; the patient must therefore use the normal protective mask and wash their hands to avoid contamination to anyone else.

When the syndrome is present, you should not worry too much and stay in bed, limit moving, if fever is present, use paracetamol, because acute rhinopharyngitis is the benign form and heal spontaneously.

XII. REFERENCE

1. Recommandations et Pratique: strategies thérapeutique, 1er Edition: Octobre 2005
2. Guide pratique des médicaments 23è Ediction: 2003
3. Respiratory Medicine; 3è Ediction 2003
4. CURRENT Medical Diagnosis and Treatment 2011
5. 10e Conférence de Consensus en Thérapeutique Anti-Infectieuse - 19 juin 1996, Lyon : Les infections ORL; Méd Mal Infect. 1997 ; 27, Spécial : 334-54

ACUTE SINUSITIS

Prof NY Chanty

I. DEFINITION

Sinusitis is inflammation of one or more sinuses (maxillary, frontal, ethmoid and sphenoid). Sinusitis is more common in children than in adults, most often of viral origin.

II. EPIDEMIOLOGY

It generally follows rhinitis or viral nasopharyngitis. Maxillary locations are the most common. Allergy, cystic fibrosis, polyposis, ciliary dyskinesia, foreign body, direct nasal trauma or barotrauma can also be causes of acute sinusitis.

III. ETIOLOGY

Most sinus infections are viral, and only a small proportion develops a secondary bacterial infection. Rhinoviruses, influenza viruses, and parainfluenza viruses are the most common causes of sinusitis.

IV. PATHOPHYSIOLOGY (SUMMARY)

The nasal cavity and sinuses are lined by the same ciliated and permeable respiratory mucosa. Nasopharyngitis is a very common viral infection that often affects the sinus mucosa, in addition to the nasal or pharyngeal mucosa.

The viral attack on the respiratory epithelium which lines the nasal cavity and sinuses leads to a modification of the relationships between the resident bacteria (pneumococcus, Haemophilus influenzae, Branhamella catarrhalis) and the mucosa. The disappearance of mucociliary movement helps to promote the adhesion of bacteria and their multiplication.

V. DIAGNOSIS

- Clinic

The arguments inflammation may be caused by viruses, bacteria, fungi, and allergens. The diagnosis of sinusitis requires the presence either of two major factors or one major plus two minor factors. The diagnostic criteria are as follows:

1. Major

- Purulence on nasal examination
- Nasal obstruction/blockage
- Nasal discharge/purulence/discolored posterior drainage
- Hyposmia/anosmia
- Fever
- Facial pain/pressure
- Facial congestion/fullness

2. Minor

- Ear pain/pressure/fullness
- Cough
- Dental pain
- Fatigue
- Halitosis
- Fever
- Headache

- Investigation

1- Biological examination

Complete blood count is not necessary because of acute viral sinusitis. The prescription can be given when there is suspicion of acute purulent sinusitis or presence of complications of orbital infection or osteomyelitis or bacterial meningitis or dental origin.

2- Radiological examination

Standard radiographic examination is not recommended routinely.

It can also be justified in the event of failure of a first antibiotic therapy and applied in the ENT specialty.

VI. DIFFERENTIAL DIAGNOSIS

The clinical presentations of several disease processes overlap with the symptoms of recurrent acute rhinosinusitis. Other diseases and conditions that should be ruled out before making the diagnosis include:

- Allergic rhinosinusitis
- Acute rhinosinusitis
- Chronic rhinosinusitis with nasal polyposis (CRSwNP)
- Chronic rhinosinusitis without nasal polyposis (CRSSNP)
- Subacute rhinosinusitis
- Viral rhinosinusitis
- Cerebrospinal fluid (CSF) rhinorrhea
- Non-Invasive fungal rhinosinusitis
- Invasive fungal rhinosinusitis

VII. MANAGEMENT

a. Objectives

- Reduction of pain, sinus drainage and improvement of nasal ventilation.
- Prevention of infectious complications (intraorbital or intracranial suppurations)

b. Critical Sign

Although rare, complications of recurrent acute rhinosinusitis include but are not limited to the following:

- Preseptal cellulitis
- Orbital cellulitis
- Subperiosteal abscess
- Orbital abscess
- Cavernous sinus thrombosis

c. Mortality risk factor

Acute invasive fungal sinusitis carries a mortality rate of up to 50%. Signs and symptoms include dark eschars on the septum, inferior and middle turbinates, and/or palate. In the late stages, signs and symptoms of cavernous sinus thrombosis and cranial neuropathies are present.

d. Outpatient Management

Consultation with an ophthalmologist may be required for additional evaluation and management of orbital complications.

5. First-line treatment

ANTIBIOTICS TO BE PRESCRIBED

Table 1: Indications and dosages of recommended antibiotics

Antibiotics			Dosage
Amoxicillin Clamoxyl®, Amoxicillin®		Acute maxillary sinusitis and other forms of sinusitis	1g x 3/day on 7-10 days orally
Amoxicillin - clavulanic acid. Augmentin®, Moxiclav®		Acute maxillary sinusitis and other forms of sinusitis	- Acute maxillary sinusitis: 1g x 2 /day - other forms of sinusitis: 1g x 3/day on 7-10 days orally
Cephalosporins	C2G (Cefuroxime) ZINNAT®	Adult sinusitis	250mg x 2/day (up to 1g/day) on 5days orally
	C3G (Cefpodoxime) ORELOX®	Acute sinusitis	200mg x 2/day on 5days orally

Fluoroquinolones (active on pneumococcus)	levofloxacin Leflox®, Tavanic®	Acute sinusitis	500mg x 1/day on 5days orally
	Moxifloxacin Avelox®, Moxiget®		400mg x 1/day On 5days orally

* Change of antibiotic can be done on the 4th day of treatment if his condition does not improve or the fever persists.

6. Treatment adjustment

7. Algorithm

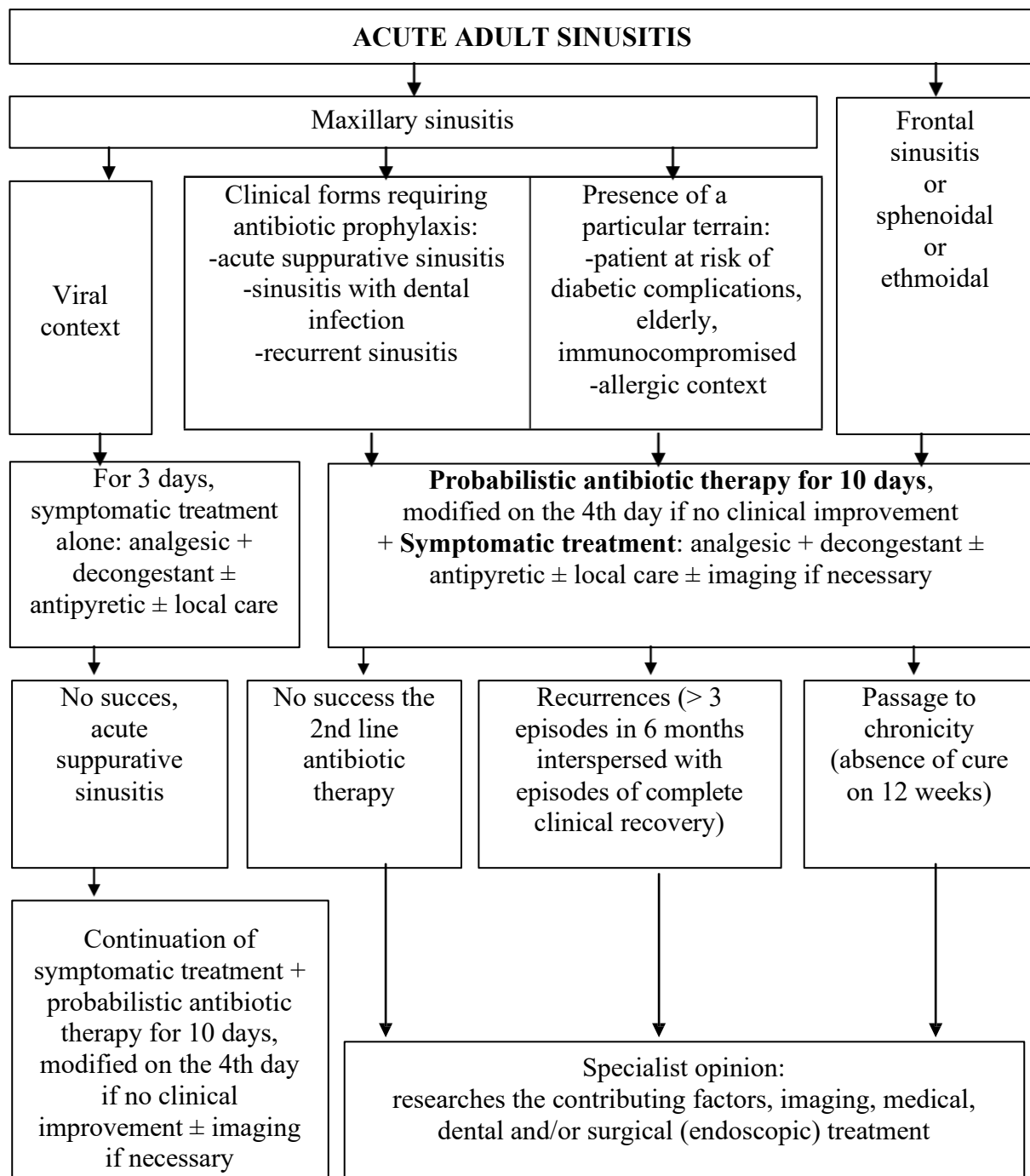


Figure 1: Therapeutic management of acute sinusitis

VIII. SPECIAL CASES

- Bacterial sinusitis of dental origin and/or sinusitis with intracranial complication require hospitalization.
- Cavernous sinus thrombosis and/or recurrent sinusitis and/or chronic sinusitis and/or dental infection must be transferred to the specialist.

IX. EVOLUTION AND PROGNOSIS

The prognosis for recurrent acute rhinosinusitis is generally good. Most patients improve with medical management, including topical nasal sprays and oral antibiotics. Patients rarely develop complications requiring hospitalization for immediate surgical intervention or intravenous antibiotics. Appropriately selected patients who undergo surgery typically experience symptomatic relief and appreciable improvement in quality of life.

X. COMPLICATIONS

Although rare, complications of recurrent acute rhinosinusitis include but are not limited to the following:

- Preseptal cellulitis
- Orbital cellulitis
- Subperiosteal abscess
- Orbital abscess
- Cavernous sinus thrombosis

XI. PREVENTION AND RECOMMENDATION

Certain measures help reduce the risk of contracting the disease for the first time, or again, or of having chronic sinusitis:

- Prevent colds by the usual means: wash your hands, avoid close contact with people with a cold.
- Prevent allergies by exposing yourself as little as possible to allergens (pets, pollen, mites, etc.) and pollutants that usually affect us.
- Have a balanced lifestyle in terms of stress, physical activity, diet, etc.
- Avoid smoking or exposure to tobacco smoke, which irritates the sinuses.

Avoid using decongestants in the form of nasal sprays (Dristan®, Otrivin®) for more than 3 days, as they can lead to habituation and “rebound” nasal congestion (which reappears when you stop). Additionally, in some people, they irritate the delicate nasal mucosa. Decongestants taken orally cause fewer problems with the nasal mucosa, but can have bothersome side effects (worsening of high blood pressure, for example), because their effectiveness is disputed.

XII. REFERENCES

1. Recommandations et Pratique: stratégies thérapeutique, 1er Edition: Octobre 2005
2. Guide pratique des médicaments 23^e Edition: 2003
3. CURRENT Medical Diagnosis and Treatment 2011
4. 10^e Conférence de Consensus en Thérapeutique Anti-Infectieuse - 19 juin 1996, Lyon : Les infections ORL; Méd Mal Infect. 1997 ; 27, Spécial : 334-54
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ACUTE AND CHRONIC DYSPNEA

Asso. Prof. SAINT Saly

I. DEFINITION

- **Acute Dyspnea:** Sudden onset of difficult or labored breathing.
- **Chronic Dyspnea:** Persistent difficulty breathing over an extended period, typically more than six weeks.

II. EPIDEMIOLOGY

- **Acute Dyspnea:** Commonly seen in emergency settings due to conditions like acute respiratory infections, exacerbations of chronic lung disease, or cardiac events.
- **Chronic Dyspnea:** Prevalence increases with age and is often associated with conditions such as chronic obstructive pulmonary disease (COPD), congestive heart failure (CHF), and interstitial lung diseases.

III. ETIOLOGY

- **Acute Dyspnea:** Causes include pulmonary embolism, pneumothorax, acute asthma exacerbation, myocardial infarction, and acute heart failure.
- **Chronic Dyspnea:** Can result from COPD, CHF, interstitial lung diseases (e.g., idiopathic pulmonary fibrosis), obesity hypoventilation syndrome, or neuromuscular disorders.

IV. PATHOPHYSIOLOGY

- **Acute Dyspnea:** Typically involves rapid changes in lung mechanics, oxygenation, or perfusion due to underlying acute conditions affecting the respiratory or cardiovascular systems.
- **Chronic Dyspnea:** Involves long-term structural changes in lung tissue, airway inflammation, impaired gas exchange, or reduced cardiac output.

V. DIAGNOSIS

Clinical Presentation and Investigation in Diagnosis

- **Acute Dyspnea:** Symptoms include sudden onset shortness of breath, tachypnea, use of accessory muscles, cyanosis, and anxiety. Diagnostic workup involves oxygen saturation monitoring, chest X-ray, electrocardiogram (ECG), and possibly arterial blood gas analysis.
- **Chronic Dyspnea:** Patients often report exertional dyspnea, orthopnea, paroxysmal nocturnal dyspnea (in CHF), or persistent breathlessness at rest (in advanced COPD). Diagnosis requires spirometry, imaging (chest X-ray, CT scan), echocardiography, and possibly pulmonary function tests.

VI. MANAGEMENT

- **Acute Dyspnea:** Immediate management focuses on supplemental oxygen, bronchodilators (if asthma or COPD exacerbation), diuretics (if heart failure), and addressing the underlying cause (e.g., thrombolysis for pulmonary embolism).
- **Chronic Dyspnea:** Treatment aims to control symptoms and manage underlying conditions with medications (e.g., bronchodilators, corticosteroids, diuretics), pulmonary rehabilitation, oxygen therapy, and in severe cases, surgical interventions (e.g., lung volume reduction surgery).

VII. SPECIAL CASES

- **Acute Dyspnea:** Considerations include pregnant patients, pediatric cases, or those with significant comorbidities affecting treatment choices.
- **Chronic Dyspnea:** Management strategies differ based on the underlying etiology (e.g., specific treatments for COPD exacerbations vs. heart failure exacerbations).

VIII. EVOLUTION AND PROGNOSIS

- **Acute Dyspnea:** Prognosis varies widely depending on the underlying cause, timeliness of intervention, and patient comorbidities. Rapid treatment can lead to significant improvement, while delayed management can result in worse outcomes.
- **Chronic Dyspnea:** Prognosis is influenced by the severity of the underlying condition, response to treatment, and patient adherence to lifestyle modifications and medications. Progression can lead to decreased functional status and increased mortality.

IX. PREVENTION AND RECOMMENDATIONS

- **Acute Dyspnea:** Preventive measures include timely management of chronic conditions (e.g., hypertension, diabetes) to reduce the risk of acute exacerbations. Smoking cessation and vaccination (e.g., influenza, pneumococcal) are crucial preventive strategies.
- **Chronic Dyspnea:** Emphasize smoking cessation, adherence to prescribed medications, regular exercise, and weight management. Patient education on recognizing exacerbation symptoms and early intervention is key.

X. REFERENCES

1. Cambodia, the National TB Control Program's Technical TB Guideline 2016, Reviewed 2022.
2. Harrison's Principle of Internal medicine, Joseph Loscalzo, 21st Edition, volume 1, 2022, P: 263-267.
3. Current Medical Diagnosis and Treatment, Maxine A, LANGE Medical book, 2023, 62nd Edition, P: 18-20.
4. The 5-Minute Clinical Consult, Frank J. Domino, 25th Edition, 2017.

ACUTE AND CHRONIC COUGH

Asso. Prof. SAINT Saly

I. DEFINITION

- **Acute Cough:** A sudden onset of coughing that typically lasts for less than 2-3 weeks and is often associated with viral upper respiratory infections or acute bronchitis.
- **Chronic Cough:** Persistent cough lasting for 3-8 weeks, chronic cough lasting for 8 weeks or longer in adult, which may be a symptom of underlying conditions such as asthma, chronic obstructive pulmonary disease (COPD), gastroesophageal reflux disease (GERD), or interstitial lung diseases.

II. EPIDEMIOLOGY

- **Acute Cough:** Common in all age groups, especially during the winter months due to viral respiratory infections. Most cases resolve without complications.
- **Chronic Cough:** Recurrence cough (incidence) has been reported at 3-40% by various population estimates. Prevalence increases with age and is more common in smokers or individuals with underlying respiratory or systemic diseases. It significantly impacts quality of life and healthcare utilization.

III. ETIOLOGY

- **Acute Cough:** Often caused by viral infections (e.g., common cold, influenza), bacterial infections (e.g., acute bronchitis), environmental irritants (e.g., smoke, pollutants), or allergic reactions.
- **Chronic Cough:** Can result from asthma, COPD, GERD, postnasal drip syndrome (due to allergic rhinitis or sinusitis), medications (e.g., ACE inhibitors), or less common causes such as bronchiectasis, lung cancer, or psychogenic factors.

IV. PATHOPHYSIOLOGY

- **Acute Cough:** Triggered by irritation or inflammation of the upper or lower airways, leading to stimulation of cough receptors. Infections cause mucosal inflammation and increased mucus production, stimulating the cough reflex.
- **Chronic Cough:** Involves persistent irritation or inflammation of the airways, which may be due to ongoing disease processes (e.g., bronchial hyperresponsiveness in asthma), chronic exposure to irritants (e.g., smoking), or mechanical stimulation (e.g., GERD causing reflux of acid into the esophagus).

V. DIAGNOSIS

Clinical Presentation and Investigation in Diagnosis

- **Acute Cough:** Presents with sudden onset of coughing, often accompanied by other upper respiratory symptoms (e.g., runny nose, sore throat). Diagnosis is clinical and may involve assessing for fever, auscultation of the lungs, and considering chest X-ray if pneumonia is suspected.

- **Chronic Cough:** Patients typically report persistent coughing, sometimes with sputum production or other symptoms related to the underlying cause (e.g., wheezing in asthma, heartburn in GERD). Diagnostic evaluation includes detailed history, physical examination, spirometry, chest X-ray, and possibly more specialized tests (e.g., CT scan, bronchoscopy) depending on suspected etiology.

VI. MANAGEMENT

- **Acute Cough:** Treatment focuses on symptomatic relief, such as hydration, rest, antitussive medications (e.g., dextromethorphan), and addressing the underlying cause if bacterial (e.g., antibiotics for bacterial bronchitis).
- **Chronic Cough:** Management depends on identifying and treating the underlying condition. Treatments may include inhaled corticosteroids (for asthma), bronchodilators, proton pump inhibitors (for GERD), allergy medications, or specific therapies for conditions like bronchiectasis or chronic bronchitis. Behavioral therapies and speech therapy may also be beneficial.

VII. SPECIAL CASES

- **Acute Cough:** Special considerations include pediatric and elderly populations, where cough reflex sensitivity and tolerance to medications may differ. Severe cases (e.g., pertussis in infants) may require hospitalization and specialized care.
- **Chronic Cough:** Evaluation in special cases involves consideration of comorbidities (e.g., immunocompromised patients, those with multiple chronic diseases) that may complicate diagnosis and management.

VIII. EVOLUTION AND PROGNOSIS

- **Acute Cough:** Most cases resolve within a few weeks without complications. Prognosis is excellent with appropriate management, though severe cases (e.g., pneumonia) can lead to prolonged recovery or hospitalization.
- **Chronic Cough:** Prognosis varies based on the underlying cause. Effective management can lead to significant improvement in symptoms and quality of life. However, chronic cough can persist despite treatment in some cases, requiring ongoing monitoring and adjustment of therapies.

IX. PREVENTION AND RECOMMENDATIONS

- **Acute Cough:** Prevention strategies include practicing good hand hygiene, avoiding close contact with sick individuals, and vaccination against influenza and pertussis.
- **Chronic Cough:** Emphasizes smoking cessation, avoiding environmental irritants (e.g., secondhand smoke, occupational exposures), managing underlying conditions promptly, and regular follow-up with healthcare providers for monitoring and adjustment of treatment.

X. REFERENCES

1. Cambodia, the National TB Control Program's Technical TB Guideline 2016, Reviewed 2022.
2. Harrison's Principle of Internal medicine, Joseph Loscalzo, 21st Edition, volume 1, 2022, p 267-268.
3. Current Medical Diagnosis and Treatment, Maxine A, LANGE Medical book, 2023, 62nd Edition, p 15-18.
4. The 5-Minute Clinical Consult, Frank J. Domino, 25th Edition, 2017, p194-195.

ASTHMA ON ADULT

Prof. Vann Mich

I- DEFINITION

Asthma is a heterogeneous disease, usually characterized by chronic airway inflammation. It is defined by the history of respiratory symptoms such as wheeze, shortness of breath, chest tightness and cough, that vary over time and in intensity, together with variable expiratory airflow limitation (1). These variations are often triggered by factors such as exercise, allergen or irritant exposure, change in weather, or viral respiratory infection.

Symptoms and airflow limitation may resolve spontaneously or in response to medication and can come and go over time.

II- EPDEMOIOLOGY

Asthma affects both children and adult and it is the most common chronic respiratory disease among children.

Asthma affected an estimated 262 million people in 2019 and caused 455 000 deaths (2).

The prevalence varies from 1 – 29% of the population in different countries (1).

III- ETIOLOGY

The cause of asthma is multifactorial. Many factors have been linked to an increased risk of developing asthma, although it is often difficult to find a single, direct cause.

- ❑ Asthma is more likely if other family members also have asthma – particularly a close relative, such as a parent or sibling.
- ❑ Asthma is more likely in people who have other allergic conditions, such as eczema and rhinitis (hay fever).
- ❑ Urbanization is associated with increased asthma prevalence, probably due to multiple lifestyle factors.
- ❑ Events in early life affect the developing lungs and can increase the risk of asthma. These include low birth weight, prematurity, exposure to tobacco smoke and other sources of air pollution, as well as viral respiratory infections.
- ❑ Exposure to a range of environmental allergens and irritants are also thought to increase the risk of asthma, including indoor and outdoor air pollution, house dust mites, moulds, and occupational exposure to chemicals, fumes or dust.
- ❑ Children and adults who are overweight or obese are at a greater risk of asthma.

IV- PATHOPHYSIOLOGY

The pathophysiology of asthma is heterogeneous and involves the following components:

- Airway inflammation
- Airway hyperplasia
- Hypersecretions
- Intermittent bronchoconstriction
- Bronchial hyperresponsiveness
- Airway remodeling

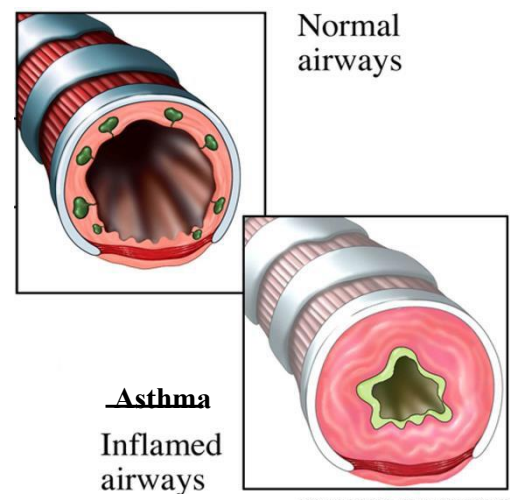
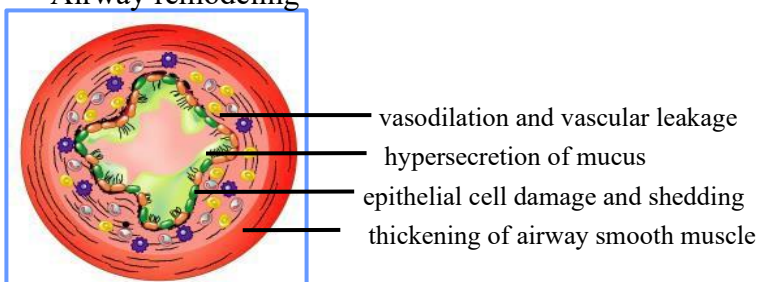


Fig.1

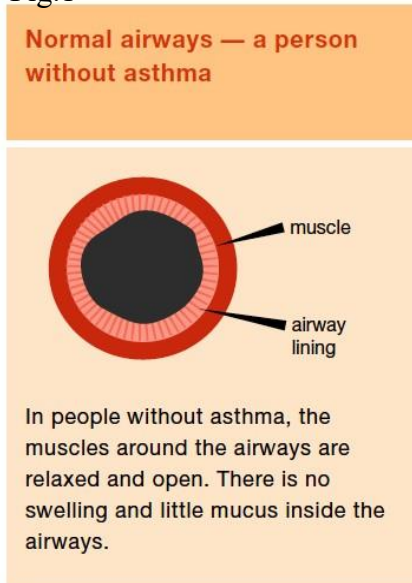


Fig.2

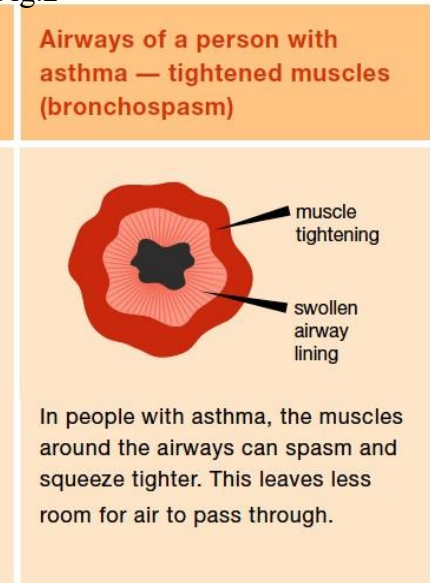
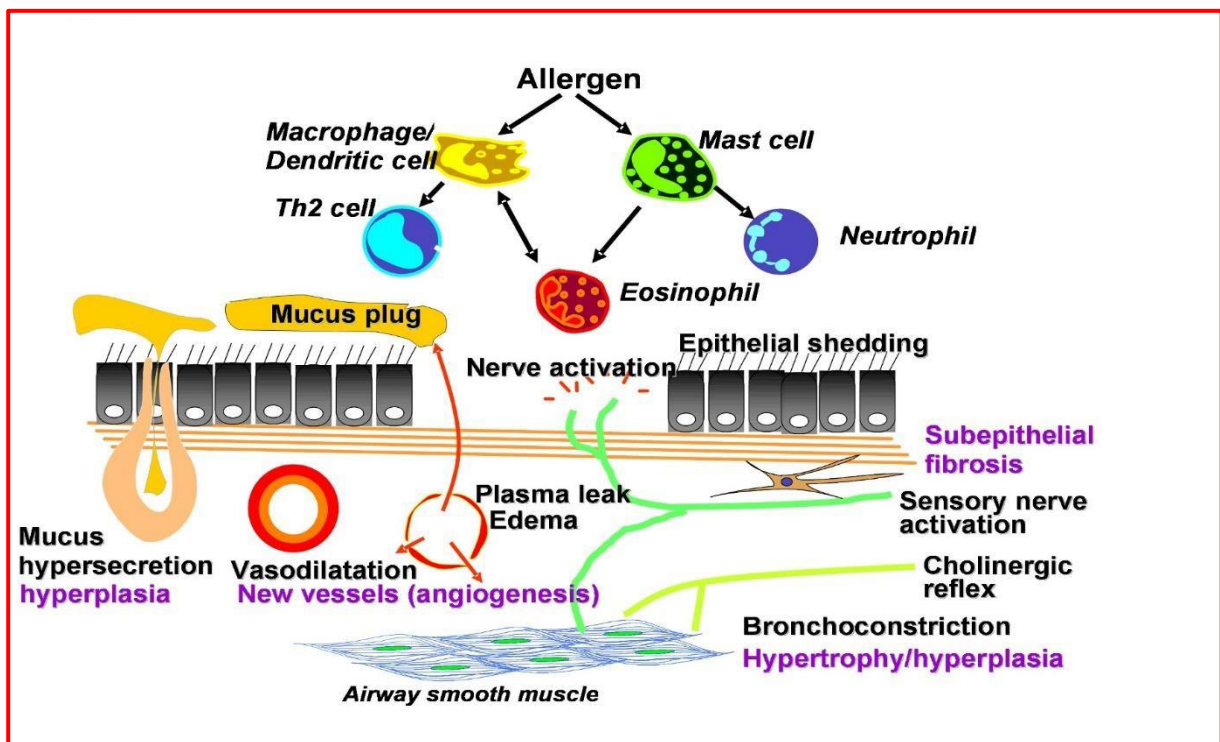


Fig.3

1- Pathogenesis

Asthma is characterized by variable airway obstruction, airway hyperresponsiveness, and airway inflammation. No single histopathologic feature is pathognomonic but common findings include airway inflammatory cell infiltration with eosinophils, neutrophils, and lymphocytes (especially T cells), goblet cell hyperplasia, plugging of small airways with mucus, collagen deposition beneath the basement membrane, bronchial smooth muscle hypertrophy, airway edema, mast cell activation, and denudation of airway epithelium.

Fig.4- Asthma inflammation: Cells and mediators



2- Asthma triggers

- ☐ Environments
- ☐ Occupational allergens
- ☐ Cold or dry air
- ☐ Infections
- ☐ Exercise
- ☐ Inhaled irritants
- ☐ Emotion
- ☐ Aspirin and other nonsteroidal anti-inflammatory drugs (NSAIDs)
- ☐ Gastroesophageal reflux disease (GERD)

V- TYPE OF ASTHMA

1- Extrinsic asthma (allergic asthma or atopic asthma):

- Allergic asthma is triggered by allergens like pollen, dust mites or pets
- About 4 out of 5 people with asthma have allergies
- It may positive for a skin prick test or blood test

2- Intrinsic asthma (non-allergic asthma):

- This form of asthma is not related to an allergy trigger like pollen, dust mites or pets
- About 1 in 5 people with asthma have non-allergic asthma
- It often develops later in life.
- Non-allergic asthma can be triggered by things as following:
 - Air pollution
 - Damp and mould
 - Stress
 - Cigarette smoke.

VI- DIAGNOSIS

1-Signs and Symptoms

-Symptoms of asthma vary overtime and in intensity.

-The following respiratory symptoms of wheeze, dyspnea, cough, and/or chest tightness are characteristic and typical of asthma:

- ☐ Patients experience more than one of these type of symptoms (episodic or chronic)
- ☐ Symptoms frequently worse at night or in the early morning
- ☐ Symptoms vary over time and in intensity
- ☐ Prolonged expiration and diffuse wheezes on physical examination
- ☐ Limitation of airflow on pulmonary function testing
- ☐ Symptoms are triggered by viral infections, exercise, allergen exposure, changes in weather, or irritants such as car exhaust fumes, smoke or strong smells
- ☐ Reversibility of airflow obstruction, either spontaneously or following bronchodilator therapy.

The most common signs of asthma are:

- Coughing, especially at night, during exercise.
- Difficulty breathing.
- Chest tightness.
- Shortness of breath.
- Wheezing (a whistling or squeaky sound in your chest when breathing, especially when exhaling)

2- Spirometry (pulmonary function testing)

The identifying of airflow obstruction of asthma is confirmed by spirometry, forced expiratory volume in 1 second [FEV₁], forced vital capacity [FVC] and FEV₁/FVC before and after the administration of a short-acting bronchodilator. These measurements help determine the presence and extent of airflow obstruction and whether it is immediately reversible.

Airflow obstruction is indicated by a reduced FEV₁/FVC ratio below 0.7. Significant reversibility of airflow obstruction is defined by an increase of 12% or more and 200mL in FEV₁ or FVC after inhaling a short-acting bronchodilator.

3- Imaging

- Chest radiography mostly is normal in un complicated asthma. In patients with more advanced asthma, varying stages of hyperinflation are reflected on chest radiographs by a flattening of the hemidiaphragm, increased retrosternal airspace.
- The value of chest radiography is in revealing complications or alternative causes of wheezing in the diagnosis of asthma and its exacerbations.
- It usually is more useful in the initial eliminated diagnosis of bronchial asthma.

4- Laboratory tests

- White blood cells, and CRP may increase in case of bacterial infection, Polynuclear eosinophil is often elevated.
- Skin testing and allergen-specific IgE to assess sensitivity to environmental allergens can identify atopy in patients with persistent asthma who may benefit from therapy directed at their allergic diathesis.

5- Diagnosis

Making the diagnosis of asthma is based on identifying both a characteristic pattern of respiratory symptoms (such as wheezing, shortness of breath, chest tightness or cough) and variable expiratory airflow limitation.

Tab.1- Diagnostic criteria for asthma

1- History of variable respiratory symptoms	
Features	Symptoms or features that support the diagnosis of asthma
Wheeze, Shortness of breath, chest tightness and cough (Descriptor may vary between cultures and by age.	- More than one type of respiratory symptom (in adults, isolated cough is seldom due to asthma). - Symptoms occur variably over time and vary in intensity. - Symptoms are often worse at night or on waking. - Symptoms are often triggered by exercise, laughter, allergens, cold air. - Symptoms are often appear or worsen with viral infections
2- Confirmed variable expiratory airflow limitation	
Features	Considerations, Definitions, Criteria
- Documented expiratory airflow limitation	At a time when FEV ₁ is reduced, confirm that FEV ₁ /FVC usually > 0.75-0.80 in adult and > 0.90 in children.
AND	
- Documented excessive variability in lung function (one or more of the following):	The greater the variations, or the more occasions excess variation is seen, the more confident the diagnosis. If initially negative, tests can be repeated during symptoms or in the early morning

Bronchodilator responsiveness test (reversibility)	-Adults: increase in FEV1 of > 12% and >200mL (greater confidence if increase is >15% and >400mL). -Children: increase in FEV1 of > 12% predicted - Measure change 10-15 minutes after 200-400 mcg salbutamol or equivalent, compared with pre-BD reading. Positive test more likely if BD withheld before test : SABA $\geq$ 4 hours, twice daily LABA 24 hours, one-daily LABA 36 hours.
Excessive variability in twice daily PEF over 2 weeks.	-Adults: average daily diurnal PEF variability >10% -Children: average daily diurnal PEF variability >13%
Significant increase in lung function after 4 weeks of anti-inflammatory treatment	-Adults: increase in FEV1 by >12% and >200mL (or PEF by >20%) from baseline after 4 weeks of treatment, outside respiratory infections
Positive exercise challenge test	-Adults: fall in FEV1 of >10% and >200mL from baseline -Children: fall in FEV1 of >12% predicted, or PEF >15%
Positive bronchial challenge test (usually only for adults)	Fall in FEV1 from baseline of $\geq$ 12% with standard dose of methacholine, or $\geq$ 15% with standardized hyperventilation, hypertonic saline or mannitol challenge.
Excessive variation in lung function between visits (good specificity but poor sensitivity)	-Adults: Variation in FEV1 of >12% and >200mL between visits outside of respiratory infections. -Children: Variation in FEV1 of >12%, or >15% in PEF between visit (may include respiratory infections)

6- CLASSIFICATION OF SEVERITY

The severity of asthma is classified as following:

Tab. 2- Classification of severity of asthma

Classification	Intermittent	Persistent		
		Mild	Moderate	Severe
Day-time symptoms	< 1/week	$\leq$ 2/month	Daily	continual
Night-time symptoms	$\geq$ 1/week but <1/day	> 2/month	>1/week	Frequent
FEV1*	$\geq$ 80%	$\geq$ 80%	60–80%	$\leq$ 30%
PEF variability	< 20%	20 – 30%	>30%	>30%
SABA use	$\leq$ 2 days/week	> 2 days/week	daily	$\geq$ twice/day

* % predicted

VII- DIFFERENTIAL DIAGNOSIS

- Chronic obstructive pulmonary disease
- Congestive heart failure
- Gastroesophageal Reflux Disease (GERD)
- Vocal cord dysfunction
- Pulmonary Embolism
- Pneumonia

- Lung tumor

VIII- MANAGEMENT OF ASTHMA

1- Based on physiopathology the objective of asthma treatment is double:

- Against bronchoconstriction
- Against bronchial inflammation

2- Goal of asthma management:

Asthma is not curable disease. The goal of asthma management is to achieve the best possible long-term outcomes for the individual patient:

- long-term asthma symptom control:
 - ☐ few/no symptoms
 - ☐ no sleep disturbance due to asthma
 - ☐ unimpaired physical activity.
- long-term asthma risk minimization:
 - ☐ no exacerbations,
 - ☐ improved or stable personal best lung function
 - ☐ no requirement for maintenance oral corticosteroids (OCS)
 - ☐ no medication side-effects.

3- Critical signs:

Critical signs of asthma suggest a severe or life-threatening exacerbation that requires Immediate medical attention:

- Shortness of breath
- Difficulty speaking in full sentences
- Use of accessory muscles
- Cyanosis
- Tachypnea
- Silence breath sound on auscultation
- Tachycardia >120bpm
- Pulsus paradoxus (a significant drop in blood pressure during inhalation [$>10\text{mmHg}$], reflecting increased respiratory effort and intrathoracic pressure changes).
- Severe hypoxemia ($\text{SpO}_2 < 90\%$) despite supplemental oxygen.
- Confusion, drowsiness
- Fatigue

4- Comorbidity:

- Allergic rhinitis and sinusitis which often coexist with asthma due to shared inflammatory pathways.
- Chronic obstructive pulmonary disease (COPD)
- Obesity associated with reduced lung function, increased inflammatory and poorer asthma control.
- Gastroesophageal Reflux Disease (GERD) which can trigger asthma symptoms due to micro-aspiration or vagal nerve stimulation.
- Psychiatric disorders: anxiety and depression are prevalent and can increase the perception of symptoms, reduce adherence to treatment, and worsen outcomes.

5- Risk factors of mortality:

- Severe asthma: persistent symptoms despite high-dose treatment.
- Frequent exacerbations leading to repeated hospitalizations.
- Nonadherence to treatment leads to uncontrolled asthma.
- Socioeconomic factors: poor access to healthcare, medication and education.

6- Pharmacologic treatment:

6-1-Asthma's medicines:

Asthma reliever medicines are used to treat asthma attacks and are considered to be bronchodilators.

Tab. 3-Quick-relief asthma drugs are available in Cambodia's market

Treatment class	Medications	Dosage
Short-acting beta-2 agonists (SABA)	-Salbutamol -Albuterol -Formoterol	200 -600mcg 1.25 - 5mg in 3mL saline every 4-8 hours. 9.2 -18.4mcg
SABA / SAMA (Short-acting muscarinic antagonist) inhalation and nebulization	Formoterol/Ipratropium bromide Combivent	2-3puffs every 6 hours 3mL every 4-6hours
Oral / IV corticosteroids	-Prednisolone -Dexamethasone -Hydrocortisone -Methyl-prednisolone	40 – 60 mg 6 – 9 mg /day 100 – 200 mg/day 40 – 60mg/day

Asthma preventer or controller medicines are used to control asthma and lower the risk of exacerbations.

Tab.4- Asthma controller drugs are available in Cambodia's market

Treatment class	Medications	Dosage
Inhaled corticosteroids (ICS)	-Beclomethasone dipropionate -Budesonide -Fluticasone propionate	200 – 1000 mcg 200 – 800mcg 100 – 500mcg
Inhaled Long-acting beta-2 agonists (LABA) - should never be used as monotherapy.	-Salmeterol -Formoterol	200 -600mcg 9.2 -18.4mcg
ICS / LABA (inhalation)	-Budesonide / Formoterol -Fluticasone / Salmeterol	100mcg – 400mcg 125 – 500mcg
Leukotrienne modifiers	Montelukast	10mg
Methylxanthines should be used with precaution because of side-effects.	Aminophyllin / Theophyllin LP	200 – 300mg

Tab.5- Estimated clinically comparable daily dosages for inhaled corticosteroids for adult with asthma

Medications	Low daily dose (mcg)	Medium daily dose (mcg)	High daily dose (mcg)
Beclometasone dipropionate (pMDI, HFA)	200 – 500	>500 – 1000	>1000
Budesonide dipropionate (DPI or pMDI)	200 – 400	>400 – 800	>800
Fluticasone propionate (DPI)	100 – 250	>250 – 500	>500
Ciclesonide (pMDI, HFA)	80 – 160	>160 – 320	>320

DPI: dry powder inhaler; HFA: hydrofluoroalkane, pMDI: pressurized metered-dose inhaler.

6-2- Stepwise pharmacologic therapy:

According to the 2024 GINA Guidelines, there are two track system depending on the choice of medications:

Track 1 is the preferred management approach with low dose ICS-formoterol taken at any step when the patient is symptomatic. The steps are as follows:

Track 1: preferred Step 1 – 2 Step 3 Step 4 Step 5

Controller and Reliever Using ICS-formoterol as the reliever reduces the risk of exacerbations compared with using a SABA reliever, and is a simpler regimen.	As-needed-only	Low dose	Medium dose	Add-on LAMA
	low dose ICS-formoterol	maintenance ICS-formoterol	maintenance ICS-formoterol	Refer for assessment of phenotype. Consider high dose maintenance ICS-formoterol, ±anti-IgE, anti-IL5//5R, anti-IL4Ra, anti-TSLP.

Track 2 is an alternative approach is tract 1 is not possible, or if a patient's asthma is stable with good adherence and no exacerbations on their current therapy. The steps are as follows:

Track 2:	Step 1	Step 2	Step 3	Step 4	Step 5
Alternative Controller and Reliever Before considering a regimen with SABA reliever, check if the patient is likely to adhere to daily controller treatment.	Take ICS whenever SABA taken	Low dose maintenance ICS	Low dose maintenance ICS-LABA	Medium dose maintenance ICS-LABA	Add-on LAMA Refer for assessment of phenotype. Consider high dose maintenance ICS-formoterol, ±anti-IgE, anti-IL5//5R, anti-IL4Ra, anti-TSLP.
	Reliever: as-needed ICS-SABA, or as-needed SABA				

Other controller options (limited indication, or less evidence for efficacy or safety...)	Low dose ICS whenever SABA taken or daily LTRA, or add HDM SLIT.	Medium dose ICS or add LTRA, or add HDM SLIT.	Add LAMA or add LTRA or switch to high dose ICS only.	Add azithromycin (adults) or add LTRA. As last resort consider adding low dose OCS but consider side-effects.
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6-3- Stepping up and down:

The treatment can be stepped up or down along one track, using the same reliever at each step, or it can be switched between tracks, according to the individual patient's need and preference.

Before stepping up, check for common problems such as incorrect inhaler technique, poor adherence, and environmental exposures, and confirm that the symptoms are due to asthma.

6-4- Assessment of the level of asthma control:

Tab. 6- Assessing the level of asthma control:

Components of asthma control	Classification of asthma control		
	Well controlled	Partly controlled	Not controlled
- Daytime symptoms > 2x/week - Nighttime awakening due to asthma - Interference with normal activity - Reliever medication needed > 2x/week	None of these components within past 4 weeks	1 or 2 of these components within past 4 weeks	≥ 3 components within past 4 weeks

7- Treatment of asthma attack and exacerbation:

The main initial therapy includes repetitive administration of rapid-acting of bronchodilators, early introduction of systemic corticosteroids, and controlled flow oxygen supplementation. The aim is to rapidly relieve airflow obstruction and hypoxemia, address the underlying inflammatory pathophysiology, and prevent relapse, infection control procedures should be controlled.

Inhaled SABA:

Treatment of asthma's exacerbation or asthma attack:

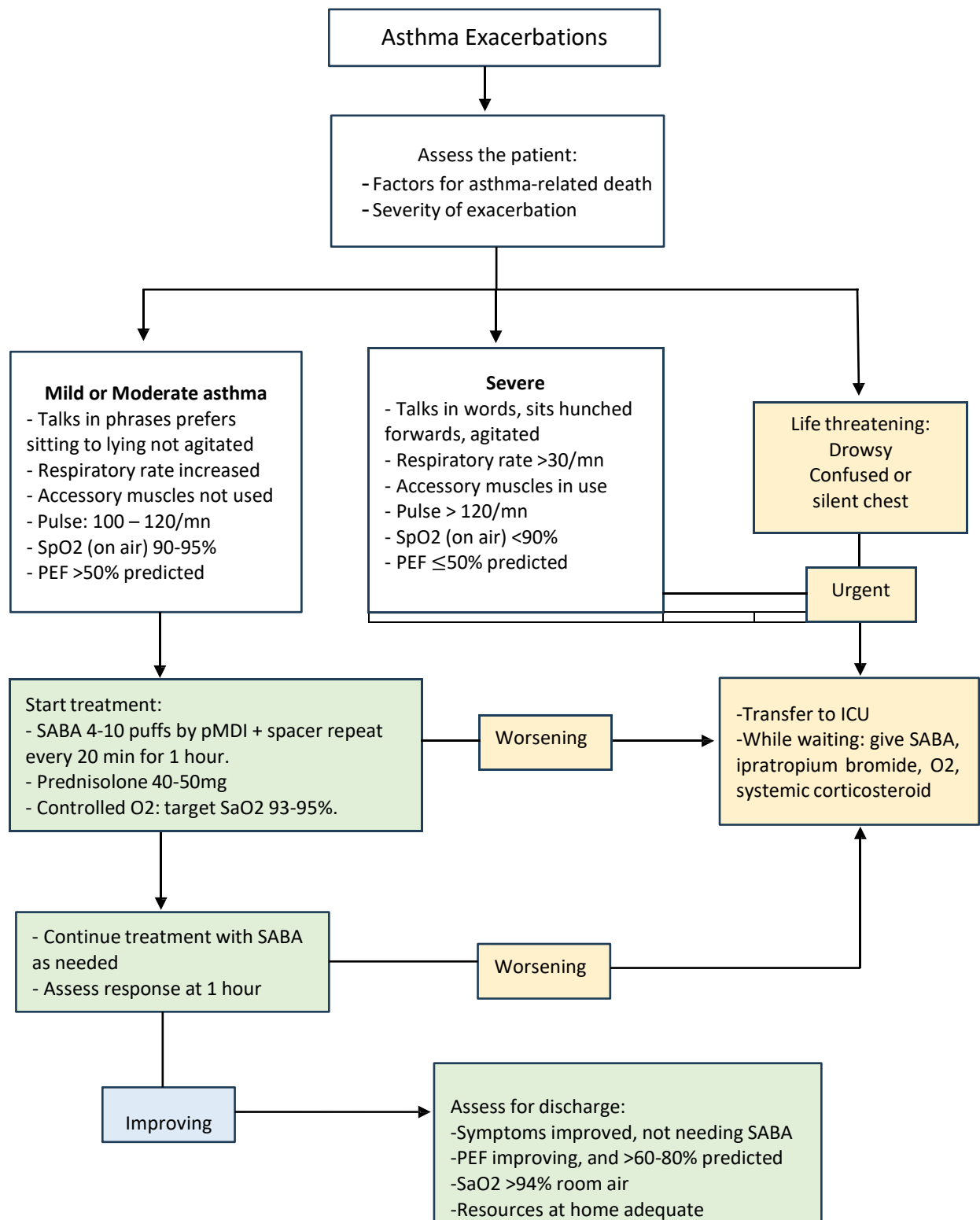
Classification of severity	Mild - moderate	Severe	Life-threatening
Signs and symptoms	-Talk in phrases, -Prefers sitting to lying, not agitated - RR increased -Accessory muscles not used -Pulse 100-120/min -SpO2 90 – 95% (on air) -PEF ≥50% predicted	-Talk in words -Sits hunched forwards, agitated -RR.30/min -Accessory muscles in use -Pulse >120/min -SpO2 <90% (on air) -PEF≤50% predicted	-Drowsy -Confused -Silent chest
Medications	-SABA 4-10 puffs by pMDI+ Spacer, or nebulized SABA 5-10mcg repeat every 20mn for 1 hour. -Prednisolone 40-50mg -Controlled O2: target SpO2 93-95%.	-Transfer to ICU - While waiting, give SABA, SAMA, O2, systemic corticosteroids	-Transfer to ICU - While waiting, give SABA, SAMA, O2, systemic corticosteroids

8-Rehabilitation Interventions for Asthma

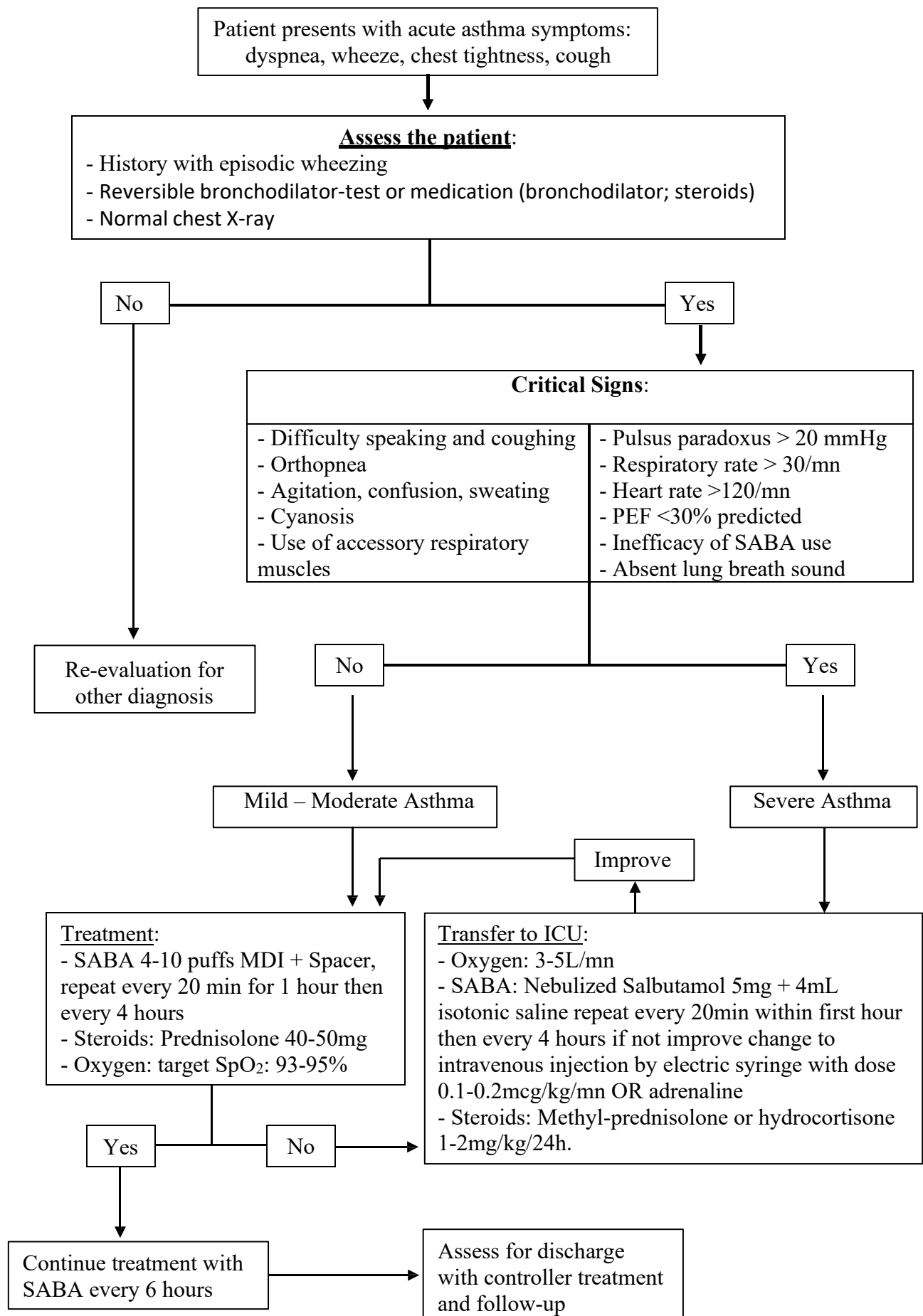
Rehabilitation include Breathing control techniques to normalize breathing patterns by stabilizing respiratory rate and increasing expiratory airflow,

Interventions

- Energy conservation techniques to improve the respiratory functions and avoid fatigue
- Muscle strengthening to address the motor functions and mobility
- Respiratory muscle strengthening exercises
- Provision and training in the use of assistive products for self-care
- Graded fitness training to enhance exercise tolerance
- Modification of the home environment (incl. environmental enrichment)



8- Algorithm:



IX- SPECIAL CASES:

1- Exercise-induced asthma:

- 9 in 10 people with asthma get tightening in their airways caused by exercise.
- Symptoms present during or after exercise: a tight chest, breathlessness, wheezing, coughing

2- Aspirin-exacerbated respiratory disease (AERD):

- Triggered by NSAIDs and characterized by severe airway inflammation, often with nasal polyps.
- Typically progressive without targeted treatment.
- Also called Samter's triad, has three features:
 - Asthma, although only a small number of people with asthma will develop AERD.
 - Nasal polyps that often come back, even after taken out by surgery.
 - Problem with taking aspirin and nonsteroidal anti-inflammatory drugs (NSAIDs)

3- Occupational asthma:

- About 1in 10 people who develop asthma as an adult have occupational asthma
- Caused by workplace exposures (e.g., chemicals, dust).
- Symptoms often improve after exposure cessation, but permanent damage can occur with prolonged exposure.

X- PROGNOSIS

One in four children with asthma, wheezing persists into adulthood or relapse occurs in later years. Female sex, earlier age of onset, smoking and sensitization to household dust mites are risk factors for persistence and relapse.

Although a significant number of deaths each year are attributable to asthma, the prognosis is good with adequate access and adherence to treatment.

Over time, the airways in some asthmatics undergo permanent structural changes (remodeling) and develop to baseline airflow obstruction that is not completely reversible. Early use of anti-inflammatory drugs (corticosteroids) may help prevent this remodeling.

XI- EVOLUTION

- 1- Stable Disease:** Many adults achieve stable asthma with proper management and adherence to therapy.
- 2- Progressive Decline:** Some individuals experience worsening symptoms and lung function over time due to factors like airway remodeling or comorbidities.
- 3- Remission:** A small subset of adults may experience remission, particularly with early intervention and trigger avoidance.
- 4- Severe Asthma:** In cases of poorly controlled inflammation or late diagnosis, asthma may evolve into a severe, debilitating condition requiring advanced therapies like biologics.

XII- COMPLICATIONS

- Bronchiectasis resulting from recurrent infection or persistent inflammation
- Pneumothorax
- COPD: Asthma-COPD overlap increases the severity of airflow obstruction.
- Hypoxia and hypercapnia caused by prolonged airway obstruction
- Arrhythmia triggered by beta-agonist medications
- Psychological and cognitive effect: anxiety and depression, cognitive decline
- Work and social impact

XIII- PREVENTION AND RECOMMENDATION

1- Prevention:

1.1-Avoid Triggers:

- Allergens: Reduce exposure to dust mites, pet dander, pollen, and mold.
- Irritant: Limit exposure to tobacco smoke, air pollution, strong odors.

1.2- Prevent Infections:

- Vaccination for influenza every year and anti-pneumococcal every 5 years.
- Control allergic: allergic rhinitis.

2- Recommendation:

2.1- Create an Asthma Action Plan

2.2- Regular monitoring (follow-up)

2.3- Environmental Control: reduce cluster where dust can accumulate and wash bedding and stuffed toys regularly in hot water.

2.4- Education:

- Learn proper inhaler technique,
- Educate family members or caregivers about asthma management.

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HEMOPTYSIS IN ADULT

Prof. Chan Sarin

I. DEFINITION (TRAUMATIC HEMOPTYSIS EXCLUDED)

The term hemoptysis typically refers to expectoration of blood originating from the lower respiratory tract or the expectoration of blood, can range from blood-streaking of sputum to gross blood in the absence of any accompanying sputum.

II. EPIDEMIOLOGY

5% of hemoptysis care in hospitalizations in pneumology unite and 1% of hemoptysis is very severe.

III. ETIOLOGY

Most episodes of hemoptysis are non-life-threatening. The most common causes of non-life-threatening hemoptysis in developed countries are acute bronchitis, bronchiectasis, and bronchial neoplasms. In contrast, in developing countries, infections due to *M. tuberculosis* and *Paragonimus westermani* and non-ctstic fibrosis-related bronchiectasis are more common causes.

Airway diseases	Trauma and bleeding disorder
Bronchitis: acute or chronic	-Bleeding disorder (eg, thrombocytopenia, DIVC,)
Bronchiectasis (eg, cystic fibrosis-related)	-Airway or parenchymal trauma
Bronchial carcinoma	-Iatrogenic injury
Bronchial carcinoid	
Metastatic cancer to bronchus or trachea	
Bronchovascular fistula	
Dieulafoy disease	
Foreign body in airway	
Pulmonary parenchymal disease	
Infection: -Necrotizing pneumonia and lung abscess -Bacterial infection -Tuberculous and NTM -Mycetoma and other fungal infections -Parasitic (eg, <i>Paragonimus</i> , <i>Strongyloides</i>) -Viral (eg, Herpes simplex, dengue virus)	Miscellaneous -Drugs and toxins (cocaine) -Nitrogen dioxide exposure in indoor ice arenas - E-cigarette or vaping product use associated lung injury
Rheumatic disease -Goodpasture disease -Granulomatosis with polyangiitis -Behçet disease -Primary antiphospholipid antibody	Idiopathic pulmonary hemosiderosis Amyloid Fibrosing mediastinitis
Catamenial hemoptysis	Pseudohemoptysis du to aspirated blood from upper airway or gastrointestinal source

Pulmonary vascular disorders -Elevation pulmonary capillary pressure -Pulmonary arteriovenous malformation -Pulmonary or bronchial artery aneurysms -Pulmonary artery pseudoaneurysms -Pulmonary embolism (thrombotic,fat,septic)	
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IV. Physiopathology – There are several sources of bleeding within the lung and endobronchial tree that can be responsible for hemoptysis. The circulation source of bleeding of life-threatening and non-life-threatening hemoptysis:

- 90% cases of hemoptysis sourced by the bronchial artery is the cause of life-threatening hemoptysis (High pressure) from thoracic aorta.
- 5% cases of hemoptysis sourced by the pulmonary vena (low pressure) but flow High (The right heart output same the left heart!)
- 5% cases of hemoptysis sourced by these alveolar capillary (Diffuse alveolar hemorrhage pulmonary capillaritis or bland hemorrhage leads to damage to basement mem- branes and leakage of erythrocytes into the alveolar space).

V. DIAGNOSIS

1. Clinical features

Initial evaluation: Once it has been determined that the patient is not experiencing respiratory compromise or immediately life-threatening hemoptysis (eg, no evidence of tachypnea, tachycardia, use of accessory muscles, or cyanosis, the goals of the initial evaluation are to determine the frequency and severity of bleeding, localize the source of bleeding.

History - The directed history for patients presenting with hemoptysis should include an assessment of the pattern and severity of hemoptysis, degree of respiratory impair, and clues to the etiology. As part of the history, it is helpful to characterize the hemoptysis, associated symptoms, cigarette smoking history, comorbidities, and family history.

2-Physical examination

The directed physical examination for the initial evaluation of hemoptysis combines assessment of the degree of respiratory compromise (if any) with examination for clues of extrapulmonary disease. The evaluation and management of life-threatening hemoptysis, which requires a prompt response to ensure adequate airway protection, maintenance of oxygenation and ventilation, and control of hemoptysis, is discussed separately.

Pseudo hemoptysis - Blood from the upper respiratory tract and the upper gastrointestinal tract can also be expectorated and, thus, mimic blood coming from the lower respiratory tract. This is called pseudo hemoptysis.

3-Investigations

Laboratory studies: majority of patients include hemoglobin and hematocrit (to assess the chronicity and magnitude of bleeding), white blood cell count and differential (evidence for infection).

Imaging - Chest radiograph, as it is usually readily available and can help direct care. Many patients will also need chest computed tomography (CT) to fully evaluate the source of hemoptysis. The CT scan should typically be done with contrast, unless there is

a strong suspicion for bronchiectasis or the patient has contrast media sensitivity or renal insufficiency.

DIRECTED EVALUATION BASED ON PRESENTATION

Patients with a normal chest radiograph — A normal chest radiograph must be evaluated in the context of the clinical presentation of hemoptysis and does not exclude the possibility of lung malignancy or bronchiectasis.

Minimal hemoptysis with likely infectious cause — For patients who present with self-limited expectoration of less than 30 ml of blood, a known or likely benign (eg, bronchitis or exacerbation of known bronchiectasis), absent risk factors for lung malignancy, and a normal chest radiograph, we typically observe (if viral infection suspected) or treat bacterial infection (if present), deferring future evaluation pending a recurrence or increase in the amount of hemoptysis.

Active hemoptysis without a clear or benign cause - For patients who present to the emergency department or outpatient office with hemoptysis that is active but not life-threatening (eg, 30 mL to 100 mL in 24 hours), without a clear or likely benign cause (eg, exacerbation of known bronchiectasis), and with a normal or nonvocalizing chest radiograph, we typically obtain a CT scan with contrast. Unless CT provides a clear diagnosis, this is often followed by flexible bronchoscopy, which is an important component of the evaluation, particularly in patients with risk factors for malignancy.

Pulmonary embolism - Rare cause of hemoptysis; when suspected, CT pulmonary angiography would be preferred.

Recurrent hemoptysis or risk factors for malignancy - For patients with recurrent hemoptysis, we typically recommend a CT scan of the chest, even if they have a normal chest radiograph. The main causes of recurrent hemoptysis in a patient with a normal chest radiograph are bronchiectasis, carcinoid tumors, catamenial hemoptysis (from endobronchial endometriosis in women), pulmonary arteriovenous malformation, pseudo hemoptysis due to extrapulmonary sources (eg, nasal or gastrointestinal), and foreign bodies.

Bronchoscopy is indicated in those in whom a CT is unrevealing or in those in whom the CT reveals an abnormality that requires a tissue diagnosis (eg, suspected malignancy). The potential advantages of bronchoscopy in this setting include identifying small endobronchial tumors or lesions (eg, Dieulafoy anomaly, bronchitis, foreign body) that may have been missed on CT scan. In addition, bronchoscopy enables samples to be obtained for microbiologic, cytologic, and histopathologic analysis and determination of whether bronchoscopic interventions (eg, laser, argon plasma coagulation, cautery) would be appropriate.

Chest radiograph suggestive of bronchogenic cancer - In addition to clinical features suggestive of malignancy, certain chest radiographic features are also suggestive of bronchogenic cancer.

These features include a pulmonary nodule that is new, enlarging, or larger than 8 mm. Nodules that are 8 to 20 mm are associated with an 18 percent risk of malignancy: those that are >20 mm have a 50 percent risk of malignancy.

Chest imaging with multiple nodules or cavitary opacities - Nodules and areas of consolidation with cavitation need evaluation for infection (eg, necrotizing pneumonia, mycobacterial, fungal, parasitic), inflammatory processes (eg. granulomatosis with polyangiitis), and malignancy (eg, lymphoma, metastatic cancer).

Chest Imaging showing diffuse opacities - The differential diagnosis of diffuse opacities in a patient with hemoptysis is broad and includes infectious, inflammatory, vascular, and malignant processes, as well as drug or toxin exposure.

VI. DIFFERENTIAL CAUSES OF HEMOPTYSIS

- Acute respiratory tract infections: Such as bronchitis and pneumonia
- Chronic obstructive pulmonary disease (COPD): Such as cystic fibrosis
- Bronchiectasis: A condition that can cause massive hemoptysis
- Lung cancer: A cause of massive hemoptysis
- Tuberculosis: The most common cause of hemoptysis worldwide.

VII. MANAGEMENT

1. Objectives

- To eliminate the clinical signs of hemoptysis.
- To cure the specific causes.
- To prevent the complications.

2. Critical Signs

- Massive Hemoptysis: Requires urgent intervention to prevent airway compromise or exsanguination.
- Severe Respiratory Distress: Oxygenation and ventilation must be stabilized immediately.
- Signs of Shock: Volume resuscitation and hemodynamic support are critical.

3. Mortality risk factor

- Advanced stage of malignancy,
- Mechanical ventilation (particularly the need for single lung ventilation),
- Hypotension, presence of cavitation or necrosis or lung densities involving two or more quadrants on chest radiograph, bleeding from the pulmonary artery, cancer, aspergillosis, alcoholism, and aspiration of blood into the contralateral lung.

4. Out patient Management

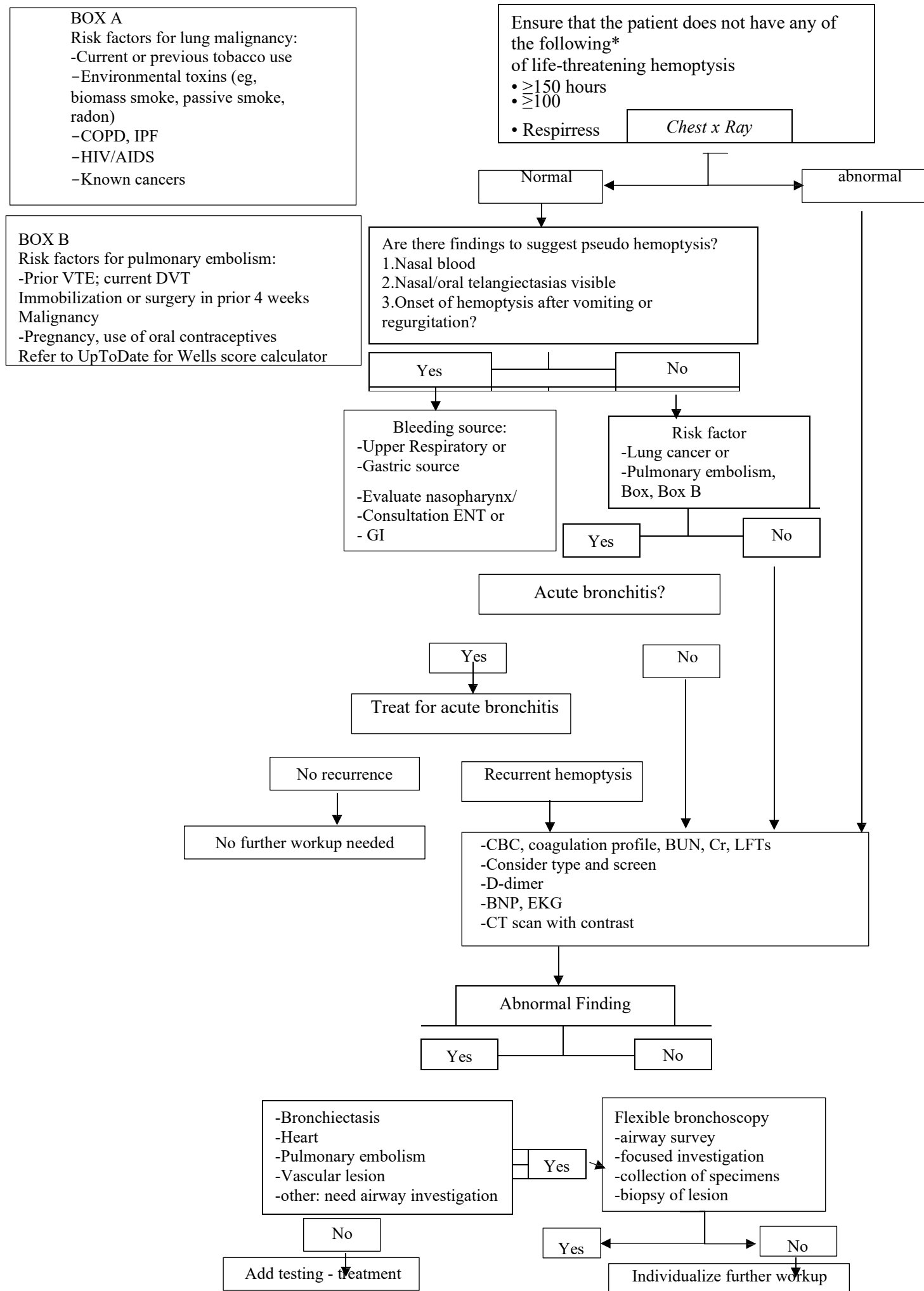
- Need to be hospitalized

5. First-line treatment

- **Initial bronchoscopy**
Blood and thrombus removal
Localization and identifying the source of bleeding
Treating the bleeding
Iced saline lavage
Topical medications
Bronchial blockade devices and balloon tamponade
Ablative therapies
- **Chest CT**
- **Triage for early surgery**

6. Treatment adjustment

- **Supportive care** (Position the patient, Ensure hemodynamic stability, Correct bleeding diathesis.....).



VII. SPECIAL CASES

Special populations - Several rare causes of life-threatening hemoptysis necessitate specific treatment.

- Patients with vascular fistulas or aneurysms should be evaluated for urgent surgical therapy.
- The threshold for surgery should be higher for those with life-threatening hemoptysis due to infection (eg, abscess) to avoid complications such as bronchopleural fistula and empyema.
- Patients with diffuse alveolar hemorrhage (DAH) are not treatable with arteriography or local therapies and continued aggressive therapy targeted the underlying cause is necessary.
- The approach to hemoptysis in the setting of life-threatening central airway obstruction (CAO) should focus on airway management and debulking of obstructing lesions, which can typically be achieved with local ablative bronchoscopic therapies or external beam radiation when CAO is due to lung cancer.

VIII. EVOLUTION AND PROGNOSIS

Outcomes: Several predictors of poor prognosis include the following:

- Poor performance status
- Advanced stage of malignancy
- Mechanical ventilation (particularly the need for single lung ventilation)
- Hypotension
- Presence of cavitation or necrosis or lung densities involving two or more quadrants on chest radiograph
- Bleeding from the pulmonary artery, cancer, aspergillosis
- Alcoholism
- Aspiration of blood into the contralateral lung

Several retrospective cohort studies have compared outcomes in patients with life-threatening hemoptysis who have non-surgical interventions with those who have undergone surgery [18,81]. Data suggest that the mortality associated with non-surgical therapies and surgery are similar among patients who qualified as surgical candidates.

IX. Complications

- Airway Obstruction: Large volumes of blood can block the airways, leading to difficulty breathing or asphyxiation.
- Aspiration: Blood may enter the lungs, causing aspiration pneumonitis or secondary infection.
- Hypoxemia: Impaired oxygen exchange due to blood in the alveoli can lead to low oxygen levels in the blood.
- Death: In severe cases, massive hemoptysis may result in fatal outcomes due to suffocation or exsanguination.

X. PREVENTION AND RECOMMENDATION

- Lifestyle Modifications
 - Avoid Smoking (Smoking damages lung tissue, increases the risk of lung cancer, and exacerbates chronic respiratory conditions).
 - Minimize Exposure to Toxins (Avoid inhaling harmful substances such as industrial fumes, pollutants, and particulate matter).

- Treat Underlying Conditions
 - Infections (Complete treatment for tuberculosis, pneumonia, bronchitis, or fungal infections to prevent recurrence).
 - Chronic Respiratory Diseases Control bronchiectasis, chronic obstructive pulmonary disease (COPD), and asthma with appropriate medications).
 - Cancer (Regular screenings and early treatment for lung or bronchial cancer in high-risk populations).

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CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

Prof. Vann Mich

I- DEFINITION OF COPD

Chronic obstructive pulmonary disease (COPD) is a heterogeneous lung condition and chronic inflammation of airway characterized by chronic respiratory symptoms such as dyspnea, cough, sputum production and/or exacerbations, due to abnormalities of the airways and/or alveoli that cause persistent, often progressive, airflow limitation.

COPD is not curable but symptoms can improve if one avoids smoking and exposure to air pollution and gets vaccines to prevent infection.

II- EPDEMOIOLOGY

COPD is the third leading cause of death worldwide, causing 3.23 million deaths in 2019.

COPD affects about 300 million people worldwide, resulting in approximately 64 million disability-adjusted life years.

Household air pollution affects almost 3 billion people worldwide and is a major risk factor for COPD.

An estimated 25% to 45% of patients with COPD worldwide have never smoked. Fourteen percent (14%) of the overall COPD burden is attributable to occupational exposures.

COPD is increasing worldwide because of increase in smoking and reproduction in mortality due to infectious disease.

It predominantly occurs in individuals aged 40 and above, with risk factors including smoking, occupational exposures, and air pollution.

III- ETIOLOGY

There are two main causes of COPD:

- Smoking and less often other inhalation exposures
- Genetic factors.

Smoking:

Cigarette smoking is the primary cause or risk factor of COPD. Overall, cigarette smoking accounts for as much as 90% of COPD risk. Although only about 15% of smokers develop clinically apparent COPD; an exposure history of 40 or more pack-years is especially predictive.

Genetic factors:

Alpha-1 antitrypsin deficiency is an important cause of emphysema in nonsmokers and markedly increases susceptibility to disease in smokers.

More than 30 genetic alleles have been found to be associated with COPD but none has been shown to be as consequential as alpha-1 antitrypsin.

Airway hyperresponsiveness (AHR)

AHR stipulates that patients who have nonspecific airway hyperreactivity and who smoke are at increased risk of developing COPD with an accelerated decline in lung function.

Nonspecific airway hyperreactivity is inversely related to FEV1 and may predict a decline in lung function.

IV- PATHOPHYSIOLOGY

Chronic exposure to irritants triggers inflammation, leading to airway narrowing, alveolar destruction (emphysema), and mucus hypersecretions (chronic bronchitis)

Key mechanisms include:

- Airflow obstruction
- Gas trapping, and
- Impaired gas exchange.

Inflammation is present in the lungs, particularly the small airways, of all people who smoke. This normal protective response to the inhaled toxins is amplified in COPD, leading to tissue destruction, impairment of the defence mechanisms that limit such destruction, and disruption of the repair mechanisms. In general, the inflammatory and structural changes in the airways increase with disease severity and persist even after smoking cessation. Besides inflammation, two other processes are involved in the pathogenesis of COPD - an imbalance between proteases and anti-proteases and an imbalance between oxidants and antioxidants

(oxidative stress) in the lungs.

Pathogenic mechanisms result in the pathological changes found in COPD:

1-Inflammation:

- Inflammatory cells increase numbers of neutrophils, macrophages, and T lymphocytes (CD8 more than CD4) in the lungs. These inflammatory cells release a variety of cytokines and mediators that participate in the disease process.

- Inflammatory mediators are increased in COPD, including:

- Leucotriene B₄, a neutrophil and T cell chemoattractant which is produced by macrophages, neutrophils, and epithelial cells.
- Chemotactic factors such as the CXC chemokines interleukin-8 and growth-related oncogene alpha, which are produced by macrophages and epithelial cells.
- Pro-inflammatory cytokines such as tumor necrosis factor α and interleukins-1 β and interleukins-6.
- Growth factors such as transforming growth factor beta, which may cause fibrosis in the airways either directly or through release of another cytokine, connective tissue growth factor.

2- Mucous hypersecretion and ciliary dysfunction:

- The hypersecretion is due to squamous metaplasia, increased numbers of goblet cells, and increased size of bronchial submucosal glands in response to chronic irritation by noxious particles and gases.
- Ciliary dysfunction is due to squamous metaplasia of epithelial cells and results in an abnormal mucociliary escalator and difficulty in expectorating.

3- Protease and anti-protease imbalance:

- Increased production of proteases and inactivation of anti-proteases results in imbalance.
- Cigarette smoke, and inflammation itself, produce oxidative stress, which primes several inflammatory cells to release a combination of proteases and inactivates several anti-proteases by oxidation.
- The main proteases involved are those produced by neutrophils (including the serine proteases elastase, cathepsin G, and protease 3) and macrophages (cysteine proteases and cathepsins E, A, L, and S), and various matrix metalloproteases (MMP-8, MMP-9, and MMP-12). The main anti-proteases involved in the pathogenesis of emphysema include alpha₁ antitrypsin, secretory leucoprotease inhibitor, and tissue inhibitors of metalloproteases.

4- Oxidative stress: The oxidative burden is increased in COPD.

- Sources of oxidants include cigarette smoke and reactive oxygen and nitrogen species released from inflammatory cells. This creates an imbalance in oxidants and antioxidants of oxidative stress.
- Many markers of oxidative stress are increased in stable COPD and are further increased in exacerbations.
- Oxidative stress can lead to inactivation of anti-proteases or stimulation of mucous production. It can also amplify inflammation by enhancing transcription factor activation (such as nuclear factor κ B) and hence gene expression of pro-inflammatory mediators.

Figure 1: The pathogenesis of COPD; dashed bars represent inhibitory effects

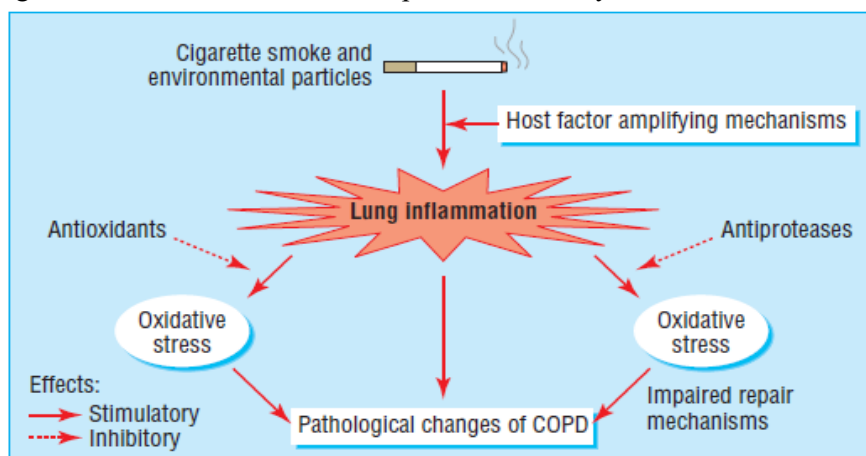
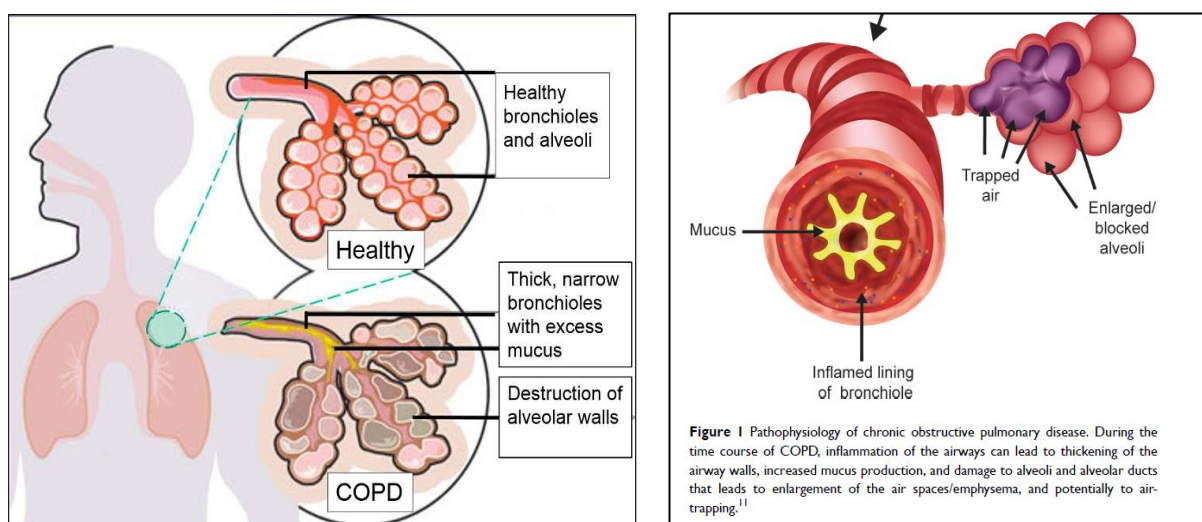


Figure 2: Pathophysiological mechanism and structural changes in COPD.



V- DIAGNOSIS

1- Clinical presentation:

- Patient with COPD typically complaint of dyspnea, activity limitation, and/or cough with or without sputum production and may experience acute respiratory events characterized by exacerbations that require specific preventive and therapeutic measures.
- Patient with COPD frequently harbor other comorbid diseases that influence their clinical condition and prognosis and require specific treatment as well. These comorbid conditions can mimic and/or aggravate an acute exacerbation.

2- Common signs and symptoms:

- Chronic cough with produce mucus
- Excess mucus production: chronic production of sputum, often worse in the morning.
- Shortness of breath particularly during physical activities or exertion.
- wheezing: A high-pitched whistling sound during breathing.
- Chest tightness: A sensation of pressure or discomfort in the chest.
- Fatigue due to decreased O₂ levels and increased effort in breathing.
- Frequent respiratory infections: Such as cold or flu, exacerbating symptoms.
- Cyanosis: A bluish tint to the lips or fingernails due to low O₂ levels (in advanced stages).
- Unintended weight loss: more common in severe cases.
- Swelling in ankles, feet, or legs indication potential right-sided heart failure (cor pulmonale)

3- Diagnostic Consideration:

- ABC symptoms of COPD stand for:
 - **A:** Age ≥ 40 years and expose to cigarette smoke (20 pack-year) or occupational dust, vapor, gases, and other chemicals.
 - **B:** Shortness of breath: dyspnea is progressive over time, worse with exercise, and persistent
 - **C:** Chronic cough may be intermittent and may be non-productive, often worse in the morning
- Diagnostic Test Confirmed:
 - Spirometry should be perform by focusing on Forced Vital Capacity (FVC) and Forced Expiratory Volume in One second (FEV₁).
 - Presence of post-bronchodilator FEV₁/FVC < 0.7 is mandatory to establish the diagnosis of COPD.
- Other investigations :

- Radiography may be presented as hyperinflated lungs, the diaphragm may look lower and flatter than usual, also may show image of emphysema bulla or fibrosis lesion or post-tuberculosis scars.
- Blood tests : complete blood count (WBC, Eosinophile, RBC, Platelete), and biomarkers (CRP, VS, PCT, Fibrinogene). These test are not used to diagnose COPD, but they may be used to find the cause of symptoms, to alert there will be an exacerbation of COPD (increased eosinophile) or rule out other conditions.
- Arterial blood gase to find out hypercapnea or hypoxemia

4- Classification of Severity:

Once the diagnosis of COPD has been confirmed by spirometry, The severity assessment of COPD must be done by focusing on the severity of airflow obstruction, nature and magnitude of current symptoms, the previous history of moderate and severe exacerbations.

Table 1: GOLD Grades and Severity of Airflow Obstruction in COPD

(based on bronchodilator FEV1):

In COPD Patients (FEV1/FVC < 0.7)		
GOLD 1	Mild	FEV1 $\geq$ 80% predicted
GOLD 2	Moderate	50% $\leq$ FEV1 < 80% predicted
GOLD 3	Severe	30% $\leq$ FEV1 < 50% predicted
GOLD 4	Very Severe	FEV1 < 30% predicted

Table 2: Modified Medical Research Council of Dyspnea Scale (mMRC):

Description	Grade
I only get breathless with strenuous exercise.	0
I get short of breath when hurrying on level ground or walking up a slight hill.	1
On level ground, I walk slower than people of my age because of breathlessness, or I have to stop for breath when walking at my own pace on the level.	2
I stop for breath after walking about 100 yards or after a few minutes on level ground.	3
I am too breathless to leave the house or I am breathless when dressing/undressing.	4

Interpretation of mMRC:

mMRC $\geq$ 2 is used as a threshold to distinguish more symptomatic patients.

COPD patients with mMRC $\geq$ 2 had a higher risk of having a moderate or severe COPD exacerbation.

Table 3: The COPD Assessment Test (CAT):

Symptom	Score						Symptom
I never cough.	0	1	2	3	4	5	I cough all the time.
I have no phlegm (mucus) in my chest at all.	0	1	2	3	4	5	My chest is completely full of phlegm (mucus).
My chest does not feel tight at all.	0	1	2	3	4	5	My chest feels very tight.
When I walk up a hill or one flight of stairs, I am not breathless.	0	1	2	3	4	5	When I walk up a hill or one flight of stairs, I am very breathless.

I am not limited doing any activities at home.	0	1	2	3	4	5	I am very limited doing activities at home.
I am confident leaving my home despite my lung condition.	0	1	2	3	4	5	I am not at all confident leaving my home because of my lung condition.
I sleep soundly.	0	1	2	3	4	5	I don't sleep soundly because of my lung condition.
I have lots of energy.	0	1	2	3	4	5	I have no energy at all.
Total Score :							

Table 4: Severity according to CAT's score

Score	Impact	Meaning
0–9	Low	You may not experience many COPD symptoms, or at least not severe enough to affect your daily activities. Most days are good, but you cough regularly and get tired easily.
10–20	Medium	COPD symptoms affect your life regularly. You have some good days, but you get breathless easily and cough up phlegm regularly. You have 1 or 2 exacerbations each year.
21–30	High	Your symptoms regularly prevent you from doing things you want to do. Regular day-to-day activities like getting dressed are tiring. You don't feel in control of your chest problem.
31–40	Very high	You never have any good days. It takes you a long time to complete even the simplest tasks. You feel like you can't even leave the house.

Generally, the GOLD guidelines suggest using a [1](#) or above to indicate symptomatic COPD.

Figure 3 : Diagnosis, assessment of the severity of airflow obstruction and proposed ABE clinical assessment tool for COPD patients.

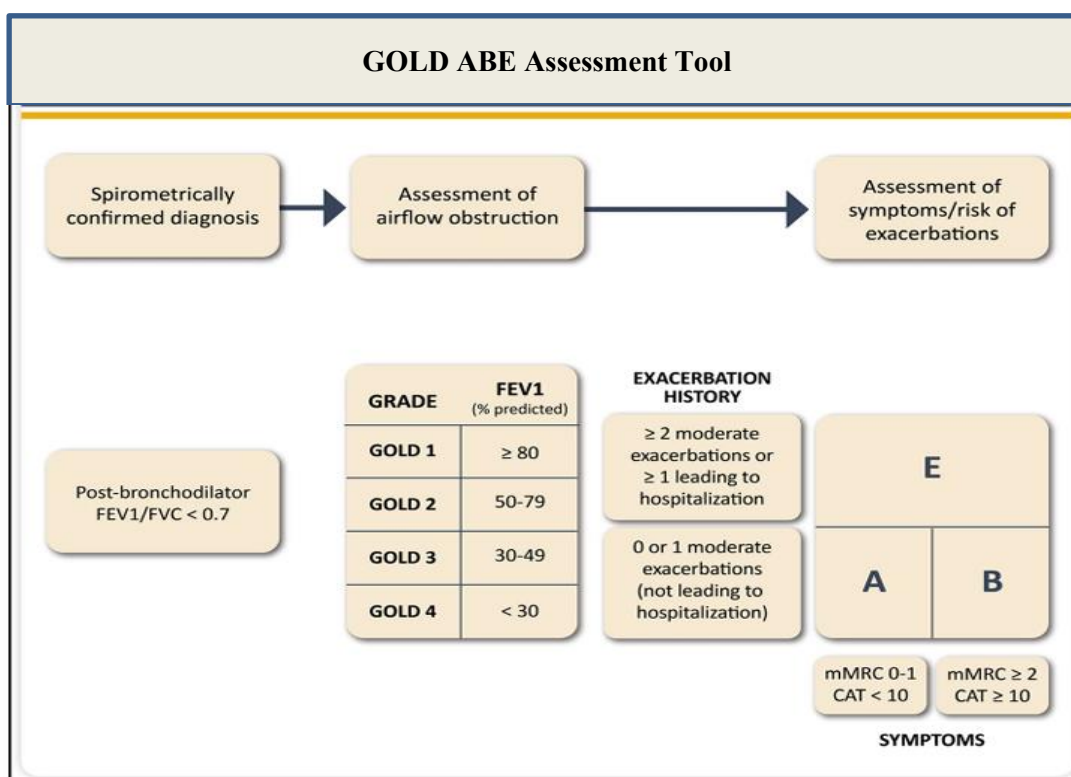


Table 5: Classification of COPD Severity Based on Spirometric Impairment and Symptom presentations:

Classification of Severity	Airflow limitation (Post-Bronchodilator Spirometric Values)		Symptoms presentations
Stage 1: Mild	FEV1/FVC<0.70	FEV1 > 80% predicted	Shortness of breath when hurrying on the level <i>or</i> walking up a slight hill (mMRC 1)
Stage 2: Moderate		50% < FEV1 < 80% predicted	Walks slower than people of the same age on the level because of breathlessness; <i>or</i> stops for breath after walking about 100 m or after a few minutes at own pace on the level (mMRC 2 to 3)
Stage 3: Severe		30% < FEV1 < 50% predicted	Too breathless to leave the house <i>or</i> breathless when dressing or undressing (mMRC 4)
stage 4: Very Severe		<i>i</i> -FEV1 < 30% predicted <i>or</i> <i>ii</i> -FEV1 < 50% predicted plus chronic respiratory failure.	Presence of chronic respiratory failure <i>or</i> clinical signs of right heart failure.

VI- DIFFERENTIAL DIAGNOSIS

The differential diagnosis of COPD involves distinguishing it from other conditions that cause chronic respiratory symptoms such as dyspnea, cough, sputum production, and airflow limitation.

Table 6: Differential diagnosis of COPD

Diagnosis	Suggestive features
Asthma	- Onset early in life (often childhood). - Symptoms vary widely from day to day. - Symptoms worse at night/early morning. - Allergy, rhinitis, and/or eczema also present. - Family history of asthma. - Obesity coexistence.
Bronchiectasis	- Large volumes of purulent sputum. - Commonly associated with bacterial infection. - Chest X-ray/CT shows bronchial dilation, bronchial thickening.
Congestive heart failure	- Chest X-ray shows dilated heart, pulmonary edema. - Pulmonary function tests indicate volume restriction, not airflow limitation.
Interstitial Lung disease	- Shortness of breath, especially with activity. - Dry cough. - Extreme tiredness and weakness. - Loss of appetite. - Unexplained weight loss. - Discomfort in the chest.
Tuberculosis	- Onset all ages - TB-related symptoms (coughing for longer than 3 weeks, loss of appetite, weight loss, fever, night sweats weakness, and hemoptysis. - Chest X-ray shows lung infiltrate

VII- MANAGEMENT

1- Objectives

The goal of treatment for COPD are :

- **To reduce symptoms:**
 - Relieve symptoms

- Improve exercise tolerance
- Improve health status
- **To reduce risk :**
 - Prevent disease progression
 - Prevent and treat exacerbations and complications
 - Reduce mortality.

2- Critical Sign

The critical signs of COPD indicate advanced disease or an acute exacerbation, requiring prompt medical attention. These signs include:

- Severe dyspnea: difficulty breathing, especially during mild activity or even at rest.
- Cyanosis: bluish discoloration of lips, tongue, fingernails, indicating low O₂ level.
- Use of accessory muscles: visible use of neck, chest, and abdominal muscles to breathe.
- Pursed-Lip breathing: a technique patients use instinctively to improve airflow during expiration.
- Barrel chest: hyperinflation of the lungs leading to an expanded, rounded chest.
- Severe fatigue and weakness: often due to poor O₂ delivery to tissues.

3- Mortality risk factors

The mortality risk in COPD is influenced by several factors related to disease severity, comorbid conditions, and patient characteristics:

i- Disease severity:

- FEV₁ decline: A lower FEV₁ is strongly associated with increased mortality risk.
- Chronic hypoxemia: persistent low O₂ levels (SpO₂ ≤ 88%) indicate a poor prognosis.

ii- Frequent exacerbations:

- Frequent or severe exacerbations, requiring hospitalization, significantly increased the risk of death.

iii- Comorbid conditions :

- Cardiovascular disease
- Lung cancer
- Diabetes, metabolic syndrome
- Osteoporosis
- Depression and anxiety
- Smoking
- Nutritional status : COPD cachexia)
- Elderly patient (≥ 65 years)

4- Management of COPD

Treatments are determined by (1) current symptoms, (2) degree of airflow obstruction, (3) history of moderate and severe exacerbations, and (4) comorbidities.

4.1- Pharmacological therapy:

Pharmacological therapy for COPD is used to reduce symptoms, reduce the frequency and severity of exacerbations, and improve exercise tolerance and health status.

- Bronchodilators:
 - Short-acting beta2-agonist (SABA) for acute relief :
Salbutamol (albuterol): 100 µg /puff, 2 puffs every 4 – 6 hours as needed, OR nebulized 2.5 – 5 mg every 4 – 6 hours.
 - Short-acting muscarinic antagonist (SAMA) for acute relief : Ipratropium bromide 20 µg/puff, 2 puffs every 6 – 8 hours, OR nebulized 0.5 mg every 6 hours as part of maintenancetherapy. During acute exacerbation, the dose can be increased to 0.5 mg every 20 mn for up to 3 doses.
 - Combined SABA+SAMA : nebulized 1.5 mL – 2.5 mL of ipratropium+Formoterol.
In severe cases, the dose can up to 4 mL.
 - Long-acting (LABA and LAMA) for regular use as maintenace therapy.
LABA: Salmeterol 50 µg twice daily ; Formterol : 12 µg twice daily.
LAMA : Tiotropium 18 mcg once daily (DPI) OR 2.5 µg /actuation, 2 inhalations once daily (Resmat).
 - Combination inhalers: are common for enhanced efficacy.
LABA + ICS (inhaled corticosteroids): indication in patients with frequent exacerbations or asthma-COPD overlap.
.Budesonide/Formoterol : 160/4.5 µg, 2 inhalations twice daily.
.Fluticasone/Salmeterol : 250/50 µg, 1 inhalation twice daily.
 - Triple therapy (LABA+LAMA+ICS): Fluticasone/Umeclidinium/Vilaterol (Trelegy):
100/62.5/25 mcg 1 inhalation once daily.
 - Methylxanthine: Theophylline, sustained-release, 200 – 400 µg once or twice daily. Rarely used due to side effect.
- Anti-inflammatory :
 - Inhaled Corticosteroids (ICS): Added for patients with frequent exacerbations or eosinophilia.
 - Systemic Corticosteroids : for short course treatment of acute exacerbations, 30 – 60 mg per day of prednisone or methylprednisolone 60 – 250 mg per day for 5 days or tapered over 5 to 10 days, depending on the clinical response.
- Roflumilast: A phosphodiesterase-4 inhbitors (PDE4 inhibitor) for severe COPD, particularly in chronic bronchitis phenotypes, 500 mcg once daily.

Table 7: Bronchodilators using in treating COPD

Bronchodilators	Mechanism of action	Duration of action	Adverse Effects
Beta2-agonists *SABA : - Salbutamol (Albuterol) - Terbutaline *LABA	-To relax airway smoth muscle by stimulating beta ₂ -adrenergic receptors. - To produce functional antagonism to bronchoconstriction.	4-6 hours	-Sinus tachycardia - Cardiac rhythm disturbances - Exaggerated somatic tremor - Hypokalemia

5-First-line treatment

Initial Pharmacological Management

A proposal for the INITIATION of pharmacological management of COPD according to the individualized assessment of symptoms and exacerbation risk following the ABE scheme, and also accounting for blood eosinophil count :

Group A:

- All group A patients should be offered bronchodilator treatment based on its effect on breathlessness. This can be either a short- or a long-acting bronchodilator. If available and affordable a long-acting bronchodilator is the preferred choice except in patients with very occasional breathlessness.

Group B:

- Treatment should be initiated with a LABA+LAMA combination. LABA+LAMA is superior to a LAMA.

- LABA+LAMA is the recommended initial pharmacological choice.

- Group B patients are likely to have comorbidities that may add to their symptomatology and impact their prognosis, and these possibilities should be investigated and treated.

Group E:

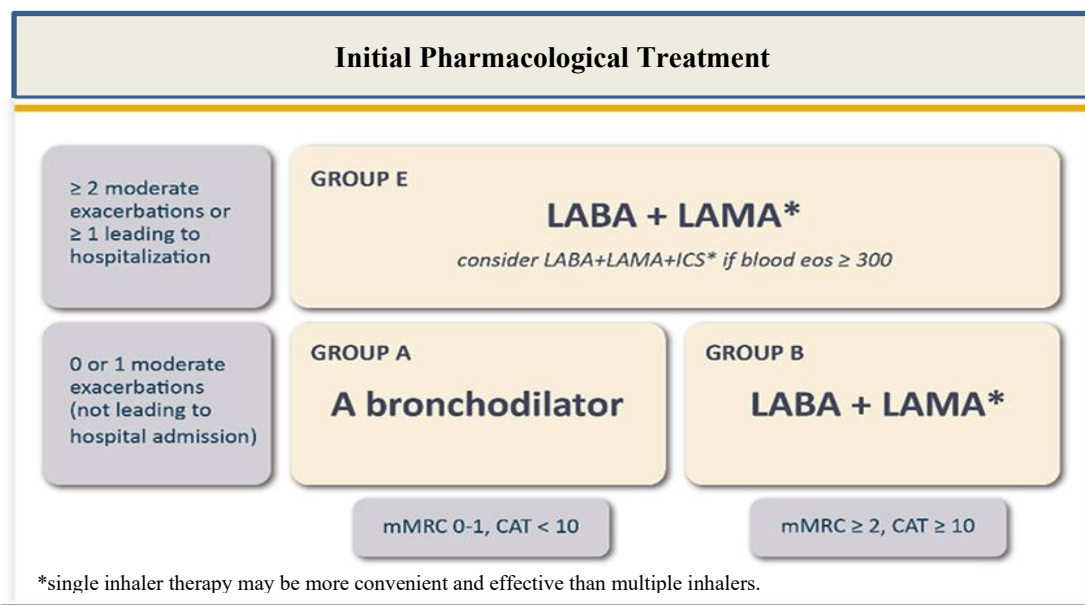
- LABA+LAMA is the preferred choice for initial therapy in Group E patients to reduce COPD exacerbations.

- Use of LABA+ICS in COPD is not encouraged. If there is an indication for an ICS, then LABA+LAMA+ICS has been shown to be superior to LABA+ICS and is therefore the preferred choice.

- Consider LABA+LAMA+ICS in group E if eosinophil ≥ 300 cells/mcg (practical recommendation).

- If patient with COPD have concomitant asthma, they should be treated like patients with asthma. The use of an ICS is mandatory.

Figure 4: Pharmacological treatment:



The modified Medical Research Council dyspnea scale (mMRC); the COPD Assessment Test (CAT); eos: blood eosinophil count in cells per microliter.

Table 9: Treatment options regarding to severity of COPD

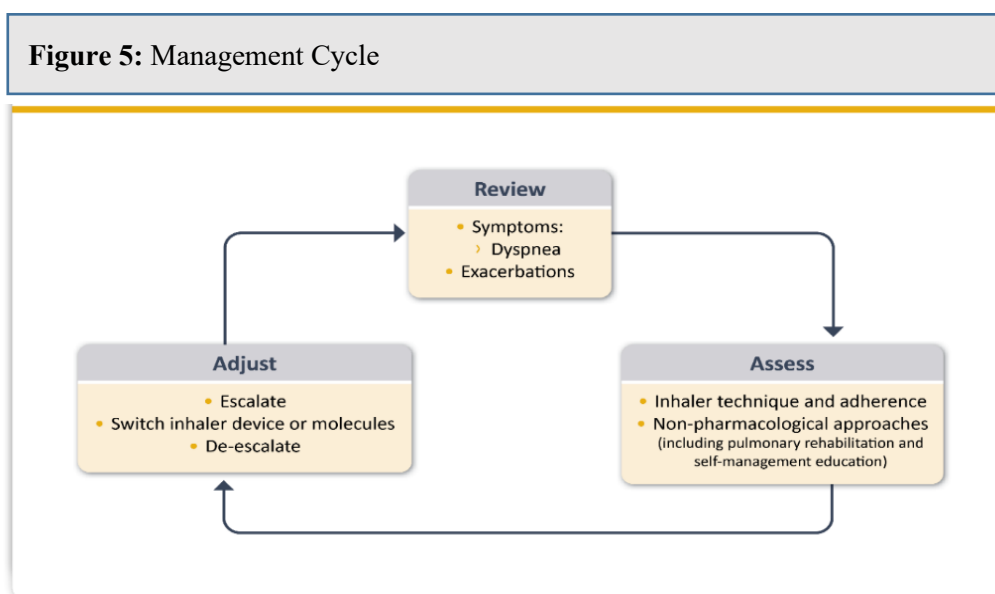
Stage	Severity	Pharmacological treatment
Stage 1	Mild obstruction	Short-acting bronchodilator as needed.
Stage 2	Moderate obstruction	Short-acting bronchodilator as needed; long-acting bronchodilator(s); cardiopulmonary rehabilitation.
Stage 3	Severe obstruction	Short-acting bronchodilator as needed; long-acting bronchodilator(s); cardiopulmonary rehabilitation; inhaled glucocorticoids if repeated exacerbations.
Stage 4	Very severe obstruction or moderate obstruction with evidence of chronic respiratory failure	Short-acting bronchodilator as needed; long-acting bronchodilator(s); cardiopulmonary rehabilitation; inhaled glucocorticoids if repeated exacerbation; long-term oxygen therapy (if criteria met); consider surgical options such as lung volume reduction surgery (LVRS) and lung transplantation.

Table 10 : Factors to consider when adding ICS to long-acting bronchodilators

Strongly favors use	- History of hospitalization(s) for exacerbation of COPD - ≥ 2 moderate exacerbations of COPD per year - Blood eosinophils ≥ 300 cells / mL - History of or, concomitant asthma
Favors use	- 1 moderate exacerbation of COPD per year - Blood eosinophils 100 to <300 / mL
Against use	- Repeated pneumonia events - Blood eosinophils < 100 cells / mL - History of mycobacterail infections

6- Treatment adjustment :

- Adjust pharmacological treatment, including escalation or de-escalation.
- Any change in treatment requires a subsequent REVIEW of the clinical response including adverse effects.



7- Non-pharmacological therapy:

Non-pharmacological treatment is complementary to pharmacological treatment and should form part of the comprehensive management of COPD.

- Smoking Cessation : Smoking cessation is the most effective intervention for slowing the progression of COPD (reduces decline FEV₁).
- Avoid exposure to occupational dust, vapor, gases, indoor biomass smoke, air pollutions, and other chemicals.
- Psychosocial support
- Vaccinations :
 - . influenza vaccination is recommended annually. It reduces mortality from influenza infection in patients over 65 years of age by 50%.
 - . pneumococcal vaccination is indicated every 5 years in patients with COPD over 65 years of age or in case of severe COPD.
- Respiratory physiotherapy is valuable aid to move the mucus from the bronchi.
- Oxygen therapy for patient with chronic hypoxemia (SpO₂ ≤88% or PaO₂ ≤55mmHg).

8- Rehabilitation :

Pulmonary rehabilitation is a key component in the management of people with the condition to achieve and maintain optimal levels of functioning. Interventions for rehabilitation are effective in reducing symptoms of breathlessness and fatigue and contribute to improving health-related quality of life and to reducing hospital readmission related to acute exacerbation. Specific interventions for rehabilitation empower people to cope with related symptoms (e.g. breathlessness or cough) and to develop strategies to perform activities despite these symptoms and to regain and maintain their exercise capacity. Furthermore, rehabilitation supports people with chronic obstructive pulmonary disease to keep engaged in meaningful activities, such as work and social life.

Rehabilitation Interventions:

- Airway clearance techniques to remove the secretions
- Energy conservation techniques to improve the respiratory functions and avoid fatigue
- Muscle strengthening to address the motor functions and mobility
- Respiratory muscle strengthening exercises
- Provision and training in the use of assistive products for self-care
- Graded fitness training to enhance exercise tolerance
- Modification of the home environment
- Assistive products for toileting, dressing and utensils for activities of daily living

Table 11 : Indication of long-term administration of oxygen in patients with COPD

Objective of oxygen therapy is to keep SaO ₂ ≥ 90%.	PaO ₂ ≤ 55 mmHg (7.3 kPa) or SaO ₂ ≤ 88% in room air
Long-term administration of oxygen > 15 hours per day with flow ≤ 3 L/mn (or depending on SaO ₂).	PaO ₂ ≥ 55 mmHg but ≤ 60 mmHg (>7.3 kPa but < 8kPa) with right heart failure or erythrocytosis.

9- Other medications :

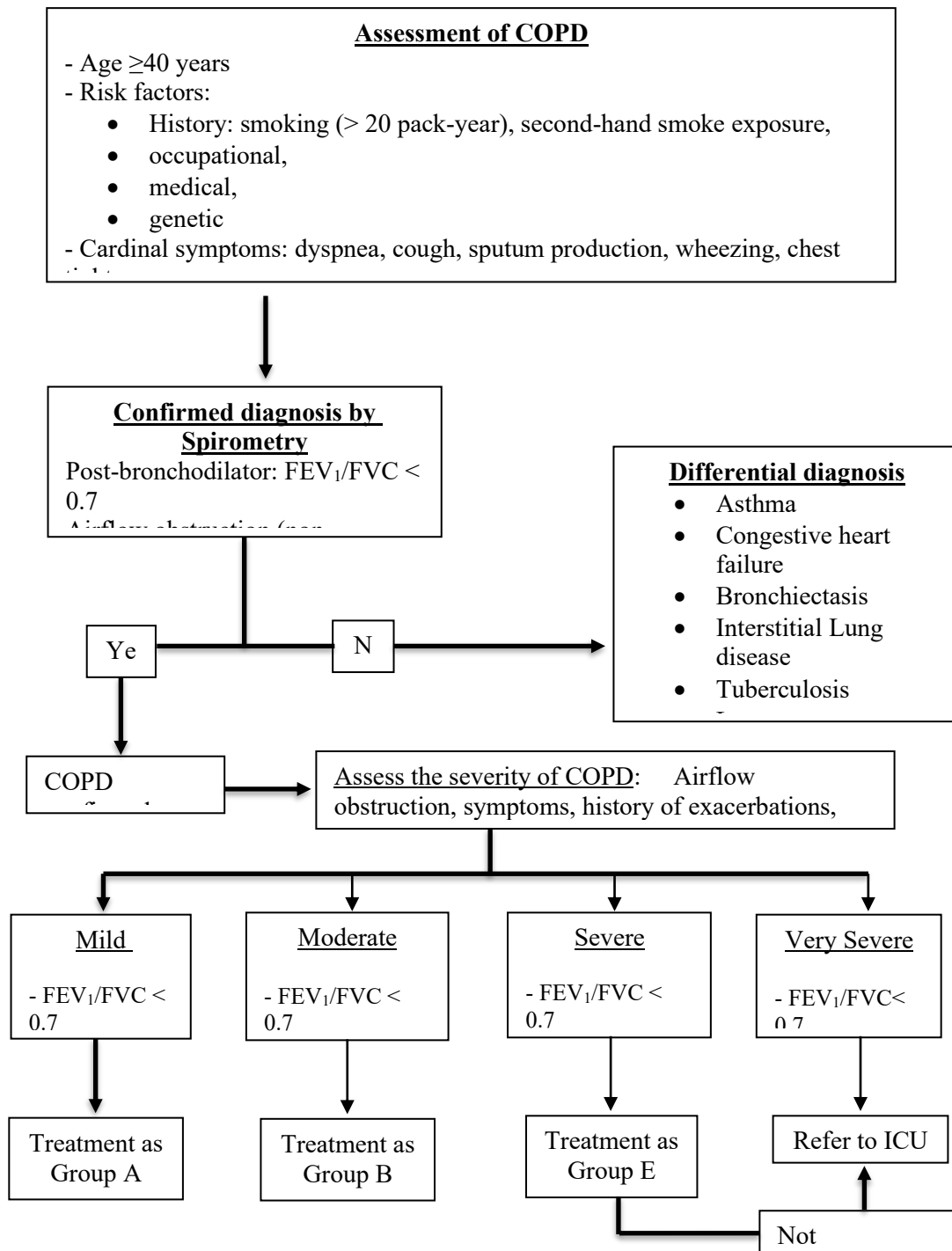
- Antitussives, such as dextromethorphan and benzonatate, have no role in treating COPD exacerbations.
- Opioids (eg, codeine, hydrocodone, oxycodone) should be used judiciously for relief of symptoms (eg, severe coughing paroxysms, pain) because these medications may suppress a productive cough, impair mental status, and cause constipation.

10- Surgery

For select patients with severe emphysema or large bullae: lung volume reduction surgery; bullectomy.

11- Algorithms

Figure 6: Management of COPD



VIII- SPECIAL CASES

i- Pre-COPD or PRISm (Preserved Ratio Impaired Spirometry):

- Individuals who have respiratory symptoms and / or other detectable structural and / or functional abnormalities, in the absence of airflow obstruction on forced spirometry.
- Normal FEV1/FVC ≥ 0.7 after bronchodilation, but impaired spirometry, FEV1 < 80% of reference after bronchodilation.
- These patients may develop persistent airflow obstruction (COPD) over time.

ii- Acute Exacerbation of COPD (AECOPD):

- Epidemiology of AECOPD: AECOPD are key events in the natural history of the disease because they impact significantly on the health status of the patient, enhance the rate of the lung function decline, worsen the prognosis of the patient and are associated with most of the healthcare costs of COPD.
- Definition of AECOPD: AECOPD is defined as an event characterized by increased dyspnea and/or cough and sputum that worsens in < 14 days which may be accompanied by tachypnea and/or tachycardia and is often associated with increased local and systemic inflammation caused by infection, pollution, or other insult to the airways.
- Classification of severity of AECOPD: AECOPD are classified into 3 stages of severity based on clinical symptoms. This classification, known as the Anthonisen Criteria, is commonly used to guide treatment decisions (Table 10).
- **Cause of acute exacerbation of COPD:**
Published data suggest that 50–70% of exacerbations are due to respiratory infections, 10% are due to environmental pollution, and up to 30% are of unknown aetiology. Commonly isolated organisms include *Haemophilus influenzae*, *Streptococcus pneumoniae*, *Moraxella catarrhalis*, *Haemophilus parainfluenzae*, *Pseudomonas aeruginosa*, and Gram negative bacteria. It appears that infections with *Pseudomonas spp* and Gram negative bacteria occur in more severe exacerbations. The data of several study in Cambodia (SARI surveillance, Caliban 2012) showed that infection with *P. aeruginosa* and Gram-negative bacilli, *Klebsiella pneumoniae*, *E. coli*, *Burkholderia pseudomallei* and Gram positive, *S. aureus* occurs in more severe respiratory infections.

Table 12: Classification of COPD's severity

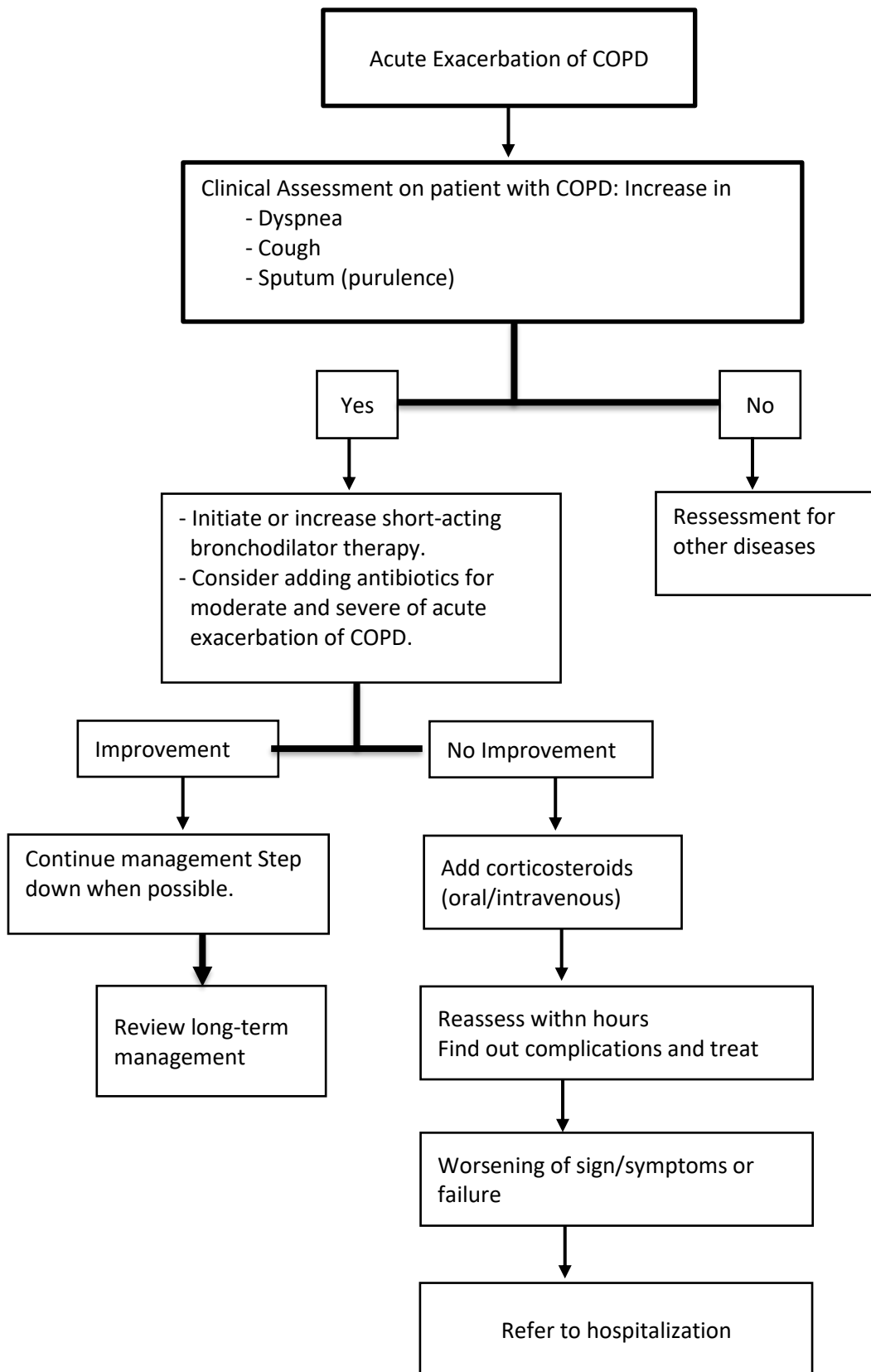
Stage	Presented Symptoms	Management
Type I (Mild exacerbation)	One of the following: - Increased dyspnea - Increased sputum production - Purulent sputum	Supportive therapy: - Bronchodilators - Adjusted maintenance medications - Antibiotic therapy is NOT recommended, except a purulent cough is present.
Type II (Moderate exacerbation)	Two of the following: - Increased dyspnea - Increased sputum production - Purulent sputum	Additional treatment: - Antibiotic - Systemic steroids - Comorbidity treatment - Antibiotic therapy is recommended.
Type III (Severe exacerbation)	All three present: - Increased dyspnea - Increased sputum production - Purulent sputum	Additional treatment and aggressive management: - Hospitalization - Oxygenotherapy - Comorbidity treatment - Non-invasive ventilation, or Mechanical ventilation. - Antibiotic therapy is recommended.

- Antibiotic Therapy on AECOPD:
 - Antibiotic therapy in patients with AECOPD should be initiated in type (moderate) and type 3 (severe), or especially when a purulent cough is present (table 13).

Table 13: Antibiotic Treatment Recommendations for Acute Exacerbations of COPD

Category	Recommended Therapy	Note
< 2 exacerbations/year and at least 2 of the following: - increased sputum purulence - increase sputum volume - increased dyspnea.	First line agent: - Amoxicillin 1g PO, TID for 5 days OR - Azithromycin (250mg) 500mg first day then 250mg for 4 days. Failure to first line agents: see below.	-Antibiotics shorten recovery time, reduce the risk of early relapse, treatment failure, and hospitalization duration. -Antibiotic treatment for ≤ 5 days had the same clinical and bacteriological efficacy to longer conventional treatment in outpatients with COPD.
≥ 2 exacerbations/year and at least 2 of the following: - increased sputum purulence - increase sputum volume - increased dyspnea OR Failure to first line agents above, OR - Antibiotics in the past 3 months.	First line agents: - Amoxicillin+Clavulanate 1g IV, BID for 5 – 10 days. OR - Cefoperazone + Sulbactam 1g IV, BID for 5 – 10 days. Alternatives: - Azithromycin 250mg PO, BID for 5 days. OR - Clarithromycin 500mg PO, BID for 5 days. OR - Levofloxacin 750mg infusion once a day for 5 days. OR - Moxifloxacin 400 mg infusion once daily for 5 days.	-Failure of first line agents: no improvement in symptoms following completion of antibiotic therapy OR deterioration after 72 hours of antibiotic therapy. -Use an antibiotic from a different class than was used in the last 3 months. -Macrolides have poor <i>Haemophilus pneumonia</i> resistance. The benefit of macrolides may be used more to anti-inflammatory properties than to antibacterial activity. -Levofloxacin has good coverage of the pathogens involved. However, because of its broad spectrum, potential for increasing resistance, risk of <i>Clostridioides difficile</i> infection, significant adverse effect profile, it should be reserved for amoxicillin and cefuroxime allergic patients or patients who have failed or cannot tolerate first line antibiotics. -Moxifloxacin does not provide <i>pseudomonal</i> coverage, has increases risk of <i>Clostridioides difficile</i> infection compared to levofloxacin, has marginal anaerobic activity which can alter oral and gastrointestinal tract flora, and has anti-tuberculosis activity possibly resulting in false negative TB cultures.

Figure 7: Algorithm of Management of AECOPD



IX- EVOLUTION

- COPD progresses gradually with worsening airflow limitation and systemic manifestations if left untreated.
- Exacerbations accelerate the decline in lung function and impact significantly on the health status and healthcare costs of the patient.

X- COMPLICATIONS

- Frequent exacerbations
- Pulmonary hypertension
- Cor pulmonale
- Respiratory failure
- Increased cardiovascular risk

XI- PROGNOSIS

Prognosis is based on a point system, with all 4 factors used to determine the score, as follows:

- Body mass index: greater than 21 = 0 points; less than 21 = 1 point
- FEV₁ (post-bronchodilator percent predicted): greater than 65% = 0 points; 50-64% = 1 point; 36-49% = 2 points; less than 35% = 3 points
- Modified Medical Research Council (mMRC) dyspnea scale: mMRC 0 = dyspneic on strenuous exercise (0 points); mMRC 1 = dyspneic on walking a slight hill (0 points); mMRC 2 = dyspneic on walking level ground, must stop occasionally due to breathlessness (1 point); mMRC 3 = dyspneic after walking 100 yards or a few minutes (2 points); mMRC 4 = cannot leave house; dyspneic doing activities of daily living (3 points)
- Six-minute walking distance: greater than 350 meters = 0 points; 250-349 meters = 1 point; 150-249 meters = 2 points; less than 149 meters = 3 points

The approximate 4-year survival based on the point system above is as follows:

- 0-2 points = 80%
- 3-4 points = 67%
- 5-6 points = 57%
- 7-10 points = 18%

XII- PREVENTION AND RECOMMENDATION

- Smoking cessation
- Reduction of occupational and environmental exposures
- Vaccinations:
 - *Influenza* vaccination annually.
 - *Pneumococcal* vaccination every 5 years in patients over 65 years-old.

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PULMONARY FIBROSIS

Asst. Prof. CHHAR Bunpaul

I. DEFINITION

Pulmonary fibrosis (PF) is a form of interstitial lung disease (ILD) in which lung parenchyma is replaced by scar tissue.

II. EPIDEMIOLOGY

This disease affects millions of people and its incidence is increasing worldwide. Some forms of PF are truly idiopathic; however, PF is also seen in a variety of common clinical situations ranging from autoimmune disease to exogenous exposures.

III. ETIOLOGY

3.1- ILDs related to distinct primary diseases

- Lymphangioleiomyomatosis
- Pulmonary alveolar proteinosis
- Eosinophilic pneumonia
- Sarcoidosis
- Langerhans-cell granulomatosis

3.2- ILDs related to environmental exposures:

- pneumoconiosis due to inhalation of inorganic substances
- Hypersensitivity pneumonitis mostly related to inhalation of organic particles (e.g., domestic or occupational exposure to mold or birds or other exposures)

3.3- ILDs induced by drugs (Cordarone, Methotrexate.) or irradiation

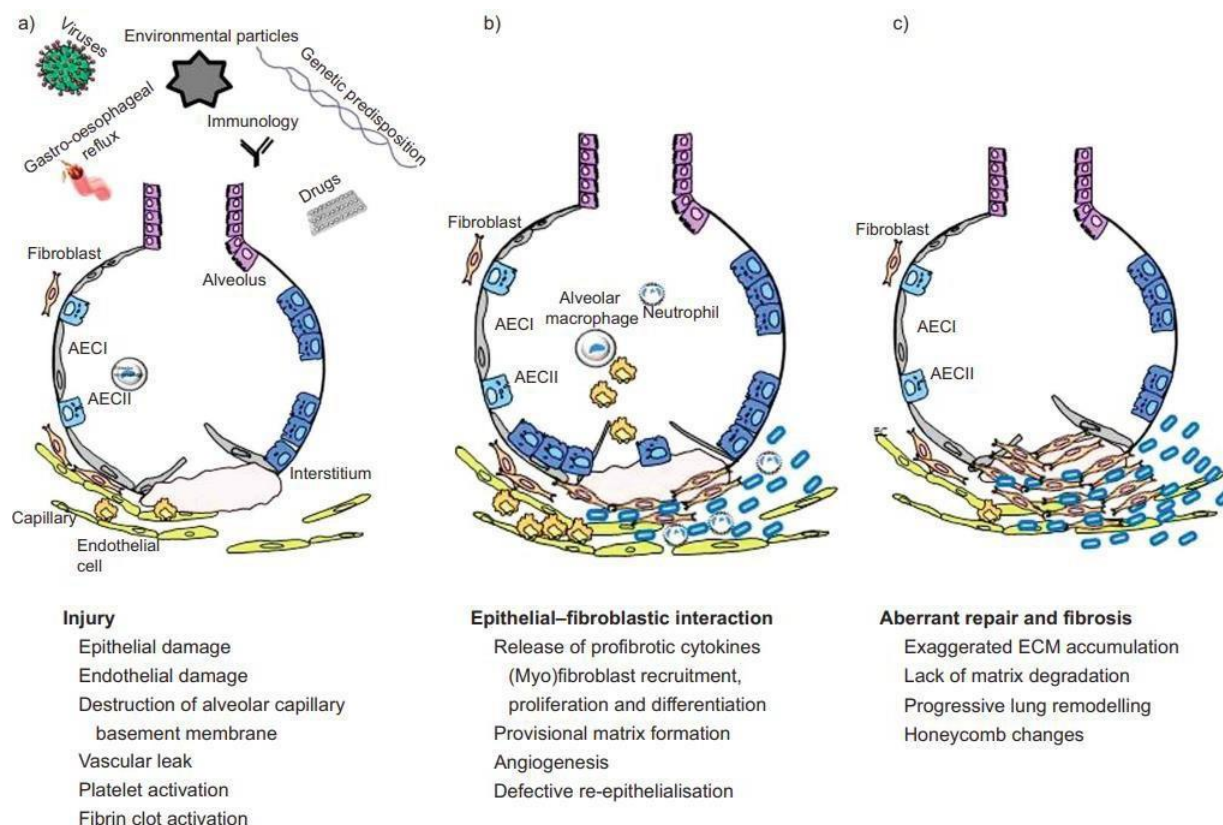
3.4- ILDs associated with connective-tissue disease (CTDs), including RA-ILD and SSc-ILD, idiopathic inflammatory myopathy and primary Sjögren's disease

3.5- Idiopathic interstitial pneumonias: IPF, Cryptogenic Organizing Pneumonia, idiopathic nonspecific interstitial pneumonia (NSIP)

3.6- ILDs post infection: TB, Covid-19...

IV.PATHOPHYSIOLOGY

Recent studies have linked their pathogenesis to predisposing genetic factors and further environmental influences.



V.DIAGNOSIS

5.1- Clinical Presentation

Table 1: Clinical presentation of Pulmonary Fibrosis

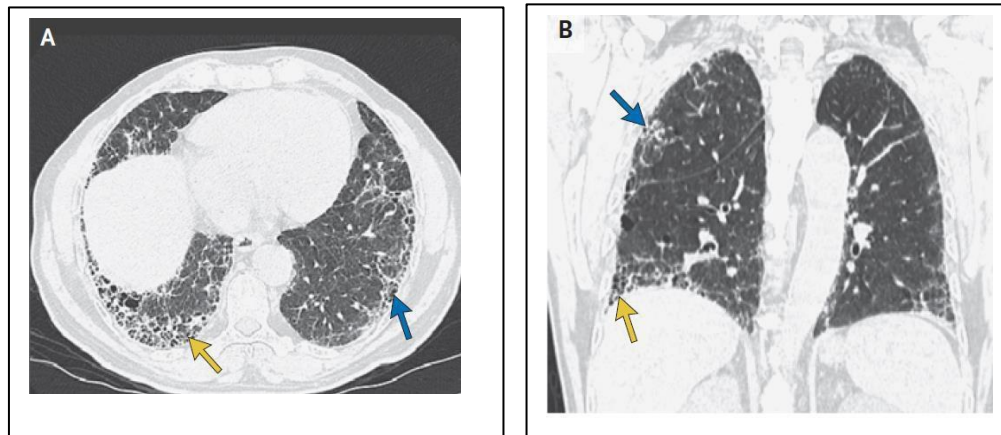
Clinical presentation of Pulmonary Fibrosis
Symptoms: • Chronic exertional dyspnea • Chronic cough without purulence • Fatigue
History taking • Environmental exposures • Medication use • Extrapulmonary signs
Physical examination • Bilateral Fine crackles (also called Bilateral Velcro-like crackles) • Clubbing • Acrocyanosis
Serologic testing: Anticardiolipin antibodies and anticitrullinated peptide antibodies

5.2- Investigation

5.2.1- Chest radiograph

→ Nonspecific changes or bilateral basal reticular abnormalities

5.2.2- High-resolution CT scan of chest: revealing reticulation, architectural distortion, and lung volume loss and may identify patterns suggestive of specific causes.



5.2.3- Lung Function Test: Restrictive lung function pattern (decreased forced vital capacity [FVC], normal or increased ratio of forced expiratory volume in 1 second to FVC, decreased total lung capacity and low residual volume).

5.2.4- Bronchoalveolar lavage +/- bronchial mucosa and lymph-node biopsies if suspected hypersensitivity pneumonitis and sarcoidosis

VI. DIFFERENTIAL DIAGNOSIS

- COPD
- Asthma
- TB sequela

VII. MANAGEMENT

1. Objectives

- Ameliorating the disease
- Slowing down disease progression
- Improving or maintaining quality of life

2. Critical Sign

- Acute exacerbation (acute respiratory symptoms: altered conscious, increased respiratory rate and heart rate)

3. Mortality risk factor: infection, exacerbation and respiratory failure.

4. Outpatient Management

- Educating the patients and sharing decisions are important, especially since there are many off-label treatment options with potentially serious side effects
- Preventing exposures and events that may drive further disease progression is essential.
- Cessation of tobacco smoking
- Pneumococcal and influenza vaccinations

5. First-line treatment

- Treatment of the underlying disorder

- Nonpharmacologic treatment

- Supplemental oxygen is indicated in patients with resting hypoxemia ($\text{PaO}_2 < 55 \text{ mmHg}$, or $\text{PaO}_2 < 60 \text{ mmHg}$ and cor pulmonale or polycythemia); ambulatory oxygen in patients with isolated exertional hypoxemia
- Psychosocial support
- Smoking cessation
- Pulmonary rehabilitation
- Symptom palliation
- End-of-life care

- Pharmacologic treatment

→ Depending on the underlying condition, antifibrotic therapy is considered in case of disease progression despite appropriate first-line therapy.

- **Antiinflammatory Agents**

Prednisolone: 1 mg/kg per day orally, followed by taper slowly over the course of weeks to months ensuring there is no recurrence with close follow up.

- **Immune Modulators:** interferon gamma-1b subcutaneous injections of interferon gamma-1b three times weekly

- **Antifibrotic agents:**

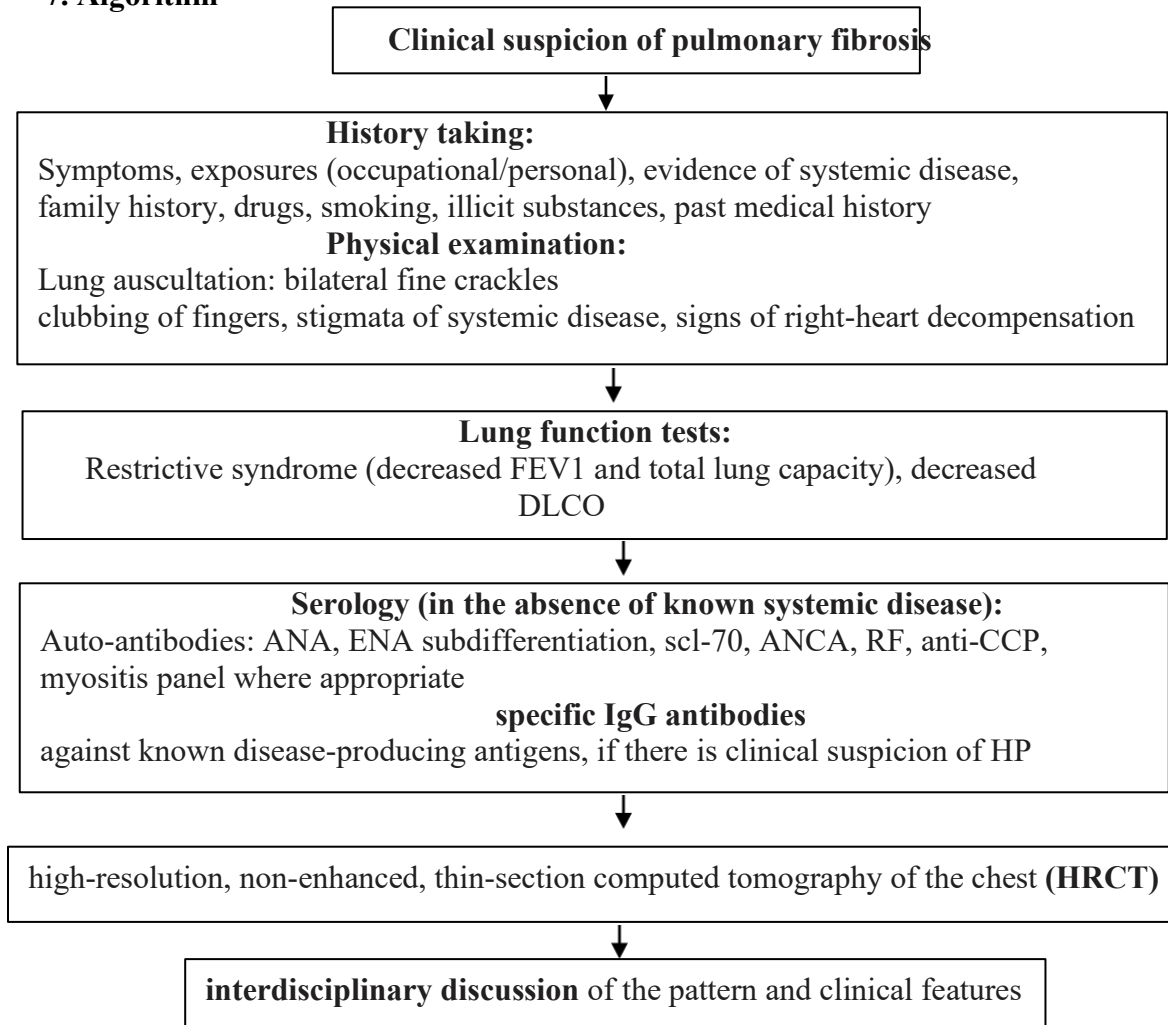
- pirfenidone 801 mg by mouth thrice daily
 - or nintedanib 150 mg by mouth twice daily

- **Lung Transplantation**

6. Treatment adjustment: According to:

- Disease progression
- Stable disease
- Amelioration

7. Algorithm



VII. SPECIAL CASES

- Nearly any systemic disease can be associated with an ILD. The differential diagnosis should always include drug toxicity, as certain immune suppressants are pneumotoxic.

VIII. EVOLUTION AND PROGNOSIS

Depend on the underlying entity

- Worsening respiratory symptoms
- Decline in lung function
- Decreased quality of life
- Risk of early death

IX. COMPLICATIONS

- Respiratory failure and infection
- Severe pulmonary hypertension with cor pulmonale
- Heart failure
- Bronchogenic carcinoma

X. PREVENTION AND RECOMMENDATION

- Preventing exposures
- Cessation of tobacco smoking are priorities.
- Pneumococcal and influenza vaccinations

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COMMUNITY ACQUIRED PNEUMONIA (CAP)

Prof. Vann Mich

I- DEFINITION

Community-acquired pneumonia (CAP) is defined as an acute infection of the pulmonary parenchyma that is acquired outside the hospital or outside the healthcare facilities or occurs within 24 hours of hospital admission.

II- EPIDEMIOLOGY

Pneumonia is one of the most common infectious diseases and an important cause of mortality and morbidity worldwide despite the availability of potent antimicrobials.

Community acquired pneumonia is one of the leading causes of hospitalization due to an infectious agent.

Community acquired pneumonia is 6th leading and the first cause of death from infectious diseases in United States of American estimated there was 4 - 5 million cases annually, 500,000 hospitalizations and 45,000 deaths with mortality rate from 2 to 40 % for hospitalized patients and less than 1% for those not requiring hospitalization.

III- ETIOLOGY

The etiology of pneumonia depends on several factors, recent antibiotic use, and location of the patient when pneumonia occurred.

According to the Caliban, community-acquired lung infection bacteria and antibiotics network 2012 and the data of US-CDC/CDC-MoH of Cambodia, Sentinental Surveillance of Severe Acute Respiratory Infections (SARI) 2009-2019:

- *S. pneumoniae* and *H. influenzae* were the two most common bacterial etiologies of pneumonia in children in Cambodia.
- The Gram-negative bacteria such as *Pseudomonas aeruginosa*, *Klebsiella pneumoniae*, *Escherichia coli*, *Burkholderia pseudomallei* were the most common bacterial etiologies of pneumonia in adult.
- *Staphylococcus aureus* was found as cause of pneumonia in both children and adult.
- The intracellular bacteria such as *Chlamydia pneumoniae*, *Mycoplasma pneumoniae* and *Legionella pneumophila* were not found as cause of pneumonia in Cambodia due to non-availability of routine test to diagnose atypical pneumonia.

These data is different from those of prospective studies in Europe that the most commonly identified cause of CAP is *S. pneumonia* and respiratory viruses and followed by *haemophilus influenzae*, *Moraxella catarrhalis*, *S.taphylococcus aureus* and *legionella spp*, *Mycoplasma pneumoniae*, *Chlamydia pneumonia*.

- Important trends: Both the distribution of pathogens that cause CAP and our knowledge of these pathogens are evolving. Key observations that have changed our understanding of CAP and influenced our approach to management include:
 - Declines in *S. pneumonia* incidence: Although *S. pneumoniae* is the most commonly detected bacterial cause of CAP in most studies, the overall incidence of pneumococcal pneumonia is decreasing. This is in part due to widespread use of pneumococcal vaccination which results in both a decline in the individual rates of pneumococcal pneumonia and herd immunity in the population.
 - The COVID-19 pandemic: SARS-CoV2 is an important cause of CAP.
 - Increased recognition of other respiratory viruses: respiratory viruses have been detected in approximately one-third of cases of CAP in adults when using molecular methods.

- Low overall rate of pathogen detection: Despite extensive evaluation using molecular methods, a causal pathogen can be identified in only half of cases of CAP.

IV- PATHOPHYSIOLOGY

In normally, LRT are sterile below the first major bronchial division. Pulmonary pathogens reach the lung by one of four routes:

- Direct inhalation of infectious respiratory droplets.
- Aspiration of oropharyngeal secretions containing bacteria.
- Direct spread along the mucosal membrane surface from the upper to lower respiratory system.
- Hematogenous spread.

CAP occurs when there is a defect in one or more of pulmonary defense mechanism, cough reflex, mucocilliary clearance system, immune responses, or when a large infectious inoculum or a virulent pathogen overwhelms the immune response.

V- DIAGNOSIS

1- Signs and Symptoms

- There are no particular symptoms and physical findings that are sensitive enough to diagnose CAP,
- Chest x-ray is required for definitive diagnosis,
- Clinical prediction rules may be employed if chest radiography is not available.
- Pneumonia should be suspected, if a patient has an acute cough and one of the following signs:
 - new focal chest signs
 - dyspnea
 - tachypnea
 - fever lasting more than 4 days
- Chest-radiograph is recommended in this setting.

2- Physical examinations

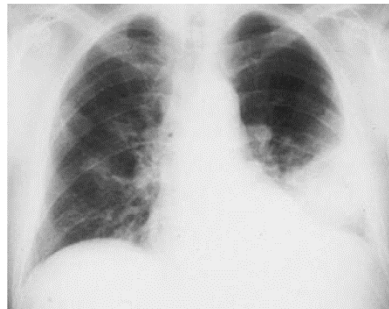
- Physical examinations may vary, depending on the type of organism, severity of infection, coexisting host factors, and the presence of complications.
- Signs of bacterial pneumonia may include the following:
 - Hyperthermia ($35^{\circ}\text{C} < T^{\circ} > 38^{\circ}\text{C}$)
 - Tachypnea (> 18 respirations/minute)
 - Use of accessory respiratory muscles
 - Tachycardia ($> 100\text{bpm}$) or bradycardia ($< 50\text{bpm}$)
 - Central cyanosis
 - Altered mental status
- Physical findings:
 - Adventitious breath sounds, such as crackles, rhonchi, or wheezes.
 - Decreased intensity of breath sounds
 - Dullness to percussion may be observed if lobar consolidation or a parapneumonic pleural effusion is present.
- The clinical evaluation is less than 50% sensitive compared to chest imaging for the diagnosis of CAP.

3- Imaging

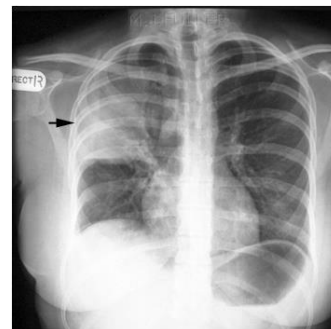
- Chest x-rays or computed topography scan are usually also done to confirm the diagnosis of pneumonia. Chest CT scan is more sensitive and specific than chest radiography but may be indicated in selected cases.
- Radiographic finding ranges from patchy airspace opacities to lobar consolidation with air bronchograms to diffuse alveolar or interstitial opacities, and can find also pleural effusions and cavitation.
- No pattern of radiographic abnormalities is pathognomonic of any infectious cause.
- Chest imaging may help assess severity and response to therapy over time.



Consolidation of the RLL



Left lower lobe infiltrate



Consolidation of the right upper lobe

4- Laboratory testing:

- Complete blood count may present increased white blood cells (neutrophil) more than $10\,000/\text{mm}^3$ or decreased less than $4\,000/\text{mm}^3$.
- Inflammatory markers such as CRP, VS, may be increased.
- Procalcitonin is released in response to bacterial toxins and inhibited by viral infection. This divergent response to bacterial and viral infection offers laboratory support for a clinical adjustment of a viral process in patients with lower respiratory symptoms; however, studies have not found a threshold at which bacterial pneumonia can be reliably distinguished from viral pneumonia. Therefore, procalcitonin is not recommended as a “**rule out**” test for bacterial pneumonia, and empiric antibacterial agents are recommended regardless of procalcitonin level at time of presentation.
- Microbiologic testing including sputum and blood cultures, at least two sets with needle sticks at separate sites, has been standard practice for patients with CAP who require hospitalization. The yield of sputum and blood culture is low, however, false- positive results are common. As result, target testing is recommended for patients with severe disease and for those treated empirically for MRSA or *P. aeruginosa* infection.
- Pleural fluid analysis may be done for patients with CAP who have significant pleural fluid collection by diagnostic thoracentesis.

5- Diagnostic testing:

- The diagnostic testing for a specific infectious cause of CAP is not generally indicated in ambulatory patients treated as outpatients because empiric antibiotic therapy is almost always effective in this population.
- The diagnostic testing is recommended in hospitalized CAP patients.
- Three widely available diagnostic tests may guide therapy:
 - The sputum Gram stain or respiratory aspiration (bronchoscopy) and culture.
 - Urinary antigen tests for *S. pneumoniae* and *Legionella species*.
 - Tests for viruses: Influenzae and SARS-CoV-2 by rapid testing or polymerase chain reaction (PCR).

- Sputum Gram stain is neither sensitive nor specific for *S. pneumoniae*. The usefulness of a sputum Gram stain lies in broadening initial coverage in patients to be hospitalized for CAP, the most commonly to cover *S. aureus* (including MRSA) or Gram-negative rods (including *P. aeruginosa* and Enterobacteriaceae).
- Urinary antigen assays for *L. pneumophila* and *S. pneumoniae* are at least as sensitive and specific as sputum Gram stain and culture. Results are not affected by early initiation of antibiotic therapy, and positive tests may allow narrowing of initial antibiotic coverage.

VI- DIFFERENTIAL DIAGNOSIS

The differential diagnoses of CAP include as follow:

- Pulmonary tuberculosis
- Acute bronchitis
- Acute exacerbation of chronic obstructive pulmonary disease
- Chronic heart failure and pulmonary edema
- Pulmonary fibrosis
- Cryptogenic organizing pneumonia
- Bronchogenic carcinoma
- Lymphoma

VII- MANAGEMENT

1- Objective: the goal of treatment for CAP is:

- to get rid of the infection, and
- to prevent complications.

2- Critical signs:

- $35^{\circ}\text{C} < T^{\circ} < 40^{\circ}\text{C}$
- Pulse > 125 bpm
- SBP < 90 mmHg
- Fast respiratory rate (> 30 per min)
- $\text{SpO}_2 < 90\%$
- Signs of severe respiratory distress (e.g. inability to speak, use of accessory muscles)
- Agitation / somnolence

3- Comorbidity:

- Signs of severe respiratory distress (e.g. inability to speak, use of accessory muscles)
- Adult age older than 65 years
- Children age younger than 2 years
- Pregnant women
- Congestive heart failure / Cardiovascular diseases/stroke
- Chronic kidney failure
- Chronic liver diseases
- Chronic respiratory diseases (COPD, Asthma, chronic respiratory failure, lung fibrosis)
- Uncontrolled diabetes mellitus, cancers
- Immunosuppression: HIV/AIDS; prolong corticosteroid use; chemotherapy
- Hospitalization during the year with history of bacterial pneumonia.
- Live in healthcare facilities

Tab.1 – Comorbidity

Patients	Comorbidities
Age > 65 years with present at least One of comorbidities	- Congestive heart failure / Cardiovascular diseases/stroke - Chronic kidney failure - Chronic liver diseases - Chronic respiratory diseases (COPD, Asthma, chronic respiratory failure, lung fibrosis ...)
Age < 65 years with present at least Two of comorbidities	- Uncontrolled diabetes mellitus - Cancers; pregnant - Immunosuppression: HIV/AIDS; prolong corticosteroid use; chemotherapy ... - Hospitalization during the year with history of bacterial pneumonia. - Live in healthcare facilities

4- Risk factors of mortality:

CURB-65 or CRB-65 score:

- CURB-65 is a severity score for CAP, comprising 5 variables, attributing 1 point for each item:
 - Confusion: new onset;
 - Urea >7 mmol/L;
 - Respiratory rate ≥ 30 /minute,
 - Blood pressure: SBP <90 mmHg and/or DBP ≤ 60 mmHg; and
 - Age ≥ 65 years.
- It has been extensively validated to predict 30-day mortality in CAP,³ and divides patients into 3 groups:
 - **score 0–1**: low risk of 30–day mortality (0.7–3.2%) and thus may be managed as outpatients;
 - **score 2**: intermediate risk of death (13%) and thus managed as inpatients (short hospitalization) in medicine ward and
 - **score 3–5**: high risk of 30–day mortality (17–57%) managed as inpatients who are admitted to an intensive care unit (ICU).

Stratification of mortality rate using the CURB-65 and CRB-65:

Mortality Risk	CURB-65 score	Observed mortality rate (%)	Recommendation	CRB-65 score	Observed mortality rate (%)	Recommendation
Low	0 or 1	1.5	Home treatment	0 - <1	1.2	Can be treated at home
Moderate	2	9.2	Hospitalization	1 or 2	8.15	Must be referred to and assessed in a patient that can accommodate inpatients as soon as possible.
High	3 – 5	22	Treatment for moderate pneumonia	3 or 4	31	Required hospitalization as soon as possible.

5- Pharmacologic treatment:

- Broad spectrum empirical antimicrobial therapy should be initiated promptly once the diagnosis of CAP is established to which the etiologic pathogen is susceptible.
- Therefore, initial antibiotic choices are typically empiric, based on available local data, patient risk factors for specific pathogens, and local antibiotic resistance patterns.
- Appropriate treatment involves starting empiric antibiotics as soon as possible, preferably ≤ 4 hours after presentation.
- As well as symptomatic and comorbidity treatment, electrolyte and fluid troublesome should be considered.

5-1-Outpatient management:

Table.2-Recommended empiric antibiotic for outpatient with pneumonia without signs of severity

Patient group	Treatment regimens
Outpatients with previously healthy outpatients with no risk factors for MRSA or <i>P. aeruginosa</i> .	- Aminopenicillin: Amoxicillin 1g orally three times daily. OR - Macrolides: Azithromycin, 500mg at first day then 250mg orally daily for 4 days or clarithromycin, 500mg orally twice a day (if local pneumococcal resistance is $< 25\%$) or doxycycline 100mg two time daily.
Outpatients with comorbid medical conditions (see risk factors of mortality)	<u>1-Combination therapy:</u> - Beta-lactam: Amoxicillin/clavulanic acid) 1.5g – 3g orally daily OR - Cefpodoxime 200mg twice daily OR - Cefuroxime 500mg orally twice daily for 7 – 10 days. Plus - an oral macrolide (as above) or doxycycline <u>2-Monotherapy:</u> - Respiratory fluoroquinolone: Levofloxacin 750mg IV infusion daily or moxifloxacin 400mg IV infusion daily.
- Re-evaluation at least for 48 hours, clinically improving based on symptoms (not worse) and vital sign (afebrile) - Duration of treatment: 7-10 days for Beta-lactam and 5 days for macrolide and fluoroquinolone and 14 days for doxycycline.	

5-2-Inpatient management:

Table.3-Recommended empiric antibiotic for inpatient with **non-severe pneumonia** not requiring ICU:

Patient group	Treatment regimens
Patients with risk factor of mortality without MDRO infection	- Non-anti-pseudomonal Beta-lactam: .Amoxicillin/clavulanic acid 1g intravenous three time daily OR .Cefoperazone/sulbactam: 1-2g IV every 12h a day or ceftriaxone 1-2g IV a day or cefotaxime 1-2g every 8 hours IV a day or ceftaroline 600mg q12h daily. Plus - Macrolide: Azithromycin 250mg twice daily or Clarithromycin 500mg twice daily. OR -Respiratory fluoroquinolone: Levofloxacin 750mg PIV a day or Moxifloxacin 400mg one hour by IV infusion daily or ciprofloxacin 400mg IV infusion q 8-12h daily.
Patients with risk factors of prior infection with MRSA (Obtain culture to confirm infection or to allow de-escalation of therapy)	- Treatment regimens above Plus -Vancomycin, 15-20mg/kg IV q12h with interval dosing based on kidney function to achieve serum trough concentration 15-20mcg/mL OR -Linezolid 600mg orally or intravenously q12h
Patients with risk factors of prior infection with <i>Pseudomonas aeruginosae</i> (Obtain culture to confirm infection or to allow de-escalation of therapy)	- Anti-pseudomonal beta-lactam: Piperacillin/tazobactam 4,5g q6h IV only or Cefepime 2g IV q8-12h or Ceftazidime 2g IV q8h or meropenem 1g IV q8h or imipenem 500mg IV q6h. Plus -Aminoglycosides: Amikacin 500mg 7.5mg/kg IV/IM twice daily or gentamycin 80mg IV/IM q12h. Plus -Macrolide: azithromycin / clarithromycin IV OR Levofloxacin 750mg IV infusion daily or Moxifloxacin 400mg IV infusion daily or ciprofloxacin 400mg infusion q8-12h daily.
- Re-evaluation at least for 48 hours, clinically improving based on symptoms (not worse) and vital sign (afebrile) - Duration of treatment: Maximum 5 days for aminoglycosides; 5-7 days for macrolide and fluoroquinolone and 7-14 days for Beta-lactam.	

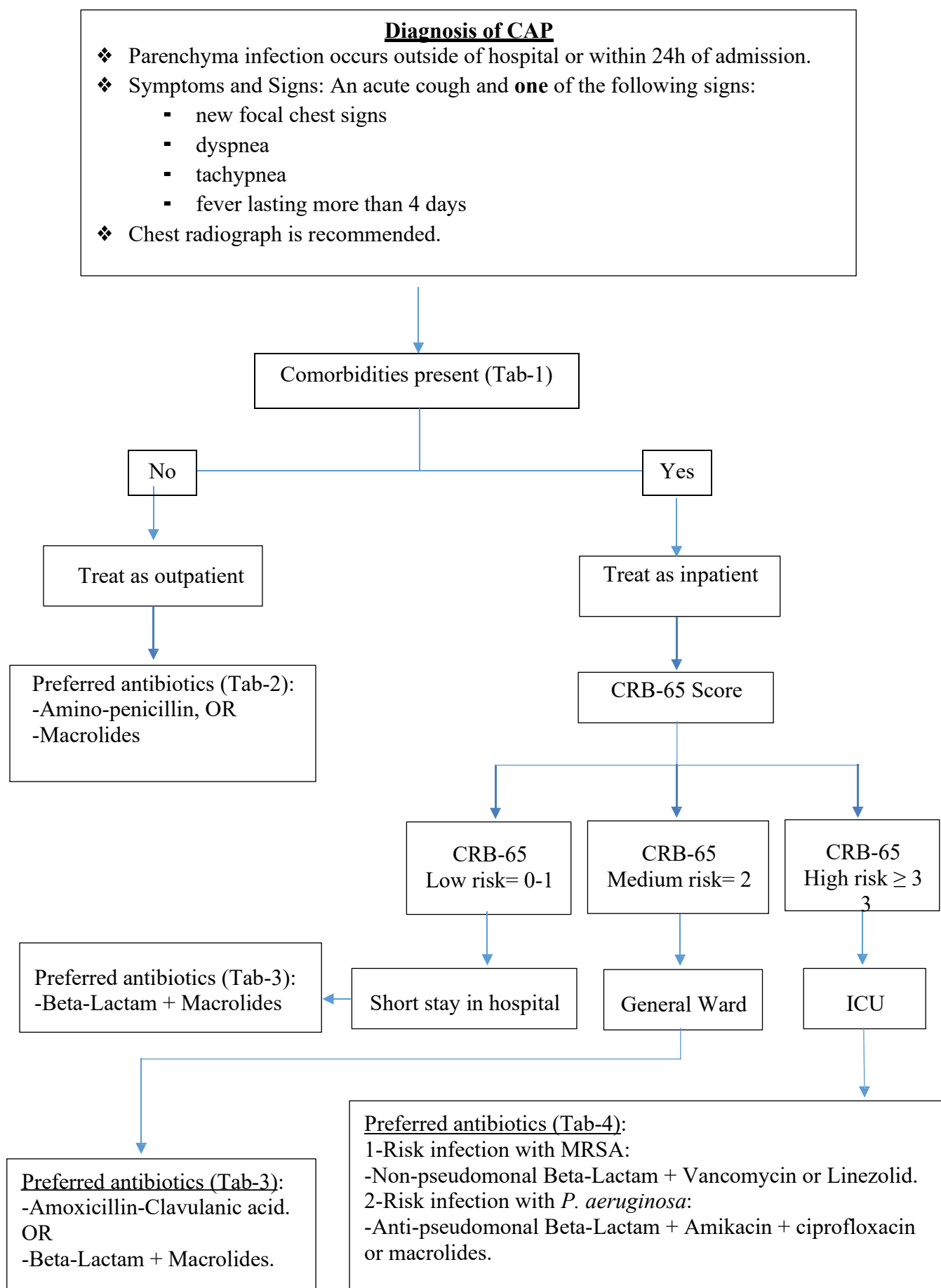
Table.4-Recommended empiric antibiotic for inpatient with **severe pneumonia** requiring ICU admission:

Patient group	Treatment regimens
Inpatients with risk factors of morbid-mortality and critical signs hospitalize in ICU	- Beta-lactam: Amoxicillin/clavulanic acid 1g intravenous three time daily or cefoperazone/sulbactam 1-2g IV every 12h daily or ceftriaxone 2g q24h daily or cefotaxime 1-2g IV q8h daily or ceftaroline 600mg IV q8h daily or piperacillin/tazobactam 4,5g IV every 8h daily or meropenem 1g IV q8h daily or imipenem 500mg IV q6h daily or ceftazidime 2g IV q8h daily. Plus - Macrolide: azithromycin 250mg twice daily or Clarithromycin 500mg twice daily. OR -Respiratory fluoroquinolone: Levofloxacin 750mg IV infusion daily or moxifloxacin 400mg IV infusion q24h daily.
Patients at risk for <i>Pseudomonas</i> infection and with critical ill, at increased risk for drug resistance (Obtain culture to confirm infection or to allow de-escalation of therapy)	- Anti-pseudomonal beta-lactam: piperacillin/tazobactam: 4,5g IV every 6-8h daily or meropenem 1g IV q8h daily or imipenem 500mg IV q6h daily or ceftazidime 2g IV q8h daily or aztreonam 2g every 8h. Plus -Aminoglycosides: amikacin 7.5mg/kg IV/IM q12h daily or gentamycin 80mg IV/IM q12h daily. Plus -Anti-pseudomonal fluoroquinolone: ciprofloxacin 400mg IV infusion q8-12h daily or levofloxacin 750mg IV infusion daily or moxifloxacin 400mg IV infusion q24h daily. OR -Macrolide: azithromycin 250mg twice daily or clarithromycin 500mg twice daily.
Patients at risk for MRSA infection with critical ill. (Obtain culture to confirm infection or to allow de-escalation of therapy)	- Non-anti-pseudomonal beta-lactam: cefoperazone/sulbactam 1-2g IV every 12h daily or cefotaxime 1-2g IV q8h daily or ceftaroline 600mg IV q8h daily. Plus - either macrolides: (azithromycin/clarithromycin) Plus -Vancomycin, 15-20mg/kg IV q12h daily with interval dosing based on kidney function to achieve serum trough concentration 15-20mcg/mL OR -Linezolid 600mg IV q12h daily.
- Re-evaluation at least for 48 hours, clinically improving based on symptoms (not worse) and vital sign (afebrile) - Duration of treatment: Maximum 5 days for aminoglycosides; 5-7 days for macrolide and fluoroquinolone and 7-14 days for beta-lactam.	

6- First line treatment

- Pneumonia with no-risk factors nor signs of severity, initial antimicrobial therapy is recommended as amoxicillin or macrolide orally [grad A], or the combination of amoxicillin and macrolide is the best choice in case of indistinguishable typical pneumonia and atypical pneumonia.
- There is no justification for the systemic use of injectable beta-lactam at home or association of amoxicillin - clavulanic acid.
- No indication in adding anaerobic coverage for suspected aspiration pneumonia beyond the standard regimens except lung abscess or empyema is suspected.
- No indication for routinely using corticosteroids in adult with pneumonia
- Pneumonia due to typical bacteria, the first antibiotic recommended is beta-lactams.
- Pneumonia due to atypical bacteria, the first antibiotic recommended is macrolides.
- In case of indetermination between typical and atypical pneumonia.

7- Algorithm: Management of Community-Acquired Pneumonia (CAP)



VIII- SPECIAL CASES

- 1- Aspirate pneumonia refers to an infection caused by bacteria, primarily oral pharyngeal anaerobes, after a repeated or a large volume aspiration event on the patient with altered consciousness or having the superior function disorders (eg. Stroke, intracranial mass, head trauma ...). Its treatment based on the anaerobic agents such as Clavulanic acid – Amoxicillin, Metronidazole ...
- 2- Lobar pneumonia: Acute exudative inflammation of the entire lobe. Uniform consolidation with a complete or near complete consolidation of a lobe of a lung. Majority of these cases are caused by *Streptococcus pneumoniae*.
- 3- Bronchopneumonia: A descending infection started around bronchi and bronchioles, which then spreads locally into the lungs. Lower lobes are usually involved. Patchy areas of consolidation which represents neutrophil collection in the alveoli and bronchi.
- 4- Parapneumonic effusion refers to exudative pleural effusion associated with ipsilateral pneumonia. Parapneumonic effusions are mainly associated bacterial infection resulting bacterial pneumonia. The treatment is based on the association of adapted antimicrobial and evacuative pleural puncture with pleural washing.

IX- PROGNOSIS AND MORTALITY:

The prognosis for CAP is determined by three main factors: the patient's age; their overall state of health (presence of any comorbidities); and the severity or seriousness of the disease presentation.

X- EVOLUTION AND COMPLICATIONS

- The evolution of most patients with community-acquired pneumonia recovers. However, pneumonia can be fatal, most often in infants and in older adults.
- Complications of pneumonia may include:
 - Parapneumonic pleural effusion / empyema which can affect 36–66% of patients with CAP.
 - Sepsis / septic shock. This is severe pneumonia and may lead to organ failure complicated by infection.
 - Acute respiratory distress syndrome (ARDS). This is a severe form of respiratory failure.
 - Lung abscesses. They may need to be drained with surgery
 - Respiratory failure. This requires the use of a ventilator.
 - Necrotizing pneumonia is characterized by multiple, small (<1cm) lung cavities with evidence of consolidation. It is a rare complication of CAP in adults and is associated with pathogens such as *Staphylococcus aureus*, *Streptococcus pyogenes*, *Nocardia spp* and *Klebsiella pneumoniae*.

Note that patients having all risk factors associated with necrotizing pneumonia such as smoking, alcohol, diabetes, chronic lung diseases or liver disease.

XI- PREVENTION AND RECOMMENDATION

- The pneumococcal vaccine is one of the most effective ways to prevent pneumonia. It is recommended every 5 years.
- The influenza (flu) vaccine is important not only for preventing influenza but also for preventing potential complications, including pneumonia. It is recommended every year.
- Hemophilus influenzae vaccine type b is available only in children.
- Avoiding smoking is another important way to prevent pneumonia as well as other health problems.
- Infection control refers to measures used to prevent the spread of any type of infection, including pneumonia. Infection control is most commonly practiced in health care settings but is useful in the community as well. Simple practices such as frequent hand washing with soap and water or alcohol-based hand rubs can be effective.

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HOSPITAL-ACQUIRED PNEUMONIA

Asst. Prof. CHHAR Bunpaul

I. DEFINITION

Hospital-acquired pneumonia (HAP) includes pneumonia that was not incubating at the time of hospital admission and develops at least 48 hours after hospital admission in patients who are not receiving mechanical ventilation.

II. EPIDEMIOLOGY

- Nosocomial pneumonia accounted for 22% of all hospital infections in the United States.
- It is the second most common infection in hospitalized patients, and the most common infection in the intensive care unit responsible for one-fourth of all ICU infections.

III. ETIOLOGY

- *P aeruginosa*
- *Staphylococcus aureus*, including methicillin-susceptible *S aureus* (MSSA) and methicillin-resistant *S aureus* (MRSA)
- *Klebsiella pneumoniae*
- *Escherichia coli*
- Non-Enterobacteriaceae bacteria such as *S marcescens*, *Stenotrophomonas maltophilia*, and *Acinetobacter* species.

IV. PATHOPHYSIOLOGY

Inhalation, aspiration, and hematogenous spread are the 3 main mechanisms by which bacteria reach the lungs.

Oropharyngeal and gastric colonization, and the subsequent aspiration of their contents into the lungs in patients with impaired mechanical, cellular and humoral defenses lead to the possible development of HAP.

V. DIAGNOSIS

5.1- Clinical Presentation

Symptoms and signs of hospital-acquired pneumonia in non-intubated patients are generally the same as those for community-acquired pneumonia and include malaise, fever, chills, rigor, cough, dyspnea, and chest pain.

5.2- Investigation:

5.2.1- Chest radiograph: a new lung infiltrate

5.2.2- Blood test: increased WBC, CRP and PCT

5.2.3- Bacterial culture: sputum, Broncho aspiration or broncho-alveolar lavage sample, blood culture.

5.2.4- Thoracentesis if pleural effusion

VI. DIFFERENTIAL DIAGNOSIS

- Congestive heart failure
- Pulmonary embolus or infarction
- Acute respiratory distress syndrome

VII. MANAGEMENT

1. Objectives

- To minimize patient harm and exposure to unnecessary antibiotics and reduce the development of antibiotic resistance,

2. Critical Sign

CRB-65 score (confusion, Respiratory rate > 30 breaths/min, Blood pressure < 90 mmHg systolic or < 60 mmHg diastolic, and age ≥ 65 yrs)

3. Mortality risk factor:

- Ventilatory support for HAP
- underlying illness
- Delays in the initiation of appropriate antibiotic therapy
- Septic shock
- Respiratory and heart Failure

4. First-line treatment

The choice of a specific regimen for empiric therapy should be based upon

- The patient's individual risk factors for multidrug resistance (MDR), including their prior microbiology data.
- Local microbiologic data

MDR risk factors:

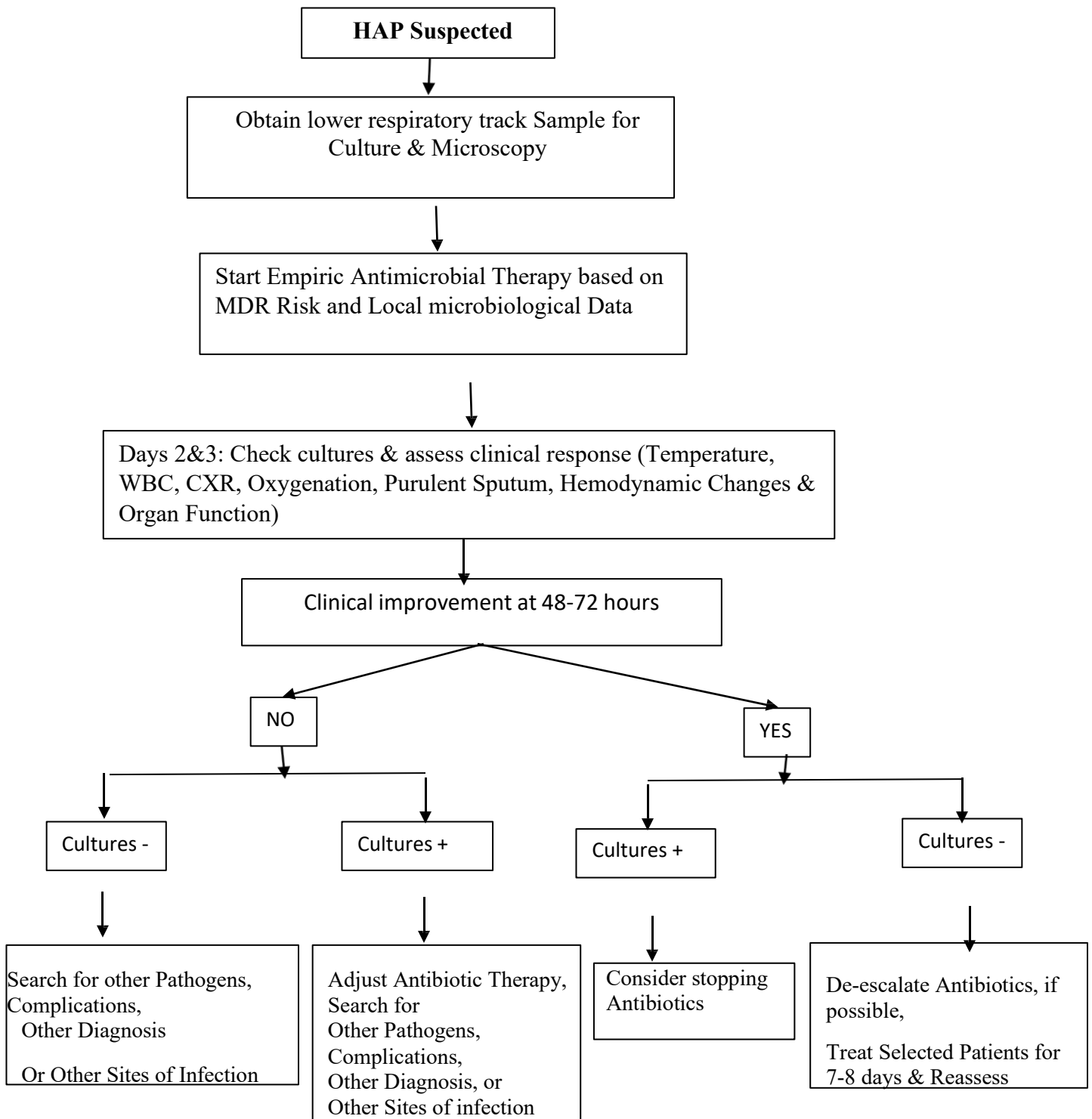
- Previous respiratory MDR pathogen isolation
- IV antibiotic use within the previous 90 days

EMPIRIC THERAPY OF HOSPITAL-ACQUIRED PNEUMONIA for 7-10 days	
No MDR risk factors	
	• Piperacillin-tazobactam 4.5 g IV every 6 hours, or • Cefepime 2 g IV every 8 hours
MDR risk factors	
	<i>ONE of the following:</i> • Piperacillin-tazobactam 4.5 g IV every six hours • Cefepime 2 g IV every eight hours • Ceftazidime 2 g IV every eight hours • Imipenem 500 mg IV every six hours • Meropenem 1 g IV every eight hours <i>PLUS, one of the following:</i> • Amikacin 15 to 20 mg/kg IV daily • Gentamicin 5 to 7 mg/kg IV daily <i>PLUS, one of the following:</i> • Linezolid 600 mg IV every 12 hours, • Vancomycin 15 mg/kg IV q8–12h with goal to target 15–20 mg/mL trough level (consider a loading dose of 25–30 mg/kg × 1 for severe illness)

5. Treatment adjustment

- The empiric regimen should be tailored to the pathogen's susceptibility pattern
- Patients who have not improved within 72 hours of starting empiric antibiotics should be evaluated for complications, other sites of infection, and alternate diagnoses.

6. Algorithm



VIII. SPECIAL CASES

- For patients with HAP who have prior intra venous antibiotic use within 90 days, a high risk for mortality, or structural lung disease (i.e., bronchiectasis or cystic fibrosis), double antipseudomonal coverage with antibiotics from two different classes should be considered.
- Either vancomycin or linezolid should be used when a patient with suspected HAP has a risk factor for MRSA infection or is at high risk for mortality.

IX. EVOLUTION AND PROGNOSIS

- The mortality associated with hospital-acquired pneumonia is high despite the availability of effective antibiotics.

X. COMPLICATIONS

- Respiratory failure
- Heart failure
- Septic shock

XI. PREVENTION AND RECOMMENDATION

- **Host related**
 - Adequate nutrition, enteral feeding via orogastric tubes
 - Reduction/discontinuation of immunosuppressive treatments
 - Prevent unplanned extubation (restraints, sedation)
 - Kinetic beds
 - Incentive spirometry, deep breathing and pain control
- **Device/treatment related**
 - Minimise use of sedatives and paralytics
 - Avoid gastric overdistention
 - Avoid intubation and reintubation
 - Expedient removal of endotracheal and nasogastric tubes
 - Semirecumbent positioning
 - Drain condensate from ventilator circuits
 - Endotracheal tube cuff pressure (>20 cmH₂O prevents leakage of bacterial pathogens around the cuff into lower respiratory tract)
 - Continuous aspiration of subglottic secretions
 - Use of heat moisture exchangers (reduces ventilator circuit colonization but not VAP incidence)

- **Environment related**

- Attention to infection-control procedures, i.e. staff education, hand washing, patient isolation
- Microbiological surveillance programmed

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OBSTRUCTIVE SLEEP APNEA

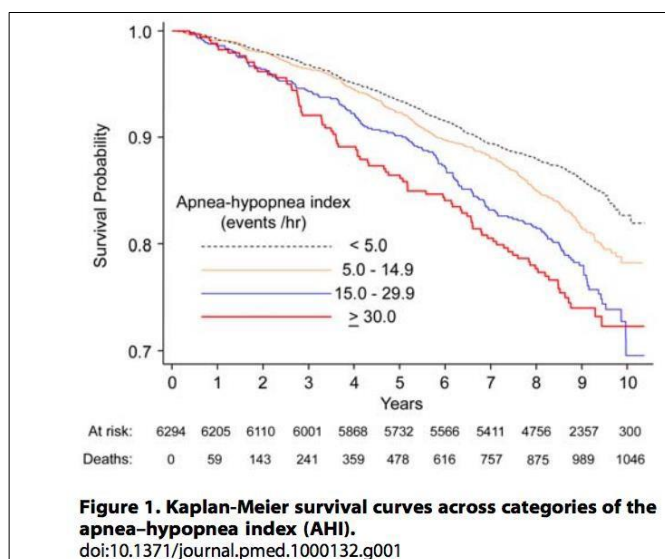
Prof. CHAN Sarin

I. DEFINITION

Obstructive sleep apnea (OSA) is a disorder that is characterized by obstructive apneas and hypopneas caused by repetitive collapse of the upper airway during sleep. The diagnosis should be considered whenever a patient presents with symptoms such as excessive daytime sleepiness, snoring, and choking or gasping during sleep, particularly in the presence of risk factors such as obesity, male gender, and advanced age. The challenge is to select the patients who are most likely to have OSA for further diagnostic evaluation, since expensive and time-consuming testing has traditionally been required to identify OSA.

II. EPIDEMIOLOGY

Obstructive sleep apnea (OSA) is a common pathology (approximately 5% of the general population). Today, in the general population, more than 2% of women and 4% of men would present with sleep apnea syndrome (SAS). The prevalence of SAS increases with age to reach a plateau after 65 years with a maximum prevalence of 10% in the general population.



Obesity is the main risk and aggravation factor for OSAS. The prevalence of OSAS increases with the severity of obesity. A weight reduction of less than 8% allows only little improvement in AHI, while greater than 10% allows an average reduction in AHI of more than 50%.

The main consequences of OSA are, in addition to excessive daytime sleepiness (part of the definition of OSA itself), associated cognitive disorders, including an increased risk of road accidents and chronic cardiovascular consequences, hypertension, coronary insufficiency, heart failure and cardiac rhythm disorders. This induces excess mortality in the general population in relation to OSA.

III. ETIOLOGY – RISK FACTORS —Most important is sleep. Clinical risk factors are advancing age, male gender, obesity, and craniofacial morphology or upper airway soft tissue abnormalities. Additional factors identified in some studies include smoking, nasal congestion, menopause, and family history. Rates of OSA are also increased in association with certain medical conditions, such as pregnancy, end-stage renal disease, congestive heart failure, chronic lung disease, post-traumatic stress disorder, and stroke.

Age — The prevalence of OSA increases from young adulthood through the sixth to seventh decade, then appears to plateau.

Gender — OSA is approximately two to three times more common in males than females, although the gap narrows at the age of menopause in women. The mechanisms for this difference are being defined.

Obesity — In both males and females, the prevalence of OSA progressively increases as the body mass index (BMI) and associated markers (eg, neck circumference, waist-to-hip ratio) increase.

Craniofacial and upper airway abnormalities — Craniofacial and upper airway soft tissue abnormalities each increase the likelihood of having or developing OSA. These factors are best recognized in Asian patients where obesity is not as obvious an issue. Examples of abnormalities include an abnormal maxillary or short mandibular size, a wide craniofacial base, tonsillar hypertrophy, and adenoid hypertrophy.

In children, adenotonsillar hypertrophy and a variety of neurological and medical conditions that can affect neural control of the upper airway or impact the collapsibility of the upper airway, such as cerebral palsy and low birth weight, are recognized risk factors for OSA.

Other risk factors — Nasal congestion confers an approximately twofold increase in the prevalence of OSA compared with controls, regardless of cause. This is probably related to increased resistance due to decreased nasal patency; however, OSA may or may not improve with correction of nasal congestion.

Medical conditions — The prevalence of OSA is increased in patients with a variety of medical conditions, including the following:

- Pregnancy
- Congestive heart failure
- End-stage renal disease
- Chronic lung disease, including asthma [20,21], chronic obstructive pulmonary disease (COPD) [22] and idiopathic pulmonary fibrosis
- Stroke and transient ischemic attacks
- Acromegaly
- Hypothyroidism
- Polycystic ovary syndrome
- Post-traumatic stress disorder

IV. PATHOPHYSIOLOGY

OSA is characterized by recurrent, functional collapse during sleep of the velopharyngeal and/or oropharyngeal airway, causing substantially reduced or complete cessation of airflow despite ongoing breathing efforts. This leads to intermittent disturbances in gas exchange (eg, hypercapnia and hypoxemia) and fragmented sleep.

V. DIAGNOSIS

1. Clinical presentation:

Most patients with OSA first come to the attention of a clinician because the patient complains of daytime sleepiness, or the bed partner reports loud snoring, gasping, snorting, or interruptions in breathing while sleeping.

2. Signs and symptoms

Daytime sleepiness, distinct from fatigue, is a common feature of OSA. Sleepiness is the inability to remain fully awake or alert during the wakefulness portion of the sleep-wake cycle, while fatigue is a subjective lack of physical or mental energy that is perceived by the individual or caregiver to interfere with usual and desired activities. It is often unclear whether a patient's complaint of daytime sleepiness represents true sleepiness or fatigue. In such cases, series of directed questions can be combined with the Epworth Sleepiness Scale to quantitatively document the patient's perception of sleepiness, fatigue, or both.

Daytime sleepiness may go unnoticed or its significance may be underestimated because of its insidious onset and chronicity. The patient may not describe the symptom as sleepiness, but may use other terms, such as fatigue, tiredness, and low energy. Careful questioning of the patient typically reveals a pattern of feeling sleepy or falling asleep in boring, passive, or monotonous situations. As an example, the patient may admit to consistently falling asleep while reading, watching television, or even while operating a motor vehicle.

In addition, embarrassing or inappropriate episodes of sleep (eg, at religious services, while driving) may be reported. Reviewing patient behavior away from the workplace is essential because daytime sleepiness can be masked by activity. Patients should always be asked about behaviors that may mask sleepiness, such as caffeine consumption.

Snoring is the other common feature of OSA. While snoring is associated with a sensitivity of 80 to 90 percent for the diagnosis of OSA, its specificity is below 50 percent. Moreover, patients with mild snoring and a body mass index (BMI) lower than 26 are unlikely to have moderate or severe obstructive sleep apnea. It is usually helpful to have the patient's bed partner or a family member present during the interview because they may have greater insight than the patient into the frequency and severity of the snoring, as well as associated events (ie, resuscitative gasping or snorting, witnessed apneic periods, periods of silence followed by loud snoring, restless or fitful sleep), since these occur during sleep and may help increase the likelihood that snoring is associated with OSA.

Other associated symptoms and historical features include the following:

- Awakening with a sensation of choking, gasping, or smothering
- Awakening with a dry mouth or sore throat
- Moodiness or irritability
- Lack of concentration
- Memory impairment
- Decreased libido and impotence
- Nocturia
- Awakening with angina pectoris
- History of hypertension, cardiovascular disease, cerebrovascular disease, or renal disease
- History of type 2 diabetes mellitus
- Depression
- Symptoms of fibromyalgia
- Gastroesophageal reflux disease (GERD)
- History of polycystic ovary syndrome.

3. Physical examination

OSA is most common among males who are 18 to 60 years old, although it is also common at other ages and in women. The physical exam is frequently normal, except for obesity (body mass index >30 kg/m²) and a crowded oropharyngeal airway. The obesity may be only moderate, since up to 40 percent of patients are less than 20 percent above their ideal body weight.

Additional physical findings that are common among patients with OSA include the following:

- Narrow airway – Numerous conditions can narrow the upper airway. These include retrognathia, micrognathia, lateral peritonsillar narrowing, macroglossia, tonsillar hypertrophy, an elongated or enlarged uvula, a high arched or narrow palate, nasal septal deviation, and nasal polyps. The modified Mallampati classification is commonly used to quantify airway narrowing, with classes 3 and 4 considered positive for airway narrowing. Both the Mallampati classification and Friedman tongue position have been shown to correlate with OSA severity.
- Large neck and/or waist circumference – OSA is more strongly correlated with an increased neck size or waist circumference than general obesity. OSA is particularly prominent among men who have a collar size greater than 17 inches and women who have a collar size greater than 16 inches.
- Elevated blood pressure – Approximately 50 percent of patients with OSA have coexisting hypertension, which is often most elevated in the morning. Similarly, patients with resistant hypertension have a very high prevalence of OSA.
- Signs of pulmonary hypertension or cor pulmonale (eg, peripheral edema, jugular venous distension) – Pulmonary hypertension and cor pulmonale are common sequelae when OSA coexists with either obesity hypoventilation syndrome or an alternative cause of daytime hypoxemia (eg, chronic lung disease).

OSA may also be associated with nocturnal cardiac dysrhythmias, including bradycardia, atrial fibrillation, and asystole.

4. Screening questionnaires

Most individual signs and symptoms have limited utility in determining the likelihood of OSA, and no one clinical feature is sufficiently sensitive or specific to effectively rule in or rule out the diagnosis.

A variety of clinical prediction rules, questionnaires, and scores have been evaluated using common signs and symptoms of OSA that are easily obtained and interpreted in the primary care setting. While these tools may have some value in detecting OSA in highly symptomatic patients in a high-risk setting (eg, preoperative evaluation), they have not been adequately tested as a screening tool in asymptomatic patients and are not recommended for diagnosis.

One major limitation is that most scores have been derived from sleep center or preoperative referral-based populations (ie, symptomatic patients), which may have a higher prevalence of OSA; for this reason, the sensitivity and specificity of most instruments are likely overestimated in relation to their value in an unselected asymptomatic community-based population.

The high false positive rate of questionnaires diminishes their diagnostic value such that when the score is high, patients do not always have OSA; in contrast, when the score is low, patients are unlikely to have OSA. Thus, questionnaires are generally only helpful when they are negative (ie, low score). Examples of such questionnaires include the following:

- STOP-Bang

- Sleep apnea clinical score (SACS)
- Berlin questionnaire
- The NoSAS score
- Multivariable Apnea Prediction (MVAP) instrument – The MVAP instrument is based on a formula consisting of three questions about the frequency of symptoms of sleep apnea, along with BMI, age, and sex. MVAP values ranged from 0 to 1, with 1 representing the highest likelihood of sleep apnea.

A. INDICATIONS FOR DIAGNOSTIC TESTING

Objective diagnostic testing is necessary to diagnose OSA because the clinical features are nonspecific and the diagnostic accuracy of clinical impression alone is poor. The American Academy of Sleep Medicine clinical practice guideline advocates that diagnostic testing be performed in conjunction with a comprehensive sleep evaluation and adequate follow-up.

Diagnostic testing for OSA should be performed on any patient with unexplained excessive daytime sleepiness, which is the clinically relevant symptom of OSA that is most responsive to treatment. In the absence of excessive daytime sleepiness, we typically pursue diagnostic testing if the patient snores and either works in a mission-critical profession (eg, airline pilots, bus drivers, and truck drivers) or has two or more additional clinical features of OSA. This approach is illustrated in the algorithm and assumes that the excessive daytime sleepiness and clinical features are not definitively explained by an alternative etiology.

For patients who are not known to snore and do not complain of excessive daytime sleepiness, we suggest a low threshold for diagnostic testing in certain high-risk groups, such as patients with resistant hypertension, unexplained pulmonary hypertension, or secondary polycythemia, particularly if they are obese or have other clinical features of OSA.

B. DIAGNOSTIC TESTS

For most patients selected for diagnostic testing, we suggest full-night (ie, diagnostic only) or split-night (ie, diagnostic and therapeutic), attended, in-laboratory polysomnography. Unattended, home sleep apnea testing (HSAT) with a technically adequate device is a reasonable alternative for patients without comorbidities (ie, uncomplicated OSA) and in whom there is a high likelihood of moderate or severe OSA. These diagnostic tests are summarized briefly below and discussed in more detail separately.

1. Polysomnography

An attended, in-laboratory polysomnography is considered the gold-standard diagnostic test for OSA. Diagnostic polysomnography, rather than home testing, is particularly preferred in patients with suspected complicated OSA (eg, patients with comorbidities such as significant cardiorespiratory disease), suspected sleep disordered breathing other than OSA (eg, narcolepsy), suspected mild OSA, and mission-critical workers (eg, airline pilots).

2. Home sleep apnea testing

There are a variety of devices that are used for in-home, unattended monitoring of cardiorespiratory parameters during sleep. HSAT should be used for the diagnosis of OSA in patients with a high pre-test probability of moderate to severe uncomplicated OSA, (ie, absence of medical comorbidities such as significant cardiorespiratory disease or other significant disorders [eg, narcolepsy and idiopathic hypersomnia]). Risk of moderate to severe OSA is indicated by the presence of daytime hypersomnolence and at least two of

the following three criteria: habitual loud snoring, witnessed apnea or gasping/choking, or diagnosed hypertension.

Choosing in-laboratory versus home testing in this population may also be influenced by environmental or personal factors (eg, inability to access a laboratory or home care company, cost, personal preference). Should a single HSAT be negative, inconclusive, or technically inadequate, and the suspicion remain for OSA, HSAT should not be repeated and in-laboratory polysomnography should be performed; it is rare that a repeat HSAT is indicated. Further details regarding the patients that are suitable or not suitable for HSAT are provided separately.

Many HSAT devices have been validated against standard polysomnography, typically by testing the same patient with both modalities in the sleep laboratory. Generally speaking, the sensitivity and specificity seem to be high in populations at high risk of moderate to severe OSA on the basis of clinical symptoms, assuming there are no significant comorbid medical disorders or sleep disorders.

3. In-hospital testing

In response to increased requests for sleep medicine consultation, many hospital-based centers and respiratory departments now provide bedside monitoring for inpatients with a high pretest suspicion for sleep apnea. However, there are few studies examining the accuracy of such studies in the hospital, given the higher rates of co-morbidities, and whether patient outcomes are modified by earlier detection of OSA in the hospital setting. One study in hospitalized cardiac patients demonstrated significantly fewer readmissions and emergency department visits at 30 days in those undergoing bedside portable studies followed by adherence to positive airway pressure after discharge.

C. DEFINITIVE DIAGNOSIS

The diagnosis of OSA is based upon the presence or absence of related symptoms, as well as the frequency of respiratory events during sleep (ie, apneas, hypopneas, and respiratory effort-related arousals [RERAs]) as measured by polysomnography or home sleep apnea testing (HSAT). In adults, the diagnosis of OSA is confirmed if either of the two conditions exists.

- There are five or more predominantly obstructive respiratory events (obstructive and mixed apneas, hypopneas, or RERAs) per hour of sleep (for polysomnography) or recording time (for HSAT) in a patient with one or more of the following:
 - Sleepiness, nonrestorative sleep, fatigue, or insomnia symptoms
 - Waking up with breath holding, gasping, or choking
 - Habitual snoring, breathing interruptions, or both noted by a bed partner or other observer
 - Hypertension, mood disorder, cognitive dysfunction, coronary artery disease, stroke, congestive heart failure, atrial fibrillation, or type 2 diabetes mellitus
- There are 15 or more predominantly obstructive respiratory events (apneas, hypopneas, or RERAs) per hour of sleep (for polysomnography) or recording time (for HSAT), regardless of the presence of associated symptoms or comorbidities.

Whether the apnea-hypopnea index ($AHI = \text{apneas} + \text{hypopneas} / \text{total sleep time in hours}$) or the respiratory disturbance index ($RDI = \text{apneas} + \text{hypopneas} + \text{RERAs} / \text{total sleep time in hours}$) should be the gold standard index remains unclear and varies across insurance carriers. Outcome studies are needed to determine which index is preferred. The main

implication of the difference is that the RDI classifies more patients as having OSA than does the AHI, using the same threshold values.

In addition, most HSAT devices do not include EEG monitoring, and therefore RERAs and hypopneas characterized by arousals cannot be identified. Accordingly, the number of respiratory events per hour of recording time rather than total sleep time is used to generate the preferred terminology, respiratory event index (REI).

In validated portable monitors, the REI correlates well with AHI and RDI. Published reports have used a cutoff REI >15 events/hour to define pathology, but outcome studies are lacking. Evidence does show that the length of the recording time used to generate the REI with HSAT should be at least 300 minutes.

Apneas, hypopneas, RERAs, AHI, and RDI are defined separately.

D. DISEASE SPECTRUM

Patients who meet criteria for a diagnosis of OSA are traditionally classified as having mild, moderate or severe disease on the basis of symptoms and the apnea-hypopnea index (AHI). Although the AHI alone is not entirely predictive of the presence and severity of symptoms and complications, the following is a general description of patients in each category.

Mild — Patients traditionally classified as having mild OSA are those with an AHI between 5 and 15 respiratory events per hour of sleep. Such patients may be relatively asymptomatic or report sedentary (ie, passive) daytime sleepiness, becoming noticeable once the patient is unstimulated. The daytime sleepiness often does not impair daily life, although it may be recognized by family members. Alternatively, daytime sleepiness may become apparent to the patient only after it improves due to weight loss, alcohol abstinence, or treatment of OSA. The sleep stages and slow wave sleep are generally preserved in mild OSA, and cardiovascular consequences such as systemic hypertension, cor pulmonale, and polycythemia is generally absent.

Moderate — Patients traditionally classified as having moderate OSA are those with an AHI between 15 and 30 respiratory events per hour of sleep. Such patients are typically aware of daytime sleepiness and take steps to avoid falling asleep at inappropriate times (eg, taking a nap or avoiding driving long distances). They are able to continue their daily activities, but at reduced levels, and they may have an increased incidence of motor vehicle violations or accidents. Systemic hypertension may co-exist. Sleep fragmentation is observed in moderate OSA, but the progression of sleep stages is better conserved than with severe disease.

Severe — Patients classified as having severe OSA are those with an AHI greater than 30 respiratory events per hour of sleep and/or an oxyhemoglobin saturation below 90 percent for more than 20 percent of the total sleep time. Such patients more often have daytime sleepiness that interferes with normal daily activities. They tend to fall asleep often during the day (and in a sitting posture) and are at risk for accidental injury from sleepiness. Patients with severe OSA are at increased risk for all-cause mortality and a variety of cardiovascular comorbidities, including hypertension, coronary artery disease, and arrhythmias.

VI. DIFFERENTIAL DIAGNOSIS

A variety of conditions must be considered in the differential diagnosis of OSA because they similarly cause excessive daytime sleepiness due to disrupted sleep. These conditions can be

distinguished from OSA via polysomnography:

- Periodic limb movements of sleep – Periodic limb movements of sleep (PLMS) are recurrent jerks of the legs and arms that can fragment sleep, leading to daytime sleepiness.
- Rotating shift workers – Night-shift workers obtain approximately seven hours per week less sleep than non-shift workers. They often revert to a daytime schedule on their leisure days, which adds to their sleep deprivation and daytime sleepiness.
- Narcolepsy – Narcolepsy is a clinical syndrome of excess daytime sleepiness with cataplexy, hypnagogic hallucinations, and sleep paralysis. "Sleep attacks" are classically described, although there is a wide spectrum of clinical presentations.
- Upper airway resistance syndrome – Upper airway resistance syndrome (UARS) was previously classified as an independent disorder, but it is now considered a type of OSA. It exists when airflow limitation due to increased upper airway resistance (ie, respiratory effort related arousals [RERAs]) induces arousals from sleep, leading to excessive daytime sleepiness. Patients with UARS have few discrete respiratory disturbances (ie, apneas or hypopneas) or episodes of desaturation. It is common in thin women with certain craniofacial abnormalities.
- Central sleep apnea – Central sleep apnea is less common than OSA, but it similarly causes symptoms of disrupted sleep due to episodic apneas and hypopneas during sleep. The primary difference between OSA and central sleep apnea is that respiratory effort is present in the former but not the latter.
- Respiratory diseases – Patients with either chronic obstructive lung disease or restrictive lung disease may have sleep-related desaturation, resulting in sudden awakenings and dyspnea that imitate OSA. Patients with neuromuscular disease depend on their accessory muscles to breathe and the inhibition of muscle activity during REM sleep can cause profound desaturations and awakenings. Poorly controlled asthma is often worse at night, with nocturnal bronchospasm and cough inducing sleep fragmentation and paroxysmal dyspnea.
- In addition, there are conditions that should be considered in the differential diagnosis of OSA because they similarly cause abrupt awakenings or abnormal sounds or sensations during sleep. These conditions can be distinguished from OSA via polysomnography:
- Primary snoring – Snoring is far more common than OSA. Therefore, even though most patients who have OSA snore, most patients who snore do not have OSA.
- Gastroesophageal reflux disease – Gastroesophageal reflux disease (GERD) can mimic OSA by producing a choking sensation and dyspnea. In addition, GERD may improve with positive airway pressure therapy, further mimicking OSA.
- Miscellaneous – Other disorders that can mimic OSA include swallowing disorders, nocturnal seizures, and psychiatric illnesses such as panic attacks.

VII. MANAGEMENT

1. **Objectives** --The treatment of OSAS aims to maintain the permeability of the upper airways during sleep.
2. **Critical sign**–Clinical symptoms, at least three of the following symptoms:
 - daytime sleepiness,
 - severe and daily snoring,
 - feelings of choking or suffocation during sleep,

- daytime fatigue,
- nocturia,
- morning headaches.

3. Mortality risk factor

- Drowsy driving and motor vehicle crashes
- Neuropsychiatric dysfunction: worse inattention, worse memory and cognitive deficits
- Cardiovascular and cerebrovascular morbidity
- Pulmonary hypertension or right heart failure
- Metabolic syndrome and type 2 diabetes
- Nonalcoholic fatty liver disease (NAFLD)
- Miscellaneous

4. Out patient management: Home sleep apnea testing

There are a variety of devices that are used for in-home, unattended monitoring of cardiorespiratory parameters during sleep. HSAT should be used for the diagnosis of OSA in patients with a high pre-test probability of moderate to severe uncomplicated OSA, (ie, absence of medical comorbidities such as significant cardiorespiratory disease or other significant disorders [eg, narcolepsy and idiopathic hypersomnia]).

Risk of moderate to severe OSA is indicated by the presence of daytime hypersomnolence and at least two of the following three criteria: habitual loud snoring, witnessed apnea or gasping/choking, or diagnosed hypertension.

Choosing in-laboratory versus home testing in this population may also be influenced by environmental or personal factors (eg, inability to access a laboratory or home care company, cost, personal preference). Should a single HSAT be negative, inconclusive, or technically inadequate, and the suspicion remain for OSA, HSAT should not be repeated and in-laboratory polysomnography should be performed; it is rare that a repeat HSAT is indicated. Further details regarding the patients that are suitable or not suitable for HSAT are provided separately.

Many HSAT devices have been validated against standard polysomnography, typically by testing the same patient with both modalities in the sleep laboratory. Generally speaking, the sensitivity and specificity seem to be high in populations at high risk of moderate to severe OSA on the basis of clinical symptoms, assuming there are no significant comorbid medical disorders or sleep disorders. The idea that HSAT is a viable alternative to in-laboratory polysomnography in selected individuals is supported by the following:

One unblinded trial randomly assigned 373 patients with suspected moderate to severe OSA to either in-laboratory polysomnography with conventional continuous positive airway pressure (CPAP) titration or HSAT with autotitrating CPAP and then transition to fixed CPAP.

Among patients who had moderate to severe OSA (ie, AHI >15 events per hour), there were no differences in the number of effective titrations, titration pressures, time to treatment, or daytime sleepiness. Patients who underwent HSAT with autotitrating CPAP were more likely to use CPAP for ≥ 4 hours at night.

5. First line treatment- Continuous positive pressure

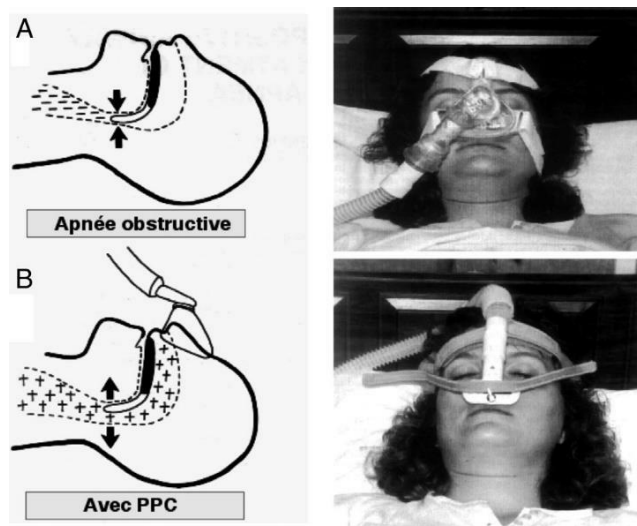
The only treatment that has demonstrated efficacy in moderate and severe sleep apnea syndrome is CPAP. This is true for drowsiness, cognitive disorders, quality of life and reduction of blood pressure. CPAP corrects nocturnal breathing abnormalities; attention disorders and restores normal vigilance.

Continuous positive airway pressure is therefore the reference treatment for obstructive sleep apnea syndrome. It consists of putting the upper airways under a pressure of around 5 to 15 cm of water, most often using a nasal, facial or nasal mask (Figure 10.4). This sub-atmospheric pressure causes an enlargement of the pharyngeal volume and prevents expiratory collapse by creating a pneumatic splint. It also increases the lung volume and therefore the body's oxygen reserves.

CPAP, by normalizing ventilation during sleep, suppresses the bursts of sympathetic hyperactivity present at the end of each respiratory event and brings the mean level of sympathetic activity closer to that found in normal subjects. Concerning hypertension, placebo-controlled studies have shown reductions in mean daytime arterial pressure of approximately 2 mmHg. These results depend on the severity of OSA, and in particular on the importance of nocturnal desaturations, the duration of CPAP use but also on the percentage of hypertensives in the population studied and the severity of hypertension.

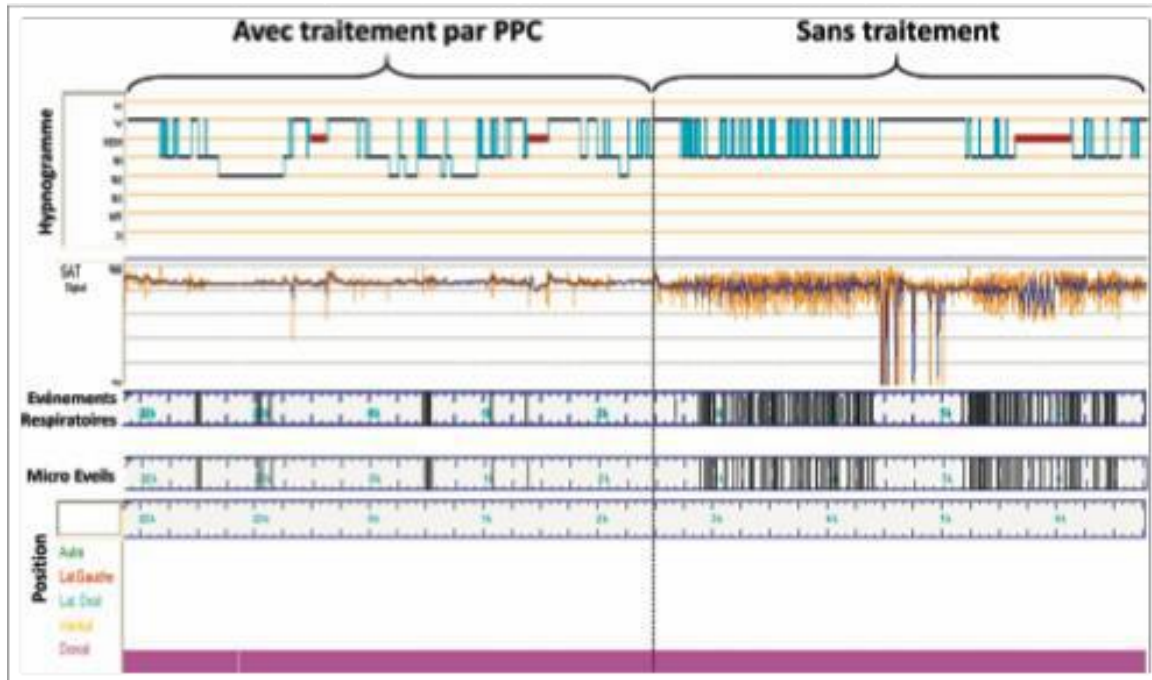
However, even in a population without comorbidity, not previously treated for hypertension, in a study comparing an angiotensin receptor blocker (valsartan) and CPAP, we demonstrated that the effect of CPAP was modest (-2 mmHg) compared to the effect of valsartan (-9 mmHg).

Initial education on treatment is one of the essential determinants of long-term compliance. Explanation of the causes of the disease and the interest in treating OSA to improve quality of life and limit comorbidities, the choice of mask and good prevention of side effects are essential elements. The majority of treatment discontinuations are early. The physicians in charge of the patient, including the general practitioner, as well as the service provider, must be available and organize facilitated exchanges during this period.



Mode action de la pression positive continue.

Un masque est relié par un tuyau à une turbine générant une pression comprise entre 5 et 15 cmH₂O le plus souvent. Cette pression constante tout au long du cycle ventilatoire est soit fixe sur l'ensemble de la nuit soit variable en fonction des événements détectés (apnées, hypopnées). Ceci permet de prévenir la fermeture du pharynx. Il convient d'appliquer ce traitement chaque nuit de sommeil et à vie, puisque l'interruption de celui-ci fait réapparaître quasi immédiatement le syndrome d'apnées.



Efficacité de la pression positive continue

L'amélioration clinique en rapport avec le traitement du SAOS est dépendante de la normalisation complète de la ventilation et de la structure du sommeil. Sur cet exemple comportant de haut en bas l'hypnogramme, la saturation en oxygène, les événements respiratoires et les micro-éveils corticaux, l'arrêt de la pression positive continue en milieu de nuit non seulement s'accompagne d'un retour des événements respiratoires mais également d'une altération de la micro et macro-structure du sommeil.

Titration or determination of the effective pressure that corrects all respiratory events is the first step in management. In France, it is most commonly done at home over several nights (1 to 4 weeks) using self-controlled positive pressure that detects residual respiratory events and delivers a variable pressure. It can also be performed in the laboratory with the same type of equipment, over a single night, under the cover of a polysomnographic recording.

Analysis of the curves provided is important because it allows invalid periods to be identified and the type of residual anomalies to be judged if they exist. Self-titration at home is effective, shortens the time to implement CPAP and does not change the failure rate or compliance in the medium term. Once the effective pressure has been determined, patients will generally continue treatment at home at a fixed pressure that corresponds to the 95th percentile of the pressure curve during titration.

Patients with positional SAS, or showing a large variability of pressure requirements during the titration period, will continue at home with a self-controlled variable pressure machine. Self-controlled machines, whether for titration or long-term treatment, should not be used in patients with heart failure or with episodes of alveolar hypoventilation during sleep (COPD, obesity class II and III). In the first case, pressure variations have potentially harmful hemodynamic consequences in heart failure but above all the risk of error in detection of central

versus obstructive events is significant. In the second case, the machines do not detect nocturnal hypoventilation and tend to lower the pressure, particularly in REM sleep.

The rate of initial refusals at the time the treatment is proposed or at the end of the titration period is 15 to 20%, including in highly trained teams. These initial refusals occurring when good education on the treatment and good medical-technical management have been carried out are most often explained by an unfavorable psychological or family context. The side effects (noise, irritation next to the mask, congestion of the machine, nasal intolerance), if they are not managed correctly, can reduce therapeutic compliance. They are extremely frequent (50% of cases).

The general practitioner is often consulted for nasal intolerance that can be confused with infectious phenomena of the ENT sphere. This involves clear rhinorrhea and bursts of sneezing predominating until the end of the morning and associated with nasal obstruction. Good management involves limiting leaks by replacing the nasal mask with a face mask or using a chin strap and, if necessary, adding a heated humidifier. It is also important that the room is sufficiently heated (at least 18 degrees) in winter to limit this side effect.

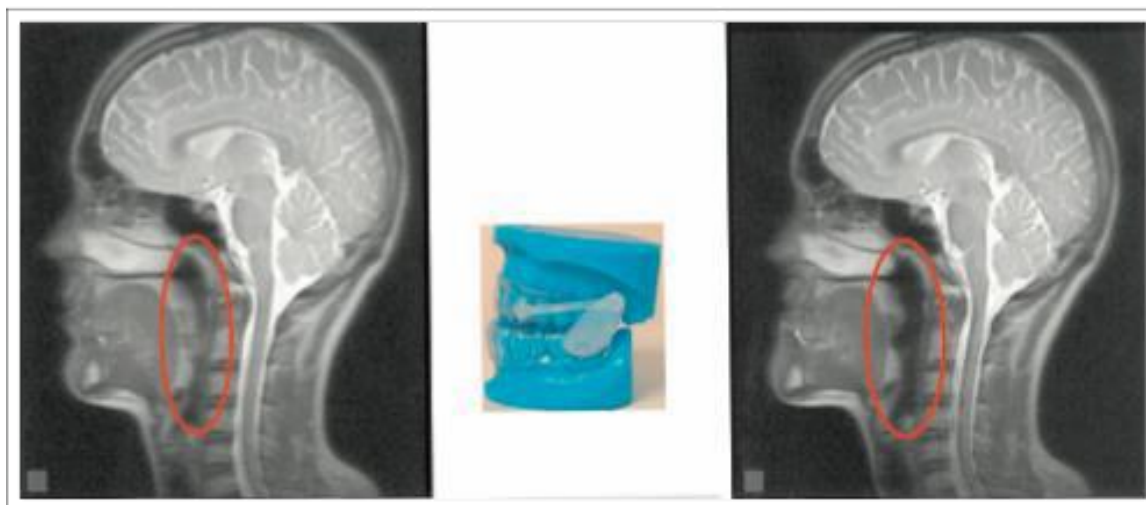
Compliance reports from the provider can help the practitioner assess the exact use of CPAP. In medicine, this is one of the only treatments for which we have a precise and objective idea of the use of the therapy. These compliance reports should be discussed with the patient: use of CPAP during naps, on weekends, when traveling, etc. The most frequently asked question is the minimum effective duration.

Mandibular advancement device

The mandibular advancement device (MAD) is the main alternative to CPAP. The mechanical principle of the MAD is to rely on the maxilla to maintain forced propulsion of the mandible. Although generating dental and temporomandibular joint constraints, this treatment allows velopharyngeal widening and therefore a reduction in upper airway resistance. This treatment, although interesting, is less effective than CPAP, including for moderate OSA. This is why the propulsion device is rather reserved for subgroups of patients (patients with moderate OSA, not overweight, in dental class II and sometimes refusing CPAP).

The opinion of the treating dentist, or even the stomatologist, is essential in order to ensure beforehand that the dentition is likely to receive this type of orthosis (sufficient number of teeth, verification of the absence of dental mobility) and that the temporomandibular joints are free of any progressive pathology.

Periodontal problems inducing dental mobility or propulsion of less than 6 mm do not allow this treatment to be implemented. Clear information on the risk of modification of the occlusion must be provided to patients.



Orthèse d'avancée mandibulaire.

Illustration de l'effet anatomique obtenue avec une orthèse d'avancée mandibulaire.

Treatment adjuvant

Weight loss

Weight loss in the field of OSA as in obesity in general is difficult to achieve. However, it allows to reduce the severity of OSA but recovery is nevertheless exceptional.

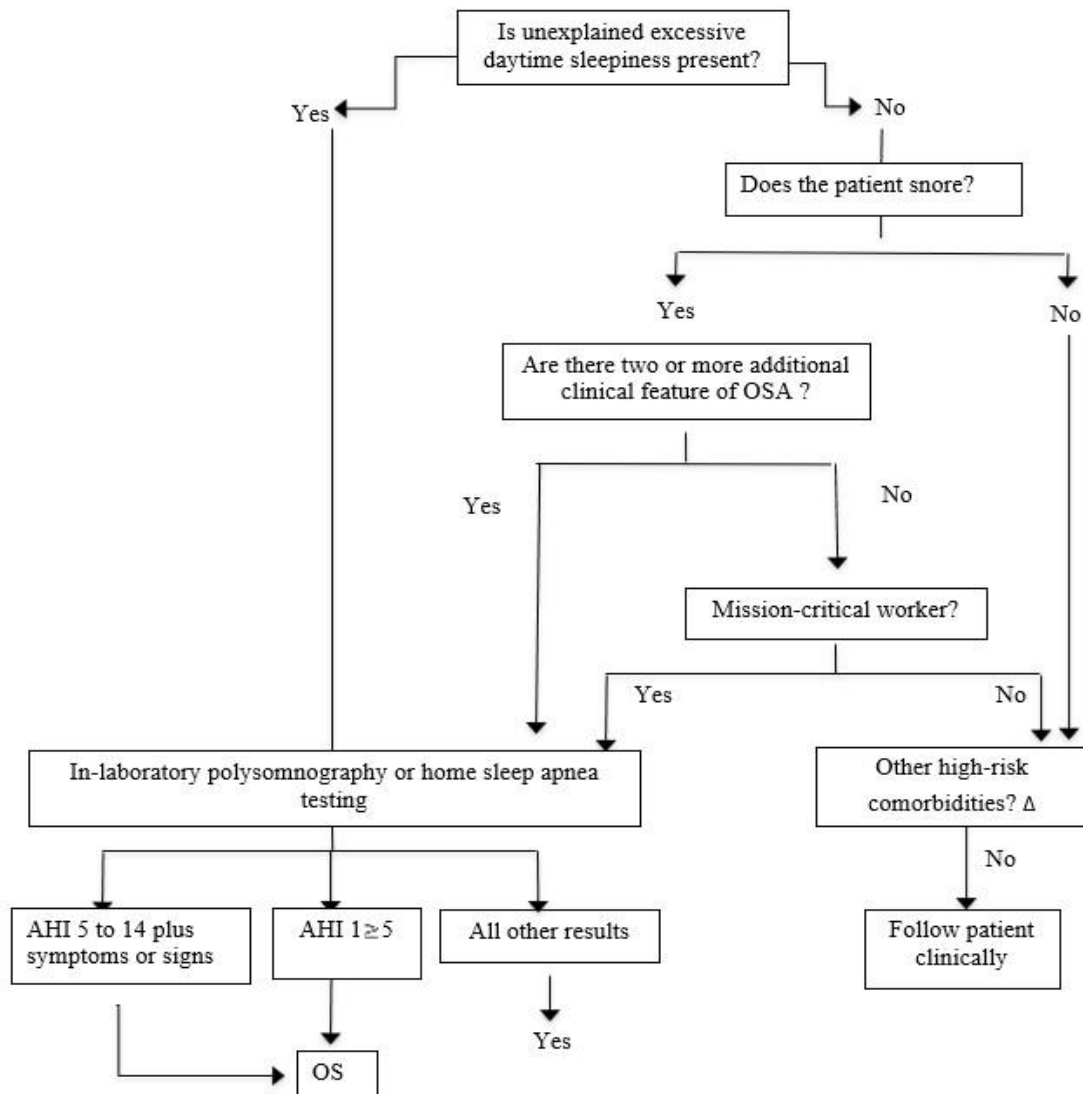
Obesity surgeries, such as gastroplasty, are sometimes indicated in cases of massive obesity ($\text{BMI} \geq 40 \text{ kg/m}^2$) or severe obesity ($\text{BMI} \geq 35 \text{ kg/m}^2$) with comorbid factors threatening the life or functional prognosis, having taken care to eliminate contraindications such as serious psychiatric disorders and compulsive eating disorders in particular.

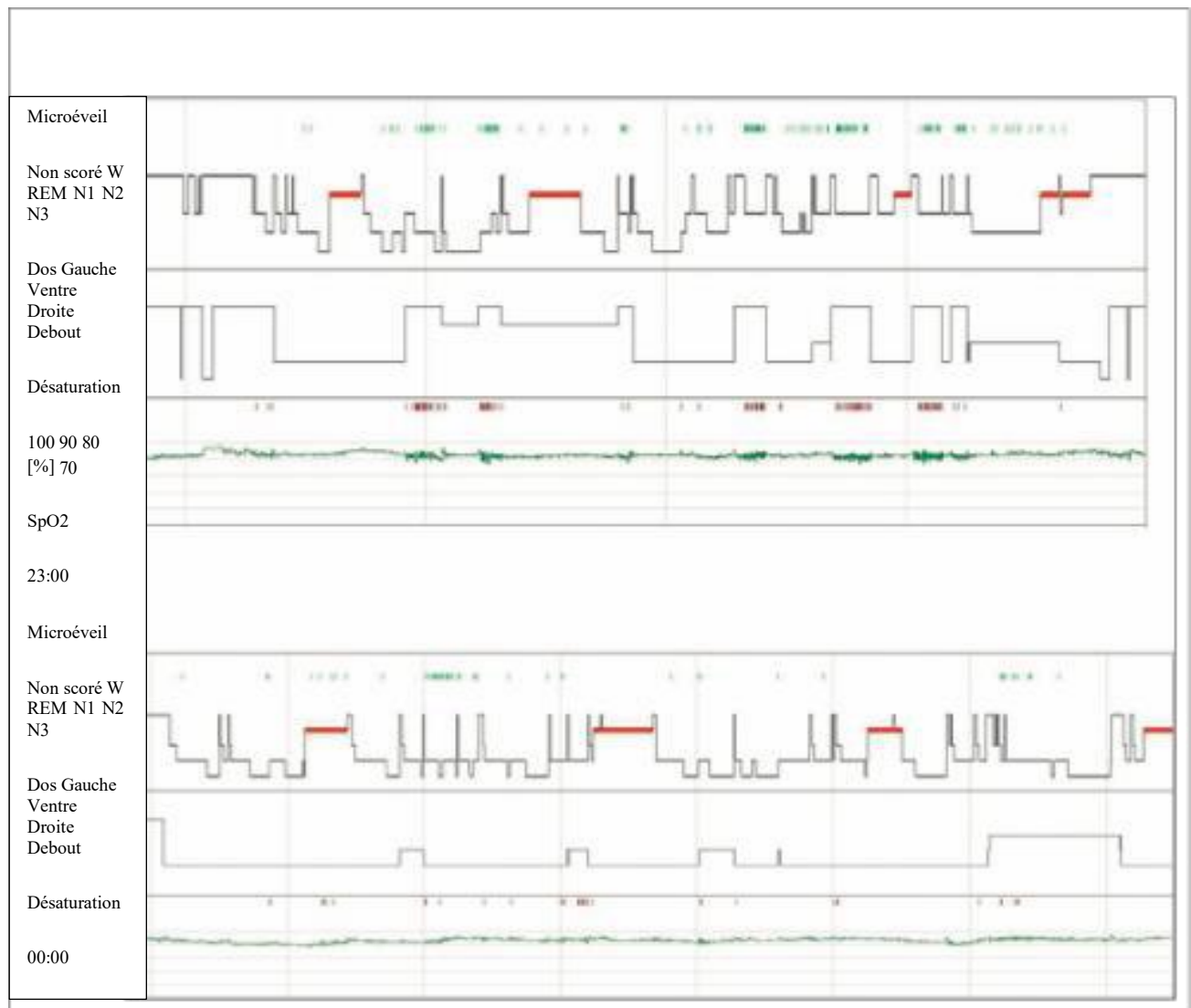
In all cases, any obese apneic patient must benefit from comprehensive nutritional management. On the other hand, before obesity surgery, sleep apnea syndrome must be sought and treated to avoid perioperative anesthetic complications. Significant weight gain in a patient with OSA treated with CPAP should lead to a reassessment of the effective pressure, which may become higher. Significant weight loss may lead to a reassessment of the severity of OSA by polysomnographic recording after a few nights of stopping CPAP treatment. In these situations, the general practitioner and the various stakeholders in the patient's care should contact the provider and the sleep specialist who are following the patient to make the necessary adjustments.

Positional treatment

It consists of imposing the lateral decubitus sleeping position. Its effectiveness is comparable to that of CPAP in cases of exclusive positional OSA proven by PSG. This treatment can be applied using vests specially designed to make the dorsal decubitus difficult or uncomfortable or by "homemade" means (design of pockets in which tennis balls are added opposite the spine). The effectiveness of this treatment must be verified by polysomnographic recording. The long-term use of this type of treatment remains difficult to maintain, being very dependent on the patient and their adherence to the treatment.

Algorithm





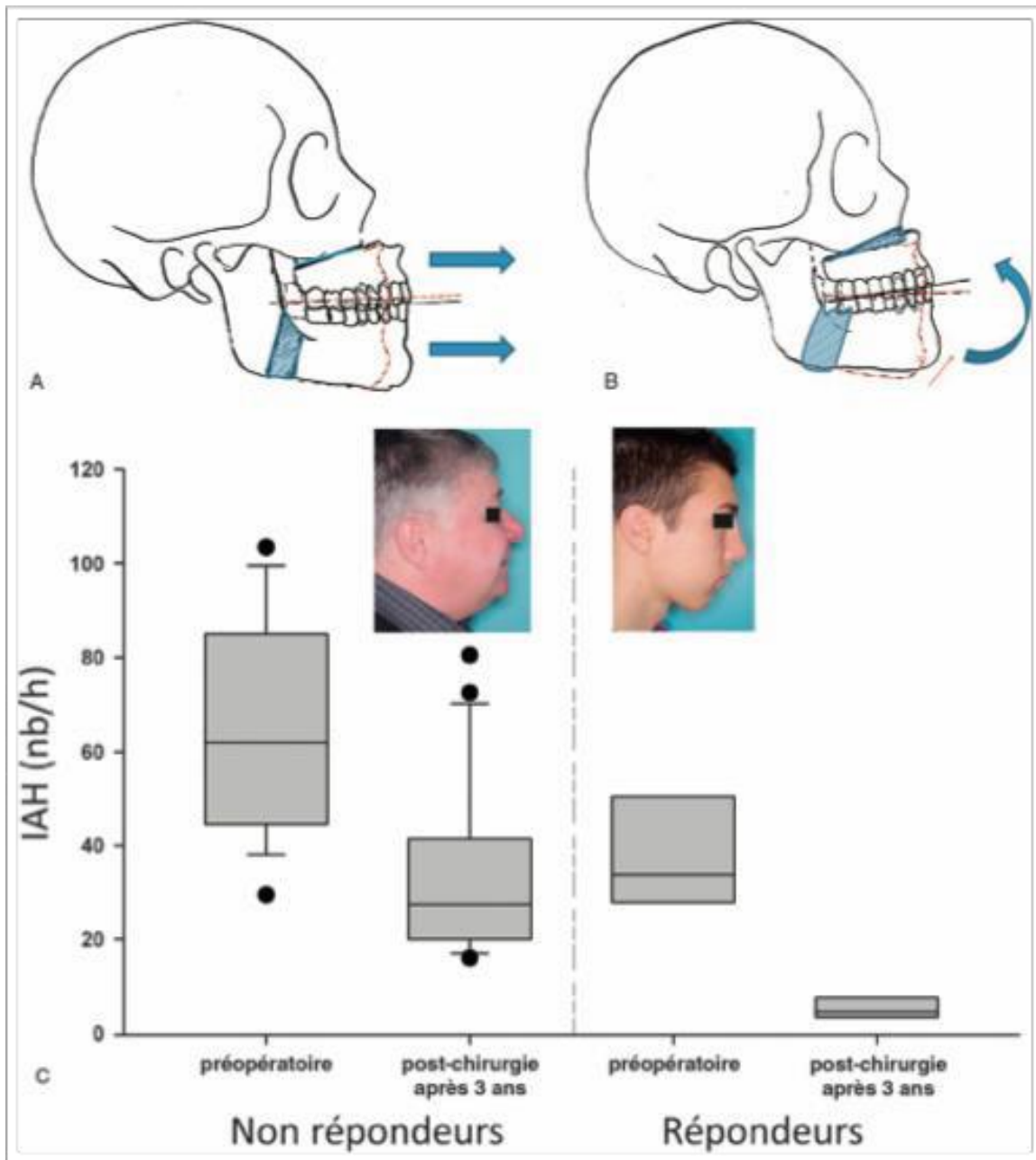
VIII. SPECIAL CASE

Surgical treatment of sleep apnea syndrome

Bimaxillary advancement osteotomy

This treatment is currently the surgical treatment providing the most guarantees in terms of efficacy (80–90%). Its indications are limited to severe OSA in young subjects without associated pathology with retrusive dysmorphosis. It is recommended to inform patients of the risk of facial morphological changes after maxillary advancement surgery.

Responding patients more than 3 years after surgery were, at the time of surgery, aged less than 45 years, had a BMI less than 25 kg/m² and a maxillofacial configuration favoring nocturnal respiratory disorders (narrow retrobasilingual space (< 8 mm) and a sella turcica-Nasion-Menton angle in lateral cephalometry less than 75°).



Bimaxillary advancement osteotomy.

A = Le Fort I and sagittal mandibular osteotomy (initial technique); B = Anterior maxillary impaction and mandibular rotation (current technique); C: Individualization of patient phenotypes according to long-term response to surgery.

Nasal surgery

Nasal surgery is not a treatment for sleep apnea syndrome. However, nasal intolerance and nasal obstruction are factors that limit the use of CPAP. After performing appropriate medical-technical management (see above) and proposing medical treatment for this nasal obstruction, an ENT opinion should be sought to indicate possible nasal surgery.

Hypoglossal nerve stimulation.

Stimulation of the hypoglossal nerve which controls the activity of the pharyngeal dilator muscles will potentially have a place in a well-selected subgroup of patients refusing CPAP.

IX. VOLUTION AND PROGNOSIS

- All apply therapy improves indices of OSA on objective sleep testing, although PAP therapy appears to be superior for this outcome (specially the AHI). Oral appliances also improve symptoms of sleepiness and blood pressure control to a degree that is similar to CPAP
- Patient who achieves gold
- Patient who does not achieve gold: Patients who fail, do not tolerate, or decline PAP
- Persistent sleepiness- Patient with persistent sleepiness who have adequately treated OSA and a whom other causes of daytime sleepiness have been reasonably excluded may be candidates for wakefulness promoting agents such as modafinil, armodafinils.

X. COMPLICATIONS

Patients with OSA are at increased risk for several adverse clinical outcomes.

- Drowsy driving and motor vehicle crashes
- Neuropsychiatric dysfunction
- Cardiovascular and cerebrovascular morbidity
- Pulmonary hypertension or right heart failure.

XI. REVENTION AND RECOMMENDATION

The use of alcohol, hypnotics or opiates is not recommended in untreated or non-adherent apneic patients. Hypnotics may be prescribed to apneic patients regularly using their CPAP. The anesthetic risk is greater in patients with sleep apnea syndrome.

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PULMONARY TUBERCULOSIS

Asso. Prof. SAINT Saly

I. DEFINITION

Pulmonary tuberculosis (TB) is a contagious bacterial infection primarily affecting the lungs caused by *Mycobacterium tuberculosis*. It mostly develops disease in the lungs called pulmonary TB which is a serious infectious disease that can spread through the air when an infected person coughs, sneezes, or speaks.

II. EPIDEMIOLOGY

TB continues to pose a significant global health challenge, especially in low- and middle- income countries. According to the World Health Organization (WHO 2021), TB ranks among the top 10 leading causes of death worldwide, with an estimated 10 million new cases and 1.6 million deaths reported annually, including 187,000 deaths among HIV-Positive. The highest burden of TB is observed in regions with limited healthcare access, poor living conditions, and high prevalence of HIV. TB can affect people of all ages, but it predominantly impacts adults, aged ≥ 15 years (89%), with greater risks observed among the elderly and those with compromised immune systems. Gender disparities in TB incidence vary across different regions and population demographics, globally 57% men, 33% women and 11% children.

In Cambodia, as of 2021, the incidence rate for all forms of TB is 288 per 100,000 population, equating to approximately 48,000 cases annually. The death rate stands at 21 per 100,000 population. In 2022, the prevalence rate of smear-positive pulmonary TB was 271 per 100,000 population. Regarding drug-resistant TB, in 2019, 1.8% of new TB cases were classified as multi-drug resistant or rifampicin-resistant (MDR/RR TB), while among TB retreatment cases, the percentage was 8.2%. (NTP Cambodia report 2023).

III. ETIOLOGY

TB is caused by *Mycobacterium tuberculosis*, a bacterium that is aerobic and acid-fast, predominantly targeting the lungs. The disease spreads through inhaling aerosolized droplets containing the bacteria released by infected individuals. Factors contributing to its transmission include close contact with people suffering from active TB. Individuals with compromised immune systems due to conditions like HIV/AIDS, diabetes, malnutrition, or the use of immunosuppressive medications are at higher risk. Additionally, social determinants such as poverty, overcrowded living conditions, and inadequate access to healthcare play significant roles in the prevalence and spread of the disease.

IV. PATHOPHYSIOLOGY

- **Primary TB Infection:** Primary tuberculosis has generally been defined as onset of clinical disease within 12 months of initial infection with *Mycobacterium tuberculosis*. Tuberculosis may involve multiple organs, such as the lung, liver, spleen, kidney, brain, and bone.
- **Post Primary TB Infection (Latent TB Infection):** Post primary TB results from either reactivation of a latent primary infection or, less commonly, from the repeat infection of a previously sensitized host. Reinfection is possible because the type of immune response required, cell-mediated immunity, needs a certain local inflammatory response to locate lesions.
- **Primary and Latent TB Infection:** Primary TB stimulates systemic immunity that kills organisms and heals granulomas resulting in both protections from disseminated TB and resistance to new infection. Post primary TB (LTBI), in contrast, requires systemic immunity that it subverts to produce local susceptibility in the apex of the lung.

V. DIAGNOSIS

5.1. Common symptoms

- **Chronic Cough:** Persistent cough lasting more than 2 weeks.
- **Fever:** Low-grade fever, especially in the afternoon or evening.
- **Night Sweats:** Profuse sweating during sleep, often drenching.
- **Weight Loss:** Unexplained weight loss despite normal or increased appetite.
- **Chest Pain:** Pain in the chest, particularly with deep breathing or coughing.
- **Haemoptysis:** Coughing up blood or blood-stained sputum.

5.2. Investigations

A. Microbiological Tests

- **Sputum Smear Microscopy:**
 - Acid-fast staining to identify *Mycobacterium tuberculosis* (M. tuberculosis).
 - Positive result shows acid-fast bacilli (AFB) in sputum.
- **Gene-Xpert MTB/RIF:**
 - PCR (polymerase chain reaction) test to detect TB DNA and rifampicin resistance.
 - Provides rapid results within a few hours.
- **TB Culture:**
 - Growth of M. tuberculosis from sputum samples.
 - Gold standard for confirming TB diagnosis.
 - Takes several weeks for results due to slow growth rate of M. tuberculosis.

B. Radiological Tests

- Chest X-ray:
 - To detect abnormalities suggestive of pulmonary TB such as:
 - Consolidation
 - Cavitation
 - Fibrosis
 - Pleural effusion
 - Helps assess the extent and severity of disease.

C. Immunological Test

- **Tuberculin Skin Test:**
 - Detects delayed hypersensitivity reaction to purified protein derivative (PPD) of *M. tuberculosis*.
 - Indicated for detecting TB infection rather than active disease.

5.3. Diagnostic Algorithms

A. Presumptive PTB among General Population

- **Clinical Presentation:** Chronic cough (> 2 weeks) with or without fever, night sweats, and weight loss.
- **Diagnostic Steps:**
 - Perform Sputum Smear Microscopy examination.
 - If sputum smear microscopy is positive, diagnosed as Smear-positive PTB.
 - If sputum smear microscopy is negative and clinical suspicion remains high, consider Empirical Anti-TB Treatment (AB treatment) is not improved, Chest X-ray (CXR) is conducted for confirming diagnosis of TB (Smear-negative PTB or Clinical PTB).

B. Presumptive PTB among High-Risk Population (example, PLWHA, Elderly > 55 years old, TB Contact)

- **Clinical Presentation:** Symptoms as mentioned (cough, fever, night sweats, weight loss).
- **Diagnostic Steps:**
 - Preferentially use **GeneXpert MTB/RIF** examination for rapid detection.
 - If GeneXpert detects MTB, diagnosed as positive PTB: (if it detects MTB resistant with rifampicin, diagnosed as positive PTB/RR).
 - If GeneXpert does not detect MTB and clinical suspicion remains high, consider Chest X-ray (CXR) is conducted for confirming diagnosis of TB (negative PTB or Clinical PTB).

VI. MANAGEMENT

- **Treatment Regimen:** Patients undergo combination therapy with antibiotics including rifampicin (R:10mg/kg/day), isoniazid (H:5mg/kg/day), pyrazinamide (Z:25mg/kg/day), and ethambutol (E:15mg/kg/day) over a period of six months to achieve complete eradication of the bacteria, typically administered as 2RHZE/4RH.
- **Directly Observed Therapy with Short-course (DOTS):** This involves supervised administration of medications to ensure patients adhere to their treatment regimen, optimizing treatment success.
- **Treatment Adjustment:** Follow-up evaluations, which may include bacteriological, clinical assessments, and/or chest X-rays, are conducted at Month 2, Month 4, and Month 6 to monitor treatment progress and adjust management as necessary.
 - **Response Monitoring:** Regularly monitor bacteriological tests (sputum conversion) and clinical response and adverse effects of medications (e.g., hepatotoxicity with isoniazid) at the end of month 2, 4, and 6.
 - **Drug Resistance Testing:** Perform drug susceptibility testing if initial treatment fails or in cases of suspected drug-resistant TB. Adjust treatment regimen accordingly (e.g., add second-line drugs for MDR/RR-TB or XDR-TB).

Table 1: Treatment regimen: 2RHZE/4RH (treatment of the positive and negative PTB)

Body weight In kg	Intensive phase: 2months, daily	Continuation phase: 4months, daily
	RHZE: 150mg+75mg+400mg+275mg (4 combination drugs)	RH: 150mg+75mg (2 combination drugs)
30-39	2 tables	2 tables
40-54	3 tables	3 tables
55-70	4 tables	4 tables
>70	5 tables	5 tables
Dose/kg/day	R:10mg, H:5mg, Z:25mg, E:15mg	

Table 2: Characteristic of Ant-TB medications

Medication	Most Common Side Effects	Tests for Side Effects	Drug Interactions	Remarks
Isoniazid	Peripheral neuropathy,	AST and ALT; neurologic examination.	Phenytoin (synergistic); disulfiram.	Bactericidal to both extracellular and

	hepatitis, rash, mild CNS effects.			intracellular organisms. Pyridoxine, 25-50 mg orally daily, is given as prophylaxis for neuropathy; 50-100 mg orally daily as treatment for it.
Rifampin	Hepatitis, fever, rash, flu-like illness, GI upset, bleeding problems, kidney failure.	CBC, platelets, AST and ALT.	Rifampin inhibits the effect of oral contraceptives, quinidine, corticosteroids, warfarin, methadone, digoxin, oral hypoglycemics; aminosalicic acid may interfere with absorption of rifampin. Significant interactions with protease inhibitors and nonnucleoside reverse transcriptase inhibitors.	Bactericidal to all populations of organisms. Colors urine and other body secretions orange. May discolor contact lenses
Pyrazinamide	Hyperuricemia, hepato-toxicity rash, GI upset, joint aches.	Uric acid, AST, ALT	Rare.	Bactericidal to intracellular organisms.
Ethambutol	Optic neuritis (reversible with discontinuance of drug; rare at 15 mg/kg); rash.	Red-green color discrimination and visual acuity.	Rare.	Bacteriostatic to both intracellular and extracellular organisms. Mainly used to inhibit development of resistant mutants. Use with caution in kidney disease or when ophthalmologic testing is not feasible.
Streptomycin	Eighth nerve damage, nephrotoxicity.	Vestibular function (audiograms); BUN and creatinine.	Neuromuscular blocking agents may be potentiated and cause prolonged paralysis.	Bactericidal to extracellular organisms. Use with caution in older Patients or those with kidney.

VII. SPECIAL CASES

- **Drug-Resistant TB:** Multi-drug-resistant/Rifampicin resistant TB (MDR/RR-TB) and extensively drug-resistant TB (XDR-TB) require prolonged treatment with second-line drugs.
- **HIV Coinfection-TB:** Increased risk of TB reactivation and severe disease; integrated management essential.

VIII. EVOLUTION AND PROGNOSIS

- **Latent TB:** May remain asymptomatic or progress to active disease.
- **Active TB:** Early diagnosis and adherence to treatment crucial for cure and prevention of complications.

IX. COMPLICATIONS

- **Pulmonary Complications:** Cavitory lesions, fibrosis, chronic respiratory impairment.
- **Extrapulmonary TB:** Spread to other organs (e.g., lymph nodes, bones, central nervous system).
- **Social and Economic Impact:** Long-term disability, loss of productivity, stigma.

X. PREVENTION AND RECOMMENDATIONS

- **TB Control Programs:** Early case detection, contact tracing, and treatment initiation.
- **Vaccination Programs:** Bacille Calmette-Guérin (BCG) vaccine in countries with high TB prevalence, though its efficacy varies.
- **Infection Control Programs:** Respiratory hygiene, ventilation, and personal protective measures in healthcare settings and households.

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PULMONARY ASPERGILLOSIS

*Dr. TANN Bunly, Dr. CHAN Sophors, Dr. CHHAR Bunpaul, Pr. CHAN Sarin, Dr. CHHORN Theary,
Pr. VAN Mich, Pr. SAING Saly, Pr. NY Chanthy*

I. Definition

Allergic bronchopulmonary aspergillosis (ABPA): a hypersensitivity reaction caused by exposure to *Aspergillus* that mostly occurs in patients with cystic fibrosis or asthma
Chronic pulmonary aspergillosis: a long-term *Aspergillus* infection of the lung which can manifest as:

- Aspergilloma: an opportunistic infection of a pre-existing cavitory lesion (e.g. from previous tuberculosis)
- Aspergillus nodule: single or multiple nodules with possible cavitation
- Chronic cavitory pulmonary aspergillosis: extensive fibrosis with tissue destruction of at least two lung lobes
- Fibrotic pulmonary aspergillosis
- Chronic necrotizing pulmonary aspergillosis: a subacute and semi-invasive form of chronic pulmonary aspergillosis characterized by localized, slowly progressive, inflammatory destruction of lung tissue and commonly associated with alcohol use disorder

Invasive aspergillosis: a severe form of *Aspergillus* infection which manifests with severe pneumonia and septicemia with potential involvement of other organs (e.g., skin, CNS) and mostly occurs in immunocompromised individuals

II. EPIDEMIOLOGY AND DEMOGRAPHICS

Incidence

- The true incidence of aspergillosis is unknown since aspergillosis is not a reportable disease in the USA.
- The annual incidence of allergic bronchopulmonary aspergillosis is estimated to be 5 million cases worldwide.
- Approximately 2.5% of adults who have asthma also have allergic bronchopulmonary aspergillosis.
- In contrast, invasive aspergillosis is much less common with an incidence of approximately 1 to 2 cases per 100,000 individuals.
- Invasive aspergillosis is the most common type of fungal infection among stem cell transplant recipients (5% to 10%) and the second most common type of fungal infection among solid organ transplant recipients (5% to 25%).

Mortality Rate

- The mortality rate of aspergillosis varies according to the clinical syndrome.
- Allergic bronchopulmonary aspergillosis and allergic *Aspergillus* sinusitis rarely result in mortality.
- In contrast, the mortality rate of either invasive aspergillosis or chronic necrotizing pulmonary aspergillosis is high and may be up to 40% to 90%.

Gender

- There is no gender predilection for the development of aspergillosis.

Age

- Individuals of all age groups may be affected with aspergillosis.
- Elderly patients with long-standing chronic lung diseases are more likely to develop aspergillosis.
- Neonates are at increased risk of cutaneous aspergillosis.

Developed Countries

- With the advancement of immunosuppressive therapy and higher rates of transplantation, the incidence and detection of aspergillosis is increasing in developed countries.

III. ETIOLOGY

- Pathogen
 - Aspergillus, a genus including over 200 species
 - Most common: Aspergillus fumigatus and Aspergillus flavus
- Transmission: airborne exposure to mold spores
 - Aspergillus spores (also referred to as conidia) are ubiquitous indoors, as they enter with the normal flow of air.
 - The spores can also settle on easily accessible sources of nutrition (e.g., water), dust, cellulose (e.g., in wallpaper), and indoor plants
 - Aspergillus spores may also be found in intensive care units
- Risk factors
 - Severe immunosuppression (e.g., due to HIV/chemotherapy, after organ transplantation) or neutropenia (e.g., due to chronic granulomatous disease) can facilitate the development of invasive aspergillosis
 - Destructive pulmonary pathology may lead to chronic pulmonary aspergillosis
 - Scar tissue or lung cavities (e.g., from tuberculosis)
 - Sarcoidosis
 - Pneumocystis pneumonia
 - COPD, emphysema
 - Aspergillus can trigger hypersensitivity reactions (ABPA) in patients with preexisting bronchopulmonary conditions (e.g., asthma, cystic fibrosis)
 - Alcohol use disorder and steroid-dependent COPD are associated with chronic necrotizing pulmonary aspergillosis

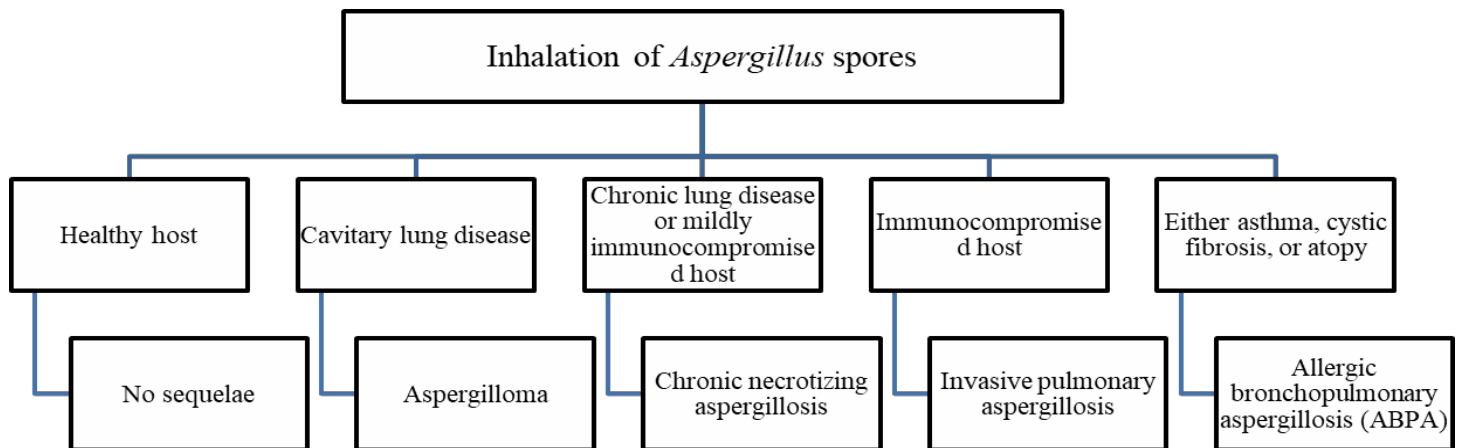
IV. PATHOPHYSIOLOGY

- Reservoir
 - Aspergillus is ubiquitous in the environment.
 - Aspergillus can be found in soil, decomposing plant matter, household dust, building materials, plants, food, and water.
- Transmission
 - Transmission of Aspergillus occurs through inhalation of airborne conidia.
 - Hospital-acquired infections may be sporadic or may be associated with dust exposure during building renovation or construction.
 - Occasional outbreaks of cutaneous infection have been traced to contaminated biomedical devices.
 - Aspergillus is not transmitted from one individual to another (no person-to-person transmission).

➤ Pathogenesis

Following the ingestion of *Aspergillus* conidia, the mechanism of disease may vary depending on the host immune factors.

- The following algorithm demonstrates common sequelae of *Aspergillus* inhalation among different patient groups:



V. CLASSIFICATION

Aspergillosis may be classified according to the clinical syndrome it causes into the following:

- Allergic bronchopulmonary aspergillosis
- Allergic *Aspergillus* sinusitis
- Aspergilloma
- Chronic pulmonary aspergillosis
- Invasive aspergillosis
- Cutaneous aspergillosis

The table below distinguishes between the different types of aspergilloses

Clinical Syndrome	Characteristics
Allergic Bronchopulmonary Aspergillosis (ABPA)	- Immune-modulated pulmonary disease due to hypersensitivity to <i>A. fumigatus</i> (not an infection) - Clinical features include persistent cough, wheezing, symptoms of asthma exacerbation or bronchiectasis - Common among patients with asthma - Recurrent pulmonary infiltrates on chest imaging - May result in development of bronchiectasis, chronic pulmonary aspergillosis, or aspergilloma - Treatment with antifungal agents, steroids, and asthma control

Allergic <i>Aspergillus</i> Sinusitis / Rhinosinusitis	• Immune-modulated sinus disease (not an infection) • Clinical features of recurrent or resistant sinusitis, such as increased sinus drainage, stuffiness, headache, facial tenderness, and eosinophilic secretions • Non-invasive • Common in healthy young adults • Diagnosed using Bent and Kuhn diagnostic criteria (see criteria here) • Unilateral, asymmetric involvement of sinuses with bone erosions on imaging • Treatment includes de-obstruction, nasal drainage, oral steroids, immunotherapy against allergens, and allergy control
Aspergilloma (Fungus Ball)	• An <i>Aspergillus</i> mass colonizes locally in either the lungs or the sinuses • May occur with chronic pulmonary aspergillosis • Clinical features vary from an asymptomatic course to fatal hemoptysis • Common among individuals with prior history of lung diseases, including tuberculosis • Chest CT in prone position shows movable fungus ball • Treatment includes surgical resection
Chronic Pulmonary Aspergillosis	• Invasion of <i>Aspergillus</i> in the lungs for > 3 months • May occur with aspergillomas (single or multiple) • Clinical features include weight loss, chronic productive cough, dyspnea, and hemoptysis • On Imaging, 1 of 3 patterns is characteristic: 1) multiple cavities (chronic cavitary pulmonary aspergillosis - CCPA, in immunocompetent patients, chronic course), 2) fibrosis (chronic fibrosing pulmonary aspergillosis - CFPA, in immunocompetent patient, chronic course, follows CCPA), or 3) subacutely enlarging single cavity (chronic necrotizing pulmonary aspergillosis - CNPA, subacute within weeks i.e. not chronic, in immunocompromised patients) • Treatment includes systemic antifungal therapy with or without intracavitary antifungal agents
Invasive Aspergillosis	• Severe infection among immunocompromised individuals (e.g. transplant recipients or neutropenic patients) • Associated with high mortality rate • May progress to involve other non-pulmonary organs • Non-specific clinical features make diagnosis difficult, may mimic asthma or bronchitis • Classical triad of fever, pleuritic chest pain, and hemoptysis • Focal nodules and infiltration common on imaging • Treatment includes antifungal agents
Cutaneous Aspergillosis	• Infection of the skin • Invasion may be either primary through a skin lesion (e.g. surgical or burn wound or at catheter site) or secondary to a distant organ with subsequent dissemination to the skin • Common among immunocompromised individuals and neonates • Clinical features usually include fever, change in lesion morphology. Lesions are commonly macules, papules, nodules, or plaques. Other lesions may also occur. • Diagnosis is by biopsy of the lesion, evaluation of other primary sources (e.g. CT scan for pulmonary lesions) is necessary • Treatment includes antifungal agents with or without surgery

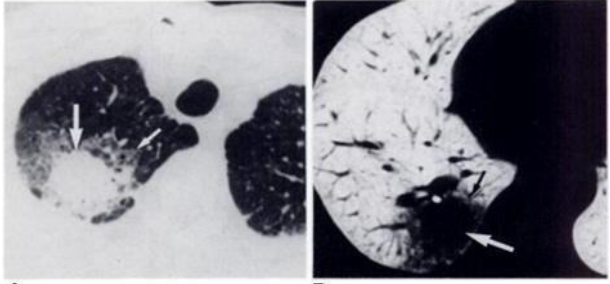
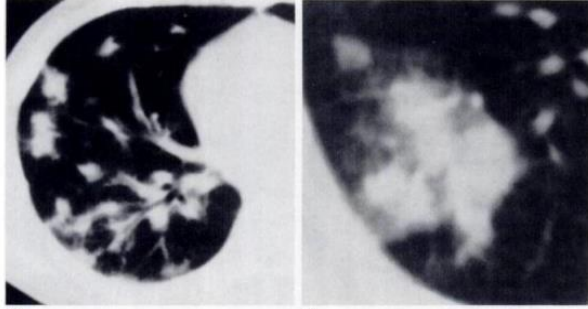
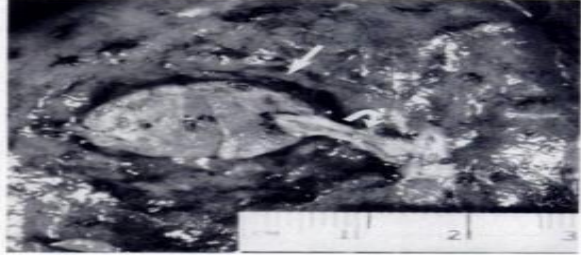
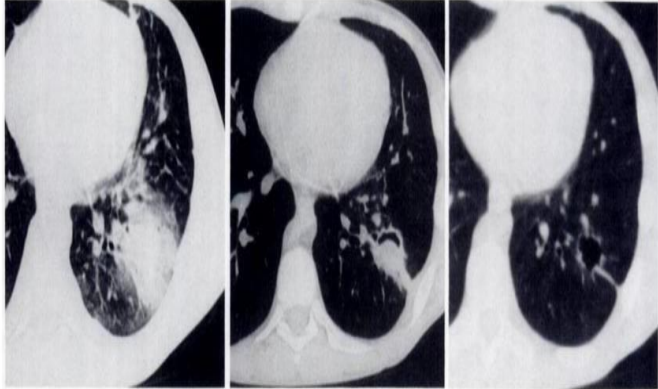
VI. DIAGNOSIS

6.1. Clinical features

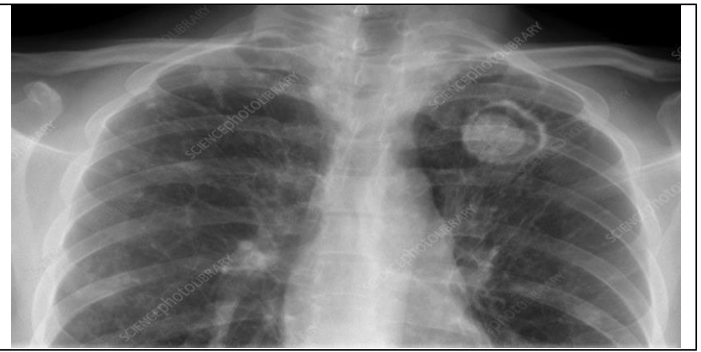
Clinical features of aspergillosis		
	<i>Lung manifestations</i>	<i>Other manifestations</i>
ABPA	• Asthmatic symptoms (e.g., shortness of breath wheezing), especially new exacerbations • Productive cough with brown bronchial mucous casts	• Nonspecific symptoms: irritation of the skin, mucus membranes, and eyes • Mucous membranes in sinuses ◦ Fungus ball in the paranasal sinuses (without tissue infiltration) with symptoms of chronic rhinosinusitis ◦ May progress in immunocompromised patients (see invasive aspergillosis)
Chronic pulmonary aspergillosis	• Possibly asymptomatic (may be an incidental finding on chest x-ray) • Hemoptysis, shortness of breath • Recurrent pneumonia caused by superinfections	• Possibly fever • Weight loss • Night sweats • Chronic fatigue • Signs of an underlying lung pathology (e.g., digital clubbing in tuberculosis)
Invasive aspergillosis	• Dry cough • In severe cases: hemoptysis, pleuritic chest pain, septic shock • Tachypnea • Recurrent pneumonia caused by superinfections	• Fever • Weight loss • Night sweats • Chronic fatigue • Mucous membranes in sinuses ◦ <i>Aspergillus</i> sinusitis with invasion of the surrounding tissue ◦ Direct invasion of the orbit can lead to reduction of visual acuity, painful exophthalmos, and/or chemosis ◦ Direct invasion of the skull can result in brain abscess, meningitis, ventriculitis, and/or venous sinus thrombosis • Skin (via direct inoculation): pustules or hemorrhagic papules/plaques, which may evolve into eschars • Disseminated infection ◦ CNS: Multiple abscess formations with varied neurological manifestations (e.g., cramps, focal neurological deficits) ◦ Heart: <i>Aspergillus</i> endocarditis

Investigation of diagnostics of aspergillosis

	<i>Laboratory tests</i>	<i>Chest x-ray and CT</i>	<i>Tissue biopsy and/or culture</i>
ABPA	• ↑ Ig E levels • Eosinophilia • ↑ ESR • Positive <i>Aspergillus</i> antigen skin test	• Bronchiectasis • Pulmonary infiltrates • Parenchymal opacities	• N/A
Chronic pulmonary aspergillosis	• Positive <i>Aspergillus</i> IgG serology	• Aspergilloma ○ Mobile fungus ball (demonstrated by moving the patient from a supine position to a prone or lateral recumbent position) ○ Monod sign: a peripheral air crescent around a fungus ball in a preexisting lung cavity ○ the upper lobe is mostly affected • <i>Aspergillus</i> nodule: multiple nodules • Cavitory pulmonary aspergillosis: one or more cavities with or without aspergilloma • Fibrotic aspergillosis: cavitaries and extensive fibrosis • Necrotizing aspergillosis: see criteria in invasive aspergillosis below	• Culture and/or histopathology ○ Specimen are collected via: - Bronchoalveolar lavage via bronchoscopy - In patients with peripheral nodular lesions on imaging: endobronchial lung biopsy via bronchoscopy or CT-guided percutaneous biopsy ○ Staining: Gomori methenamine silver stain or PAS stain ○ Findings: monomorphic and/or septate hyphae branching dichotomously at 45°
Invasive aspergillosis	• Positive galactomannan enzyme immunoassay: an EIA that identifies galactomannan antigen • Galactomannan antigen: a heteropolysaccharide component of the <i>Aspergillus</i> cell wall that is shed during the phase of hyphal growth. • Positive 1,3-β-D glucan test	• Multiple nodules • Halo sign: hemorrhagic ground glass opacities around nodules • Air crescent sign: a radiolucent halo around a radiopaque nodule that is characteristic of invasive aspergillosis • Cranial MRI - Perform if CNS infection is suspected - Most commonly shows abscess formation	

A and B, Early CT findings of invasive pulmonary aspergillosis in two patients with leukemia and aplasia. The CT halo sign represents a discrete zone of lower attenuation (small arrows) surrounding a pulmonary mass (large arrows).	 A B
Early CT findings of invasive pulmonary aspergillosis. A, Multiple small inflammatory masses. B, Clusters of fluffy nodules.	 A B
Early invasive pulmonary aspergillosis Central nodule of necrosis is surrounded by ring of haemorrhage corresponding to CT halo (straight arrow). Thrombosed artery is seen at edge of lesion (curved arrow)	
Characteristic pattern of resolution of invasive pulmonary aspergillosis on CT. A. Active early lesion of invasive pulmonary aspergillosis with CT halo sign during week 2 of aplasia. High-dose amphotericin was begun and infection was stabilized. B. Air-crescent formation 4 weeks later at time of bone-marrow recovery. C. 2 months later. Lesion is healed. CT shows residual thin-walled cyst.	 A B C

- Multiple nodules
- Halo sign: haemorrhagic ground glass opacities around nodules
- Air crescent sign: a radiolucent halo around a radiopaque nodule that is characteristic of invasive aspergillosis



VII. DIFFERENTIAL DIAGNOSES

The differential diagnosis of a space-occupying lesion within a lung cavity includes:

1. Primary lung malignancy
2. Metastatic disease
3. Aspergilloma
4. Hydatid cyst
5. Lung abscess

The differential diagnosis of CCPA includes:

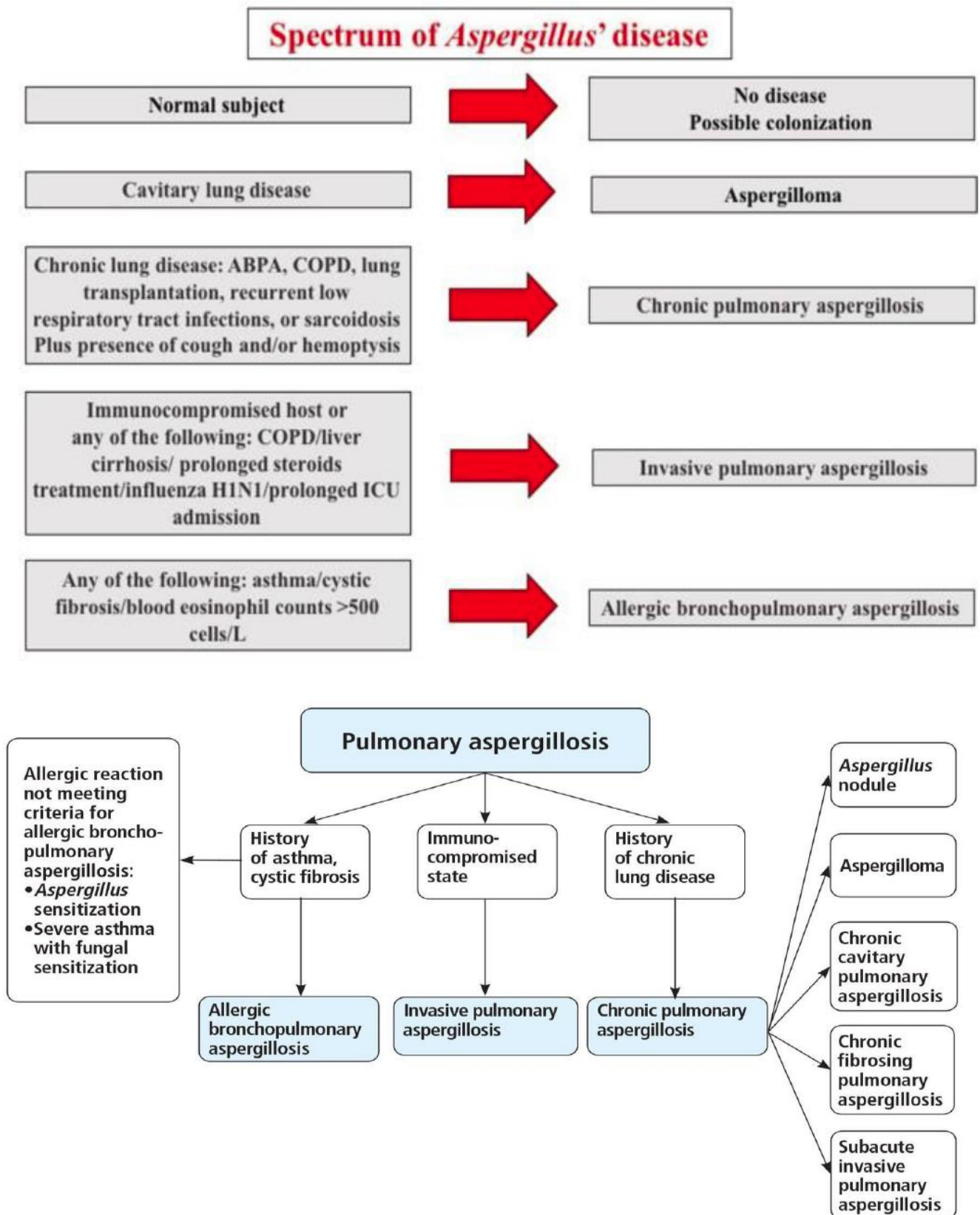
1. Active TB
2. Non-tuberculous mycobacteria
3. Histoplasmosis
4. Coccidioidomycosis
5. Actinomycosis
6. Neoplasm

VIII. MANAGEMENT

Pulmonary Aspergillosis: An Evolving Challenge for Diagnosis and Treatment

<i>Aspergillus</i> lung disease	First-line treatment	Duration of therapy	Alternative treatment	Comments
ABPA	Prednisolone 0.5 mg/kg/day for 4 weeks followed by 0.25 mg/kg/day for 4 weeks followed by 0.125 mg/kg/day for 4 weeks Itraconazole 200 mg twice daily	3–5 months	Oral voriconazole Posaconazole	Glucocorticoids are the first-line treatment for exacerbations Antifungal therapy has a corticosteroid-sparing effect and could be considered in patients who fail to show improvement after steroid treatment
Aspergilloma	No therapy Surgical resection when appropriate	–	Itraconazole Voriconazole Bronchial artery embolization	Anti-fungal therapy may be considered in cases of lung invasion or when there is the possibility of peri-operatively dissemination
CPA	Itraconazole 200 mg twice daily Voriconazole 6 mg/Kg po/IV q12 h × 1 day followed by 4 mg/Kg po/IV q12 h	6 months	Posaconazole Liposomal Amphotericin B Caspofungin Micafungin	Prolonged treatment may be necessary Surgery has a limited role and may be associated with complications Anti-fibrinolytic agent or bronchial artery embolization could be considered for management of haemoptysis
IPA	Voriconazole 6 mg/Kg po/IV q12 h × 1 day followed by 4 mg/Kg po/IV q12 h Isavuconazole 372 mg po/IV q8 h × 6 doses followed by 372 mg po/IV daily	6–12 weeks	Liposomal Amphotericin B Caspofungin	Combination therapy is not routinely recommended, but may be considered in selected refractory cases Surgical resection is considered in selected situations Empiric treatment could be considered in critically-ill patients with severe liver cirrhosis and/or end-stage chronic obstructive pulmonary disease and/or clinical worsening despite broad-spectrum antibiotics

Algorithm



IX. PECIAL CASES

Endobronchial aspergilloma (EBA) is an extremely rare presentation of pulmonary involvement of *Aspergillus*. It is a non-invasive form of pulmonary aspergillosis where the fungus overgrows and colonizes the bronchial lumen. The patient may present with chronic cough, dyspnoea, haemoptysis, or wheezing.

X. PROGNOSIS

The prognosis of aspergilloma and CCPA is impossible to quantify given its unpredictable history, variables such as associated lung diseases, and the functional status of the patient. Spontaneous resolution occurs in approximately 10% of cases of simple aspergilloma. The recent availability of triazole and echinocandin antifungal agents in symptomatic patients suggests that benefits can be achieved in about 65% of those treated. Similar benefits have been attributed to the intracavitary instillation of antifungals. The impact of these modalities on long-term prognosis has not been evaluated, and symptomatic benefits do not necessarily correlate with life expectancy. The mortality estimates for those experiencing massive haemoptysis from any cause has historically been reported to be as high as 38%. It is unclear if the figure is relevant to current management strategies for aspergilloma and CCPA. After successful bronchial artery embolization, the recurrence of haemoptysis occurs in 19% to 55% of cases. The impact of new antifungal agents and non-surgical management on the prognosis of aspergilloma and CCPA needs further clarification.

XI. OMPLICATIONS

aspergillosis

Bleeding. Both aspergillomas and invasive aspergillosis can cause severe, and sometimes fatal, bleeding in your lungs.

Systemic infection. The most serious complication of invasive aspergillosis is the spread of the infection to other parts of your body, especially your brain, heart and kidneys.

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PLEURAL EFFUSION

Dr CHHORN Sotheary, Pr. NY Chanty, Pr. VAN Mich, Pr. SAING Saly, Pr. CHHAR Bunpaul

I- DEFINITION

Pleural effusion is an accumulation of fluid in the pleural space that is classified as transudate or exudate according to its composition and underlying pathophysiology.

II- PIDEMIOLOGY

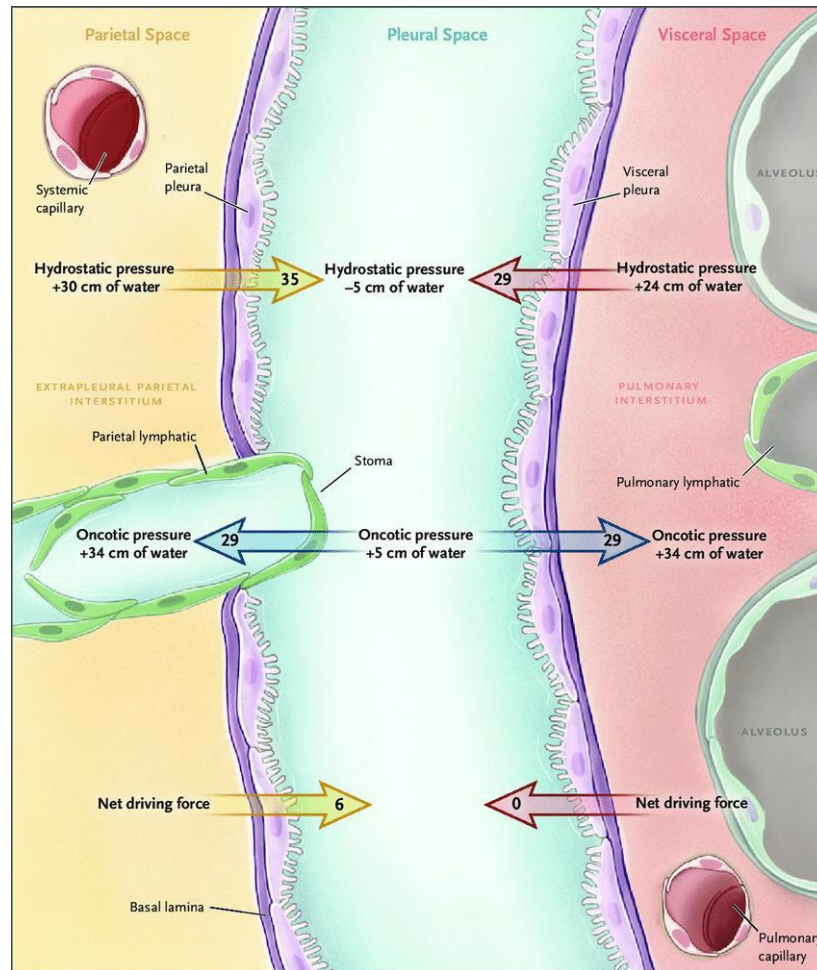
Pleural effusion is diagnosed in about 1.5 million individuals in the United States annually. Among the causes, pleural infection, heart failure, and malignancy are the most common. Nonmalignant pleural effusions (NMPEs) have a wide variety of etiologies [2].

III- PHYSIOLOGY AND PATHOPHYSIOLOGY

Both the visceral and the parietal pleura play a role in fluid homeostasis in the pleural space. The mean rate of both the production and the absorption of pleural fluid is normally 0.2 mL/kg/hour, which implies that the entire volume of the pleural fluid normally turns over within one hour [3].

The volume of the pleural fluid is determined by the balance of the hydrostatic and oncotic pressure differences that are present between the systemic and pulmonary circulation and the pleural space. Pleural fluid is resorbed via lymphatic vessels in the parietal pleura. In health, the production and resorption of pleural fluid are at equilibrium.

A pleural effusion represents a disturbance of this equilibrium, probably because of both increased production and decreased resorption. Low oncotic pressure (e.g., in hypoalbuminemia), elevated pulmonary capillary pressure, increased permeability, lymphatic obstruction, and diminished negative intrapleural pressure are all pathophysiological components that lead to the clinically relevant and distinguishing features of a pleural effusion transudate vs. exudate.



IV- TIOLOGY WITH DIAGNOSIS

Pleural effusion has a wide differential diagnosis.

Differentiation between transudative and exudative effusion is considered the initial step in the etiological diagnosis of any PE. Transudates are most often caused by heart failure (80%) and then by hepatic cirrhosis. Conversely, exudates require more extensive diagnostic evaluation, since they have numerous etiologies and it's in 80% of cases.

1.1. Transudate causes:

- Heart failure is the most common cause. NT-proBNP in PF for the diagnosis of HF has a diagnostic sensitivity and specificity of 94%
- Hypoalbuminemia: Liver cirrhosis, Nephrotic syndrome ...
- Hypothyroidism/ Meigs syndrome ...

1.2. Exudate

- Parapneumonic pleural effusions, Tuberculosis ...
- Pulmonary embolism
- Connective tissue diseases, vascularitis
- Drugs (Methotrexate, Amiodarone, phenytoin, nitrofurantoin...)

V- DIAGNOSIS

1- Clinical presentation

The presenting manifestations of pleural effusion are largely determined by the underlying disease. The common symptoms: cough, fever, pleuritic chest pain, dyspnea.

2- Diagnosis investigation

2.1- Radiology

- A postero-anterior chest x-ray reveals effusions of volume 200 mL or larger.

2.2- Chest (transthoracic) ultrasound

- It is very useful for diagnosis and punctures guide. It provides guidance for the best location for drainage placement, improved performance, and reduces the risk of complications.
- helps identify small PE
- The presence of septa on ultrasonography suggests complicated PPPE and hyperechogenicity is associated with pus in the pleural cavity.

2.3- Chest CT scan

- Distinguish malignant, complicated cases and before some procedures or surgery indication.
- Contrast-enhanced CT is useful in differentiating lung consolidation, with contrast medium uptake, from a hypodense PE. Interlobular, mediastinal pleural and small-sized paravertebral collections are identified on CT [1].
- However, CT has low sensitivity for displaying septa in the pleural cavity.

2.4- Pleural fluid analyses by pleural aspiration (thoracentesis) with or without ultrasound guide. Use **Light's** criteria for transudates and exudates, with sensitivity of 98% to identify exudates [1]: (1) The ratio of pleural fluid protein to serum protein is greater than 0.5, (2) The ratio of pleural fluid lactate dehydrogenase (LDH) to serum LDH is greater than 0.6, (3) LDH content in PF is greater than 2/3 of the upper limit of normal serum levels of LDH.

- pH of pleural fluid
- Microbiology, Xpert of TB, fluid culture, PCR of pleural fluid...
- blood-cell count: an elevated concentration of neutrophils is often in acute parapneumonic effusion, emphyema, a pulmonary embolism. A predominant lymphocytic in tuberculosis, longstanding pleural effusion, congestive heart failure, malignant.
- Non routine tests such as cytology 40-50% of lung cancer metastasis, cholesterol (cholesterol concentration above 55mg/dL), triglyceride, LDH (concentration above 200 U/mL), glucose (low glucose concentration in emphyema, tuberculosis, malignancy, and rheumatoid arthritis), amylase (in acute pancreatitis), or rheumatoid factor, haematocrit (> 50% of peripheral blood haematocrit) were performed should also be determined.

VI- DIAGNOSIS SPECIFIC

Parapneumonic pleural effusion (PPPE)

- It is associated with lung infection, usually pneumonia, abscess, or infected bronchiectasis.
- No responsible microorganism is found in over 40% of empyema.
- In PPPE associated with community-acquired pneumonia, the most frequently isolated Gram-positive bacteria are aerobic, such as streptococci (*Streptococcus milleri* and *Streptococcus pneumoniae*) and *Staphylococcus aureus*, followed by anaerobic microorganisms (common in aspiration pneumonia), and a small group of Gram-negative

bacteria (Enterobacteriaceae, Escherichia coli and Haemophilus influenzae) in patients with comorbid conditions, especially diabetes or alcoholism [1].

Pleural empyema

- Is a fibrinopurulent phase, and present issues such as turbid pleural fluid with predominantly polymorphonuclear cells (PMNs) and microorganisms, pH <7.20, low glucose and high LHD, and the presence of fibrinous septa in the pleural space.
- Normally associated with pneumonia, lung abscess, bronchiectasis, COPD, lung cancer, tuberculosis, aspiration or immunodeficiency...
- The microorganism is found in over 40%. It can be Gram-positive, anaerobic, Gram-negative in diabetes or alcoholism...
- The management require not only antibiotic treatment but also chest drainage or surgery for resolution.

Tuberculous pleural effusion (TBPE)

- Typically, TBPE is an exudate with a nucleated cell count ranging in most cases from 1000 to 6000cells/mm, usually with a predominantly lymphocytic content of >50% (sometimes even >90%) or a lymphocyte/neutrophil ratio >0.75.
- Adenosine deaminase (ADA) remains the primary biomarker for TBPE with an overall sensitivity of 92% and a specificity of 90%.
- Hydropneumothorax due to pleural rupture of a parenchymal cavitory lesion, and may become chronic and require chest drainage or surgery.
- Levels of IFN gamma (T-lymphocyte CD4 increased with sensitivity 89-99%, specificity 92-98% [1])
- Transmural pleural biopsy shows granulomas, with sensitivity > 80% and presence of acid-fast bacilli (30%-50%). Diagnostic thoracoscopy if negative results of other performed.

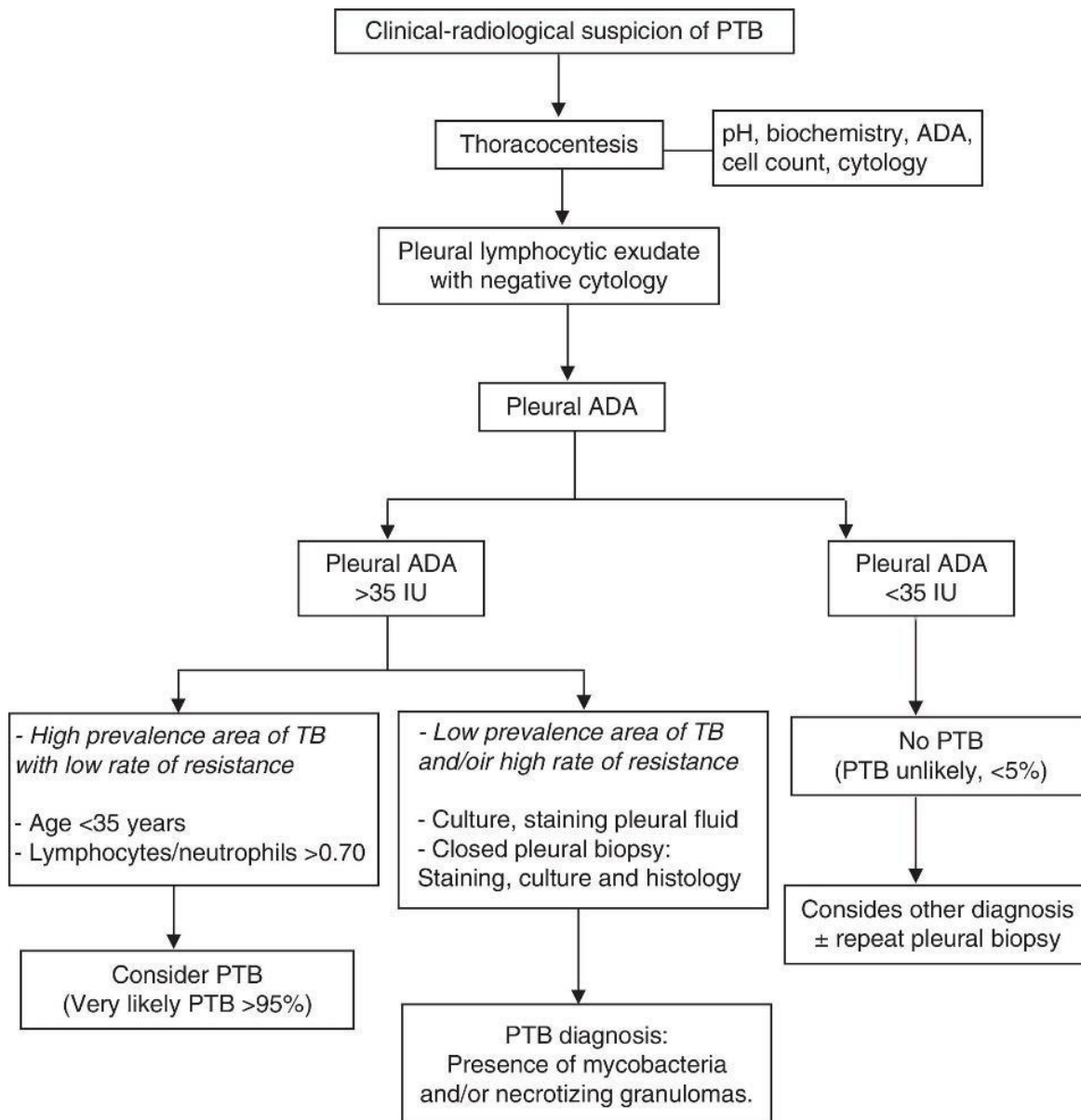
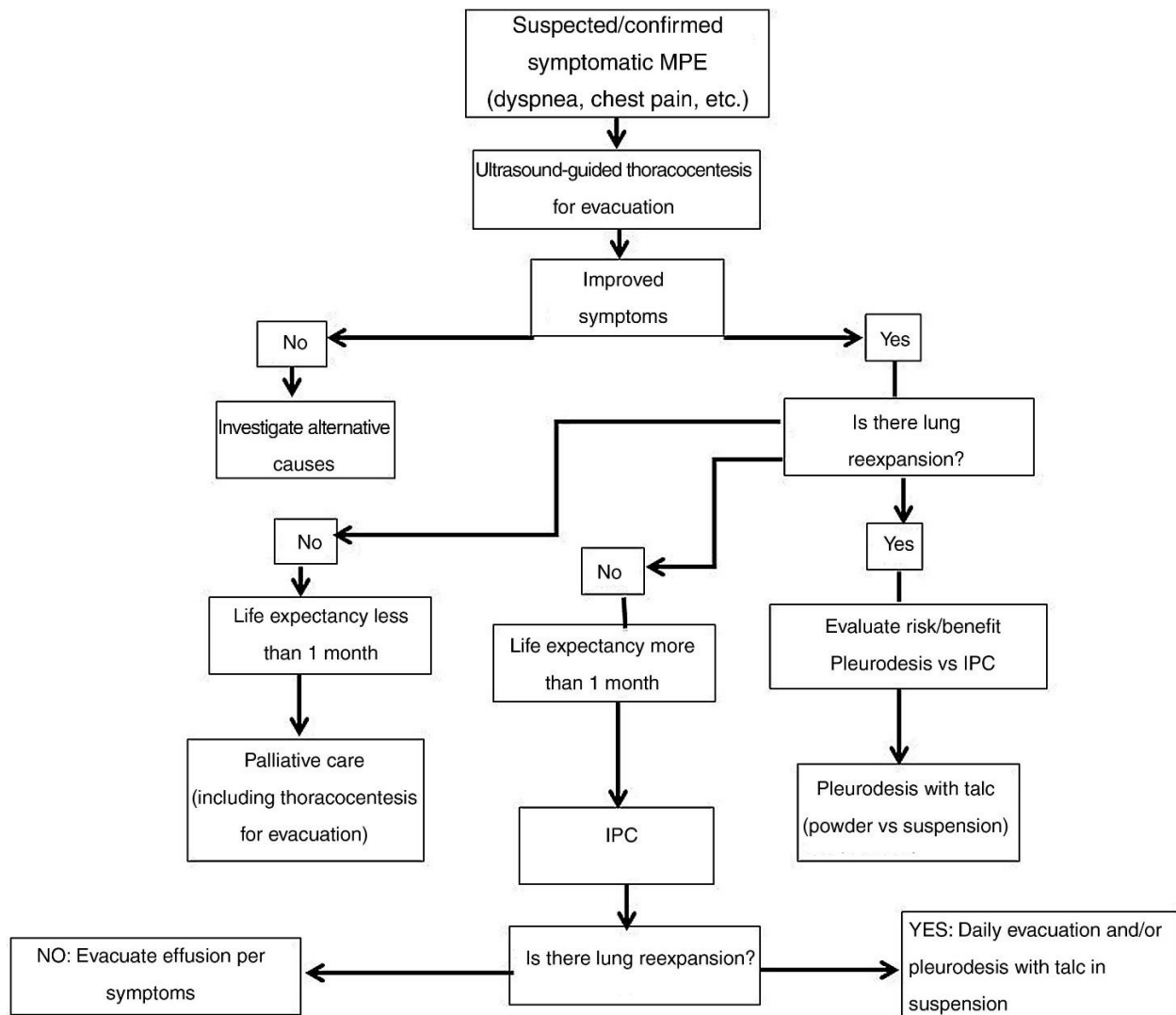


Fig. Diagnostic algorithm for patients with suspected pleural tuberculosis (PTB). TB, tuberculosis; ADA, adenosine deaminase.

Malignant pleural effusion

- It is the result of direct infiltration of the pleura by neoplastic cells. It represents between 15% and 35% of all PE exudate.
- Ultrasound-guided pleural biopsy is especially indicated in the presence of pleural thickening or masses or nodules on the diaphragm surface are observed [1]. Reported sensitivity ranges from 76% to 95%.
- CT with contrast is recommended. Malignancy should be suspected if pleural nodules, diffuse thickening of the parietal pleura.
- Pleural fluid analysis: is an exudate in > 95% with hemorrhagic. More than 80% are predominantly lymphocytic. Studies of biopsy and cytology (sensitivity 60%).
- Pleural needle biopsy with sensitivity 50%

- Thoracoscopy is the gold standard in the diagnosis of MPE, and it has a diagnostic yield of between 90% and 100%.
- tumor markers in PF is not recommended as a routine method



VII- MANAGEMENT

Treatment varies depending upon the etiology.

1. Antibiotherapy

- For pleural effusion infection. First choice is Co-amoxiclav 1.2g TID then guided by bacterial culture. Total antibiotics therapy duration will be depending on severity of infection from 3 weeks to 6 weeks.
- In emphyema, a combination of a third generation cephalosporin with Clindamycine or Metronidazol.

2. Thoracentesis

- To find diagnosis, relieve shortness of breath and reduce the risk of complications.

3. Chest tube

- Chest tubes are indicated when sepsis persists despite antibiotic therapy, in all cases of complicated empyema. If pH is not available, glucose levels of less than 60mg/dL and LDH greater than 1000U/L are useful for identifying the need for pleural drainage [1].

4. Surgery

- Specifically, video-assisted thoracoscopy, is an option for patients with pleural effusion resistant to medical treatment and drainage with fibrinolytics fail. Its allows pleural debridement with the subsequent lung reexpansion, pus evacuation and drainage placement.
- In the case of highly organized empyema with extensive pleural fibrosis, decortication may be necessary.

5. Algorithm

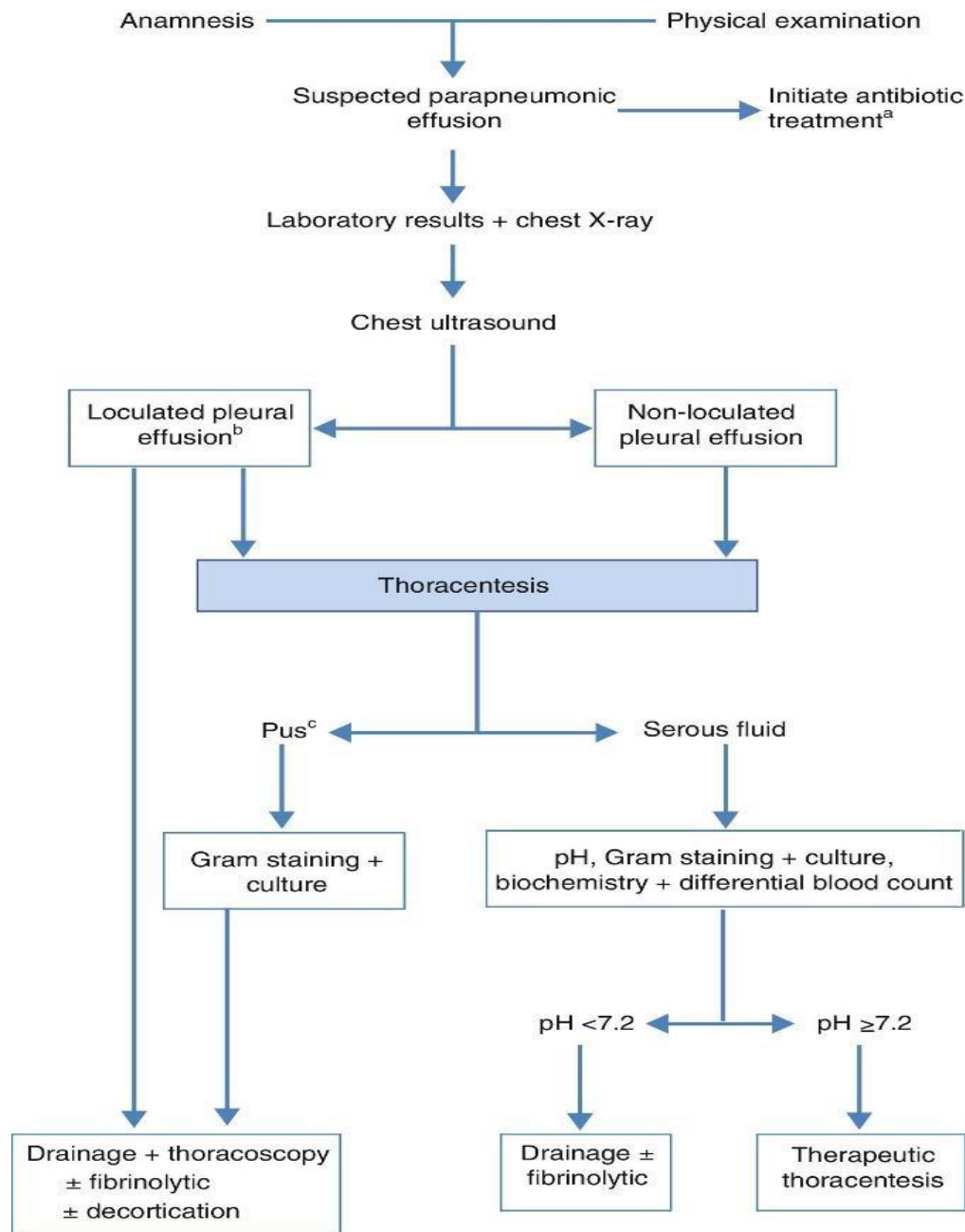


Fig. Algorithm for the treatment of parapneumonic effusion.

VIII. SPECIAL CASES

1. Tuberculous pleural effusion
 - The treatment is 2RHZE/4RH. It can be dominated with chest drainage if complicated and decortication if not improve
 - Tuberculous empyema is often requiring pleural decortication due to functional disability and treatment failure.

2. Parapneumonic pleural effusion (PPPE)

- Treated with appropriate antibiotic. A combination of a third-generation cephalosporin with clindamycin or metronidazole, or monotherapy with amoxicillin-clavulanate may be used. If the source is a nosocomial infection, recommended antibiotics include vancomycin or linezolid (for methicillin-resistant *S. aureus*), antipseudomonal penicillins (piperacillin-tazobactam), carbapenem, or third-generation cephalosporins with metronidazole.
- Chest tube is indicated to have failed when sepsis persists despite antibiotic therapy, in all cases of empyema or complicated PPPE (pH less than 7.20, loculated PPPE or bacteria in the PF). When pH determination is not available, glucose levels below 60mg/dL and LDH levels above 1000U/L are useful for identifying the need for pleural drainage.
- Early thoracoscopy is an option for patients with loculated PPPE. It allows pleural debridement with the subsequent lung reexpansion, pus evacuation and drainage placement. Surgical treatment is indicated for control of sepsis when medical treatment and drainage fail.

3. Malignant pleural effusion

- Chemotherapy can be effective in controlling MPE when associated with lymphoma, small-cell lung carcinoma, or breast cancer with therapeutic Thoracentesis
- Pleurodesis is indicated when PE is symptomatic (dyspnea) with a tendency to recur after therapeutic thoracentesis. Talc is the agent of choice for pleurodesis and, in order to minimize the risk of complications.

4. Pleural effusion due cardiovascular disease

- 60% are bilateral.
- Thoracentesis is only indicated in unilateral PE, patients with fever or pleuritic pain and persistent PE despite diuretic therapy.

5. Postoperation pleural effusion

- After complicated abdominal or thoracic surgery (day 1-4 after surgery), usually small and asymptomatic, and resolve spontaneously within the first month.
- In cardiac or thoracic surgery, postoperative PE is neutrophilic exudate during the first month.
- Postsurgical PE does not require specific treatment.

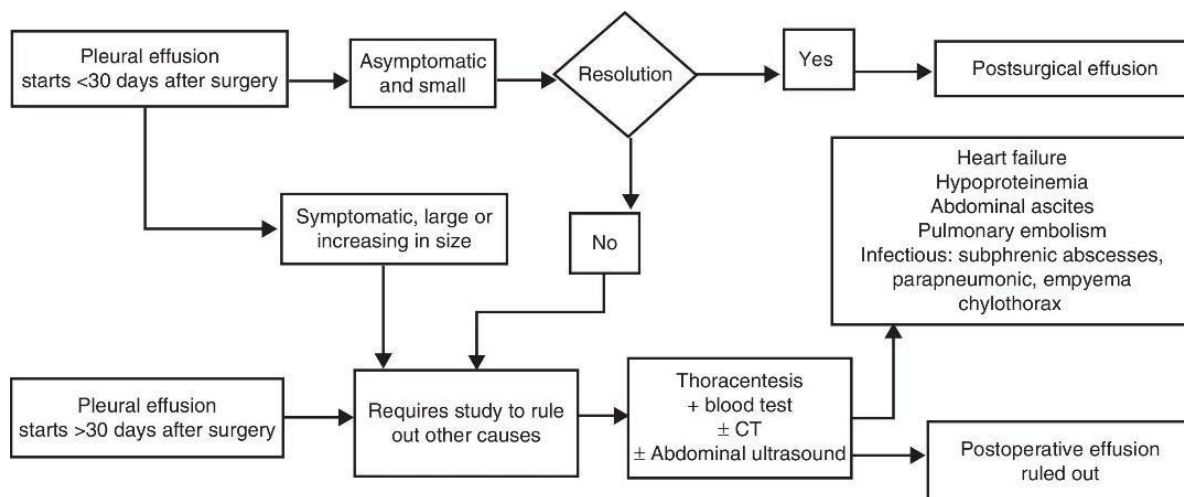


Fig. Diagnostic algorithm of pleural effusion after thoracic or abdominal surgery.

6. Pleural effusion due to drug
 - Usually associated with parenchymal involvement. It can be eosinophilic.
 - Frequency drug: Amiodarone, Nitrofurantoin, methysergide...
7. Chylothorax
 - The accumulation of lymph fluid from intestinal origin (chyle) in the chest cavity as a result of the rupture or obstruction of the thoracic duct. The most common causes are malignancy and trauma.
 - Diagnosis is established by triglyceride levels in PF greater than 110mg/dl.
 - This drainage is more effective than thoracentesis, to close the fistulous area, and continuous drainage also provides accurate monitoring of chyle leak.
 - Can add Somatostatin, octreotide (decreases splanchnic, hepatic and portal blood flow and the lymph volume)
8. Hemothorax
 - Hemothorax is the presence of blood in the pleural cavity with hematocrit exceeds 50% of that in peripheral blood.
 - The initial drainage is greater than 1500ml/24h (20ml/kg body weight) or the drainage rate is greater than 200ml/h (3ml/kg body weight) for more than three consecutive hours, thoracotomy is indicated

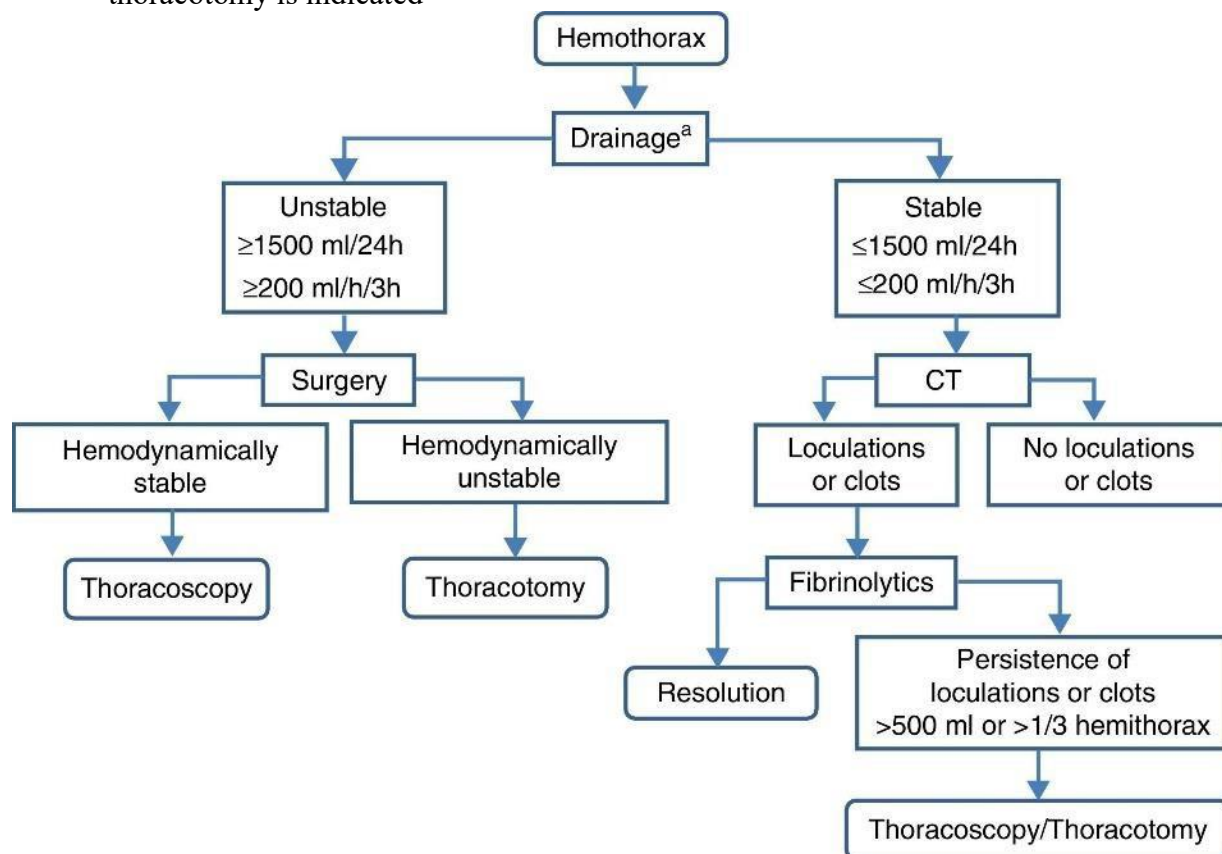


Fig. Treatment algorithm for hemothorax. Excludes aortic dissection.

IX- COMPLICATIONS

- Pleural sequelae or scars
- Lungs damage

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PNEUMOTHORAX

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Asst. Prof. CHHAR Bunpaul*

I. DEFINITION

Pneumothorax is defined as air in the pleural space. Primary pneumothoraces arise in healthy people without any lung disease and the secondary pneumothoraces arise in people with underlying lung disease.

II. EPIDEMIOLOGY

Primary spontaneous pneumothorax mainly occurs at 20-30 years of age. The incidence of PSP in the United States is 7 per 100,000 men and 1 per 100,000 women per year. Secondary spontaneous pneumothorax is seen more in old-age patients 60-65 years. The incidence of SSP is 6.3 and 2 cases for men and women per 100,000 patients, respectively. The male-to-female ratio is 3:1. COPD has an incidence of 26 pneumothoraces per 100,000 patients.

III. ETIOLOGY

Regard to the etiology of pneumothorax, anatomical abnormalities have been demonstrated, even in absence of over underlying lung disease. Subpleural blebs and bullae are found at the lung apices at thoracoscopy and on CT scanning in up to 90% cases of primary spontaneous pneumothorax (PSP). Smoking has been implicated in this etiological pathway; the smoking habits being associated with a 12% risk of developing pneumothorax in healthy smoking men compared with 0.1% in non-smoking. The risk of recurrence of PSP is as high as 54% within the first 4 years, with isolated risk factors including smoking, height and age > 60 years. Risk factors for recurrence of secondary pneumothorax (SSP) include age, pulmonary fibrosis and emphysema.

3.1. Risk factors for primary spontaneous pneumothorax

- Smoking
- Tall thin body habitus in an otherwise healthy person
- Marfan syndrome, familial pneumothorax

3.2. Diseases associated with secondary spontaneous pneumothorax

- Asthma, COPD
- Necrotizing pneumonia, tuberculosis, Severe ARDS
- HIV with pneumocystis pneumonia
- Other uncommon cause: sarcoidosis, cystic fibrosis, bronchogenic carcinoma, idiopathic pulmonary fibrosis, langerhans cell histiocytosis, lymphangioleiomyomatosis, collagen vascular disease, inhalational drug use like cocaine or marijuana, thoracic endometriosis

3.3. Causes of iatrogenic pneumothorax:

- Pleural biopsy, transbronchial lung biopsy, transthoracic pulmonary nodule biopsy, central venous catheter insertion, tracheostomy, intercostal nerve block, positive pressure ventilation

3.4. Causes of traumatic pneumothorax

IV. CLINICAL EVALUATION

The most common symptoms are chest pain and shortness of breath. The chest pain is pleuritic, sharp, severe, and radiates to the ipsilateral shoulder. In general, the clinical symptoms associated with secondary pneumothoraxes are more severe than those associated with primary pneumothoraxes, and most patients with a secondary pneumothorax complain of breathlessness which is out of proportion to the size of the pneumothorax. In defining a management strategy, the size of a pneumothorax is less important than the degree of clinical compromise.

In tension pneumothorax clinical symptoms:

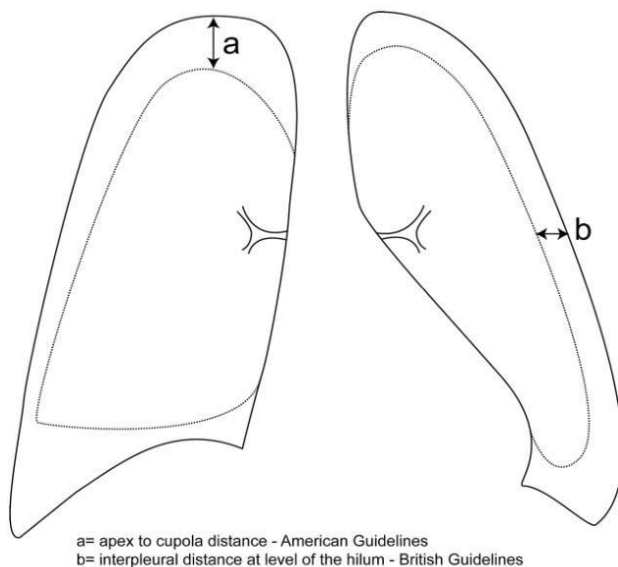
- Tachycardia of more than 130 beats per minute
- Hypotension
- Jugular venous distension
- Cyanosis
- Respiratory failure and cardiac arrest

V. DIAGNOSIS

In both primary and secondary spontaneous pneumothorax, the diagnosis is normally established by plain, chest radiography and other modalities such as Ultrasound scanning, CT scanning.

5.1. Radiologic

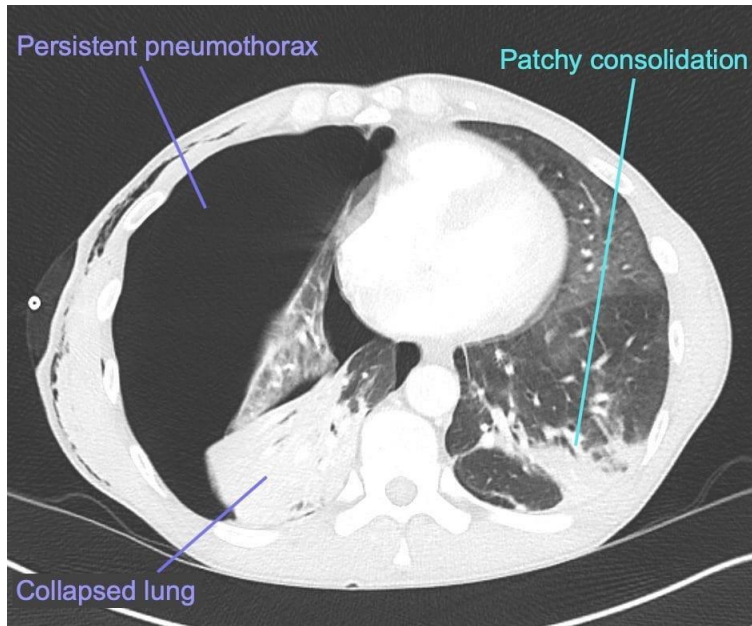
The size of a pneumothorax is divided into “small” or “large” depending on the presence of a visible rim of <2 cm or >2 cm between the lung margin and the chest wall



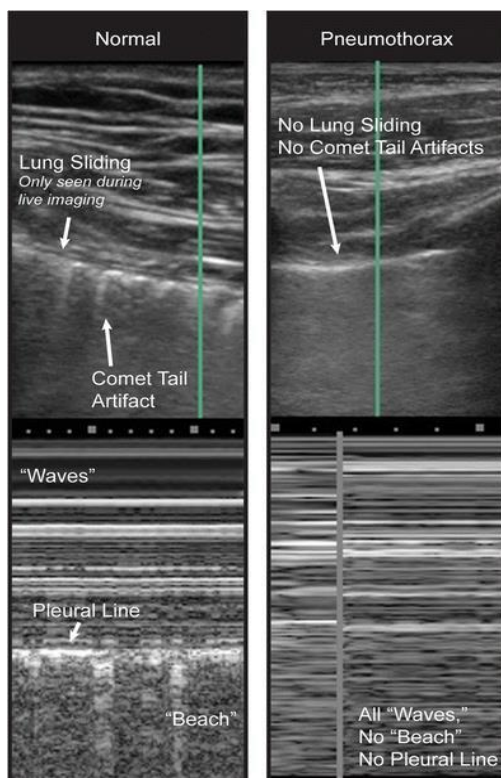
The differentiation of a 'large' from a 'small' pneumothorax is the presence of a visible rim of more than 2 cm between the lung margin and the chest wall, at the level of the hilum as defined in the BTS Guidelines, (b) in figure. A 2cm radiographic pneumothorax approximates to a 50% pneumothorax by volume.

5.2. CT scan

Accurate pneumothorax size calculations are best achieved by CT scanning. CT scanning is recommended when differentiating a pneumothorax from complex bullous lung disease, when aberrant tube placement is suspected, and when the plain chest radiograph is obscured by surgical emphysema.



5.3. Ultrasonography has up to a 94% sensitivity and 100% specificity (better than chest x-ray).



VI. MANAGEMENT

Management may need clarification include:

- (1) Observation for stable cases
- (2) Simple aspiration (thoracentesis) versus intercostal tube drainage (tube thoracostomy) as the second step in the management of primary and secondary pneumothoraxes in instable cases.
- (3) Treatment of elderly pneumothorax patients or patients with underlying lung disease,
- (4) When to refer patients with a difficult pneumothorax to a chest physician or thoracic surgeon for persistent air leak or failure of re-expansion of the lung, and
- (5) Treatment of the recurrent pneumothorax.

In case of stability:

- Respiratory rate <24 breaths per minute.
 - Heart rate <120 and >60 beats per minute
 - Normal blood pressure (not defined)
 - Room air oxygen saturation >90 percent
 - Ability to speak in whole sentences
- ⇒ All other patients are considered unstable.

Observation is the treatment of choice for small PSP without significant breathlessness or an asymptomatic small primary spontaneous pneumothorax (depth less than 2cm), with follow-up in outpatient after 2-4 weeks.

The patients with associated symptoms and showing signs of instability, needle decompression is the treatment of a pneumothorax. This usually is performed with a 14 to 16 gauge and 4.5 cm in length. Using supplemental oxygen can increase this reabsorption rate. Both tension pneumothorax and bilateral pneumothorax are potentially life-threatening events that require chest drain insertion.

6.1.Spontaneous pneumothorax

Stable patients

- It depends on the size of the pneumothorax, associated symptoms, the event is the first or a recurrent even, uni or bilateral pneumothorax, with or without the pleural effusion and presence of complex loculations.

6.1.1. Observation

- It is the treatment of choice for small closed pneumothoraxes, first episode without significant breathlessness.
- Patients with small primary pneumothoraxes and minimal symptoms do not require hospital admission.

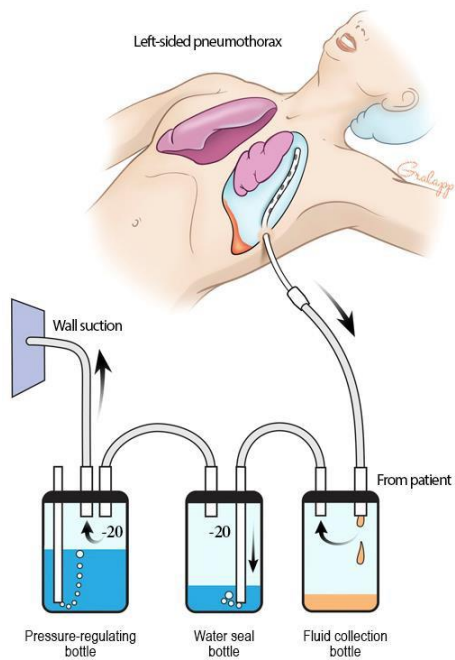
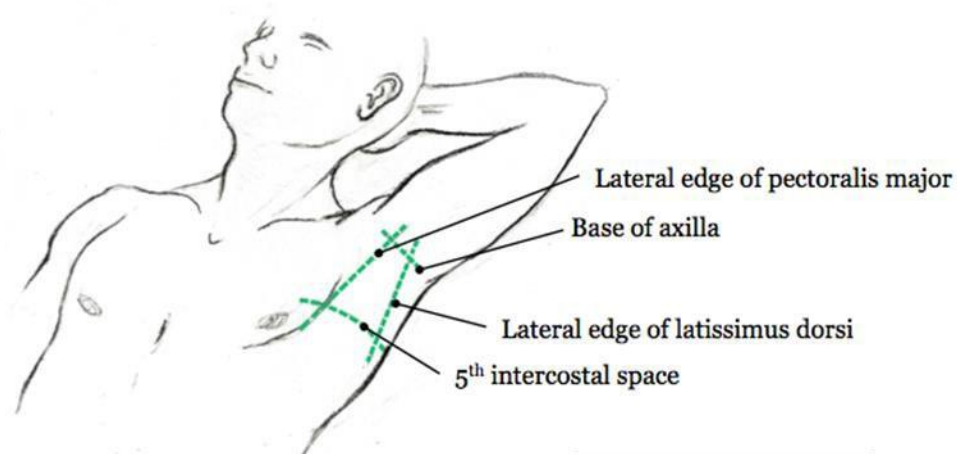
6.1.2. Procedures

The large pneumothorax: needle or catheter aspiration, chest tube

- Needle aspiration (14-16 G) of intrapleural air



- Catheter thoracostomy or Chest tube drainage without suction or with suction if pneumothorax not improved in 24h. Large-bore chest drains are not needed for pneumothorax.



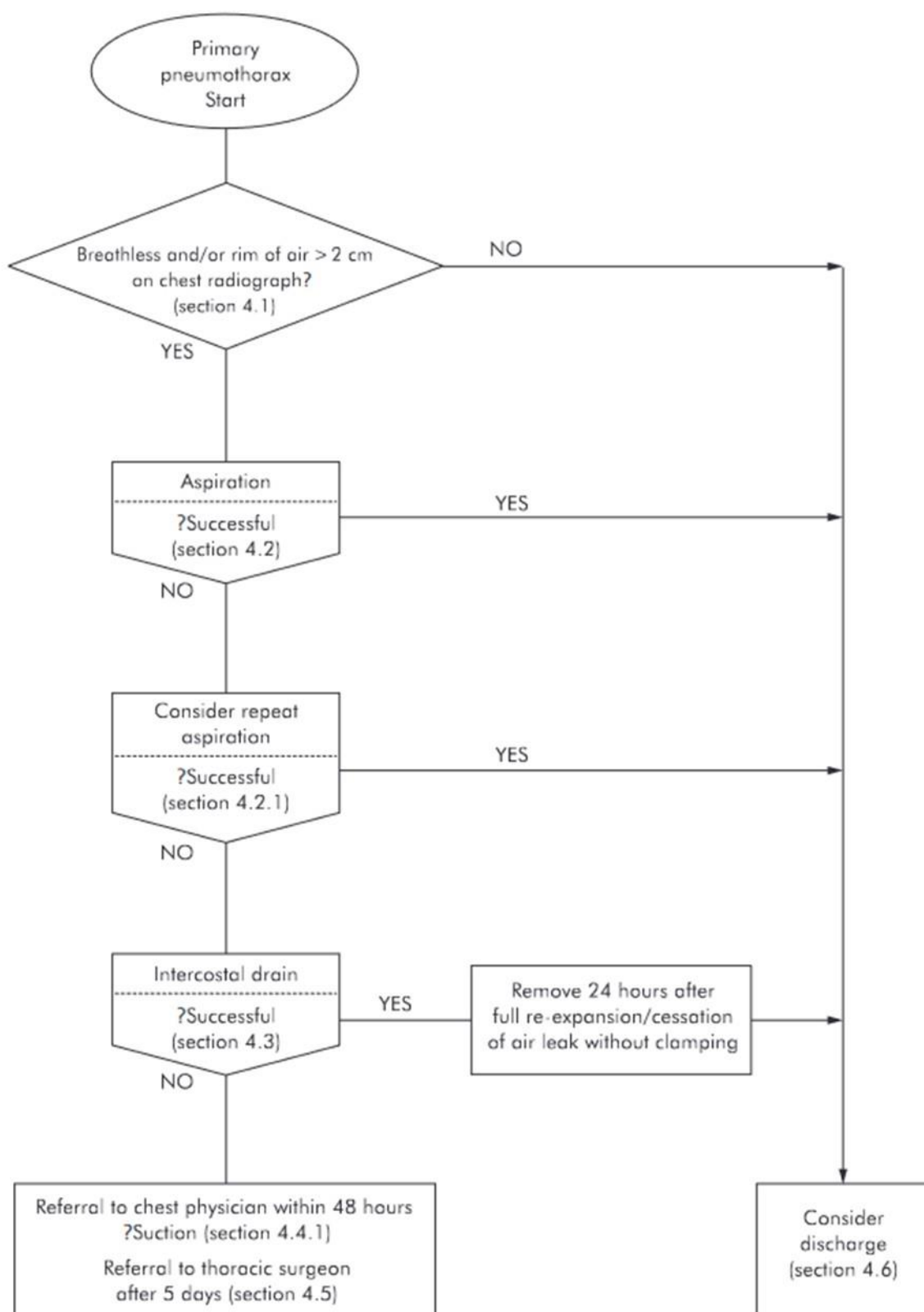


Figure 1 Recommended algorithm for the treatment of primary pneumothorax.

6.2.Secondary pneumothoraxes

- 6.2.1. Observation alone is only recommended in patients with small secondary pneumothoraxes.
 - Inhalation of high concentrations of oxygen may reduce the total pressure of gases in pleural capillaries.
 - Those with a persistent air leak should be discussed with a thoracic surgeon at 48 h.
- 6.2.2. Simple aspiration
 - Patients with secondary pneumothoraxes treated successfully with simple aspiration should be admitted to hospital and observed for at least 24 hours before discharge.
- 6.2.3. Intercostal tube drainage
 - If simple aspiration or catheter aspiration drainage of any pneumothorax is unsuccessful, large secondary pneumothoraxes (>2 cm), particularly in patients over the age of 50, should be considered a high risk of failure for simple aspiration and recurrence and therefore tube drainage is recommended as appropriate initial treatment with small size of chest tube (10–14 F) should be used initially. The use of catheter over guidewire systems (Seldinger technique) may prove to be as safe and effective as small calibre tubes.

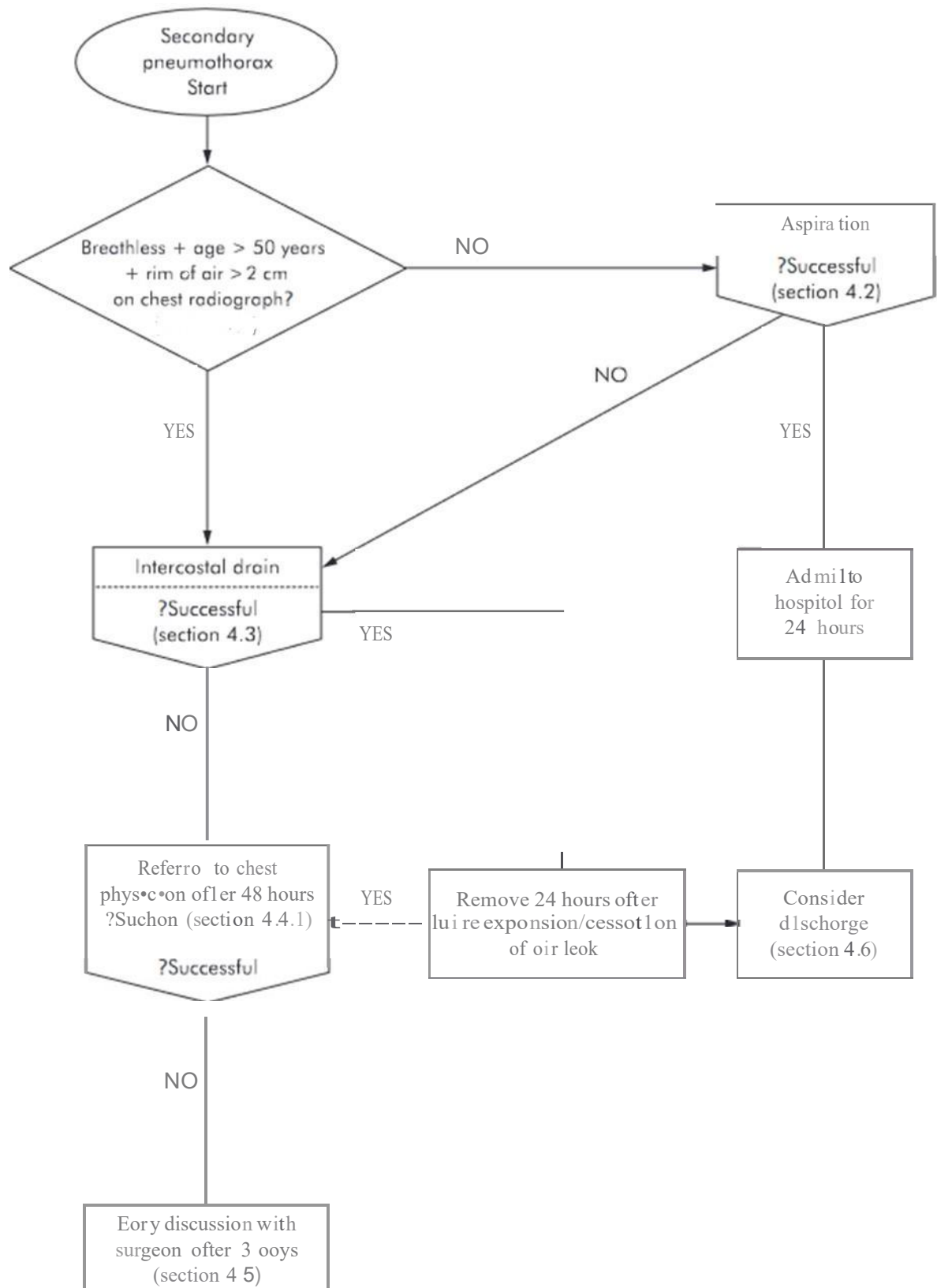


Figure 2 Recommended algorithm for the treatment of secondary pneumothorax.

**** Referral to respiratory specialists** in case allure of a pneumothorax to re-expand or a persistent air leak exceeding 48 hours duration should prompt to refer. Such patients may require sustained chest drainage with complex drain management (suction, chest drain repositioning) and thoracic surgery decisions.

6.2.4. Chest drain suction

- Suction to an intercostal tube should not be applied directly after tube insertion, but can be added after 48 hours for persistent air leak or failure of a pneumothorax to re-expand.
- Low pressure (–10 to –20 cm H₂O) suction systems are recommended.
- The surgical referral for a persistent air leak in those without pre-existing lung disease should be made at 5–7 days.

6.2.5. Chemical pleurodesis

- Chemical pleurodesis can control difficult or recurrent pneumothorax but should only be attempted if the patient is either unwilling or unable to undergo surgery.
- Talc may be used on the grounds that it is the most effective agent in malignant effusion.

**** Referral to thoracic surgeons:**

- In cases of persistent air leak or failure of the lung to re-expand, the managing respiratory specialist should seek an early (5-7 days) thoracic surgical opinion.

Open thoracotomy and pleurectomy

- The procedure with the lowest recurrence rate for difficult or recurrent pneumothoraxes. Minimally invasive procedures, thoracoscopy (VATS), pleural abrasion, and surgical talc pleurodesis are all effective alternative strategies.

Indications for surgical intervention (VATS vs. thoracotomy) [5]

- Continuous air leak for longer than seven days
- Bilateral pneumothoraxes
- The first episode in high-risk profession patients, i.e., Divers, pilots
- Recurrent ipsilateral pneumothorax
- Contralateral pneumothorax
- Patients who have AIDS

The accepted indications for operative intervention are as follows:

- Second ipsilateral pneumothorax
- First contralateral pneumothorax
- Bilateral spontaneous pneumothorax
- Persistent air leak (>5–7 days of tube drainage; air leak or failure to completely re-expand)
- Spontaneous hemothorax
- Professions at risk (e.g. pilots, divers)

6.3.Surgical strategies

There are two objectives in the surgical management of a primary pneumothorax. The first objective is resection of blebs or the suture of apical perforations to treat the underlying defect. The second objective is to create a pleural symphysis to prevent recurrences.

6.3.1. Open thoracotomy:

- Open thoracotomy yields the lowest postoperative recurrence results. Bulla ligation/excision, thoracotomy with pleural abrasion, and either apical or total parietal pleurectomy all have failure rates under 0.5%.
- It may be associated with increased postoperative respiratory dysfunction and hospital stay compared with VATS procedures.

6.3.2. Surgical chemical pleurodesis with or without Video assisted thoracoscopic surgery (VATS)

- It is best achieved with 5 g sterile talc with successful pleurodesis (85–92%) that is similar to alternative thoracoscopic techniques for complicated pneumothorax.

6.4. Special cases management

6.4.1. Tension pneumothorax

Treatment is with high concentration oxygen and emergency needle decompression. A cannula usually being introduced in the second anterior intercostal space in the mid-clavicular line.

6.4.2. Pneumothorax and pregnancy

The modern and less invasive strategies of simple observation and aspiration are usually effective during pregnancy, with elective assisted delivery and regional anesthesia at or near term. A corrective surgical procedure (VATS) should be considered after delivery.

6.4.3. Catamenial pneumothorax

A combination of surgical intervention and hormonal manipulation requires cooperation with thoracic surgeons and gynecologists.

6.4.4. Pneumothorax and AIDS

The combination of pneumothorax and HIV infection requires early intercostal tube drainage and surgical referral, in addition to appropriate treatment for HIV and PJP infection.

6.4.5. Pneumothorax and cystic fibrosis

The development of a pneumothorax in a patient with cystic fibrosis requires early and aggressive treatment with early surgical referral.

VII. EVOLUTION AND PROGNOSIS

- Patients discharged without intervention should avoid air travel until a chest radiograph has confirmed resolution of the pneumothorax and until 5 weeks interval.
- Pneumothorax patients who are successfully treated with simple aspiration should be admitted for 24 hours before discharge to ensure no recurrence.
- Return for a follow up chest radiograph after 2 weeks.
- Smoking influences the risk of recurrence, Smoking cessation.

VIII. COMPLICATIONS

- Pneumopericardium, Pneumoperitoneum, Pneumohemothorax
- Bronchopulmonary fistula, Pyopneumothorax,
- Re-expansion pulmonary edema
- Respiratory failure,
- Cardiac arrest
- Damage to the neurovascular bundle during tube thoracostomy
- Pain and skin infection at the site of tube thoracostomy

IX. PROGNOSIS

Recurrence can occur for up to three years period. The recurrence rate in the following five years is 30% for PSP and 43% for SSP [5]. The risk of recurrence increases with each subsequent pneumothorax; it is 30% with the first, 40% after a second, and more than 50% after the third recurrence. SSPs are more lethal depending on underlying lung disease and the size of the pneumothorax. Patients with COPD and HIV have high mortality after pneumothorax. The

mortality of SSP is 10%. Mortality of tension pneumothorax is high if appropriate measures are not taken.

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BRONCHIECTASIS

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I. DEFINITION

Bronchiectasis: an irreversible and abnormal dilation of the bronchial tree that produces chronic respiratory symptoms (e.g., chronic productive cough)

Acute exacerbation of bronchiectasis: a deterioration in the symptoms of bronchiectasis that requires a change in the regular treatment (e.g., adding antibiotics, increasing airway clearance techniques) For research purposes, the definition of acute exacerbations specifies that at least three symptoms (cough, quality and quantity of sputum, fatigue, haemoptysis, or dyspnea) are worsened for ≥ 48 hours. In clinical practice, there is a lower threshold regarding the number and duration of worsening symptoms.

II. EPIDEMIOLOGY

- An estimated 350,000 to 500,000 adults have bronchiectasis in the United States.
- The prevalence of bronchiectasis increases with age with an 8- to 10-fold difference in prevalence after the age of 60 years (300 to 500 per 100,000) as compared to ages <40 to 50 years (40 to 50 per 100,000). Reports from the United Kingdom and Germany describe similar marked age differences in prevalence.

III. ETIOLOGY

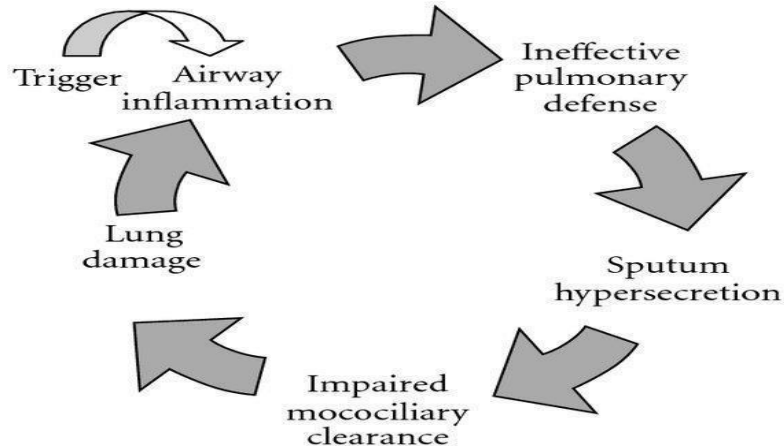
These processes result in the permanent dilation of airways.

- Pulmonary infections (i.e., bacterial, viral, fungal), especially severe or chronic infections
- Disorders of secretion clearance or mucous plugging
 - Cystic fibrosis (CF) most common congenital cause of bronchiectasis
 - Primary ciliary dyskinesia (PCD)
 - Allergic bronchopulmonary aspergillosis (ABPA)
 - Kartagener syndrome
 - Smoking: associated with poor ciliary motility
- Bronchial narrowing or other forms of obstruction
 - COPD
 - Aspiration
 - Tumors
 - $\alpha 1$ -antitrypsin deficiency
 - Other congenital and acquired conditions (e.g., congenital bronchiectasis, tracheomalacia, bronchogenic cyst)
 - Immunodeficiency (e.g., common variable immunodeficiency, hypogammaglobulinemia, HIV)
 - Chronic inflammatory diseases (e.g., rheumatoid arthritis, Sjogren syndrome, Crohn disease)

IV. PHYSIOPATHOLOGY

Bronchiectasis requires the combination of two important processes taking place in the bronchi: either local infection or inflammation along with either inadequate clearance of secretions, airway obstruction, or impaired host defences.

Pathogenesis: The common thread in the pathogenesis of bronchiectasis consists of difficulty clearing secretion and recurrent infections with a (Vicious Circle) of infection and inflammation resulting in airway injury and remodelling.



3 Mechanisms of bronchiectasis

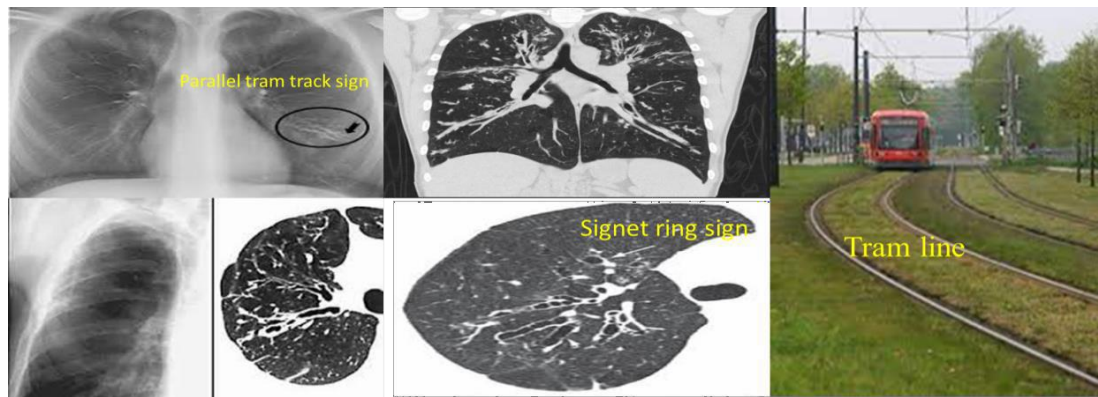
1. Obstruction: Can occur because of tumor, foreign body, impacted mucus due to poor muco-ciliary clearance, external compression, bronchial webs and atresia.
2. Infections by *Bordetella pertussis*, Measles, Rubella, Adenovirus and mycobacterium tuberculosis induce chronic inflammation.
3. Chronic inflammation contributes to the mechanism by which obstruction leads to bronchiectasis.

V. CLASSIFICATION OF BRONCHIECTASIS

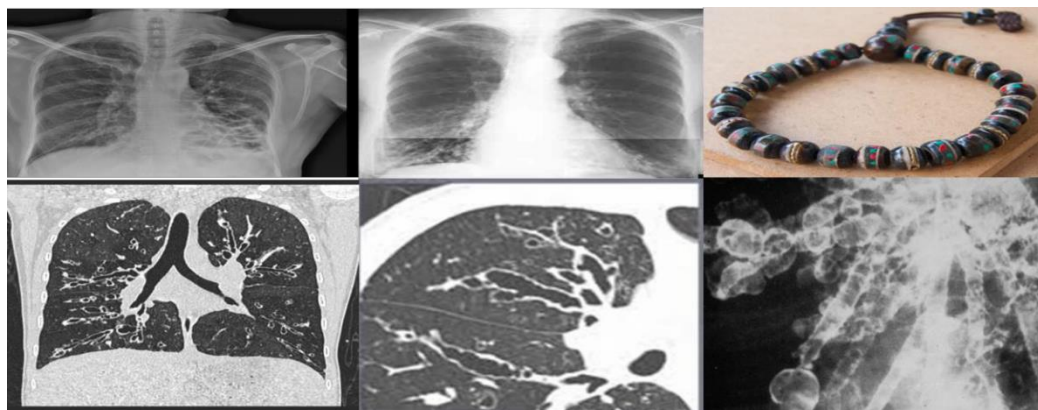
- Saccular or Cystic Bronchiectasis:
 - Bronchial dilatation progresses and results in ballooning of bronchi that end in fluid or mucous filled sacs.
 - Radio: Dilated airways form grape-like or ballooning structures with possible air-fluid levels.



- Cylindrical or Tubular of bronchiectasis:
 - Bronchial outlines are regular, but there's diffuse dilatation of the bronchial unit.
 - Bronchial lumen ends abruptly because of mucous plugging.
 - Parallel tram track sign or Tramline appearance on CT scan.



- Varicose Bronchiectasis:
 - Degree of dilatation is greater, local constrictions cause irregularity of outline resembling that of varicose veins.
 - Dilated airways with areas of focal constriction (Beaded Contour on CT scan)



VI. DIAGNOSIS

6.1 Clinical failures

- Chronic productive cough (lasting months to years) with copious mucopurulent sputum
- Bronchophony
- Dyspnea
- Hemoptysis: usually mild or self-limiting, but severe hemorrhage that requires embolization may occur
- Clubbing of nails (uncommon)
- Auscultation: Crackles and rhonchi or Wheezing (due to obstruction from secretions, airway collapsibility, or a concomitant condition)

6.2 Investigation Imaging

- Imaging modalities
 - X-ray chest: best initial exam (may not show mild disease)
 - High-resolution computed tomography (HRCT) chest: confirm diagnosis
- Findings
 - Bronchial dilation
 - Cylindrical or tubular (most common): parallel tram track sign and signet ring sign
 - Varicose
 - Saccular or cystic (most severe form)
 - Thickened bronchial walls, mucus plugging, honeycombing (suggests late-stage bronchiectasis)
 - Specific findings suggestive of the underlying etiology
- Focal distribution: localized disease (e.g., tumors, foreign bodies, strictures)
 - Diffuse distribution: fibrosing diffuse lung diseases
 - Lower lung predominance (e.g., recurrent aspiration, pulmonary fibrosis, α 1-antitrypsin deficiency)
 - Upper or middle lung predominance (e.g., CF, sarcoidosis, ABPA, tuberculosis)
 - ✓ Perihilar lymphadenopathies (e.g., sarcoidosis, tuberculosis)
 - ✓ Situs inversus (seen in primary ciliary dyskinesia)
- Laboratory studies
 - CBC: Depend on the underlying etiology.
 - Decrease of Hb
 - Variable leukocyte levels
 - Increase of Thrombocytes
 - Consider ABG (e.g. in patients with decrease of SaO₂ on room air): may show hypercapnia
 - Quantitative measurement of serum immunoglobulins and electrophoresis (IgA, IgG, IgM, and IgG subclasses): to exclude immunodeficiencies
 - Aspergillus fumigatus IgG and IgE: to exclude ABPA
- Sputum culture and smear: culture induced sputum or bronchoalveolar lavage for all of the following pathogens to detect infection or colonization
 - Bacterial cultures: e.g., H. influenzae or P. aeruginosa
 - Fungal cultures: Candida spp. or Aspergillus spp.
 - Mycobacterial culture (with AFB microscopy): tuberculosis or nontuberculous mycobacteria smear
- Spirometry: not needed to establish a diagnosis but useful to monitor disease progression
 - Mixed obstructive/restrictive pattern
 - Restrictive pattern: Decrease FEV1 and FVC with normal or increase FEV1/FVC ratio)

- Obstructive pattern (decrease FEV1/FVC ratio)
- Additional diagnostics

These studies may be considered depending on clinical suspicion if the initial workup was not diagnostic.

 - Laboratory studies
 - Specific antibodies (e.g., rheumatoid factor, ANAs, anti-CCP antibodies, ANCA): suspected connective tissue or autoimmune diseases
 - Sweat chloride test and/or genetic testing for CFTR mutations: suspected CF
 - HIV testing, serum α 1-antitrypsin levels, nasal nitric oxide testing for primary ciliary dyskinesia
 - Further studies
 - Bronchoscopy
 - ✓ Indicated to visualize tumors, foreign bodies, or other lesions (most useful in patients with localized disease)
 - ✓ May also be used in combination with bronchoalveolar lavage (BAL) to obtain specimens for staining and culture
 - Upper GI studies (e.g., upper endoscopy, barium swallow, esophageal manometry): consider for suspected gastroesophageal reflux disease or recurrent pulmonary aspiration
 - Colonoscopy with biopsies: if associated inflammatory bowel disease is suspected (e.g., GI bleeding).

VII. DIFFERENTIAL DIAGNOSES

- Aspiration Pneumonitis and Pneumonia
- Asthma
- Bacterial Pneumonia
- Bronchitis
- Chronic Obstructive Pulmonary Disease (COPD)
- Cystic Fibrosis
- Emphysema
- Gastroesophageal Reflux Disease
- Tuberculosis (TB)

VIII. MANAGEMENT

8.1. The objectives of care in bronchiectasis:

- Treat identifiable underlying causes
- Control symptoms
- Reduce the number and severity of exacerbations
- Prevent deterioration in pulmonary function
- Improve quality of life
- Minimise complications
- Reduction in hospital admissions, length of stay, and disease and health status decline.

8.2. Critical Sign

- Exacerbations of bronchiectasis
 - Recurrent bouts of pneumonia and acute bacterial infection of sections of dilated bronchi
 - Frequently due to *Pseudomonas aeruginosa*

8.3. Mortality risk factor:

Frequent infection and Exacerbation

8.4. Management of acute exacerbations

- Approach
 - Provide supportive treatment and oxygen therapy as needed.
 - Optimize mucoactive agents (Pharmacotherapy for airway clearance in bronchiectasis).
 - Optimize airway clearance techniques.
 - Obtain a new sputum culture and start empiric antibiotic therapy, based on the meet recent sputum culture.
 - Tailor antibiotics to the most recent sputum culture once available and preferably complete 14 days of therapy.
- Monitoring and disposition
 - Outpatient treatment: for hemodynamically stable patients with mild to moderate symptoms
 - Consider inpatient treatment in the following cases:
 - Clinical deterioration or no clinical improvement following completion of an oral antibiotic regimen (e.g., lasting 10–14 days)
 - Sepsis or severe pneumonia
 - Haemoptysis ≥ 10 mL in 24 hours and clinical deterioration
 - New isolation of *P. aeruginosa*
 - Common empiric antibiotic regimens for bronchiectasis exacerbations
 - Outpatient treatment
 - No previous culture data available: Consider fluoroquinolones.
 - ✓ Ciprofloxacin 500–750 mg PO every 12 hours
 - ✓ Levofloxacin 500 mg PO every 12 hours
 - Previous culture data and antibiogram available
 - ✓ Colonization with sensitive pathogens
 - Amoxicillin 500 mg PO every 8 hours
 - OR Doxycycline 100 mg PO every 12 hours
 - OR Clarithromycin 500 mg PO every 12 hours
 - ✓ Colonization with beta-lactamase positive organisms
 - Amoxicillin/clavulanic acid 625 mg PO every 8 hours
 - OR Doxycycline 100 mg PO every 12 hours
 - ✓ Colonization with *P. aeruginosa*: Ciprofloxacin 500–750 mg PO every 12 hours
 - Inpatient treatment
 - ✓ Colonization with *P. aeruginosa*
 - Piperacillin- tazobactam 4.5 g IV every 6–8 hours
 - OR Ceftazidime 2 g IV every 8 hours
 - OR Meropenem 2 g IV every 8 hours
 - ✓ Colonization with MRSA
 - Vancomycin 15–20 mg/kg every 8–12 hours; adjust based on therapeutic monitoring (maximum dose of 3000 mg/day)
 - OR Linezolid 600 mg IV every 12 hours

8.5. Long-term management

Management goals are to stop or delay disease progression, reduce exacerbation frequency (goal ≤ 2 per year), achieve symptom control, and improve the patient's quality of life.

8.5.1. Approach

- **General measures**
 - Educate the patient regarding prognosis and the use of long-term medications.
 - Promote lifestyle changes like regular exercise and smoking cessation
 - Educate the patient on airway clearance techniques.
 - Bronchopulmonary hygiene and chest physiotherapy: **e.g., cupping and clapping, postural drainage**, directed cough, hydration
 - Pulmonary rehabilitation: may improve exercise capacity and respiratory symptoms
 - Administer vaccinations (i.e., seasonal influenza vaccine, pneumococcal vaccine).
 - Consider treatment with mucoactive agents, bronchodilators, or corticosteroids if airway clearance is difficult.
 - Provide specific treatment for the underlying cause if identified.
- Follow-up: Perform follow-ups every 6–12 months to identify disease progression.
 - Bacterial and fungal sputum culture: every 6–12 months
 - Spirometry: every 12 months
- Disease progression
 - Consider long-term antibiotic therapy for bronchiectasis with ≥ 3 exacerbations per year.
 - Repeat diagnostic studies to assess disease progression.
- Screen for complications.
 - ECG
 - Consider echocardiogram.
 - Consider CT chest with contrast.
- Advanced disease
 - Ensure respiratory support as needed.
 - Long-term oxygen therapy for chronic respiratory failure (indications are the same as for COPD)
 - Consider oxygen with humidification for patients with respiratory failure with hypercapnia.
 - Invasive procedures: not routinely indicated

8.5.2. Pharmacotherapy for airway clearance in bronchiectasis

- **Mucoactive agents**

Consider in patients with difficulty expectorating or persistent symptoms despite adequate airway clearance techniques; continue use based on clinical benefits. There is a paucity of evidence for the benefit of mucoactive agents in bronchiectasis.

 - Nebulized mucoactive agents: Trial for at least 3 months.
 - Hypertonic saline (3–7% NaCl)
 - Mannitol
- **Oral mucolytics**: Trial for at least 6 months, e.g., N-acetylcysteine.
- **Bronchodilators**

Consider on a case-by-case basis. Evidence to support the use of bronchodilators in bronchiectasis is limited.

 - Short-term bronchodilators (e.g., SABA): in patients with airway reactivity and bronchospasm or prior to inhaled mucoactive treatment [15]
 - Long-term bronchodilators (e.g., LABA, LAMA): may be used in patients with severe dyspnea
 - Regular use of bronchodilators (e.g., for asthma or COPD): Continue as usual.

- **Corticosteroids**
Corticosteroids are not routinely recommended because of the limited benefit and side effects of corticosteroid therapy.
 - Inhaled corticosteroids: Continue if already indicated for asthma or COPD.
 - Systemic corticosteroids: indicated in patients with ABPA
- **Long-term antibiotic therapy for bronchiectasis**
 - The goal is to suppress bacterial growth and to reduce symptoms and exacerbations as a measure of secondary prevention in patients with ≥ 3 exacerbations per year. Antibiotic therapy should be administered for at least 3 months and may be extended based on clinical response and tolerability.
 - No colonization with *P. aeruginosa*
- **First-line: oral macrolides: A macrolide antibiotic that is typically used to treat pneumonia, gram-positive infections and some sexually transmitted diseases (e.g., chlamydial infections).**
 - Azithromycin 250 mg PO daily OR Azithromycin 250 mg PO 3 times/week OR Azithromycin 500 mg PO 2–3 times/week (is indicated for at least six months, and it may be continued indefinitely if tolerated).
- Alternative: inhaled antibiotics (e.g., tobramycin 160–300 mg nebulized every 12 hours) colonization with *P. aeruginosa*
- Growth suppression
 - Tobramycin 160–300 mg nebulized every 12 hours
 - Aztreonam 75 mg nebulized every 8 hours
 - Alternative: long-term treatment with macrolides (e.g., azithromycin)
- Eradication therapy (weak evidence)
 - E.g., 2 weeks of oral fluoroquinolones (e.g., ciprofloxacin) followed by 3 months of inhaled antibiotics (e.g., tobramycin, colistin)

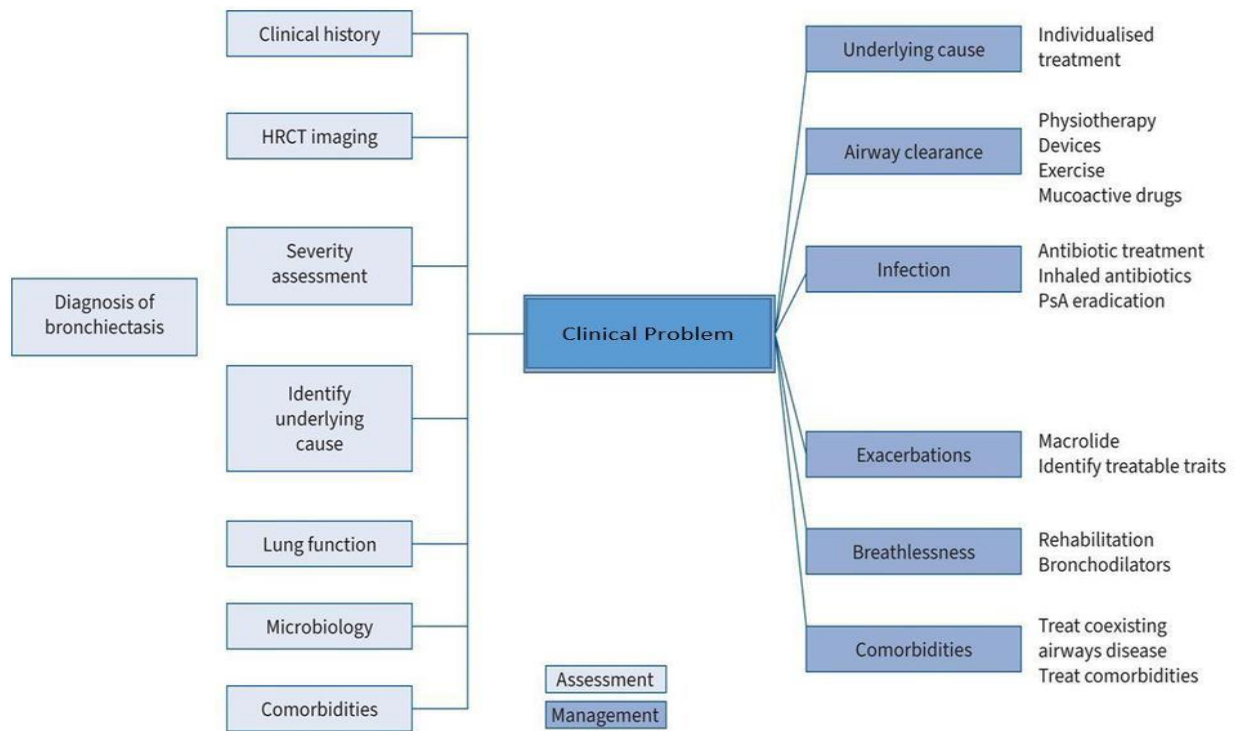
8.5.3. Isolation of other pathogens

- Nontuberculous mycobacteria: Treatment usually includes three drugs (macrolide PLUS rifampicin PLUS ethambutol) for at least one year.
 - MRSA: Consider eradication therapy based on local protocols.
 - Methicillin-resistant *Staphylococcus aureus* (MRSA): An isolate of *Staphylococcus aureus* that is resistant to certain beta-lactam antibiotics as a result of a penicillin-binding protein (PBP)

8.5.4. Invasive procedures

- Surgical resection of bronchiectatic lung or lobectomy: indicated in pulmonary hemorrhage, inviable bronchus, and poor control of symptoms despite optimal medical therapy in unilateral bronchiectasis with well-localized disease
- Pulmonary artery embolization: indicated in pulmonary haemorrhage.
- Lung transplantation: The transplantation of a lung from a deceased donor. Indicated in patients with end-stage lung disease who are predicted to survive < 2 years under medical and/or surgical management. Complications include pneumonia, graft rejection, airway anastomotic complications, and infection as a result of immunosuppressive therapy.
 - Consider for severe disease
 - Rapid disease progression.

Algorithm



IX. SPECIAL CASES

In patients with suspected bronchiectasis, diagnosis is confirmed using imaging studies, preferably a HRCT scan. Additional diagnostic studies are useful to identify the underlying cause and possibly provide specific treatment.

X. PROGNOSIS

The disease is progressive when associated with ciliary dysfunction and cystic fibrosis, and eventually causes respiratory failure in other patients. The prognosis can be relatively good if postural drainage is performed regularly, and antibiotics are used judiciously.

XI. COMPLICATIONS

- Recurrent bronchopulmonary infections → chronic obstructive pulmonary disease → respiratory failure and cor pulmonale
- Pulmonary hemorrhage (massive hemoptysis)
- Lung abscess.

XII. REFERENCE

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6. 2019 British Thoracic Society Guideline for bronchiectasis in adults

LUNG ABSCESS

Prof. CHAN Sarin

I. DEFINITION

Lung abscesses are pus-containing necrotic lesions of the lung parenchyma that result from aspiration of bacteria-laden secretions and show an air-fluid level. Lung abscesses are distinct from, and may follow, necrotizing pneumonia, in which multiple small cavities develop in contiguous areas of the lung [1,2]. Lung abscesses must be distinguished from septic pulmonary emboli, which are often multiple and bilateral, involve the lower lobes, and are secondary to an endovascular infection.

Lung abscesses can be classified as primary or secondary. Primary lung abscesses result from direct infection of the pulmonary parenchyma in an otherwise healthy person. Most result from aspiration of oral or gastric contents. Primary lung abscesses can also arise within an area of pneumonia caused by necrotizing organisms (eg, *S. aureus*). Secondary lung abscesses occur when there is a predisposing condition such as bronchial obstruction (eg, foreign body, neoplasm), hematogenous spread (eg, right-sided endocarditis), thoracic surgery, or impaired host defenses.

The terms "necrotizing pneumonia" or "lung gangrene" are used to describe pneumonia that is complicated by necrosis and numerous small abscesses.

II. EPIDEMIOLOGY

Lung abscess affects one woman four/ man ratio, between 40-50 years old. In children, the abscess has a particular entity which is pleuropulmonary staphylococcal disease of infants. An increasing incidence of serious forms.

- Staphylococcus is the germ often responsible,
- Gram negative bacillus, increasingly discovered,
- Lung cancer and tuberculosis are still being sought.

III. ETIOLOGY - MICROBIOLOGY

The microbiology of lung abscesses varies with the route of infection.

Aspiration — Lung abscesses that result from aspiration are typically polymicrobial, reflecting oral and gingival flora.

The most commonly isolated pathogens are microaerophilic *Streptococcus anginosus*, *Streptococcus milleri*, *Streptococcus mitis*, and anaerobes. The most common anaerobes include *Peptostreptococcus*, *Prevotella*, *Bacteroides* and *Fusobacterium*. Gram-negative organisms may be present in up to one-third of cases. These organisms may be a part of a polymicrobial infection or may be the sole pathogens.

Pneumonia caused by pyogenic bacteria — Aerobic bacteria that can cause monomicrobial lung abscess include *Staphylococcus aureus* (eg, in a patient with influenza), *Klebsiella pneumoniae* (eg, in a patient with structural lung disease), other Gram-negative bacilli (eg, *Pseudomonas aeruginosa*), *Streptococcus pyogenes*, *Burkholderia pseudomallei*, *Haemophilus influenzae* type b, *Legionella*, *Nocardia*, and *Actinomyces*. There have been occasional case reports of lung abscess caused by *Streptococcus pneumoniae*, although superinfection by anaerobic bacteria may be the cause in such cases.

The bacteriology of lung abscess may be subject to geographic variability. For example, in a retrospective evaluation of 90 Taiwanese adults with lung abscess, Gram-negative bacilli (especially *K. pneumoniae*) accounted for 36 percent of isolates [16]. Other organisms included anaerobes (34 percent), Gram-positive cocci (26 percent), and Gram-positive bacilli (4 percent).

Nonbacterial pathogens — Selected nonbacterial pathogens can generate or colonize cavitory lesions and radiographically resemble lung abscess. Culprit pathogens include many fungi (eg, *Aspergillus* spp, *Cryptococcus* spp, *Histoplasma capsulatum*, *Blastomyces dermatitidis*, *Coccidioides* spp, the agents of mucormycosis), *Mycobacterium tuberculosis*, nontuberculous mycobacteria (eg, *M. avium*, *M. kansasii*, *M. abscessus*), and parasites (eg, *Entamoeba histolytica*, *Paragonimus westermani*, *Echinococcus* [hydatid cyst]). Pyogenic bacteria can also superinfect cavities caused by mycobacterial, fungal, and parasitic infections, leading to accumulation of liquid in an otherwise empty cavity.

Opportunistic infections — In the immunocompromised host, the most common causes of lung abscess are *Pseudomonas aeruginosa* and other aerobic Gram-negative bacilli, *Nocardia* spp, and fungi (*Aspergillus* and *Cryptococcus* spp). A number of other organisms can occasionally cause lung abscess in immunocompromised hosts, such as the agents of mucormycosis, *Rhodococcus equi*, *Mycobacterium tuberculosis*, and nontuberculous mycobacteria (table 1).

An overview of pulmonary infections in immunocompromised patients is presented separately.

Infectious etiology	Non-Infectious etiology
• Cocci Gram (+): Staphylococcus Aureus +++ Peptostreptococcus, and Streptococcus A endotoxin (+) (rare) • Gram (-) bacilli: Klebsiella pneumoniae; Bacteroides, Fusobacterium nucleatum (Lemierre syndrome), Burkholderia Pseudomallei (For more information, consult the practical guide “Meliodiosis” (Fig. 4 and 5)) • Anaerobic germ • Champions: Aspergillus (Fig 6) • Amoebiasis, ++	• Lung cancer • Others: Cyst (bronchiectasis, cystic form) second infection • Pulmonary infarction (see Embolism pulmonary) • Systemic disease: Vasculitis of Wegener, Pan arteritis nodosa
• Tuberculosis +++.	.

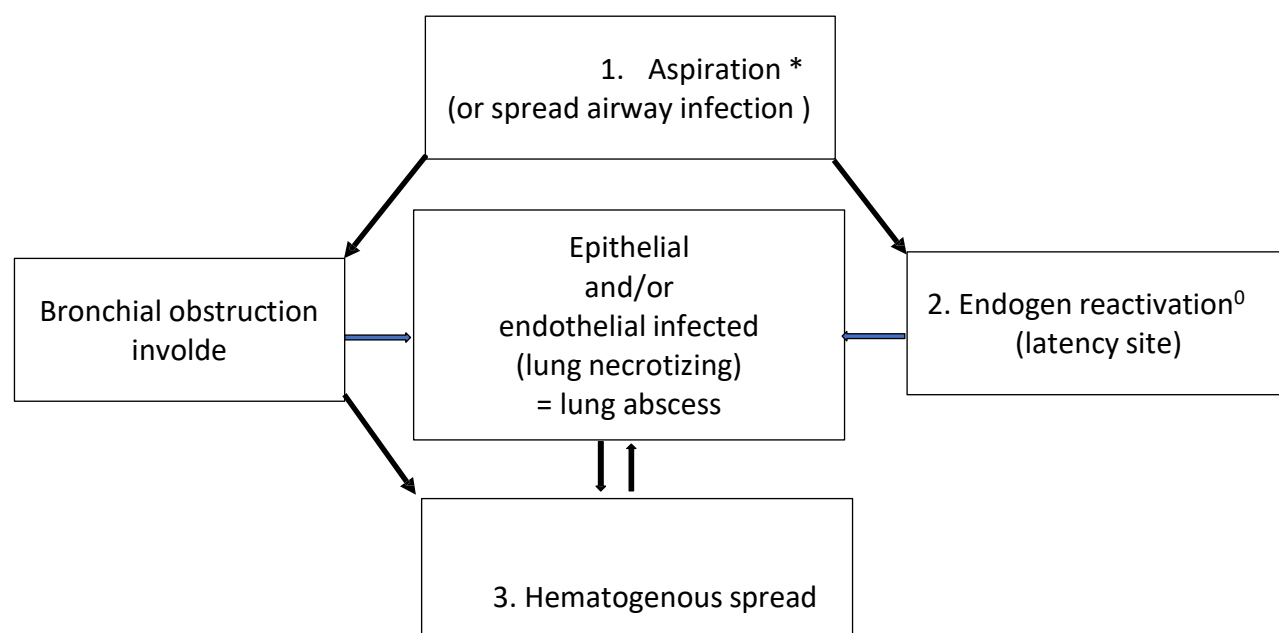
IV. PHYSIOPATHOLOGY

The three pathologies of infections, explained by:

• **Aspiration** —The airway pathway can occur during an inhalation syndrome characterized or more often* by micro-aspiration of oropharyngeal secretion. Lung abscess is classically caused by aspiration of oropharyngeal secretions especially in patients with recurrent aspiration who have dental, gingival, or periodontal infection or paranasal sinusitis. Anaerobes from the gingival crevice or sinuses reach the lower airways [3]. Impaired swallowing and enteral feeding while supine are risk factors. Infection is initiated either because the secretions are not cleared due to suppressed consciousness, because the inoculum size is large, or due to impaired host defenses. Pneumonitis arises first and then progresses to tissue necrosis after 7 to 14 days. Risk factors for aspiration are discussed separately.

- **Hematogenous spread** – Lung abscesses can occur as a consequence of septic embolization during bacteremia (eg, tricuspid valve endocarditis, intravascular catheters, intravenous (IV) drug use, or Lemierre syndrome [ie, jugular vein suppurative thrombophlebitis]). Which still remains a public health problem in developing countries (hygiene, sanitation), occurs in the context of sepsis or septic catheter embolism, in drug addicts, venous injection and dental infection.

- **Endogenous pathway or direct extension** – More specific, can be the consequence of propagation from a latency site⁰ (TB, Melioidosis). A lung abscess may develop by direct extension of an empyema, subphrenic, or mediastinal abscess or a tracheo- or broncho-esophageal fistula.



* Strok and alcohol risk factors

⁰ Immune host impairment

V. DIAGNOSIS

1. Clinical features

The clinical features of a lung abscess are typically nonspecific and mimic those of pneumonia, albeit with a more subacute presentation. It is typically the identification of one or more cavities filled with fluid or with an air-fluid interface on chest imaging that raises the suspicion for lung abscess. The entry point must be carefully sought: dental, sinus and cutaneous.

Symptoms and signs — The presenting signs and symptoms include fever and chills (80 percent), productive cough (often putrid, sour-tasting; 55 to 90 percent), dyspnea (10 percent), chest pain when the pleural space is involved (20 to 35 percent), and hemoptysis (10 percent) [17,24,32-34]. Since abscesses of the lung are usually chronic, unlike pneumonia, most patients have symptoms that evolve over weeks to months. Systemic symptoms such as night sweats, weight loss, anorexia, and fatigue may also be present.

More rapid presentations with fulminant disease (eg, shock) may also occur with acute abscesses (eg, *Staphylococcus aureus*, *Klebsiella pneumoniae*), particularly in those who are immunocompromised, and those with septic emboli.

Typical findings on physical examination are fever, gingival crevice disease (eg, gingivitis, dental caries), associated conditions that reduce consciousness or cause dysphagia (eg, facial or tongue

weakness, bulbar speech, neurologic disease with deconditioning, alcohol intoxication). Chest auscultation may be normal or may demonstrate egophony or increased vocal fremitus, reflecting consolidation, particularly if the abscess is large and subpleural.

Evolution in three phases:

- 1- Phase of and more marked deterioration of the general condition (Fig.1 and 2).
- 2- Vomica phase: vomit and main symptom of the abscess. It is vomit, rarely frank, abundant, asphyxial, more often fractionated. Vomiting is accompanied by thermal defervescence.
- 3- Open hearth phase: very marked alteration of the general condition. The retention phases with low sputum volume and fever are followed by periods of debridement with defervescence and vomit: thus, the sputum and fever curves intersect (Fig.3).

It can direct towards the causal germ:

- *Staphylococcus aureus* (infant, young child, diabetic, co-incidence of flu),
- *Klebsiella pneumoniae* (alcoholic),
- Enterobacteriaceae (digestive operation, patient in intensive care for anti-ulcerative by anti-H2),
- *Pseudomonas aeruginosa* (bronchiectasis, cystic fibrosis).
- Poor oral health increases the risk of contamination by flora anaerobic and that of a polymicrobial infection.
- States of immunosuppression contribute to the occurrence of often stale suppuration. In this regard, it must be remembered that a loss of substance in pneumonia is equivalent to an abscess in neutropenia, which does not produce pus.

We must always look for a locoregional cause: a bronchoscopy systematically can visualize:

- a bronchial tumor or
- a foreign body;
- among other locoregional causes; we must mention esophageal pathology (cancer, stenosis, fistula), bronchiectasis, pulmonary embolism, subphrenic abscess. We must be careful of the pitfalls: hydro-pneumothorax and hiatal hernia.

Chest radiograph — Lung abscesses are often visible on the chest radiograph, manifest as a fluid-filled space, typically with an air-fluid interface, within an area of consolidation, mass, or nodule (image 1 and image 2), although more subtle forms of cavitation may only be appreciated on computed tomography of the chest. Most often, lung abscesses are unilateral.

2. Investigations

2.1. Hemato-Biology:

- Routine laboratory tests such as complete and differential blood count, chemistries, and liver and kidney function tests should be obtained. Laboratory findings are generally non-specific and include leukocytosis and anemia of chronic disease.
- Very accelerated sedimentation rate
- Inflammatory syndrome
- Search for an associated metabolic defect such as diabetes.

For patients who are immunocompromised, the potential causes of infection are broader and additional testing is often needed

Patients with concomitant skin, nasal, kidney, or neurologic disease that suggests granulomatosis with polyangiitis should be tested for antineutrophil cytoplasmic antibodies (ANCA).

2.2. Bacteriology

All patients should have two sets of blood cultures for both aerobic and anaerobic culture. Additional cultures from any suspected source are also appropriate (eg, intravascular catheter or wound). Sputum should be obtained for Gram stain and culture, ideally before antibiotics are started; however, antimicrobial therapy should not be withheld for those with signs of sepsis or septic shock. For patients who do not expectorate sputum, bronchoscopic sampling and/or percutaneous needle aspiration or (rarely) biopsy may be indicated. (See 'Additional testing' below.)

A micro-organism present on Gram stain and abundant in culture is often a pathogen. However, care must be exercised in interpreting the results, particularly when antimicrobials have already been started, since sputum samples can yield contaminants from upper airway flora that do not need to be treated (eg, *Corynebacterium* spp) and may not reveal the presence of anaerobes that do need to be treated.

Regardless of culture results, a putrid odor of expectorated sputum suggests an anaerobic infection and should be treated with an agent that includes activity against anaerobes.

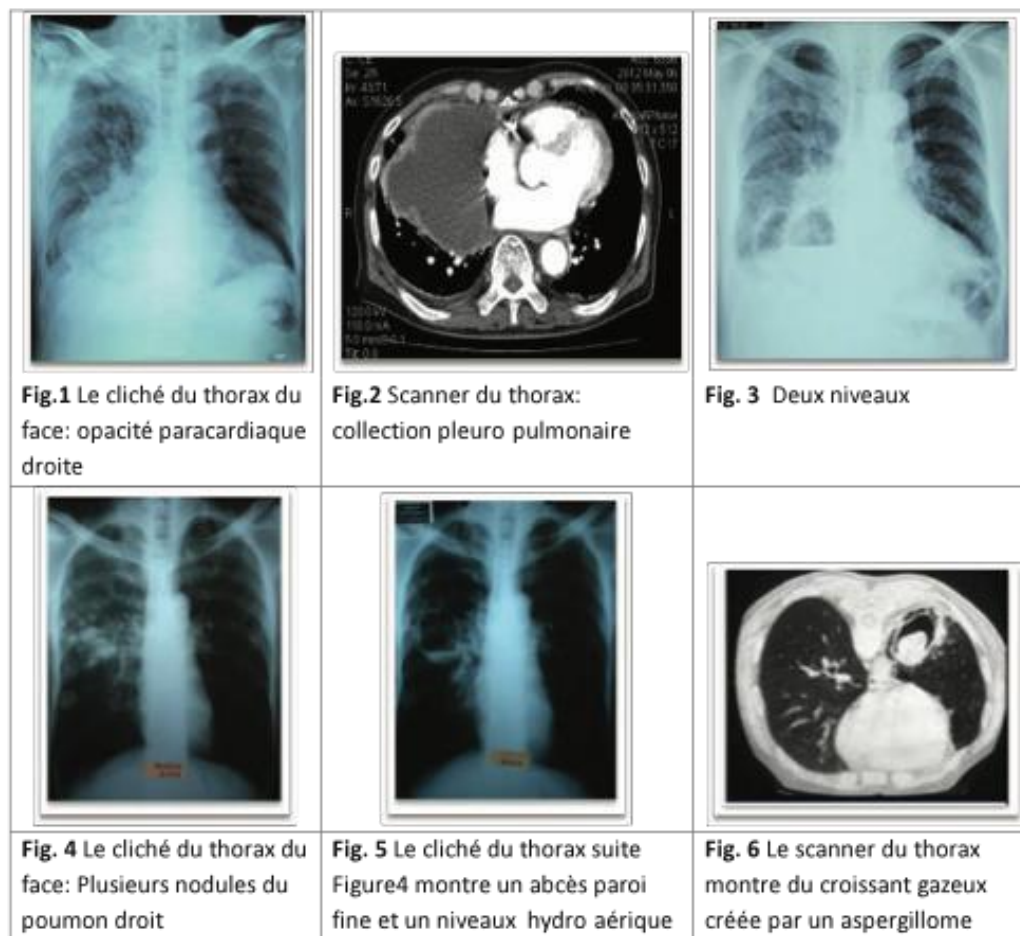
Despite obtaining respiratory cultures, up to 50 percent of patients do not have a positive culture and so antimicrobial therapy is empiric.

As bacterial infection can complicate mycobacterial cavities leading to an air-fluid interface, acid fast stains and mycobacterial cultures of sputum are prudent, especially if the cavity walls are thin or the patient has risk factors for *M. tuberculosis*.

The bacteriological examination is fundamental in the choice of antibiotic therapy.

Take bacteriological samples before antibiotic therapy but do not wait for the results to start treatment. Before antibiotic therapy, as close to the vomit as possible.

- Blood cultures +++
- Pleural puncture or cystic pleurisy
- Respiratory samples (Cyto Bacteriology Study of Sputum or during endoscopy using protected brushes, BAL, or broncho aspiration).



2.3. Imaging

Chest computed tomography — When a lung abscess is suspected, chest computed tomography (CT) should be performed. CT provides more precise anatomic definition than chest radiography (image 1 and image 2) and can identify other abscesses and thoracic lesions (eg, associated malignancy, lymphadenopathy, bullae) not clearly delineated on the chest radiograph.

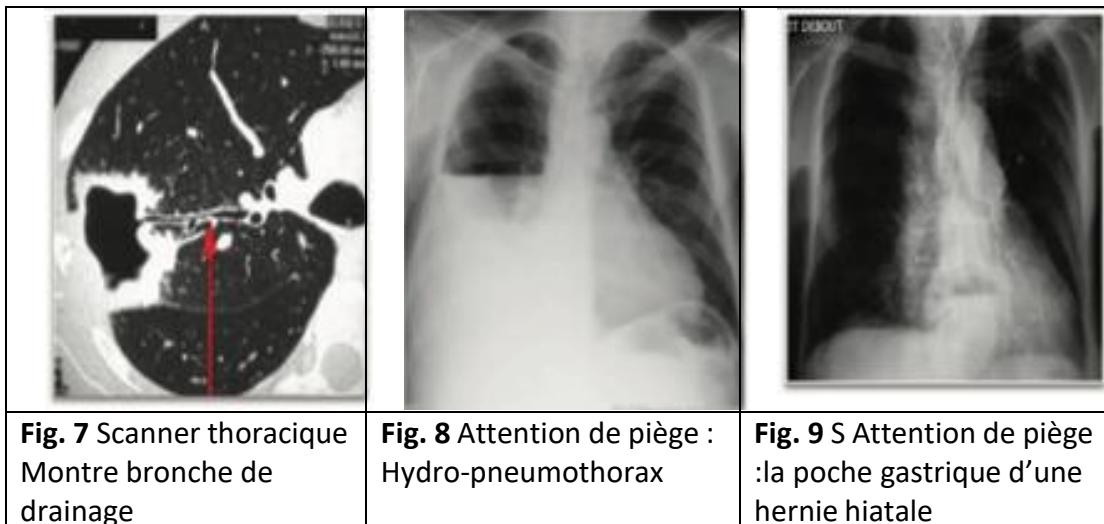
CT findings of acute lung abscess include an area of cavitation surrounded by consolidation. The necrotic region tends to be irregular in shape. The wall of an acute abscess is often thick but can vary widely; the cavity wall typically becomes thin and uniform as the abscess heals. Air fluid levels within the abscess are common.

A CT can also distinguish between a parenchymal lesion and a pleural collection (eg, parapneumonic effusion or empyema), which are managed very differently and determine whether the abscess is due to an infected cyst or bulla, underlying congenital lesion (eg, infected sequestration), airway obstruction (eg, tumor, inhaled foreign body), or sub-diaphragmatic or pleural infection. Last, if surgery is indicated, CT facilitates decision-making regarding resection.

Differentiating between peripheral lung abscesses and empyemas can occasionally be difficult. Abscesses tend to be more irregular in shape, spherical, and cavitated whereas empyemas tend to be better defined, elliptical, homogeneous, and have a smooth and uniform wall. Empyema may displace the blood vessels leading from the hilum to the periphery whereas abscesses typically do not.

The four major radiological signs are

1. Hydro-aerial image
2. Thin wall
3. Drainage bronchus (fig. 7)
4. Variability multiple exams



Additional testing — Additional testing may be warranted for patients who are unable to produce sputum, have an atypical presentation (eg, suspected cancer, noninfectious granulomatous disease), are immunocompromised, or do not respond to empiric antibiotics. As an example, bronchoscopy may be pursued more quickly in patients who are unable to raise sputum or have sputum that is not putrid and those who are at increased risk for opportunistic organisms due to immune deficiency or underlying lung disease, as early diagnosis and specific therapy of opportunistic infections are the cornerstone of successful management in such patients.

- **Bronchoscopy** — Flexible bronchoscopy is not routinely needed but is reserved for patients without a microbiologic diagnosis after noninvasive testing (sputum and blood cultures) and those with atypical presentations, uncertain diagnosis, or immunocompromised state.

When performing bronchoscopy, the clinician should evaluate the airways in the areas of lung abscesses for endobronchial stenosis, foreign bodies, and lesions suspicious for malignancy that may be responsible for or mimic lung abscess. Samples of secretions observed in the affected region of the lung can be aspirated directly or by gentle washing of the area. However, caution is required as aggressive suctioning and sampling, including brushing, bronchoalveolar lavage, and transbronchial biopsy of the cavity, can cause sudden spillage of the abscess contents and acute respiratory distress syndrome.

Samples should be sent for cytology, routine Gram stain, aerobic culture and anaerobic culture, and special stains for *Nocardia*, *Actinomyces*, fungi, *Pneumocystis*, mycobacteria, and antigen testing for galactomannan. Visible airway lesions should be biopsied or brushed per usual practice. Further details on bronchoscopy are provided separately.

Obtaining samples using a protected specimen brush and subjecting them to quantitative cultures may reduce the chances of oral contamination and provide a better idea of the degree of growth; however, this approach is inconsistently available, costly, and may still be of low yield for anaerobic pathogens.

- **Transthoracic needle aspiration or biopsy** – Transthoracic needle aspiration of the abscess or needle biopsy of the abscess wall under ultrasound, fluoroscopic, or CT guidance may be indicated for microbiologic sampling in rare patients who fail to improve with empiric antibiotic therapy and in whom sputum cultures and bronchoscopy are not revealing. Careful review of the CT scan is needed to ensure that the suspected abscess is not an infected bulla, since the risk of pneumothorax with needle aspiration may be high. Pneumothorax and pleural seeding appear less likely to occur if the abscess abuts the pleura and the needle or catheter can be passed through an area of pleural thickening into the cavity. Transthoracic drainage of lung abscess is described below.
- **Echocardiography** – For patients with multiple lung abscesses, transthoracic echocardiography is advised to assess for right-sided valve infective endocarditis [43-46]. If transthoracic echocardiography is negative when the suspicion for endocarditis is high, a transesophageal echocardiogram should be performed.
- **Thoracentesis** – If pleural fluid is present on the chest radiograph, a thoracentesis is performed to obtain a sample for microbiologic testing and to exclude empyema (fluid appearance, cell count and differential, chemistries including LDH and glucose, pH).
- **Transtacheal aspiration** - While popular in the past, transtracheal aspiration is no longer performed due to safety concerns.

VI. DIFFERENTIAL DIAGNOSIS

Lung abscess may also complicate pulmonary infarction, primary lung cancer (central carcinoma with necrosis), metastatic malignancies, and the necrotic conglomerate lesions of silicosis and coal miners' pneumoconiosis. Lesions in diseases such as granulomatosis with polyangiitis (formerly termed Wegener granulomatosis) and rheumatoid arthritis with rheumatoid nodules may also mimic lung abscess.

The key processes to consider in the differential diagnosis of lung abscess are other causes of a fluid or air-fluid filled mass in the lung parenchyma (eg, lung cancer, granulomatosis with polyangiitis, infected lung cyst or bulla, hydatid cyst, or superinfected fungal or mycobacterial cavity). Additionally, an empyema with an air-fluid interface can mimic a lung abscess on conventional chest radiographs. While less common in adults, intralobar pulmonary sequestrations can develop abscesses.

- **Malignancy** – Primary lung cancer and metastatic cancer can present with cavitary lesions that mimic lung abscess or become superinfected.

Additionally, endobronchial cancer can cause a postobstructive lung abscess. A solitary thick-walled cavity (wall >15 mm diameter) without surrounding consolidation is concerning for lung cancer. Diagnostic cytology and/or biopsy are warranted if the patient does not have systemic features suggestive of infection (fever, productive cough, leukocytosis) or if the lesion fails to improve despite antibiotics. Note, however, that endobronchial tumor can lead to postobstructive pneumonia and lung abscess, so it is possible for both malignancy and infection to be present.

- **Tuberculosis with or without bacterial superinfection** – Postprimary *Mycobacterium tuberculosis* in the lung is associated with thin-walled cavities or cavities on a background of patchy airspace opacity. An air-fluid interface may be present in the cavity in as many as 20 percent of cases, due to bleeding or bacterial superinfection.

● **Chronic pulmonary aspergillosis** – Chronic cavitary aspergillosis can present with several months of weight loss, productive cough, hemoptysis of variable severity, shortness of breath, and sometimes fever. Cavities may contain fungus balls, debris, or fluid, and the walls are usually thin. Aspergillus antibody (precipitins) is elevated in over 90 percent of patients.

● **Hydatid cyst** – For patients with travel or residence in an area endemic for Echinococcus, computed tomography (CT) imaging can identify single or multilocular cysts that may have an air crescent sign, germinative membranes floating in cyst fluid (water lily sign), or are empty. Liver cysts are also present in about 20 percent of patients with lung cysts and provide an etiologic clue. Peripheral blood eosinophilia is uncommon and serologic tests are variably helpful.

● **Empyema with air-fluid interface** – It can sometimes be difficult to determine from a chest radiograph whether an air-fluid interface is in the pleural space or lung parenchyma. Empyema can be associated with chest pain, but a lung abscess abutting the pleura can also cause chest pain. Chest CT is the procedure of choice for making this differentiation and guiding therapy. Abscesses tend to be more irregular in shape, spherical, and cavitated whereas empyema's tend to be better defined, elliptical, homogeneous, and have a smooth and uniform wall. Empyema may displace the blood vessels leading from the hilum to the periphery whereas abscesses typically do not. Empyema's will need to be drained via chest tube.

● **Infected sequestration** – Infection is a well-described complication of pulmonary intralobar sequestration and may undergo abscess formation. Diagnosis of sequestration is important because sequestration infections tend to be recurrent and generally require surgical resection.

VII. MANAGEMENT

1. Objectives

- Sterilization of infectious site
- Avoid complications

2. Medications

- Cloxacillin (SA-Methi S) or Vancomycin (SA Methi R) and aminoglycoside (Gentamycin or Amikacin) in case of Staphylococcus germ
- Amoxicillin-clavulanic acid PO or injection in case of suspicion of Oropharyngeal
- 3rd generation cephalosporin combined with an aminoglycoside, in case of suspicion of enterobacterium with possibly metronidazole in addition if the cephalosporin is little or no active on anaerobes.
- Metronidazole in case of amoebic abscess

** dose to be adapted according to renal insufficiency and/or hepatic insufficiency

Table 1 Medication list

Medication list	Dosage	Administration form	Duration
Cloxacilline	50-100 mg/kg TID q8hour	PO, infusion	4-8 weeks
Vancomycine**	2g x 2-4 daily	infusion	4-8 weeks

Amoxicilline-ac clavulanic	1g625 TID q8hour	PO, injection	4-8 weeks
Gentamycine		injection	3-5 days
Amikacine*	15kg BID q12hour	infusion	4-8 weeks
Ceftriaxone	1-2g/daily	injection	4-8 weeks
Métronidazole	500mg TID q8hour	PO, infusion	4-8 weeks

^ The optimal duration of antibiotic treatment has not been determined, although treatment for 6 to 8 weeks is commonly employed.

- The duration of antibiotic treatment: It must be sufficiently prolonged (4 to 8 weeks). The re-assessment in 48-72 hours after start of antibiotic treatment, with the microbiological results, and adapted. In the absence of a specific orientation in favor of a microorganism, and monitoring: Fever, CRP, NFS and chest X-ray or chest ultrasound.
- **Where to treat the patient**
 - Outpatient (OPD) : If there are no signs of severity and no comorbidity
 - In hospital (IPD)/continuous care/resuscitation: if BP <9cmHg, tachycardia (>120/min), tachypnea (>25/min) and/or confusion.
- Drainage of pus
 - It is essential +++
 - It will be promoted by respiratory kinesia drainage, possibly fibro-aspiration (bronchoscopy).
- **Treatment of a concomitant underlying disease**
 - It will be continued and adapted (balancing of diabetes, correction of immunoglobulin deficiency for example associated with a high-calorie diet.

VIII. TREATMEN

Treatment of lung abscess is based on prompt initiation of an empiric antibiotic regimen, adjustment of the regimen based on culture results, and a prolonged course of therapy (algorithm 1). This approach is successful in most patients, while a few require a drainage procedure or surgical resection (about 10 percent).

The role of postural drainage and chest percussion (chest physiotherapy) is less clear; it has been used with the rationale that it would facilitate intrabronchial drainage. However, efficacy has not been demonstrated and fatal cases of intrabronchial abscess rupture have been reported. We reserve cautious chest physiotherapy for patients with cystic fibrosis and noncystic fibrosis bronchiectasis who are on a routine program of bronchial hygiene.

Initial empiric antibiotics — Empiric antimicrobial therapy should be started promptly in patients with lung abscess to reduce the risk of abscess rupture into the airways or pleural space or invasion of local structures (eg, blood vessels).

The initial regimen can be adjusted, if needed, based on culture results and response to therapy, remembering that anaerobic organisms are difficult to grow in culture and susceptibility testing is rarely performed.

For patients with suspected lung abscess from aspiration, empiric intravenous (IV) regimens should penetrate the lung parenchyma and target both strict anaerobes and microaerophilic streptococci.

- For most patients, acceptable regimens include any combination of a beta-lactam–beta-lactamase inhibitor (eg, ampicillin-sulbactam 3 g intravenously [IV] every six hours) or a carbapenem (eg, imipenem, meropenem). This choice is based upon in vitro data, our experience, and the increasing rates of penicillin-resistance among anaerobes including *Bacteroides rumenicola*, black-

pigmenting strains of *Prevotella* (formally *Bacteroides melaninogenicus* group), *Bacteroides gracilis*, *Bacteroides ureolyticus*, *Fusobacteria*, and others.

- For patients who are allergic to penicillin, we generally use moxifloxacin 400 mg orally daily, or combination therapy with levofloxacin (750 mg orally daily) plus metronidazole (500 mg orally three times daily).

For hospitalized patients, ceftriaxone (or other third-generation cephalosporin) plus metronidazole is a reasonable option. For outpatients who are allergic to penicillin, we favor a fluoroquinolone-based regimen when starting treatment because these regimens have better bioavailability.

Clindamycin monotherapy (600 mg IV every eight hours followed by 300 mg orally four times daily) is also a potential alternative. However, clindamycin has the potential to fail, as abscesses are caused by gram negatives or resistant gram positives in 20 to 30 percent of patients.

Adjusting antimicrobial regimen — Once culture results are available, empiric antimicrobial therapy can be tailored to cover identified pathogens, although the empiric regimens described above remain appropriate for most patients with aspiration-related mixed anaerobic and streptococcal lung abscess (see 'Initial empiric antibiotics' above). When a single pathogen that is not a typical part of the oral flora is detected (eg, *K. pneumoniae*, *S. aureus*), it is often appropriate to tailor the regimen to target that pathogen based upon the results of in vitro susceptibility testing. However, as culture of anaerobes from the lung is difficult, continuing empiric anaerobic coverage in addition may be prudent, particularly in patients with suspected aspiration who have putrid sputum.

- For suspected methicillin-susceptible *S. aureus* (MSSA), the agents of choice are cefazolin (2 g IV every eight hours), nafcillin (2 g IV every four hours), or oxacillin (2 g IV every four hours).
- For methicillin-resistant *S. aureus* (MRSA), linezolid (600 mg IV every 12 hours) or vancomycin are acceptable. Other options are ceftaroline, trimethoprim-sulfamethoxazole, and telavancin.

Duration of antibiotics — In general, we switch from IV to oral agents once the patient has defervesced and become clinically stable; for some patients this takes a few days while in others it may be one to three weeks. Appropriate oral regimens depend upon the identified or suspected infecting pathogens. For patients with a mixed anaerobic and streptococcal infection, amoxicillin-clavulanate is an appropriate regimen, but the choice of regimen should be guided by the causative pathogen(s) and susceptibility results when available.

The optimal total duration of therapy (IV plus oral) is unknown; reported ranges include 21 to 48 days. Some experts treat for three weeks as a standard and others treat based upon the response. Our practice is to continue antibiotic treatment until chest imaging (preferably computed tomography [CT]) shows a small, stable residual lesion or is clear. This generally requires several weeks of treatment, most of which can be accomplished with an oral regimen on an outpatient basis.

Response to therapy — Most patients are followed clinically, but repeat imaging is prudent for new onset or recurrent fevers, chest pain, hemoptysis, dyspnea, or failure to improve.

Patients with aspiration-related lung abscesses usually show clinical improvement with decreased fever and leukocytosis within three to four days after beginning antibiotic treatment. Defervescence is expected in 7 to 10 days. The disappearance of putrid sputum may take longer.

Persistent fevers beyond one to two weeks may indicate delayed response, and such patients should undergo further diagnostic tests to better define the underlying anatomy and microbiology of the infection; typically, another chest CT is performed to look for progression of the abscess, development of complications (eg, empyema or new abscesses),

or an underlying precipitant that may not have been picked up on the initial CT scan (algorithm 1).

In addition, further respiratory sampling via flexible bronchoscopy is usually indicated. Major considerations in patients with a delayed response include:

- An associated condition that precludes response, such as obstruction with a foreign body or neoplasm, or a pulmonary sequestration.
- Erroneous microbiologic diagnosis with infection due to bacteria, mycobacteria, or fungi that are not being treated.
- Failure of medical therapy despite appropriate antibiotics. Drainage may be required to facilitate recovery. This is more typical in patients with large cavities but there is no clear size cut-off that mandates intervention, and drainage has been reported to be helpful even for some smaller abscesses that fail to improve with antibiotics.
- An alternative, nonbacterial cause of cavitary lung disease, such as cavitating neoplasm, vasculitis. (See 'Differential diagnosis' above.
- Conditions that mimic an abscess such as an infected cyst or empyema.
- Other causes of persistent fever, such as drug fever or *Clostridioides difficile*–associated colitis.

Patients who fail to improve by 7 to 10 days despite appropriate antimicrobial therapy may need a drainage procedure or surgery.

Failure of antimicrobial therapy — Most lung abscesses respond to a prolonged course of antimicrobials.

However, a small proportion fail antimicrobials and require a drainage procedure or surgical therapy (approximately 10 percent).

Needle or catheter drainage — For patients who fail to improve after 7 to 10 days of antibiotic therapy that has been adjusted based on microbiologic results, drainage of the abscess via transthoracic or transbronchial catheters can often lead to clinical improvement.

Among these options, percutaneous drainage is preferred since experience with this option is more extensive than with bronchoscopic drainage. Patients continue with their systemic antimicrobial until the abscess cavity closes or is small and stable.

The typical duration of drainage catheter placement is approximately three days to three weeks and patients generally continue with their systemic antimicrobial.

- **Transthoracic (percutaneous) catheter drainage** – Transthoracic (percutaneous) catheter drainage can be performed under imaging guidance; depending on the available facilities, computed tomography, ultrasound, or fluoroscopy may be utilized. The transthoracic route is generally preferred when the abscess abuts or is close to the chest wall.

Transthoracic drainage procedures carry a small risk of infecting the pleural space. In addition, they are also associated with a small risk of hemorrhage or pneumothorax.

- **Transbronchoscopic catheter drainage** – The bronchoscopic approach may be preferred for more centrally located lesions and for those that do not abut the pleura; ultrasound-guided bronchoscopy may be used for localization. Bronchoscopic drainage is performed by placing a pigtail catheter into the abscess cavity via the nose under bronchoscopic visualization, leaving the catheter in place until the cavity has drained. In some cases, the cavity is rinsed daily with sterile saline and directed antibiotics (eg, gentamicin or antifungal).

One of the risks associated with bronchoscopic drainage is spillage of infected material to other parts of the lung. Intracavitary fibrinolytic therapy is not recommended due to the potential development of a bronchopleural fistula.

This procedure is not universally available and is restricted to centers with pulmonary interventional expertise. Bronchoscopy for drainage, in the absence of catheter placement, is

of little use and can result in spillage of abscess contents into other airways.

Treating endobronchial obstruction — For post-obstructive lung abscess due to bronchial stenosis or tumor, efforts should be directed at relieving the obstruction or possibly curative surgical resection of the tumor (typically lobectomy) (algorithm 1). Foreign body removal is often achieved with flexible or rigid bronchoscopy. For bronchial stenosis or tumor, endobronchial balloon dilation with stent placement may be an option pending definitive therapy. Obstruction from mediastinal lesions (eg, aneurysms or lymphadenopathy) require individualized assessment and treatment. The management of airway obstruction is discussed separately.

fail to improve clinically or radiographically with antibiotic therapy with or without catheter drainage or who develop complications during therapy (eg, significant hemorrhage, bronchopleural fistula), surgery may very rarely be indicated.

Factors that may contribute to a slow response or no response are obstruction of the bronchus, an extremely large size abscess, and abscesses involving antibiotic-resistant organisms, such as *P. aeruginosa*.

The usual procedure in such cases is a lobectomy or pneumonectomy but small peripheral abscesses may be amenable to segmentectomy or wedge resection.

Video-assisted thoracic surgery (VATS) is the procedure of choice and attention should be paid to removing all infected material in order to avoid stump infection and resultant bronchopleural fistula.

Mortality with surgery is as high as 15 to 20 percent although the high mortality rate likely reflects the severity of lung abscess in patients referred for lung surgery as well as their underlying conditions, in addition to the risk of surgery itself.

IX. SPECIAL CASES

- Surgical case (is rare). Surgical indication of excision may be indicated in the case:
 - Chronic lung abscess
 - Bronchial stenosis which promotes the formation of the abscess
 - Decortication in the case of retractile encysted pleurisy.

X. EVOLUTION AND PROGNOSIS

Patients with primary lung abscess (about 60 percent overall) with typical aspiration-related lung abscesses, including those with alcohol or injection drug use, generally do well with antibiotic treatment with cure rates of 90 to 95 percent.

Patients who require surgery, are immunocompromised, or have malignancy or irreversible bronchial obstruction have higher mortality rates.

Associated disease in the host is an important factor in determining the outcome of a lung abscess.

Patient information:

- Universal hygiene (oral and cutaneous-mucous condition) and especially pay attention to false route
- Provide adequate treatment for broncho pneumonia by avoiding these automedication without medical advice or a peripheral abscess or furunculosis

XI. COMPLICATIONS

- Acute complications:
 - Sepsis and/or shock
 - Pleural empyema
 - Lung abscess extent
 - Sepsis to others organs
- Long-term complications:
 - Chronic forms
 - Bronchial dilatations

Re commendation: Very important any lung abscess OR pleural empyema JUSTIFIES a bronchial endoscopy to assessment and evaluation FOR a lung tumor.

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LUNG CANCER

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I. DEFINITION

Lung cancer is defined by the presence of cancer cells that develop in the lung tissue. The histological classification recognizes 2 main histological types of primary bronchial tumors: non-small cell lung cancers and small cell lung cancers.

- 1- Primary bronchopulmonary cancers
 - Small cell lung carcinoma (SCLC)
 - Non-small cell bronchial carcinoma (NSCLC)
 - o Adenocarcinomas
 - o Squamous cell carcinomas (localization proximal)
 - o Large cell undifferentiated carcinoma,
- 2- Secondary lung cancers
 - Melanoma, choriocarcinoma, sarcomas, kidney cancer, breast cancer, uterine cancer.
 - Lymphangitis carcinomatosa

II. EPIDEMIOLOGY

The average age at diagnosis is estimated at 65 in men and 64 in women. It is the second leading cause of cancer deaths in women. Incident of bronchial cancer in Cambodia ranges from 23 in men and 5 in women per 100,000 inhabitants in 2008.

Small cell lung cancer currently represents 15 to 20% of primary bronchial tumors, and occurs preferentially between the ages of 55 and 65, especially in women.

Non-small cell lung cancer is the leading cause of cancer death in the male population. Its incidence is estimated between 40 and 70 per 100,000 in French.

III. ETIOLOGY

- Tobacco
- Occupational exposure: silicosis, asbestosis (fiber cement used in house roofs), radon at home hydrocarbons aromatiques polycycliques, diesel, mines.
- Patients suffering from COPD, fibrosing pulmonary pathologies, tuberculosis, having certain gene mutations or having family history of lung cancer
- Other factors: atmospheric pollution, cooking smoke in the closed room, breathing secondhand smoke.

IV. PATHOPHYSIOLOGY

Carcinogenesis evolves in several stages, involving a series of chromosomal modifications and mutations leading to the activation of oncogenes (KRAS, EGFR) and/or inactivation of tumor suppressor genes, leading to the acquisition of the cancer cell's fundamental properties.

In lung cancer, the "multi-step" process is suggested by the fact that exposure to an agent such as tobacco or asbestos often occurred more than 20 years before diagnosis. The order in which alterations occur is not completely known.

In some cases, lung cancer is dependent on a single mutation of an oncogene: this is known as "oncogenic addiction". This is the case for cancers carrying Epidermal growth factor receptor (EGFR) mutation, or a rearrangement ALK (Anaplastic Lymphoma Kinase) or ROS1.

V. DIAGNOSIS

1. Clinic

Early symptoms may be mild or dismissed as common respiratory issues (cough, dyspnea), leading to delayed diagnosis, associated altered general status: weight loss with no known cause, fatigue.

The tumor can also cause pneumonia, which is accompanied respiratory symptoms, also associated biological inflammatory syndrome.

Table 1: Respiratory symptoms

Respiratory symptoms	Dyspnea, cough, wheezing, hemoptysis
	Recurring respiratory infection
Local extension	Superior cave syndrome
	Dysphonia
	Chest pain
	Pancoast syndrome
	Recurring respiratory infection
Paraneoplastic syndrome	Finger clubbing
	Hypertrophic osteopathy
	Inappropriate secretion of anti-diuretic hormone
	Long-term fever
	Pseudo myasthenia, peripheral neuropathies
	Hypercalcemia in blood (secretion of pseudo-PTH)
	Cushing's syndrome (secretion of pseudo-ACTH)

2. Investigation

Chest X-rays are most often performed initially, in the face of respiratory symptoms, However, chest X-Ray cannot give a definitive diagnosis lung cancer.

CT-scan for extension assessment and TNM staging (and should be performed as a first-line procedure, with injection, at thoracic, abdominal (including adrenal) and encephalic levels to seek or eliminate extra thoracic metastases.

Imaging suspected lung cancer:

- Rounded opacity nodular of different size is present with central necrosis,
- Micro or macronodular interstitial image, balloon release image,
- Multiple mediastinal lymphadenopathies, i.e. suprasternal lymphadenopathy
- Either presence of metastatic image: liver, adrenal or pleural.

Flexible fiberoptic bronchoscopy allow the exploration or biopsy a centrally located tumors (should be greater than 5 biopsy specimens)

Endoscopic ultrasound (EUS) / Endobronchial ultrasonography with transbronchial (EBUS) allows the exploration of mediastinal lymph node staging in early-stage lung cancer

Lung Ultrasonography, for specify the parietal extension of a tumor, or biopsy under pleural ultrasound guidance.

PET-Scan, for staging of the mediastinum and screening for metastases that might not be detected by CT alone; for radiotherapy planning. Fixation must therefore always be histologically proven, as it may be a false positive due to infectious pathology.

Blinded pleural biopsy is not recommended to identify nodular lesion localized pleural,

3. Histology

- The diagnosis of bronchopulmonary cancers is based on cytology.
- The diagnosis based on immunohistochemistry of small cell cancer is optional but can help in the differential diagnosis using Neuro-Endocrine markers (Chromogranin A, Synaptophysin, CD 56) and TTF1.
 - Adenocarcinomas: architecture glandular, TTF1 (+)
 - Squamous cell carcinomas: p40 (+)

Classification and extension assessment: Based on the TNM classification

V. DIFFERENTIAL DIAGNOSIS

- Benign pleural mass
- Lipoid pneumonia
- Lung abscess
- Lung cyst
- Lung sequelae
- Tuberculoma

VI. MANAGEMENT

1. Objectives

- Avoid complications linked to treatment and minimize therapeutic after-effects.
- Preserve quality of life and offer support to the patient and those around them.

2. Critical sign

- Acute respiratory failure
- Hemoptysis
- Bleeding
- Neutropenia febrile
- Tumor lysis syndrome
- Superior cave syndrome
- Intracranial hypertension
- Spinal cord compression

3. Mortality risk factor

Factors associated with mortality is age, histologic cancer type (undifferentiated carcinoma), metastatic disease at diagnosis, poorer performance status, poorer nutrition, concomitant pulmonary diseases (chronic bronchopathy, pulmonary fibrosis, tuberculosis).

4. Outpatient Management

Nutritional evaluation, supportive care (symptomatic treatment, provide pain relief, and give emotional support) it helps to increase quality of life for patient with lung cancer.

5. First-line treatment

- **Non-Small cell lung cancer**

- **Stage I and II**

- If the patient is not operable, stereotactic radiotherapy is recommended.
- If the patient is operable, resection surgery is proposed. Adjuvant chemotherapy may be offered, depending on the stage of the disease postoperatively (pTNM).
- An adjuvant chemotherapy protocol is **Cisplatin combined with Vinorelbine** (4 cycles). If contraindication to Cisplatin (age, PS, comorbidities, renal failure), chemotherapy with **Carboplatin Paclitaxel** may be recommended.

- **Stage III (locally advanced disease)**

They involve a combination of local treatment (surgery or radiotherapy) combined with chemotherapy. If patient PS 0-1, no comorbidity, age < 70, concomitant chemo-radiotherapy and once this procedure has been completed, consolidation treatment with immunotherapy is recommended.

Chemotherapies concomitant with radiotherapy are **Cisplatin and Vinorelbine or Cisplatin and Etoposide or Carboplatin and Paclitaxel**.

- **Stage IV**

The treatment depends on the patient's age, PS, PD-L1 and the presence of oncogene mutation.

- Tyrosine kinase inhibitors (TKI) in case of addictive molecular alterations (including EGFR, ALK, ROS1)
- Combination chemotherapy + immunotherapy in PS0-1 and absence of alterations in oncogenesis
 - Adenocarcinomas: **Carboplatin-Pemetrexed Pembrolizumab**
 - Squamous cell carcinoma: **Carboplatin Paclitaxel Pembrolizumab**

Depending on the response to induction chemotherapy (4 cycles) and its toxicities, maintenance or second-line treatment, or discontinuation, will be **discussed at the multidisciplinary meeting**,

- **Small cell lung cancer**

Surgery is not the rule, even in the localized stage, treatment is essentially based on chemotherapy and immunotherapy and should be initiated as early as possible (oncological emergency).

- Stage I - III: combination of chemotherapy and thoracic radiotherapy (**Cisplatin/Carboplatin combined with Etoposide**).
- Stage IV: platinum-based doublet combined with immunotherapy for 4 cycles, then maintenance with immunotherapy alone (**Cisplatin/Carboplatin combined with Etoposide and Durvalumab**).

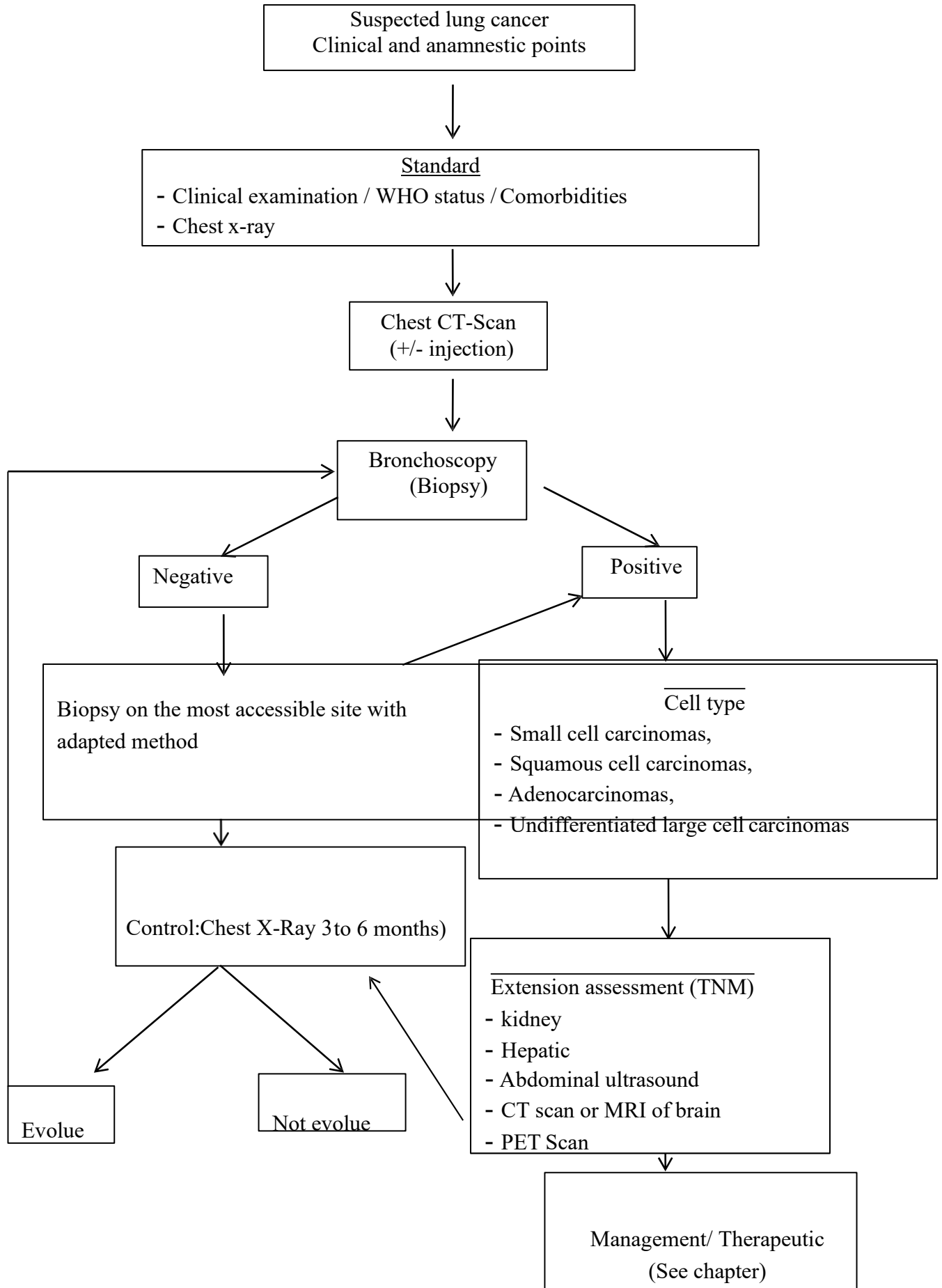
When to evaluate?

- For chemotherapy: evaluated after 6 to 9 weeks (2 to 3 cycles).
- For immunotherapy: evaluated after 6 to 9 weeks, depending on the immunotherapy molecule used.

Treatment adaptation

- In case of genetic mutations, treatment with a tyrosine kinase inhibitor is preferred, if mutation in EGFR gene is found (**Erlotinib or Gefitinib or Afatinib**), if mutation in ALK gene is found (**Crizotinib**), indicated after multidisciplinary meeting advice.
- Conventional chemotherapy if the patient is not a candidate for immunotherapy: **Cisplatin-Vinorelbine, Cisplatin-Gemcitabine, Carboplatin-Paclitaxel**.
- If the patient is contraindicated for Cisplatin, Carboplatin can be replaced.

Algorithm



VII. SPECIAL CASES

- **Pulmonary sarcomatous carcinoma**
 - poorly differentiated carcinoma with components of sarcoma or sarcoma-like differentiation
 - poor prognostic, aggressive tumors with early systemic metastases (esophagus, peritoneum, stomach, pancreas, gums, subcutaneous tissue)
- **Pleural Mesothelioma**
 - Primarily linked to asbestos exposure, not linked with smoking
 - Usually progressively increasing dyspnea and chest pain and unintentional weight loss, associated with lung effusion.
 - BAP1 (+), MTAP or homozygous deletion of CDKN2A by FISH

VIII. EVOLUTION AND PROGNOSIS

Tumor grade was found to be associated with survival after adjusting for the effects of age, gender, smoking history, tumor stage, histological cell type, and treatment modality.

NSCLC patients had a lower risk of mortality with a longer duration of smoking abstinence.

IX. COMPLICATIONS

- Lung or pericardium effusion due to metastasis
- Multiorgan failure
- Renal failure from Cisplatinum
- Pancytopenia from chemotherapy

X. PREVENTION AND RECOMMENDATION

- Smoking cessation
- Annual screening for lung cancer with low-dose computed tomography in asymptomatic individual aged 50 to 80 years who currently smoked or formerly smoked and have ≥ 20 pack-year a smoking history
- Surveillance every 6 months for 2 years with a visit including history, physical examination and preferably chest CT scan at least at 12 and 24 months in order to detect second primary tumors.

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ANNEXES

ECOG PERFORMANCE STATUS*	
Grade	ECOG
0	Fully active, able to carry on all pre-disease performance without restriction
1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
2	Ambulatory and capable of all selfcare but unable to carry out any work activities. Up and about more than 50% of waking hours
3	Capable of only limited selfcare, confined to bed or chair more than 50% of waking hours
4	Completely disabled. Cannot carry on any selfcare. Totally confined to bed or chair
5	Dead

T	Tumeur primitive
Tx	Tumeur primaire non connue ou tumeur prouvée par la présence de cellules malignes dans les sécrétions broncho-pulmonaires mais non visible aux examens radiologiques et endoscopiques.
To	Absence de tumeur identifiable
Tis	Carcinome <i>in situ</i>
T1	Tumeur ≤ 3 cm dans sa plus grande dimension, entourée par le poumon ou la plèvre viscérale et sans signe endoscopique d'envahissement au-delà de la bronche lobaire (c'est-à-dire dans la bronche souches) • T1a(mi) Adénocarcinome minimalement-invasif • T1a ≤ 1 cm • T1b > 1 cm et ≤ 2 cm • T1c > 2 cm et ≤ 3 cm
T2	Tumeur de plus de 3 cm, mais de moins de 5 cm OU avec un quelconque des éléments suivants ○ envahissement d'une bronche souche quelle que soit sa distance par rapport à la carène mais sans envahissement de la carène, ○ envahissement de la plèvre viscérale, ○ existence d'une atélectasie ou pneumonie obstructive
T3	Tumeur de plus de 5 cm et de moins de 7 cm OU associée à un(des) no- dule(s) tumoral(aux) distinct(s) et dans le même lobe, OU envahissant directement : ○ la paroi thoracique (incluant les tumeurs du sommet), ○ le nerf phrénique, ○ la plèvre pariétale ou le péricarde pariétal.

T4	Tumeur de plus de 7 cm OU associée à des nodules tumoraux séparés dans deux lobes différents du même poumon, OU envahissant directement : • le médiastin, • le cœur ou les gros vaisseaux, • la trachée, ou la carène • le diaphragme, • le nerf récurrent, • l'œsophage, • un(des) corps vertébral(ux).
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N	Adénopathies locorégionales
Nx	Envahissement locorégional inconnu
N0	Absence de métastase dans les ganglions lymphatiques régionaux.
N1	Métastases ganglionnaires péribronchique et/ou hilaires homolatéraux y compris par une extension directe de la tumeur primitive
N2	Métastases ganglionnaires médiastinales homolatéraux ou dans les ganglions sous-carénaux
N3	Métastases ganglionnaires médiastinales controlatérales ou hilaires controlatérales ou scaléniques, sus-claviculaires homo- ou controlatérales.
M	Métastases
M0	Pas de métastase à distance.
M1	Existence de métastases : • M1a Nodule(s) tumoral(ux) séparés dans un lobe controlatéral, ou nodules pleuraux ou pleurésie maligne ou péricardite maligne • M1b Une seule métastase extra-thoracique dans un seul organe • M1c Plusieurs métastases extrathoraciques dans un seul ou plusieurs organes
Stades	
Stade IA	T1 N0 M0
Stade IB	T2 N0 M0
Stade IIA	T1 N1 M0
Stade IIB	T2 N1 M0
Stade IIIA	T1 N2 M0, T2 N2 M0, T3 N1-2 M0
Stade IIIB	Tous T N3 M0, T4 Tous N M0
Stade IV	Tous T Tous N M1

Classifications TNM 8ème édition

	N0	N1	N2	N3	M1a tout N	M1b tout N	M1c tout N
T1a	IA1	IIB	IIIA	IIIB	IVA	IVA	IVB
T1b	IA2	IIB	IIIA	IIIB	IVA	IVA	IVB
T1c	IA3	IIB	IIIA	IIIB	IVA	IVA	IVB
T2a	IB	IIB	IIIA	IIIB	IVA	IVA	IVB
T2b	IIA	IIB	IIIA	IIIB	IVA	IVA	IVB
T3	IIB	IIIA	IIIB	IIIC	IVA	IVA	IVB
T4	IIIA	IIIA	IIIB	IIIC	IVA	IVA	IVB

CHAPTER IV HEPATO-GASTROENTEROLOGY

1. Non-traumatic acute abdominal pain
2. Dyspepsia
3. Dysphagia
4. Gastroesophageal reflux disease
5. H. pylori infection
6. Nonsteroidal anti-inflammatory drugs, Low-dose aspirin and Gastroduodenal toxicity
7. Management of peptic ulcer disease
8. Management of upper gastrointestinal bleeding
9. Management of lower gastrointestinal bleeding
10. Caustic ingestion
11. Acute diarrhea
12. Constipation
13. Irritable bowel syndrome
14. Jaundice
15. Metabolic dysfunction-associated steatotic liver disease
16. Cirrhosis and Its Complications
17. Liver abscess
18. Hepatocellular carcinoma
19. Acute pancreatitis
20. Biliary tract infection
21. Anal pain
22. Hemorrhoids

NON-TRAUMATIC ACUTE ABDOMINAL PAIN

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Asst. Prof. OUK Saram, Dr. SANN Channa,*

I. DEFINITION

Acute abdominal pain (AAP), conventionally defined as pain of non-traumatic origin with a maximum duration of 7 days. AAP may indicate a potentially life-threatening condition that requires urgent surgical intervention and urgent medical attention.

II. PIDEMIOLOGY

AAP is a common complaint in emergency departments (EDs), accounting for 7–10% of all ED visits. About 15% of all emergency hospitalizations are due to AAP.

III. IOLOGY

- The causes of surgical and nonsurgical
- The commonest causes of acute abdomen are:
 - renal colic
 - biliary colic
 - cholecystitis
 - appendicitis
 - diverticulitis
 - nonspecific abdominal pain
- Likely etiology varies according to age. Renal colic and appendicitis are more common in patients < 60 years, while gallbladder disease and diverticulitis are more common in older patients.

IV. PATHOPHYSIOLOGY

1. **Visceral pain** comes from the abdominal viscera, which are innervated by autonomic nerve fibers and respond mainly to the sensations of distention and muscular contraction—not to cutting, tearing, or local irritation.
 - Visceral pain is typically vague, dull, and nauseating. It is poorly localized and tends to be referred to areas corresponding to the embryonic origin of the affected structure.
 - Foregut structures (stomach, duodenum, liver, and pancreas) cause upper abdominal pain.
 - Midgut structures (small bowel, proximal colon, and appendix) cause periumbilical pain.
 - Hindgut structures (distal colon and genitourinary tract) cause lower abdominal pain.
2. **Somatic (Parietal) pain** comes from the parietal peritoneum, which is innervated by somatic nerves, which respond to irritation from infectious, chemical, or other inflammatory processes. Somatic pain is sharp and well localized.
3. **Referred pain** is pain perceived distant from its source and results from convergence of nerve fibers at the spinal cord. Common examples of referred pain are scapular pain due to biliary colic, groin pain due to renal colic, and shoulder pain due to blood or infection irritating the diaphragm.

V. DIAGNOSIS

1. History related to etiological suspected

- Personal data: age, sex, and habits of medical importance
- History of presenting illness
 - Analysis of pain: SOCRATES
 - Associated symptoms
 - Review of other systems
- Past personal history
 - Medical and surgical
 - Gynecological history
 - Traveling history
- Past family history (Hepato-Gastrointestinal cancer...)

2. Physical examination

- General examination
- Abdominal examination in all 9 regions of the abdominal cavity and other important examinations: rectal, pelvic, and scrotal if necessary.

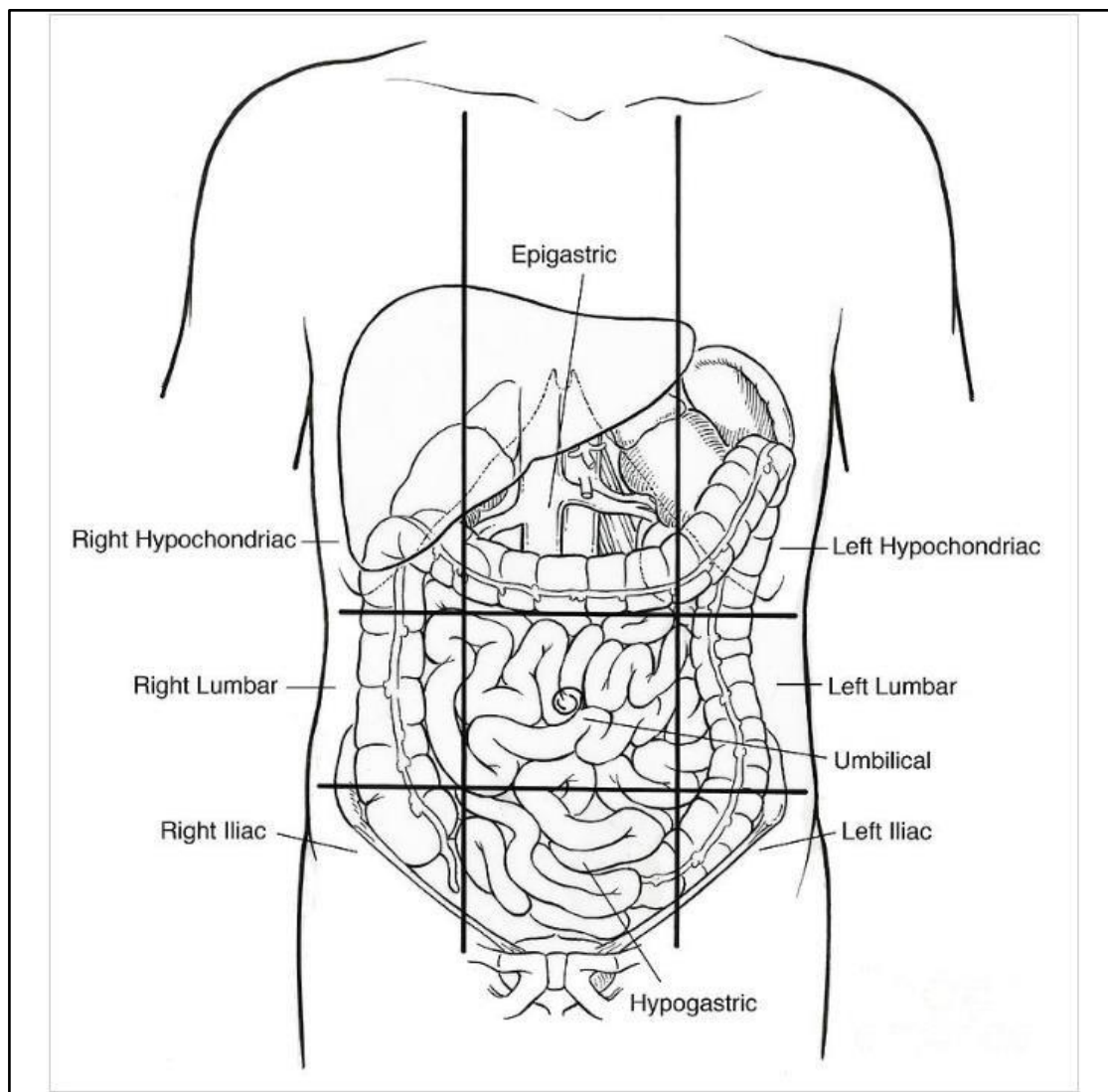


Figure 1: Regions of the abdominal cavity

Table 1: Non-Traumatic Acute Abdominal Pain Related to Diseases

Right hypochondriac pain	Epigastric pain	Left hypochondriac pain
<u>Biliary:</u> 1. Biliary colic 2. Acute cholecystitis 3. Acute cholangitis 4. Sphincter of Oddi dysfunction <u>Hepatic:</u> 1. Acute hepatitis 2. Perihepatitis (Fitz-Hugh-Curtis syndrome) 3. Liver abscess 4. Budd-Chiari syndrome 5. Portal vein thrombosis	1. Acute myocardial infarction 2. Acute pancreatitis 3. Chronic pancreatitis 4. Peptic ulcer disease 5. Gastroesophageal reflux disease 6. Gastritis/gastropathy 7. Functional dyspepsia 8. Gastroparesis	1. Splenomegaly 2. Splenic infarct 3. Splenic abscess 4. Splenic rupture
Right lumbar pain	Umbilical pain	Left lumbar pain
1. Kidney stones 2. Urine infection 3. Constipation 4. Lumbar hernia	1. Pancreatitis 2. Early appendicitis 3. Inflammatory bowel 4. Small bowel 5. Umbilical hernia	1. Kidney stones 2. Diverticular disease 3. Constipation 4. Inflammatory bowel
Right iliac pain	Hypogastric pain	Left iliac pain
1. Appendicitis 2. Constipation 3. Pelvic pain 4. Inguinal hernia	1. Urine infection 2. Appendicitis 3. Diverticular disease 4. Inflammatory bowel 5. Pelvic pain	1. Diverticular disease 2. Pelvic pain 3. Inguinal hernia

3. Laboratory testing and imaging

- **Full blood count:** leukocytosis is often (but not invariably) present in conditions such as appendicitis, cholecystitis.....
- **Serum electrolytes panel:** sodium, potassium, chloride, bicarbonate, urea, creatinine,
- **Pregnancy test:** all women of reproductive age. Important in ruling out ectopic pregnancy
- **Comprehensive metabolic panel:** with liver function tests
- **C-reactive protein:** non-specific marker of inflammation.
- **Coagulation studies:** carried out in patients with suspected vascular causes of abdominal pain (including aortic dissection, ruptured aortic aneurysm, or acute mesenteric ischemia), and in unstable patients, especially if surgery is indicated.
- **Serum amylase and lipase levels.**
- **Ultrasound:** rapid and can be performed at the bedside by emergency physicians, it is part of the initial evaluation in unstable patients with concern for traumatic hemoperitoneum, ruptured ectopic pregnancy, and AAA leak or rupture.
- **Computed tomography scan:** evaluation of undifferentiated abdominal pain.

VI. DIFFERENTIAL DIAGNOSIS OF NON-TRAUMATIC ACUTE ABDOMINAL PAIN

1- Extra-abdominal (less common) involves

- Cardiopulmonary (Acute myocardial infarction, Pulmonary embolism...)
- Abdominal wall (Hernia....)
- Neurogenic pain (Zoster...)

2- Nonspecific abdominal pain – not well explained or described

- Toxic-metabolic (DKA....)
- Abdominal aortic aneurysm (AAA)
- Dissection aortic abdominal
- Psychic (Anxiety, Depression, etc.)

VII. MANAGEMENT

1. Objectives

- To relieve the pain in cases acute abdominal pain.
- To assess the etiological diagnosis, the management and the follow up patients by depending on its etiology.

2. Critical Signs

- Severe pain
- Signs of shock (eg, tachycardia, hypotension, diaphoresis/excessive sweating, confusion)
- Abdominal distention
- Signs of peritonitis

3. Mortality risk factor

- Geriatric patients (age > 60 years)
- Comorbidity (Heart failure, chronic kidney failure.....)
- Sepsis
- Immunosuppression / immunocompromised

4. Outpatient Management

- It is only for mild to moderate pain and has a specific diagnosis, such as food poisoning, acute gastroenteritis....

5. First line treatment

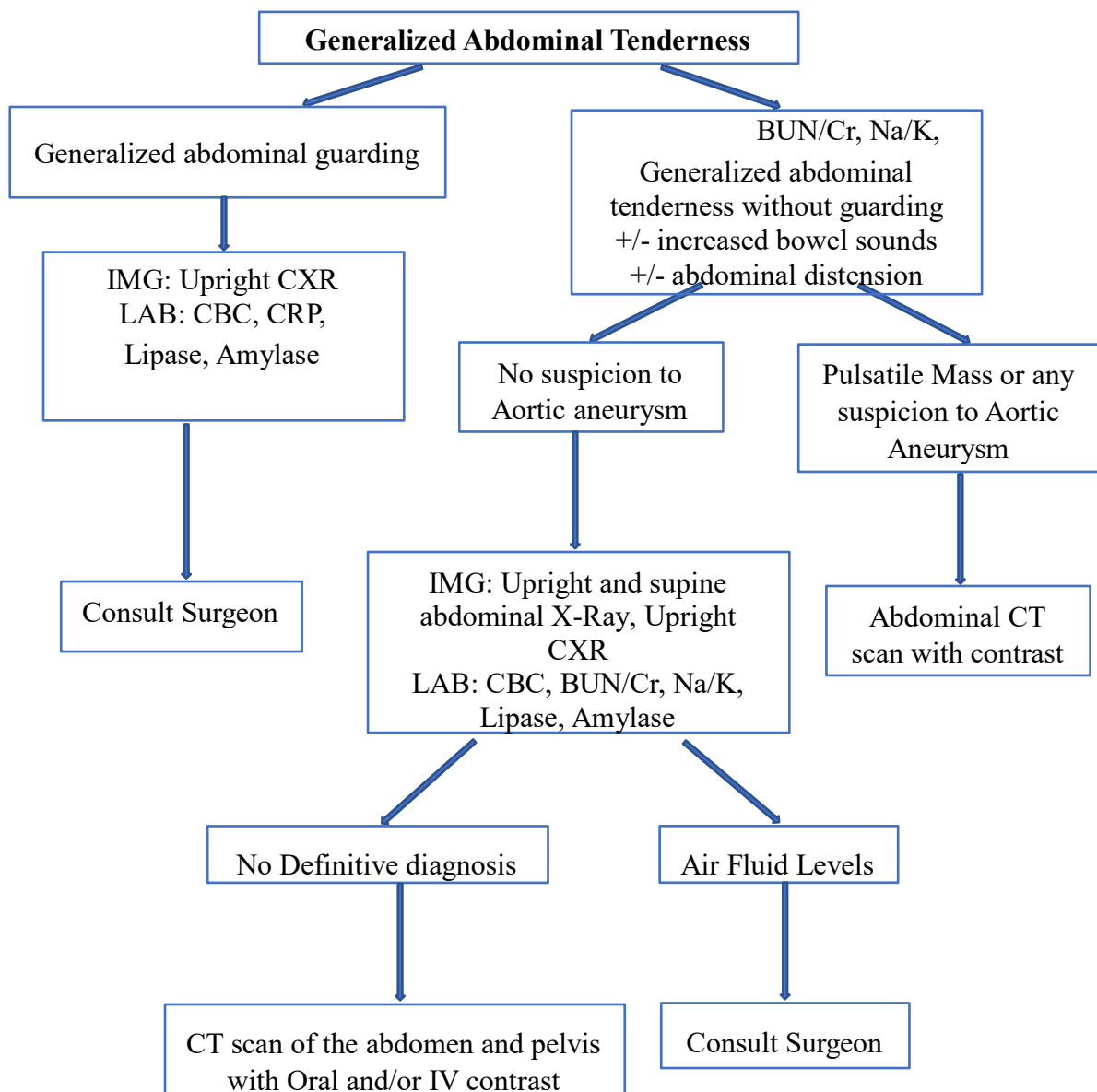
- Painkillers such as: spasfon, tramadol, acetaminophen or morphine (inject).

6. Treatment adjustment

- Based on etiological diagnoses

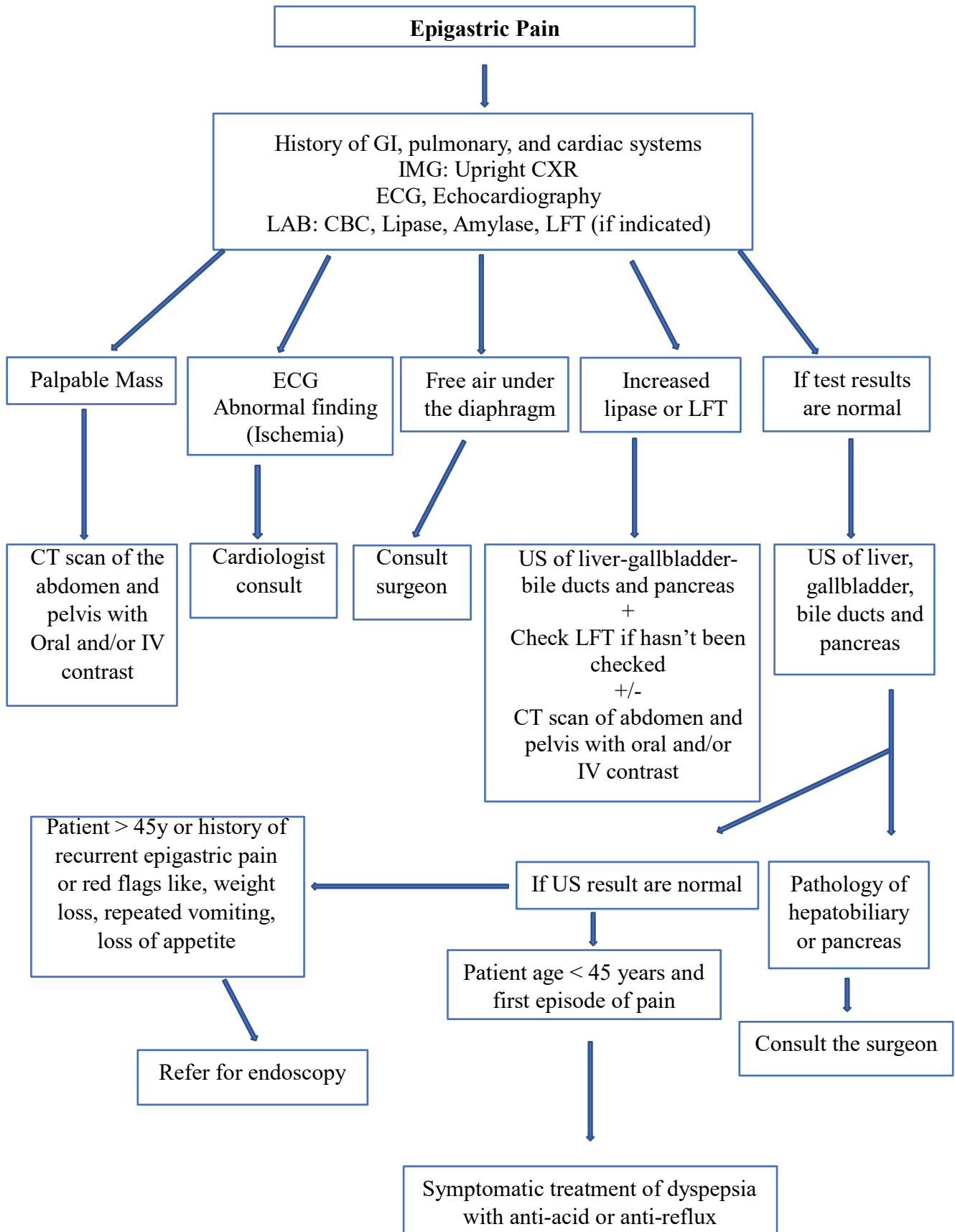
7. Algorithm

Approach to Non-Traumatic Acute Abdominal Pain (1)



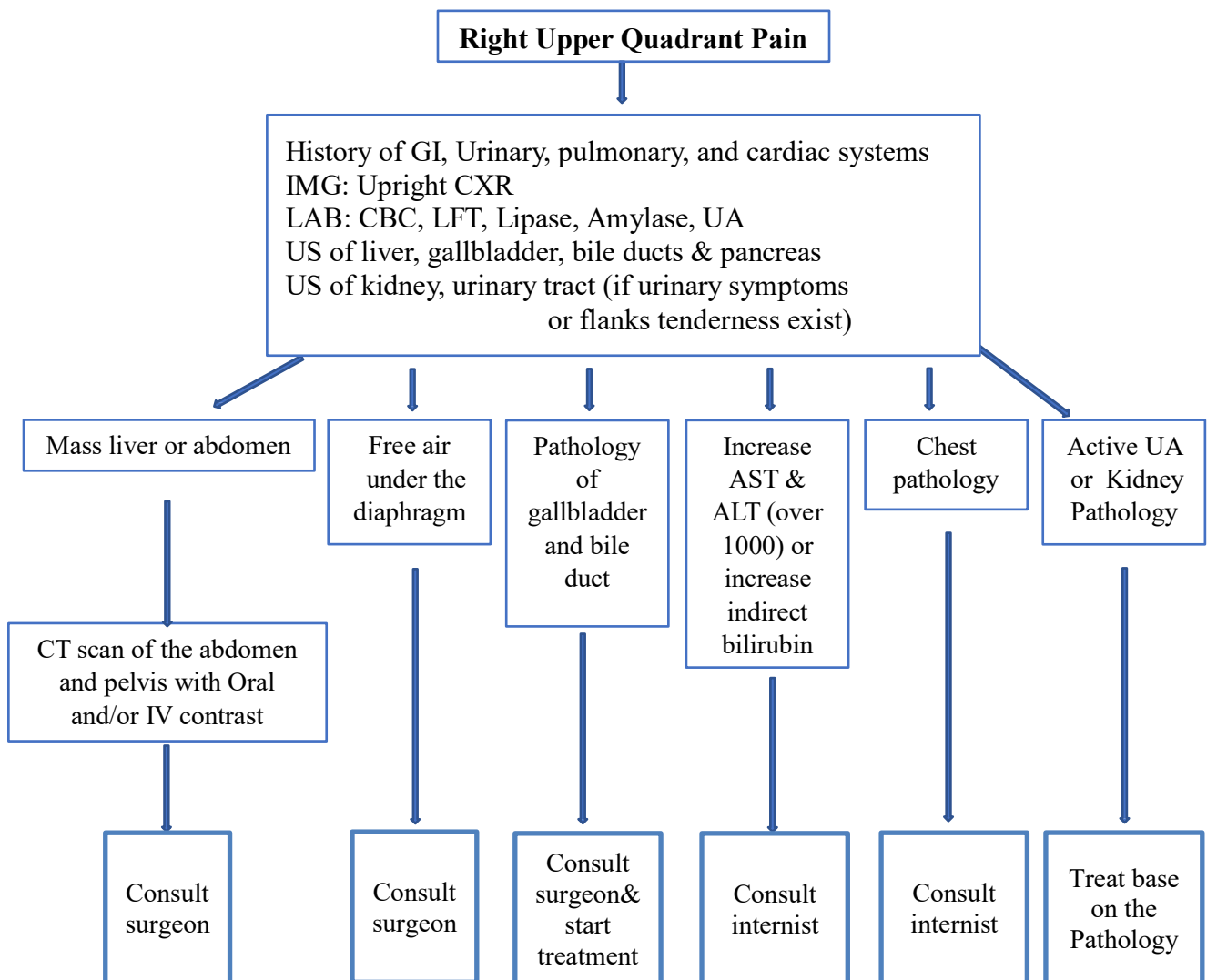
IMG: Imaging; CXR: Chest X-Ray; LAB: Laboratory;

Approach to Non-Traumatic Acute Abdominal Pain (2)



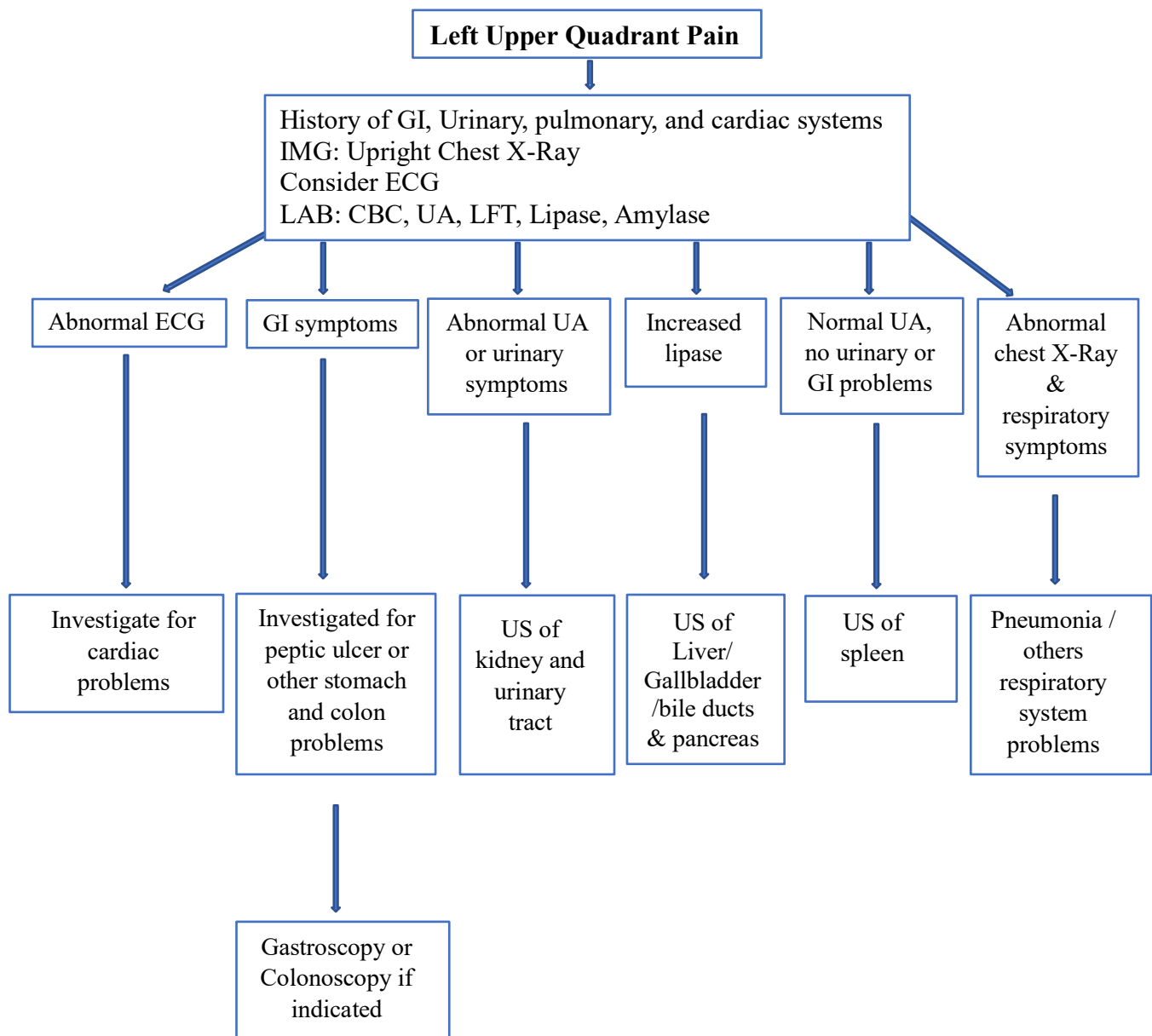
GI: Gastrointestinal; IMG: Imaging; US: Ultrasonography; CXR: Chest X-Ray; LAB: Laboratory; LFT: Liver function test;

Approach to Non-Traumatic Acute Abdominal Pain (3)



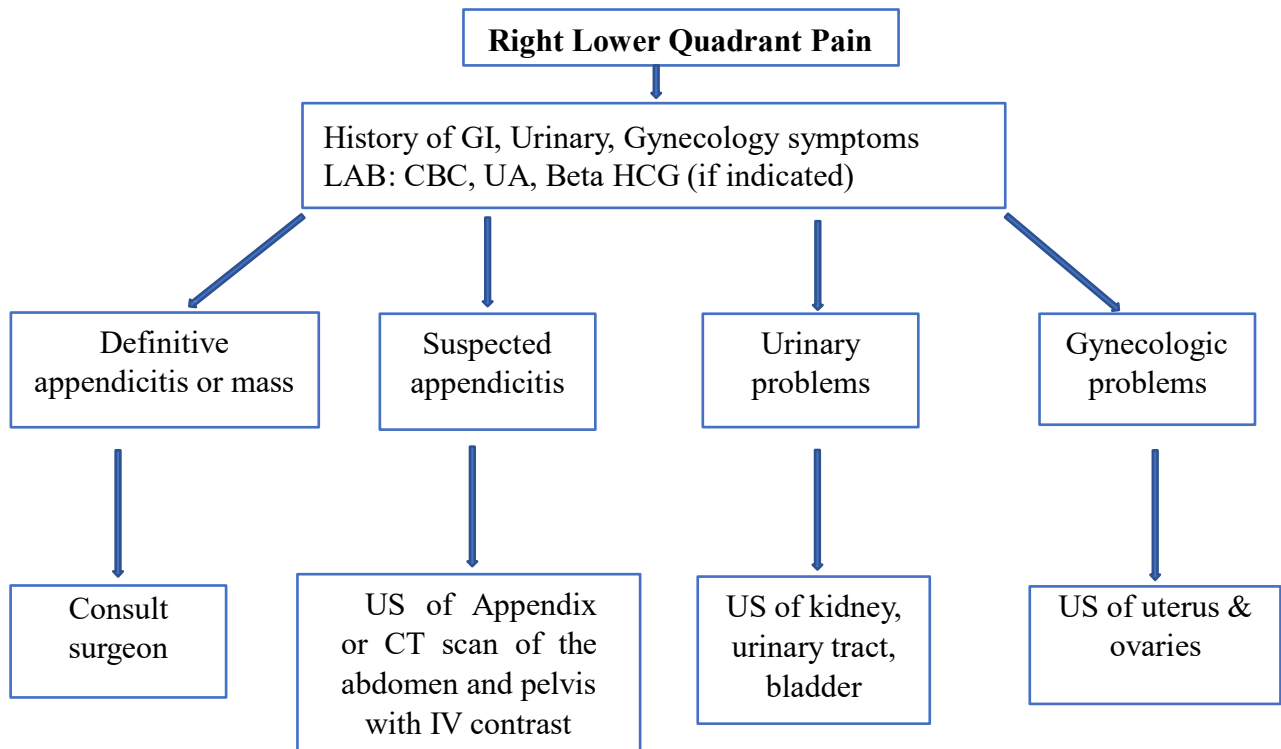
GI: Gastrointestinal; IMG: Imaging; CXR: Chest X-Ray; US: Ultrasonography; LAB: Laboratory;
LFT: Liver function test; UA: Urine analysis;

Approach to Non-Traumatic Acute Abdominal Pain (4)



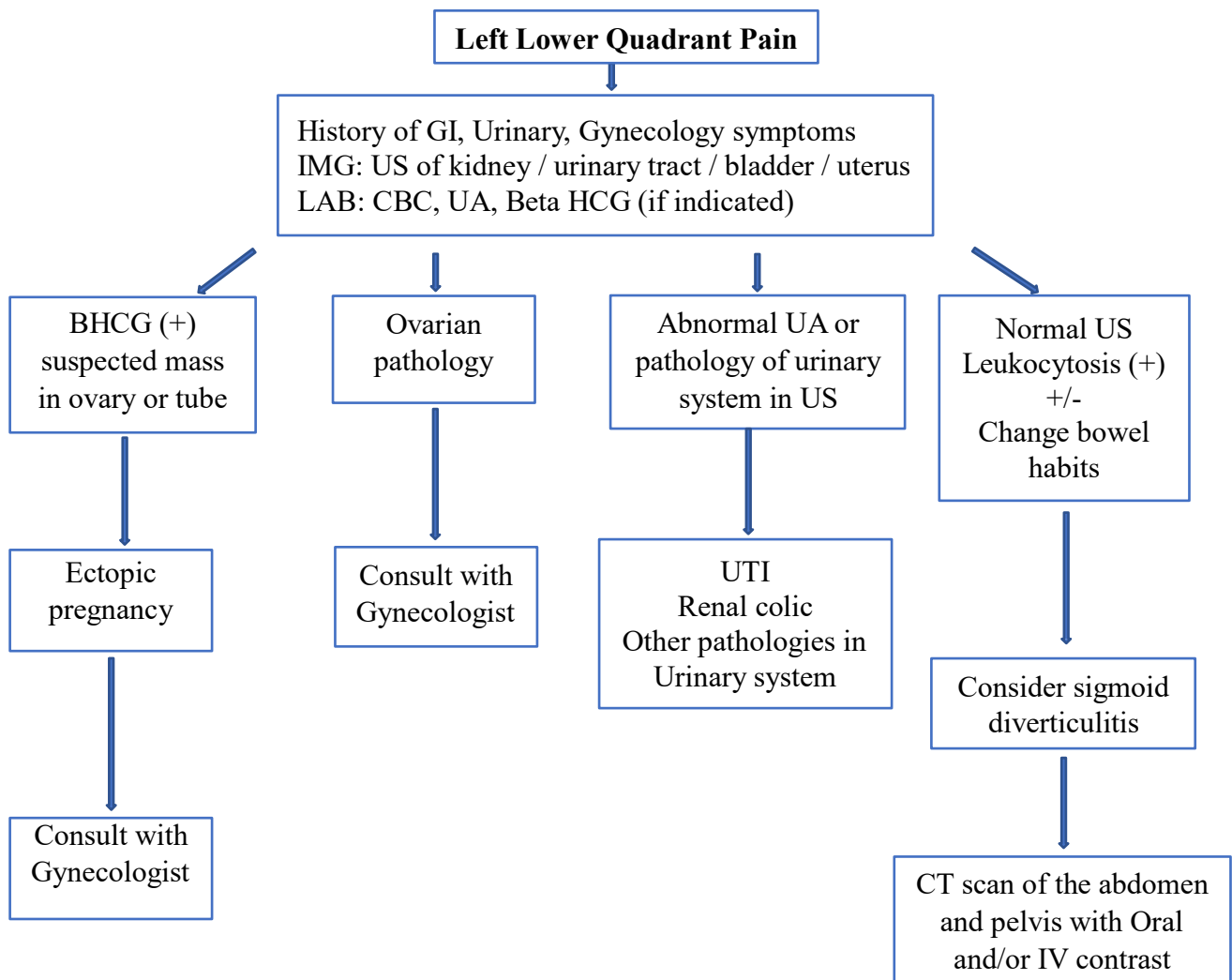
*GI: Gastrointestinal; IMG: Imaging; US: Ultrasonography; LAB: Laboratory;
LFT: Liver function test; UA: Urine analysis;*

Approach to Non-Traumatic Acute Abdominal Pain (5)



GI: Gastrointestinal; US: Ultrasonography; LAB: Laboratory; LFT: Liver function test; UA: Urine analysis;

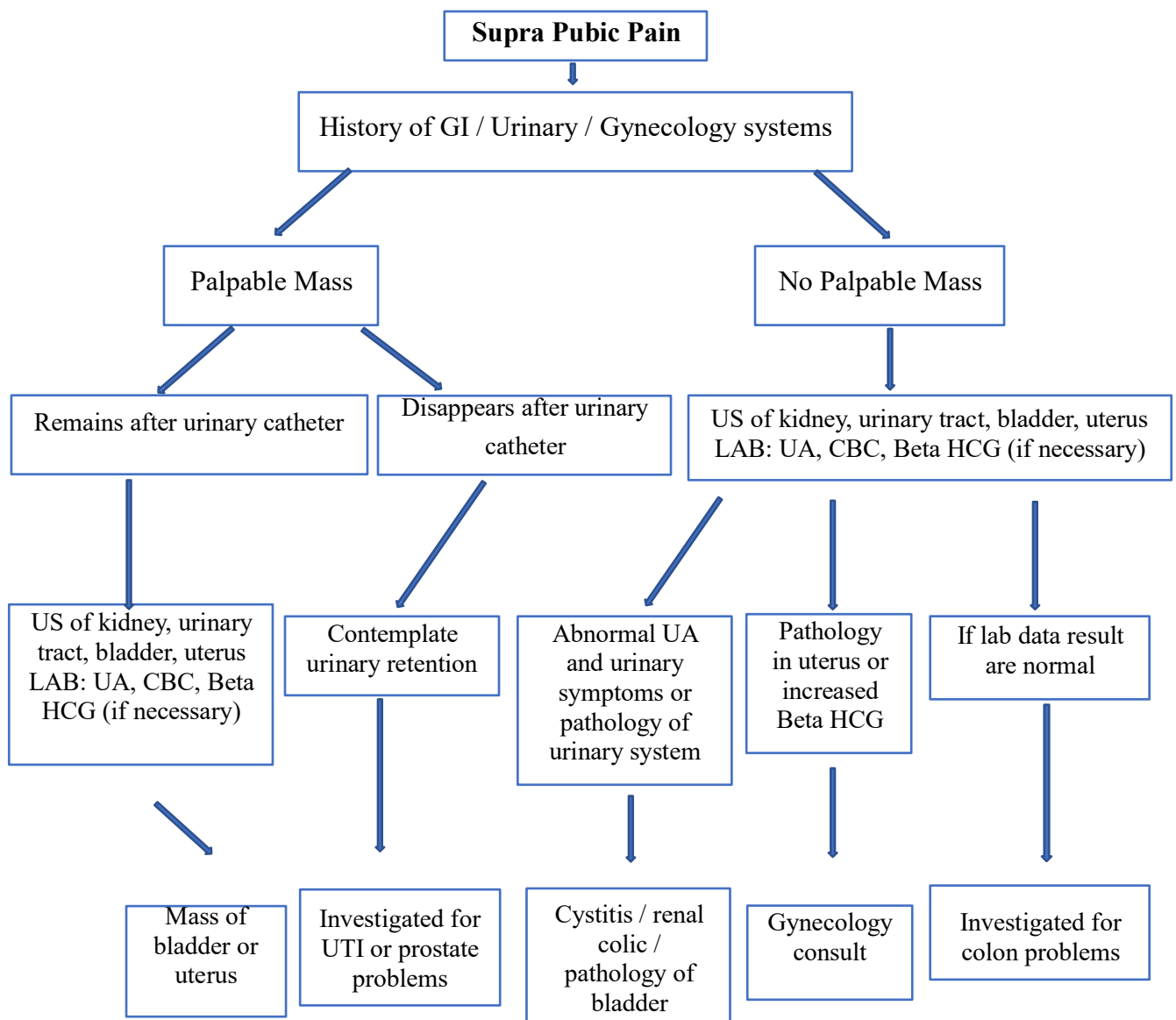
Approach to Non-Traumatic Acute Abdominal Pain (6)



GI: Gastrointestinal; IMG: Imaging; US: Ultrasonography; LAB: Laboratory;

UA: Urine analysis; UTI: Urinary tract infection;

Approach to Non-Traumatic Acute Abdominal Pain (7)



GI: Gastrointestinal; IMG: Imaging; US: Ultrasonography; LAB: Laboratory;

UA: Urine analysis; UTI: Urinary tract infection;

VIII. SPECIAL CASES

1. Older adults

- Older adult patients often do not present with the same signs and symptoms of disease characteristic of younger individuals.
- Older patients may not have fever or abnormal laboratory values with infectious etiologies for abdominal pain.
- The frequency of misdiagnosis of the acute abdomen in older patients is high and associated with higher mortality rates than in younger patients.

2. HIV-infected patients

- Diagnostic evaluation of abdominal pain in the HIV-infected patient is similar to that in the general population, but it is also guided by the immunologic function as represented by the CD4 cell count.
- Differential diagnosis includes common etiologies seen in the general population (eg, appendicitis, diverticulitis) but also opportunistic infections (eg, cytomegalovirus [CMV], Mycobacterium avium complex [MAC], cryptosporidium) and neoplasms (eg, Kaposi sarcoma, lymphoma) if there is evidence of advanced immunodeficiency (CD4 cell count <100 cells/microL).
- In this context, there should be a lower threshold for radiologic imaging and obtaining tissue culture and/or biopsy where appropriate.

IX. EVOLUTION AND PROGNOSIS

- The evolution and prognosis of non-traumatic acute abdominal pain varies widely based on the cause of acute abdominal pain.

X. COMPLICATIONS

- In addition, many of the underlying conditions that cause abdominal pain can lead to serious complications, including the possibility of structural damage to the digestive system from infection, cancer and its treatments, and inflammatory bowel disease.

XI. PREVENTION AND RECOMMENDATION

- Eating enough fiber and doing regular exercise can help prevent constipation and keep your bowels working well, which will prevent some forms of abdominal pain. It will also reduce your long-term risk of some diseases, such as diverticular disease or bowel cancer.
- Drinking plain water but not carbonated or fizzy drinks will reduce the chance of pain from bloating as well as keeping your body healthy.
- Following a specific diet can help reduce pain due to lactose intolerance, gallstones, irritable bowel syndrome and trapped gas, or wind.

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DYSPEPSIA

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I. DEFINITION

Dyspepsia is defined as any symptoms referable to the upper gastrointestinal tract. It consists of epigastric pain or burning, sensation of fullness, early satiation, anorexia, belching, nausea, vomiting, bloating, heartburn, and regurgitation.

Dyspepsia can be divided into 2 categories: functional dyspepsia and dyspepsia secondary to other organic diseases.

In this guideline, functional dyspepsia will be elaborated in detail. However, for dyspepsia secondary to organic diseases, a brief introduction related to its causes and epidemiology will be given, whereas its management will be elaborated separately elsewhere.

II. EPIDEMIOLOGY

The global prevalence of dyspepsia varies from 10 to 30%, in which underlying organic diseases were found in 25% of the patients. The variation seems to be related to the difference in population and definition criteria of studies. In Southeast Asian, a pooled prevalence of uninvestigated dyspepsia (UD) of 21 studies are 21.6%.

In Asian population, the risk of malignancy associated with UD is approximately 1.3% (95% CI, 0.80-2.10) and 18% of UD has underlying organic disease. In Thailand, the prevalence of dyspepsia is 66%. Of those, 60 - 90% are eventually diagnosed with FD.

III. ETIOLOGY

- Functional dyspepsia: no identified causes (the most common)
- Dyspepsia secondary to organic diseases
 - Upper gastrointestinal diseases such as gastritis, peptic ulcer diseases, gastroparesis, gastroesophageal malignancy, eosinophilic gastroenteritis...
 - *Helicobacter pylori* (*H. pylori*) infection
 - Biliary pain
 - Drug-induced dyspepsia: NSAIDs, aspirin, calcium channel blocker, alendronate, potassium supplements, acarbose, dabigatran, iron ...
 - Chronic pancreatitis
 - Parasitic infection: *Cryptosporidium*, *Giardia*
 - Other rare causes: Celiac disease, carbohydrate malabsorption, infiltrative diseases (Cronh, sarcoidosis...), myocardial ischemia (MI), renal insufficiency, Diabetes mellitus, thyroid disease and pregnancy.

IV. PATHOPHYSIOLOGY

The physiopathology of secondary dyspepsia can be explained by the mechanism of each pathology that causes the changes in the structure and the motility of gastro-intestinal tract. However, for FD, investigation reveals no structural abnormality to explain the symptoms.

The most recent iteration, Rome IV published in 2016, advocates rethinking these conditions as disorders of gut-brain interaction, acknowledging the complex interaction of biological, psychological, and social factors in their pathogenesis. They are characterized as complex bidirectional dysregulation of brain-gut interaction, via the brain-gut axis, rather than diseases (Figure 1).

Visceral hypersensitivity, abnormal gastrointestinal motility, and psychological disturbances have been recognized to contribute to the pathogenesis for decades, but more recently low-grade intestinal inflammation, increased intestinal permeability, immune activation, and disturbances in the microbiome have been identified, challenging the idea that structural changes are absent entirely.

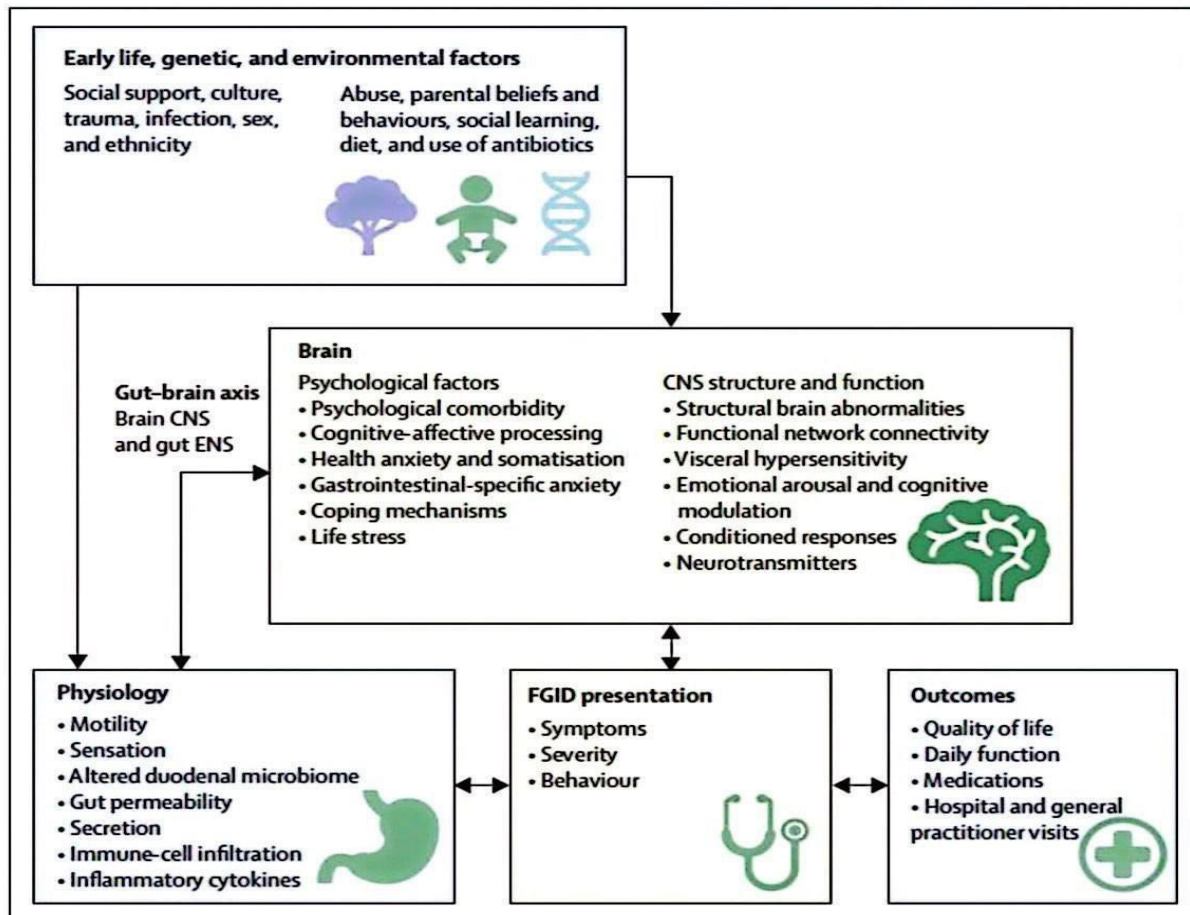


Figure 1: A biopsychosocial model of functional gastrointestinal disorders

Adapted from van Oudenhove and colleagues.³⁵ FGID=functional gastrointestinal disorders. ENS=enteric nervous system.

V. DIAGNOSIS

1. Clinical Manifestations: any dyspeptic symptoms that appeared for at least 4 weeks are considered as dyspepsia.
 - Dyspeptic symptoms in patients who have not yet undergone specific investigations are defined as UD.
 - FD refers to dyspeptic symptoms that fulfil the Rome Criteria and that cannot be explained by a routine clinical evaluation, including endoscopy and there is no evidence of *H. pylori* infection.

* Based on Rome Criteria IV: FD are divided into 2 subcategories: epigastric pain syndromes (EPS) and postprandial distress syndromes (PDS).

 - EPS consists of intermittent pain or burning in the epigastrium, occurring at least once per week. Postprandial epigastric bloating, belching, and nausea can also be present, but are not specific to dyspepsia.
 - PDS is defined as bothersome postprandial fullness after normal-sized meals, or else early satiety that prevents someone finishing a regular-sized meal, with such episodes occurring at least several times per week. Postprandial epigastric pain burning, bloating, belching, or nausea may also be present, but are not specific to dyspepsia.
 - The criteria should be fulfilled for the last three months with symptom onset at least six months before diagnosis.
2. Investigation for UD
 - Routine tests: CBC, parasite stool test, *H. pylori*, abdominal ultrasound.
 - Endoscopy is indicated in patient with alarm symptoms such as:
 - Age ≥ 45 years
 - Dysphagia, odynophagia
 - Unexplained iron deficiency anemia,
 - Unexplained weight loss ($> 10\%$ of body weight),
 - Persistent vomiting (> 10 times/24h or after each meal),
 - Family history of GI cancer at first degree relative
 - Palpable mass or lymphadenopathy
 - Overt gastro-intestinal bleeding (GIB): hematemesis, melena, hematochezia
 - *H. pylori* test:
 - Invasive tests (Gastroscopy): if gastroscopy is indicated.
 - Histology (gold standard)
 - Rapid urease testing
 - Culture
 - Polymerase chain reaction
 - Noninvasive tests: if endoscopy is not indicated.
 - Urease breath test (UBT): best option.
 - Fecal or stool antigen test (SAT): alternative.
 - Serological test (ELISA): not recommended for diagnostic test, but it can serve as screening test in specific clinical situations (in which false negative of other tests is high) such as:

- Bleeding peptic ulcers
- Gastric MALT Lymphoma
- Gastric cancer
- Gastric atrophy
- Recent use of antibiotics or PPI.

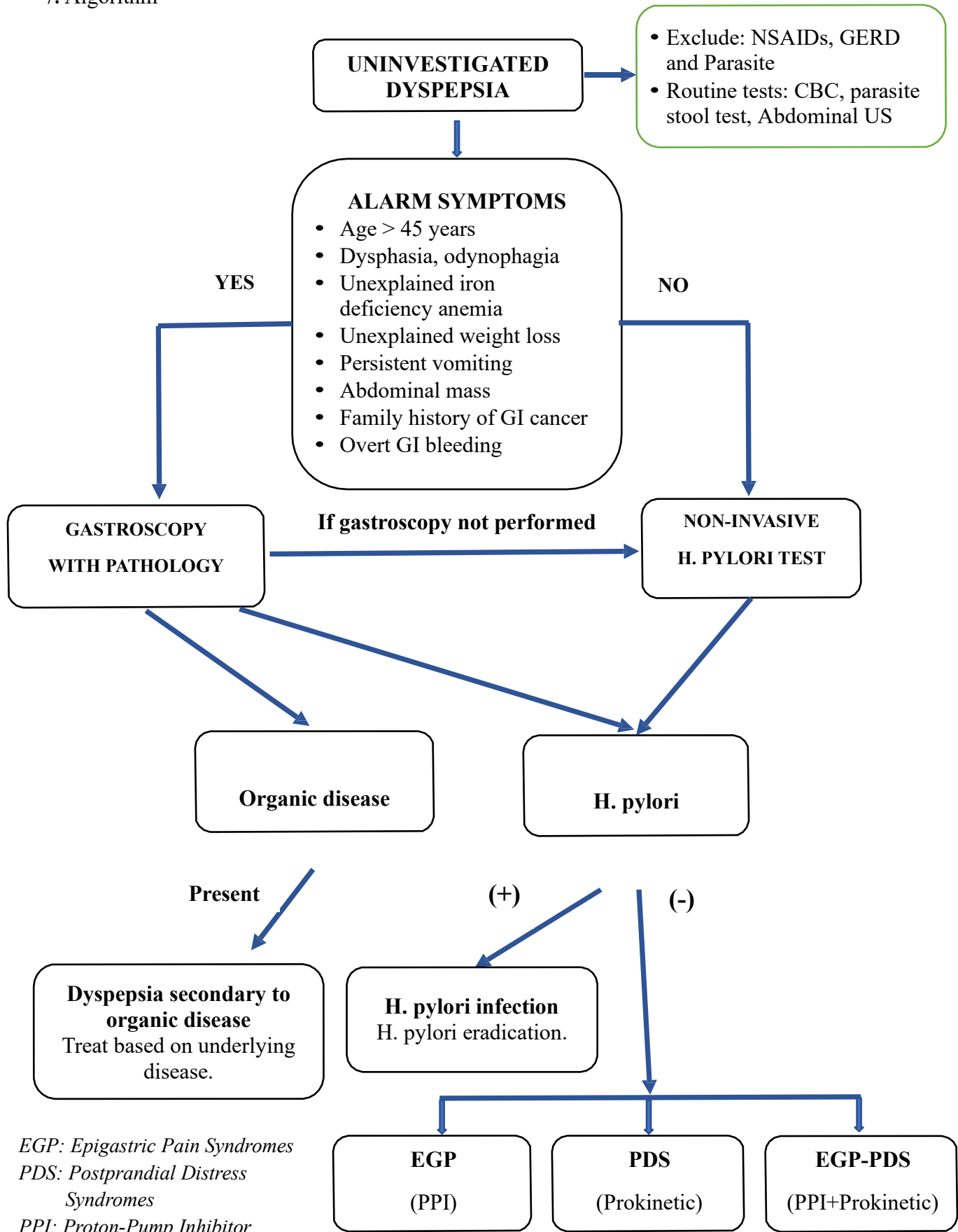
VI. DIFFERENTIAL DIAGNOSIS

- Gastroesophageal reflux disease
- Gastroparesis
- Irritable bowel syndrome

VII. MANAGEMENT

1. Objectives: eliminating the causes and symptoms controlled.
2. Critical Signs:
 - Severe upper abdominal pain can rarely be seen in functional dyspepsia. However, if it appears, other causes should be ruled out.
 - Over GI bleeding with hemodynamic shock
3. Mortality Risk Factors:
 - Functional dyspepsia is a benign disease without any mortality risk.
4. Outpatient Management:
 - FD is usually treated in outpatient settings. However, secondary dyspepsia sometimes need hospitalization (eg. Overt GI bleeding caused by ulcer diseases...)
5. First-Line Treatment
 - For FD
 - If H. pylori is positive: H. pylori eradication
 - If H. pylori is negative or persisting symptoms after H. pylori eradicated
 - In case of EGP syndrome
 - Proton pump inhibitors (PPI: Lansoprazole 30mg, Omeprazole 20mg, Pantoprazole 40mg, Rabeprazole 20mg, Esomeprazole 20mg) for 4 to 8 weeks It can be repeated if symptoms recur.
 - H2-receptor antagonists (cimetidine, famotidine...) can also be an option.
 - In case of PDS
 - Prokinetics (metoclopramide, domperidone, cisapride, itopride, mosciapride...) for 4 to 8 weeks. It can be repeated if symptoms occur.
 - Buspirone: 10 mg, 3 times daily for 4 weeks, may help relax the gastric fundus.
 - In case overlap EGP-PDS: PPI + Prokinetic.
6. Treatment Adjustments
 - If symptoms not improved after 8 weeks of PPI.
 - Tricyclic antidepressant (TCA: Amitriptyline 10mg, nortriptyline 10mg, desipramine 25mg at night) for 8 to 12 weeks and continued for 6 months if patients respond appropriately.
 - TCA should be started with low dose and be increased at 1 to 2 weeks intervals. It can be resumed if symptoms recur (e.g. the maximal dose of amitriptyline and nortriptyline is 150mg/d and desipramine 300mg/d).

7. Algorithm



VIII. SPECIAL CASES

Patients with psychological factors associated functional dyspepsia and who fail medical therapy, psychological therapy should be recommended.

Patients with FD and weight loss, Mitazapine should be recommended. The dose is titrated from 7.5 mg an hour before bedtime to 45 mg daily.

IX. VOLUTION AND PROGNOSIS

FD is relapsing and remitting. Population studies have shown, over a follow-up of 10 to 12 years, approximately 15 to 20% of individuals had persistent symptoms and 40 to 52% had symptom resolution. In 30 to 35% of patients, symptoms fluctuated over time and patients met criteria for another functional gastrointestinal disorder. No data shows an association with decreased survival.

X. COMPLICATIONS

Although the condition is not associated with increased mortality, FD does cause physical and mental distress affecting the patient's quality of life. FD can lead to physical and mental distress and score higher on psychometric tests for symptoms of anxiety, depression, and somatization. About 10% to 25% of patients report the social impact of their symptoms is significant enough to seek a physician, leading to more healthcare visits, greater health impairment, and a worsening quality of life.

XI. REVENTION AND RECOMMENDATION

Having healthy diets, doing enough exercise, and keeping good mental health can reduce the risk of functional dyspepsia.

Patients are recommended to:

- Avoid fatty foods (red meat, butter, fried foods, and cheese) and foods that seem to make their symptoms worse.
- Eat a bunch of small meals each day, rather than 2 or 3 big meals.
- Avoid or limit alcohol intake.
- Avoid taking gastro-toxic drugs (aspirin or ibuprofen....)

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DYSPHAGIA

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I. DEFINITION

Dysphagia comes from the Greek words, dys (difficulty) and phagia (to eat), refers to the subjective sensation of difficulty or abnormality of swallowing that interferes with the passage of solids or liquids from the oral cavity to the stomach.

II. EPIDEMIOLOGY

According to recent population-based research focusing on dysphagia, the incidence of dysphagia in adults was up to 17% and the highest age group range for both men and women was observed between 40-49 years.

The prevalence of dysphagia is 10% to 22% in Americans aged 50 and over.

The estimated prevalence of oropharyngeal dysphagia involving the general population has ranged from 2-16%.

In Cambodia, according to our study of 168 patients admitted to Gastroenterology department at Khmer Soviet Friendship Hospital, Phnom Penh, from 2020 to 2022, esophageal dysphagia is predominantly observed among the male population (69.6%) and among individuals in older age groups, >50 years (74.4%).

III. ETIOLOGIES

Dysphagia is usually sub-classified into oropharyngeal and esophageal dysphagia.

- Oropharyngeal dysphagia (High dysphagia): difficulty initiating a swallow or passing food through the mouth or throat
- Esophageal dysphagia (Low Dysphagia): difficulty swallowing several seconds after initiating a swallow and a sensation of food getting stuck.

1. Oropharyngeal dysphagia (High Dysphagia)

The majority of cases of oropharyngeal dysphagia occur as a result of neuromuscular degeneration or damage such as cerebral vascular accident (stroke).

Table 1. Causes of Oropharyngeal dysphagia (High Dysphagia)

Neuromuscular disturbances	Structural (mechanical) disorders
Central nervous system diseases • Stroke • Cranial nerve palsy • Bulbar palsy Peripheral Nervous system • Amyotrophic lateral sclerosis • Guillain-Barré syndrome • Bulbar poliomyelitis • Drugs (botulinum toxin, procainamide...)	Intrinsic lesion: • Oropharyngeal malignancies and neoplasms • Zenker's diverticulum • Cricopharyngeal achalasia, bar, rings and stenosis • Foreign body • Proximal esophageal web (Plummer-Vinson) • Inflammatory disease (pharyngitis, tonsillar abscess, eosinophilic esophagitis...)

Myogenic • Myasthenia gravis • Hyperthyroidism, hypothyroidism • Dermatomyositis • Amyloidosis • Oculopharyngeal muscular dystrophy	• Head and neck malignancies and the consequences of surgical and/or radiotherapeutic interventions on these tumors Extrinsic lesions: • Thyromegaly • Cervical lymphadenopathy • Vascular anomalies • Cervical osteophytes • Skeletal abnormalities
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2. Esophageal dysphagia (Low Dysphagia)

Most Common Causes of Low dysphagia are GERD-related strictures, esophageal cancer, achalasia, Eosinophilic esophagitis (young patients), Schatzki ring (intermittent history) and peptic stricture (progressive course).

Table 2. Causes of Esophageal dysphagia (Low Dysphagia)

Structural disorders	Motility disorders
Intrinsic Encroachment: • Stricture: Peptic stricture, caustic, pill or radiation-induced • Mucosal rings and webs: Schatzki, Plummer-Vinson, eosinophilic esophagitis • Esophageal tumors: adenocarcinoma, squamous cell carcinoma, metastatic, leiomyoma, lymphoma or granular cell tumor • Systemic diseases: scleroderma, Crohn's disease, pemphigus/pemphigoid or lichen planus • Post-surgery (laryngeal, esophageal or gastric cancers), acute esophageal infection, esophageal diverticular or foreign bodies. Extrinsic compression: • Mediastinal masses: lung cancer, lymphoma, lymph node or thyromegaly • Vascular compression: dysphagia aortic, cardiomegaly, dysphagia lusoria (aberrant right subclavian artery) • Miscellaneous: cervical spine osteophytes/spondylosis or fundoplication.	• Primary: Achalasia, diffuse esophageal spasm, hypertensive lower esophageal sphincter, ineffective esophageal motility disorder • Hypomotility (secondary to connective tissue disorders)

Table 3. Medications Associated with Dysphagia

Medications that can cause direct esophageal mucosal injury
• Antibiotics
• Doxycycline (Vibramycin)
• Tetracycline
• Clindamycin (Cleocin)
• Trimethoprim-sulfamethoxazole (Bactrim, Septra)
• Nonsteroidal anti-inflammatory drugs
• Alendronate (Fosamax)
• Zidovudine (Retrovir)
• Ascorbic acid
• Potassium chloride tablets (Slow-K)
• Theophylline
• Quinidine gluconate
• Ferrous sulfate

IV. PATHOPHYSIOLOGY

The role of the esophagus is to transport food bolus from the oropharynx to stomach. Difficulty in bolus transport due to either abnormal structures or functions can arise anywhere in the upper gastrointestinal tract from the oropharynx, upper esophageal sphincter, esophageal body, and lower esophageal sphincter.

Dysphagia can result from neuromuscular or structural causes, which produce narrowing of the esophagus, lack of salivary secretion, weakness of the muscular structures that propel the food bolus towards the stomach, or disruption of the neural networks coordinating the swallowing mechanism.

- **Disorder of the oral preparatory phase:**
 - Inadequate mastication or salivary production: Sjögren's disease, some medications reducing salivary flow (anticholinergics, antihistamines, antihypertensive agents).
 - Neuromuscular incoordination: neural or muscular disorders that interfere with voluntary swallowing or peristalsis.
 - Disruption of the oropharyngeal mucosa (inhibit normal processing of the bolus): mucositis, aphthous ulcers, herpetic lesions or trauma.
- **Disorder of the pharyngeal phase:**
 - Neurologic disorders (involve the central or peripheral nervous system): stroke, motor neuron disease, myasthenia gravis.
 - Poor compliance of the upper esophageal sphincter (UES) relaxation: Parkinson disease, primary cricopharyngeal dysfunction
- **Disorder of the esophageal phase:**
 - Related to structural disorders, obstruction to passage of the bolus: stricture or or diverticular herniations or tumor (intrinsic/ extrinsic obstruction).

- Related to esophageal motility disorders (affect peristaltic wave): failure of relaxation of the lower esophageal sphincter (LES) like achalasia, distal esophageal spasm.

V. DIAGNOSIS

The diagnosis of dysphagia requires a stepwise approach, including clinical evaluation, imaging, and specialized tests to determine whether the problem is High dysphagia or Low dysphagia.

1. CLINICAL APPROACH

1.1. Clinical manifestation

Patients who have dysphagia may present with a variety of complaints, but they usually report coughing or choking, or the abnormal sensation of food sticking in the back of the throat or upper chest when they are trying to swallow. A carefully conducted patient history will enable the physician to identify 80 to 85 percent of the causes of dysphagia. Specific questions or Key summary in table 3,4,5 may help narrow the differential diagnoses to a specific diagnosis or to an anatomic or pathophysiologic-related diagnosis.

Clinical evaluation and history are critically important in evaluating dysphagia.

- ***Onset and progression:***
 - Sudden: stroke, neurological cause
 - Gradual and progressive: malignancy or stricture
- ***Types of Food Affected***
 - Solids first, then liquids → Mechanical obstruction (stricture, cancer)
 - Both solids & liquids from onset → Motility disorder (achalasia, spasm)
- ***Associated Symptoms***
 - Coughing, choking, nasal regurgitation, at the initiation of swallowing or after a few seconds of swallowing → High Dysphagia
 - Globus sensation (feeling of food stuck in throat/chest) → Low Dysphagia
 - Regurgitation of undigested food → Zenker's diverticulum, achalasia
 - Heartburn & reflux symptoms → GERD-related dysphagia
 - Weight loss → Concern for malignancy
 - Odynophagia → Esophagitis (infectious, pill-induced, GERD, cancer)
 - Neurological symptoms: voice change, drooling, aspiration and/or weakness.
 - Prior history of any caustic ingestion
 - Self-imposed dietary changes regarding food consistency, preference for liquids or semisolids
- ***Risk Factors***
 - Neurological disease: stroke, Parkinson's, Amyotrophic Lateral Sclerosis (ALS)
 - GERD (risk for strictures, Barrett's esophagus)
 - Smoking, alcohol use (risk for head & neck or esophageal cancer)
 - Recent radiation therapy (post-radiation fibrosis)
 - Immunosuppression (risk for infectious esophagitis)
 - Social isolation, frustration, or depression.

It is necessary to carefully establish the location of the perceived swallowing problem: oropharyngeal or esophageal dysphagia.

1.2. Physical examination

- **Neurological evaluation:**
 - Assessment of the patient's mental status
 - Motor and sensory functioning
 - Deep tendon reflexes and cranial nerves
 - Cerebellar examination
- **Oral & throat examination:** Bimanual palpation of the floor of the mouth, tongue and lips with a gloved hand: → masses, dental issues (signs of inflammation or other structural disorders), mucosal lesions and abnormal motor function.
- **Examination of the neck:** masses or organomegaly
- **Signs of systemic disease:** scleroderma skin changes, myositis weakness

2. INVESTIGATION

2.1. Laboratory Evaluation:

Laboratory tests are not always necessary for diagnosing dysphagia but can help identify underlying systemic conditions, infections, nutritional deficiencies, or autoimmune diseases contributing to swallowing difficulty.

When to order Labs in Dysphagia

- Suspected malignancy or anemia → CBC, Iron studies
- Suspected infectious esophagitis → Candida/HSV/CMV tests
- Suspected eosinophilic esophagitis → Eosinophil count, IgE levels
- Suspected autoimmune disease → ANA, Anti-Scl70, Anti-MuSK, CK
- Suspected myasthenia gravis → Anti-AChR, Anti-MuSK
- Goiter, hypothyroidism: Thyroid Function Tests (TSH, Free T4) (Detects thyroid dysfunction causing compression or neuromuscular dysfunction)

2.2. Imaging studies

Imaging studies play a crucial role in diagnosing the cause of dysphagia by evaluating the oropharyngeal and esophageal anatomy, detecting structural abnormalities, and assessing motility disorders. The choice of imaging depends on whether the dysphagia is high dysphagia or low dysphagia.

- Modified Barium Swallow (Videofluoroscopy) → Best for High dysphagia (Assess the degree of dysfunction and severity of aspiration).
- Barium Swallow Esophagram → Best for Low dysphagia (structural and motility disorders: rings and stricture (post caustic injury or radiation therapy, achalasia. Achalasia: => Image: "Bird's beak" narrowing of LES, aperistalsis).
- EGD with biopsy → Best for cancer, esophagitis, strictures, eosinophilic esophagitis; also, can perform therapy (dilation of an esophageal ring).
- Manometry → Best for achalasia, esophageal spasm, scleroderma.

- Esophageal manometry:
 - High-resolution esophageal manometry (HROM) is the gold standard test for diagnosis of esophageal motility disorder.
 - Indicated in dysphagia patients with prior non diagnostic EGD and/or barium esophagram.
- Functional lumen imaging probe (FLIP)
 - To assess distensibility and esophageal contractions
 - A complementary testing in patients with a suspected but unconfirmed diagnosis of achalasia or with esophagogastric junction outflow obstruction.
- Nasopharyngeal laryngoscopy:
 - Detail examination of the oropharynx, hypopharynx, larynx and proximal esophagus to rule out a structural lesion
- Fiberoptic endoscopic evaluation of swallowing (FEES) / N Esoendoscopy
 - Optimal method of identifying structural abnormalities in the oropharynx
 - Obtain biopsies if needed
 - Penetration or aspiration risk is more severe on FEES as compared with videofluoroscopy assessment.
 - Require specialized training

Key Summary for Diagnosis of Dysphagia:

Table 4. Presenting history and symptoms suggestive of structural and motility disorders

	Structural disorders	Motility disorder
Symptoms	• Regular • Short duration, rapid progression • Solid (may progress to liquid)	• Intermittent • Long duration • Solid and liquid
History	• Alarm features - o Onset age >50 y - o Bleeding - o Odynophagia - o Weight loss - o Vomiting	• Connective tissue diseases • Diabetes mellitus • Non-cardiac chest pain

Table 5. Key Clinical Clues for Differentiation between High and Low Dysphagia

Feature	High Dysphagia	Low Dysphagia
Swallow initiation difficulty	Yes	No
Coughing/choking while eating	Common	Rare
Nasal regurgitation	Possible	No
Globus sensation	Yes	Yes
Progression pattern	Sudden or gradual	Progressive
Associated neurological signs	Common	Rare
Regurgitation of undigested food	Yes (Zenker's diverticulum)	Yes (Achalasia)
Odynophagia (painful swallowing)	Possible (infection, inflammation)	Common (esophagitis, malignancy)

Table 6. Diagnostic Approach Based on Symptoms

Clinical Feature	First-line Test	Confirmatory Test
Difficulty initiating swallowing, choking	VFSS / Modified Barium Swallow	Neurological evaluation if needed
Food stuck in throat or chest	Barium Swallow	EGD for further evaluation
Solids > Liquids (progressive)	Barium Swallow	EGD + biopsy for cancer/stricture
Both solids & liquids from onset	Barium Swallow	Esophageal Manometry for motility disorder
Regurgitation of undigested food	Barium Swallow	EGD or Manometry (Zenker's or Achalasia)
Heartburn, reflux symptoms	Empiric PPI trial	pH monitoring if needed

VI. DIFFERENTIAL DIAGNOSIS

Thorough History taking can help distinguish some conditions that are similar to dysphagia.

- Globus
- Food aversion
- Voluntary anorexia
- Anorexia nervosa
- Xerostomia
- Odynophagia

VII. MANAGEMENT**1. Objectives**

- Restore Safe and Efficient Swallowing
- Improve Nutritional Status and Hydration
- Improve Quality of Life
- Treat Underlying Cause ((high dysphagia or low dysphagia)
- Prevent Complications

A tailored treatment plan, often developed by a multidisciplinary team (MDT), is essential to address the specific needs and underlying causes of dysphagia in each patient.

2. Critical Signs

- Acute dysphagia in context of caustic ingestion with sepsis or perforation
- Dysphagia on severe stroke
- Cancer-related Dysphagia

3. Mortality risk factor

- Aspiration and aspiration Pneumonia
- Malnutrition and dehydration
- Underlying Medical Conditions: Stroke, cancers, ...
- Higher age
- Delay in Diagnosis and Management

4. Outpatient Management

As dysphagia is a symptom related to multiple diseases, some mild diseases could be managed as outpatients such as functional dysphagia, GERD-related dysphagia, eosinophilic esophagitis (Table 7).

5. First-line Treatment:

5.1. Diet modification:

- Swallowing rehabilitation and re-education along with modifications to the consistency of the food supervised by SLTs and dieticians (See Chapter Stroke).
- Proper oral hygiene to prevent bacterial colonization and pneumonia.
- Oral feeding is preferable
- Nasogastric tube feeding, Percutaneous endoscopic gastrostomy: need MDT.

5.2. Etiological Treatment

Table 7. Key Summary Treatment for Dysphagia

Dysphagia Type	Treatment Approach
Oropharyngeal Dysphagia (Neurological, Muscle-related)	Swallowing therapy, texture-modified diet, enteral feeding if needed
Esophageal Strictures	Endoscopic dilation + PPIs
Schatzki ring or esophageal web	Endoscopic dilation
Esophageal Cancer	Surgery, chemotherapy, radiation, stenting for palliation.
Achalasia	Balloon dilation, Botox, myotomy (POEM)
GERD-related dysphagia	PPIs, lifestyle changes, fundoplication if refractory, dilation if strictures present
Eosinophilic Esophagitis (EoE)	Diet modification, steroids (swallowed budesonide), PPIs
Esophageal Spasm	CCBs (diltiazem), nitrates, botulinum toxin
Scleroderma esophagus	High-dose PPI (for severe reflux), promotility agents
Infectious Esophagitis: Candida HSV CMV	Antifungal (fluconazole), antiviral (acyclovir) antiviral (ganciclovir)

CCBs: Calcium channel blockers; BT: botulinum toxin, AI

6. Treatment adjustment:

There might be some treatment adjustment according to specific causes of dysphagia.

7. Algorithms for Dysphagia

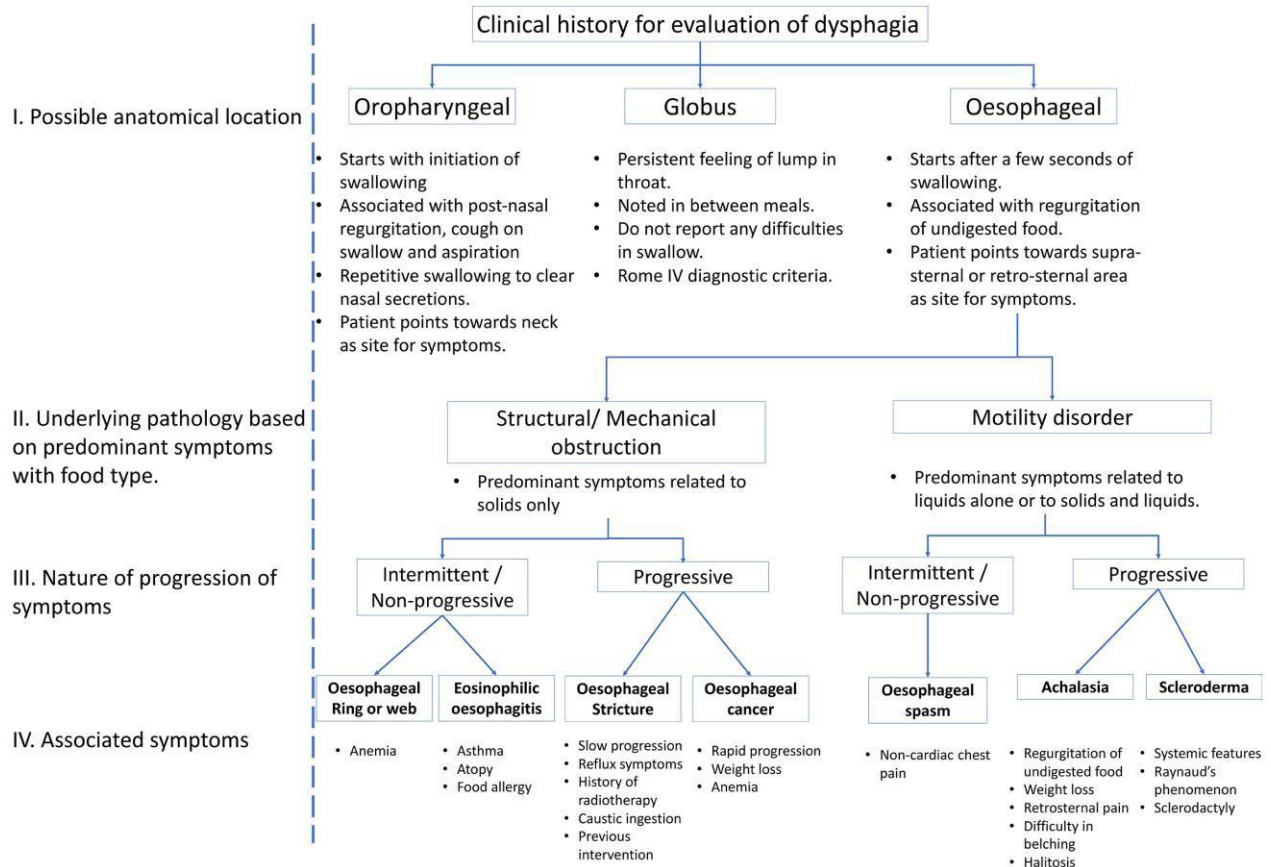


Figure 1. Approach to dysphagia.

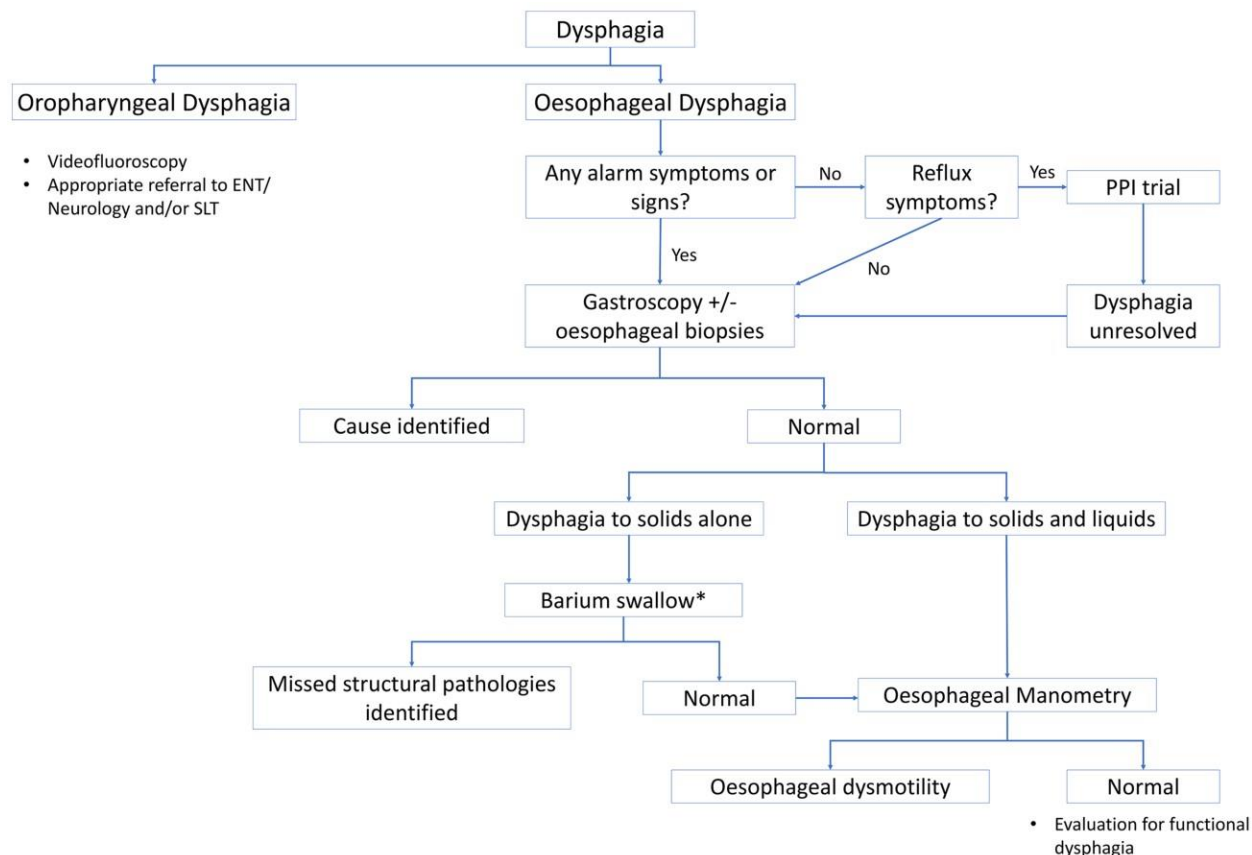


Figure 2. Selection of appropriate investigation.

ENT, ear, nose and throat surgeon; PPI, proton pump inhibitor; SLT, speech and language therapist.

*Barium swallow may be considered earlier in the algorithm when gastroscopy is not possible and/or where structural disorders require further scrutiny (prior surgery, radiotherapy, caustic injury, complex stricture and so on).

VIII. SPECIAL CASES

1- **Achalasia** (Figure 3).

2- **Dysphagia in caustic ingestion** (acute period) (See the Chapter Caustic Ingestion)

3- **Functional dysphagia:**

- **Functional dysphagia** is a diagnosis of exclusion after ruling out structural, motility, or inflammatory causes.
- **Key Diagnostic Criteria (Rome IV Criteria):**
 - Sensation of food sticking or passing abnormally through the esophagus
 - Symptoms present for at least 3 months
 - No structural abnormalities on endoscopy, barium swallow, or CT scan
 - Normal esophageal motility on high-resolution manometry
- **Treatment:**

- Lifestyle and Dietary Modifications:
 - ✓ Eat slowly and chew food thoroughly
 - ✓ Avoid dry, sticky, or hard foods
 - ✓ Drink water between bites
 - ✓ Avoid large meals, caffeine, and alcohol
- Pharmacologic Therapy:
 - Proton Pump Inhibitors (PPIs)
 - Tricyclic Antidepressants (TCAs): low dose
 - Avoid large meals, caffeine, and alcohol
- Psychological and Behavioral Therapy:
 - Cognitive Behavioral Therapy (CBT) – Helpful for anxiety-driven dysphagia.

4- Post Stroke-related Dysphagia (See Chapter Stroke).

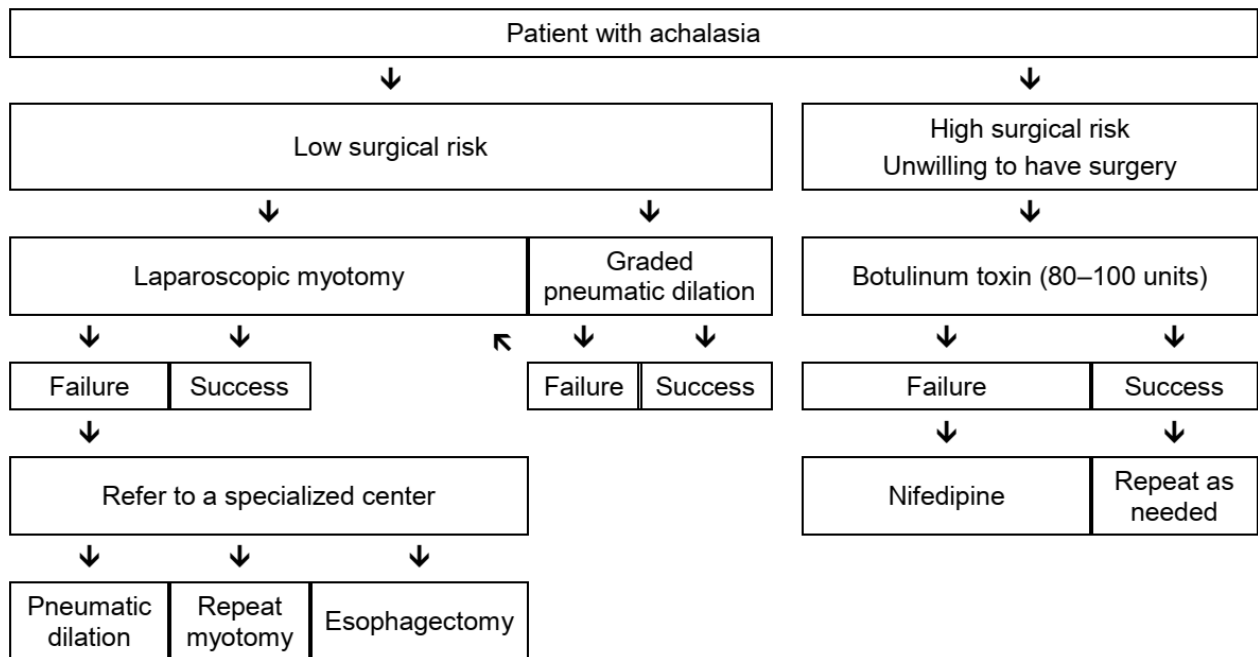


Figure 3. Management options in patients with achalasia.

IX. EVOLUTION AND PROGNOSIS

Normally, the evolution and prognosis vary through underlying causes.

- Dysphagia in stroke patients normalizes gradually from 3 weeks to approximately 6 months or longer.
- Dysphagia due to achalasia or mechanical obstruction may resolve with surgical intervention.
- Dysphagia by foreign body completely recovers after extraction.
- Dysphagia worsens over time, leading to aspiration pneumonia and malnutrition.

X. COMPLICATIONS

- Risk of chronic aspiration and pneumonia (High dysphagia)
- Malnutrition
- Impair the quality of life
- Social isolation due to cough, embarrassment and difficulties of swallowing

XI. PREVENTION AND RECOMMENDATION

- Control risk factors for stroke (e.g., hypertension, diabetes, smoking, atrial fibrillation).
- Avoid smoking and alcohol (risk of esophageal cancer)
- Prevention of Infectious Esophagitis
 - Maintain Good Oral Hygiene
 - Strengthen the Immune System: Healthy lifestyle, healthy food, Avoid stress
 - Stay Hydrated and Avoid Irritants
- Treat Underlying Conditions that might cause dysphagia:
 - Manage neurological disorders (e.g., Parkinson's, stroke, multiple sclerosis) with the help of a doctor or speech therapist.
 - Treat GERD to prevent esophageal strictures.
- Early dysphagia screening (especially in stroke, ICU, and elderly patients).
- Practice Safe eating habits to avoid impaction of food in the esophagus:
 - Take small bites and chew food thoroughly.
 - Eat slowly and avoid talking while eating.
- Prevention of caustic ingestion (accidental or intentional)

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GASTROESOPHAGEAL REFLUX DISEASE

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I. DEFINITION

Gastroesophageal Reflux Disease (GERD) is a condition in which the reflux of gastric contents into the esophagus results in symptoms and/or complication. GERD is objectively defined by the presence of characteristic mucosal injury seen at endoscopy and/or abnormal esophageal acid exposure demonstrated on a reflux monitoring study. GERD is notable for its high prevalence, variety of clinical presentations, under-recognized morbidity, and substantial economic consequences.

GERD is classified based on the appearance of the esophageal mucosa on upper endoscopy:

- Erosive esophagitis: Erosive esophagitis is characterized by endoscopically visible breaks in the distal esophageal mucosa with or without troublesome symptoms of GERD.
- Nonerosive reflux disease: Nonerosive reflux disease or endoscopy negative reflux disease is characterized by the presence of troublesome symptoms of GERD without visible esophageal mucosal injury.

II. EPIDEMIOLOGY

The prevalence of gastroesophageal reflux disease was found to be 10 to 20 percent in the western world and less than 5 percent in Asia. The incidence in the Western world was approximately five per 1000 person-years or 0.5 percent per year.

There are limitations in the epidemiologic estimates of the prevalence of GERD, as they are based upon the assumption that heartburn and/or regurgitation are the only indicators of the disease. However, patients with objective evidence of GERD (e.g., esophagitis or Barrett's esophagus) do not always have heartburn and heartburn is not always sufficiently severe to be indicative of GERD.

III. ETIOLOGY

The etiology of GERD involves a complex interplay of factors:

1. Transient lower esophageal sphincter relaxation
2. Hiatal hernia
3. Impaired esophageal clearance
4. Delayed gastric emptying.
5. Obesity
6. Dietary factors: Fatty foods, caffeine, alcohol, and spicy foods
7. Pregnancy
8. Smoking

IV. PATHOPHYSIOLOGY

The pathophysiology of GERD includes a poorly functioning esophagogastric junction; the anti-reflux barrier composed of the LES and crural diaphragm, coupled with impaired esophageal clearance and alterations in esophageal mucosal integrity. Reflux esophagitis develops when refluxed gastric juice triggers the release of cytokines and chemokines that attract inflammatory cells and that also might contribute to symptom. Other contributors to GERD symptoms may include decreased salivary production, delayed gastric emptying, and esophageal hypersensitivity. As such, GERD can no longer be approached as a single disease, but one with multiple phenotypic presentations and different diagnostic considerations.

V. DIAGNOSIS

The presence of heartburn (reflux) and regurgitation is characteristic of most cases of GERD; however, there is significant variability across studies. It is estimated that heartburn is experienced at least daily in 24% of people with GERD and once or twice weekly in 43%. It is generally accepted that predominance of either heartburn or regurgitation is sufficient for a GERD diagnosis.

Endoscopy investigation can classify GERD pathology as erosive reflux disease (ERD) or non-erosive reflux disease (NERD). Studies indicate that erosion can only be found in up to 25% of people with GERD symptom. Therefore, the other 75% are classified as having NERD. If available, the diagnosis of NERD can be verified by performing gastric pH studies, which can identify excess acid and correlate symptoms with episodes of reflux.

VI. DIFFERENTIAL DIAGNOSIS

- Esophageal cancer
- Infectious esophagitis
- Pill esophagitis.
- Eosinophilic esophagitis

VII. MANAGEMENT

1. Objectives
 - Symptom relief
 - Healing the esophagitis
 - Prevention of complication
 - Improvement in health-related quality of life
 - Prevention of symptom recurrence
2. Critical signs
 - New onset atypical symptoms at age 45-55 years
 - Dysphagia
 - Odynophagia
 - Gastrointestinal tract bleeding
 - Frequent nausea and/or vomiting.
 - Iron-deficiency anemia
 - Epigastric mass
 - Family history of either esophageal or gastric adenocarcinoma

3. Mortality risk factor

Gastroesophageal Reflux Disease (GERD) itself is typically not considered a direct cause of mortality. However, severe complications of GERD, if left untreated or poorly managed, can increase the risk of mortality. Some potential mortality risk factors associated with GERD include:

- Barrett's Esophagus
 - Esophageal cancer
 - Respiratory complication
 - Stricture and bleeding
- ### 4. Outpatient Management

Lifestyle interventions are part of therapy for GERD. Counseling is often provided regarding weight loss, head of bed elevation, tobacco and alcohol cessation, avoidance of late-night meals, and cessation of foods that can potentially aggravate reflux symptoms including caffeine, coffee, chocolate, spicy foods, highly acidic foods such as oranges and tomatoes, and foods with high fat content.

5. First-line treatment

Diet and lifestyle modification

- Eating smaller meals
- Avoiding problematic foods
- Avoiding high-fat foods
- Drinking fluids between and not during meals
- Avoiding lying down, bending/stooping, going to bed soon after meals (within 2-3 hours)
- Avoiding eating before exercise
- Elevating the head of the bed
- Lying on the left side of the body
- Quitting smoking
- Weight loss

Pharmacotherapy of GERD

a. Antacids and alginate

- Antacids neutralize hydrochloric acid leading to increase in pH of the gastric contents.
- Alginate
 - Precipitates to form a gel, which entraps carbon dioxide to develop foam.
 - The foam floats on the stomach contents, displaces the postprandial gastric acid pocket, and physically blocks refluxate from entering the esophagus.
 - Alginates may also coat and protect the esophageal mucosa.

b. Prokinetic agents

Considering GERD is a disorder of intestinal motility and that delayed gastric emptying has been identified in some patients with GERD, it follows that prokinetic agents may be of benefit.

c. Histamine 2 receptor antagonist (H2RA)

H2RA decrease the secretion of acid by inhibiting the histamine 2 receptor on the gastric parietal cells. H2RAs have a slower onset of action, reaching peak concentrations 2.5 hours after dosing, but a significantly longer duration of action of 4 to 10 hours. H2RAs are also more effective in decreasing the frequency and

severity of heartburn symptoms as compared with antacids and placebo.

Table 1: Histamine 2 receptor antagonists

Medication	Low dose	Standard dose
Famotidine	10 mg twice daily	150 mg twice daily
Cimetidine	200 mg twice daily	400 mg twice daily
Nizatidien	75 mg twice daily	150 twice daily

d. Proton Pump Inhibitors (PPIs)

PPIs are the most potent inhibitors of gastric acid secretion by irreversibly binding to and inhibiting the hydrogen-potassium (H-K) ATPase pump. PPIs are most effective when taken 30-60mn before the first meal of the day because the amount of H-K- ATPase present in the parietal cell is greatest after a prolonged fast. PPIs should be administered daily rather than on demand because continuous therapy provided better symptom control, quality of life and higher endoscopic remission rates.

6. Treatment adjustment

a. Partial response to PPI therapy:

For patients with partial response to once-daily PPI therapy, tailored therapy with adjustment of dose timing and/or consideration of twice-daily dosing should be evaluated, especially in patients with nighttime symptoms, variable schedules, or sleep disturbance.

Table 2: Proton Pump Inhibitors (PPIs)

Medication	Low dose	Standard dose
Omeprazole	10mg daily	20 mg daily
Lansoprazole	15 mg daily	30 mg daily
Esomeprazole	10 mg daily	20 mg daily
Pantoprazole	20 mg daily	40 mg daily
Dexlansoprazole	Not available	30 mg daily
Rabeprazole	10 mg daily	20 g daily

b. Non-responders to PPI:

Patients who do not respond to PPI therapy should be referred for further evaluation.

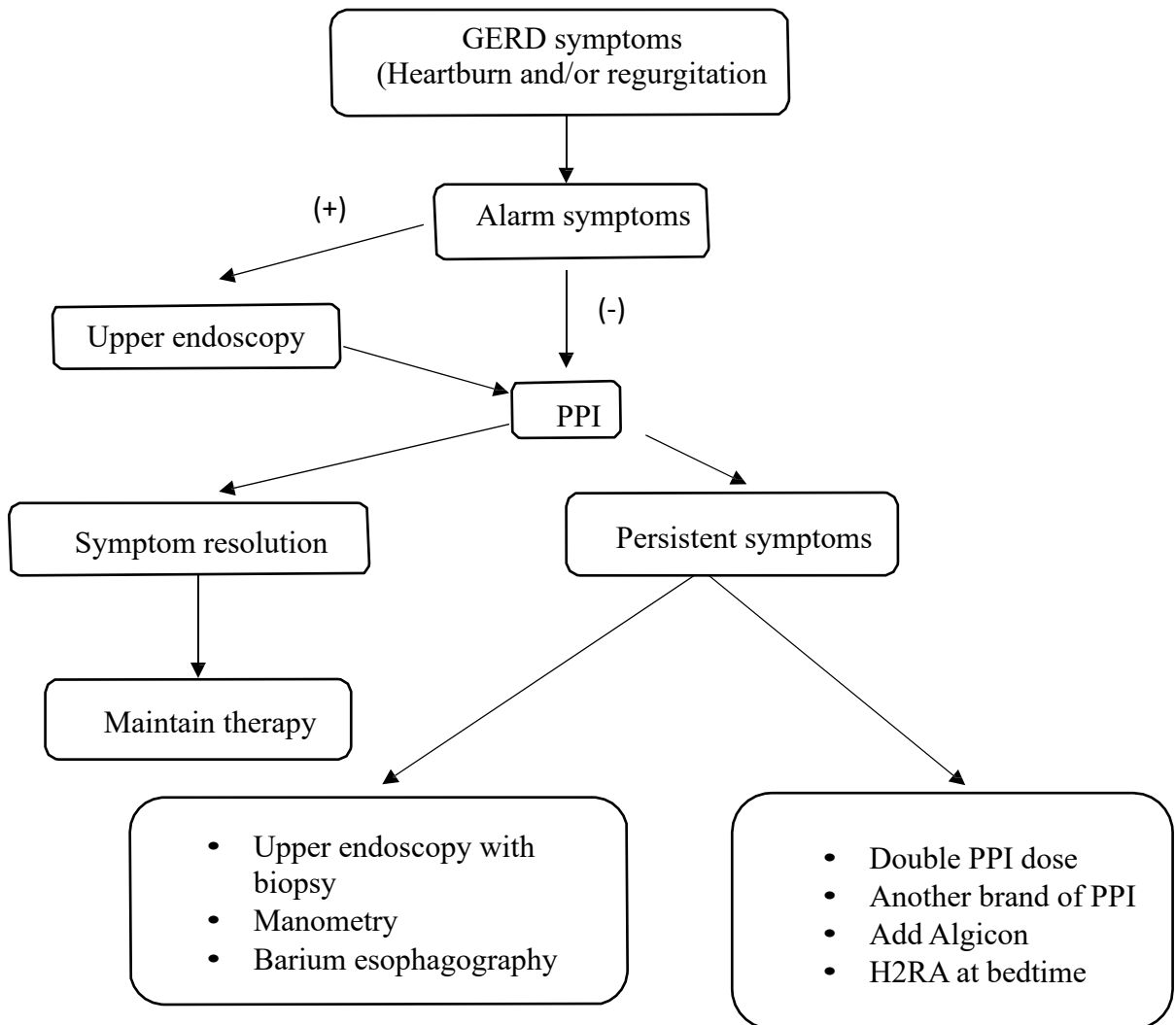
c. Maintenance therapy:

Maintenance PPI therapy should be considered for GERD patients who continue to experience symptoms after discontinuation of PPI therapy, and for patients with complications such as erosive esophagitis and Barrett's esophagus.

d. Switching PPIs:

In cases of partial response to PPI therapy, increasing the dose to twice daily or switching to a different PPI may provide additional symptom relief.

7. Algorithm



VIII. SPECIAL CASES

1. **Pregnancy:** GERD is common during pregnancy due to hormonal changes and the pressure of the growing uterus on the stomach. Management of GERD in pregnant women typically involves lifestyle modifications, such as eating smaller meals, avoiding trigger foods, and elevating the head of the bed. Antacids that are safe for use during pregnancy may be recommended for symptom relief.

2. **GERD in the Elderly:** Older adults may have atypical symptoms of GERD, such as cough, hoarseness, or throat clearing, which can be mistaken for other conditions. Due to age-related changes in the digestive system, older adults may be more prone to GERD. Careful evaluation and management are essential in this population to prevent complications.

IX. EVOLUTION AND PROGNOSIS

The prognosis of GERD is generally good with appropriate management. Up to 90 % of patients with GERD can achieve good symptom control with optimal treatment. However, complications may occur in some cases. It is important to monitor and manage GERD effectively to prevent the progression of the disease and the development of complications. Regular follow-up and adherence to treatment recommendations can help improve the long-term prognosis of GERD.

X. COMPLICATIONS

- Esophagitis
- Esophageal stricture
- Barrett's esophagus
- Esophageal ulcers
- Respiratory complication
- Dental problems
- Esophageal cancer

XI. PREVENTION AND RECOMMENDATION

To help prevent Gastroesophageal Reflux Disease (GERD), consider the following preventive measures:

- a. Maintain a Healthy Weight
- b. Dietary Modifications.
- c. Elevate the head of the bed.
- d. Avoid tight clothing.
- e. Quit smoking.
- f. Limit alcohol and caffeine.
- g. Manage stress.
- h. Medication reviews.

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HELICOBACTER PYLORI INFECTION

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I. DEFINITION

Helicobacter pylori (*H. pylori*) infection refers to the infection of the stomach caused by *H. pylori* (discovered by Dr. Barry J. Marshall and Dr. Robin Warren in 1980s), and is associated with gastritis, gastroduodenal ulcer, and gastric malignancy.

II. EPIDEMIOLOGY

H. pylori is the most common chronic bacterial infection in humans (about 50% of the world population). Infection is more frequent and acquired at an earlier age in resource-limited countries (most children infected before age 10) compared with industrialized nations. The prevalence of *H. pylori* infection in Thailand is around 28.4%. In Vietnam, the seroprevalence of *H. pylori* infection, varying between 50 and 80% in adults.

In Cambodia, a study of seroprevalence of 109 dyspeptic patients conducted by Khun, P. et al. showed that more than 90% of the patients had IgG anti *H. pylori* (+). Moreover, 2 retrospective studies of histological prevalence of *H. pylori* infection (non-published) conducted by Chhut, S.V. and Mak, S. showed that 48% of 2270 patients and 57% of 115 patients were infected with *H. pylori*, respectively.

III. ETIOLOGY

Infection caused by *H. pylori*, a spiral shaped microaerophilic gram (-) bacterium causing gastroduodenal diseases.

IV. PATHOPHYSIOLOGY

H. pylori is transmitted via oral ingestion of the bacterium (saliva, vomitus, feces predominates and water). After *H. pylori* enters the host stomach, four steps are critical for bacteria to establish successful colonization, persistent infection, and disease pathogenesis [Figure 1]:

- Survival in the acidic stomach (produces urease neutralizing gastric acid)
- Movement toward epithelium cells by flagella-mediated motility
- Attachment to host cells by adhesins/receptors interaction
- Causing tissue damage by toxin release

As a result, it can cause acute gastritis, chronic gastritis and then lead to gastric ulcer, gastric cancer, and MALToma [Figure 2].

H. pylori infection is reported to be associated with iron deficiency anemia (IDA). Possible pathogenetic mechanisms involved in IDA is occult blood loss secondary to chronic

erosive gastritis, decreased iron absorption secondary to chronic gastritis of the corpus causing hypo- or achlorhydria and increased iron uptake and use by bacteria .

H. pylori infection can also cause idiopathic thrombocytopenic purpura (ITP). Possible mechanisms include molecular mimicry due to production of autoantibodies against *H. pylori* surface virulence factors (Cag A) and cross reactivity of these antibodies with platelet surface antigens (GP IIb/IIa, GP Ib/IX, and GP Ia/IIa), phagocytic perturbation due to enhanced phagocytic activity of monocytes, enhanced dendritic cell numbers and response, platelets aggravation due to presence of anti-*H-pylori* IgG and von Willebrand factors (vWf) and finally host immune response against *H-pylori* virulent factors CagA and VacA leading to ITP.

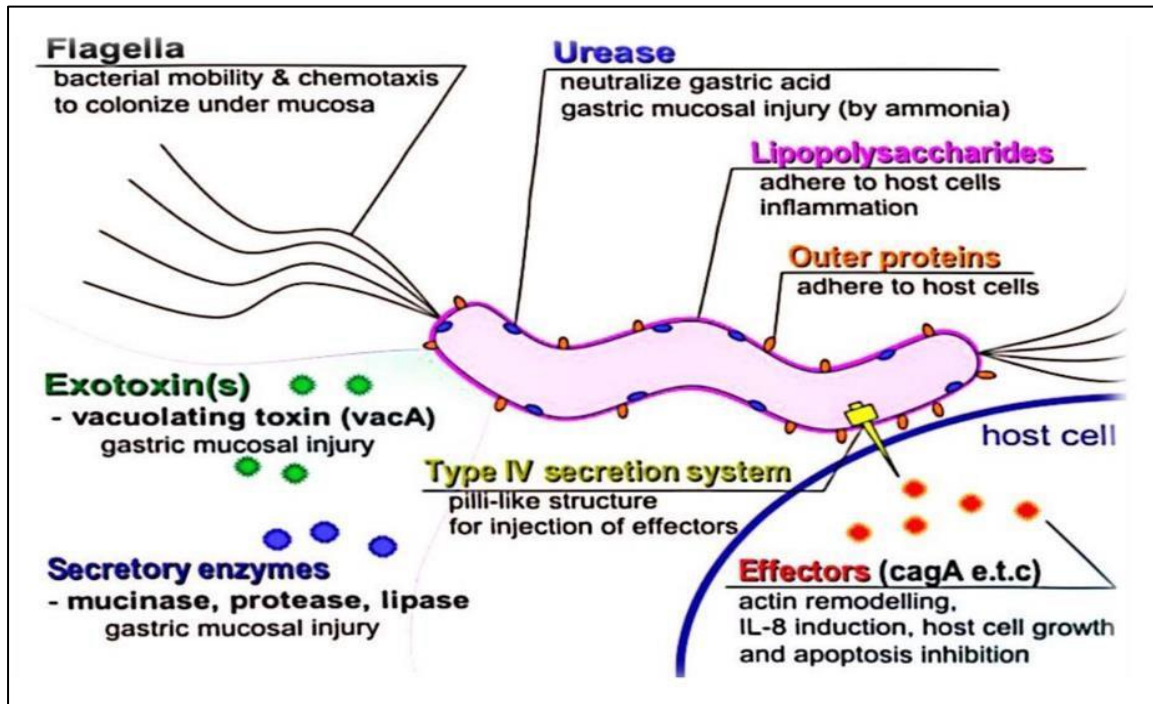


Figure 1: Diagram of *H. pylori* and associated virulence factors

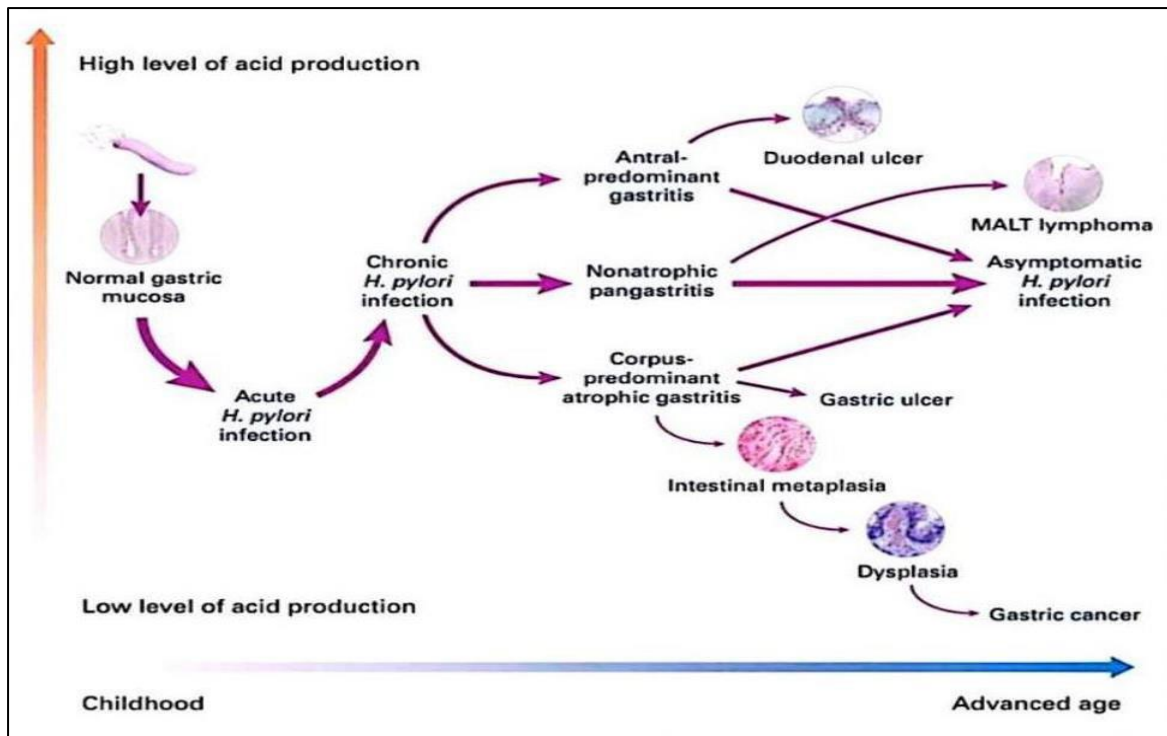


Figure 2: Natural history of *H. pylori* infection

V. DIAGNOSIS

The clinical course of *H. pylori* infection is highly variable and is influenced by both microbial and host factors.

1. Clinical Manifestations:

- Asymptomatic
- Dyspeptic symptoms: epigastric pain or burning, nausea, vomiting, belching, bloating, early satiation.
- Melena or and hematemesis (signs of gastroduodenal ulcer)

2. Investigation:

- There 2 types of tests for *H. pylori* infection
 - Invasive tests (endoscopy): if endoscopy is indicated .
 - Histology (gold standard)
 - Rapid urease testing
 - Culture
 - Polymerase chain reaction
 - Noninvasive tests: if endoscopy is not indicated.
 - Urease breath test (UBT): best option.
 - Fecal or stool antigen test (SAT): alternative.
 - Serological test (ELISA): not recommended for diagnostic test, but it can serve as screening test in specific clinical situations (in which false negative of other tests is high) such as:

- Bleeding peptic ulcers
- Gastric MALT Lymphoma
- Gastric cancer
- Gastric atrophy
- Recent used of antibiotics or PPI
- PPI and bismuth should be stopped for at least 2 weeks and antibiotics for at least 4 weeks before the tests.
- Patients with dyspeptic symptoms, patient with family history of gastric cancer, patients with long term PPI, or long term NSAIDs should be tested for H. pylori infection.
- The choice of diagnostic methods depends on the clinical manifestation, patients' preference, availability, and cost of the test.
- Invasive tests should be indicated in patients with alarm symptoms:
 - Age > 50 years with new onset of dyspepsia
 - Unintentional weight loss
 - Dysphasia
 - Overt GI bleeding
 - Abdominal mass
 - Iron deficient anemia.
 - Previous gastric surgery

VI. DIFFERENTIAL DIAGNOSIS

- Dyspepsia
- Gastroesophageal reflux disease
- Other causes of gastritis (NSAID induced gastritis, stress induced gastritis, autoimmune gastritis...)
- Peptic ulcer disease
- Gastric cancer
- Gastrinoma
- Non-hodgkin lymphoma
- Lactose intolerance
- Celiac disease
- Upper gastrointestinal Crohn disease

VII. MANAGEMENT

1. Objective:
 - To cure infection and to prevent complications (gastric ulcer and cancer).
2. Critical Signs:
 - Upper gastrointestinal bleeding: hematemesis, melena
 - Alter general condition: asthenia, anorexia, and unintentional weight loss.
3. Mortality Risk Factors:
 - Comorbidity: ischemic heart disease, severe chronic kidney disease...

4. Outpatient Management

- H. pylori eradication
- Symptomatic treatment

5. First-Line Treatment:

The selection of recommended first-line treatments varies regionally, geographically, and per individual patient depending on the known or anticipated pattern of antimicrobial resistance (Mahachai et al., 2018).

In Cambodia, prevalence of clarithromycin resistance seems to be high. According to a study conducted by Mak, S. et al. in 115 patients with dyspepsia, showed that the resistance rate of H. pylori to amoxicillin, clarithromycin and metronidazole were 55.6%, 33.3% and 94.4%, respectively. Another study conducted by Aumpan, S. et al. in 58 patients in Koh Kong, also showed that the resistance rate of H. pylori to clarithromycin, metronidazole and fluoroquinolone were 27.8%, 78.6% and 50%, respectively; however, there was no amoxicillin and tetracycline resistance (Aumpan et al., 2020).

6. First line therapy:

- Bismuth quadruple: best regiment
- Concomitant therapy: alternative
- Sequential therapy (lower efficacy than concomitant therapy, but lower side effects): alternative
- Hybrid therapy alternative
- Triple therapy (in case of susceptible to clarithromycin): not recommended.
- Fluoroquinolone triple therapy (in case of susceptible to fluoroquinolone): not recommended.

Salvage therapy:

- Furazolidone quadruple therapy
- Rifabutin triple therapy
- High dose PPI – Amoxicillin dual therapy
- Tailored therapy (based on antibiogram = susceptible tests)

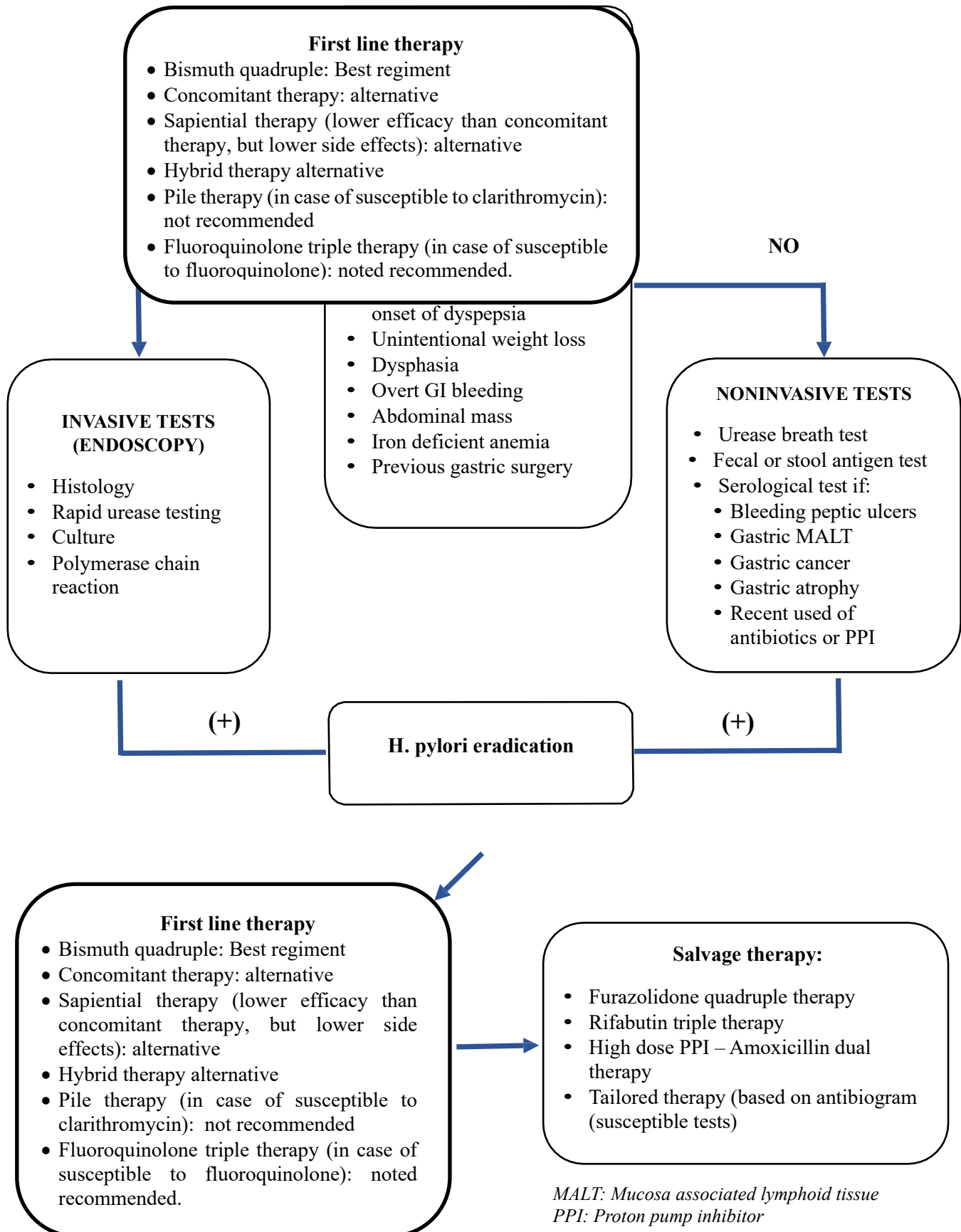
H. PYLORY ERADICATION REGIMENS

Bismuth quadruple (14 days)	• PPI or P-CAB: 1-0-1 (before meals) • Bismuth subsalicylate 524 mg: 1-1-1-1 (after meals) • Tetracycline 500 mg : 1-1-1-1 or Amoxicillin 1g: 1-0-1 (after meals) • Metronidazole 500 mg: 1-1-1 (after meals)
Concomitant therapy (14 days)	• PPI or P-CAB: 1-0-1 (before meals) • Amoxicillin 1g: 1-0-1 (after meals) • Tinidazole or Metronidazole 500mg: 1-0-1 (after meals) • Clarithromycin 500mg: 1-0-1 (after meals)
Sequential therapy (14 days)	First 7 days • PPI or P-CAB: 1-0-1 (before meals) • Amoxicillin 1g: 1-0-1 (after meals) Then for another 7days • PPI or P-CAB: 1-0-1 (before meals) • Tinidazole or Metronidazole 500mg: 1-0-1 (after meals) • Clarithromycin 500mg: 1-0-1 (after meals)
Hybrid therapy (14 days)	First 7 days • PPI or P-CAB: 1-0-1 (before meals) • Amoxicillin 1g: 1-0-1 (after meals) Then for another 7days • PPI or P-CAB: 1-0-1 (before meals) • Amoxicillin 1g: 1-0-1 (after meals) • Tinidazole or Metronidazole 500mg: 1-0-1 (after meals) • Clarithromycin 500mg: 1-0-1 (after meals)
Triple therapy (14 days)	• PPI or P-CAB: 1-0-1 (before meals) • Amoxicillin 1g: 1-0-1 (after meals) • Clarithromycin 500mg 1-0-1 (after meals) • Tinidazole/metronidazole 500mg: 1-0-1 (after meals) can be used to replace amoxicillin or clarithromycin.
Fluoroquinolone triple therapy (14 days)	• PPI or P-CAB: 1-0-1 (before meals) • Amoxicillin 1g: 1-0-1 (after meals) • Levofloxacin 500mg 0-0-1 (after meals)
Furazolidone quadruple therapy (14 days)	• PPI or P-CAB: 1-0-1 (before meals) • Bismuth subsalicylate 524 mg: 1-1-1-1 (after meals) • Tetracycline 500 mg : 1-1-1-1 or Amoxicillin 1g: 1-0-1 (after meals) • Furazolidone 100mg: 1-0-1 (after meals)
Rifabutin triple therapy (14 days)	• PPI or P-CAB: 1-0-1 (before meals) • Amoxicillin 1g: 1-1-1 (after meals) • Rifabutin 150mg : 1-0-0 (after meals)
High dose PPI – Amoxicillin dual therapy (14 days)	• PPI: 1-1-1-1 (before meals) • Amoxicillin 1g: 1-1-1-1 (after meals)

PPI: proton pump inhibitor (omeprazole 40 mg, lansoprazole 60 mg, pantoprazole 20 mg, rabeprazole 20 mg, or esomeprazole 40 mg)

P-CAB: potassium competitive acid-blocker

7. Algorithm



VIII. SPECIAL CASES

- H. pylori infection should be tested and treated in patients with unexplained IDA and in those with ITP.
- H. pylori eradication reverses IDA in patients with asymptomatic gastritis and improves oral iron absorption and induces a significant positive platelet response in a proportion of patients with ITP.

IX. VOLUTION AND PROGNOSIS

- The evolution and prognosis of H. pylori infection is usually good even in patients with complications such as MALT lymphoma.
- The prognosis becomes poor for patients who develop gastric carcinoma.

X. COMPLICATIONS

- Acute or chronic gastritis
- Gastric ulcer
- MALT lymphoma
- Gastric carcinoma

XI. REVENTION AND RECOMMENDATION

H. Pylori infection can be passed from one person to another. Some ways to help keep others safe:

- Avoid touching the saliva, vomit, or bowel movements of anyone who has H. pylori infection.
- Wash your hands with soap and water. This is especially important after going to the bathroom or changing diapers and before cooking or preparing food.
- Only drink water from safe sources. Avoid bathing or swimming in contaminated water.

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NONSTEROIDAL ANTI-INFLAMMATORY DRUGS, LOW-DOSE ASPIRIN AND GASTROINTESTINAL TOXICITY

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I. DEFINITION

Nonsteroidal anti-inflammatory drugs (NSAIDs) is well recognized as causing upper gastrointestinal (GI) complications, ranging from dyspeptic symptoms to life-threatening complicated ulcers (hemorrhage, perforation, or pyloric obstruction).

There are two main types of NSAIDs, Nonselective and Selective. The terms nonselective and selective refer to different NSAIDs ability to inhibit specific types of Cyclooxygenase (COX) enzymes:

- Nonselective NSAIDs – Nonselective NSAIDs inhibit both COX-1 and COX-2 enzymes to a significant degree.
Rank of NSAIDs drugs in Peptic Ulcer:
 - Highest risk: Ketoprofen, Piroxicam, Azapropazone, Tolmetin
 - Moderate risk: Aspirin, Indomethacin, Naproxen, Sulindac
 - Lowest risk: Ibuprofen, Diclofenac
- Selective NSAIDs – Selective NSAIDs only inhibit COX-2 enzyme.

Low-dose Aspirin for cardiovascular protection even at doses as low as 75/day.

The risk of gastroduodenal toxicity is particularly high among 'at-risk patient' (aged > 65 years, previous ulcer or upper gastrointestinal bleeding and users of another antiplatelets or NSAIDs).

II. EPIDEMIOLOGY

In the United States, an estimated 13 million people use NSAIDs on a regular basis. Approximately 70 million prescriptions are written each year, and 30 billion over the counter (OTC) NSAIDs are sold annually.

The risk of hospitalization for serious upper GI complications (bleeding or perforation) from NSAIDs use is 1–2% per year. The prevalence of peptic ulceration in chronic NSAIDs users is 20–30%, with gastric ulcers approximately six times more common than duodenal ulcers.

III. ETIOLOGY

NSAIDs are popular and important for the treatment of inflammation and pain. However, conventional NSAIDs are intrinsically toxic to the gastroduodenal mucosa and even low doses of oral aspirin to prevent cardiovascular risk can also cause the injury.

IV. PATHOPHYSIOLOGY

NSAIDs disrupt the normal gastric mucosal barrier of bicarbonate and hydrophobic mucus by two mechanisms:

- 1) The principal mechanism results from disturbance of prostaglandin synthesis by inhibition of the constitutive COX-1 enzyme.
In the stomach, prostaglandins have a vital protective role, maintaining mucosal blood flow, stimulating secretion of bicarbonate and mucus and regulating mucosal cell turnover and repair.

- 2) Loss of protection by inhibition of prostaglandins renders the stomach vulnerable to damage by gastric acid.

The NSAIDs, including aspirin, cause gastric mucosal damage in two specific ways:

- Systemic inhibition of endogenous mucosal prostaglandin synthesis.
- Direct or topical irritation of the gastric epithelium.

Using it during or after meals cannot prevent gastrointestinal toxicity.

V. DIAGNOSIS

1. Clinical manifestation

- Often with atypical symptoms such as dyspepsia, epigastric burning, atypical localization or irradiation, irregular times, associated with nausea, vomiting,
- Sometime: symptomatic ulcers and/or complications such as GI bleeding,
 - Epigastric pain, without irradiation,
 - Hematemesis and/or melena,

2. Endoscopic finding

- The clear diagnosis is essentially based on esophagogastroduodenoscopy.

VI. DIFFERENTIAL DIAGNOSIS

- Gastroesophageal Reflux Disease
- Cholecystitis
- Pancreatitis
- Myocardial infarction

VII. MANAGEMENT

1. Objectives

- To relieve pain and heal ulcers.
- To prevent recurrences and complications.

2. Critical Sign

- Gastrointestinal bleeding (hematemesis and/or melena/rectal bleeding).
- General peritonitis

3. Mortality risk factor

- Use of multiple NSAIDs; concomitant administration of anticoagulant
- Hepatic-renal dysfunction
- Serious systemic illness

4. Outpatient Management

A- Primary prevention of GI toxicity

*According to 2009 of American College of Gastroenterology, patients taking NSAIDs were classified as being at **high**, **moderate**, or **low risk** for gastroduodenal toxicity:*

- **High risk** was defined as a history of a complicated ulcer or ≥ 3 risk factors.
- **Moderate risk** was defined as the presence of one or two risk factors.
- **Low risk** was defined as none of the four risk factors.

The four risk factors are:

- History of an uncomplicated ulcer
- Age > 65 years
- High-dose NSAID therapy
- Concurrent use of aspirin (including low dose), glucocorticoids, or anticoagulants

Prevention Strategies

Options for reducing the risk of Gastroduodenal lesions include:

- (a) Test and treat *H. pylori* (*See in H. pylori topic*).
H. pylori are an independent and additive risk factor, need to test and treat before using NSAIDs (including Aspirin).
- (b) Use NSAID together with a proton pump inhibitor (PPI) or Potassium-competitive acid blockers (P-CAB) or Misoprostol or
- (c) Use COX-2 inhibitors with or without a PPI or P-CAB.
The use of high-dose H2 blockers is reserved for patients who cannot tolerate PPIs or misoprostol.

Strategies for prevention of NSAIDs related GI Toxicity

1) Low risk

- Use NSAID alone.

2) Moderate risk

- Use NSAID + PPI / P-CAB / Misoprostol or
- COX-2 inhibitor alone

3) High risk

- Alternative therapy if possible, or
- COX-2 inhibitor + PPI / P-CAB / Misoprostol

B- Secondary prevention of GI toxicity

- For patients with symptomatic gastroduodenal diseases or multiple erosions who must continue NSAIDs or low-dose aspirin therapy, or both, we recommend treatment with PPI / P-CAB. We continue PPI / P-CAB therapy for as long as the NSAID and/or aspirin are used.
- For patients who are at high risk for gastroduodenal toxicity on low-dose aspirin for cardiovascular prophylaxis, selecting celecoxib (or another COX-2) plus a PPI / P-CAB may be more effective than using a nonselective NSAID plus PPI / P-CAB in reducing recurrent bleeding ulcers. Nevertheless, recurrent GI bleeding remains a concern in such patients.

5. First line treatment

- The primary prevention of GI toxicity should be tailored as first line treatment.

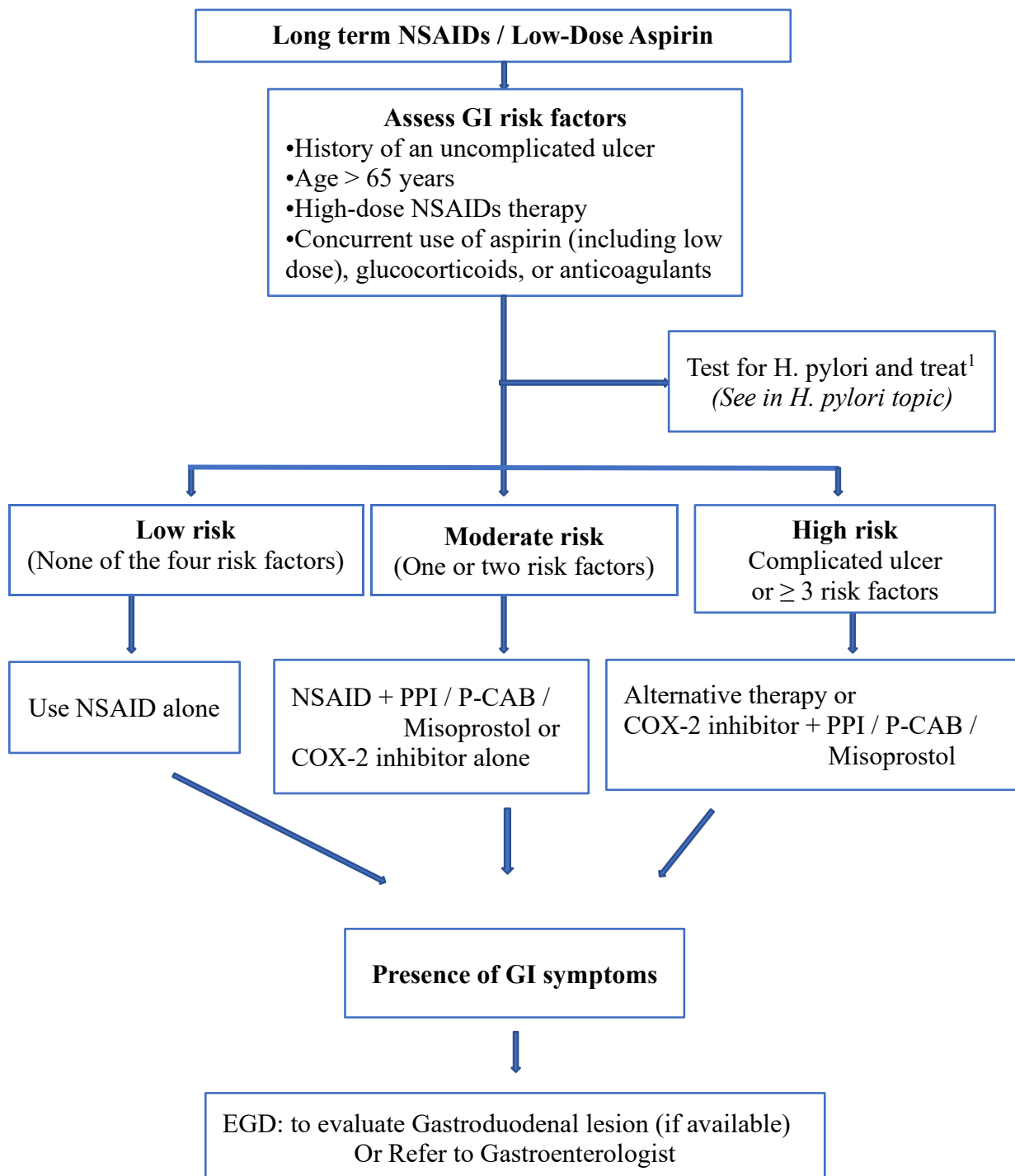
6. Treatment adjustment

- Stop smoking.
- Avoid alcohol consummation.

Follow-up:

- Clinical signs: find complications of GI toxicity (*see in clinical manifestation*)
- Most patients have no symptoms until shortly before the adverse event occurs. Complete Blood Count (CBC) to find unexplained blood loss anemia, iron deficiency, or exhibits overt GI bleeding.

7. Algorithm



¹ H. pylori is an independent and additive risk factor, need to test and treat before using NSAIDs

NSAIDs: Nonsteroidal anti-inflammatory drugs

PPI: Proton pump inhibitor; P-CAB: Potassium-competitive acid blockers;

COX: Cyclooxygenase

GI: Gastrointestinal

EGD: Esophagogastroduodenoscopy

VIII. SPECIAL CASES

1- Use of NSAIDs according to Gastrointestinal and Cardiovascular Risks

Table 1: NSAIDs according to GI and CV risks

CV Risks ¹	Gastrointestinal Risks		
	Low risk	Moderate	High risk
Low CV risk <i>And</i> Moderate CV risk <i>Aspirin Low-dose not required</i>	NSAIDs	NSAIDs + PPI / P-CAB / Misoprostol or COX-2 inhibitor	COX-2 inhibitor + PPI / P-CAB / Misoprostol
High CV risk <i>And</i> Very high CV risk <i>Aspirin Low-dose required</i>	Naproxen + PPI / P-CAB / Misoprostol or low dose COX-2 inhibitor ²	Naproxen + PPI / P-CAB / Misoprostol	Avoid NSAIDs and COX-2 inhibitors if possible; Naproxen + IPP / P-CAB / Misoprostol if CV risk > GI risk Low dose COX-2 inhibitor + PPI / P-CAB / Misoprostol if GI risk > CV risk

CV: Cardiovascular; NSAIDs: Nonsteroidal anti-inflammatory drugs

PPI: Proton pump inhibitor; P-CAB: Potassium-competitive acid blockers; COX: Cyclooxygenase;

GI: Gastrointestinal; EGD: Esophagogastroduodenoscopy

¹ Easy Calculation by App (application program);

e.g. ESC (European Society of Cardiology) or ACC/AHA CV Risk Calculator (2013)

² Low dose of COX-2 inhibitor, for example, celecoxib ≤ 200mg per day. So low dose of COX-2 inhibitors are safe in patients with high CV risk requires new assessment.

2- Use of Low-Dose Aspirin according to Gastrointestinal and Cardiovascular Risks

Table 2: Low-Dose Aspirin according to GI and CV risks

CV Risks ¹	Gastrointestinal Risks		
	Low risk	Moderate risk	High risk
Low CV risk <i>And</i> Moderate CV risk	Aspirin alone	Aspirin + H. pylori eradication	Aspirin + H. pylori eradication + PPI / P-CAB
High CV risk <i>And</i> Very high CV risk <i>Clonidogrel</i> <i>required</i>	Aspirin + Clonidogrel + PPI / P-CAB	Aspirin + Clonidogrel + H. pylori eradication + PPI / P-CAB	Aspirin + Clonidogrel (short duration) + H. pylori eradication + PPI / P-CAB

CV: Cardiovascular; PPI: Proton pump inhibitors; P-CAB: Potassium-competitive acid blockers;

¹ Easy Calculation by App (application program);

e.g. ESC (European Society of Cardiology) or ACC/AHA CV Risk Calculator (2013)

Note: Interactions between antiplatelet agents (Clopidogrel) and Omeprazole remain discussed.

We can give other molecules like Lansoprazole, Pantoprazole, Rabeprazole, and Esomeprazole if we doubt this interaction.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis of gastro-duodenal toxicity by NSAIDs (including Aspirin) according to:

- Terrain (elderly, comorbidities...).
- Multiple NSAIDs; concomitant administration of Anticoagulant.
- Severity of gastro-duodenal lesion.

X. COMPLICATIONS

The complications of nonselective NSAIDs-induced in Gastrointestinal toxicity

Table 3: Nonselective NSAIDs and Gastroduodenal injury

Injury type	Gastro-duodenal lesions	Frequency
Acute (1 to 2 weeks)	Mucosal erythema, superficial erosions, Submucosal hemorrhage,	60 – 100%
Chronic (> 4 weeks)	Gastric erosions and ulcers, Duodenal erosions and ulcers,	5 – 30%

The use of NSAIDs is an important cause of upper gastrointestinal adverse event,

NSAIDs-induced gastrointestinal toxicity can also involve the lower gastrointestinal tract.

- The dose ingested (Aspirin dose from 30 mg can cause gastrointestinal diseases),
- The severity of gastrointestinal toxicity varies considerably, ranging from dyspepsia to bleeding and perforation depending on the terrain (risk factors), the duration of treatment, the combination of different molecules (NSAIDs-NSAIDs, NSAIDs-Aspirin, NSAIDs-corticoids, NSAIDs-anticoagulant, etc.).

XI. PREVENTION AND RECOMMENDATION

The prevention of gastrointestinal toxicity by NSAIDs (including Aspirin), we could inform the population about the risk of NSAIDs (including Aspirin) and avoid self (auto)- medication.

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MANAGEMENT OF PEPTIC ULCER DISEASE

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I. DEFINITION

Peptic ulcer disease is a problem of the gastrointestinal tract characterized by mucosal damage secondary to pepsin and gastric acid secretion. It usually occurs in the stomach and proximal duodenum; less commonly, it occurs in the lower esophagus, the distal duodenum, or the jejunum. Peptic ulcers may present with dyspeptic or other gastrointestinal symptoms or may be initially asymptomatic and then present with complications such as hemorrhage or perforation.

II. EPIDEMIOLOGY

The prevalence of peptic ulcer disease has been reported to be between 4% and 10% in the general population worldwide. Data specific to Cambodia is limited; however, like many developing countries, it is likely that the prevalence of *H. pylori* infection is high. Local studies suggest that gastrointestinal issues, including peptic ulcers, are common due to dietary and lifestyle factors. Socioeconomic status plays a significant role in the prevalence of PUD. In Cambodia, limited access to healthcare, especially in rural areas, may contribute to the delay in diagnosis and treatment of peptic ulcers.

III. ETIOLOGY

The etiology of PUD

- *Helicobacter pylori* infection: *H. pylori* infection is the leading cause of peptic ulcer disease, responsible for about 70-90% of all duodenal ulcers and 60-70% of gastric ulcers.

- Nonsteroidal anti-inflammatory drugs (NSAIDs): NSAIDs are the second most common cause of peptic ulcers, especially gastric ulcers.

- Lifestyle factors

- Smoking
- Alcohol
- Diet: Although diet has a less direct role in PUD etiology, spicy foods, caffeine, and high-fat foods can exacerbate symptoms but are not causative factors.

- Genetic factors

There is evidence suggesting a genetic predisposition to PUD, with a higher

prevalence among first-degree relatives of affected individuals. Blood type O has been associated with an increased risk of duodenal ulcers, possibly due to differences in mucosal glycoprotein structure.

IV. PATHOPHYSIOLOGY

The pathophysiology involves several factors, primarily the disruption of the mucosal defense and repair mechanisms, which leads to damage from gastric acid and pepsin.

- Helicobacter pylori infection:
 - Pathogenesis: H. pylori disrupts the mucous layer, producing inflammation (gastritis) that can lead to mucosal damage. The bacterium's urease enzyme produces ammonia, which neutralizes gastric acid, allowing the bacteria to survive. The chronic inflammation can lead to the breakdown of the mucosal defenses, exposing the epithelium to acid and pepsin.
- Nonsteroidal anti-inflammatory drugs (NSAIDs)
 - Prevalence: NSAIDs are the second most common cause of peptic ulcers, especially gastric ulcers.
 - Pathogenesis: NSAIDs inhibit cyclooxygenase (COX) enzymes, particularly COX-1, which is crucial for the production of prostaglandins that protect the gastric mucosa by stimulating mucus and bicarbonate secretion, promoting mucosal blood flow, and maintaining the integrity of the mucosal lining. The inhibition of these protective mechanisms makes the mucosa more susceptible to damage by gastric acid.
- Lifestyle factors
 - Smoking: Smoking is associated with an increased risk of peptic ulcers and impairs healing, it is thought to increase gastric acid secretion, reduce bicarbonate production, and decrease mucosal blood flow.
 - Alcohol: Chronic alcohol consumption can damage the gastric mucosa, leading to ulceration. However, alcohol is less strongly associated with PUD compared to H. pylori and NSAIDs.

V. DIAGNOSIS

Diagnosis peptic ulcer disease involves several steps and can include a variety of tests to confirm the presence of ulcers and identify potential causes

1. Medical history and symptom assessment:
 - Evaluation of symptoms such as burning epigastric, nausea, bloating or early satiety.
 - Assessment of risk factors, including use of NSAIDs, smoking, alcohol consumption, and family history
2. Physical examination:
 - A physical exam may be conducted to check for abdominal tenderness, distension, or signs of gastrointestinal bleeding like pallor or heme-positive stools.
3. Esophagogastroduodenoscopy:
 - The most definitive diagnostic test for peptic ulcers. An endoscope is used to directly visualize the stomach and duodenal lining, allowing for detection of ulcers and biopsy for malignancy exclusion or H. pylori testing.
4. Tests for Helicobacter pylori:
 - Urea breath test: Measures carbon dioxide in the breath following ingestion of a urea solution labeled with an isotope. Presence of H. pylori is indicated by urease activity breaking down the urea.

- Stool antigen test: Detects H. pylori antigens in the stool, useful for diagnosis and confirming eradication post-treatment
- Serology: Blood tests for antibodies against H. Pylori, though less specific as it cannot distinguish between active and past infection.

5. Laboratory tests: Basics blood tests to check for anemia (indicating possible bleeding) or signs of infection/inflammation

VI. DIFFERENTIAL DIAGNOSIS

- Gastroesophageal reflux disease
- Gastritis
- Gastric and duodenal cancer
- Pancreatitis
- Biliary colic or Cholecystitis
- Functional dyspepsia
- Abdominal aortic aneurysm

VII. MANAGEMENT

1. Objectives
 - Symptom relief: Alleviating pain and discomfort associated with ulcers.
 - Healing of the ulcer: Promoting the healing of the mucosal lining of the stomach or duodenum
 - Reducing recurrence: Preventing the recurrence of ulcers by addressing underlying causes
 - Eradication H. pylori: If the ulcer is related to H. pylori infection, eradicating this bacterium is a primary goal
 - Preventing complication: Avoiding serious complications, such as bleeding or perforation of the ulcer.
2. Critical sign
 - Hematemesis
 - Melena
 - Weight loss
 - Anemia
3. Mortality risk factor
 - Age: Older patients have a higher risk of complications and mortality due to decreased physical reserve
 - Comorbidities: The presence of other health issues such as liver disease, renal failure, or cardiovascular disease can significantly increase the risk
 - Ulcer type: Gastric ulcers tend to have a higher risk of complications compared to duodenal ulcers
 - Bleeding: Patients who experience significant gastrointestinal bleeding have a higher mortality risk
 - Perforation: A perforated ulcer can lead to peritonitis, which is a life-threatening condition

- Chronic NSAID use: Long term use of nonsteroidal anti-inflammatory drugs increases the risk of ulcer formation and complications.

4. Outpatient Management

Lifestyle modifications

- Dietary changes: Avoid food that can irritate the stomach lining, such as spicy foods, acidic foods, and caffeine. Eating smaller, more frequent meals
- Smoking cessation
- Alcohol reduction
- Stress management

5. First-line treatment

The first-line treatment of PUD depends on the underlying cause of the ulcer, particularly whether it is associated with *H. pylori* infection or the use of NSAIDs

a. *H. pylori*-associated ulcers

For ulcers caused by *H. pylori* infection, the first-line treatment is PPI with antibiotic

- Proton Pump Inhibitor (PPI): Standard-dose PPI (Omeprazole, Lansoprazole, Pantoprazole, Esomeprazole, Dex-lansoprazole and Rabeprazole) twice daily
- Antibiotics: Amoxicillin, Metronidazole, Levofloxacin or Bismuth

b. NSAID-induced ulcers

For ulcers related to NSAID use

- PPI: Administered for at least 8 weeks. The exact duration may vary depending on the severity and location of the ulcer.
- Discontinue of NSAIDs: If possible, NSAIDs should be stopped. If NSAD therapy must continue, the patient may need to remain on a PPI or switch to a COX-2 selective inhibitor, which has a lower risk of causing ulcers.

c. Non-*H. pylori*, non-NSAID ulcers

If the ulcer is not associated with *H. pylori* or NSAIDs use

- PPI: Administered for 4-8 weeks, depending on the ulcer's location (duodenal or gastric) and severity.

Table 1: Proton Pump Inhibitors (PPIs)

Medication	Low dose	Standard dose
Omeprazole	10mg daily	20 mg daily
Lansoprazole	15 mg daily	30 mg daily
Esomeprazole	20 mg daily	40 mg daily
Pantoprazole	20 mg daily	40 mg daily
Dexlansoprazole	30 mg daily	60 mg daily
Rabeprazole	10 mg daily	20 mg daily

6. Treatment adjustment

Treatment adjustments for PUD may be necessary based on several factors, such as persistent symptoms, incomplete healing, or complications like bleeding

a. Incomplete healing of ulcers:

- Extended PPI therapy: Continue or increase the dose of PPI for an additional 4-8 weeks.
- Evaluate for Zollinger-Ellison syndrome: If ulcers are resistant to treatment, testing for hypersecretory conditions like Zollinger-Ellison syndrome may be warranted.

b. Recurrent ulcers

- Maintenance therapy: Long-term, low-dose PPI therapy may be necessary, especially in patients with a history of complicated ulcers, ongoing NSAID use, or those who cannot eradicate *H. pylori*.
- *H. pylori* retesting and treatment: Ensure that *H. pylori* eradication was successful and retreat if necessary.
- Review medications: Assess the patient's medication list for NSAIDs or other ulcerogenic drugs and discontinue or replace them if possible.

c. Complicated ulcers

For patients with complications such as bleeding, perforations, or gastric outlet obstruction

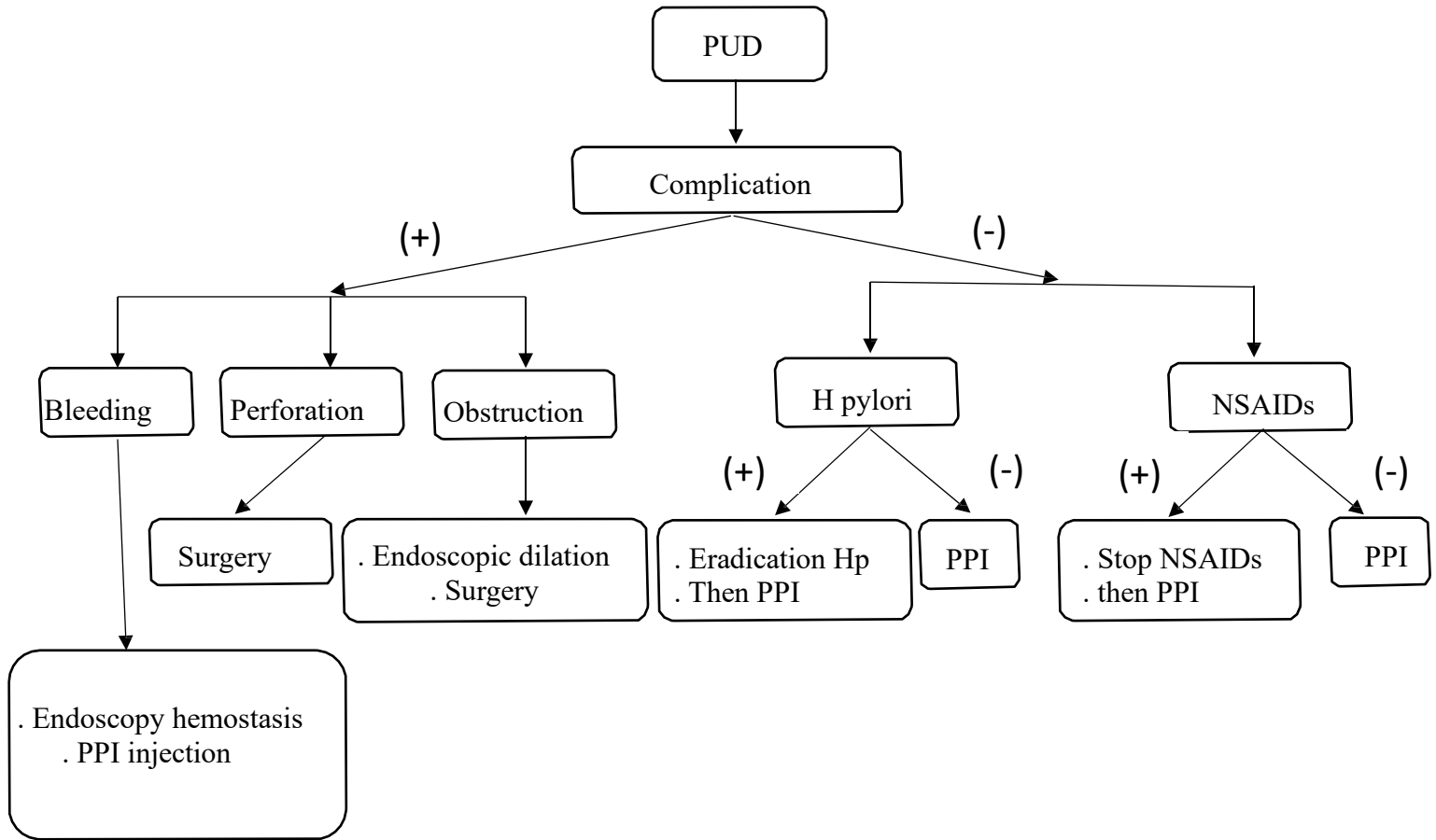
- Hospitalization and IV PPI therapy: High-dose intravenous PPI therapy (e.g., Esomeprazole 80mg IV bolus followed by continuous infusion) is used initially for bleeding ulcers.
- Endoscopic treatment: Endoscopic intervention may be required for hemostasis in bleeding ulcers
- Surgery: In cases of perforation or uncontrolled bleeding, surgical intervention may be necessary.

d. Adjustments for NSAID-induced ulcers

If NSAID therapy cannot be stopped

- Co-therapy with PPIs: Continue PPI therapy indefinitely while NSAIDs are used.
- Switch to COX-2 inhibitors: Use selective COX-2 inhibitors, which are less likely to cause ulcers, along with a PPI.

7. Algorithm



VIII. SPECIAL CASES

1. Giant ulcers:

- Overview: Ulcers larger than 2 cm are classified as giant ulcers and may be more difficult to treat.

- Management:

- Extended PPI therapy: Longer courses of PPI therapy (8-12 weeks or more) may be required to ensure complete healing.
- Endoscopic surveillance: Repeat endoscopy may be necessary to confirm healing and rule out malignancy
- Surgical consideration: In cases of non-healing ulcers or suspicious of malignancy, surgical intervention might be necessary.

2. PUD in the elderly

- Overview: Elderly patients may have atypical presentations and a higher risk of complications like bleeding or perforation.

- Management:

- Careful medication review: Avoid NSAIDs and carefully evaluated the use of anticoagulants.

- Lower-dose PPI: Start with lower doses of PPIs and titrate as needed due to altered pharmacokinetic in older adults.
 - Close monitoring: Regular follow-up to monitor for adverse effects of medications and healing status.
3. PUD in patients on anticoagulants
- Overview: Patients on anticoagulants (e.g., warfarin, direct oral anticoagulants) have a higher risk of bleeding from ulcers.
 - Management:
 - PPI prophylaxis: Strongly consider PPI therapy to reduce the risk of gastrointestinal bleeding.
 - Close monitoring of INR: For patients on warfarin, careful monitoring of INR levels to maintain a safe therapeutic range.
 - Coordinated care: Close collaboration with the patient's cardiologist or primary care physician to balance ulcer prevention with the risk of thromboembolic events.
4. PUD in Pregnancy
- Overview: PUD can occur during pregnancy, though it is less common due to physiological changes that reduce acid production.
 - Management:
 - Lifestyle modifications: Emphasize non-pharmacological measures such as diet changes and stress reduction.
 - Safe medication use: Use antacids and H₂-receptor antagonists (like ranitidine) as first-line treatments. PPIs may be used, if necessary, but the safety profile should be carefully considered.
 - Avoid NSAIDs: NSAIDs should be avoided due to the risk of complications.

IX. EVOLUTION AND PROGNOSIS

The prognosis of PUD has improved dramatically with modern treatment options, but it can depend on several factors:

- Uncomplicated PUD
 - Good prognosis with treatment: Most patients with uncomplicated PUD respond well to medical treatment, particularly when *H.pylori* is eradicated, and recurrence rates are low if the underlying cause is addressed.
 - Risk of recurrence: Recurrence is more likely in patients with persistent *H.pylori* infection, continued NSAID use, or non-compliance with treatment.
- Complicated PUD
 - Bleeding ulcers: The prognosis depends on the severity of bleeding and the patient's overall health. Endoscopic therapy and PPIs have significantly reduced the mortality associated with bleeding ulcers. However, re-bleeding can occur, particularly in high-risk patients.

- Perforated ulcers: Perforation is a serious complication with a high mortality rate, especially in elderly patients or those with comorbid conditions. Prompt surgical intervention is often required.

X. COMPLICATIONS

- Gastrointestinal bleeding
- Perforation
- Gastric outlet obstruction
- Malignancy

XI. PREVENTION AND RECOMMENDATION

Preventing PUD involves several strategies aimed at reducing the risk factors and addressing underlying causes

- Cautious use of NSAIDs
- H. pylori screening
- Avoid smoking
- Limit alcohol consumption
- Stress management
- Regular medical check-ups

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MANAGEMENT OF UPPER GASTROINTESTINAL BLEEDING

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I- DEFINITION

Upper gastrointestinal bleeding (UGIB) is a term used to describe any bleeding that occurs above the ligament of Treitz. Patients with acute UGIB commonly present with hematemesis and/or melena, though patients with large-volume UGIB may also present with hematochezia.

II- EPIDEMIOLOGY

The annual incidence of hospitalization for acute UGIB in the United States is approximately 65 per 100,000 individuals and is more common than lower GIB (LGIB). The hospitalization rate for UGIB is estimated to be six-fold higher than for LGIB. The incidence of UGIB is higher in men than in women (128 versus 65 per 100,000 in one study) and increases with age.

The reported frequencies of specific causes of UGIB vary and have changed over time.

III- ETIOLOGY

UGIB can be caused by several pathologies that affect the upper gastrointestinal tract (GIT) and broadly can be divided into non-variceal and variceal etiologies. Historically, most cases of severe UGIB are related to peptic ulcer disease (PUD) (50–60%) followed by variceal bleeding (4–16%). However, more recent data suggest that the epidemiology is changing with a decreasing incidence of PUD (20–40%) while other causes are being encountered more commonly.

In Cambodia, a retrospective study of UGIB (1,000 cases) by Sidet C in Calmette Hospital from 2002 to 2012, revealed variceal bleeding (VB) 48.6% and PUD 46.3% and by Aurorereaksme TH (Cambodia-China Friendship Preah Kossamak Hospital, 120 cases), revealed VB 48.3% and non-VB 51.6%.

Table 1. Causes of upper gastrointestinal bleeding (UGIB)

Etiology	Approximate prevalence (%)
Peptic ulcer disease (PUD)	40–63
Esophageal/gastric varices (EV/GV)	4–16
Gastritis/duodenitis	18–22
Esophagitis	8–20
Angiodysplasia	4–6
Mallory–Weiss tear	5.0–7.4
Gastric antral vascular ectasias (GAVE)	2.3–4.0
Malignancy	2.6–4
Dieulafoy's lesion	1.5–2.3
Cameron lesion	<1
Hemobilia	<1
Hemosuccus pancreaticus	<1
Aorto-enteric fistula	<1
No lesions identified	10–15

IV- PHYSIOPATHOLOGY

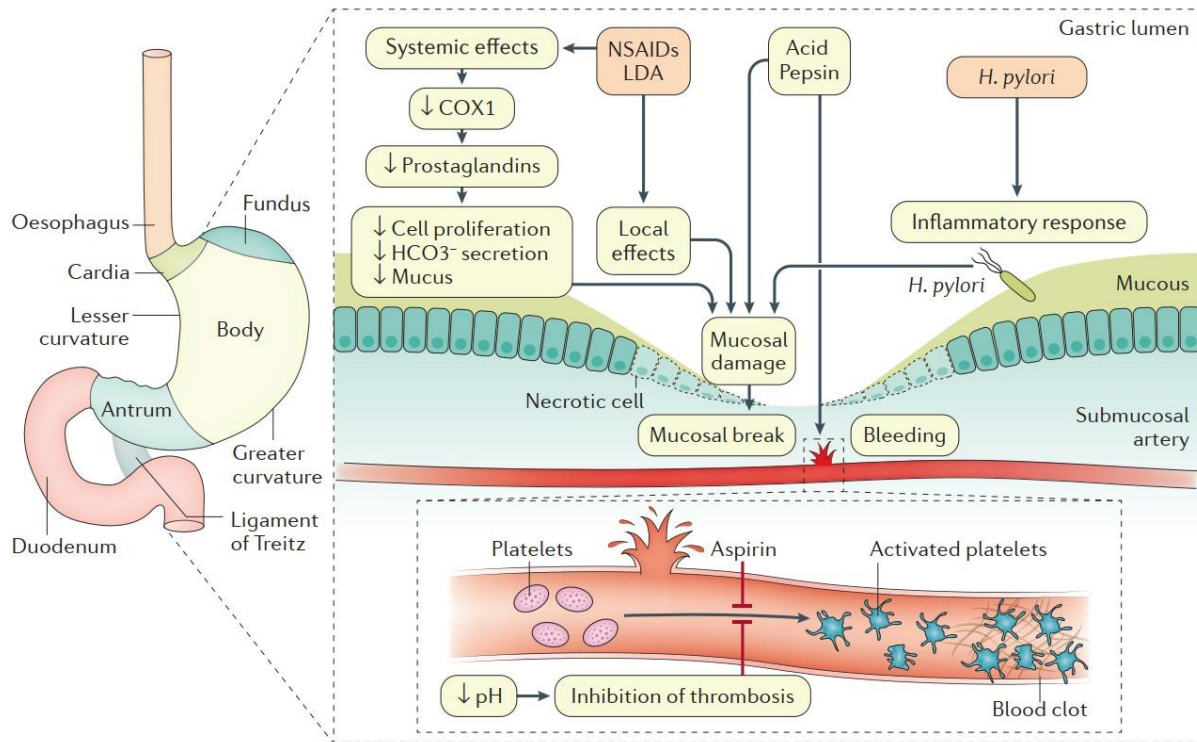


Figure 1. Mechanisms of UGIB induced by NSAIDs, Low dose aspirin or *H. pylori* infection. NSAIDs (including aspirin doses of $\geq 500\text{mg}$) or low-dose aspirin (LDA; which is a dose of $\leq 325\text{mg}$ but is typically 75–100mg) induce a break in the mucosa by acting both locally and systemically, whereas *Helicobacter pylori* damage the epithelial barrier by inducing a cytokine-mediated host immune response. Once the mucosal barrier is disrupted, the underlying blood vessels are exposed to gastric acid and pepsin, which affect blood vessel integrity and can induce bleeding. Platelets and the coagulation system, which should be activated to stop the bleeding, are inhibited by the low pH and pepsin. In addition, platelet cyclooxygenase 1 (COX1) activity suppression by LDA inhibits thromboxane formation and, therefore, platelet aggregation and clot formation. HCO₃⁻, bicarbonate.

The mechanisms involved in the development of NVUGIB are not well understood. The mucosal barrier and submucosal blood vessels must be damaged for the bleeding event to occur. Disruption of the mucosal barrier is commonly associated with *H. pylori* infection and/or NSAID or low dose aspirin use (figure 1), but exposure of the blood vessels to the luminal content can also be the consequence of causes such as Mallory–Weiss syndrome, vascular lesions and neoplastic lesions. Whatever the cause, exposure of the underlying blood vessels to the action of acid and pepsin further erodes the vessel wall and interferes with blood coagulation (Figure 1).

Pathophysiology of PHT (See Figure 2)

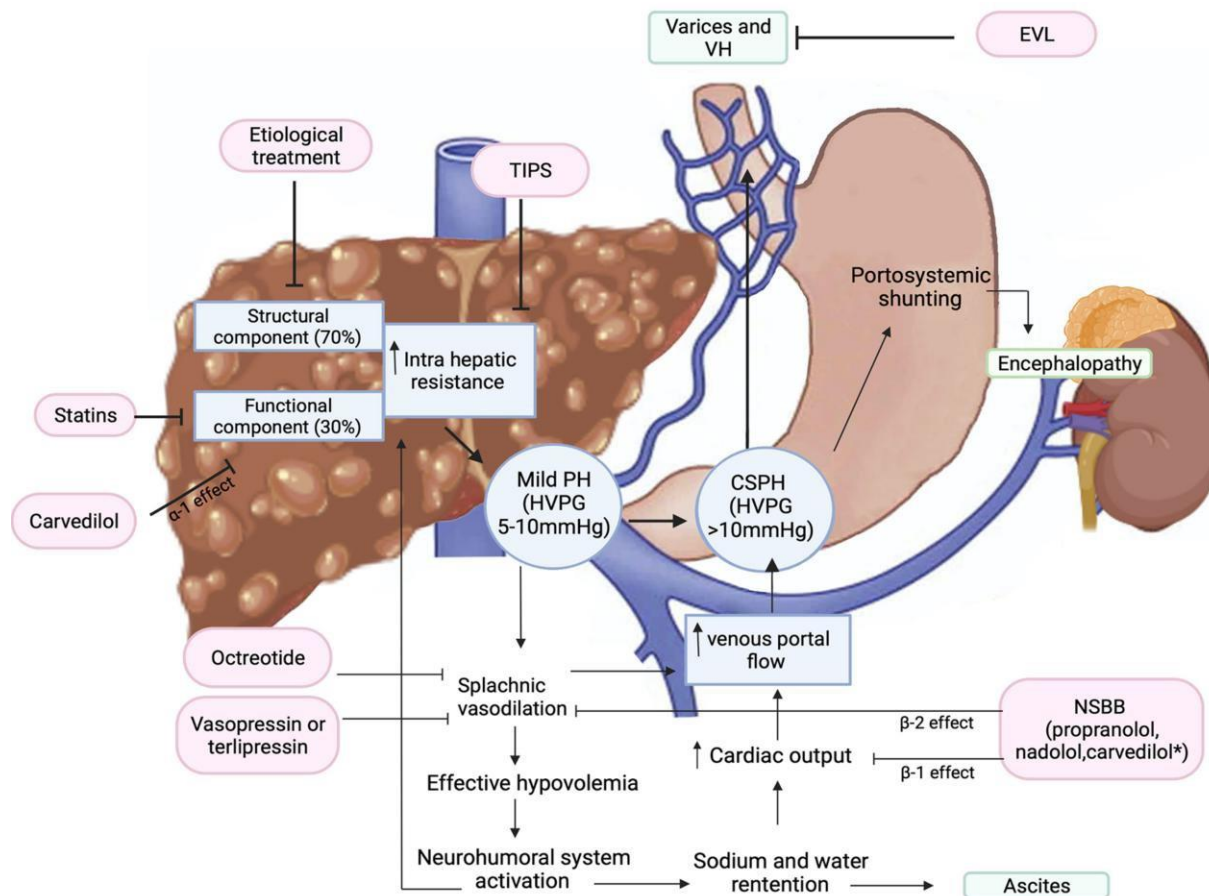


Figure 2. Pathophysiology of portal hypertension and mechanism of action of various therapies used in the management of portal hypertension and variceal hemorrhage.

CSPH: Clinically significant portal hypertension; EVL: endoscopic variceal ligation; HVPg: hepatic venous portal gradient; NSBB: nonselective beta-blockers; PH: portal hypertension; TIPS: transjugular intrahepatic portosystemic shunt; VH: variceal hemorrhage. *Carvedilol has additional α -1 blockade effect.

V- DIAGNOSIS

1. CLINICAL APPROACH

• Clinical feature of UGIB

- Hematemesis: vomiting of blood, which may be obviously red or have an appearance similar to coffee grounds.
- Melena: passage of black, tarry stools, always confirm by digital rectal exam (DRE).
- Hematochezia: passage of fresh blood or maroon blood per anus, usually in or with stools.

• Past medical history

- Prior episodes of UGIB
- GERD
- HBV, HCV, Alcohol, smoking
- Comorbid illnesses
- Medication history (NSAIDs, ...)
- Abdomen surgery/ trauma
- Previous procedures: ERCP, EUS, RFA for HCC

- **Physical examination**

- Clinical findings of cirrhosis
- Clinical findings of cancer (stomach, pancreatic, biliary tract)

Table 2. Causes of UGIB and their Key Features

Cause Key	Clinical Features	Risk Factors
PUD	Hematemesis (bright red or coffee-ground), melena, epigastric pain	H. pylori, NSAIDs, smoking, alcohol, stress
Esophageal Varices	Massive, painless hematemesis, signs of liver disease (jaundice, ascites)	Cirrhosis, portal hypertension, alcohol use
Gastric Varices	Massive bleeding, often with splenomegaly	Portal hypertension (e.g., splenic vein thrombosis), cirrhosis
Mallory-Weiss Tear	Hematemesis after repeated vomiting, self-limited bleeding	Alcohol use, severe vomiting (e.g., pregnancy, bulimia)
Dieulafoy's Lesion	Sudden massive hematemesis, no prior symptoms	Unknown, but associated with hypertension
Erosive Esophagitis	Coffee-ground emesis, GERD symptoms	GERD, NSAIDs, alcohol, infections (e.g., candida, CMV in immunocompromised)
Erosive Gastritis/Duodenitis	Coffee-ground emesis, epigastric pain	NSAIDs, alcohol, severe illness (stress ulcers in ICU patients)
Gastric or Esophageal Cancer	Occult bleeding, IDA, weight loss, early satiety	H. pylori, chronic gastritis, Smoking, alcohol use
Angiodysplasia	Occult or overt GIB, often recurrent	Older age, chronic kidney disease, von Willebrand disease
Aortoenteric Fistula	Massive, life-threatening bleeding, history of aortic aneurysm repair	Prior vascular surgery, aortic aneurysm
Hemobilia	UGIB with jaundice and RUQ pain	Cancers: Primary or metastatic hepatobiliary malignancy, pancreatic cancer, gallbladder cancer Percutaneous liver biopsy, PTC, and PTBD placement. ERCP, ES, EUS Laparoscopic and laparotomic surgeries (cholecystectomy, LT, and pancreaticoduodenectomy) RFA used for early-stage HCC Portal biliopathy, with or without preceding biliary tract intervention. Chronic obstruction of the pancreato-biliary Infectious hemobilia is "tropical hemobilia"
Cameron Lesions	Chronic GIB, IDA	Large hiatal hernia

IDA: Iron-deficiency anemia; PUD: Peptic Ulcer Disease; PTC: Percutaneous transhepatic cholangiography; EUS: Endoscopic ultrasound; RFA: Radiofrequency ablation; LT: liver transplantation; ES: endoscopic sphincterotomy

2. Laboratory tests

It will be required according to clinical features.

- Venous Blood gas can help to assess hemoglobin level quickly
- Full blood count (FBC), Blood group, Cross match, Coagulation screen
- Electrolytes, urea, creatinine
- Glycemia
- Liver function tests (identifying liver disease)
- HBV, HCV

3. Imaging

- Esophagogastroduodenoscopy (EGD): Gold Standard for diagnosis and treatment.
- EKG and heart ultrasound
- Chest X ray
- Abdominal X ray (perforation?)
- Abdominal ultrasound
- Whole body CT scan or CT angiography (CTA)
- MRI of the abdomen if necessary

4. Assessment of severity of UGIB

- Vital signs
- Altered mental Status (AMS)
- Shock Index (SI) > 1 suggests significant instability and possible shock.
- Scores: Glasgow-Blatchford Score (GBS), Rockall score, AIMS65

Table 3. Summary of Critical Features in UGIB

Category	Critical Feature	Significance
Clinical Signs	Massive hematemesis, syncope, altered mental status	Indicates severe bleeding
Hemodynamics	SBP < 90, HR > 100, cold extremities, low urine output	Suggests hypovolemic shock
Labs	Hb < 8, INR > 1.5, platelets < 50k, high lactate	Suggests severe anemia or coagulopathy
Endoscopy	Spurting artery, oozing vessel, visible clot, varices	High risk of rebleeding and mortality
Risk Scores	GBS > 6, Rockall > 8, AIMS65 ≥ 2	Identifies high-mortality patients

VI- DIFFERENTIAL DIAGNOSIS

The differential diagnosis of UGIB:

- Hemoptysis
- Bleeding from Ear, Nose, Throat (ENT)
- Dental bleeding
- Vomiting of some food, fruits (with red color).

VII- MANAGEMENT

1. Objectives

- Stop the bleeding and prevent the recurrence.
- Prevent the complications of bleeding
- Treat the underlying causes

2. Critical Signs

- Hypotension: Systolic blood pressure < 90 mmHg suggests significant blood loss.
- Tachycardia: Heart rate > 100 bpm, a compensatory response to volume loss.
- Signs of Shock/ Orthostatic hypotension
- Shock Index (SI) > 1
- Large volume of bleeding/Recurrent Hematochezia.
- Dyspnea or Chest pain: Particularly in patients with comorbid cardiac conditions.
- Severe abdominal pain: ischemic colitis, perforation, or other serious conditions
- Abdominal distension or Tenderness: bowel obstruction or perforation.
- Confusion, lethargy, or reduced level of consciousness: profound hypovolemia or shock.
- See also Table 3.

3. Mortality risk factors

Overall Mortality Rates:

- Non-variceal UGIB: ~2–10%
- Variceal UGIB: ~15–30%

Table 4. Summary of High Mortality Risk Factors

Category	Risk Factor
Patient Factors	Age >65, cirrhosis, CKD, heart disease, malignancy
Clinical Signs	Shock (hypotension, tachycardia, AMS) Massive Bleeding Rebleeding/Early Recurrent Bleeding Failure of Initial Hemostasis
Lab Markers	Hemoglobin <8 g/dL, INR >1.5, platelets <50k, high lactate
Endoscopy Findings	Active arterial bleeding, large ulcers, varices, malignancy
Scoring Systems	High Glasgow-Blatchford, Rockall, or AIMS65 scores

AMS: altered mental status; CKD: Chronic Kidney Disease

4. Outpatient management

- Hemodynamic stability
- Absence of significant comorbidities
- Glasgow-Blatchford Score (GBS): A GBS ≤ 1
- AIMS65=0/1

- Non-acute bleeding: No active vomiting of blood or passing of large volumes of bright red blood or clots.
- No criteria of High Mortality Risk Factors (table 4)

5. Treatment

Pre-endoscopic treatment

First-line treatments vary based on the bleeding's severity, etiology, and patient stability.

- Systematic approach: The ABCDE approach (Airway, Breathing, Circulation, Disability, Exposure), identify the bleeding source, and administer appropriate treatment.
- Patients who are hemodynamically unstable with active bleeding should be admitted to the medical intensive care unit (MICU)
- Use Restrictive RBC transfusion policy:
 - If Hb ≤ 7 g/dL: target Hgb 7-9g/dL post-transfusion if no CVD.
 - If Hb ≤ 8 g/dL: target Hgb ≥ 10 post-transfusions if known CVD.
- Give plasma for coagulopathy or after transfusing four units of RBCs; give platelets for thrombocytopenia (platelets $<50,000$) or platelet dysfunction (eg, chronic aspirin therapy) or after transfusing four units of RBCs.
- Risk stratification (Clinics, Scores)
- Medications:
 - Proton pump inhibitor (PPI) therapy in case of PUD suspected/ active bleeding (eg, hematemesis, hemodynamic instability)
 - High-dose bolus (eg, esomeprazole 80 mg) then 8mg/h (see algorithm)
 - Somatostatin or somatostatin analogues in case variceal bleeding suspected
 - Octreotide is given as a 50 mcg bolus then 25-50 mcg/h per for two to five days.
 - Prokinetic medications: Erythromycin: 250 mg, infused 30–120 min prior to EGD.
 - The goal: Improve gastric visualization at the time of endoscopy by clearing the stomach of blood, clots, and food residue.
 - Patients receiving erythromycin need to be monitored for QTc prolongation.
- Nasogastric/orogastric tube aspiration and lavage: Not recommend as the routine use in patients presenting with acute UGIB.
- Monitoring: Signs of non-response of treatment (pulses, blood pressor, AMS), complications of GIB, current disease deteriorated or rebleeding in order to justify treatment or change treatment strategies or require MDT.

Endoscopic treatment (in specialized center)

- Timing of EGD
 - Urgent: ≤ 12 hours
 - Early: ≤ 24 hours
 - Delayed: > 24 hours
 - If hemodynamic stable, early EGD should be performed.
 - Unstable patients: urgent EGD is not recommend since as compared to early EGD, patient outcomes are not improved, and additionally this may be associated with worse outcomes.
- **Endoscopic hemostasis for peptic ulcer bleeding (PUB):** combined therapy is recommended.
 - Injection therapy

- Several agents can be used including diluted epinephrine, sclerosing agents (e.g. ethanol and ethanolamine), and tissue adhesives (e.g. thrombin and cyanoacrylate). Diluted epinephrine (1:10,000 or 1:20,000) is the most commonly used and works mostly by causing tissue tamponade and partly by vasoconstriction.
- Thermal coagulation
 - Contact thermal therapy: Thermal therapies achieve hemostasis by generating heat that leads to coagulation of tissue and contraction of the blood vessels. Several contact thermal devices are available including bipolar electrocoagulation, heater probes, and soft monopolar electrocoagulation. The current recommendation is to use the large 3.2-mm probe with firm pressure over the vessel and the application of heat energy for 8–10 seconds.
 - Non-contact thermal therapy: Argon plasma coagulation (APC) is a non-contact thermal modality that produces a superficial tissue coagulation (1–2 mm). It is frequently used to manage superficial vascular lesions such as angiodysplasia and GAVE (see other causes of UGIB section).
- Endoscopic clips
 - Through-the-scope clips: TTS clips are typically applied at the bleeding site, resulting in mechanical compression, achieving hemostasis. As discussed previously, this mode of hemostasis can be combined with epinephrine injection, achieving superior outcomes compared with epinephrine monotherapy.
 - OTSC: The OTSC (or cap-mounted clip) technology has increasingly been used for many different endoscopic therapeutic indications. It has been shown to be efficacious as rescue compared to standard therapy (TTS clips, principally) in decreasing persistent bleeding among patients presenting with high-risk endoscopic bleeding ulcer lesions who have experienced recurrent bleeding.
- Alternative approaches
 - Topical hemostatic agents: The most commonly studied agent is TC-325 (Hemospray VR, Cook Medical, Bloomington, USA), which is a biologically, silica-based, inert powder that creates a mechanical barrier over bleeding sites when it comes in contact with moisture in the GIT.
- **Endoscopic hemostasis for esophageal/gastric variceal bleeding (EVB, GVB)**
 - Endoscopic variceal ligation or endoscopic band ligation (EBL) for EVB.
 - Glue injection – Endoscopic sclerotherapy using *n*-butyl-2-cyanoacrylate (glue) is a safe and effective hemostatic method for GVB.
- **Balloon tamponade tubes:** effective as a temporizing measure in cataclysmic bleeding or endoscopic hemostatic failure (Sengstaken-Blakemore tube for EVB or Linton for GVB)
- **Second-look endoscopy**
 - Second-look endoscopy refers to the practice of performing a planned follow-up endoscopy, generally within 24 h of the initial endoscopy. Situations that might warrant a second-look endoscopy:
 - If visualization during the initial endoscopy was limited by blood or debris
 - If there is concern on the part of the endoscopist/instruments that the prior endoscopic hemostasis was suboptimal.

Post-endoscopic treatment

- **Post-endoscopic treatment of PUB**
 - PPI therapy: iv bolus followed by continuous infusion (e. g., 80 mg iv bolus then 8 mg/h) for 72 h post endoscopy (see algorithm)
 - High dose PPI therapies given as iv bolus, bid or PO, bid can be considered as alternative regimens.
 - Management of recurrent bleeding: → Repeat EGD with hemostasis if indicated.
 - If failure of second EGD: → Transcatheter angiographic embolization (TAE) should be considered.
 - Surgery if TAE not available or after failed TAE.
- **Post-endoscopic treatment of EVB/GVB**
 - Patient who has undergone EBL for acute EVB should be scheduled for EBLs at 1 to 4 weekly intervals to eradicate EV (secondary prophylaxis).
 - NSBBs (Carvedilol or Propranolol) in combination with endoscopic therapy for secondary prophylaxis of variceal bleeding (VB).

6. Treatment adjustment

It varies through clinical setting, severity, underlying causes and comorbidities.

- PUB: look for and treat H. pylori infection; NSAIDs use (→ avoid/alternative)
- Variceal bleeding (VB):
 - look for and prevent or treat other complications of cirrhosis (ascites, hepatic encephalopathy (HE), hepatorenal syndrome, ...)
 - Lactulose to prevent HE.
 - Antibiotic 7 days (ceftriaxone 1g/j or ciprofloxacin iv 200mg bid; or PO 400mg bid) to prevent infection especially spontaneous bacterial peritonitis (risk factor of rebleeding)
 - PPI (iv/PO standard dose) to prevent EBL-induced ulcer (risk of rebleeding)
 - Chronic GIB => IDA: → treat cause, Iron supplements.
- Monitoring

7. Algorithm (See Figure 3).

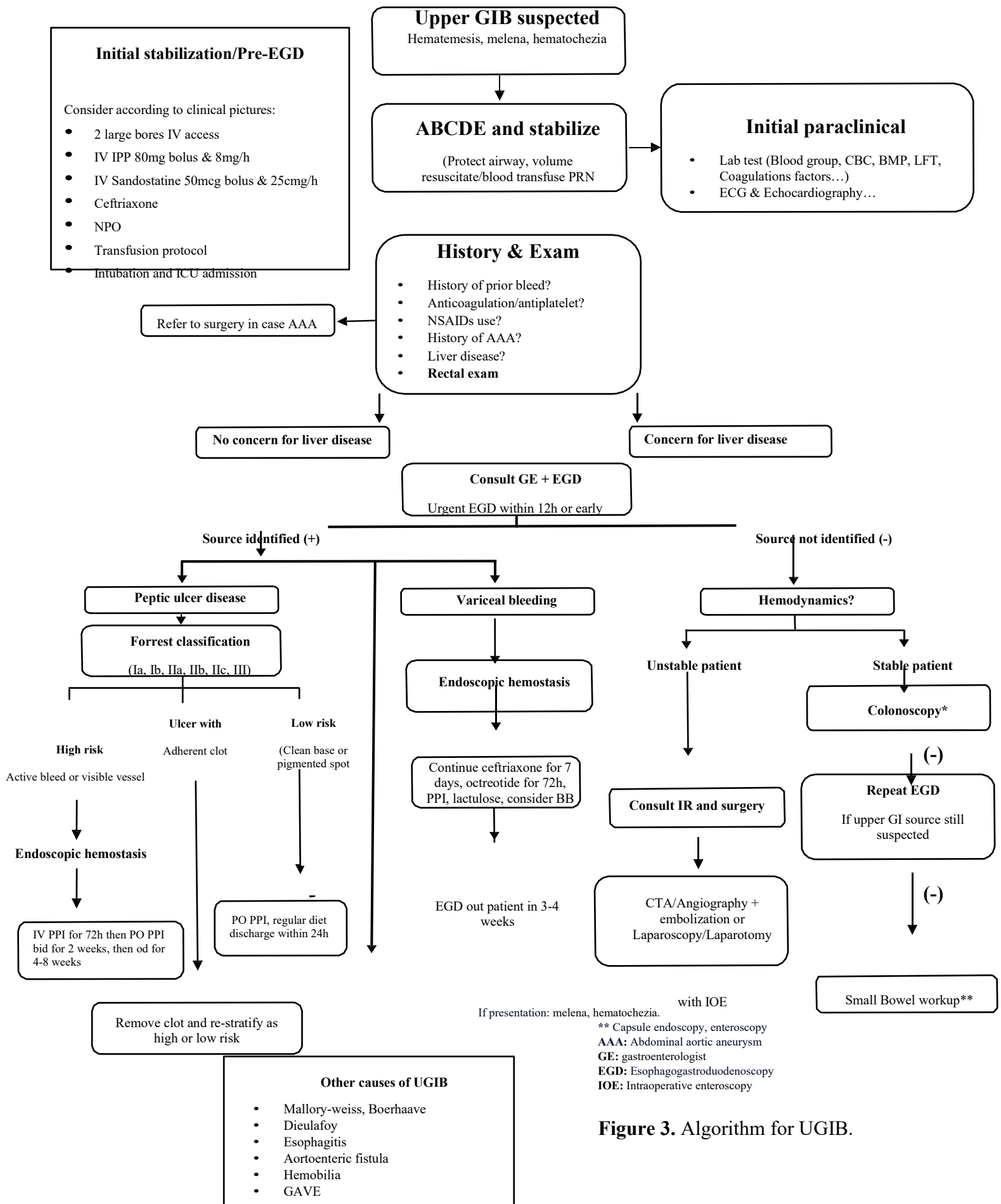


Figure 3. Algorithm for UGIB.

VIII- SPECIAL CASES

1. **UGIB in patients with antithrombotic drugs** (see Ch. Lower gastrointestinal bleeding)
2. **UGIB on old patients/concomitant NSAIDs-steroid** with sudden severe abdominal pain (peritonitis): → Rule out perforations (Abdominal Rx/ CT)
3. **Hematemesis + normal EGD:** → Consider Dieulafoy's lesion
 - Dieulafoy's lesion is a rare but potentially life-threatening cause of UGIB caused by an abnormally large, dilated artery within the submucosa that erodes through the mucosa, leading to sudden and severe bleeding.
 - Endoscopic Detection Rate: missed (10-30%), due to:
 - Intermittent bleeding (lesion may not be actively bleeding at the time of endoscopy).
 - Small mucosal defect (~2–5 mm), making visualization difficult.
 - Blood clots obscuring the lesion, requiring careful irrigation and a second-look.
 - Factors increasing Detection:
 - Urgent endoscopy (<12–24 h) improves detection.
 - Cap-assisted endoscopy or magnification endoscopy may enhance visualization.
 - Adjunctive imaging (CTA, angiography, or capsule endoscopy) can help if endoscopy fails to identify the source.

IX- EVOLUTION AND PROGNOSIS

It depends on the cause, severity, response to treatment, and underlying comorbidities.

Table 5. Summary of Prognosis in UGIB

Factor	Good Prognosis	Poor Prognosis
Cause	Peptic ulcer (treated), mild erosions	Varices, malignancy, large ulcers
Clinical Stability	No shock, stable vitals	Hypotension, tachycardia, altered mental status
Endoscopic Findings	Clean ulcer base, no active bleeding	Spurting artery, varices, tumor bleeding
Response to Treatment	Hemostasis achieved, no rebleeding	Rebleeding despite therapy
Comorbidities	None or well-controlled conditions	Cirrhosis, CKD, malignancy, heart disease

X- COMPLICATIONS

The UGIB can lead to serious complications (acute/chronic) if not promptly managed.

- Hemodynamic Shock (Hypovolemic Shock) – Life-Threatening
- Rebleeding (10–30% of cases)
- Aspiration Pneumonia
- Perforation (Rare but Severe – especially in Large Ulcers)
- Anemia (IDA) – Common in chronic UGIB

- Infection & Sepsis (Common in Cirrhosis Patients with Variceal Bleeding)
- Multi-Organ Failure (MOF) in Severe Cases.

Table 6. Summary of Key Complications of UGIB

Category	Complication	Risk Factors	Management
Immediate	Hypovolemic shock	Massive bleeding, cirrhosis, delayed resuscitation	IV fluids, blood transfusion, EGD. ABCDE bundle
	Rebleeding	Spurting ulcers, varices, coagulopathy	Repeat EGD, angiography, surgery Prevention: ABT in case of VB
	Aspiration pneumonia	Hematemesis, AMS	Airway protection, ABT
	Perforation	Large ulcers, NSAID	Surgery, ABT
Delayed	IDA	Chronic slow bleeding	Iron supplements, treat cause
	Infection (SBP, sepsis)	Cirrhosis, prolonged hospitalization	IV ABT, prophylaxis
	MOF	Persistent shock, comorbidities	ICU care, organ support

AMS: altered mental status; ABT: Antibiotic therapy; IDA: Iron deficiency anemia

XI- PREVENTION AND RECOMMENDATION

Preventing UGIB focuses on reducing risk factors, managing underlying conditions, and using gastric protection in high-risk patients.

To avoid cirrhosis (PHT => EV, GV):

- Healthy lifestyle (food, exercise) to avoid metabolic syndrome or NAFLD/MASLD.
- Avoid alcohol intake, smoking.
- Hepatitis B, HCV: understanding mode of transmission, Vaccination (for HBV), treatment

Table 7. Summary of UGIB Prevention Strategies

Category	Prevention Strategy
Peptic Ulcers	PPI therapy, treat GERD, H. pylori eradication, NSAID modification*
Variceal Bleeding	Beta-blockers, EBL, TIPS in high risk
ICU/Stress Ulcers	IV PPIs, H2 blockers for high-risk ICU patients
Anticoagulation-Related UGIB	PPIs with NSAIDs/aspirin, careful anticoagulant use

(*See Ch. Nonsteroidal anti-inflammatory drugs, low-dose aspirin and gastrointestinal toxicity)

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MANAGEMENT OF LOWER GASTROINTESTINAL BLEEDING

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I. DEFINITION

Gastrointestinal bleeding (GIB) is a term used to describe any bleeding that occurs within the GI tract from mouth to the anus. Any bleeding that occurs at the proximal to the ligament of Treitz is upper gastrointestinal bleeding (UGIB) whereas at the distal (jejunum, ileum, colon and rectum) is lower gastrointestinal bleeding (LGIB). Categorization into one of the two groups is important as it is different in epidemiology, management, and prognosis.

II. EPIDEMIOLOGY

LGIB encompasses a wide clinical spectrum ranging from trivial bleeding to massive hemorrhage with shock. It is approximately one-fifth as common as UGIB and accounts for approximately 20.5 to 27 hospitalizations per 100,000 adults per year, in the United States. The incidence increases substantially with age, presumably due to the high incidence of diverticulosis and vascular disease in this group. Prognosis in LGIB varies. However, since most acute LGIB is self-limited, outcomes are typically favorable. Indeed, the mortality associated with LGIB is generally considered to be less than 5% and when it occurs, is often a result of comorbid conditions.

III. ETIOLOGY

Causes of LGIB:

- Proctological (during stools) [4], (5-20%): internal hemorrhoids and anal fissure.
- Non proctological (non during stools): abundance of three types: massive, moderate, and occult bleeding.

The sources of acute LGIB not related to proctological diseases and their frequency are listed:

Table 1. Sources of hematochezia reported in the literature

Source/finding	Frequency (%)
Diverticular bleeding	17-40
Angiodysplasia	9-21
Colitis (ischemic, infectious, chronic IBD, radiation injury)	2-30
Neoplasia, post-polypectomy bleeding	11-14
Anorectal disease (including rectal varices)	4-10
Upper gastrointestinal bleeding (UGIB)	0-11
Small bowel bleeding	2-19
Stercoral ulceration	0-5
Radiation proctopathy	0-2
NSAID-induced colopathy	0-2
Dieulafoy's lesion	Rare

IBD : Irritable Bowel Syndrome ; NSAID : Non-steroidal anti-inflammatory drug

IV. PATHOPHYSIOLOGY

The causes of LGIB may be grouped into categories based on the pathophysiology: anatomic, vascular, inflammatory and neoplastic.

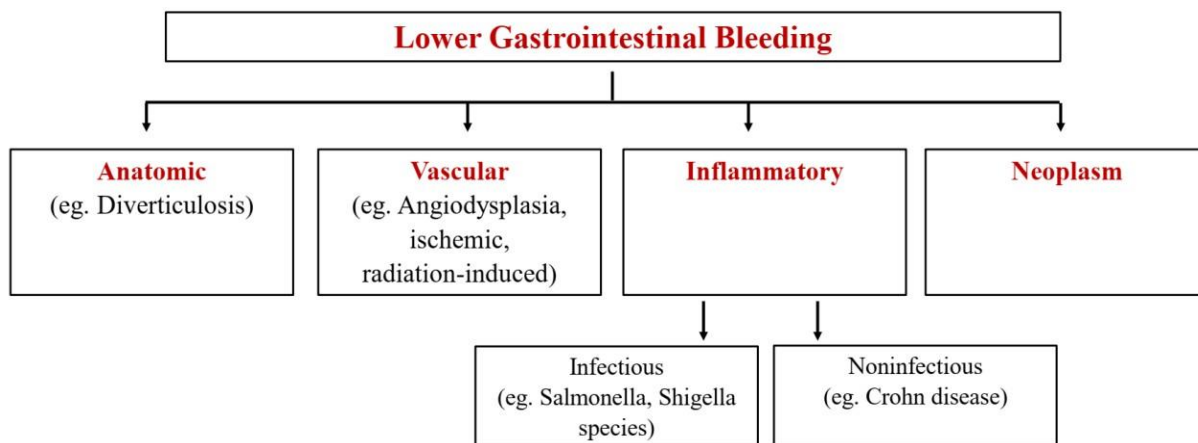


Figure 1. Pathophysiology of Lower Gastrointestinal Bleeding (LGIB).

Diverticular bleeding

Mucosal injury and vessel exposure: Factors such as mechanical trauma (e.g., hard stool) or chemical causes or inflammation can cause erosion or ulceration of the mucosa within the diverticulum. This erosion may expose the underlying blood vessels.

Gastrointestinal Angiodysplasias (GIADs)

Pathophysiology: proposed risk factors and associations:

Impaired arterial flow, intermittent submucosal venous obstruction due to increased contractility at the level of muscularis propria, and local hypoxemia that leads to vascular degeneration and formation of small arterio-venous collaterals.

Gastro-intestinal ischemic injuries

Vascular occlusion > Splanchnic hypoperfusion, Angiotensin activation > Intestinal hypoxia > Intestinal barrier injury, Translocation, Local inflammatory pathway > Systemic inflammatory pathway > Necrosis, Organ failures

Infectious colitis causes GIB, due to pathogenic microorganisms:

- Mucosal Inflammation and Ulceration: Shigella, Salmonella, Campylobacter, and certain strains of Escherichia coli (e.g., EHEC)
- Toxin Production: Clostridioides difficile releases toxins A and B (PMC)
- Vascular Injury: Inflammation can extend to the submucosal layers, affecting blood vessels and increasing their fragility.
- Immune-mediated damage: The host's immune response to infection involves the release of cytokines and recruitment of inflammatory cells. While aimed at eliminating pathogens, this response can inadvertently damage the colonic epithelium, leading to ulceration and bleeding.

V. DIAGNOSIS

1- Clinical approach

Patients presenting with LGIB can have varying symptoms and signs. Therefore, a thorough history and physical examination are hallmarks to make a diagnosis.

- Time of onset, character of the bleeding (e.g. hematochezia, melena),
- Associated symptoms: abdominal pain, fever
- Previous similar bleeding events
- Past medical history [comorbid medical conditions, prior intra-abdominal surgeries (e.g. vascular surgery/stenting)] and current medications (antithrombotic agents, NSAIDs).
- Cardiopulmonary examination and abdominal palpation: any focal tenderness, guarding, rebound, and/or mass.
- Digital rectal examination: looking for any anorectal lesions, fresh blood or melena.
- The family history of colorectal cancer/polyps or IBD.

Table 2. Key Clinical scenarios for the causes of LGIB.

Diagnosis	Distinguishing features
Diverticular bleeding	Acute, severe, painless bleeding in the setting of known or suspected diverticular disease.
Angiodysplasia	Recurrent, painless bleeding episodes; can be chronic, leading to iron deficiency anemia (IDA).
Colitis:	
<i>Ischemic colitis</i>	Self-limited, bloody diarrhea followed by acute lower abdominal pain in patients with cardiac risk factors
<i>Infectious colitis</i>	Bloody diarrhea with fever and high-risk diet or previous antibiotic use [pseudomembranous colitis (PMC)].
<i>IBD</i>	Bloody diarrhea associated with recurrent abdominal pain and weight loss
Colon cancer	Slow, chronic blood loss with change in bowel habit or IDA.
Post-polypectomy or post-endoscopic biopsy bleeding	Self-limited bleeding occurring within 30 days of a previous polypectomy or biopsy.
Hemorrhoids	Bleeding associated with bowel movements and anal pruritus; usually painless, but pain may be present with thrombosed hemorrhoids.
UGIB	Severe hematochezia: Decompensated liver disease or prior PUD along with epigastric discomfort on examination. BUN/Cr>30, aspirin/NSAID use, and positive NG lavage.

BUN, blood urea nitrogen; Cr, creatinine serum; PUD, peptic ulcer disease; NG, nasogastric

2- Investigations

The objectives are for urgent treatment, causes, appropriate treatment and prognosis.

- Blood tests: CBC, ABO group and Rh type, prothrombin time, INR, glycemia, electrolytes, renal function, liver profile, C-reactive protein (CRP).
- EKG, Echocardiography

- Colonoscopy, abdominal scan, and/ or angiography (based on clinical findings).
- Oakland score (clinical hematological score): The Score (table 4) is a risk assessment tool to predict safe discharge for LGIB. It is recommended for LGIB risk-stratification by World Society Guidelines (ACG, ESGE and ESG). It should be used to supplement but not replace clinician judgment.

Colonoscopy

It is the diagnosis test of choice, excellent safety profile; high diagnostic (48-90%) and therapeutic yield, so can provide a definitive diagnosis even if active bleeding has stopped.

Condition for the procedure:

- Patients hemodynamically stable
- After colon preparation with 4 liters of polyethylene glycol (PEG) solution over 3-4 h.

Timing of colonoscopy: Urgent or early (< 24 h) and elective colonoscopy (24–96 h).

The optimum time to perform colonoscopy for acute LGIB remains uncertain.

By ACG Guideline 2023, there is no clinical benefit (rebleeding, diagnostic yield, blood transfusion, length of stay or mortality) of urgent as opposed to elective colonoscopy in the setting of LGIB. However, to minimize length of stay, if an inpatient colonoscopy is planned, it should be performed at the next available nonurgent opportunity.

Esophagogastroduodenoscopy (EGD)

EGD is recommended in patients presenting with acute LGIB (severe or massive hematochezia) and hemodynamic instability unless CTA has already been performed showing a definitive bleeding source in the LGI tract.

CT Angiography (CTA)

In patients presenting with acute hematochezia and hemodynamically unresponsive to fluid resuscitation or who are unable to receive preparation for colonoscopy, initial evaluation with CTA is recommended. CTA provides the fastest and least invasive means to localize the site of blood loss (sensitivity of 79–95% and specificity of 95–100%).

Positive CTA (extravasation): Refer to interventional radiology for transcatheter arteriography and possible embolization (TAE) or GI endoscopy centers for endoscopic hemostasis.

Contrast enhanced CT (CECT):

CECT is often used as a first-line imaging study for LGIB, especially in hemodynamically stable patients. In cases of positive findings, interventional radiology (angiography and embolization) or surgery may follow.

Intraoperative Enteroscopy (IOE)

Video capsule endoscopy (VCE):

In cases of LGIB excluded by repeated endoscopy and contrast enhanced CT abdomen, consider:

- Stable patient: VCE for surgery with or without IOE.
- Unstable patient: CTA for TAE. If it fails, perform surgery with or without IOE.

3- Assessment for the severity of LGIB

There a lot of scores to predict the adverse outcomes of LGIB.

Risk stratification scores to predict adverse outcomes:

- ABC score
- Strate score/ Strate Prediction Model
- NOBLADS
- Sengupta score
- BLEED,
- Birmingham score
- Severe Acute LGIB (SALGIB) score
- HAKA score
- Glasgow–Blatchford bleeding score (GBS) (for both UGIB & LGIB)
- Rockall score (for both UGIB & LGIB)

Risk stratification scores to predict low risk of adverse outcomes:

- Oakland score
- SHA2PE

Table 3. Risk Stratification Scores for LGIB.

Score	Best For	Key Predictors	Pros	Cons
Oakland Score	Safe discharge decision	Age, sex, HR, SBP, hemoglobin, bleeding severity	Simple, validated for outpatient use	Less predictive for severe bleeding
GBS	Need for hospital admission (originally for UGIB but sometimes used in LGIB)	Hb, BUN, BP, HR, melena, syncope	High sensitivity, widely used	Less specific for LGIB
BLEED Criteria	Predicts in-hospital complications	BP, INR, mental status, active bleeding	Quick bedside tool	Less predictive for long-term risk
Strate Score	Severe LGIB (e.g., transfusion, intervention needed)	HR, SBP, hemoglobin, aspirin use, comorbidities	More specific for LGIB than GBS	Requires more variables

No risk score has been directly compared with clinician judgment, therefore the clinical data available at the time of initial patient presentation is the best option to identify patients at high risk for severe bleeding and other adverse outcomes.

Table 4. Oakland score (a risk assessment tool to predict safe discharge for LGIB)

Predictors	Score component value
Age, yr	
> 40	0
40–69	1
> 70	2
Sex	
Female	0
Male	1
Previous LGIB admission	
No	0
Yes	1
DRE findings	
No Blood	0
Blood	1
Heart rate (bpm)	
< 70	0
70–89	1
90–109	2
> 110	3
Systolic Blood Pressure (mm Hg)	
50–89	5
90–119	4
120–129	3
130–159	2
> 160	0
Hemoglobin (Hb) (g/dL)	
3.6–6.9	22
7.0–8.9	17
9.0–10.9	13
11.0–12.9	8
13.0–15.9	4
> 16.0	0

Comments: Oakland score ≤ 8 predicted Safe discharge.

Safe discharge was defined as the absence of all of the following after presentation:

rebleeding, defined as additional blood transfusion requirements or a further decrease in hematocrit (Ht) concentration of 20% or more after 24 h of clinical stability; Red Blood Cell (RBC) transfusion; therapeutic intervention to control bleeding, defined as endoscopic, radiologic, or surgical hemostasis; in-hospital death; and readmission with further LGIB within 28 days.

bpm, beats per minute; DRE, Digital Rectal Examination; LGIB, lower gastrointestinal bleeding.

ACG, American College of Gastroenterology; ESGE, European Society of Gastrointestinal Endoscopy;

BSG, British Society of Gastroenterology

Most Practical Score for Clinical Use

- For Safe Discharge? Oakland Score (≤ 8 predicts safe outpatient management)
- For Admission Decision? GBS (≥ 2 suggests admission)
- For Predicting Severe Bleeding? Strate Prediction Model

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of LGIB is the abnormal coloration of stools (red or black) due to some foods, fruits or medicine (iron). Precise history taking, thorough physical examination including DRE and the characters of stools including its smell can help to rule out/in.

VII. MANAGEMENT

Initial assessment of patients presenting with acute LGIB by ESGE 2021:

- Hemodynamic parameters;
- Thorough physical examination (including DRE);
- Laboratory markers
- History of co-morbidities and medications that promote bleeding;
- Risk score (Oakland score)

1. Objectives

- To stop the bleedings
- To find the sources of bleedings
- To assess the severity of the bleeding (time, type and abundance of bleeding)
- To assess the risk stratification using clinician judgment plus Oakland score (supplement)
- To rule out the bleeding from the UGI tract (more serious)
- To rule out/in the conditions that may affect subsequent management
- To prevent the complications of bleedings and the rebleeding.

2. Critical Signs

- Hypotension: Systolic blood pressure < 90 mmHg suggests significant blood loss.
- Tachycardia: Heart rate > 100 bpm, a compensatory response to volume loss.
- Signs of Shock/ Orthostatic hypotension
- Shock Index (SI) > 1 (SI = pulse divided by Systolic blood pressure) at the time of patient presentation: considered as hemodynamically unstable.
- Large volume of bleeding/Recurrent Hematochezia.
- Dyspnea or Chest pain: Particularly in patients with comorbid cardiac conditions.
- Severe abdominal pain: ischemic colitis, perforation, or other serious conditions.
- Abdominal distension or Tenderness: bowel obstruction or perforation.
- Confusion, lethargy, or reduced level of consciousness: profound hypovolemia or shock.

3. Mortality risk factors

- Hemodynamic instability
- Massive or Persistent bleeding (requiring transfusion of > 4 units of packed RBC).
- Recurrent bleeding
- Hb < 7 g/dL, indicating significant blood loss.
- Coagulopathy: Elevated INR or prolonged PT/PTT (e.g., due to anticoagulants, liver disease).
- Platelet count < 50,000/ μ L.
- Elevated Serum Lactate/Acidosis

- Diverticular Disease
- Ischemic colitis: severe ischemia or necrosis.
- Aorto-enteric Fistula: Rare but catastrophic cause of massive LGIB
- Age > 60 Years
- Co-morbidities: Cardiovascular diseases (CVD) (e.g., coronary artery disease, heart failure), CKD, cirrhosis, type 2 Diabetes mellitus (T2D), anticoagulation/antiplatelet therapy, malignancy.
- Immunosuppression: (e.g., chemotherapy, HIV/AIDS) may have worse outcomes.
- Delay in diagnosis or treatment
- Hospital-Related Factors: Inadequate monitoring/Resource limitations.

4. Outpatient Management

Patients must meet the following criteria to qualify for outpatient care:

- Oakland score: ≤ 8
- Stable Hemodynamics: Normal blood pressure and heart rate/No orthostatic hypotension.
- Mild Bleeding
- No High-risk features:
- Hb > 10 g/dL.
- Normal coagulation profile (INR, PT, PTT).
- No active anticoagulant use or corrected coagulopathy.
- Absence of significant comorbidities (e.g., cardiovascular instability, advanced liver disease).
- Normal initial evaluation: No alarming findings on DRE or initial diagnostic tests.
- Good clinical follow-up capability

5. First-line Treatment

First-line treatments vary based on the bleeding's severity, etiology, and patient stability. Systematic approach: The ABCDE approach (Airway, Breathing, Circulation, Disability, Exposure), identify the bleeding source, and administer appropriate treatment. Patients who are hemodynamically unstable with active bleeding should be admitted to the medical intensive care unit (MICU).

- Hemodynamic stabilization: Immediate resuscitation with iv fluids (normal saline or lactated Ringer solution) and blood transfusions is essential for patients presenting with hemodynamic instability (two large-bore peripheral iv catheters).
- Use Restrictive RBC transfusion policy:
 - If Hb ≤ 7 g/dL: target Hgb 7-9g/dL post-transfusion if no CVD.
 - If Hb ≤ 8 g/dL: target Hgb ≥ 10 post-transfusion if known CVD.
- Airways: Oxygen therapy if needed.
- Risk stratification (Oakland score, ...)
- Temporarily withhold antiplatelet agents and anticoagulants
- Colonoscopy (stable patients): the gold standard test.

Endoscopic hemostatic modalities:

- Mechanical therapy (clips),
- Thermal coagulation (Argon Plasma Coagulation),

- Injection therapy (sclerotherapy)
- Radiological intervention for acute LGIB using TAE is indicated for patients not amenable or refractory to endoscopic hemostasis therapy.
- Surgical intervention (rare): patients with refractory bleeding unresponsive to endoscopic and radiologic measures or those with indications for surgery, such as malignancy.

Table 5. Endoscopic management of specific lesions: a summary of case studies

Method	Diverticulum	Angiodysplasia	Dieulafoy	Neoplasia	Postpolypectomy	Colitis	NSAID	Hemorrhoids	Rectal varices
BICAP	++ ^{12,13}	++ ⁸⁶	++	+	+	+ ²⁸	+	++ ⁹¹	–
Heater	+	++ ⁸⁶	++	+	+	+	+	++ ^{91,94}	–
Laser	–	+ ³²	+	++	–	–	–	–	–
Argon plasma coagulation	++	++; P ⁸⁷	++; P	++; P	+	+	+	–	–
Epinephrine injection	++ ^{12,13}	+	+ ⁹¹	+	++ ¹⁴	++ ²⁸	++	+	–
Sclerosant injection	–	+	+	++ ⁸⁸	+	+ ⁸⁹	–	++ ⁹²	+
Acryl glue injection	–	–	–	–	–	–	–	–	++; P
Clip	++; P	+	++ ⁹¹	+	++ ^{14,87} ; P	++ ⁹⁰ ; P	++; P	++	+
Rubber band	+ ⁸⁵	–	+	–	–	–	–	++ ⁹²	+
Cryotherapy	–	–	–	–	–	–	–	+ ⁹³	–
Infrared coagulation	–	–	–	–	–	–	–	+ ^{92,94}	–
Low voltage current	–	–	–	–	–	–	–	+ ^{84,93,94}	–

Abbreviations: +, can be done; ++, recommended; –, not recommended; P, personal preference of the authors.

6. Treatment adjustment

- Octreotide in Gastrointestinal angiodysplasia (GIADs): if endoscopic treatment is insufficient or unfeasible or recurrent bleeding.
 - Dose: Octreotide 50 microg, sc/12h then long-acting octreotide 20 mg, IM, bid; every 4 weeks for 52 weeks.
- Antibiotic therapy (ABT): clinical settings of infection or ischemic colitis.
- Ischemic colitis: iv fluid, ABT, conservative therapy, revascularization or surgery
- Symptomatic based treatment: painkiller, antipyretic
- ESGE does not recommend the use of tranexamic acid (TXA) in patients with LGIB (iv TXA => Risk of venous thromboembolic events).
- NG tube and antiemetics can be used if needed.

7. Algorithms for LGIB

See Figure 3 and Figure 4.

VIII. SPECIAL CASES

1. Patients on warfarin therapy: Interrupt warfarin therapy at presentation.
In cases of unstable GIB, anticoagulation should be reversed with prothrombin complex concentrate and vitamin K. For patients with low thrombotic risk, warfarin should be restarted at 7 days after GIB.
2. In patients with high thrombotic risk (ie, prosthetic metal heart valve in mitral position, atrial fibrillation with prosthetic heart valve or mitral stenosis, <3 months after venous

thromboembolism), low molecular weight heparin treatment is considered at 48 h after GIB.

3. Aspirin for primary prophylaxis of cardiovascular events, it should be permanently discontinued for these patients.
Aspirin for secondary prevention is not routinely stopped. If it is stopped, it should be restarted as soon as hemostasis is achieved.
4. Dual antiplatelet therapy with a P2Y₁₂ receptor antagonist and aspirin is not routinely stopped in patients with coronary stents in situ, and the management should be in liaison with a cardiologist. In unstable GIB, continue aspirin if the P2Y₁₂ receptor antagonist is interrupted. P2Y₁₂ receptor antagonist therapy should be reinstated within 5 days.
5. Patients on direct oral anticoagulant (DOAC) therapy, interrupt it at presentation.
Consider treatment with inhibitors such as idarucizumab or andexanet for life-threatening hemorrhage on DOAC. Restart DOAC at a maximum of 7 days after hemorrhage.
6. Rectal Variceal bleeding: band ligation, octreotide (see Management of UGIB)
7. LGIB by the complication of Typhoid fever (ABT, colonoscopy, see chapter typhoid fever)
8. Hemorrhoids, anal fissures: Refer to proctologist (lifestyle, correct risk factors, phlebotonics, instrumental treatment or surgery guided by severity)

IX. EVOLUTION AND PROGNOSIS

LGIB has a variable clinical course depending on factors such as the bleeding source, severity, patient comorbidities, underlying causes and promptness of diagnosis and management. The evolution and prognosis can range from self-limited cases to life-threatening events requiring aggressive interventions.

- In most cases, bleeding will stop spontaneously.
- Diverticular bleeding: Spontaneously stop 80–85%
Persistent or recurrent bleeding: 10–25%
- GIADs: Follow up for 6–55 months: rebleeding rate 34–45%
- Recurrent episodes may indicate conditions like GIADs, IBD, or malignancy.
- Chronic anemia and iron deficiency may develop due to slow, occult blood loss over time.
- Overall mortality for LGIB is low, ranging from 2–4% in general populations, but it can rise to 10–20% in high-risk groups with severe bleeding, comorbidities, or delayed care.

X. COMPLICATIONS

LGIB can lead to various complications depending on the severity, underlying cause, and timeliness of management. Complications can arise from the bleeding itself, the underlying pathology, or the treatment process.

- Hemorrhagic or septic shock
- Multi-Organ Dysfunction
- Perforation and peritonitis
- Anemia, ...

XI. PREVENTION AND RECOMMENDATION

LGIB can be caused by various conditions. Prevention focuses on mitigating risk factors, managing underlying conditions, and promoting lifestyle modifications (multifaceted approach).

Below are prevention strategies and key recommendations:

1. Lifestyle Modifications

- Dietary Adjustments:
 - High-Fiber Diet: Helps prevent constipation and reduces the risk of diverticular disease and hemorrhoids.
 - Hydration: Adequate fluid intake prevents hard stools and reduces straining.
- Limit Alcohol: Excessive alcohol use may exacerbate mucosal injury and bleeding risks.
- Exercise: Regular physical activity improves bowel motility and overall GI health.
- Avoid straining: Reduces the risk of hemorrhoids and anal fissures.

2. Medication Management

- Avoid non-steroidal anti-inflammatory drugs (NSAIDs): NSAIDs can increase the risk of GIB. Use acetaminophen or other alternatives when possible.
- Monitor Anticoagulation Therapy: Patients on anticoagulants (e.g., warfarin, DOACs) or antiplatelets (e.g., aspirin, clopidogrel) should have therapy closely monitored to balance bleeding and thrombotic risks.
- Proton Pump Inhibitors (PPIs): Consider using PPIs as gastroprotection in patients at high risk for GIB, especially those on dual antiplatelet therapy or corticosteroids.

3. Screening and prevention of Colorectal cancer

- Routine Screening for Colorectal cancer: Regular colonoscopies for individuals aged 45 and older, or earlier for those with a family history or other risk factors.

4. Managing Underlying conditions

- Diverticular Disease
 - Prevent complications with a high-fiber diet and prompt treatment of diverticulitis.
 - Avoid triggers like NSAIDs or constipation.
- Angiodysplasia: Regular monitoring and therapeutic interventions (e.g., endoscopic cauterization) in patients with recurrent bleeding.
- Inflammatory Bowel Disease: Optimize medical therapy
- Ischemic colitis: Control risk factors (hypertension, DM, and atherosclerosis) to prevent recurrent ischemia.

5. Healthy lifestyle

- To avoid metabolic syndrome (risk of multiple diseases including colorectal cancer)

6. Regular Health Check-up as advice from doctors

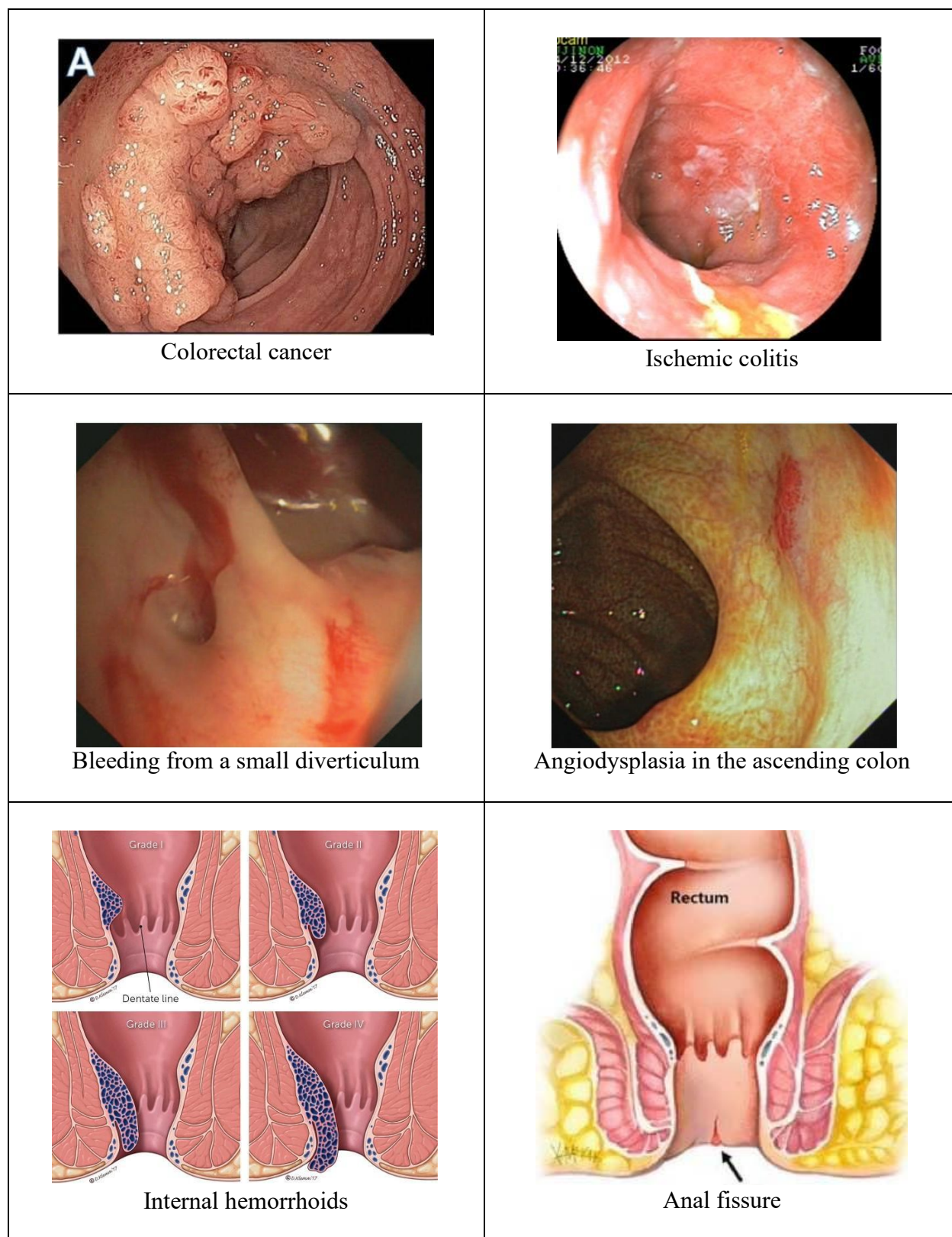


Figure 2. Images of Lower Gastrointestinal Lesions.

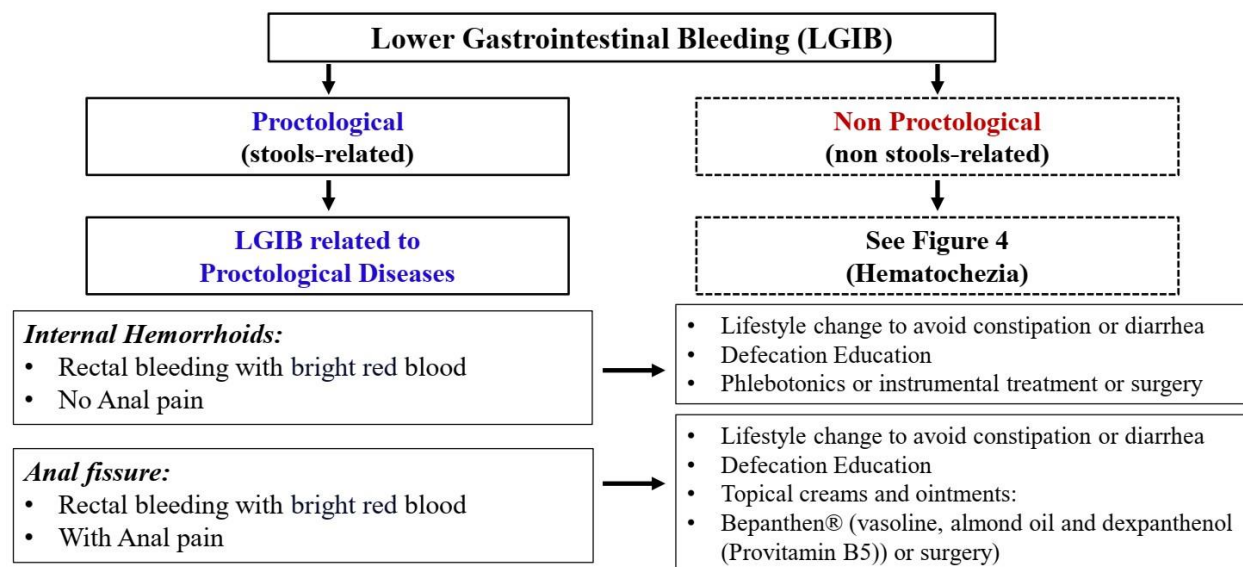
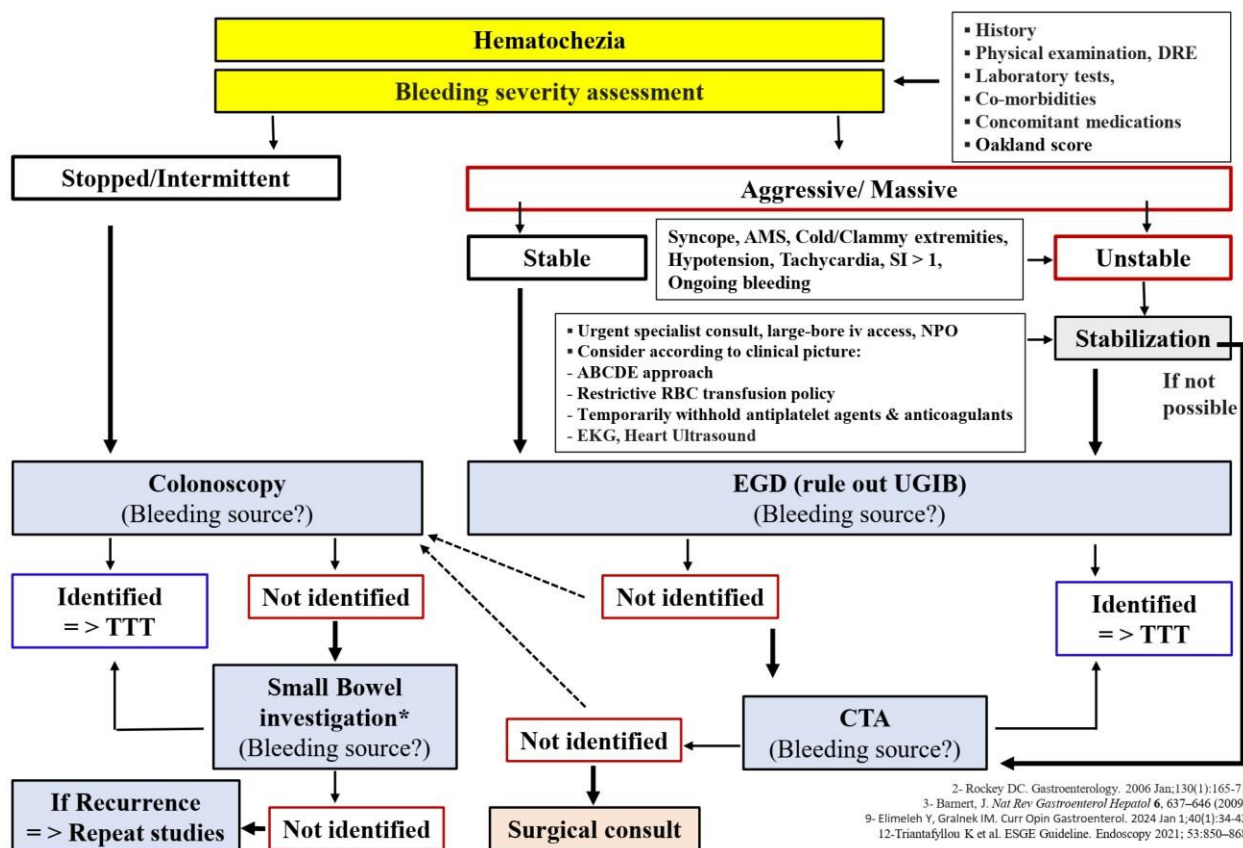


Figure 3. Algorithm for LGIB (related to proctological Diseases).



*Capsule endoscopy and/or double-balloon enteroscopy.

Figure 4. Algorithm for LGIB (not related to proctological Diseases).

ABCDE: Airway, Breathing, Circulation, Disability, Exposure; AMS: Altered mental status; CTA: CT Angiography; DRE: Digital Rectal Examination; EGD: Esophagogastroduodenoscopy; NPO: Nothing by Mouth; RCB: Red Blood Cells; SI: Shock Index (SI = pulse divided by Systolic blood pressure); TTT: Treatment

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CAUSTIC INGESTION

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I. DEFINITION

Caustic ingestion refers to the act of swallowing or ingesting substances that are capable of causing corrosion or tissue damage to the gastrointestinal tract (GIT).

II. EPIDEMIOLOGY

Worldwide, children represent 80% of the ingestion injury population, primarily due to accidental ingestion. Adults usually ingest caustic agents with suicidal intent. Occasionally, ingestion of caustic agents is seen in psychiatric or alcoholic patients. There is a higher prevalence in developing countries, where the lack of education and prevention contributes to the increasing number of reported cases. In Korea, while detailed statistics are limited, it is known that 60% of incidents involving the consumption of corrosive materials are the result of suicide attempts, and the remaining 40% occur due to accidental ingestion. The U.S. poison control centers received reports of over 200,000 cases each year where individuals have consumed cleaning agents. 10% of adults referring with caustic ingestions expire and 1-2% of ingestions lead to stricture formation.

III. ETIOLOGY

Various factors, including the availability of corrosive substances, socioeconomic conditions, and cultural differences in suicide attempts in different regions and age groups influence the etiology of caustic ingestion.

a. Caustic agents

Table 1. Examples of commonly ingested caustic/corrosive agents.

Caustic substance	Commercially available form	Type	Example product	Concentration
Acid	Cleaning agents	Sulfuric Mister Hydrochloric	Mister Plumber	99.5%
			SnoBol toilet cleaner	15.0%
			Lysol toilet cleaner	8.5%
			Cost Cutter toilet cleaner	9.5%
Alkali	Drain cleaners	Na hydroxide	Drano	9.5%
			Drano Professional	32.0%
			Dow oven cleaner	4.0%
			Crystal Drano	54.0%
	Bleaches	Na hypochlorite Hydrogen peroxide Calcium hypochlorite	Clorox	5.2%
			peroxide	3.0%
			Minute mildew remover	48.0%
	Detergents	Na tripolyphosphate Trisodium phosphate	Oxide	25–49.0%
			Comet	14.5%

b. Causes of ingestion

- accidental ingestion (often the case in children, less severe)
- intentional (as suicidal acts or underlying psychiatric illness in the adult, more severe)

IV. PATHOPHYSIOLOGY

The Pathophysiology of caustic ingestion involves immediate damage to the mucosal lining of the esophagus, stomach, and other parts of the gastrointestinal tract. The caustic agents, classified as acidic or alkaline, lead to tissue destruction. Risks of mortality and perforation seem dose-dependent for acids and pH-dependent for alkalis.

- **Alkali ingestion**, typically from household cleaners containing sodium and potassium hydroxide, can cause liquefactive necrosis in the esophagus. This condition rapidly destroys the mucosal layer and may penetrate deeper, leading to serious complications like perforation, mediastinitis, and death.
- **Acidic ingestion** causes coagulation necrosis, typically resulting in more superficial damage, but can still cause significant harm if ingested in large quantities. Acid ingestion in sufficient concentration can cause esophageal and gastric perforation with peritonitis. Upper airway injuries are more common with ingestion of acid, perhaps related to their bad taste, which tends to stimulate gagging, choking, and attempts to spit out the ingested material.

The extent of injury depends on several factors, including the pH, concentration, form (solid or liquid), and duration of mucosal contact with the caustic substance.

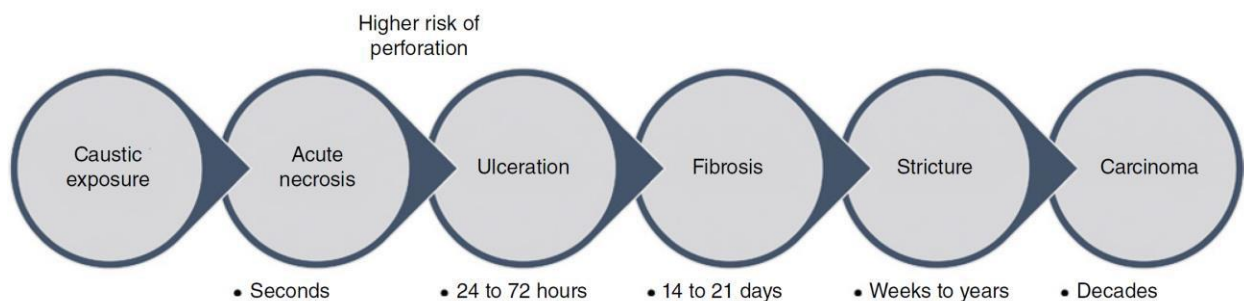


Figure 1. Timing of tissue damage and repair after caustic injury of the esophagus.

V. DIAGNOSIS

The diagnosis of caustic ingestion involves a multidisciplinary approach, including endoscopy, imaging, clinical assessment, and laboratory studies. These diagnostic tools are essential for accurately assessing the extent of injury, guiding treatment decisions, and predicting potential complications.

1. Clinical Assessment

History taking

- Individuals with caustic injuries may present unresponsive, breathless, and pulseless. After a quick primary survey, resuscitation must be administered promptly for patients with signs of cardiorespiratory arrest, regardless of the cause. Airway, breathing, and circulation must be stabilized. Patients should be monitored continuously on telemetry, with frequent assessments of vital signs. Once stabilized, the secondary survey may be initiated.
- When taking the history of individuals with suspected caustic ingestion, 5 significant details (the "5 Ws") must be elicited:

- Who - patient's personal information, including age and weight; details about accompanying family, friends, or witnesses are also important
- What - characteristics of the substances ingested
- When - time and date of ingestion
- Where - route of poisoning and geographic location of the incident
- Why - determines if the caustic ingestion is intentional

Signs and Symptoms

The clinical features of caustic ingestions vary widely. Early signs and/or symptoms may not correlate with the severity and extent of tissue injury. Symptoms may include (table 2):

- oropharyngeal, retrosternal or epigastric pain, dysphagia/odynophagia, or hypersalivation.
- vomiting and hematemesis.
- pain and inability to clear pharyngeal secretions with persistent drooling, hoarseness, stridor, aphonia, and respiratory difficulties are less common and result from caustic burns of the epiglottis and larynx.
- Fever, tachycardia, and shock (Severe extensive injury)
- persistent severe retrosternal or back pain (Esophageal perforation with mediastinitis)
- the presence of persistent, localized abdominal tenderness, rebound, and rigidity (Esophageal or gastric perforation with peritonitis)

Table 2. Signs and symptoms of caustic ingestion

Injury Localization	Signs and symptoms	
	Mild to moderate injury	Severe injury
Oropharyngeal/ Larynx	- Odynophagia - Mucosal ulceration - Drooling - Hoarseness	- Hypoxia - Aphonia - Stridor - Cyanosis
Esophageal	- Dysphagia - Chest pain - Odynophagia	- Fever - Shock/hypotension - Leukocytosis
Gastric	- Abdominal pain - Vomiting - Hematemesis	- Rebound tenderness - Abdominal rigidity/ guarding

Physical examination

- Inspect the oro- and hypopharynx, neck, chest, and abdomen.
- Examination of the oropharynx may reveal edema, erosions, or deep necrosis with grayish pseudo-membranes.
- Respiratory distress: laryngoscopy to evaluate the need for intubation.
- Oropharyngeal injury: monitor closely for the possible development of airway obstruction. Note: the absence of oropharyngeal burns does not preclude the presence of significant esophageal or gastric injury.
- Signs of GI perforation (eg, rebound tenderness, subcutaneous emphysema) and hemodynamic instability should prompt immediate surgery.

2. Investigation

Laboratory studies

- Complete blood count (CBC), serum C-reactive Protein (CRP),
- Serum electrolytes, glycaemia, calcemia
- Liver/ kidney function tests,
- Serum lactate,
- Toxicology screening.

Correlation between laboratory values and the severity/ outcome of injury is poor. A high white blood cell count (> 20000 cells/mm³), elevated serum C-reactive protein, age, and the presence of an esophageal ulcer have been considered predictors of mortality in adults; an arterial pH less than 7.22 or a base excess (BE) lower than -12 have been considered indications of severe esophageal injury and of emergency surgery.

Essentially, laboratory studies are more useful in monitoring and guiding patient management than in predicting morbidity or mortality.

Imaging

1. Chest and abdomen X-ray:

- Helpful to rule out pneumonia or signs of perforation (pneumomediastinum or subdiaphragmatic air).

2. Computed tomography (CT) scan of the chest and abdomen:

- Hold significant potential for initial examinations of corrosive substance injuries as it is non-invasive.
- Determine the severity of the caustic injury and the potential need for emergent surgical intervention.
- Provide a comprehensive assessment of the full-thickness damage to the walls of the esophagus and stomach, the risk or presence of a rupture in the stomach wall as well as determine the severity of tissue death (more effective than early endoscopy).
- Provide information about the involvement of adjacent organs *e.g.*, lung and pleural cavity. A CT-based classification system has been suggested as a predictive tool for the development of narrowing in the esophagus (table 3).

3. Esophagogastroduodenoscopy (EGD)

- Recommended within the first 12-48 hours, safely performed up to 96 hours after caustic ingestion (the injured esophagus enters the phase of ulceration and granulation, and the esophagus becomes fragile and easily perforated after safe period)
- Essential for assessing the endoscopic classification of the injuries (Zagar classification, Table 3), the extent of injury, determining the severity of esophageal burns, and guiding subsequent management.

Table 3. Assessment of severity and prognosis: endoscopic score and CT score

Grade	Endoscopic score	CT score	Prognosis
I	Edema and hyperemia of the mucosa	No definite swelling of esophagus wall (< 3 mm, within normal limit)	Complete recovery
II	IIA: Friability, hemorrhages, erosion, blisters, whitish membranes, exudates and superficial ulcerations IIB: IIA with deep or circumferential ulceration	Edematous wall thickening (> 3 mm) without periesophageal soft tissue infiltration	Stricture unlikely High risk of stricture, low risk of perforation
III	IIIA: Small scattered areas of necrosis IIIB: Extensive necrosis	Edematous wall thickening with periesophageal soft tissue infiltration plus well-demarcated tissue interface	High risk of stricture, greater risk of perforation than with grade IIB injury High risk of perforation and stricture
IV	Perforation	Edematous wall thickening with periesophageal soft tissue infiltration plus blurring of tissue interface or localized fluid collection around the esophagus or the descending aorta	Often fatal

❖ EGD Contraindication

- patients with hemodynamic instability,
- possible perforation,
- severe respiratory insufficiency,
- and severe laryngopharyngeal edema or necrosis.

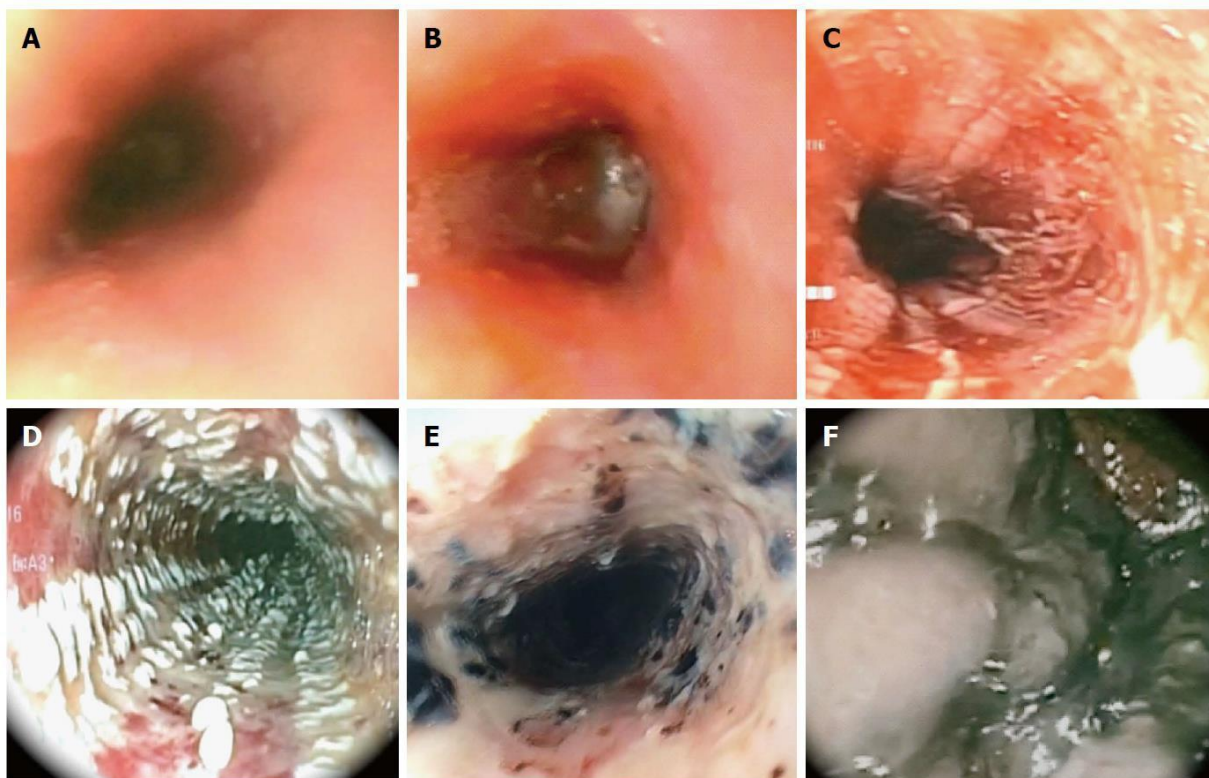


Figure 2: Endoscopic pictures of Zargar classification 0 to IIIb.

A: Zargar Grade 0: Normal mucosa; B: Zargar Grade I : Edema and erythema of the mucosa, C:Zargar Grade IIA: Hemorrhage, erosions, blisters, superficial ulcers; D: Zargar Grade IIB: Circumferential bleeding, ulcers. Exudates; E: Zargar Grade IIIA: Focal necrosis, deep gray or brownish black ulcers; F: Zargar Grade IIIB: Extensive necrosis, deep gray or brownish black ulcers.

4. Endoscopic ultrasound (EUS)

- can delineate the layers of the esophageal wall
- predict stricture formation following caustic injury by visualizing the structure of the esophageal wall
- Predict the prognosis of the esophageal dilatation: if caustic injury is confined to the submucosa in the EUS, the injured esophagus requires fewer sessions of esophageal dilatation than those with muscularis propria involvement.

VI. DIFFERENTIAL DIAGNOSIS

The differential diagnosis of caustic ingestion includes Trauma, Congenital abnormalities, Direct burns, Multiple drug ingestion etc. A complete evaluation can help distinguish between these conditions.

VII. MANAGEMENT

1. Objectives

- Distinguish between the degrees of severity, identifying cases as mild, moderate, or severe according to the extent of tissue damage and the symptoms present.
- Begin prompt management by undertaking urgent actions for safeguarding the airway, ensuring stability, and providing treatment for symptoms.
- Determine the cause of ingestion, details of the corrosive substance, the volume, timing before admission, and pre-hospital treatment.
- Execute treatment plans that are specific to the gravity of the injury, which may involve supportive care, procedures using an endoscope, or surgical operations.
- Work in conjunction with a multidisciplinary team to improve patient results following the ingestion of corrosive substances.

2. Critical Sign

- Airway obstruction: stridor, change in the voice.
- Perforation: mediastinitis, peritonitis
- Bleeding (erosion of a vessel)
- Systemic toxicity

3. Mortality risk factor

- Metabolic acidosis
- Perforation
- Old age
- Higher comorbidity score,
- Intensive care unit admission during management,
- Emergency surgery for digestive necrosis.

4. Outpatient Management

- Patients with mild symptoms may be managed on an outpatient basis, while those with severe ingestion require hospitalization.
- Asymptomatic patients without significant ingestion: No oral burns, history of a low volume, accidental ingestion of low concentration acid or alkali, upper endoscopy is not necessary.
- Symptomatic patients or those with significant ingestion: Symptomatic patients, oral burns, or are known to have ingested a high-risk substance (high - concentration acid or alkali or high volume (> 200 mL) of a low-concentration acid or alkali) should be hospitalized.

5. First-line Treatment:

- Initial management: Supportive care, observation, respiratory support, fluid replenishment, anti-emetics, and pain management.
- Airway assessment:
 - o Urgent laryngoscopy for adults with voice changes, breathing noise, or severe breathlessness;
 - o Avoid "blind" nasotracheal intubation due to the risk of further damage;
 - o Consider emergency tracheostomy for severe and rapidly worsening throat burns.
- Monitoring and fasting: Observe in intensive care and withhold food until assessment is complete.

- Diagnostic imaging and endoscopy (EGD): Perform CT for suspected severe injury or perforation and conduct EGD within 12-48 hours if no perforation is evident.
- Treatment decisions: Based on the patient's condition and findings from EGD/CT;
 - o start oral fluids for minor injuries, discharge if tolerated,
 - o use broad-spectrum antibiotics for severe injuries or suspected perforation,
 - o consider acid suppression (PPI) for esophageal damage.
- Mental health care: Psychiatric evaluation for ingestions related to suicide attempts.
- Avoid nasogastric intubation, gastric lavage, emetic medication, activated charcoal, and the use of diluting or neutralizing substances, as these methods are not advised due to the risk of causing additional harm from caustic agents refluxing into the esophagus.
- Decontamination:
 - o Any dermal exposures: remove clothing, and wash with soap and water.
 - o Ocular exposures require irrigation with a minimum of 2L NSS until pH is within normal limits (once globe perforation is ruled out).

6. Treatment adjustment:

- **Sucralfate:** achieves its therapeutic effect by maintaining mucosal vascular integrity and blood flow, hasten mucosal healing by providing a physical barrier between the harmful effects of the corrosive substance and the gastroesophageal mucosa [22-24]. It may reduce the development of esophageal stricture in children when used to manage IIB esophageal burns due to ingestion of caustic agents. However, the use of sucralfate in caustic ingestion, particularly for stricture prevention, is not widely established as a standard treatment and may vary based on clinical practice and guidelines. [29] Sucralfate can be used 2g every 2hours for the first 3 days. In the next 21 following days 2g every 2hr (between 8am to 12pm) and 2g every 4hr (between 12pm to 8am) and in another 45 days, it can be used 2g every 8hr.
- **PPI:** Reduce the risk of short-term developing esophageal strictures following grade 2b and 3a corrosive esophageal injuries. Dosage: 80 mg/day iv x 3 days or 80 mg Iv bolus followed by 8 mg/hr 72h then maintained at 40 mg/day PO for 4 weeks.
- **Antibiotics:** For patients with injuries higher than grade 3 that are observed by using endoscopy or those suspected to have esophageal perforation, broad-spectrum IV antibiotics such as third generation cephalosporins should be administered.
- **Antiemetics and analgesia:** in case of vomiting and pain.
- **Emetics, neutralizing agents, or corticosteroids:**
 - o Emetics: contraindicated: vomiting re-exposes the esophagus and the oropharynx to the caustic agent, further aggravating injury.
 - o Neutralizing agents (weakly acidic or basic substances) should not be administered: neutralization releases heat that adds thermal injury to the ongoing chemical destruction of tissue.
 - o Corticosteroids: not recommended for adults with caustic esophageal injuries, because of the lack of demonstrable efficacy.

7. Algorithm

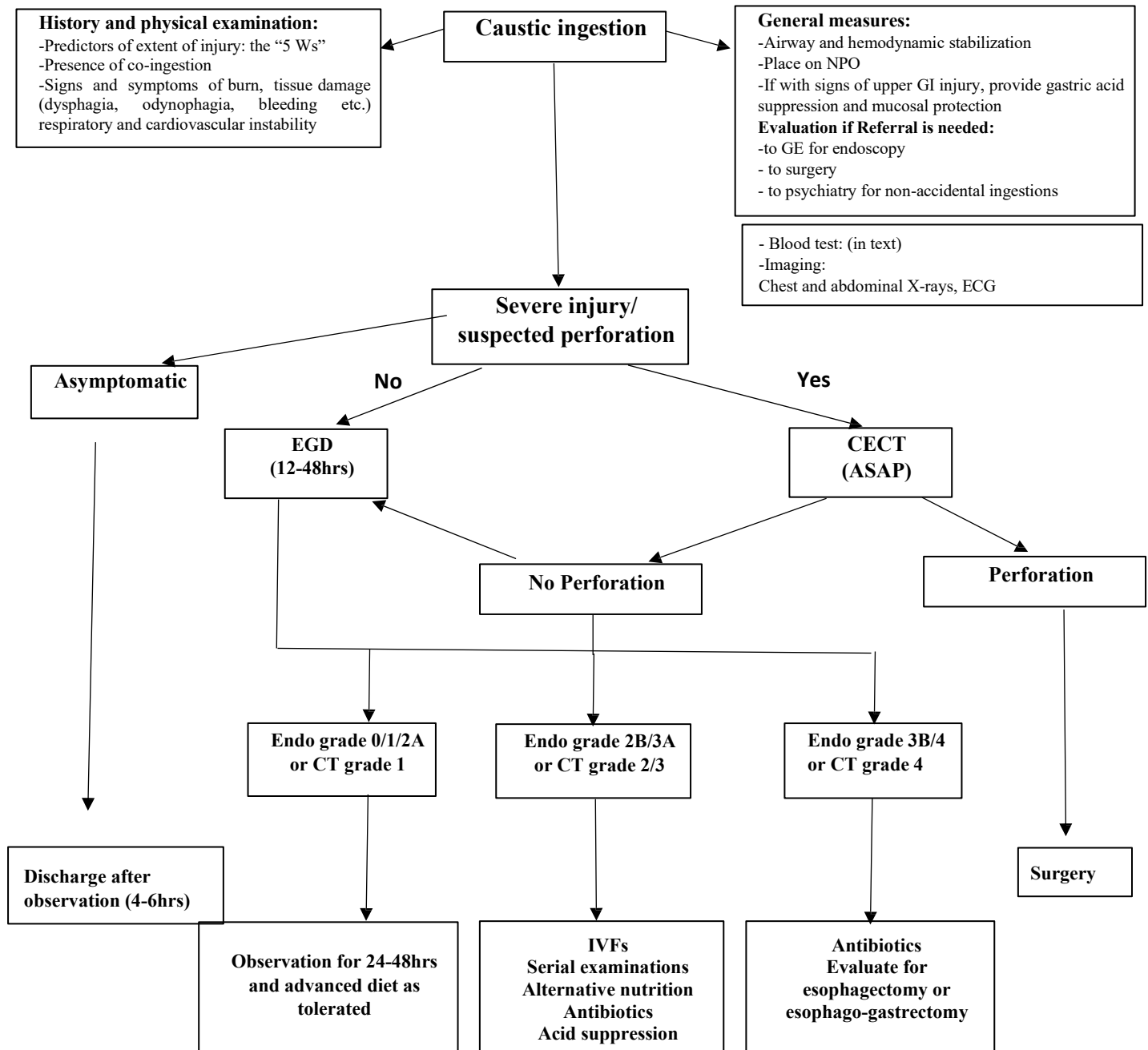


Figure 3. Algorithm for Evaluation and Management of patients with Caustic Ingestion.

Abbreviation: ASAP: as soon as possible; CECT: Contrast-Enhanced Computed Tomography;
EGD: Esophagogastroduodenoscopy; Endo: endoscopic; GE: Gastroenterologist;
IVFs: Intravenous fluids; NPO: Nothing by Mouth

VIII. SPECIAL CASES

- Sign of perforation needs a multidisciplinary approach (MDT) for evaluation and the decision of surgery.
- Hydrofluoric acid exposure may lead to extreme decreases in blood calcium levels.
- Administer calcium empirically in cases of major exposure or if there are signs of QT interval extension or irregular heart rhythms.
- Consumption of Tide pods could result in intense metabolic acidosis and chemical burns.
- Ingesting hydrogen peroxide may produce gas emboli.
- Administer oxygen and perform diagnostic imaging appropriate to the symptoms, such as a brain CT for changes in consciousness.
- Esophageal stricture: treatment relies mainly on endoscopy (dilation, Stenting) which can be done at least 3 weeks after ingestion) or complex surgical reconstructive procedures. After 8 weeks, scar tissue is completely formed and the result of endoscopic dilatation is poor.

IX. EVOLUTION AND PROGNOSIS

- The severity of tissue injury from acidic and alkaline substances is determined by the duration of contact; the amount and state (liquid, solid) of the substance involved; and the substance's physical properties, such as its pH, concentration, ability to penetrate tissue, and its treatable reserve.
- The latter reflects the amount of tissue required to neutralize a given amount of the involved substance and is particularly useful for measuring the amount of damage that can be caused by caustics, such as phenol, that have a near-neutral pH.
- The prognosis depends upon the extent of esophageal injury and the underlying medical condition of the patient. In general, the prognosis based on endoscopic grading is as follows:
 - o Patients with grades 1 and 2A have an excellent prognosis without significant acute morbidity or subsequent stricture formation.
 - o Patients with grades 2B and 3A develop strictures in 70 to 100 percent of cases.
 - o Grade 3B injuries are associated with an early mortality rate of 65 percent, and esophageal resection with colonic or jejunal interposition is required in most cases.
 - o Most deaths are due to the sequelae of perforation and mediastinitis.
 - o For patients who require pharyngeal reconstruction (colopharyngoplasty) during esophageal reconstruction because of severe pharyngoesophageal caustic injuries, long-term functional outcomes are poor.
- Adults with psychiatric comorbidities have poorer prognosis. A retrospective study of 839 adults treated for caustic ingestion injuries treated at a single hospital reported that those with confirmed psychiatric diagnoses had a predicted 5-year survival of < 50%.

X. COMPLICATIONS

- GI bleeding
- Esophageal or gastric perforation
- Fistula formation (tracheobronchial, aortoenteric),
- Esophageal strictures, gastric outlet obstruction, and esophageal cancer (in the long term)
- Pulmonary complications

XI. PREVENTION AND RECOMMENDATION

- Prevention strategies focus on the safe storage of caustic substances, public education, and the use of child-resistant packaging.
- Psychiatry consultation is important in case there is a suspicion of a mental health problem.
- Recommendations for the management of caustic ingestion are based on current evidence and expert consensus.

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ACUTE DIARRHEA

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I. DEFINITION

Diarrhea is defined as the passage of loose or watery stools, typically at least 3 times in a 24 hour period. Acute diarrhea is defined as diarrhea ≤ 14 days in duration.

II. EPIDEMIOLOGY

The Global Burden of Disease Study found that in 2019, diarrhea contributed to over 1.5 million deaths around the world. In the United States, an estimated 48 million foodborne diarrheal illnesses occur annually, resulting in more than 128,000 hospitalizations and 3,000 deaths. In the developing world, infectious causes of acute diarrhea are largely related to contaminated food and water supplies.

III. ETIOLOGY

The most common causes of acute diarrhea are infections, travelers' diarrhea, and side effects of medicines.

1. Infections

- Viral infections: Rotavirus and Viral gastroenteritis,
- Bacterial infections: Campylobacter, Escherichia coli (E. coli), Salmonella and Shigella
- Parasitic infections: Entamoeba histolytica and Giardia lamblia,

2. Travelers' diarrhea

- Travelers' diarrhea is caused by eating food or drinking water contaminated with bacteria, viruses, or parasites.

3. Side effects of medicines

- Purgatives like magnesium sulfate or magnesium containing antacids especially associated with excessive intake of Sorbitol and Mannitol.

IV. PATHOPHYSIOLOGY

Secretory Diarrhea

- Active intestinal secretion of fluid and electrolytes as well as decreased absorption.

Forms: Large volume, painless, persist with fasting

Inflammatory Diarrhea

- Damage to the intestinal mucosal cell leading to a loss of fluid and blood

Forms: pain, fever, bleeding, inflammatory manifestations

Osmotic Diarrhea

- Osmotic diarrhea results from the presence of osmotically active, poorly absorbed solutes in the bowel lumen that inhibits normal water and electrolyte absorption.

Forms: cease with fasting, discontinue oral agents

Abnormal Motility Diarrhea

- Increased frequency of defecation due to underlying diseases

Forms: The amount of stool is not much but frequency.

V. DIAGNOSIS

1. History related to etiological suspected

- Food history (Ingestion of raw or contaminated food is a common cause of food poisoning diarrhea include:
 - Oysters - *Vibrio* *Calicivirus* and *Plesiomonas* and *Calicivirus* species
 - Dairy food - *Campylobacter* and *Salmonella* species
 - Eggs - *Salmonella* species
 - Meats - *C perfringens* and *Aeromonas*, *Campylobacter*, and *Salmonella* species
 - Ground beef - Enterohemorrhagic *E coli*
 - Seafood - *Astrovirus* and *Aeromonas*, *Plesiomonas*, and *Vibrio* species
 - Vegetables - *Aeromonas* species and *C perfringens*.
- Foreign travel history: *in Asia* (*E. coli*, *Giardia*, *Salmonella*, *Shigella* and *V. cholerae*).
- Medications (most common): Colchicine, NSAIDs, Digoxin, Cisapride.
- Elderly used antibiotics (antibiotic-associated colonic inflammatory complication) by *Clostridium difficile* infection: Pseudomembranous colitis.

2. Clinical evaluation

- Type of diarrhea: watery diarrhea or bloody diarrhea (dysentery),
- Associated symptoms: Fever, Nausea, Vomiting, Abdominal pain,
- Abdomen examination, rectal examination (presence of blood, mucus),
- Dehydration evaluation (Table 1),
- Signs of severity: severe dehydration, shock

Table 1: Dehydration Evaluation

Symptoms	None to Mild (<i><5% loss of BW</i>)	Mild to Moderate (<i>5-10% loss of BW</i>)	Severe (<i>>10% loss of BW</i>)
Mental status	Well; alert	Fatigue, or irritable	Lethargic, unconscious
Thirst	Drinks normally	Thirsty; eager to drink	Drinks poorly or unable
Heart rate	Normal	Increase	Tachycardia
Pulses	Normal	Decrease	Weak, or impalpable
Breathing	Normal	Fast	Deep
Eyes	Normal	Slightly sunken	Deeply sunken
Tears	Present	Decrease	Absent
Mouth	Moist	Dry	Parched
Skin fold	Instant recoil	Recoil in <i><2</i> seconds	Recoil in <i>>2</i> seconds
Extremities	Warm	Cool	Cold; mottled; cyanotic
Urine	Normal	Decreased	Minimal
Water Loss	<i>< 50ml/Kg</i>	<i>50 – 100ml/Kg</i>	<i>> 100ml/Kg</i>

BW: Body Weight

3. Investigation

- Routine: CBC, CRP, Electrolytes, Urea, Creatinine,
- Stool Testing (Not routinely): mostly expective management without testing
 - Indicate for patients with acute community-acquired diarrhea and the following
 - *Severe illness* (Fever $\geq 38,5^{\circ}\text{C}$, symptoms of hypovolemia, severe abdominal pain).

- *High-risk patients* (Age > 65 years, Comorbidities: heart disease, chronic renal failure, cirrhosis, and advanced HIV infection).
- *Bloody diarrhea (Dysentery)*
- **Imaging Study (Not routinely):** mostly management do not need imaging.
 - Indicated in patient with peritoneal signs or ileus, abdominal imaging (most typically computed tomography) => potential complications:
 - Bowel perforation, abscess,
 - Fulminant colitis,
 - Intestinal obstruction.
- **Endoscopy (Not routinely):** discuss in cases,
 - Diagnosis is unclear after routine blood and stool tests,
 - Empiric therapy is ineffective, or if symptoms persist,

VI. DIFFERENTIAL DIAGNOSIS OF NONINFLAMMATORY & INFLAMMATORY DIARRHEAL SYNDROMES

Table 2: Noninflammatory & Inflammatory Diarrheal Syndromes

Factor	Noninflammatory	Inflammatory
Etiology	Usually viral, but can be bacterial or parasitic	Generally invasive or toxin-producing bacteria
Pathophysiology	More likely to promote intestinal secretion without significant disruption in the intestinal mucosa	More likely to disrupt mucosal integrity, which may lead to tissue invasion and destruction
History and examination findings	Nausea, vomiting; normothermia; abdominal cramping; larger stool volume; non-bloody, watery stool	Fever, abdominal pain, tenesmus, smaller stool volume, bloody stool
Laboratory	Absence of fecal leukocytes	Presence of fecal leukocytes
Common pathogens	Enterotoxigenic Escherichia coli, Clostridium perfringens, Bacillus cereus, Staphylococcus aureus, Rotavirus, Norovirus, Giardia, Cryptosporidium, Vibrio cholerae	Salmonella (non-Typhi species), Shigella, Enteroinvasive E. Coli, Clostridium difficile, Entamoeba histolytica, Yersinia
Other	Generally milder disease. Severe fluid loss can still occur, especially in malnourished patients	Generally more severe disease

VII. MANAGEMENT

1. Objectives

- To prevent the dehydration.
- To cure the specific condition.
- To avoid the recurrence and the contamination relatives.

2. Critical Sign (Severe dehydration or shock). Table 1.

General Management in Severe dehydration or shock

Table 3: Rehydration Therapy in Critical Phase

Rehydration Therapy	Replacement of Ongoing Losses	Nutrition
Lactated Ringers or Normal Saline* intravenously in boluses of 20 mL/Kg BW until mental status, vital sign improves, then administer 100 mL/Kg BW of ORS over 4 hours or 5% Dextrose ½ Normal Saline intravenously to maintenance fluid rates.	Administer ORS: 120-240 mL for each diarrhea or vomiting episode. or administer 5% dextrose ¼ Normal Saline with 20 mEq/L potassium chloride intravenously,	- Normal diet after initial rehydration, including adequate caloric intake for maintenance. - Avoid milk and foods potentially sources of reinfection (raw vegetables, fruits).

ORS: Oral Rehydration Solution.

* In severe dehydrating diarrhea, Normal Saline is less effective for treatment because it contains no bicarbonate or potassium. Use Normal Saline only if Ringer Lactate is not available, and supplement with ORS as soon as the patient can drink. Plain glucose in water is ineffective and should not be used.

3. Mortality risk factor

- Geriatric patients (age > 60 years)
- Comorbidity (Heart failure, chronic kidney failure.....)
- Severe dehydration, shock
- Sepsis

4. Outpatient Management

Table 4: Mild and Moderate Rehydration

Degree	Rehydration Therapy	Replacement of Ongoing Losses	Nutrition
None to Mild	Not applicable	ORS: 120-240 mL for each diarrhea or vomiting episode. <i>AND/OR</i> COCA-COLA nonlight (as much as possible)	- Normal diet after initial rehydration, including adequate caloric intake for maintenance. - Avoid milk and foods potentially sources of reinfection (raw vegetables, fruits).
Mild to Moderate	ORS : 50-100 mL/Kg body weight over 3-4 hours	ORS: 120-240 mL for each diarrhea or vomiting episode. <i>AND/OR</i> COCA-COLA nonlight (as much as possible)	- Normal diet after initial rehydration, including adequate caloric intake for maintenance. - Avoid milk and foods potentially sources of reinfection (raw vegetables, fruits).
	If the patient cannot drink	Hospitalization: Lactated Ringers or Normal Saline intravenously to maintenance fluid rates.	

Empiric Antibiotics

- Indicate in cases :
 - Severe illness (fever $\geq 38.5^{\circ}\text{C}$, hypovolemia, ≥ 6 unformed stools per 24 hours, severe abdominal pain)
 - Features of inflammatory diarrhea (bloody diarrhea, small volume mucous stools, fever)

- High-risk host features (age $\geq$ 65 years, cardiac disease, immunocompromising condition, inflammatory bowel disease, pregnancy)

5. First line treatment

The first line and alternative antibiotic treatments. Table 5

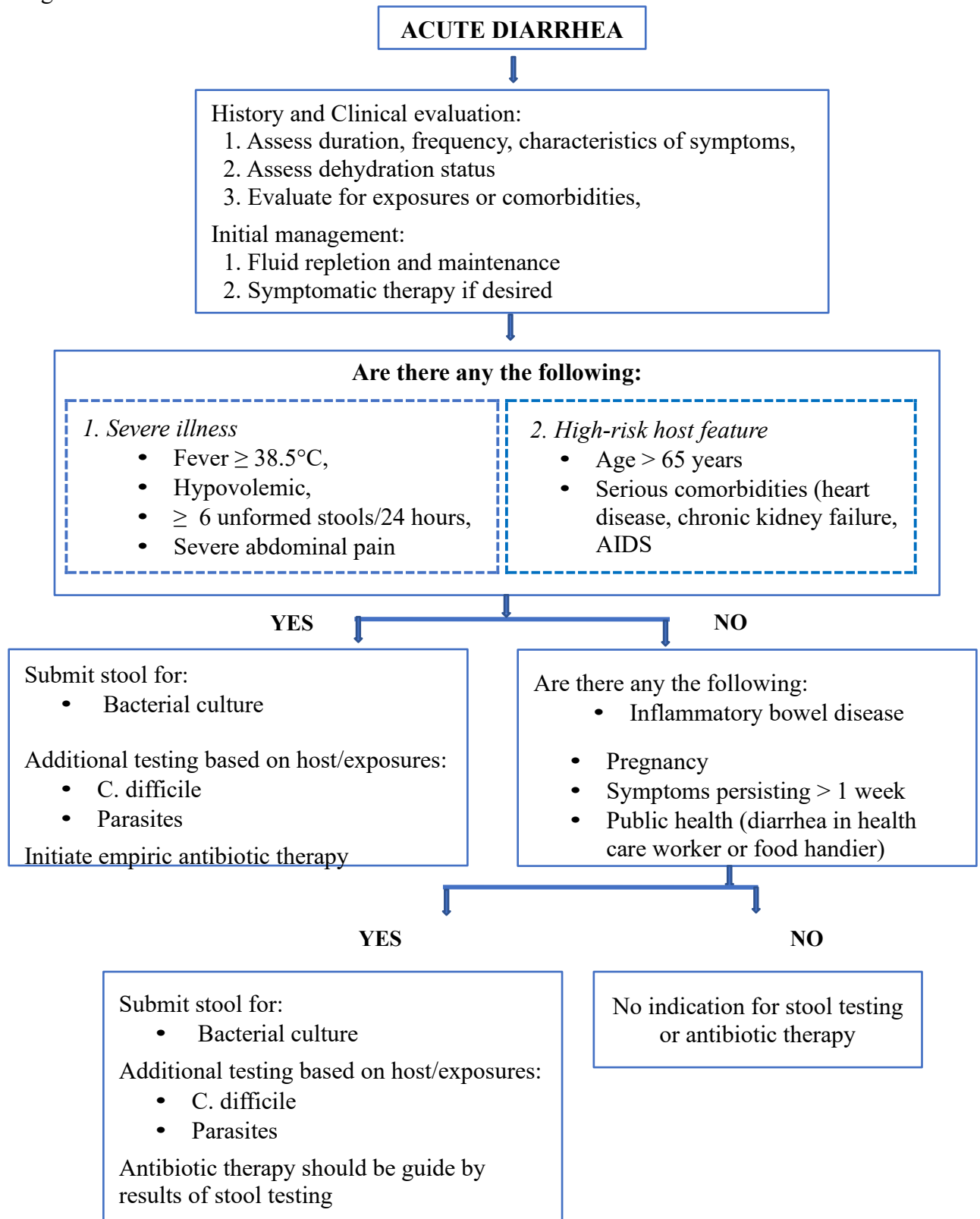
Table 5: First line and Alternative antibiotics

Bacteries	First line	Duration	Alternative
Salmonella, Shigella	<i>Fluoroquinolone</i> Ciprofloxacin 500 mg x 2/day (PIV) switch to 500 mg x 2/day (PO) OR Levofloxacin 500 mg x 1/day (PIV) or (PO)	3 to 5 days	Ceftriaxone 1g/day, Azithromycin
Campylobacter jejuni	Azithromycin (500 mg Day 1, then 250 mg Day 2 to day 5, (PO)	5 days	<i>Fluoroquinolone</i>
Yersinia enterocolitica	<i>Fluoroquinolone</i>	10 days	Doxycycline
Clostridium difficile	Metronidazole 250 mg x 4/day (PO)	10 to 14 days	Vancomycine 125 mg x 4/day (PO)
Vibrio cholerae	Doxycycline 200 mg/day (PO) OR <i>Fluoroquinolone</i>	1 to 3 days	Erythromycin 2-3 g/day (2 to 3 times)

6. Treatment adjustment

- Antidiarrheal agents: antidiarrheal agents use only for patients with watery diarrhea (nonbloody diarrhea), absent or low-grade fever), the antimotility agent, loperamide:
 - may be used cautiously for $\leq$ 2 days.
 - may reduce diarrheal output and cramps, but do not accelerate cure.
 - generally contraindicated for children.
 - Bismuth salicylate is an alternative.
- Probiotics
 - Work by stimulating the immune system and competing for binding sites on intestinal epithelial cells.
 - In children with acute diarrhea is associated with reduced severity and duration of illness.
- Diet (feeding)
 - Early refeeding decreases intestinal permeability caused by infections,
 - Reduces illness duration,
 - Improves nutritional outcomes.
 - The BRAT diet (bananas, rice, apples, and toast), and takes adequate oral fluid contenting carbohydrate and electrolyte,
 - Avoidance of high fiber food, fat, milk product, caffeine, alcohol and spicy,
 - Usually within 2-3 hours of starting rehydration.
 - Food should be given every 2-3 hours, day and night.

7. Algorithm



VIII. SPECIAL CASES

1. Clostridium difficile associated diarrhea (C. Difficile-Associated Diarrhea)

- Clostridium difficile associated diarrhea is a serious condition with a mortality of up to 25% in frail elderly people. It affects older, frailer, hospitalized patients and younger patients who are immunosuppressed.
- Cross infection by C difficile is common in neonatal units, but neonates do not seem to develop C difficile associated diarrhea.
- C difficile associated diarrhea is classically associated with clindamycin, but it may occur after exposure to a wide range of antibiotics.
- Symptoms usually start during antibiotic treatment or shortly afterwards. Symptoms can be delayed by a few weeks, so it is worth asking patients whether they are currently taking, or have recently taken, antibiotics.

2. Diarrhea in immunocompromised patients

- Patients with Immunocompromised states (eg, HIV infection, solid organ transplantation, stem cell transplantation, hematologic malignancies) are at a high risk for severe gastrointestinal infections caused by viruses, bacteria and parasites. Compared with healthy hosts, these infectious diseases frequently run a more severe clinical course in immunocompromised patients and are associated with significant morbidity and mortality throughout the world.

IX. EVOLUTION AND PROGNOSIS

Acute diarrhea is usually self-limiting and has good prognosis, when the underlying cause is identified and treated.

X. COMPLICATIONS

- Watery Diarrhea (nonbloody diarrhea)
 - Dehydration / shock
 - Electrolyte imbalances (Hypokalemia)
 - Metabolic acidosis
 - Chronic diarrhea
- Bloody Diarrhea
 - Perforation (peritonitis)
 - Sepsis

XI. PREVENTION AND RECOMMENDATION

Acute diarrheal diseases can be prevented with a variety of measures focused on preventing the spread of organisms from person to person and within the community:

- Hand washing before meals; drink boiled or drinking water.
- The fruits and vegetables must be washed with safe water.

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CONSTIPATION

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I. DEFINITION

Constipation is a symptom not a disease. Constipation generally means passing fewer than three stools a week. Stools are often dry, hard and may be abnormally large or abnormally small.

II. EPIDEMIOLOGY

The prevalence of chronic constipation varied between 2 to 27 percent with a more prevalent in women, nonwhites, and those aged over 60 years old.

Constipation appears to correlate with decreased caloric intake in the elderly but not with either fluid or fiber intake.

III. ETIOLOGY

The etiology of constipation is usually multifactorial, but it can be broadly divided into two main groups: primary constipation and secondary constipation.

1. Primary constipation

Primary (idiopathic, functional) constipation can generally be subdivided into the following three types:

- Normal-transit constipation (NTC): NTC is the most common subtype of primary constipation. Although the stool passes through the colon at a normal rate, patients find it difficult to evacuate their bowels. Patients in this category sometimes meet the criteria for IBS with constipation (IBS-C), *see IBS topic*.
- Slow-transit constipation (STC): STC is characterized by infrequent bowel movements, decreased urgency, or straining to defecate. It occurs more commonly in female patients. Patients with STC have impaired phasic colonic motor activity. They may demonstrate mild abdominal distention or palpable stool in the sigmoid colon.
- Outlet delay or Pelvic floor dysfunction (ie, pelvic floor dyssynergia): Pelvic floor dysfunction is characterized by dysfunction of the pelvic floor or anal sphincter. Patients often report prolonged or excessive straining, a feeling of incomplete evacuation, or the use of perineal or vaginal pressure during defecation to allow the passage of stool, or they may report digital evacuation of stool.

2. Secondary constipation

- Dietary issues that may cause constipation include inadequate water intake; inadequate fiber intake; overuse of coffee, tea, or alcohol; a recent change in bowel habit paralleled by changes in the diet; and ignoring the urge to defecate. Reduced levels of exercise may play a role as well.
- Structural causes of secondary constipation include anal fissures, thrombosed hemorrhoids, colonic strictures, obstructing tumors, volvulus, and idiopathic megarectum.
- Systemic diseases that may cause constipation include the following:

- Endocrinologic and metabolic disorders - Hypercalcemia, hyperparathyroidism, hypokalemia, hypothyroidism, pregnancy, and diabetes mellitus.
- Neurologic disorders - Stroke, Parkinson disease, diabetic autonomic neuropathy, spinal cord lesion, head injury, cerebrovascular accident.
- Connective-tissue disorders – Scleroderma.
- Medications that may contribute to constipation include the following:
 - Many psychotropic drugs.
 - Antidepressants (eg, cyclic antidepressants and monoamine oxidase inhibitors).
 - Opioids (eg, codeine and morphine).
 - Antacids (eg, aluminum and calcium compounds).
 - Metals (eg, iron and bismuth).
 - Anticholinergics (eg, benztropine and trihexyphenidyl).
 - Calcium channel blockers (eg, verapamil).
 - NSAIDs (eg, ibuprofen and diclofenac).
 - Sympathomimetics (eg, pseudoephedrine).
 - Inadequate thyroid hormone supplementation.
 - Cholestyramine and stimulant laxatives (long-term use) - Although laxatives are frequently used to treat constipation, chronic laxative use becomes habituating and may lead to the development of a dilated atonic laxative colon, which necessitates increasing laxative use with decreasing efficacy.

IV. PATHOPHYSIOLOGY

The pathophysiology of constipation groups the disorder into primary and secondary causes. Primary causes are intrinsic problems of colonic or anorectal function, whereas secondary causes are related to organic disease, systemic disease, or medications.

V. DIAGNOSIS

1. History related to etiological suspected
 - An important part of history includes defining the nature and duration of constipation.
 - History should also focus upon identifying secondary causes of constipation. Most patients with idiopathic constipation are otherwise asymptomatic.
 - A careful drug history is important, particularly the temporal relationship between starting a particular drug and the onset of constipation.
 - Many systemic or neurologic disorders that impair colonic motility affect organs outside of the gastrointestinal tract; thus, patients with these disorders may have other symptoms in addition to constipation.
 - Local processes (eg, tumors) often present with alarm symptoms (*Red Flag*):
 - Patients > 45 years
 - Family history of colorectal cancer.
 - Obstructive symptoms.
 - Persistent unexplained change in bowel habits.
 - Unexplained weight loss.
 - Rectal bleeding or melena.
 - Heme-positive stool.
 - Chronic anemia (Iron-deficiency anemia).

- Palpable epigastric mass or lymphadenopathy.
2. Physical examination
 - The general physical examination is not helpful in most patients presenting with chronic constipation. In contrast, an anorectal and digital rectal examination may be quite useful.
 - The anorectal examination provides information about the following:
 - Anal fissure, particularly in children who retain their feces to avoid painful defecation.
 - Anal stenosis.
 - Partially obstructing rectal masses.
 - Rectal prolapse.
 - Digital rectal examination provides information about the following:
 - Anorectal masses.
 - Tone of the internal anal sphincter.
 - Strength of the external anal sphincter and puborectalis muscle.
 - Presence of a rectocele (an outpouching usually presents in the anterior rectal wall).
 - Presence of gross blood or occult bleeding by checking the stool guaiac.
 - Presence of fecal impaction.
 3. Laboratory evaluation
 - Laboratory evaluation does not play a large role in the initial assessment of the patient.
 - Occult blood tests, Complete blood cell count: exclude organic diseases.
 - Thyrotropin (T4, TSH): exclude thyroid disease.
 - Calcium levels exclude metabolic disease.
 4. Endoscopy finding
 - Flexible sigmoidoscopy and colonoscopy can identify lesions that narrow or occlude the bowel; they also permit biopsy specimens to be obtained and polypectomy to be performed.
 - We suggest diagnostic colonoscopy in patients with constipation in the following settings:
 - Patients present with alarm symptoms (*Red Flag*).
 - Prior to surgery for constipation.
 5. Radiography
 - Plain films of the abdomen can detect significant stool retention in the colon and suggest the diagnosis of megacolon. They also are used to monitor bowel cleansing in patients with fecal retention. However, a diagnosis of constipation should not be made based on a single radiograph alone.
 6. Computed Tomography Scanning (CT scan)
 - CT scan may be indicated to further evaluate the possibility of an intra-abdominal abscess or mass.
 7. Others physiological tests (only for specialized center)
 - Colonic transit study with radiopaque markers: Evaluates the presence of slow, normal, or rapid colonic transit (classifying patients according to pathophysiologic subtypes)
 - Anorectal manometry: Identifies evacuation disorder (rectal hyposensitivity or hypersensitivity, impaired compliance, Hirschsprung disease).

- Balloon expulsions test: Assessment of the ability to expel a simulated stool and identify evacuation disorder.
- Barium defecography: evaluate lower bowel disorders (rectocele, enterocele and contraction of puborectalis muscle).

VI. DIFFERENTIAL DIAGNOSIS OF PRIMARY & SECONDARY CONSTIPATION

Before making the diagnosis of primary (functional, idiopathic) constipation, these possible etiologies should be excluded:

- Mechanical obstruction: colorectal tumor, diverticulosis, external compression from tumor/other, large rectocele, megacolon, postsurgical abnormalities.
- Systemic conditions, psychosocial issues, drugs, or toxins need to be identified.
- Other conditions: spinal cord anomalies, cystic fibrosis, connective tissue disorders, hypothyroidism, diabetes mellitus, hypercalcemia, hypokalemia.

VII. MANAGEMENT

1. Objectives

- To relieve the symptoms of constipation.
- To treat the causes when possible.
- To prevent the complications of constipation especially in older patients.

2. Critical Signs

- Constant persistent pain
- Fever
- Immediate obstructive symptoms
- Associated rectal bleeding
- Associated weight loss

3. Mortality risk factor

- Constipation status and laxative use are independently associated with higher risk of all-cause mortality and incident coronary heart diseases and ischemic stroke.

4. Outpatient Management

- The initial management of primary (functional, idiopathic) constipation includes

Non-Pharmacologic Treatment:

- Dietary Measures (The key to treating most patients with constipation is correction of dietary deficiencies. This generally involves increasing intake of fiber and fluid and decreasing the use of constipating agents, such as milk products, coffee, tea, and alcohol):
 - Increase dietary fiber (20 to 35 g/d) by starting with small amounts and slowly increasing fiber intake according to tolerance and efficacy. (*See the Quantity of Fiber in Foods. Annex 1*).
 - Ensure adequate fluid intake (up to 1.5 to 2L/d).
- Regular physical exercise
 - 200 to 300mn/week (30 to 45mn/day), 4 to 7days per week.
 - Run (slowly) or Walk (quickly): 20 000 steps/week (2 to 2.5 Km/day).
- Change their toileting routines (to defecate after meals, because of increased post-prandial colonic motility especially in the morning).
- Medications inducing constipation should be stopped.

Pharmacologic Treatment:

When non-pharmacologic management does not improve symptoms, then laxatives should be added to the management of constipation.

(See the Main Types of Laxatives. Annex 2).

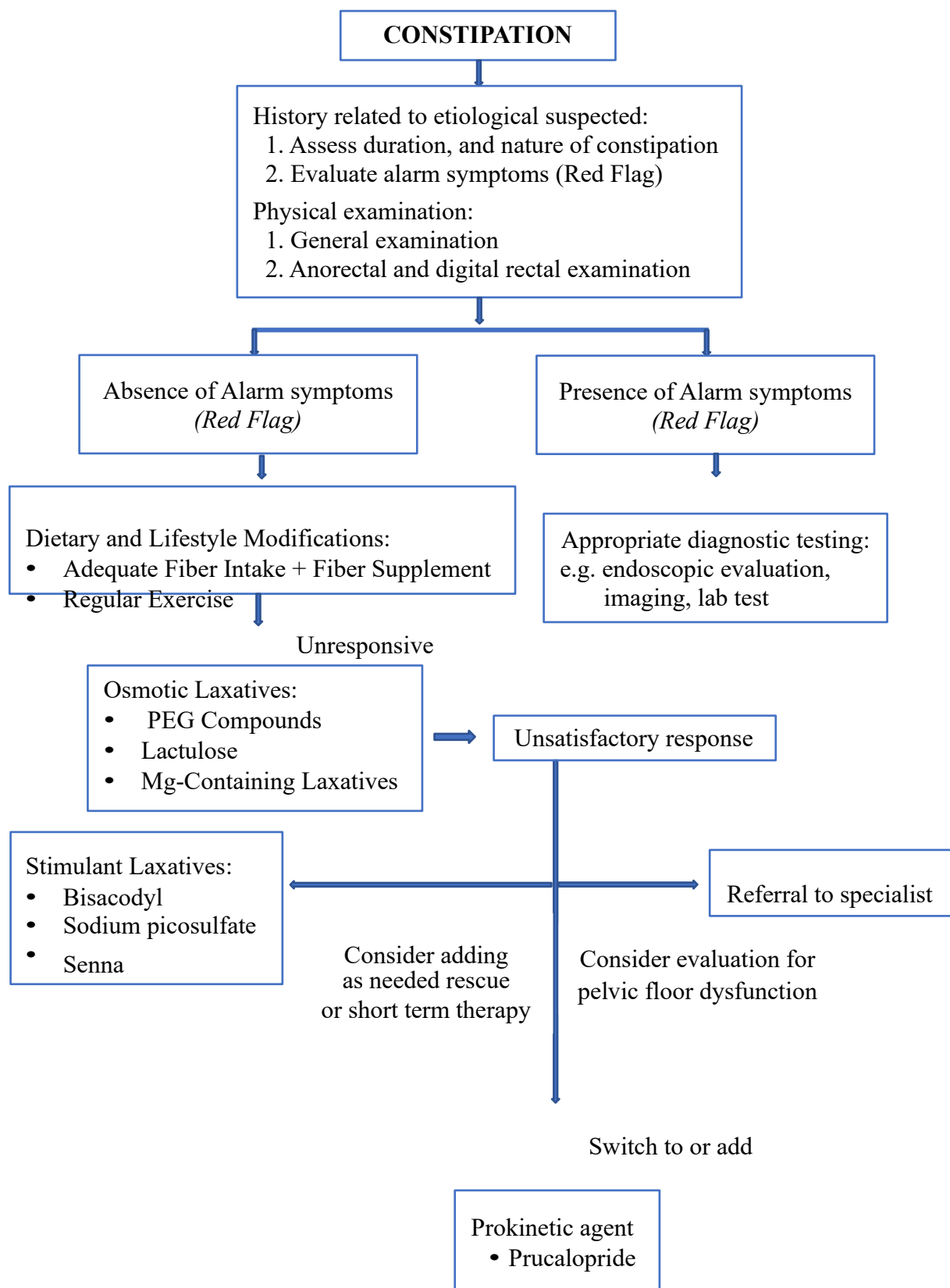
5. First line treatment

- It's difficult to know whether a particular laxative will work better than another. It depends on the person.
- Unless there's a reason why a type of laxative may be more suitable for use than another:
 - Start with bulk-forming/Fiber-based laxative (Fiber Supplement).
 - If the stool remains hard, try using an osmotic laxative in addition to, or instead of, a bulk-forming laxative.
 - If the stool is soft but is still difficult to pass, try taking a stimulant laxative in addition to a bulk-forming laxative.

6. Treatment adjustment

- The outpatient management should be tailored for primary (idiopathic, functional) chronic constipation.
- The additional treatment should discuss case by case:
 - Behavioral approaches
 - Habit training: successfully used in children with severe constipation.
 - A modified program may also be helpful in adults with neurogenic constipation, dementia, or those with physical impairments.
 - Biofeedback
 - is used to correct inappropriate contraction of the pelvic floor muscles and external anal sphincter during defecation in patients with functional defecation disorder such as dyssynergic defecation.

7. Algorithm



VIII. SPECIAL CASES

- **Laxatives induced disorders:** Long term use of stimulants laxatives can cause laxatives dependence. Patients feel that interruption of laxatives is impossible due to the recurrence of constipation. Moreover, it can cause organic diseases (organic colonopathy, hypokaliemia...). In these cases, patients should refer to specialization center for the management.
- **Dyschezia:** Evacuating stools from the rectum is inadequate or difficult with sometimes need for digital manipulation to facilitate evacuation. Its causes consist of anal lesions. (hemorrhoidal disease, anal fissure), functional disorders (abnormal anorectal reflexes...). These patients should refer to specialization center for the management because other specific examinations and treatment are needed (Colonic transit study with radiopaque markers, anorectal manometry, balloon expulsion...).
- **The elderly:** Digital rectal examination is important to eliminate impact. Treatment is the same as for younger adults, with an emphasis on changing lifestyle and diet. For immobility, it is better to use stimulant laxatives instead of bulking agents. Senna-fiber combinations are more effective than lactulose. In case of fecal impaction, enema alone is usually not enough a large, hardened impaction. Manual removal is sometime necessary.
- **Pregnancy:** Dietary fiber, increased fluid intake, and exercise as the main treatment options. Use drugs only for short periods. Bulking agents are safer than stimulants. Senna is considered safe at normal doses, but caution is necessary when it is used near term or if the pregnancy is unstable. Bulking agents and lactulose will not enter breast milk. Senna in large doses will enter breast milk and may cause diarrhea and colic in infants.
- **Diabetic patients:** Bulk-forming laxatives are safe and useful for those unable or unwilling to increase dietary fiber. Diabetics should avoid stimulant laxatives such as lactulose and sorbitol, since their metabolites may influence blood glucose levels, especially in patients with brittle type 1 diabetes.

IX. EVOLUTION AND PROGNOSIS

Most patients with constipation are managed well with holistic treatment, and most improve. A small percentage of adult patients are severely debilitated due to constipation. At least 30% of children with constipation will remain symptomatic until adulthood.

In many patients, recurrence of constipation is common, chiefly due to a lack of compliance with the diet. For the significant number of people who become disabled because of chronic constipation, their quality of life can be poor.

A few patients who fail to respond to medical management may need to undergo total abdominal colectomy; however, patient selection is vital for a good outcome. The most difficult patients are those addicted to laxatives, who will not change their lifestyle and continue to use a variety of laxatives.

Factors associated with a worse prognosis are female gender, older age of onset, longer time between symptom presentation and starting therapy, and longer colonic transit time.

X. COMPLICATIONS

- Abdominal discomfort or cramps.
- Poor quality of life.
- Hemorrhoids.
- Anal fissures.

- Fecal impact.
- Rectal ulcers.
- Rectal prolapse.

XI. PREVENTION AND RECOMMENDATION

Primary care providers including pharmacists and nurse practitioners play an important role in advising patients on lifestyle change: increased fiber intake, water 1.5 to 2 liters per day and regular physical exercise are essential. In addition, patients must be advised to refrain from using laxatives, avoid drugs that cause constipation and avoid drinking too much alcohol or coffee. For those habituated to laxatives, a gastroenterology consult is necessary.

If laxatives are prescribed, the pharmacist should assist the team by educating the patient regarding the risks and benefits.

A dietary consult may be necessary to educate the patient on what foods to eat. If constipation persists, patients may be tried on pharmacological therapy. However, it is vital that a serious organic disorder is ruled out first.

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Annex 1: Quantity of fibers in foods chart

Food	Serving size	Fiber (grams per serving)
Vegetables		
Lentils	½ cup	8
Kidney beans	½ cup	6
Green peas	½ cup	4
Broccoli	½ cup	3
Brussels sprouts	½ cup	2
Green beans	½ cup	2
Spinach	½ cup	2
Carrot	1 medium	2
Potato	1 medium	2
Onions	½ cup	1
Cauliflower	½ cup	1.7
Cucumber	½ cup	0.5
Lettuce	½ cup	0.5
Mushrooms	½ cup	0.4
Fruits		
Pear	1 medium	6
Apple	1 medium	4
Banana ripe	1 medium	3
Orange	1 medium	3
Strawberries	1 cup	3.4
Pineapple	1 piece	2
Grapes	1 cup	1.2
Watermelon	1 piece	0.8
Grains / Cereals		
Brown rice	1 cup	4
White rice	1 cup	1
Wheat bran, dry	¼ cup	6
Corn Pop	1 cup	3
All-Bran	½ cup	10
Fiber One, General Mills	½ cup	13
Breads/Grains		
Wheat, Bakery Light	1 slice	2.5
Whole wheat	1 slice	2
Multi-grain	1 slice	1.5
French bread	1 slice	0.5

Annex 2: Main Types of Laxatives

Mode of Use	Types of Laxatives
ORAL LAXATIVES	Bulk-forming laxatives / Fiber-based laxatives: Mode of Action: These laxatives are effective in increasing the frequency and softening the consistency of stool with a minimum of adverse effects. But patients who respond poorly to fiber, or who do not tolerate it, may require laxatives other than bulk forming agents. Potential Limitations: Flatulence and abdominal distension. Not recommended in frail, immobile, or palliative care patients • Psyllium, polycarbophil, methylcellulose, carboxymethylcellulose <i>They take 24 or 72 hours to work.</i>
	Osmotic laxatives: Mode of Action: Osmotic laxatives draw water from the rest of the body into your bowel to soften poo and make it easier to pass. Potential Limitations: Bloating, flatulence, Electrolyte imbalance (must be used with caution in patients with compromised renal or cardiac function). • Saline laxatives: Magnesium, sulfate, potassium, phosphate salts, • Poorly absorbed sugars: Lactulose, sorbitol, mannitol, lactose, Polyethylene glycol (PEG) <i>They take 24 or 72 hours to work.</i>
	Stimulant laxatives: Mode of Action: They act as hyperosmolar agents, increasing the water content of stool and therefore making stool softer and easier to pass, as well as increasing colonic peristalsis. Potential Limitations: Abdominal discomfort and cramps. • Dulcolax (bisacodyl) • Senna (Senokot) • Sodium picosulfate <i>They take 6 to 12 hours to work.</i>
	Surfactant laxatives / Stool-softener laxatives: Mode of Action: This type of laxative works by letting water into poo to soften it and make it easier to pass. Potential Limitations: Abdominal discomfort and cramps. • Docusate (colace) <i>They take 24 to 48 hours to work.</i>
RECTAL LAXATIVES	Lubricant laxatives: Mode of Action: They work by causing retention of water and indirectly increasing bowel movements. Potential Limitations: Dehydration, Electrolyte imbalance • Glycerin suppositories. • Lubricant laxatives (Arachis oil enema, feet enema). <i>They take 1 to 15 minutes to work.</i>

IRRITABLE BOWEL SYNDROME

Dr. UNN Keoseyla, Asst. Prof. KY Vutha, Dr. MAK Sopheak, Dr. VONG Chanlina, Asst. Prof. CHHAY Kimpav, Dr. TOCH Sorina, Asst. Prof. OUK Saram, Asst. Prof. CHEA Ong, Dr. SANN Channa, Asst. Prof. SOU Syphanna, Asst. Prof. CHHEY Vithiarithy

I. DEFINITION

Irritable bowel syndrome (IBS) is a common, chronic, relapsing and lifelong functional disorder of the gastrointestinal tract characterized by chronic abdominal pain and altered bowel habits which has a significant negative impact on quality of life and social functioning in many patients

II. EPIDEMIOLOGY

The prevalence of IBS ranges from 2.5% to 37% if using Manning criteria (one to six) and around 10% if using Rome I or II criteria. The incidence is similar in many countries despite substantial differences in lifestyle.

The frequency of IBS symptoms peaks in the third and fourth decades with a female predominance of approximately 2:1 in the 20s and 30s.

IBS is found in 51% of patients with chronic fatigue syndrome, in 64% of patients with temporomandibular joint disorder, and in 50% of patients with chronic pelvic pain. However, it is not associated with the long-term development of any serious disease.

III. ETIOLOGY

The etiologies of IBS are not well elaborated. There is more and more evidence that IBS can be associated with:

- Infections
- Genetic
- Psychological disorders
- Some diseases that can cause low grade inflammation in the gut.

IV. PATHOPHYSIOLOGY

The pathophysiology of IBS is not yet well established. It is characterized as complex bidirectional dysregulation of gut-brain interaction, via gut-brain axis, rather than disease (Figure 1). Visceral hypersensitivity, abnormal gastrointestinal motility, and psychological disturbances have been recognized to contribute to the pathogenesis of functional gastrointestinal disorders (FGIDs) for decades, but more recently low-grade intestinal inflammation, increase intestinal permeability, immune activation, disturbance in the microbiome and genetic factor have been identified.

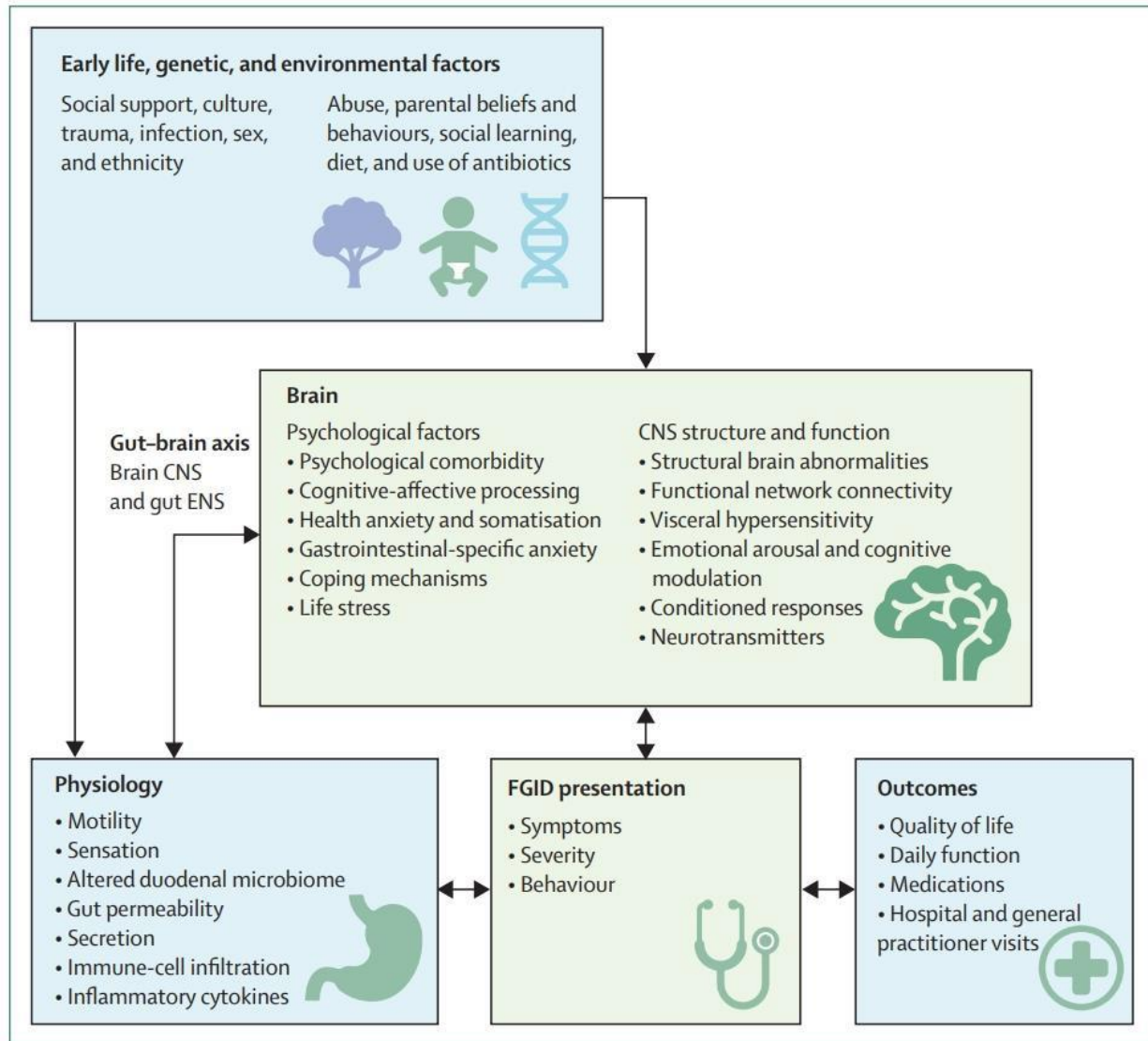


Figure 1: A biopsychosocial model of functional gastrointestinal disorders

Adapted from van Oudenhove and colleagues. FGID=functional gastrointestinal disorders. ENS=enteric nervous system.

V. DIAGNOSIS

1. Clinic

The diagnosis of IBS depends on the Rome criteria derived from the Manning criteria. The last Rome criteria (the Rome IV criteria) was refined in 2016 which is shown in table I.

Table I. Rome IV diagnostic criteria for irritable bowel syndrome
Recurrent abdominal pain on average at least 1 d*/W** in the last 3M***, associated with 2 or more of the following criteria:
• Related to defecation
• Associated with a change in frequency of stool
• Associated with a change in form (appearance) of stool
These criteria should be fulfilled for the past 3 months with symptom onset at least 6 months before diagnosis. *d = day **W = Week ***M = Month

IBS has been divided into **4 subtypes** according to the varying symptoms profile:

- Diarrhea predominant IBS (IBS-D): defined as patients with loose stools more than 25% of the time and hard stools less than 25% of the time.
- Constipation predominant IBS (IBS-C): defined as patients with hard stools more than 25% of the time and loose stools less than 25% of the time.
- Mixed bowel pattern (IBS-M): defined as both hard and soft stools more than 25% of the time.
- Unclassified IBS (IBS-U): with neither loose nor hard stools more than 25% of the time.
- Moreover, some patients (now called “alternators”) with 1 subtype can switch to another over a period of time: IBS-D or IBS-C to IBS-M, or IBS-D to IBS-C.

Other investigations for organic disease are needed if any of the alarm symptoms exist (table II)

Table II. Lower gastrointestinal alarm symptoms or signs that are referral criteria for suspected colorectal cancer
Definite referral criteria 1. Aged ≥ 40 years with unexplained weight loss and abdominal pain. 2. Aged ≥ 50 years with unexplained rectal bleeding. 3. Aged ≥ 60 years with: a. Iron deficiency anemia b. Change in bowel habit 4. Positive fecal occult blood test.
Probable referral criteria 1. Adults of any age with an abdominal or rectal mass. 2. Aged < 50 years with rectal bleeding and any of the following unexplained symptoms or findings: a. Abdominal pain b. Change in bowel habit c. Weight loss d. Iron deficiency anemia

2. Laboratory examination

a. Initial investigation:

- Full blood count (FBC), erythrocyte sedimentation rate (ESR) or plasma viscosity
- C-reactive protein (CRP),
- Thyroid function: T4, TSH,
- Stool examination,
- Gastroscopy with duodenal biopsy may be undertaken to eliminate giardiasis or other parasites and celiac disease in case of diarrhea as symptoms.
- HIV if suspected and abdominal ultrasound.

b. Second level investigations are based on specific conditions and the likely differential diagnosis. These investigations include:

- **Colonoscopy** is recommended for the patients who have alarm symptoms or in whom the onset of symptoms at the age greater than 50-year-old or earlier is a first degree relative affected by colorectal cancer (age less than 45 years), or two affected first-degree relatives. *Colonoscopy with biopsy* should be done to eliminate microscopic colitis in IBS-D patients.
- **CT scan:** for detection of liver and pancreatic cancer whose symptoms can be confused with IBS.

VI. DIFFERENTIAL DIAGNOSIS

- Food allergies
- Lactose intolerance
- Parasites (Giardiasis...)
- Colon cancer
- Biliary and pancreatic diseases: bile stone disease, chronic pancreatitis...
- Anxiety disorders
- Gastroenteritis (Bacterial, viral)
- Endometriosis
- Hypercalcemia
- Endocrine disorders (hypothyroidism, hyperthyroidism, diabetes, Addison's disease...)
- Certain rare endocrine tumors (Gastrinomas, carcinoid tumors, pheochromocytoma, somatostatinomas)
- HIV infection
- Malabsorption syndromes (celiac disease or pancreatic insufficiency...)
- Bacterial overgrowth syndrome
- Microscopic colitis (Collagenous and lymphocytic colitis)
- Inflammatory bowel disease (ulcerative colitis, Crohn's disease...)
- Chronic mesenteric ischemia
- Mesenteric artery or venous thrombosis
- Lead toxicity
- Malignant neoplasms of the small intestine
- Postcholecystectomy Syndrome
- Acute intermittent porphyria

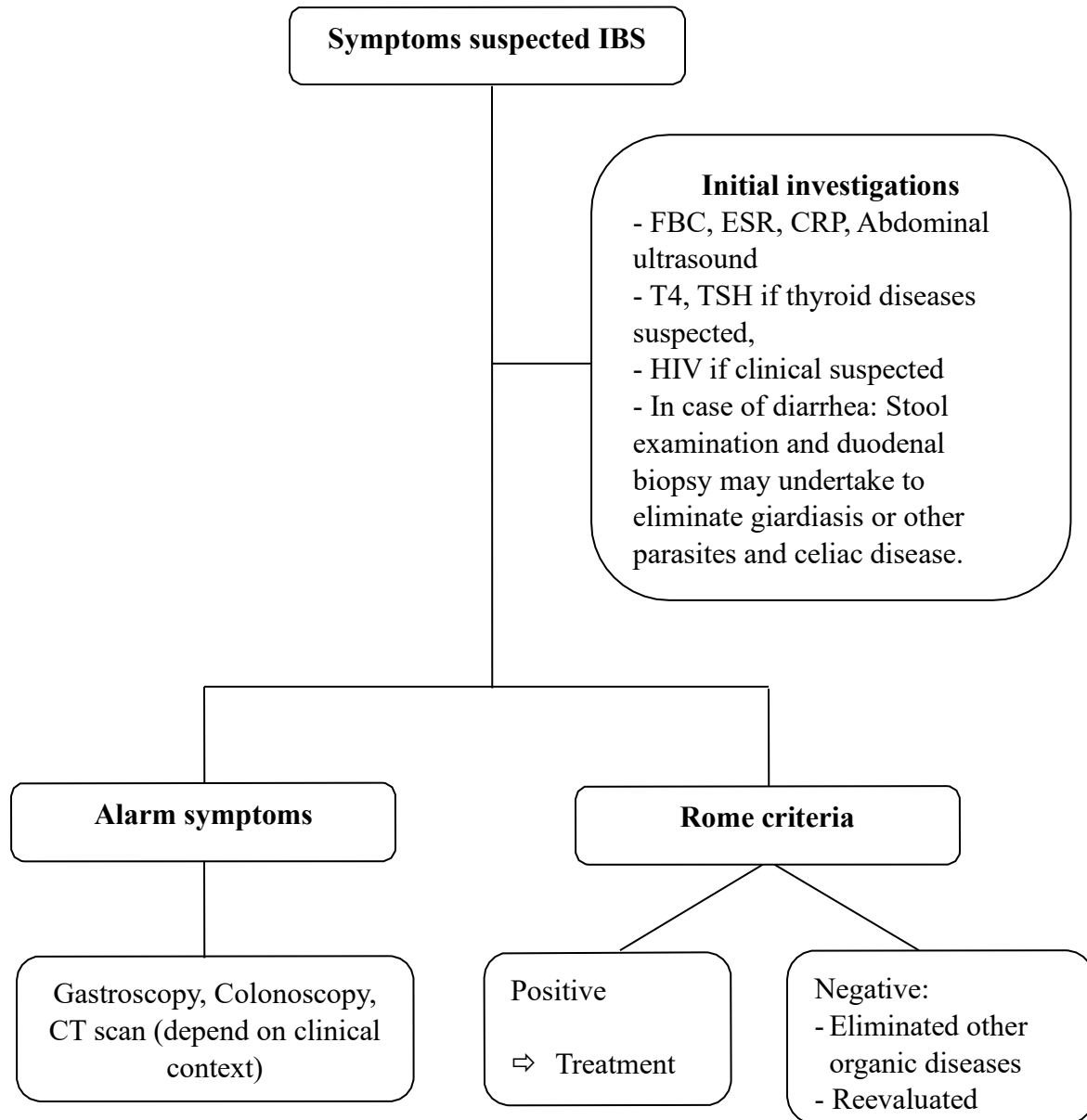
VII. ANAGEMENT

1. Objectives
 - help the patients to improve and cope with their symptoms, and to improve their quality of life.
2. Critical Signs
 - Severe upper abdominal pain can rarely be seen in IBS. However, if it appears, other causes should be ruled out.
 - Over GI bleeding with hemodynamic shock
 - Alter general condition: asthenia, anorexia, and unintentional weight loss.
3. Mortality Risk Factors
 - IBS is a benign disease without any mortality risk.
4. Outpatient Management
 - IBS is usually treated in outpatient settings.
5. First-Line Treatment
 - Initial therapy
 - Establishment of clinician-patient relationship (the most important component of treatment) and continuity of care:
 - The doctor should be non-judgmental, establish realistic expectations with consistent limits, and involve the patient in treatment decisions.
 - Patients with established, positive physician interactions have fewer IBS-related follow-up visits.
 - Education and assurance: A simple clear explanation and reassurance (advice about lifestyle, including diet and stresses) and symptom control (chronic disease, no increased risk of malignancy...)
 - Patients with mild and intermittent symptoms (not impair quality of life):
 - Lifestyle and dietary modification
 - **Food:** no specific food restrictions are recommended; each patient has his own experience to individual food trigger of their symptoms (especially the **gas-producing foods**) and should be avoid on individual basis.
 - **Soluble fiber** (ispaguhula, husk, and psyllium) should be recommended and insoluble fiber (wheat bran) should be avoided.
 - **Physical activity:** Increased physical activity helps to improve the symptoms.
6. Treatment adjustment: for patients who fail to respond to initial management.
 - Pharmacological therapies are chosen according to the predominant symptoms.
 - For IBS-C
 - Soluble fiber: ispaguhula, husk, psyllium
 - Osmotic laxatives: Macrogol, polyethylene glycol or magnesium-containing products (safe and well tolerated)
 - Acting chloride channel activator agent: Lubiprostone
 - Guanylate cyclase agonists: Linaclotide, Plecanatide (if osmotic laxatives are failed)
 - Sodium/hydrogen exchanger 3 (NHE3): Tenapanor (if other therapies are failed)

- For IBS-D
 - Antidiarrheal agents: loperamide
 - Bile acid sequestrants: cholestyramine, colestipol, colesevelam
 - 5-hydroxytryptamine (serotonin) 3 receptor antagonists: alosetron, cilansetron, ondansetron and granisetron
 - Combines a mu-opioid receptor agonist and a delta-opioid receptor: eluxadoline (if all other agents are failed)
- Abdominal pain and bloating
 - Antispasmodics: mebeverine, pinavarine, trimebutine, dicyclomine and hyoscyamine
 - Antidepressants
 - Tricyclic antidepressants: amitriptyline, nortriptyline, desipramine, and imipramine
 - Selective serotonin reuptake inhibitors: fluoxetine and citalopram
- **Psychological therapy:** if psychological symptoms are associated.
- **Behavioral treatments** may be considered for motivated patients who associate symptoms with stressors, although their benefits remain controversial.
- Hypnosis, biofeedback, and psychotherapy help to reduce anxiety levels, encourage health-promoting behavior, increase patient responsibility and involvement in the treatment, and improve pain tolerance.

*** **Therapeutic relationship:** The most important component of treatment.

7. Algorithm



FBC: full blood count, ESR: erythrocyte sedimentation rate or plasma viscosity and CRP: C-reactive protein, HIV: human immunodeficiency virus.

VIII. SPECIAL CASES

- Overlap between IBS and functional dyspepsia are commons. Management should be considered according to predominant symptoms.
- Difficult cases with psychiatric symptoms: a small subset of patients with IBS present to tertiary care centers with severe, unrelenting symptoms that are often associated with underlying psychiatric impairment and frequent health care utilization. This kind of patient be referred to psychiatric specialists.

IX. VOLUTION AND PROGNOSIS

- Most current IBS patients will have symptoms for some years; however, it is not associated with the long-term development of any serious disease.
- There is no evidence that IBS is linked to excess mortality.
- IBS has a good prognosis, and the diagnosis is unlikely to change on follow-up. The use of ambulatory health services by IBS patients can be reduced if a positive physician-patient interaction is developed.

X. COMPLICATIONS

- Significant negative impact on quality of life and social functioning in many patients.
- IBS can cause psychological disorders.

XI. REVENTION AND RECOMMENDATION

- There are no specific measures for prevention of IBS.
- **Dietary modification:** Patients with IBS may benefit from exclusion of gas-producing foods; a diet low in fermentable oligo-, di-, and monosaccharides and polyols (FODMAPs); and in select cases, lactose avoidance.
- **Physical activities:** Increased physical activity.

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JAUNDICE

Asst. Prof. CHEA Ong

I. DEFINITION

Jaundice is defined as accumulation of yellow (jaune) pigment in the skin and other tissues (Bilirubin);

- Level bilirubin over 3 mg/dL (51 μ mol/L); normal bilirubin < 1.0 mg/dL (17 μ mol/L).

II. EPIDEMIOLOGY

The prevalence of jaundice differs among patient populations; newborns and elderly more commonly present with the disease. Bile duct stones, drug-induced liver disease, and malignant biliary obstruction occur in the elderly population.

III. ETIOLOGY

- **Pre-Hepatic Jaundice (Hemolytic Jaundice)**

HEMOLYTIC DISORDERS

- Inherited (spherocytosis, elliptocytosis, glucose 6 phosphate dehydrogenase, pyruvate kinase deficiency, sickle cell anemia).
- Acquired (microangiopathic hemolytic anemias, spur cell anemia, immune hemolysis, parasitic infections: malaria...).

INEFFECTIVE ERYTHROPOIESIS

- Cobalamin, Folate deficiency (megaloblastic anemia)
- Severe iron deficiencies (microcytic hypochromic anemia)
- Thalassemia

INCREASED BILIRUBIN PRODUCTION

- Massive blood transfusion
- Resorption of hematoma

DRUGS

- Rifampicin, Ribavirin, Probenecid.

- **Hepatic Jaundice (Hepatocellular Jaundice)**

- Liver's ability to conjugate or excrete bilirubin is affected.
- Increased level of conjugated and unconjugated bilirubin present.
- Alcoholic hepatitis
- Viral hepatitis
- Drug toxicity
- Environmental toxins
- Autoimmune hepatitis

- **Post-Hepatic Jaundice (Cholestatic or Obstructive Jaundice)**

MALIGNANT CONDITIONS

- Gallbladder cancer
- Cholangiocarcinoma
- Gallbladder cancer
- Pancreatic cancer
- Ampullary carcinoma
- Malignant involvement of porta hepatis lymph nodes

BENIGN CONDITIONS

- Choledocolithiasis
- Post operative biliary strictures

- Primary sclerosing cholangitis
- Chronic pancreatitis
- AIDS Cholangiopathy

IV. PATHOPHYSIOLOGY

A- Pre-Hepatic (Hemolytic) Jaundice

- Excess production of bilirubin due to excess breakdown of hemoglobin.
- Indirect bilirubin (insoluble in water since unconjugated).

B- Hepatic Jaundice

- Mixed hyperbilirubinemia (unconjugated and conjugated bilirubin).
- Hepatocyte damage disrupts normal bilirubin processing.

C- Post-Hepatic (Obstructive) Jaundice

- Elevated conjugated bilirubin due to impaired bile excretion.
- Bile salts and cholesterol accumulate, leading to pale stools and dark urine.

V- DIAGNOSIS

1. History related to etiological suspected

- Patients usually present with varying symptoms apart from yellowish discoloration of skin along with pruritus, thus providing clues to narrow down the etiology or can also be asymptomatic.
- A thorough questioning regarding the use of drugs, alcohol or other toxic substances, risk factors for hepatitis (travel, unsafe sexual practices).

2. Physical examination

- Physical examination begins by evaluation of body habitus and nutritional status. Temporal and proximal muscle wasting suggests malignancy or cirrhosis. It is worth reviewing the well-known stigmata of chronic liver disease, which includes spider nevi, palmar erythema, gynecomastia, caput medusae, parotid gland enlargement.
- Palpable lymph nodes can also direct the clinician towards malignancy (left supraclavicular & periumbilical). Increased volume status of the patient evidenced by jugular venous distension can be a sign of right-sided heart failure, suggesting hepatic congestion.
- The abdominal examination should provide information on the presence of hepatosplenomegaly, or ascites. Jaundice with ascites indicates either cirrhosis or malignancy with peritoneal spread.
- Right upper quadrant tenderness with palpable gallbladder (Courvoisier sign) suggests obstruction of the cystic duct due to malignancy.
- Associated signs: Pruritus, Dark urine, Fat malabsorption (steatorrhea, weight loss).

3. Laboratory evaluation

- Total and direct (conjugated) bilirubin, AST, ALT, ALP, TP, GGT, Albumin
- Hepatocellular workup: viral serologies, autoimmune antibodies, serum ceruloplasmin, ferritin.
- Cholestatic workup: Additional tests include abdominal ultrasound or ultrasonography), computed topography (CT), magnetic resonance cholangiopancreatography (MRCP), endoscopic retrograde cholangiopancreatography (ERCP), percutaneous transhepatic cholangiography (PTC), endoscopic ultrasound (EUS).

VI. DIFFERENTIAL DIAGNOSIS

Yellow skin by other causes:

- Medicines (ointment).
- Cosmetic agents.
- Traditional medicine in Cambodia (LMEAT or PUNLEY).

VII- MANAGEMENT

1. Objectives

- To reduce the clinical signs of pruritus, if present.
- To cure the specific causes.
- To prevent the complications.

2. Critical Signs

- Severe sepsis.
- Bilirubin encephalopathy.

3. Mortality risk factor

- Elderly patient (> 65 years).
- Immune-compromised states.
- Comorbidity (chronic liver disease, chronic heart diseases, and chronic kidney diseases).

4. Outpatient Management

- It is only for mild to moderate jaundice and has a specific diagnosis, such as Alcoholic hepatitis, Viral hepatitis, Drug toxicity....

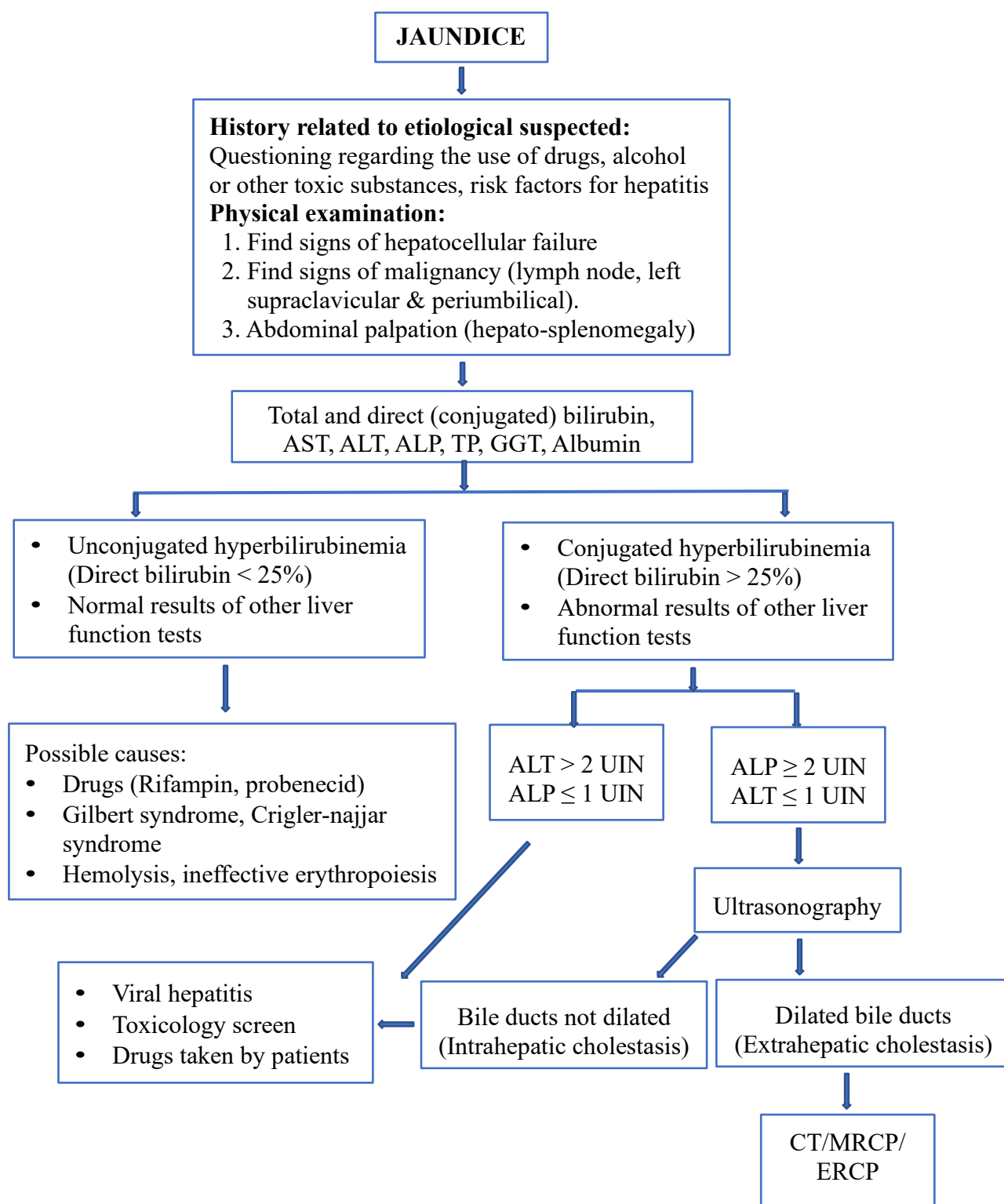
5. First line treatment

The first line of treatment depends on its causes.

6. Treatment adjustment

- Pruritus associated with cholestasis can be managed based on the severity. For mild pruritus, warm baths or oatmeal baths can be relieving. Antihistamines can also help with pruritus.
- Patients with moderate to severe pruritus respond to bile acid sequestrants such as cholestyramine or colestipol. Other less effective therapies include rifampin, naltrexone, sertraline, or phenobarbital.

7. Algorithm



CT: Computed Topography

MRCP: Magnetic Resonance Cholangiopancreatography

ERCP: Endoscopic Retrograde Cholangiopancreatography

VIII. SPECIAL CASES

1- Acute cholangitis

Acute cholangitis is a clinical syndrome characterized by fever, jaundice, and abdominal pain that develops as a result of stasis and infection in the biliary tract.

The most frequent causes of biliary obstruction in patients with acute cholangitis without bile duct stents are biliary calculi (28 to 70%), benign biliary stricture (5 to 28%), and malignancy (10 to 57%).

Malignant obstruction may be due to the presence of tumor in the gallbladder, bile duct, ampulla, duodenum, or pancreas.

Benign biliary strictures may be congenital, post-infectious or inflammatory.

2- Sepsis and low perfusion states

Bacterial sepsis is very often accompanied by cholestasis. Multiple factors including hypotension, drugs, and bacterial endotoxins are responsible for the jaundice in these patients

On the other hand, hyperbilirubinemia can promote bacterial sepsis by increasing intestinal wall permeability and altering mucosal immunity.

Signs of cholestasis can also be found in other low perfusion states of the liver (heart failure, hypotension) and hypoxemia that is not profound enough to produce hepatic necrosis.

3- Postoperative jaundice

Jaundice occurs commonly in postoperative patients and is usually multifactorial in origin. Serum unconjugated bilirubin levels may increase because of blood transfusions, hematoma resorption, and hemolysis after heart surgery.

Other contributing factors include sepsis, TPN, the administration of hepatotoxic drugs during surgery (such as halothane), postoperative hypoxia, hypotension, or a newly acquired viral hepatitis.

Concomitant renal failure will exaggerate the hyperbilirubinemia. Surgery also can exacerbate a preexisting hemolytic disease or unmask an underlying genetic disorder (eg, Gilbert syndrome or glucose-6-phosphate dehydrogenase deficiency).

4- End-stage liver disease

Chronic hepatocellular injury, the biochemical profile changes over time as the liver injury progresses to cirrhosis and liver failure.

The elevation of liver enzymes, a marker of active liver injury, becomes less prominent or even disappears.

The primary manifestations at this time result from impaired hepatic protein synthesis (eg, hypoalbuminemia and a prolonged prothrombin time) and impaired excretory function, leading to jaundice.

The hallmarks of end-stage liver disease and cirrhosis, regardless of its etiology, include jaundice with an elevation of both conjugated and unconjugated bilirubin as well as portal hypertension and decreased hepatic synthetic function.

5- Drugs and toxins

Drugs and toxins causing hepatocellular injury may eventually present as a predominantly cholestatic syndrome.

Drug-induced cholestasis can occur in a dose-related fashion (eg, methyltestosterone) or allergic reaction in a minority of subjects (eg, chlorpromazine).

Certain plants used in natural medicines (eg, Jamaican bush tea) contain pyrrolizidine alkaloids, which may cause veno-occlusive disease of the liver.

Arsenic can cause cholestasis. It has recently gained increased attention because of its contamination of ground water in various parts of the world.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis of jaundice depend on its underlying causes, complications, and comorbidities.

- Etiologies of jaundice with excellent prognosis include jaundice from resorption of hematomas, physiologic jaundice of newborn, breastfeeding, breast milk jaundice, Gilbert syndrome, choledocholithiasis.
- As a general rule, malignant biliary obstructions and cirrhosis with jaundice predict a poorer prognosis.

X. COMPLICATION

- Indirect (insoluble) bilirubin is harmful to cells and cellular structures, the toxic effects are limited to neonates due to the poorly developed blood-brain barrier. High levels of bilirubin are neurotoxic and can lead to permanent neurologic injury.

XI- PREVENTION AND RECOMMENDATION

- The biggest challenges in improving patient health outcome include a healthcare delivery system that focuses more on treating as opposed to preventing disease. Most cases of jaundice are preventable by vaccinations, safe sex practices, health education, risk factor modification.

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METABOLIC DYSFUNCTION ASSOCIATED STEATOTIC LIVER DISEASE

Dr. SANN Channa, Asst. Prof. CHEA Ong

I. DEFINITION

Metabolic dysfunction-associated steatotic liver disease (MASLD), previously termed non-alcoholic fatty liver disease (NAFLD), is defined as steatotic liver disease (SLD) in the presence of one or more cardiometabolic risk factor(s) and the absence of harmful alcohol intake. The spectrum of MASLD includes steatosis, metabolic dysfunction-associated steatohepatitis (MASH, previously NASH), fibrosis, cirrhosis and MASH-related hepatocellular carcinoma (HCC).



Figure 1. Fatty liver disease timeline. New nomenclatures: SLD, MASLD and MASH replace the term FLD, NAFLD and NASH.

II. EPIDEMIOLOGY

MASLD, worldwide, is prevalent and responsible for 25 % of chronic liver disease (CLD) and it is a leading cause of cirrhosis and HCC and the most common cause of Liver transplantation. In the most recent meta-analysis, 38% of adults around the world were shown to have MASLD (years 2016-2019). The prevalence is expected to grow as the global prevalence is forecasted to reach 55.4% by 2040.

MASLD is silent and negligent for years but it is rising and represents a significant health threat worldwide. Up to now, India is the first country globally to integrate NAFLD into the National Non-Communicable Disease (NCD) program in 2021 (the forefront of the global fight against NAFLD).

III. ETIOLOGIES

SLD (accumulation of fat in hepatocytes > 5%) includes several conditions that might be primary in the setting of the metabolic syndrome (MetS) (i.e. obesity, T2D, HTN ...) or secondary (i.e. alcohol, HCV, drugs, ...)

MetS is a group of five conditions that can lead to heart disease, diabetes, stroke and other health problems. MetS is diagnosed when someone has three or more of its risk factors (table 2).

The definition of obesity, metabolic syndrome, cardiometabolic risk factors, NAFLD and risk factors for MAFLD are detailed in table 1, 2, 3, 4 and 5.

Table 1. Working definition of overweight/obesity and central obesity for Asian adults.

Lean (normal range)	BMI 18.5–22.9 kg/m ²
Overweight	BMI 23.0–24.9 kg/m ²
Obesity	BMI > 25.0 kg/m ²
Central obesity	Waist circumference (measured at the top of the iliac crest) > 90 cm for males and > 80 cm for females

Table 2. Metabolic syndrome

▪ High blood glucose (sugar)
▪ Low levels of HDL (“good”) cholesterol in the blood
▪ High levels of triglycerides in the blood
▪ Large WC or “apple-shaped” body
▪ High blood pressure

Positive diagnosis: ≥ 3 Risk factors; WC: Waist circumference

Table 3. Cardiometabolic Risk Factors (CMRF)

▪ BMI ≥ 25 kg/m ² OR WC > 94 cm (M) 80 cm (F) OR ethnicity adjusted equivalent
▪ Fasting serum glucose ≥ 5.6 mmol/L [100 mg/dL] OR
▪ 2-hour post-load glucose levels ≥ 7.8 mmol/L [≥140 mg/d] OR HbA1c ≥ 5.7% [39 mmol/L] OR T2D OR treatment for T2D
▪ Blood pressure ≥ 130/85 mmHg OR specific antihypertensive drug treatment
▪ Plasma triglycerides ≥ 1.70 mmol/L [150 mg/dL] OR lipid lowering treatment
▪ Plasma HDL-cholesterol ≤ 1.0 mmol/L [40 mg/dL] (M) and ≤ 1.3 mmol/L [50 mg/dL] (F) OR lipid lowering treatment

T2D: Type 2 diabetes; M: Male; F: Female; BMI: Body Mass Index (Asia, normal: 23)

Table 4. Definitions of NAFLD, Hepatic Steatosis, and NASH

Term	Definition	Comments
NAFLD	Fatty infiltration of the liver in the absence of other causes of steatosis, such as alcohol	Consider after an incidental discovery of unexplained elevation of liver enzyme levels, or when hepatic steatosis is noted on imaging (e.g., ultrasonography)
Hepatic steatosis	Excessive fat accumulation in the liver	Intracellular fat in more than 5% of hepatocytes
NASH	Steatosis with liver cell injury and inflammation	Hepatocyte ballooning, Mallory hyaline, lymphocytic and neutrophilic inflammatory infiltrate in perivenular areas; hepatocyte necrosis, apoptosis, and fibrosis may be present

NASH: Nonalcoholic Steatohepatitis; MASH: Metabolic Dysfunction-Associated Steatohepatitis

Table 5. Risk factors (RF) for MAFLD by APASL 2020

Major risk factor	Common and uncommon risk factor
Overweight/obesity	Gut microbioa
Central obesity	Hyperuricemia
Type 2 Diabetes (T2D)	Hypothyroidism
Dyslipidemia	Sleep apnoea syndrome (SAS)
Hypertension (HTN)	Polycystic ovary syndrome (POS)
Metabolic syndrome (MetS)	Polycythaemia
Insulin resistance	Hypopituitarism
Dietary factors: high-calorie diets rich in saturated fats and cholesterol, soft drinks high in fructose, highly processed foods	Genetic variations: PNPLA3, TM6SF2, GCKR, MBOAT7, and HSD17B13
Sedentary lifestyle or sedentary occupation, low level of physical activity	Epigenetic factors: microRNAs (miR), DNA methylation, histone modification, and ubiquitination alterations
Sarcopenia	A personal or family history of T2D, premature vascular disease, atherogenic dyslipidemia and HTN (MetS), fatty liver

IV. PATHOPHYSIOLOGY

The pathophysiology of MASLD is related to multiple simultaneous risk factors such as the genetic background, the environmental, metabolic, demographic, medication, psychiatric illness and gut dysbiosis that cause fat accumulation (steatosis), inflammation, and, variably, fibrosis (figure 2,3).

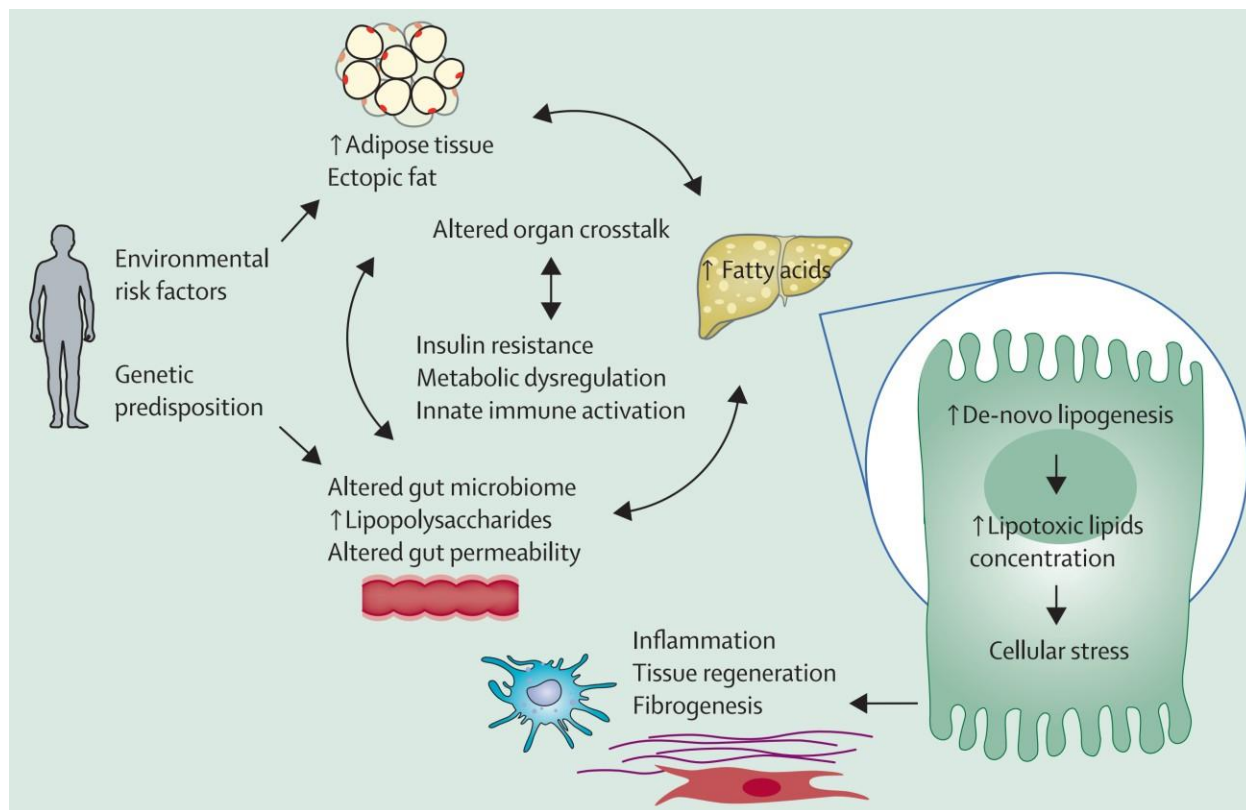


Figure 2. Multiple pathways and interactions between different organs, affect the pathogenesis of NAFLD.

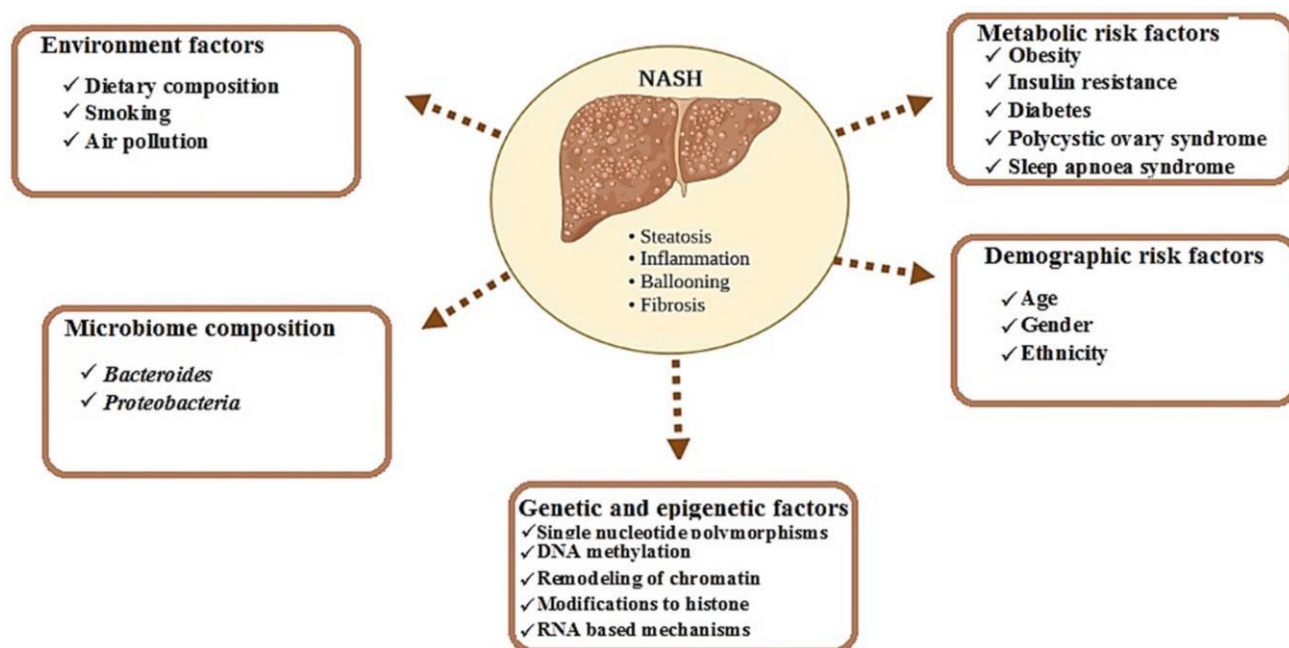


Figure 3. Pathophysiology of MASLD.

V. DIAGNOSIS

1- CLINICAL APPROACH

Symptoms and Signs of MASLD

- Most patients are asymptomatic unless it becomes serious or complicated so the diagnosis should be suspected in patients with CMRF and/or in patients with unexplained laboratory abnormalities suggesting liver disease.
- Some patients : fatigue, malaise, or right upper quadrant abdominal discomfort.
- Hepatomegaly: 75%
- Splenomegaly may develop if advanced hepatic fibrosis is present and is usually the first indication that portal hypertension has developed.
- MASH-related cirrhosis can be asymptomatic and may lack the usual signs of CLD.
- Body fat distribution (eg, android versus gynoid, lipodystrophic),
- Features of insulin resistance (eg, dorsal-cervical fat pad, acanthosis nigricans)
- Looking for signs of associated MetS

2- INVESTIGATIONS

Objectives are for diagnosis and grading MASLD and Assess the comorbidity.

a- Blood tests:

- Cell Blood Counts with platelets, hepatic panels, lipid profile, glycemia and A1c, creatinine and urine microalbumin or protein to creatinine ratio; HBsAg, HBsAb, HBcAb total and Anti-HCV

b- Imaging tests

- Abdominal Ultrasound:
 - the first-line imaging test for NAFLD assessment due to its low-cost, wide commercial availability and safety.
 - evidence of steatosis (sensitivity of 85%; specificity of 94%)
- Vibration-controlled Transient Elastography (VCTE) (FibroScan) gives a 'liver stiffness measurement' (LSM) using pulsed-echo ultrasound as a surrogate marker of fibrosis.
- Controlled Attenuation Parameter (CAP) is a new ultrasound-based technique to measure steatosis with assessment of liver stiffness using transient elastography.
- Other ultrasound-based methods implementable with common ultrasound devices are two-dimensional shear wave elastography and point shear wave elastography.
- MR elastography (MRE)
- Other imaging modalities such as CT or MRI can also detect hepatic steatosis, but they are not routinely used in the assessment of steatosis.

The sensibility and specificity of imaging Tests for NALFD are shown in table 6.

Table 6. Imaging Tests for the Evaluation of NAFLD

Test	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	LR+*	LR-*
Contrast-enhanced CT	84 to 87	75 to 86	59 to 72	92 to 94	1.4 to 2.7	0.06 to 0.09
Magnetic resonance imaging	96	93	85	98	5.9	0.02
Ultrasonography	60 to 100	77 to 95	52 to 89	82 to 100	1.1 to 8.6	∞ to 0.2
Unenhanced CT	88 to 95	90 to 99	79 to 98	95 to 98	3.7 to 40.7	0.02 to 0.05

CT: computed tomography; **LR-:** negative likelihood ratio; **LR+:** positive likelihood ratio; **NPV:** negative predictive value; **PPV:** positive predictive value. *—Calculated using a prevalence of 30%.

c- Liver biopsy (LB)

LB remains the gold standard for characterizing liver histological alterations in patients with NAFLD. However, LB is expensive, requires expertise for interpretation, and carries some morbidity and very rare mortality risk. Thus, it should be performed in those who would benefit the most from diagnostic, therapeutic guidance, and prognostic information.

Indications of Liver biopsy (LB):

- LB should be considered in patients with MASH who are at increased risk of having MASH and/or advanced fibrosis.
- LB should be considered in patients with suspected NAFLD in whom competing etiologies for MASH and the presence and/or severity of coexisting CLDs cannot be excluded without a LB.

3- LIVER FIBROSIS ASSESSMENT:

- The presence of MetS, NFS or FIB-4, or liver stiffness measured by Fibroscan or MRE may be used for identifying patients who are at risk for MASH and/or advanced fibrosis (figure 6)

There are a lots of Non-invasive fibrosis scores for liver fibrosis assessment but FIB-4 is the most commonly used (table 6).

Table 7. Noninvasive Tests (NITs) for Advanced Fibrosis in Patients with NAFLD.

Test	Cutoff	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	LR+*	LR-*
AST/ALT ratio ²²	0.8	74	78	59	88	1.4	0.1
AST/platelet count ratio ²²	1	27	89	51	49	1.1	0.4
BARD score ²²	2	89	44	41	90	0.7	0.1
Enhanced liver fibrosis panel ²³	10.5	100	98	96	100	21	0 to ∞
FIB-4 ²²	1.3	85	65	51	91	1.0	0.1
FibroTest (FibroSure) ²⁴	0.7	15	98	76	73	3.2	0.3
Fibrometer ²⁵	0.8	79	96	89	91	8.5	0.09
Magnetic resonance elastography ²⁶	2.74 kPa	94	73	60	97	1.5	0.04
Nonalcoholic fatty liver disease fibrosis score ²²	-1.5	78	58	44	86	0.8	0.2

ALT: alanine transaminase; **AST:** aspartate transaminase; **BARD:** Body mass index, AST/ALT ratio, diabetes mellitus; **FIB-4,** Fibrosis Index Based on 4 Factors (Age, AST, ALT and Platelets) **kPa:** kilopascal; **LR-:** negative likelihood ratio; **LR+:** positive likelihood ratio; **NPV:** negative predictive value; **PPV:** positive predictive value; *—Calculated using a prevalence of 30%.

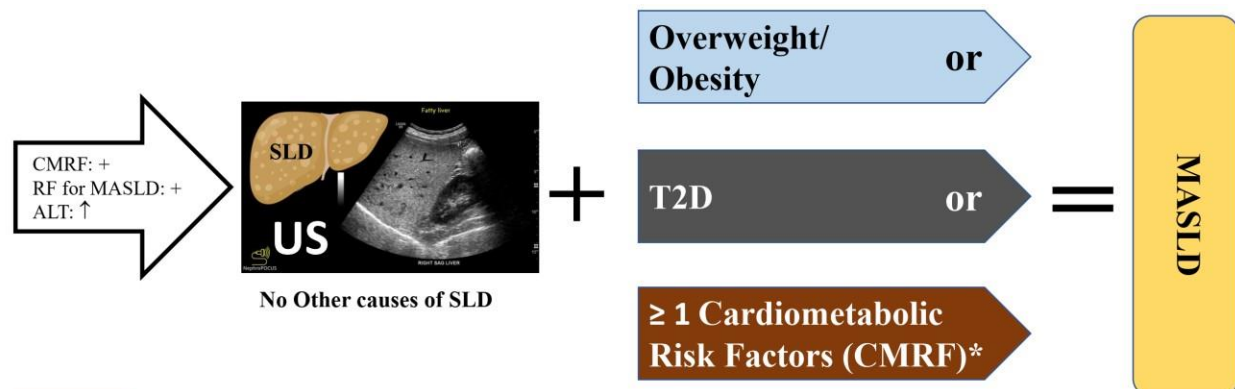
- Assess the coexistence and severity of CMRF and Risk factors of MASLD

Risk factors and Comorbidities that are the greatest impact on MASLD => Risk of progressive Fibrosis and the development of Cirrhosis and its Complications, HCC.

- T2D and Obesity (particularly abdominal obesity)
- Individuals with multiple CMRF
- Concurrent alcohol intake and/or smoking
- Individuals with FIB-4 >3.25/ LSM >10 kPa and increasing change in LSM over time
- Environmental factors : diet, gut microbiota
- Males aged >50 years, postmenopausal women

In summary: The diagnosis and staging of MASLD

- History (presence of risk factors, absence of excessive alcohol intake)
- Weight history; medical comorbidities; recent & current medications;
- Screening for Obstructive Sleep Apnea (OSA)
- Family history of T2D, NAFLD, or cirrhosis
- Blood tests mentioned above
- To rule out differentials based on clinical settings.
- Ultrasound evidence of steatosis or MRE with fat fraction
- Liver fibrosis assessment (Step 1: FIB-4, or + Fibroscan if FIB-4 ≥ 1.3)
- MetS component assessment
- Looking for Risk factors and comorbidities that are greatest impact on MASLD.



CMRF:*

- BMI ≥ 25 kg/m² [23 Asia] OR WC > 94 cm (M) 80 cm (F) OR ethnicity adjusted equivalent
- FSG ≥ 5.6 mmol/L [100 mg/dL] OR 2-hour post-load glucose levels ≥ 7.8 mmol/L [≥ 140 mg/dL] OR HbA1c $\geq 5.7\%$ [39 mmol/L] OR T2D OR ttt for T2D
- Blood Pressure $\geq 130/85$ mmHg OR specific antihypertensive drug ttt
- Plasma TG ≥ 1.70 mmol/L [150 mg/dL] OR lipid lowering ttt
- Plasma HDL-cholesterol ≤ 1.0 mmol/L [40 mg/dL] (M) and ≤ 1.3 mmol/L [50 mg/dL] (F) OR lipid lowering treatment.

Kanwal F, et al. AASLD. Hepatology. 2024
APASL CPG. Hepato Inter 2020. 14:889–919
Cameron Gofton, et al. Clin Mol Hepatol 2023;29(Suppl):S17-S31

BMI: Body mass index
FSG: Fasting plasma glucose
US: Ultrasound
SLD: Steatotic liver disease
TG: triglycerides
T2D: Type 2 Diabetes;
ttt: treatment
WC: Waist circumference

Figure 4. Diagnosis of MASLD.

WHO TO SCREEN FOR MASLD?

- Patients with CMRF/ Risk factors for MASLD.
- and/or in patients with unexplained laboratory abnormalities suggesting liver disease.

VI. DIFFERENTIAL DIAGNOSIS

Not all SLD are related to MASLD because it might be due to secondary causes whose prognosis and treatment are different so it is important to rule them out.

Table 8. Differentials of MASLD

▪ Excessive alcohol consumption ▪ Hepatitis C (genotype 3) ▪ Lipodystrophy ▪ Acute weight loss (bariatric surgery and starvation) ▪ Malnutrition ▪ Parenteral nutrition ▪ Endocrine diseases (hypothyroid, PCOS)² ▪ Coeliac disease² ▪ Abetalipoproteinemia ▪ Reye syndrome ▪ Pregnancy associated: ○ <i>HELLP syndrome</i> ○ <i>Acute fatty liver of pregnancy</i> ▪ Medications (eg, corticosteroids, mipomersen, lomitapide, amiodarone, methotrexate, tamoxifen, valproate, and antiretroviral medicines) ▪ Rare causes: AIH, A1AT deficiency, Wilson syndrome, and other
--

AIH: Autoimmune hepatitis; AMA: Antimitochondrial antibodies; ANA: antinuclear antibodies;

ASMA: Anti-smooth muscle antibody; A1AT: alpha-1 antitrypsin;

HELLP: Hemolysis, Elevated Liver enzymes and Low Platelets; PCOS: Polycystic ovary syndrome

VII. MANAGEMENT

1- Objectives

- To stop or slow down the progression of MASLD
- To Assess Risk stratification of patients with MASLD
- To cure the patients with MASLD
- To screen and treat the components of MetS
- To avoid the complications of MASLD

2- Critical Signs

- Advanced fibrosis/ cirrhosis/ HCC
- Multiple CMRF

3- Mortality risk factor?

- Old age
- Uncontrolled comorbidity (T2D, HTN,...)
- Late diagnosis, noncompliance.

4- Outpatient Management

As the majority of individuals with MASLD are asymptomatic, most of them are managed as outpatients unless they become decompensated cirrhosis or complicated course.

5- First-line Treatment:

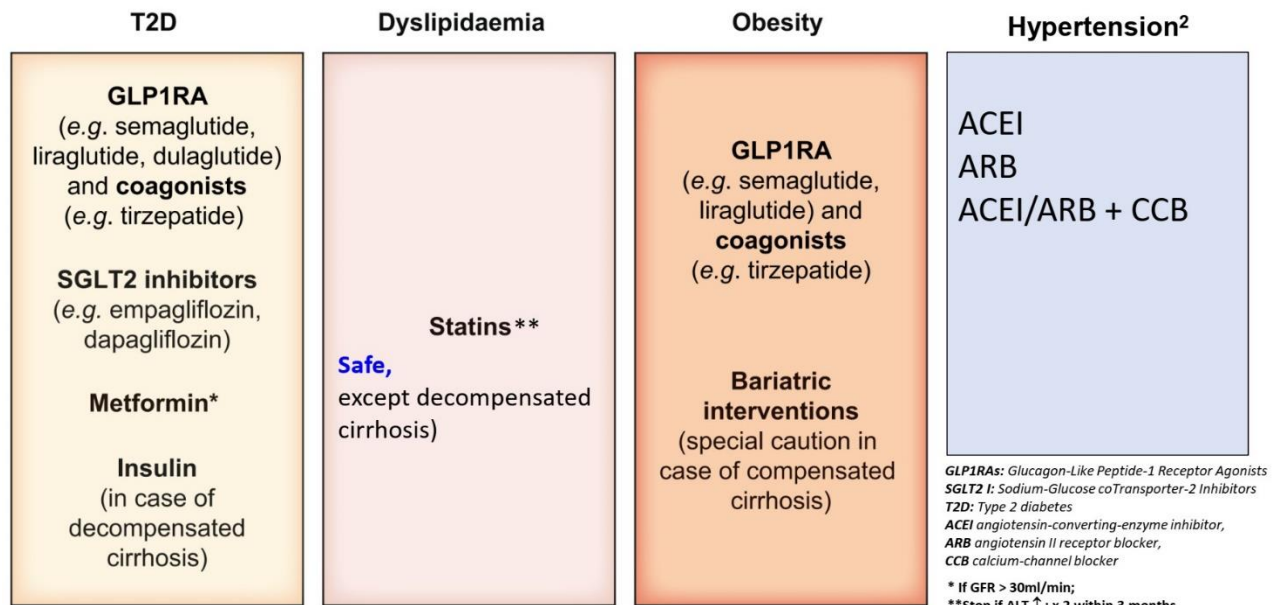
a- Lifestyle modification

Lifestyle and dietary modifications are the cornerstone of managing and even reversing MASLD.

- Dietary modifications
 - Weight loss and energy restriction: Diets with less calorie for a goal of 500 fewer calories daily (normal, around 1,600 to 2,000 calories), beneficial for steatosis and fibrosis.
 - Avoid highly processed snacks, fried foods, and fatty foods.
 - Avoid juice or sugary drinks. Water is best.
 - Avoid foods with added sugar including candy, desserts, and soda pop, especially high fructose corn syrup.
 - Coffee consumption (≥ 3 cups): prevent or revers liver fibrosis.
 - Avoid alcohol intake/smoking
 - Mediterranean diet.
- Exercise (30–45 mins thrice weekly/ 150 to 300 mins of exercise per week/ a walk of 10,000 steps a day)
- Reduce sedentary lifestyle
- Avoid too rapid weight loss ($>1.6\text{kg/week}$) (to prevent worsening liver damage)

b- Pharmacological Therapy for Metabolic syndrome (figure 5)

- Antidiabetic, Antihypertensive, Lipid lowering and Anti-obesity agents.



*if glomerular filtration rate >30 ml/min

2- Philipp Kasper et al. *Clinical Research in Cardio* (2021) 110:921–937

Figure 5. Treatment of co-morbidities (MetS) with secondary benefits on the liver

c- MASH-targeted Therapy

▪ Resmetirom

Resmetirom is the first drug that the U.S. Food and Drug Administration approved (march 14, 2024) for the treatment of adults with MASH with moderate to advanced liver scarring (fibrosis F2-F3), to be used along with diet and exercise.

It is a selective thyroid hormone receptor-beta (THR- β) agonist that targets liver fat metabolism.

Mechanism of Action

- Activates thyroid hormone receptor-beta (THR- β) in the liver.
- Increases fat metabolism, reducing liver fat accumulation.
- Decreases liver inflammation and fibrosis, slowing disease progression.
- Improves cholesterol levels, lowering LDL (“bad” cholesterol).

Indications

- NASH with moderate or advanced liver scarring (Not allowed in decompensated cirrhosis or Child Pugh B or C)

Dosage and duration (12-54 months):

- Body weight <100 kg: 80mg, PO, once daily
- Body weight $\geq$ 100 kg: 100mg, PO, once daily

6- Treatment adjustment:

There are lots of drugs with off label uses/or current clinical trials for MASLD:

- a- Ursodeoxycholic acid (UDCA): 15mg/kg/day

MASLD. Beneficial for liver enzymes.

- b- Vitamin E (Anti-oxidant): 800 IU daily

NASH without D2T or cirrhosis

Side Effect: Cardiovascular mortality, Stroke, Prostate cancer

- a- Omega-3 (PUFAs): Hepatic anti-inflammatory and insulin-sensitizing effects

MASLD + • Triglyceride. Randomized controlled trials: not improve Fibrosis)

- b- Metformin: • Insulin sensitivity, • ALT.

T2D with MASLD.

Benefit: • Transplant-free survival and • risk of HCC and extrahepatic cancers. c-

- c- Pioglitazone (PPARs): 30-40 mg, PO

NASH with and without T2D

Side Effect: Weight gain, Heart failure exacerbation, bone loss

- d- Dapagliflozin (SGL-2i):

D2T and MASLD

Side Effect: Genitourinary yeast, volume depletion, bone loss

- e- Silymarin: a potent liver-tropic antioxidant extracted from milk thistle (*Silybum marianum*). Silymarin has multiple hepatoprotective effects, including antioxidant activity, cell regeneration, increased protein synthesis, and anti-inflammatory and antifibrotic properties.

Dosage: 140–420 mg per day, divided into 2–3 doses.

Beneficial for liver enzymes and improve liver histology

None of these medications are FDA approved for treatment of MASH but for associated comorbidities with potential benefit in MASLD.

7- Algorithm for the Risk stratification and Treatment of MASLD.

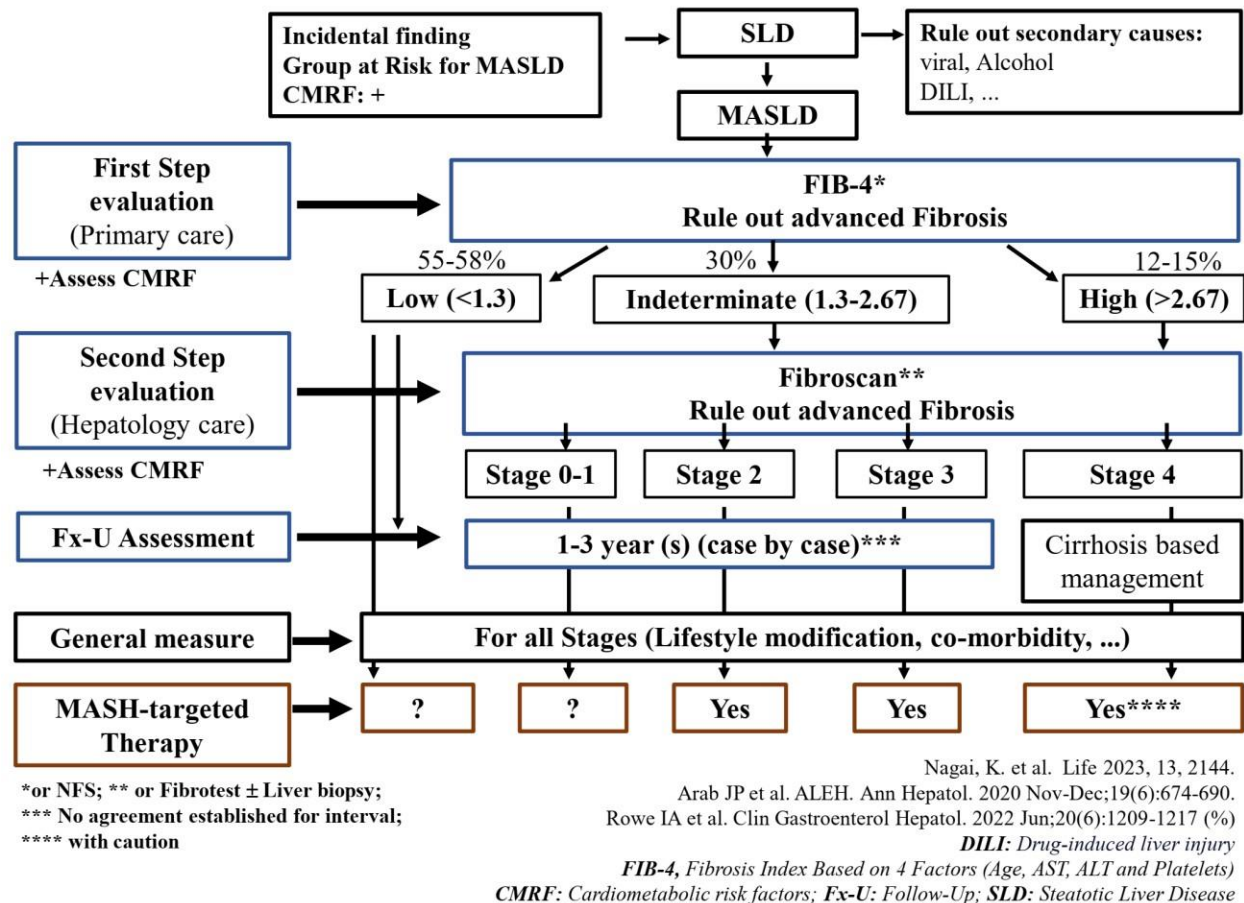


Figure 6. Algorithm for the risk stratification and treatment of MASLD.

VIII. SPECIAL CASES

- 1- MASLD with cirrhosis: Refer to hepatologist (gastroscopy, screening for HCC, ...)
- 2- MASLD with Morbid obesity: MDT is needed including dietician.
- 3- MASLD with multiple and severe MetS: MDT is needed (hepatologist, endocrinologist, cardiologist, dietician)
- 4- Obstructive Sleep Apnea (OSA): Continuous positive airway pressure (CPAP).

IX. EVOLUTION AND PROGNOSIS

The evolution of MASLD varies through individuals or comorbidity.

Regarding liver prognosis, most MASLD (80%) progress slowly or stably and only 20% of cases might develop serious or life-threatening (MASLD can progress to MASH, liver cirrhosis, and even HCC) but NASH or early fibrosis might be reversible with appropriate care (figure 7).

In addition to liver-related complications, patients with SLD are also at increased risk for developing MetS, CV complications, T2D, chronic kidney disease (CKD) and malignancy (hepatic and extrahepatic cancers).

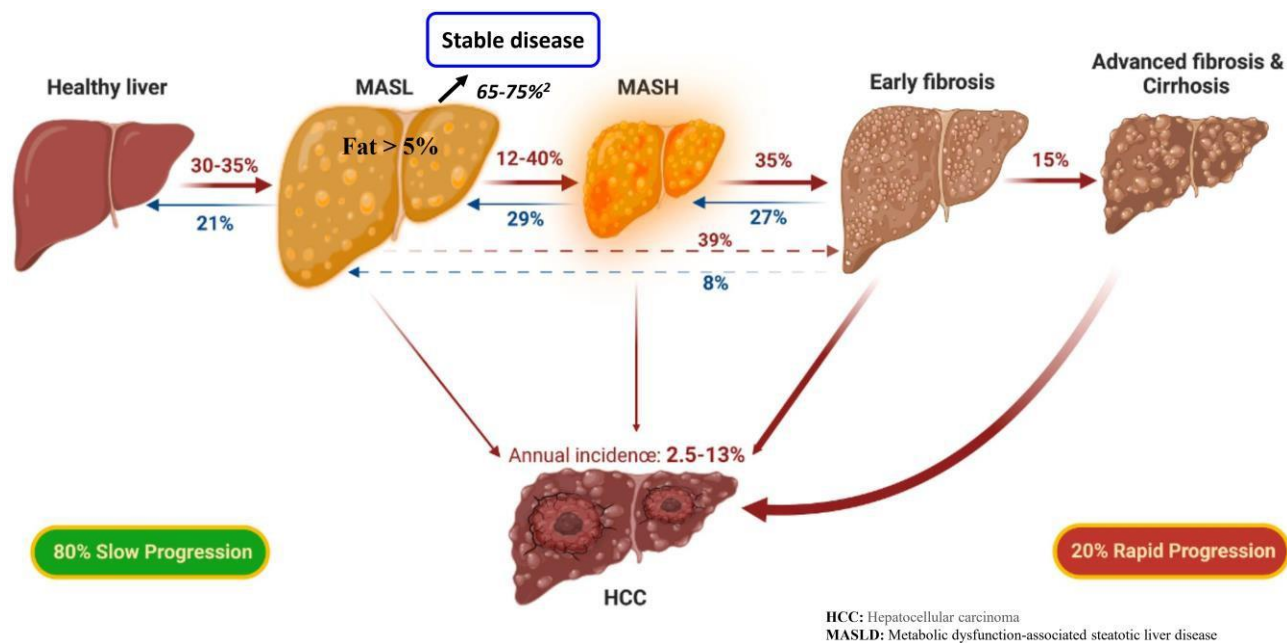


Figure 7. Natural history of MASLD

X. COMPLICATIONS

As MASLD is a multisystem disease so its complications are multiple (MASLD, a Wolf in Sheep's Clothing) (figure 8).

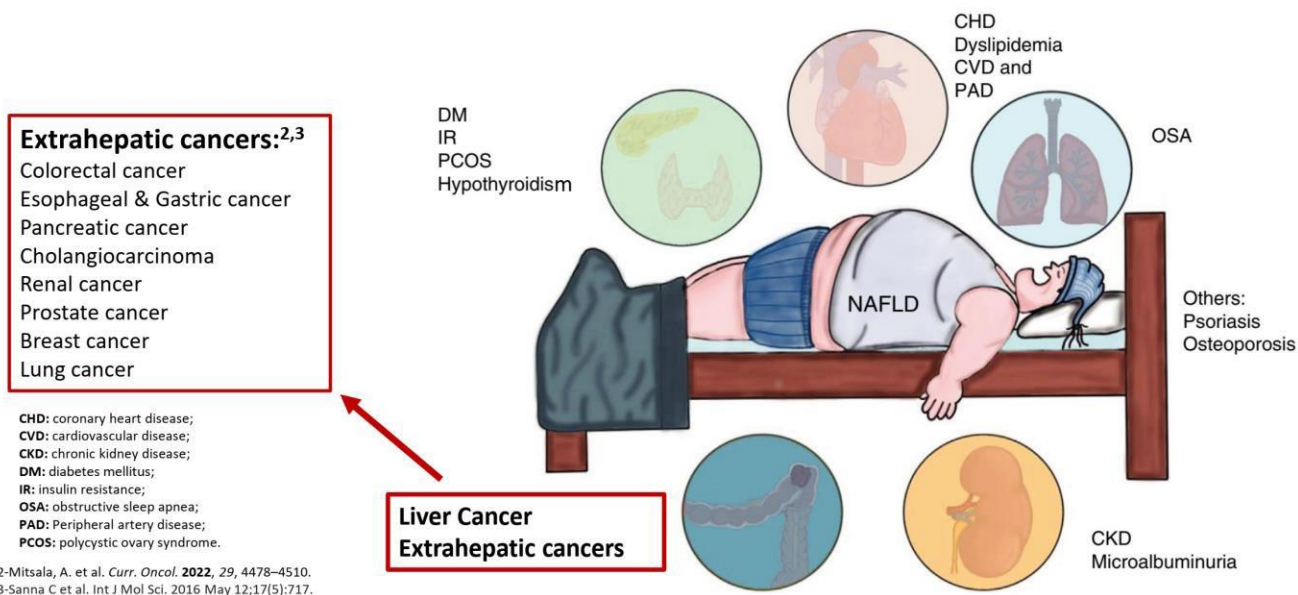


Figure 8. Diseases associated with the multi-systemic involvement present in NAFLD.

1- Hepatic complications

- Cirrhosis
- HCC
- Liver failure

2- Extrahepatic complications

- Cardiovascular disease
- Chronic kidney disease
- T2D, Hypothyroidism
- Extrahepatic cancers: cancer of digestive tract, breast, lung, kidney and prostate.

XI. PREVENTION AND RECOMMENDATION

Healthy lifestyle and Healthy diets and strategies to avoid or control metabolic syndromes.

1- Focus on Weight Loss (If overweight)

2- Follow a Liver-Friendly Diet

- Fresh fruits, vegetables, whole grains, nuts, seeds, and legumes.
- Lean proteins like fish, chicken, eggs, and plant-based sources (tofu, lentils, beans).
- Healthy fats like olive oil, avocado, and omega-3-rich foods (salmon, walnuts).

3- Exercise Regularly

4- Manage Blood Sugar, Cholesterol and HTN

- T2D or prediabetes: keep blood sugar under control.
- Dyslipidemia: monitor and manage through diet and medication if needed.
- HTN: monitor and manage through diet and medication.

5- Alcohol and Smoking cessation:

- Individuals with significant hepatic fibrosis ($\geq$ F2) or CMRF: abstain from alcohol use completely.

6- Get Enough Sleep

- Aim for 7–9 hours of quality sleep per night.
- Poor sleep can worsen insulin resistance and liver fat accumulation.

7- Reduce Stress

8- Avoid Unnecessary Medications

9- Regular Liver Check-Ups

10- Be Consistent & Patient

- NAFLD takes time to improve—stick to lifestyle changes consistently.
- Even small improvements in diet and activity can make a big difference.

Summarized advice:

- Eat clean, move more, and maintain a healthy weight.
- Stay away from sugar, salt, alcohol, and processed foods.
- Monitor liver health regularly.

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CIRRHOSIS AND ITS COMPLICATIONS

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I. DEFINITION

Cirrhosis is defined as a late stage of progressive hepatic fibrosis characterized by distortion of the hepatic architecture and the formation of regenerative nodules.

II. EPIDEMIOLOGY

The worldwide prevalence of cirrhosis is unknown; however, it has been estimated to be between 0.15% and 0.27% in the United States. Mortality and morbidity across the world. It is the 11th leading cause of death, but it is the 4th most common cause of death in Europe, and 15th leading cause of morbidity, accounting for 2.2% of deaths and 1.5% of disability-adjusted life years worldwide in 2016.

III. ETIOLOGY

The etiology of cirrhosis can usually be identified by the patient's history combined with serologic evaluation. The most common causes are chronic hepatitis B, C, alcohol, and Metabolic dysfunction associated fatty liver disease (MAFLD) which represent more than 80%. The other rare causes are auto-immune hepatitis, iron overload...

IV. PATHOPHYSIOLOGY

- 1) Irreversible chronic injury of hepatic parenchyma (in consequence of an exogenous/toxic, infectious, toxic/allergic, immunopathological/autoimmune, or vascular process or an inborn error of metabolism).
- 2) Extensive fibrosis - distortion of the hepatic architecture (repair cellular proliferation and regeneration then formation of fibrous scar).
- 3) Formation of regenerative nodules (process of fibrosis becomes irreversible and leads to cirrhosis).

V. DIAGNOSIS

Cirrhosis can be classified as compensated and decompensated.

Compensated cirrhosis is an early-stage cirrhosis, this stage is often asymptomatic, unsuspected. The diagnosis is made only by noninvasive test (Fibrotest or Fibroscan) or invasive test (Liver biopsy). Ultrasonography cannot be used for positive diagnosis.

Decompensated cirrhosis can be diagnosed by combining clinical, biological, endoscopic, and imaging findings. Liver biopsy is no longer used.

1. Clinical manifestation

Clinical manifestation of decompensated cirrhosis related to impaired hepatocellular function (Liver failure) and portal hypertension.

- *Impaired hepatocellular function (Liver failure):* Asthenia, Jaundice, Spider angiomas, Palmar erythema, Digital clubbing, White nails, Gynecomastia, Testicular atrophy, Feter hepaticus.
- *Portal hypertension:* Feet edema, Ascites, Collateral veins visible or Cruveilhier-Baumgarten Syndrome (Caput medusae), Splenomegaly, Upper gastrointestinal bleeding (esophageal or gastric varices bleeding).

2. Laboratory finding
 - Anemia, leucopenia, especially thrombocytopenia, AST/ALT>1, ↑ Bilirubin. Prothrombin time <70%, hypoalbuminemia.
3. Imaging finding
 - Ultrasonography showed liver dysmorphia (atrophic right lobe and hypertrophic left lobe), ascites, dilated portal vein and sometime, liver nodules
4. Endoscopic finding
 - Most presence of esophageal varices and/or gastric varices, portal hypertensive gastropathy.

Assessment of the Severity of Cirrhosis

For many years, the severity and most common prognosis of decompensated cirrhosis by Child-Pugh Score. Table 1.

Table 1: Child-Pugh Score

Clinical Variable	1 Point	2 Points	3 Points
Encephalopathy	None	Grade 1-2	Grade 2-3
Ascites	Absent	Slight	Moderate or large
Bilirubin (mg/dL)	< 2	2-3	> 3
Albumin (g/dL)	> 3.5	2.8-3.5	< 2.8
Prothrombin time (INR)	INR < 1.7	INR 1.7-2.3	INR > 2.3

- Class A = 5 to 6 points
 - least severe liver disease
 - one to five-year survival rate: 95%
- Class B = 7 to 9 points
 - moderately severe liver disease
 - one to five-year survival rate: 75%
- Class C = 10 to 15 points
 - most severe liver disease
 - one to five-year survival rate: 50%

VI. DIFFERENTIAL DIAGNOSIS

Differential diagnosis of decompensated cirrhosis according to the consequences that occur in each category.

VII. MANAGEMENT

Currently, the only curative treatment for decompensated cirrhosis is a **liver transplant**, but this procedure cannot be carried out in all patients. Liver transplantation cannot be performed in all cases for two main reasons: (1) the number of donors is limited and (2) the possible existence of other associated diseases that contraindicate transplantation.

Our goal of treatment is to prolong survival time.

1. Objectives

- To cure the causes of cirrhosis and to reverse the compensated cirrhosis.
- To do the primary preventions of its complications.
- To find and treat early the complications.

2. Critical Sign

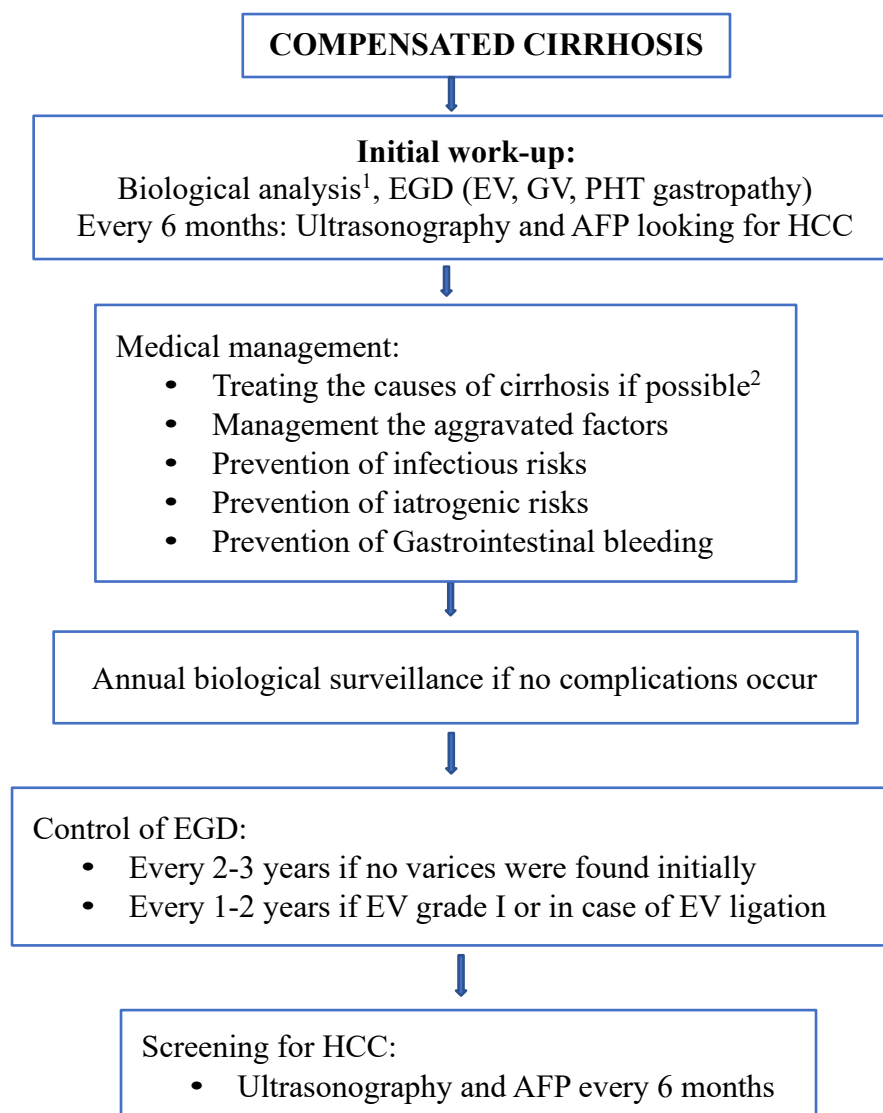
- Refractory ascites
- Hypervolemic / Dilutional hyponatremia
- Somnolent or Coma

- Upper Gastrointestinal bleeding
3. Mortality risk factor
- Elderly patient (> 65 years).
 - Child–Pugh C
 - Presence of hepatocellular carcinoma
 - Presence of hepatorenal syndrome
4. Outpatient Management
- Managing the causes, if possible, to slow or reduce the cellular damage
 - Abstinence alcohol consumption
 - Treatment hepatitis B, C
 - Managing metabolic factors like cholesterol, blood sugar and overweight
 - Prophylactic treatment
 - Vaccinations: Hepatitis A and B vaccine, Pneumococcal vaccine, Influenza vaccine.
 - Avoidances: NSAIDs and Nephrotoxic drugs, Herbal medications.
5. First line treatment
- The choice of a specific regimen for empiric therapy should be based upon:
- Etiology of cirrhosis
 - Primary prevention strategy
 - Complication of cirrhosis
6. Treatment adjustment
- Salt restriction $\leq 3\text{g/day}$ (1 coffee spoon)
 - Maintain healthy foods. In addition, some patients need dietary supplements to treat nutritional deficiencies.

Follow-up:

- Clinical signs: Find Complications.
- Biology: Child-Pugh every 3 to 6 months
- Screening for HCC: Ultrasonography and AFP every 6 months
- Control of Esophagogastroduodenoscopy (EGD):
 - Every 2 - 3 years, if no varices were found initially.
 - Every 1 - 2 years, if Esophageal Varices (EV) Grade I or in case of EV ligation.

7. Algorithm



¹ Complete blood count, protein C reaction, fasting blood sugar, transaminases, alkaline phosphatase, gamma GT, total bilirubin, prothrombin time, albumin, urea, creatinine, and electrolytes

²Treating hepatitis B, Hepatitis C, MAFLD, stop alcohol ...; treat aggravating causes like obese, diabetes... etc.

EGD: Esophagogastrroduodenoscopy

EV: Esophageal varices

GV: Gastric varices

PHT: Portal hypertension

AFP: Alpha-fetoprotein

HCC: Hepatocellular carcinomas

VIII. SPECIAL CASES

Once a patient develops complications of cirrhosis, they are considered to have decompensated disease. The high morbidity and mortality of cirrhosis is secondary to these devastating complications.

The quality of life and survival of patients with cirrhosis can be improved by the prevention and treatment of these complications.

1- Ascites

Ascites is the accumulation of fluid within the peritoneal cavity. It is the most common complication of cirrhosis. The first step leading to fluid retention and ultimately ascites in patients with cirrhosis is the development of portal hypertension.

Patients without portal hypertension do not develop ascites or edema. Those with ascites have several circulatory, vascular, functional, and biochemical abnormalities that contribute to the pathogenesis of fluid retention. We recommend the treatment of ascites as follow:

- Education about the importance of dietary sodium restriction is a central component of the management of cirrhotic ascites. Those who understand and follow the diet usually do not require readmission for fluid overload; in comparison, noncompliant patients often require multiple hospitalizations.
- Patients should be instructed to avoid any nephrotoxic drugs: nonsteroidal anti-inflammatory drugs (NSAIDs), which can cause sodium retention.
 - We suggest **3g per day sodium diet** (Grade 2C) (1 coffee spoon of powder salt).
 - In patients who require diuretic therapy, we suggest combination therapy with a single morning oral dose of **spironolactone** and **furosemide**, beginning with **100mg and 40mg**, respectively (Grade 2B).
 - In small patients with a small volume of ascites, we use lower doses (eg, 50mg of spironolactone and 20mg of furosemide).
 - This combination in this dose ratio usually maintains normokalaemia. The doses can be doubled if a clinical response is not evident.
 - **The maximum recommended doses are spironolactone 400mg/day and furosemide 160mg/day.**
 - The long half-life of spironolactone makes single-daily dosing most appropriate. Giving both drugs once per day in the morning maximizes compliance and minimizes nocturia.
 - In some cases, we cannot use any spironolactone, especially when the glomerular filtration rate is very low, or the patient develops hyperkalemia.
 - We suggest that intravenous furosemide be avoided unless there is no other option, (eg, pulmonary edema in a patient taking nothing by mouth).
 - Oral furosemide is absorbed well in the setting of cirrhosis, and intravenous furosemide may cause an acute reduction in renal function and can lead to crescendo azotemia, which can then masquerade as hepatorenal syndrome.
 - **Patients must be reevaluated rapidly with daily weights, urinary volume/24h, abdominal perimeter in the hospital or at home.** After discharge, the first clinic visit should be in less than one week. Once patients lose their sodium avidity (usually because of abstinence of healing of the reversible component of their liver disease), we begin tapering diuretics, reducing to a final dose of 50 mg of spironolactone plus 20 mg of furosemide before discontinuing altogether.

- **Paracentesis** must be done when ascites become refractory to optimal medical treatments and when the patient becomes intolerated. The volume can be extracted as much as possible. We must give albumin or other colloid when the volume extracted is over 2L (**500 ml of colloid/L of ascites or 10 g of albumin/L of ascitic fluid removed**). Spontaneous bacterial peritonitis must be diagnosed or ruled out each time when perform paracentesis.

2- Spontaneous bacterial peritonitis

Spontaneous bacterial peritonitis (SBP) is an ascitic fluid infection without an evident intra-abdominal surgically treatable source; it primarily occurs in patients with advanced cirrhosis.

The syndrome arises most commonly in patients whose low-protein ascites (< 1 g/dL) contains low levels of complement, resulting in decreased opsonic activity. SBP appears to be caused by the translocation of gastrointestinal tract bacteria across the gut wall and by the hematogenous spread of bacteria. The most common causative organisms are *Escherichia coli*, *Streptococcus pneumoniae*, *Klebsiella* species, and other gram-negative enteric organisms.

Clinical manifestations of SBP include fever, abdominal pain, abdominal tenderness, and altered mental status. Some patients are asymptomatic and present with only mild laboratory abnormalities.

The index of suspicion for SBP must be high with a low threshold for diagnostic paracentesis. The diagnosis is established by a positive ascitic fluid bacterial culture and/or an elevated ascitic fluid absolute polymorphonuclear leukocyte (PMN) count (≥ 250 cells/mm³).

Without early treatment, mortality is high. Efforts to prevent SBP should be made in high-risk patients.

The most used regimen in the treatment of SBP is a 5-day course of **cefotaxime** at 1 - 2g intravenously every 8 hours, or **ceftriaxone** (1g two times /day IV).

Alternative treatments include **ofloxacin** (200mg two times/day IV), **ciprofloxacin** (500 mg, two times /day IV), **amoxicillin–clavulanic acid** (1g, three times/day IV) can be used immediately after having the result of ascitic fluid analysis (**PMN > 250 /mm³**).

The duration of treatment is 7 days and must be continued by oral treatment to prevent reinfection (secondary prevention). Repeat paracentesis in 48-72 hours to document a decrease in the ascites PMN count 25-50% to ensure efficacy of therapy.

Albumin must be administered on day 1 and day 3 for renal protection (**1.5g/kg on day 1 and 1g/kg on day 3**).

3- Hepatic encephalopathy

Hepatic encephalopathy (HE), a syndrome observed in some patients with cirrhosis, is marked by personality changes, intellectual impairment, and a depressed level of consciousness.

Evaluation for precipitating causes — There are several conditions that may precipitate an episode of hepatic encephalopathy in patients with liver disease or a portal-systemic shunt. These include:

- Gastrointestinal bleeding
- Infection (including spontaneous bacterial peritonitis and urinary tract infections)
- Hypokalemia and/or metabolic alkalosis
- Renal failure
- Hypovolemia
- Hypoxia
- Sedative or benzodiazepine use

- Hypoglycemia
- Constipation
- Rarely, hepatocellular carcinoma and/or vascular occlusion (hepatic vein or portal vein thrombosis).

Clinical manifestations and grading

Table 2: Clinical manifestations and grading of HE

Grade	Mental Status	Neurologic Signs
1	Inverted sleeping habits, Irritability, Mild confusion	Incoordination, Tremor
2	Drowsiness, Lethargy or Apathy, Disorientation, Mild amnesia, Moderate confusion	Asterixis, Ataxia, Dysarthria
3	Somnolence, Severe confusion Gross disorientation	Muscle rigidity, Clonus, Hyperreflexia
4	Coma	No response to pain, Decerebrate posturing

Management of hepatic encephalopathy includes:

- Treatment of precipitating causes,
- Symptomatically treatment: *Agitation*; hepatic encephalopathy patients are always agitated. Management may include judicious use of restraints, which may be a safer option than pharmacologic treatment since patients with advanced liver disease and HE may be particularly vulnerable to oversedation with medications. Should medications be required, **haloperidol (Hadol®: 5 -20 mg IM or IV), Hydroxyzine (Atarax®:100 -300 mg IM or IV after diluted with 10 ml of Normal Saline Solution)** is a safer option than benzodiazepines based mainly on clinical experience and some limited data.
- Treatment is aiming at reducing **elevations of ammonia**
- Lactulose (Duphalac®): 1 sachet = 10g/15ml
 The dose of **lactulose (40 to 90 g/day) (Duphalac®)** should be titrated to achieve two to three soft stools per day with a pH below 6. Approximately 70 to 80 percent of patients with hepatic encephalopathy improve on lactulose treatment. Treatment is usually well tolerated, and the principal toxicity is abdominal cramping, diarrhea, and flatulence.
 The treatment can be given orally (if possible) or by rectal route (Enema).
 - Oral route or by Nasogastric tube: Duphalac®:6 -10 sachets diluted in water.
 - Rectal route: 20 sachets of Duphalac® diluted in 700 ml of water and kept 20 mn to 1 hour, repeated every 2 hours until the episode resolves.
- The patients should eat high protein food, especially vegetable proteins.
- **Antibiotherapy:** non absorbable antibiotics: metronidazole, oral vancomycin (be careful nephrotoxic), and rifaximin (400 mg taken orally three times daily or 550 mg taken orally two times a day, **NOT Rifampicine for anti-TB drug!!**).
- In case of hepatic encephalopathy by benzodiazepines **IV flumazenil (Anexate®: 0,2mg IV within 15 second)** might be beneficial. It is not indicated systematically for other types of HE.
- Treatment of HE with **L-ornithine-L-aspartate (Ornicétil® 5g: 4 -10 vials per 24 hours after diluting in NSS or D5%)**. The maximum infusion rate is 5 g of L-Ornithine L-Aspartate (corresponding to content of 1 ampoule) per hour.). in patients with cirrhosis markedly improved quality of life and was well tolerated.

- **Chronic therapy:** In patients with recurrent type chronic encephalopathy, we suggest continuous administration of lactulose (Grade 2B). Protein restriction is not needed unless encephalopathy is resistant to lactulose. In patients whose symptoms worsen with protein intake, substitution of proteins from fish, milk, or meat with vegetable proteins may improve nitrogen balance. Another alternative for patient intolerant to protein is the addition of branched-chain amino acids to a low protein diet. Branched-chain amino acids (BCAA) supplementation is indicated only in severely protein-intolerant patients.

4- Hepatorenal syndrome

Hepatorenal syndrome (HRS) is one of many potential causes of acute kidney injury in patients with acute or chronic liver disease. Affected patients usually have portal hypertension due to cirrhosis, severe alcoholic hepatitis, or (less often) metastatic tumors, but can also have fulminant hepatic failure from any cause. The hepatorenal syndrome represents the end-stage of a sequence of reductions in kidney perfusion induced by increasingly severe hepatic injury.

Precipitant factors

- Low mean arterial blood pressure (< 80 mm Hg),
- Low serum sodium (Dilutional hyponatremia),
- Severe urinary sodium retention (Urine sodium < 5 mEq/L),
- Spontaneous Bacterial Peritonitis (SBP),
- Acute alcoholic hepatitis,
- Large-volume paracentesis without albumin replacement,

Based upon the rapidity of the decline in kidney function, two forms of HRS have been described:

- **HRS-AKI (type 1 hepatorenal syndrome)** – This more serious type of hepatorenal syndrome is referred to as hepatorenal syndrome-acute kidney injury (HRS-AKI) or, traditionally, type 1 hepatorenal syndrome. It is defined as at least a twofold increase in serum creatinine (reflecting a 50 percent reduction in creatinine clearance) to a level greater than 2.5 mg/dL (221 micromol/L) during a period of less than two weeks. At the time of diagnosis, some patients with type 1 hepatorenal syndrome have a urine output of less than 400 to 500 mL per day.
 - HRS-AKI has a median survival of 2 weeks, few patients surviving more than 10 weeks.
- **Diuretic-resistant ascites (type 2 hepatorenal syndrome)** – Diuretic-resistant ascites, or type 2 hepatorenal syndrome, is defined as kidney function impairment that is less severe than that observed with HRS-AKI/type 1 disease. The major clinical feature in patients with type 2 hepatorenal syndrome is ascites that are resistant to diuretics.
 - Diuretic-resistant ascites have a median survival of 3-6 months.

The diagnosis is one of exclusion, being made when other causes of renal dysfunction have been excluded. Volume depletion (as with overly rapid diuresis) can mimic all the findings of hepatorenal syndrome. The prognosis is poor unless hepatic function improves, or a liver transplant is performed. No medical treatment is available for cure of this syndrome.

IX. EVOLUTION AND PROGNOSIS

The evolution and prognosis of cirrhosis depend on Child Pugh Score, underlying causes, its complications, and comorbidity.

X. COMPLICATIONS

Asymptomatic cirrhosis will evolve to decompensated cirrhosis or complicated cirrhosis. Those complications are:

- Ascites; *see in special cases above*
- Spontaneous bacterial peritonitis (SBP); *see in special cases above*
- Hepatic encephalopathy (HE); *see in special cases above*
- Hepatorenal syndrome (HRS); *see in special cases above*
- Upper gastrointestinal bleeding (UGIB) related to portal hypertension (PHT) *see in Upper gastrointestinal bleeding topic.*
- Hepatocellular Carcinoma (HCC); *see in Hepatocellular Carcinoma topic.*

XI. PREVENTION AND RECOMMENDATION

Lower risk of cirrhosis by taking these steps to care for liver:

- Avoid alcohol intake.
- Take healthy food. Choose a diet that's full of fruits and vegetables. Select whole grains and lean sources of protein. Cut down on the amount of fatty and fried foods.
- Maintain a healthy lifestyle. Too much body fat can damage the liver.
- Avoid the risk of contaminating viral hepatitis B, C. No sharing needles and performing safe sex.

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LIVER ABSCESS

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I. DEFINITION

Liver abscess can be defined as purulent collections in the liver parenchyma that result from parasitic, bacterial, or fungal infection. Infection can spread to the liver through the biliary tree, hepatic vein, or portal vein, by extension of an adjacent infection, or because of trauma. One or multiple abscesses can be present. Fungal liver abscess can occur in immunocompromised hosts. Amoebic liver abscess is a complication of amoebiasis.

II. EPIDEMIOLOGY

Liver abscesses are the most common type of visceral abscess (an abscess within an intra-abdominal organ). The annual incidence of liver abscess has been estimated at 2.3 cases per 100,000 people and is higher among men than women (3.3 versus 1.3 per 100,000), substantially higher rates have been reported in East Asian countries (up to 17.6 cases per 100,000).

III. ETIOLOGY

The etiology of liver abscess is usually multifactorial, but it can be broadly divided into two main groups: Amebic liver abscess and Pyogenic liver abscess.

1. Amebic liver abscess

- Amebic liver abscess is caused by a parasite called *Entamoeba histolytica*. This parasite causes amoebiasis, an intestinal infection that is also called amebic dysentery. After an infection has occurred, the parasite may be carried by the bloodstream from the intestines to the liver.
- The infection occurs worldwide. It is most common in tropical areas where crowded living conditions and poor sanitation exist. Africa, Latin America, Southeast Asia, and India.

2. Pyogenic liver abscess

The source of infection often remains unclear, but if the etiology is identified, the cause is most often found in:

- Diseases of the biliary tract (40-60%),
- Portal route (20%): infectious gastrointestinal disorders,
- Hematogenous spread via the hepatic artery (10%),
- Direct extension from an intra-abdominal infection (subphrenic abscess, perinephric abscess, cholecystitis).
- Penetration trauma, Iatrogenic (liver surgery....).

Potential pathogen:

- Enteric gram-negative bacilli: *Escherichia coli* and *K. pneumoniae* are the most identified pathogens. In East Asia, *K. pneumoniae* is an important cause of primary liver abscess.
- Streptococci: *Streptococcus milleri* group (including *Streptococcus anginosus*, *Streptococcus constellatus*, and *Streptococcus intermedius*) is an important cause of liver abscess in the United States.

- *Staphylococcus aureus*, *Streptococcus pyogenes* and other gram-positive cocci: in children, liver transplantation, underwent transarterial embolization for hepatocellular carcinoma.

IV. PATHOPHYSIOLOGY

The liver receives its blood circulation from the systemic and portal circulations; it is more susceptible to getting infections and abscesses from the bloodstream.

The usual pathophysiology of amebic liver abscess by *Entamoeba histolytica*, however the pathophysiology of pyogenic liver abscesses by bacteria. Bacteria travel to the liver via the portal vein and reside there. Infection can also originate in the biliary system.

V. DIAGNOSIS

1. History related to etiological suspected

- Personal history, occupation, travel, place of origin, recent infections, or treatments.
- Certain risk factors promote the development of liver abscesses, such as diabetes, cirrhosis, alcoholism, steroid abuse, male gender, elderly, immune-compromised states, and people with proton pump inhibitor usage.

2. Clinical manifestation

(Summary; 3H: *Hyperthermia, Hepatalgia, Hepatomegaly*)

- Fever (90%)
- Right upper quadrant pain (usually dull) may refer to the epigastrium, right chest, or the right shoulder (50-75%)
- Anorexia, Malaise, and Weight loss
- Nausea and Vomiting
- Symptoms of diaphragmatic irritation (hiccough).
- Tender hepatomegaly (50% of case)
- Jaundice (10%)

3. Laboratory evaluation

- CBC: neutrophilic leukocytosis (1H: *Hyperleukocytosis*)
- Inflammatory marker: ↑ CRP
- Liver chemistries: ↑ alkaline phosphatase (67-90%), ↑ AST and ALT, hyperbilirubinemia
- Serology for *Entamoeba histolytica*
- Blood culture (positive cultures of pyogenic liver abscess are obtained in over ninety percent of the cases)

4. Imaging

- **Ultrasound** and **CT scan** are the first-choice imaging studies for identification of liver abscess. CT scan is slightly more sensitive than ultrasound (approximately 95% vs 85%).
- **Magnetic resonance imaging (MRI)** is sensitive; it is not widely available outside well-resourced health services.

VI. DIFFERENTIAL DIAGNOSIS OF AMEBIC & PYOGENIC LIVER ABSCESS

Table 1: Differential diagnosis between Amebic and Pyogenic liver abscess

Characteristics	Amebic liver abscess	Pyogenic liver abscess
Age	Younger adults	Older adults
Gender distribution	Usually men affected	Both sexes affected
History	History of living in endemic area	Prior history of gallstone, Diabetes mellitus
Left shift on white blood cell count	Usually absent	Usually present
Serum bilirubin concentration	Usually normal	Often elevated
Entamoeba histolytica serology*	Positive	Negative

*Serology tests are based on immunoglobulin G enzyme-linked immunosorbent assay (ELISA).

Approximately 99% of patients with amebic liver abscess develop detectable antibodies.

In endemic areas, up to 35% uninfected individuals have anti-amebic antibodies due to previous infection with *E. histolytica*. Therefore, negative serology is helpful for exclusion of the disease.

VII. MANAGEMENT

1. Objectives

- To eradicate parasites and/or bacteria.
- To heal the liver abscess.
- To prevent the complications of liver abscess.

2. Critical Signs

- Severe sepsis
- Extension of the other organs neighbors (lungs, peritoneum, viscera, and skin)
- General peritonitis

3. Mortality risk factor

- Elderly patient (> 65 years).
- Immune-compromised states.
- Comorbidity (chronic liver disease, chronic heart diseases, and chronic kidney diseases).

4. Outpatient Management

- The patient presenting diagnosis of liver abscess must be hospitalized.

5. First line treatment

The first line treatment for Amebic and Pyogenic liver abscess. Table 2.

Table 2: First line treatment for Amebic & Pyogenic liver abscess

Amebic liver abscess	Pyogenic liver abscess
Preferred • <i>Tissue agents:</i> ○ Metronidazole* (500mg orally three times daily for 10 - 14 days) <i>OR</i> ○ Tinidazole (2g once daily for 5 days).	Preferred • <i>Beta-lactam/Beta-lactamase inhibitor</i> ○ Piperacillin-tazobactam: 3.375 to 4.5 g IV every six hours <i>OR</i> ○ Ticarcillin-clavulanate: 3.1 g every four hours. • <i>Third generation cephalosporin PLUS metronidazole</i> ○ Ceftriaxone: 2 g IV one daily <i>PLUS</i> ○ Metronidazole: 500mg IV or orally every eight hours. • <i>Ampicillin PLUS Gentamycin PLUS Metronidazole</i> ○ Ampicillin: 2 g IV every four to six hours <i>PLUS</i> ○ Gentamycin: 5 to 7 mg per kg IV daily <i>PLUS</i> ○ Metronidazole: 500mg IV or orally every eight hours
Alternative treatment ○ Ornidazole (500mg twice daily for 10 days).	Alternative treatment • <i>Fluoroquinolone PLUS metronidazole</i> ○ Ciprofloxacin: 400mg IV every 12 hours or 750mg orally twice daily ○ <i>OR</i> Levofloxacin: 500 or 750mg IV or orally one daily <i>PLUS</i> ○ Metronidazole: 500mg IV or orally every eight hours. • <i>Carbapenem</i> ○ Imipenem-cilastatin: 500mg IV every six hours. <i>OR</i> ○ Meropenem: 1 g IV every eight hours.
<i>If no improvement or worsening of symptoms under treatment for more than 5 days. → Reconsider the diagnosis of pyogenic abscess or other pathologies.</i>	
• <i>Luminal agents (if available)</i> After finished tissue agents ○ Paromomycin (25-30mg/kg/day orally three times daily for 7 days) <i>OR</i> ○ Diloxanide furoate (500mg orally three times daily for 10 days).	

* The oral route is rapidly absorbed; the intravenous route is reserved for digestive intolerance.

6. Treatment adjustment

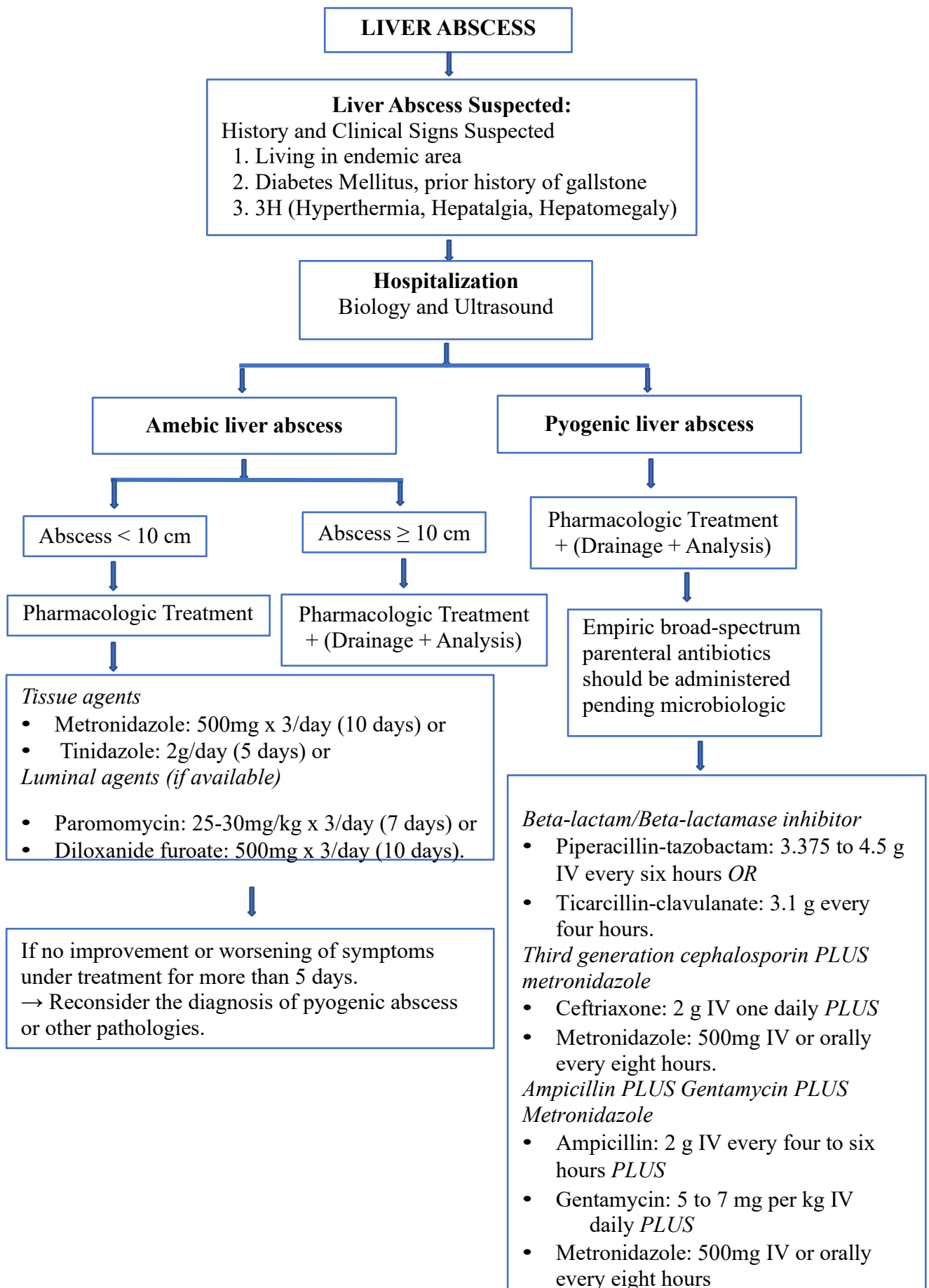
Table 3: Drainage indicated for Amebic & Pyogenic liver abscess

Amebic liver abscess	Pyogenic liver abscess
Drainage • Needle aspiration under Ultrasound or CT scan guidance is not routinely required but may be warranted if: ○ The cyst ≥ 10 cm in diameter, ○ Appears to be at imminent risk of rupture (particularly for lesion in the left lobe), ○ Clinical deterioration or lack of response to empiric therapy, ○ Exclusion of alternative diagnoses is needed.	Drainage • Strongly recommend drainage whenever practical and feasible; it is both diagnostic and therapeutic. • Percutaneous drainage should be CT scan or Ultrasound guidance. • Simple aspiration may be sufficient for some small abscess but, where possible, placement of drainage catheter is preferred. • Surgical drainage is indicated for patients who have an inadequate response to percutaneous drainage after 7 days or viscous contents obstructing the drainage catheter. • Empiric broad-spectrum parenteral antibiotics should be administered pending aspiration of abscess and microbiologic analysis. • Should cover streptococci, gram-negative bacilli, anaerobes, and <i>E. histolytica</i>.

Follow-up:

- Clinical signs:
 - Fever and pain in the right hypochondrium disappear within a few days.
 - Regression of hepatomegaly is slower.
- Biological profiles:
 - CBC and CRP normalize in a few days.
- Ultrasound and Serology are not recommended, in this case they can persist for several months.

7. Algorithm



VIII. SPECIAL CASES

Hepatic Melioidosis

Melioidosis is due to the Gram-negative bacterium, *Burkholderia pseudomallei*, which is an environmental saprophyte found in wet soils and is endemic to Southeast Asia and parts of Australia. In rural areas muddy standing water in drains and ditches are commonplace. It occurs most commonly in those who are immunosuppressed in some form, particularly in those with diabetes mellitus.

Although a wide range of organ systems can be involved in the lungs, liver and spleen are the most common sites. It is common to observe involvement of various organs at the same time. A “honeycomb” appearance has been described as characteristic for large melioidosis hepatic abscesses.

The large lesions are typically multi-loculated and multi-septated, giving rise to the “honeycomb” appearance. This has been reported to be characteristic for melioidosis liver abscesses that are larger than 2 cm.

IX. EVOLUTION AND PROGNOSIS

With new techniques available for drainage and antibiotics specific for appropriate organisms, liver abscesses have a much better prognosis now. In-hospital mortality is estimated at 2.5% to 19%.

The mortality rate is higher in elderly, ICU admissions, shock, cancer, fungal infections, cirrhosis, chronic renal failure, acute respiratory failure, severe disease, and biliary origin of an abscess. Liver abscess recurrence is frequent in patients who are present with biliary tract disease.

X. COMPLICATIONS

Complications of liver abscess may include the following:

- Sepsis.
- Empyema resulting from contiguous spread or intrapleural rupture of abscess.
- Rupture of abscess with resulting peritonitis.
- Septic pulmonary embolism may occur in patients with a liver abscess caused by *K pneumoniae*.

XI. PREVENTION AND RECOMMENDATION

- Individual prophylaxis is based on hygiene measures:
 - Wash hands regularly, especially after touching soil.
 - Wash raw vegetables well.
 - Water hygiene: filter or boil.
- Collective prophylaxis aims to prevent the dissemination from stools:
 - Ensure a clean environment.
 - Installation of the toilets.
 - Treatment of waste and drinking water.

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HEPATOCELLULAR CARCINOMA

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I. DEFINITION

Hepatocellular carcinoma (HCC) is a primary tumor of the liver that usually develops in the setting of chronic liver disease, particularly in patients with cirrhosis due to alcohol use, chronic hepatitis B or C virus infections, or metabolic dysfunction-associated steatohepatitis.

II. EPIDEMIOLOGY

HCC is one of the most common cancers globally, ranking fifth worldwide and representing the second leading cause of cancer-related mortality, with an increasing incidence of approximately "> 500,000 new cases per year".

In 2020, liver cancer stood as the "sixth most commonly diagnosed cancer" and the "third most common cause of cancer death," affecting an estimated "905,700 people" worldwide, with "830,200 deaths" attributed to it. Global age-standardized incidence and mortality rates for liver cancer were "9.5 and 8.7" for new cases and deaths, respectively, per "100,000 people." Liver cancer incidence and mortality rates were most prominent in Eastern Asia, Northern Africa, and South-Eastern Asia, accounting for more than half of the world's estimated cases and deaths (54.3% and 54.1%, respectively), despite comprising only "21.5% of the world's population in 2020."

In the case of Cambodia, one study done in Calmette hospital (455 cases) from 2016-2023 revealing that most cases were linked to viral hepatitis B and C, with tumors often diagnosed at advanced stages. Limited treatment options, late diagnosis, and economic barriers contributed to high mortality, highlighting the need for early screening and improved healthcare access in Cambodia.

III. ETIOLOGY

The etiologies (risk factors) of HCC are associated with:

- Cirrhosis
- Viral hepatitis (hepatitis B, C and D)
- Environmental toxins (aflatoxin B1, betel nut chewing, iron overload, contaminated drinking water)
- Lifestyle factors (alcohol, tobacco, dietary factors: sugar-sweetened beverages)
- Metabolic factors (nonalcoholic fatty liver disease, diabetes mellitus, obesity)
- Genetic susceptibility
- Other factors (porphyria cutanea tarda)

IV. PATHOPHYSIOLOGY

In healthy conditions, hepatocytes have a low turnover rate and typically live for more than a year. When parenchymal cells are lost due to events like partial hepatectomy or toxic injury, the liver regenerates by proliferating its main epithelial cells (hepatocytes and cholangiocytes), followed by the growth of mesenchymal cells (such as hepatic stellate cells) and

endothelial cells. If the damage is severe or the hepatocytes become senescent-often after decades of chronic disease-hepatic progenitor cells (HPCs) are activated and proliferate. This process, known as the ductular reaction, involves the expansion of a small biliary cell population in the Hering canal, which contains stem cell progeny destined for terminal differentiation. These biliary cells can become either cholangiocytes or hepatocytes, depending on the type of liver damage.

As HPCs differentiate into hepatocytes, they gradually lose biliary characteristics, shedding markers like Keratin 19 (K19), SOX9, and TROP2, followed by K7 and EPCAM, while activating the HIPPO pathway and NUMB (a Notch pathway inhibitor). The regenerating liver operates in an environment filled with inflammation, stress, and various signals, which guide HPCs toward specific cell fates while also managing the apoptosis or senescence of hepatocytes and the development of scar tissue. This complex environment sets the stage for the development of hepatocellular carcinoma (HCC).

V. DIAGNOSIS

1. Clinical features

- **Patterns of Presentation**
 - **Asymptomatic Presentation:** Common in patients under regular surveillance, where HCC is often detected early.
 - **Cirrhosis as a Risk Factor:** Present in 90% of HCC cases in Western countries, with symptoms often reflecting underlying cirrhosis rather than the tumor.
 - **Liver Decompensation:** Initial symptoms may include variceal bleeding or ascites due to HCC extension into hepatic or portal veins.
- **Symptomatic Presentation**
 - **Advanced Lesions:** May present with upper abdominal pain, weight loss, early satiety, or a palpable abdominal mass.
 - **Paraneoplastic Syndromes** (indicating poor prognosis in most cases):
 - **Hypoglycemia:** Common in advanced HCC, due to the tumor's metabolic activity or secretion of insulin-like growth factor-II.
 - **Erythrocytosis:** Rare, linked to erythropoietin (EPO) secretion.
 - **Hypercalcemia:** Caused by parathyroid hormone-related protein, even without bone metastases.
 - **Diarrhea:** Rare, due to tumor-secreted peptides causing intestinal secretion.
 - **Cutaneous Features:** Rare and include dermatomyositis, pemphigus foliaceus, Leser-Trélat sign, and pityriasis rotunda.
- **Other Clinical Presentations**
 - **Intraperitoneal Bleeding:** From tumor rupture, presenting with sudden abdominal pain, distension, hypotension, and anemia. Requires urgent management (e.g., angiography, embolization, or surgery).
 - **Obstructive Jaundice:** Due to biliary tree invasion or hemobilia.
 - **Fever:** May occur with central tumor necrosis.
 - **Rare Presentations:** Pyogenic liver abscess or hemobilia.
- **Extrahepatic Metastases**
 - Found in 10–15% of cases at diagnosis (common in advanced tumors).

- Common sites:

1. Lungs
 2. Intra-abdominal lymph nodes
 3. Bones
 4. Adrenal glands
- Brain metastases are rare (0.2–2%).
 - Note: Perihepatic lymphadenopathy in cirrhosis may be benign and unrelated to metastases. (9)

2. Laboratory tests

Initial investigations

- Full blood count (FBC), blood group
- ASAT/ALAT (aspartate aminotransferase/alanine aminotransferase), prothrombin time, albumin, gamma glutamyl transferase (GGT), alkaline phosphatase (ALP), total bilirubin/conjugated bilirubin
- Electrolytes, urea, creatinine
- Glycemia, glycated hemoglobin (HbA1c)
- Lipid profile: triglycerides, total cholesterol, HDL cholesterol, LDL cholesterol, VLDL
- Tumor markers: AFP (alpha feto-protein), CEA (carcinoembryonic antigen), CA19-9 (cancer antigen 19-9)
- Erythrocyte sedimentation rate (ESR) or plasma viscosity, C-reactive protein (CRP),
- Etiology of HCC: HBsAg, anti-HBs antibody, anti-HBc antibody, anti-HCV antibody, anti-HDV antibody, anti-HIV antibody
- Child Pugh score, MELD-Na score and ALBI score

3. Imaging

- Chest X ray
- Abdominal ultrasound
- Whole body CT scan
- EKG and heart ultrasound
- MRI of the abdomen if necessary
- Gastroscopy
- CT scan of the head if suspected brain metastasis

4. Others

- Liver biopsy

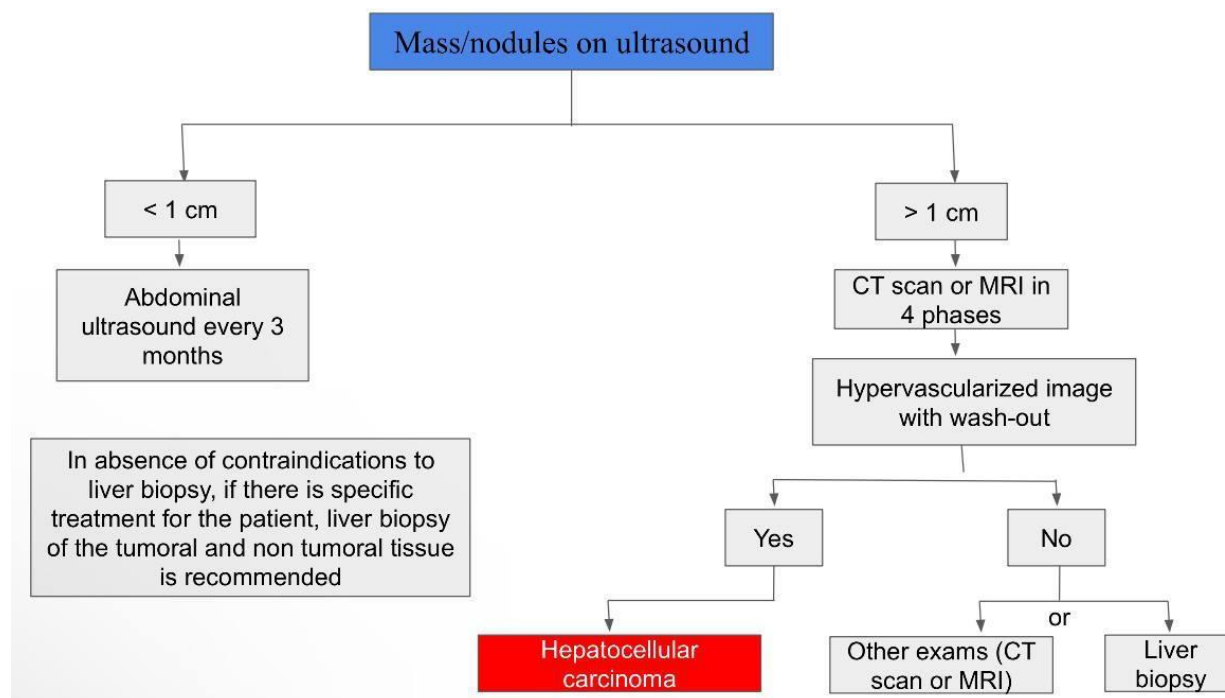


Figure 1: Diagnostic algorithm for masses/nodules of the liver on ultrasound

VI. DIFFERENTIAL DIAGNOSIS

- Cholangiocarcinoma
- Hepatoblastoma
- Mucinous cystic neoplasms (biliary cystadenoma and cystadenocarcinoma)
- Mesenchymal tumors
- Metastatic liver cancer
- Fibrous nodular hyperplasia
- Primary hepatic lymphoma
- Dysplastic nodules in cirrhosis
- Hepatocellular adenoma (hepatic adenoma)

VII. MANAGEMENT

1. **Objectives:** to cure the disease, improve quality of life, and extend survival
2. **Management plan:** All treatment plan must be discussed in a multidisciplinary team (hepatologist, onco-hepatologist, medical oncologist, surgeon and interventional radiologist)
 - **Curative treatment (in specialized hepato-gastroenterology center)**
 - a. **Liver transplantation:** a curative treatment for patients with HCC who are not surgical candidates with underlying liver dysfunction (Child Pugh B or C) who meets the Milan criteria (a single tumor < 5 cm, up to three tumors, each < 3 cm, no vascular invasion and no evidence of extrahepatic disease)
 - b. **Resection:** is a potentially curative treatment and the preferred treatment for eligible patients (solitary potentially resectable HCC confined to the liver without any invasion of the hepatic vasculature, and well-preserved hepatic function)

(Child Pugh class A) without evidence of portal hypertension and with satisfactory liver function reserve.

- c. **Percutaneous Ablation (radiofrequency (RF) or microwave (MW) or percutaneous ethanol injection):** Most patients eligible for resection can also undergo thermal ablation. Although thermal ablation generally has fewer complications and may offer comparable long-term outcomes, especially for tumors smaller than 2-3 cm, we recommend resection for most patients. This is particularly the case if a laparoscopic approach can be used, as it shortens recovery time.
- d. **Stereotactic external-beam radiation (6-20 Gy/session):** it bears antitumoral activity but more prospective studies are required to define its role.
- **Adjuvant treatment (in specialized hepato-gastroenterology center)**
 - a. **Systemic treatment:** The combination of atezolizumab with bevacizumab (Atezo-Bev) is currently the first choice first-line treatment, as it confers a superior survival benefit compared to sorafenib
 - b. **Treatment of the underlying liver disease:** The parallel management of chronic liver disease improves its prognosis and can help reduce the risk of tumor recurrence. This management involves treating the underlying cause, addressing comorbidities, and, in cases of cirrhosis, preventing complications related to portal hypertension.
 - Etiological treatment: stopping alcohol consumption, eradicating HCV, controlling HBV...
 - Managing comorbidities: alcohol, overweight, diabetes...
 - In cases of cirrhosis, check for clinically significant portal hypertension and, if necessary, initiate treatment with carvedilol.
- **Palliative treatment (in specialized hepato-gastroenterology center)**
 - a. **TACE (Transarterial chemoembolization):** it might be performed using chemotherapy emulsified in lipiodol followed by gelfoam or any other material injection (conventional TACE) or using drug-eluting microspheres (DEB-TACE). TACE is restricted to those with “well-defined nodules with preserved portal flow,” with systemic chemotherapy or immunotherapy for those with “diffuse, infiltrative, extensive bilobar involvement (“Up-to-seven” criteria-out nodules)”.
 - b. **TARE (Transarterial radioembolization):** Using intra-arterial injection of yttrium-90 (90Y)-labeled glass or resin microspheres induces extensive tumor necrosis with an acceptable safety profile. TARE is preferred over TACE in the setting of an HCC complicated by a malignant main or lobar branch portal vein thrombus.

3. Algorithm

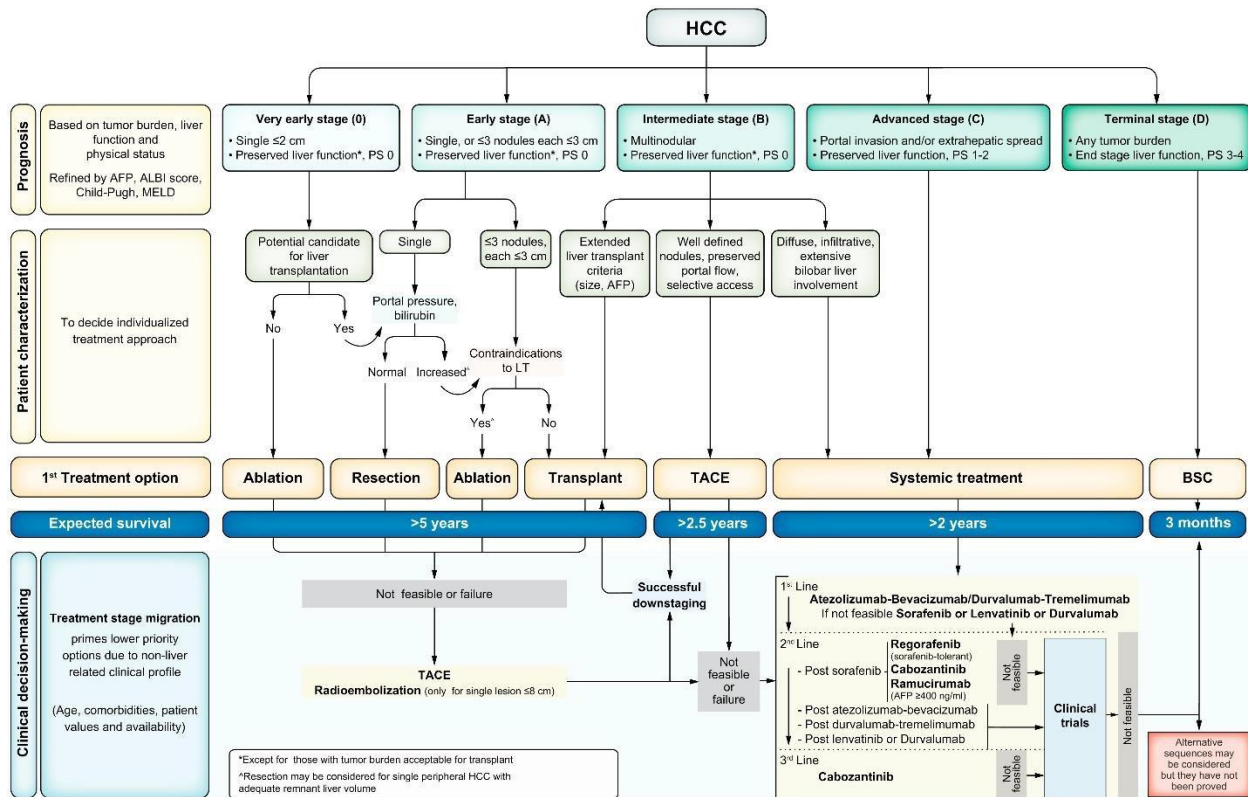


Figure 2: BCLC staging and treatment strategy in 2022

VIII. SURVEILLANCE

For HCC surveillance in most at-risk patients, it is recommended to do an ultrasonography (US) and a-fetoprotein (AFP) every 6 months.

IX. COMPLICATION

Tumor-Related Complications

- Portal Vein Thrombosis
- Biliary Obstruction
- Rupture of HCC (Spontaneous Rupture)
- Extrahepatic Metastasis

Liver Dysfunction-Related Complications

- Liver Failure (Hepatic Decompensation)
- Hepatic Encephalopathy
- Ascites and Spontaneous Bacterial Peritonitis (SBP)
- Hepatorenal Syndrome (HRS)

Bleeding and Coagulation Disorders

- Gastrointestinal Bleeding (Esophageal and gastric varices rupture)
- Disseminated Intravascular Coagulation

Treatment-Related Complications

- Post-Treatment Liver Failure (Resection or trans arterial therapies can worsen liver function, leading to hepatic decompensation).
- Adverse Effects of Systemic Therapy (Sorafenib, Lenvatinib, Immunotherapy)

X. PATIENT COUNSELING

Prevention of HCC:

- Alcohol use: Heavy use of alcohol (>40 g/day for males and >20 g/day for females) has been associated with worsening liver disease and an increased risk of HCC. Although the exact amount of alcohol that can be safely consumed is unclear, advising patients to be completely abstinent is reasonable in those who have cirrhosis.
- Immunizations of HBV if the patients are not yet vaccinated and/or in high-risk population (healthcare workers, people who inject IV drugs, etc.)
- Screening, diagnosis and treatment for HBV and HCV infection according to the national guidelines for hepatitis B and C
- Preventing transmissions to others- Carriers of HBV should be counseled regarding the risk of transmission to others. Patients should be advised regarding prevention of sexual transmission (ie, vaccination of spouses and steady sex partners in individuals with monogamous partners and safe sex practice including use of condoms in individuals with multiple partners), perinatal transmission, and risk of environmental exposure from blood.
- Healthy lifestyle- Patients should be counseled about the importance of a healthy diet and regular exercise to maintain normal weight and metabolism. With the increasing prevalence of obesity, many patients with chronic HBV infection have concomitant hepatic steatosis (Metabolic syndrome associated steatohepatitis-MASH), which may accelerate progression to cirrhosis and increase risk of HCC.

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ACUTE PANCREATITIS

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I. DEFINITION

Acute pancreatitis is an acute inflammatory process of the pancreas.

II. EPIDEMIOLOGY

The reported annual incidence of acute pancreatitis ranges from 4.9 to 35 per 100,000 population. The incidence of acute pancreatitis is increasing worldwide due to increased rates of obesity and gallstones. Smoking may increase the risk for non-gallstone-related pancreatitis by mechanisms that are unclear and may potentiate alcohol-induced damage to the pancreas.

Overall mortality was approximately 5 percent, with mortality rates in patients with interstitial, and necrotizing pancreatitis, of 3 percent, and 17 percent, respectively. Mortality in acute pancreatitis is usually due to systemic inflammatory response syndrome and organ failure in the first two-week period, while after two weeks it is usually due to sepsis and its complications.

III. ETIOLOGY

The etiology of acute pancreatitis consists of:

- Gallstones
- Alcohol
- Hypertriglyceridemia
- Post-endoscopic retrograde cholangiopancreatography (ERCP)
- Genetic
- Medications: mechanisms of drug-induced pancreatitis include
 - Immunologic reactions: 6-mercaptopurine, aminosaliclates, sulfonamides...
 - Direct toxic effect: diuretics, sulfonamides ...
 - Accumulation of a toxic metabolite: valproic acid, pentamidine, tetracycline...
 - Overstimulation of acinar cells: GLP-1 receptor agonists...
 - Ischemia: diuretics, azathioprine...
 - Intravascular thrombosis: estrogen...
 - Increased viscosity of pancreatic juice: diuretics and steroids
- Pancreatic duct injuries
- Biliary sludge and microlithiasis
- Conditions causing obstruction of the ampulla: periampullary diverticula or tumors, pancreatic tumors, duodenal inflammation and papillary stenosis secondary to celiac disease (recurrent episodes of acute pancreatitis)
- Hypercalcemia
- Infection and other toxins:
 - Virus: *Mumps, coxsackievirus, hepatitis B, cytomegalovirus, varicella-zoster, herpes simplex, human immunodeficiency virus (HIV), COVID-19*

- Bacteria: *Mycoplasma*, *Legionella*, *Leptospira*, *Salmonella*
- Fungi: *Aspergillus*
- Parasite: *Toxoplasma*, *Cryptosporidium*, *Ascaris*
- Vascular diseases: pancreatic ischemia (uncommon)
 - Systemic lupus erythematosus
 - Polyarteritis nodosa
 - Atheroembolism,
 - Intraoperative hypotension
 - Hemorrhagic shock
- Anatomic or physiological pancreatic anomalies:
 - Choledochal cysts
 - Large juxta-ampullary diverticula
 - Annular pancreas
 - Sphincter of Oddi dysfunction
 - Pancreas divisum
- Idiopathic: 15 – 25% of acute pancreatitis

IV. PATHOPHYSIOLOGY

Acute pancreatitis is an inflammatory condition of the pancreas characterized by activation of pancreatic enzymes within the gland, leading to autodigestion, inflammation, and damage to pancreatic tissue and surrounding structures. The physiopathology involves the following key steps:

1. Initiation: Premature Activation of Pancreatic Enzymes

- Normally, digestive enzymes (e.g., trypsinogen, lipase, amylase) are stored in zymogen granules in an inactive form and activated in the duodenum.
- In acute pancreatitis, **inappropriate activation of trypsinogen to trypsin** occurs within acinar cells.
- Common triggers:
 - Gallstone-induced obstruction of the pancreatic duct.
 - Alcohol abuse causing direct toxicity and oxidative stress.
 - Hypertriglyceridemia or hypercalcemia.

2. Intracellular Injury

- The premature activation of trypsin initiates a cascade of enzyme activation (e.g., chymotrypsin, elastase), leading to:
 - **Autodigestion of acinar cells** and nearby tissues.
 - Disruption of intracellular calcium signaling, causing sustained elevations in calcium that perpetuate enzyme activation and cell injury.

3. Inflammatory Response

- Damaged pancreatic tissue releases pro-inflammatory cytokines (e.g., IL-1, IL-6, TNF-alpha).
- Recruitment of neutrophils and macrophages amplifies local inflammation.
- Increased vascular permeability and edema further exacerbate tissue damage.

4. Systemic Effects

- If inflammation extends beyond the pancreas:

- Release of cytokines and inflammatory mediators leads to systemic inflammatory response syndrome (SIRS).
- Multi-organ dysfunction can result from hypovolemia, ischemia, and activation of coagulation pathways.

5. Complications

- Local complications:
 - Necrosis of pancreatic or peripancreatic tissue.
 - Formation of pseudocysts or abscesses.
- Systemic complications:
 - Acute respiratory distress syndrome (ARDS).
 - Hypotension and shock due to third-spacing of fluids.
 - Renal failure secondary to hypoperfusion.

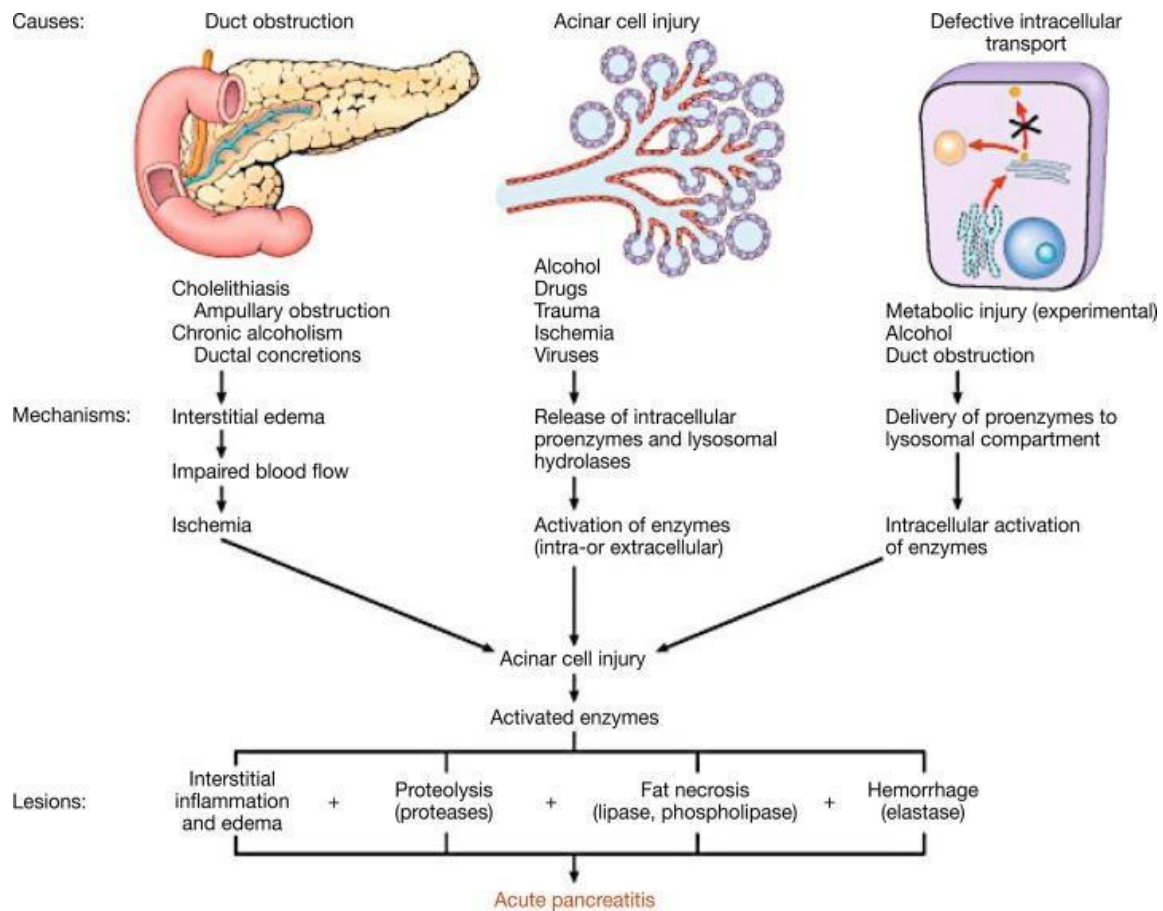


Figure 1: Pathophysiology of Acute Pancreatitis

V. DIAGNOSIS

1. Clinical Manifestations:

- Most patients have acute onset of persistent, severe epigastric, irradiated to the back and may be partially relieved by sitting up or bending forward and could also be associated with nausea and vomiting. In some patients, the pain may also be in the right upper quadrant or, rarely, confined to the left upper quadrant.
- In patients with gallstone pancreatitis, the pain is well localized, and the onset of pain is rapid, reaching maximum intensity in 10 to 20 minutes. In contrast, in patients with pancreatitis due to hereditary or metabolic causes or alcohol, the onset of pain may be less abrupt, and the pain may be poorly localized.
- Patients with severe acute pancreatitis may have dyspnea due to diaphragmatic inflammation secondary to pancreatitis, pleural effusions, or acute respiratory distress syndrome.
- Approximately 5 to 10% of patients with acute severe pancreatitis may have painless disease and have unexplained hypotension (eg, postoperative and critically ill patients, patients on dialysis, organophosphate poisoning, and Legionnaire's disease).

2. Physical examination

- Physical findings vary depending upon the severity of acute pancreatitis.
 - For mild acute pancreatitis, the epigastrium may be minimally tender to palpation. In contrast, for severe pancreatitis, there may be significant tenderness to palpation in the epigastrium or more diffusely over the abdomen.
 - Patients with severe pancreatitis may have fever, tachypnea, hypoxemia, and hypotension.
 - Ecchymotic discoloration may be observed in the periumbilical region (Cullen's sign) or along the flank (Grey Turner sign) (picture 1) can be seen in approximately 3% of the patients suggesting the present of retroperitoneal bleeding in the setting of pancreatic necrosis.
- Patients may have abdominal distention, and hypoactive bowel sounds due to an ileus secondary to inflammation.
- Patients may have scleral icterus due to obstructive jaundice due to choledocholithiasis or edema of the head of the pancreas.
- In rare cases, patients may have subcutaneous nodular fat necrosis or panniculitis (picture 2). These lesions are tender red nodules that frequently occur on the distal extremities but may occur elsewhere. There may also be associated polyarthritis (wrist and ankles), so called PPP (pancreatitis, panniculitis, polyarthritis) syndrome.
- Patients may also have findings suggestive of the underlying etiology. As examples, hepatomegaly may be present in patients with alcoholic pancreatitis, xanthomas in hyperlipidemic pancreatitis, and parotid swelling in patients with mumps (picture 3 and picture 4).

3. Investigation :

- **Laboratory finding**

- **Amylasemia** rise within 6 to 12 hours of the onset of acute pancreatitis with a short half-life of approximately 10 hours. In uncomplicated attacks, the serum amylase returns to normal within three to five days.

- Amylasemia ≥ 3 times upper normal limit (UNL) $\Rightarrow$ acute pancreatitis (sensitivity: 67 – 83%, specificity: of 85 – 98%).

Amylasemia ≥ 3 times UNL may not be seen in approximately 20% of patient with alcoholic pancreatitis, 50% triglyceridemia-associated pancreatitis (triglyceride interfere with the amylase assay).

- Diagnostic may be missed in patients who present > 24 hours after the onset of pancreatitis.
- **Lypasemia** rises within 4 to 8 hours of the onset of symptoms, peak at 24 hours and returns to normal within 8 to 14 days.
 - Lipase elevations occur earlier and last longer as compared with elevation of amylasemia and are therefore especially useful in patients who present >24 hours after the onset of pain.
 - Serum lipase is also more sensitive as compared with amylase in patients with pancreatitis secondary to alcohol.



Pic. 1. Grey Turner Sign



Pic. 2. Inflammatory nodules on the distal leg



Pic. 3 Eruptive Xanthomata



Pic. 4 Mump

- **Other enzymes and products**
 - Trypsinogen activation peptide (TAP): elevated in acute pancreatitis (early even in the pathogenesis of acute pancreatitis), may be useful in detection of early acute pancreatitis as a predictor of its severity.
 - Urinary and serum trypsinogen-2 levels are elevated in early acute pancreatitis. However, additional studies are needed to determine their role in the diagnosis of acute pancreatitis.
 - **Markers of immune activation:** Activation of granulocytes and macrophages in acute pancreatitis results in release of a number of cytokines and inflammatory mediators.
 - Acute pancreatitis is associated with elevations in C-reactive protein (CRP), interleukin (IL)-6, IL-8, IL-10, tumor necrosis factor (TNF), and PMN elastase.
 - CRP level above 150 mg/L at 48 hours is associated with severe pancreatitis.
 - **Other laboratory findings:**
 - Patients with pancreatitis may have leukocytosis and an elevated hematocrit from hemoconcentration due to extravasation of intravascular fluid into third spaces.
 - Metabolic abnormalities include elevated blood urea nitrogen (BUN), hypocalcemia, hyperglycemia, and hypoglycemia may also occur.
 - **Imaging:**
 - **Abdominal ultrasound:**
 - Pancreas diffusely enlarged and hypoechoic
 - Gallstones may be visualized in the gallbladder or the bile duct
 - **Contrast-enhanced abdominal CT scan:**
 - Acute interstitial edematous pancreatitis: focal or diffuse enlargement of the pancreas with heterogeneous enhancement with intravenous contrast.
 - Necrotizing pancreatitis: necrosis of pancreas tissue is recognized as lack of enhancement after intravenous contrast administration.
 - A common bile duct stone may occasionally be visualized on contrast-enhanced abdominal CT scan. A pancreatic mass may be seen in patients with an underlying pancreatic cancer, and diffuse dilation of the pancreatic duct or a cystic lesion may be seen in patients with an intraductal papillary mucinous neoplasia or cystic neoplasm.
 - **Magnetic resonance imaging (MRI):**
 - focal or diffuse enlargement of the pancreas
 - Necrotizing of pancreatic tissue
4. Diagnosis of acute pancreatitis
- requires the presence of two of the following three criteria:
 - Acute onset of persistent, severe, epigastric pain often radiating to the back,
 - Serum lipase or amylase > 3 UNL,
 - Characteristic findings of acute pancreatitis on imaging (contrast-enhanced computed tomography [CT scan], magnetic resonance imaging [MRI], or transabdominal ultrasonography [abdo. US])
5. Classification of acute pancreatitis and assessment of its severity:

- Classification:
 - According to morphology, acute pancreatitis can be divided into two broad categories:
 - **Interstitial edematous acute pancreatitis:** acute inflammation of the pancreatic parenchyma and peripancreatic tissues, but without recognizable tissue necrosis
 - **Necrotizing acute pancreatitis:** inflammation associated with pancreatic parenchymal necrosis and/or peripancreatic necrosis
 - According to severity, acute pancreatitis is divided into the following:
 - **Mild acute pancreatitis:** absence of organ failure and local or systemic complications
 - **Moderately severe acute pancreatitis:** transient organ failure (resolves within 48 hours) and/or local or systemic complications without persistent organ failure (>48 hours)
 - **Severe acute pancreatitis:** persistent organ failure that may involve one or multiple organs
- Parameters help predict the severity of acute pancreatitis:
 - Persisting of systemic inflammatory response syndromes (SRIS) > 48H (Table 1)
 - CRP > 150 at 48H
 - Elevated hematocrit > 44%,
 - Blood urea nitrogen (BUN) > 20mg/dL
 - Creatinine > 1.8mg/dL
 - Age > 60 years
 - Obesity or underlying cardiac or pulmonary disease

Table 1: SIRS Criteria

Present 2 or more of the following conditions

- Temperature > 38.3 °C or < 36 °C
- Heart rate (HR) > 90/mn
- Respiratory rate (RR) > 20/mn or PaCO₂ < 32 mmHg
- WBC > 12,000 cells/mm³, < 4,000 cells/mm³, or > 10% or immature (band) forms

* SIRS (+) > 48H => acute severe pancreatitis

VI. DIFFERENTIAL DIAGNOSIS

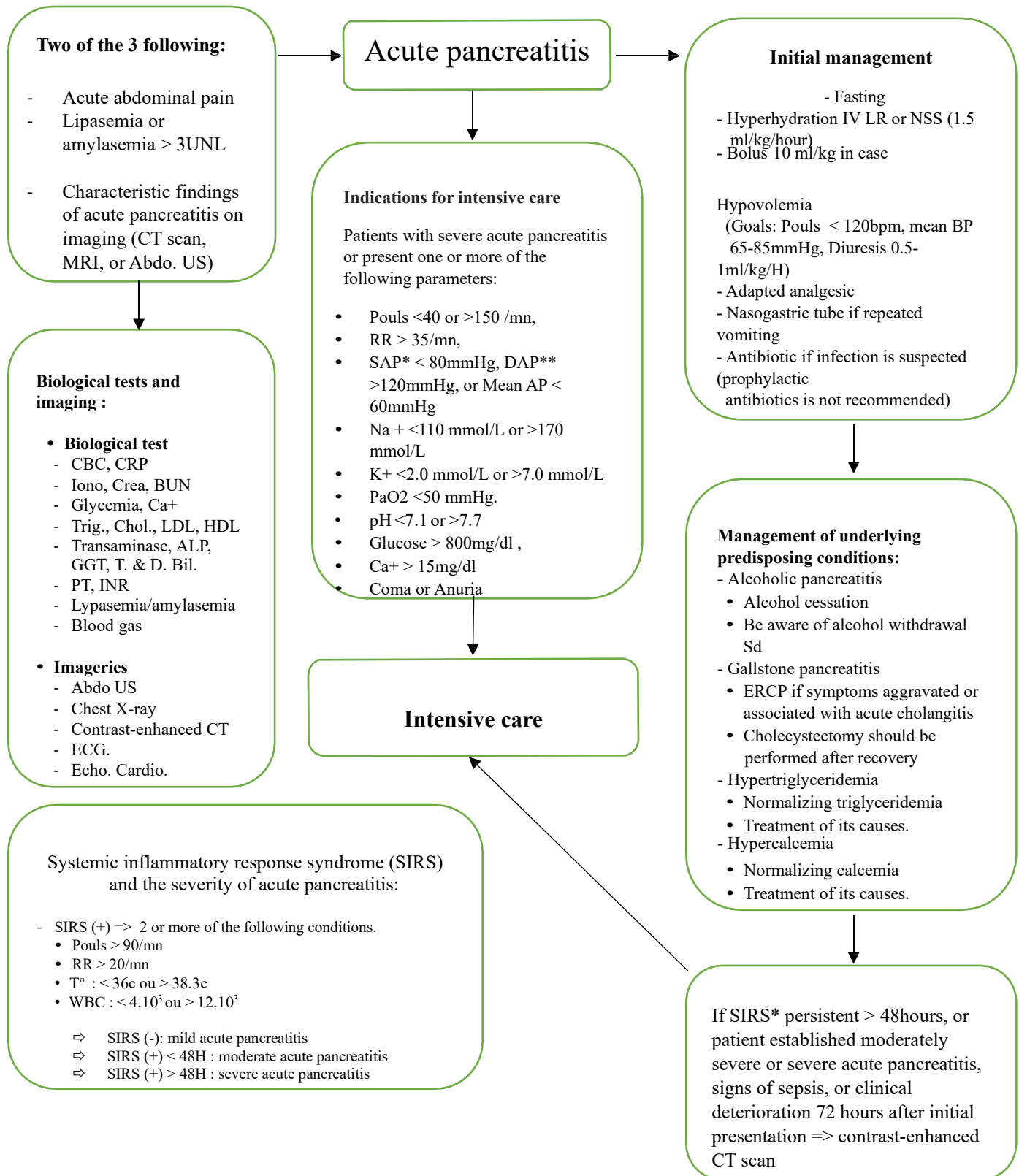
- Peptic ulcer disease
- Gallstone and its complications
- Perforated viscus
- Intestinal obstruction
- Mesenteric ischemia
- Hepatitis

VII. ANAGEMENT

1. Objectives: To allow pancreas to rest and heal from inflammation.
2. Critical Signs:
 - Multiorgan failure (cardiovascular, respiratory, renal...).
 - Persistence SRIS > 48H
 - Necrotizing pancreatitis
3. Mortality Risk Factors:
 - Comorbidity: obesity, age > 60 years, underlying cardiac or pulmonary diseases...
4. Outpatient Management
 - Not recommended
5. First-Line Treatment: consists of supportive care with fluid resuscitation, pain control, and nutritional support.
 - **Fluid replacement:** 1.5ml/kg/hour with a 10ml/kg bolus in patient with hypovolemia (for the 1st 24 to 48 hours)
 - Fluid requirements are reassessed at frequent intervals in the first six hours of admission and for the next 24 to 48 hours (especially in older adults and those with a history of cardiac and/or renal disease, and adjusted based on clinical assessment, hematocrit, blood urea nitrogen (BUN), and creatinine values.
 - Adequate fluid replacement can be assessed by an improvement in vital signs (goal heart rate <120 beats/minute, mean arterial pressure between 65 to 85 mmHg), urine output (>0.5 to 1 cc/kg/hour) and reduction in hematocrit (goal 35 to 44 percent) and BUN over 24 hours, particularly if they were high at the onset. Monitoring the BUN may be particularly important, as both the BUN at the time of admission and the change in BUN during the first 24 hours of hospitalization predict mortality)
 - **Pain control:** uncontrolled pain can contribute to the hemodynamic instability.
 - Opioids: safe and effective (hydromorphone or fentanyl (intravenous) may be used for pain relief)
 - a. Fentanyl is being increasingly used due to its better safety profile, especially in renal impairment.
 - **Nutrition:**
 - Mild pancreatitis: recovery generally occurs quickly, making it unnecessary to initiate supplemental nutrition. A soft diet can be started early (within 24 hours) as tolerated. We usually start with a low-residue, low-fat, and soft diet, provided there is no evidence of ileus or significant nausea and/or vomiting.
 - Moderately severe and severe pancreatitis: enteral nutrition through a nasogastric (or nasojejunal) tube placed endoscopically or radiologically. If the target rate is not achieved within 48 to 72 hours and if severe acute pancreatitis is not resolved, supplemental parenteral nutrition should be provided.
 - **Antibiotics:**
 - Prophylactic antibiotics and selective decontamination of the gut are not recommended in patients with acute pancreatitis, regardless of the type (interstitial or necrotizing) or disease severity (mild, moderately severe, or severe).
 - AB is used when infection is suspected.

6. Treatment adjustment and management of underlying and predisposing conditions
- Patients with predicted or established moderately severe or severe acute pancreatitis, signs of sepsis, or clinical deterioration 72 hours after initial presentation should undergo a contrast-enhanced CT scan to assess the presence of pancreatic or extrapancreatic necrosis and local complications.
 - Patients with persistent organ failure and extensive local complications should be transferred to centers of expertise.
 - In case of gallstone pancreatitis:
 - Endoscopic retrograde cholangiopancreatography (ERCP): if associated with cholangitis
 - Cholecystectomy to prevent relapses is recommended after recovery in all patients.
 - In case of alcoholic pancreatitis: stop alcohol use, treatment of alcohol withdrawal Sd.
 - In case of hypertriglyceridemia: normalizing triglyceridemia, treatment of its underlying etiology.
 - In case of hypercalcemia: normalizing calcium levels, treatment its underlying etiology.

7. Algorithm



* SAP: systolic arterial pressure, ** DAP: diastolic arterial pressure

VIII. SPECIAL CASES

Management of local and systemic complications of acute pancreatitis are discussed in detail elsewhere.

IX. VOLUTION AND PROGNOSIS

- Mild acute pancreatitis: excellent prognosis with a mortality rate of <1%.
- Severe acute pancreatitis: mortality rate ranges from 15% to 30%.
- Up to 20% of patients experience recurrent episodes, often due to unresolved causes like gallstones or alcohol use.
- Approximately 10% progress to chronic pancreatitis, especially with recurrent or severe disease.
- Post-pancreatitis diabetes occurs in 20-30% of severe cases.

X. COMPLICATIONS

- Local complications:
 - Acute peripancreatic fluid collection
 - Pancreatic pseudocyst
 - Acute necrotic collection and walled of necrosis (infected and sterile necrosis)
- Peripancreatic vascular complications
 - Splanchnic venous thrombosis
 - Pseudoaneurysm
- Abdominal compartment syndrome
- Systemic complications
 - Organ failure
 - Diabetes
 - Exacerbation of underlying comorbidities

XI. REVENTION AND RECOMMENDATION

Patients with acute pancreatitis should be advised:

- To avoid alcohol consumption and smoking.
- To have a balanced, low-fat diet to prevent recurrence.

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BILIARY TRACT INFECTION

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I. DEFINITION

Biliary tract infection is an infectious process in the biliary system, including the bile ducts, gallbladder, or both, often arising from bacterial proliferation in the setting of biliary stasis or obstruction.

II. EPIDEMIOLOGY

- **Prevalence:**
 - Biliary infections are among the main causes of sepsis and emergency department admissions, especially in elderly patients with comorbidities.
 - The mortality rate of severe biliary infections ranges from 1% to 6%. However, when combined with bacteremia (which occurs in 10% of cholecystitis and 50% of cholangitis cases), mortality can rise to 10–20%.
 - Acute cholecystitis affects approximately 20% of patients with symptomatic gallstones.
 - Cholangitis occurs in 10–15% of patients with biliary obstruction.
- **Risk Factors:**
 - Gallstones (the most common cause in both conditions).
 - Advanced age, female sex (for cholecystitis), obesity, and diabetes.
 - Post-surgical biliary alterations (e.g., biliary stents or prior ERCP).
- **Global Distribution:** More prevalent in regions with high gallstone incidence.

III. ETIOLOGY

- **Obstruction of the Biliary System:**
 - Gallstones: most common cause.
 - Malignant or benign biliary strictures.
 - Biliary stents, tumors, or parasites (*Ascaris lumbricoides*, liver flukes).
- **Bacterial Infections:**
 - Organisms typically ascend from the duodenum into the biliary system.
 - Common pathogens:
 - Gram-negative bacteria: *Escherichia coli* (25–50%), *Klebsiella spp*, *Enterobacter spp*.
 - Gram-positive cocci: *Enterococcus spp*.
 - Anaerobes: *Clostridium spp.*, *Bacteroides spp.* (in polymicrobial infections).

IV. PATHOPHYSIOLOGY

1. Mechanism of Infection:

- Biliary obstruction:
 - Bacteria ascend from duodenum into the bile duct due to a blockage, like gallstones, causing bile to stagnate and creating an environment conducive to

bacterial growth. This obstruction leads to increased pressure within the biliary system, facilitating bacterial invasion and infection.

- Hematogenous spread:
 - Bacteria can also reach the biliary tract through bloodstream, particularly from a portal vein infection originating in the gastrointestinal tract.
- Surgical manipulation
 - Procedure like endoscopic retrograde cholangiopancreatography (ERCP) can introduce bacterial into the biliary tract if proper precautions are not taken.

2. Host Response:

- Inflammation of bile ducts or gallbladder causes local tissue damage.
- Severe cases can lead to systemic inflammatory response syndrome (SIRS) and sepsis.

3. Key Differences:

- Acute cholangitis involves systemic infection (fever, jaundice, hypotension).
- Acute cholecystitis primarily involves localized inflammation of the gallbladder.

V. DIAGNOSIS

The clinical course of *biliary* infection is variable and is influenced by the localization of the biliary tract obstruction and infection.

1. Clinical Manifestations for:

- Acute cholecystitis
 - Biliary colic: right upper quadrant (RUQ) pain, sometimes associated with nausea or vomiting.
 - Local signs of inflammation: Murphy's sign positive or RUQ mass/pain/tenderness.
 - Systemic sign: fever and/or shaking chills
- Acute Cholangitis:
 - Signs of cholangitis and jaundice (Charcot's Triad: Fever, RUQ pain, jaundice).
 - Severe cases: Reynolds' Pentad (Charcot's Triad + altered mental status, hypotension).

2. Investigation:

- Laboratory tests
 - Elevated WBC, CRP, and procalcitonin.
 - Elevated liver enzymes: AST, ALT (suggest cytolysis), ALP, GGT, bilirubin (suggest cholestasis).
 - Blood cultures for suspected sepsis.
- Imaging
 - Abdominal Ultrasound (abdo. US):
 - In acute cholecystitis:
 - Stone in gallbladder.
 - Gallbladder wall thickening > 5 mm
 - Pericholecystic fluid
 - Gallbladder distention > 5 cm
 - Sonographic Murphy sign
 - In acute cholangitis
 - Dilatation of intrahepatic and common bile duct

- Abdominal CT scan:
 - Less sensitive than abdo. US for gallbladder stones but more sensitive for gallstones in the distal common bile duct.
 - Useful for detecting complications of acute cholecystitis (eg. abscess, perforation), intrahepatic stones or recurrent pyogenic cholangitis.
- MRCP: more accurate than abdominal US in detecting stone in common bile ducts (reserved for cases in which choledocholithiasis is suspected).
- ERCP: reserved for therapeutic procedure.
 - Usually performed in conjunction with endoscopic retrograde sphincterotomy and gallstone extraction or stent insertion.

VI. DIFFERENTIAL DIAGNOSIS

- Hepatitis (viral or toxic).
- Acute pancreatitis.
- Pyelonephritis or renal colic.
- Peptic ulcer disease.
- Right lower lobe pneumonia (referred pain).
- Upper gastrointestinal Crohn disease

VII. MANAGEMENT

1. Objective: to eradicate the infection, to relieve biliary obstruction, and to prevent and treat its complications.
2. Critical Signs:
 - Signs of severe and moderate acute cholecystitis (Table 1) or cholangitis (Table 2)
3. Mortality Risk Factors:
 - Comorbidity: obesity, age > 75 years, underlying cardiac or pulmonary diseases...
4. Outpatient Management
 - Not recommended

Table 1: severity assessment criteria for acute cholecystitis

Severe acute cholecystitis

- Associated with dysfunction of any one of the following organs/systems:
 1. Cardiovascular dysfunction: hypotension requiring dopamine ≥ 5 lg/kg per min, or any dose of norepinephrine
 2. Neurological dysfunction: disturbance of consciousness
 3. Respiratory dysfunction: PaO₂/FiO₂ ratio <300
 4. Renal dysfunction: oliguria, serum creatinine >2.0 mg/dl
 5. Hepatic dysfunction: PT-INR >1.5
 6. Hematological dysfunction: platelet count <100,000/mm³

Moderate acute cholecystitis

- Associated with any one of the following conditions:
 1. Elevated white blood cell count (>18,000/mm³)
 2. Palpable tender mass in the right upper abdominal quadrant
 3. Duration of complaints >72 h
 4. Marked local inflammation (gangrenous cholecystitis, pericholecystic abscess, hepatic abscess, biliary peritonitis, emphysematous cholecystitis)

Mild acute cholecystitis

- Does not meet the criteria of “Severe” or “Moderate” acute cholecystitis. Mild acute cholecystitis can also be defined as acute cholecystitis in a healthy patient with no organ dysfunction and mild inflammatory changes in the gallbladder, making cholecystectomy a safe and low risk operative procedure

Table 2: severity assessment criteria for acute cholangitis

Sever acute cholangitis

- Associated with dysfunction of any one of the following organs/systems:
 1. Cardiovascular dysfunction: hypotension requiring dopamine ≥ 5 lg/kg per min, or any dose of norepinephrine
 2. Neurological dysfunction: disturbance of consciousness
 3. Respiratory dysfunction: PaO₂/FiO₂ ratio <300
 4. Renal dysfunction: oliguria, serum creatinine >2.0 mg/dl
 5. Hepatic dysfunction: PT-INR >1.5
 6. Hematological dysfunction: platelet count <100,000/mm³

Moderate acute cholangitis

- Associated with any two of the following conditions:
 1. Abnormal WBC count (>12,000/mm³, <4,000/mm³)
 2. High fever ($\geq 39^{\circ}\text{C}$)
 3. Age ≥ 75 years
 4. Hyperbilirubinemia (total bilirubin ≥ 5 mg/dl)
 5. Hypoalbuminemia (<ULN x 0.7)

Mild acute cholangitis

- “Mild” acute cholangitis does not meet the criteria of “Severe” or “Moderate” acute cholangitis at initial diagnosis

-
5. First line treatment: consists of fasting, sufficient fluid resuscitation, the administration of antibiotics and pain control, along with the monitoring of respiratory hemodynamic conditions in preparation for emergency drainage.
 - Fasting
 - Adequate fluid replacement and electrolytes correction
 - Analgesics for pain control
 - Antibiotics:
 1. Empiric regimens for intra-abdominal infections (antimicrobials with activity against enteric streptococci, coliforms and anaerobes). (Table 3)
 2. Subsequently, it will be tailored to culture and susceptibility results when they become available.

Table 3: Community-acquired biliary infections

Severity	Community-acquired biliary infections			Health-care-associated biliary infections*
	Grade I	Grade II	Grade III*	
Antimicrobial agents	Cholangitis and cholecystitis	Cholangitis and cholecystitis	Cholangitis and cholecystitis	Health-care-associated cholangitis and cholecystitis
Penicillin-based therapy	- Ampicillin/sulbactam [¶] is not recommended if >20% resistance rate	- Piperacillin/Tazobactam	- Piperacillin/tazobactam	- Piperacillin/tazobactam
Cephalosporin-based therapy	One of the following with or without metronidazole ^Δ : - Ceftriaxone - Cefotaxime - Cefazolin [◇] - Cefotiam [◇] - Cefuroxime [◇]	One of the following with or without metronidazole ^Δ : - Ceftriaxone - Cefotaxime - Cefepime - Cefozopran - Ceftazidime	One of the following with or without metronidazole ^Δ : - Cefepime - Ceftazidime - Cefozopran	One of the following with or without metronidazole ^Δ : - Cefepime - Ceftazidime - Cefozopran
	- Cefmetazole, [◇] - Cefoxitin, [◇] - Flomoxef, [◇] - Cefoperazone/Sulbactam	- Cefoperazone/Sulbactam		
Carbapenem-based therapy	- Ertapenem	- Ertapenem	- Imipenem/cilastatin, - Meropenem, - Doripenem, - Ertapenem	- Imipenem/cilastatin, - Meropenem, - Doripenem, - Ertapenem
Monobactam-based therapy	–	–	- Aztreonam ± metronidazole ^Δ	- Aztreonam ± metronidazole ^Δ
Fluoroquinolone-based therapy [§]	One of the following with or without metronidazole ^Δ : - Ciprofloxacin - Levofloxacin - Pazufloxacin	One of the following with or without metronidazole ^Δ : - Ciprofloxacin - Levofloxacin - Pazufloxacin	–	–
	- Moxifloxacin	- Moxifloxacin		

* Vancomycin is recommended to cover *Enterococcus* spp for grade III community-acquired acute cholangitis and cholecystitis and health-care-associated acute biliary infections. Linezolid or daptomycin is recommended if vancomycin-resistant *Enterococcus* (VRE) is known to be colonizing the patient, if previous treatment included vancomycin, and/or if the organism is common in the community.

¶ Ampicillin/sulbactam has little activity left against *Escherichia coli*. It has been removed from the North American guidelines.

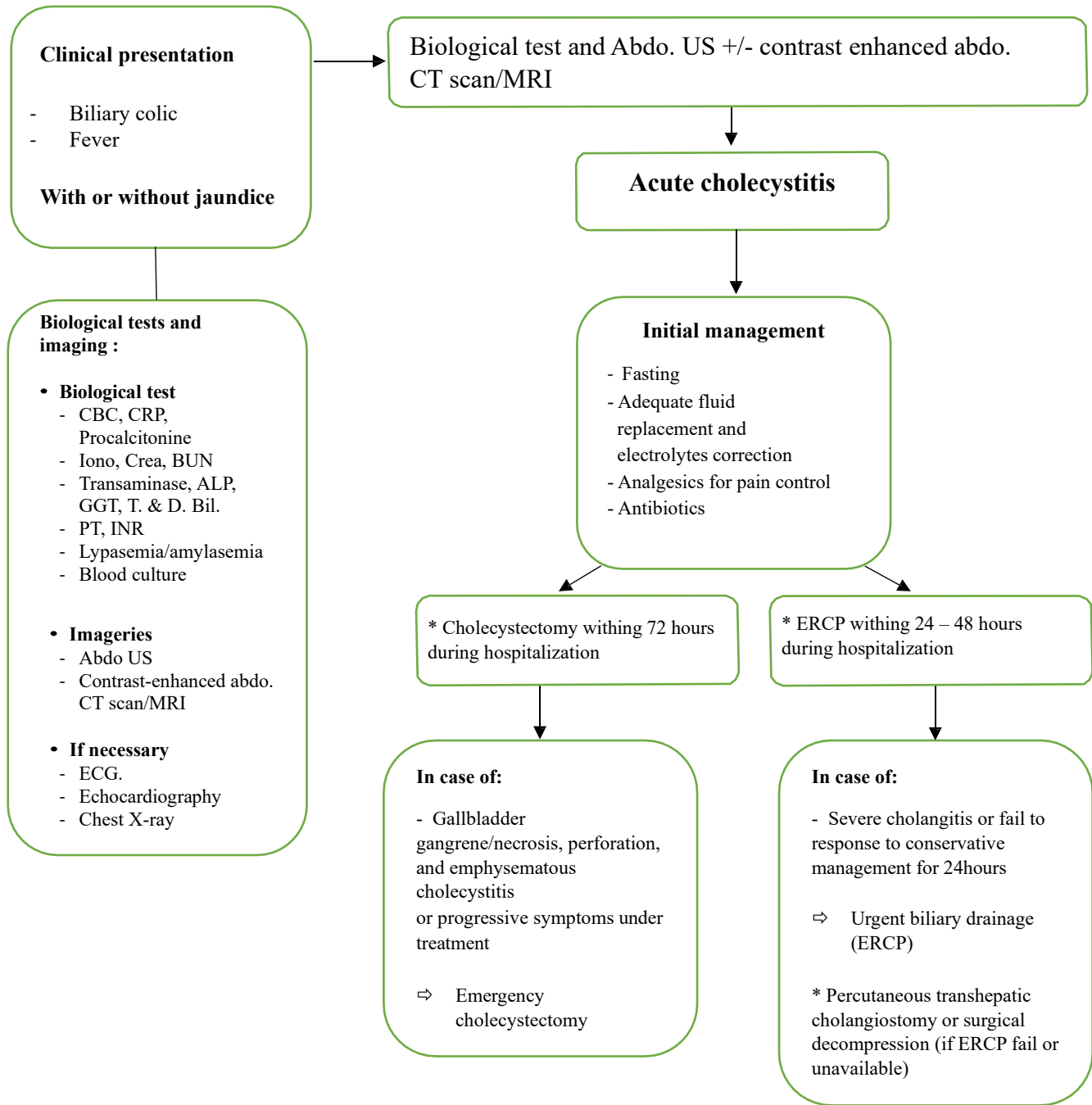
Δ Antianaerobic therapy, including use of metronidazole, tinidazole, or clindamycin, is warranted if a biliary-enteric anastomosis is present. The carbapenems, piperacillin/tazobactam, ampicillin/sulbactam, cefmetazole, cefoxitin, flomoxef, and cefoperazone/sulbactam have sufficient antianaerobic activity for this situation.

◇ Local antimicrobial susceptibility patterns (antibiogram) should be considered for use.

§ Fluoroquinolone use is recommended if the susceptibility of cultured isolates is known or for patients with beta-lactam allergies. Many extended-spectrum beta-lactamase (ESBL)-producing gram-negative isolates are fluoroquinolone resistant.

6. Treatment adjustment and management of underlying and predisposing conditions
- For acute cholecystitis:
 - Cholecystectomy within 72 hours during hospitalization is recommended.
 - In case of complicated acute cholecystitis (gallbladder gangrene/necrosis, perforation, and emphysematous cholecystitis), emergency cholecystectomy is recommended.
 - In case of progressive symptoms and signs such as high fever, hemodynamic instability, or intractable pain in spite of best supportive care (including antibiotics and gallbladder drainage) indicate disease progression, which is a sign of gallbladder gangrene and an indication for emergency cholecystectomy to prevent further complications.
 - For acute cholangitis:
 - In patients with mild to moderate cholangitis: biliary drainage by endoscopic retrograde pancreatography (ERCP) within 24 to 48 hours is indicated.
 - In patients with severe cholangitis or fail to response to conservative management for 24 hours, urgent biliary drainage is recommended.
 - Percutaneous transhepatic cholangiostomy or surgical decompression are rarely performed and are reserved for cases when endoscopic drainage fails or is unavailable.
 - In case of gallstone, elective cholecystectomy after the resolution of cholangitis is recommended to prevent future attacks of biliary colic and complications of gallstone disease.

7. Algorithm



CBC: complete blood count, CRP: C reactive protein, Iono: ionogram, Crea: creatinine, BUN: blood urea nitrogen, ALP: alkaline phosphatase, GGT: gamma-glutamyl transferase, T & D bili: total and direct bilirubin, PT: prothrombin time, INR: international normalized ratio, Abdo US: abdominal ultrasounds, CT scan: computed tomography scan, ECG: electrocardiogram, ERCP: endoscopic retrograde cholangiopancreatography.

VIII. SPECIAL CASES

- **For pregnant woman**
 - Managed the same way as patients who are not pregnant, with antibiotics and biliary drainage. However, antibiotic choices should take into account potential fetal toxicity.
- **Acalculous cholecystitis**
 - The management is the same as acute calculous cholecystitis.
 - Although cholecystectomy is preferred for definitive treatment, biliary drainage could be a safe, effective and alternative to surgery.
 - Once the episode of acalculous cholecystitis has been resolved, cholecystectomy is typically not required (the risk of recurrent cholecystitis appears to be low among those treated with cholecystostomy drainage).

IX. EVOLUTION AND PROGNOSIS

- The prognosis of biliary tract infection depends heavily on the severity of the infection, with early diagnosis and treatment leading to a good outcome, while advanced stage with complications like cholangitis can be life-threatening, especially in immunocompromised individuals.

X. COMPLICATIONS

- Acute pancreatitis
- Liver abscess
- Portal vein thrombosis
- Bacteremia and sepsis
- Multiorgan failure
- Endocarditis

XI. PREVENTION AND RECOMMENDATION

To prevent biliary tract infection, we must focus on managing risk factors like obesity, high fat diet and existing gallstones.

- Lifestyle modification:
 - Maintain a healthy weight.
 - Reduce saturated fat intake
 - Regular physical activity
 - Balance diet with fiber
- Patients who have gallstones should be informed about the sign and symptoms of biliary colic and biliary tract infection for early diagnosis and treatment.

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ANAL PAIN

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I- DEFINITION

Anal pain is pain in or around the anus or rectum (the perianal region). and the pain itself can be severe because of the many nerve endings in this region. It is a symptom which encompasses a broad spectrum of diagnoses from the benign and self-limited to the neoplastic and life-threatening and the pain itself can be severe because of the many nerve endings in this region.

II- EPIDEMIOLOGY

In 2014, Abramowitz et al. evaluated the prevalence of anal symptoms in general practice. Among 1079 patients who consulted their GP in France, 2% consulted spontaneously for an anal symptom. However, after systematic questioning of all patients, the prevalence was in fact 14.2%.

III- ETIOLOGY

Anal pain can occur before, during, or after a bowel movement or may not be associated with feces.

Table 1. Causes of Anal pain

• Most common causes ○ Anal fissure ○ Anorectal abscess/fistula ○ Thrombosed external hemorrhoid (TEH) ○ Thrombosed internal hemorrhoid (TIH) ○ Proctalgia fugax ○ Pruritus ani ○ Postoperative (anal procedures) infrequent
• Infrequent causes ○ Rectal prolapse ○ Foreign body ○ Fecal impaction ○ Hidradenitis suppurativa ○ Crohn's disease ○ Low pilonidal abscess ○ Bartholin's abscess
• Rare causes ○ Anal tumors (squamous cell carcinoma, melanoma, basal cell carcinoma) ○ Rectal cancer ○ Tumors of pelvic bones or organs (prostate, cervix, vulva, vagina or urethra) ○ HIV and other sexually transmitted infections ○ Tuberculosis and actinomycosis

IV- PATHOPHYSIOLOGY

1. Nociceptive Pain

- Caused by stimulation of nociceptors in the anal and perianal tissues due to injury, inflammation, or irritation.
- Common conditions:
 - **Anal fissure:** A tear in the anoderm, often caused by trauma during defecation or constipation, leading to severe, sharp pain.
 - **Hemorrhoids:** Engorged or thrombosed veins that become inflamed, causing pain, swelling, and bleeding.
 - **Perianal abscess:** Infection in the anal glands resulting in localized swelling, redness, and throbbing pain.

2. Neuropathic Pain

- Results from damage or irritation to the nerves supplying the anal region.
- Common causes:
 - **Pudendal neuralgia:** Compression or irritation of the pudendal nerve.
 - **Chronic anal fissures:** Persistent irritation of nerve endings.
 - **Trauma or surgery:** Nerve injury during procedures like hemorrhoidectomy.

3. Inflammatory Mechanisms

- Inflammation in the anal region triggers the release of cytokines, prostaglandins, and other mediators that sensitize pain receptors.
- Common conditions:
 - **Inflammatory bowel diseases (IBD):** Crohn's disease or ulcerative colitis can involve the anorectal area, causing pain, ulcers, or abscesses.
 - **Proctitis:** Inflammation of the rectal lining, often due to infections (e.g., sexually transmitted infections) or radiation therapy.

4. Spasm of Anal Sphincter

- Hypertonicity or spasm of the internal or external anal sphincter can cause severe, cramping pain.
- Common conditions:
 - **Levator ani syndrome:** Spasms in the levator ani muscle group.
 - **Proctalgia fugax:** Sudden, severe, and episodic pain due to transient spasm of the anal sphincter.

5. Ischemic Mechanisms

- Poor blood flow to the anal region can result in ischemia, leading to pain.
- Common causes:
 - **Thrombosed external hemorrhoid:** Blood clot in a hemorrhoidal vein, causing ischemia and pain.
 - **Ischemic proctitis:** Rare, but can occur with vascular compromise.

6. Malignancy-Related Pain

- Tumors in the anal canal or surrounding structures can cause direct tissue damage, inflammation, and nerve compression.
- Common malignancies:
 - Anal squamous cell carcinoma.
 - Rectal cancer.

7. Referred Pain

- Pain originating from nearby structures, such as the pelvis or lower gastrointestinal tract, can be perceived in the anal area.
- Common sources:
 - Pelvic inflammatory disease.
 - Gynecological conditions: endometriosis.

V- DIAGNOSIS

Many anorectal diagnoses share similar symptoms, and benign anorectal conditions share common symptoms with cancers of the anus and rectum. As such, a thorough evaluation is critical toward making the correct diagnosis and directing further evaluations.

1. History oriented toward the etiologies suspected

- The clinician should first elicit a history of anorectal bleeding, anorectal or perianal pain, perianal drainage, anal pruritus, or the prolapse of tissue through the anal orifice.
- Additionally, the clinician needs to ask:
 - Duration (constant, few minute...),
 - Location (anal, perianal),
 - Character (burning, sharp, dull),
 - Causative agents (bowel movement, diarrhea, hard stool, exercise, fecal incontinence, drainage),
 - Associated signs and symptoms (fever, chills, weight loss, change in bowel habits), and items that provide any relief (warm water bath, bowel movement, topical creams),
 - Men who have sex with men (MSM)...
 - Personal and family history: Diabetes Meletus, IBD, colorectal cancer (CRC).
- Elements of the patient history can provide some guidance:
 - DM may suggest an anal abscess or Fournier's gangrene.
 - IBD may hint at an anal fissures, fistulae, or abscess.
 - Infliximab or etanercept use may point to psoriasis as a cause for pruritus.
 - Family history of CRC may lead to consideration to rule out rectal cancer.
 - Anoreceptive intercourse may raise the concern about sexually transmitted infections (STIs), anal dysplasia, or anal cancer.

The patient response will help direct the subsequent evaluation.

Anorectal diseases could be organic or functional (e.g. Proctalgia fugax)

• **Proctalgia fugax** is a functional anorectal disorder characterized by severe, intermittent episodes of rectal pain that are self-limited. The diagnosis requires exclusion of other causes of rectal or anal pain. The patients have recurrent attacks of severe anorectal pain that can occur during the day or night. Each episode may last from a few seconds to minutes, but the duration does not exceed 30 minutes.

2. Physical examination

- **Anorectal examination:** Inspection, palpation and Digital rectal examination (DRE). The DRE should be performed with adequate lubrication, and the clinician should communicate to the patient throughout the examination to optimize patient comfort and compliance.

3. Anoscopy, proctoscopy, and endoscopy

- An anoscope is helpful for evaluating for anal condyloma, anal neoplasms, internal hemorrhoids, and anal fissures.
- A rigid proctoscopy is appropriate for identifying the location of rectal neoplasms as well as evaluating for proctitis.
- Endoscopy (sigmoidoscopy or colonoscopy): an anorectal examination is not complete without endoscopy. Clinicians should never make assumptions that anal bleeding is

secondary to benign anorectal diseases (eg, hemorrhoids), which could lead to a missed diagnosis of colorectal cancer.

VI- DIFFERENTIAL DIAGNOSIS

To distinguish anal pain with sources outside the anorectal region

- Chronic Pelvic Pain Syndrome
- Irritable Bowel Syndrome (IBS)
- Endometriosis.

VII- MANAGEMENT

1. Objectives

- To stop pain in order to improve the quality of life
- To cure the underlying causes
- To prevent the complications.

2. Critical Signs

- Severe and Sudden Onset of Pain.
- Fever and Systemic Symptoms.
- Bleeding.
- Swelling or Mass.

3. Mortality risk factor

- Delayed medical care.
- Immunosuppression (e.g., diabetes, cancer, HIV).
- Advanced age or comorbidities.
- Untreated infections or malignancies.
- Systemic complications like sepsis or organ failure.

4. Outpatient Management

- It is only for mild to moderate anal pain and has a specific diagnosis, such as Proctalgia fugax, Thrombosed external hemorrhoid, Pruritus ani....

5. First line treatment

The first line of treatment depends on its causes.

- **TEH:** Lifestyle change to avoid constipation, Pain killer (NSAID) or surgery (see Chapter hemorrhoids)
- **TIH:** Surgery (see Chapter hemorrhoids)
- **Anal fissure:** Lifestyle change to avoid constipation, bepanthen cream ointment or surgery (chronic and refractory case)
- **Proctalgia fugax**

For most patients, symptomatic episodes are brief and infrequent and require no specific treatment. Only a small percentage of patients have more frequent and troubling symptoms.

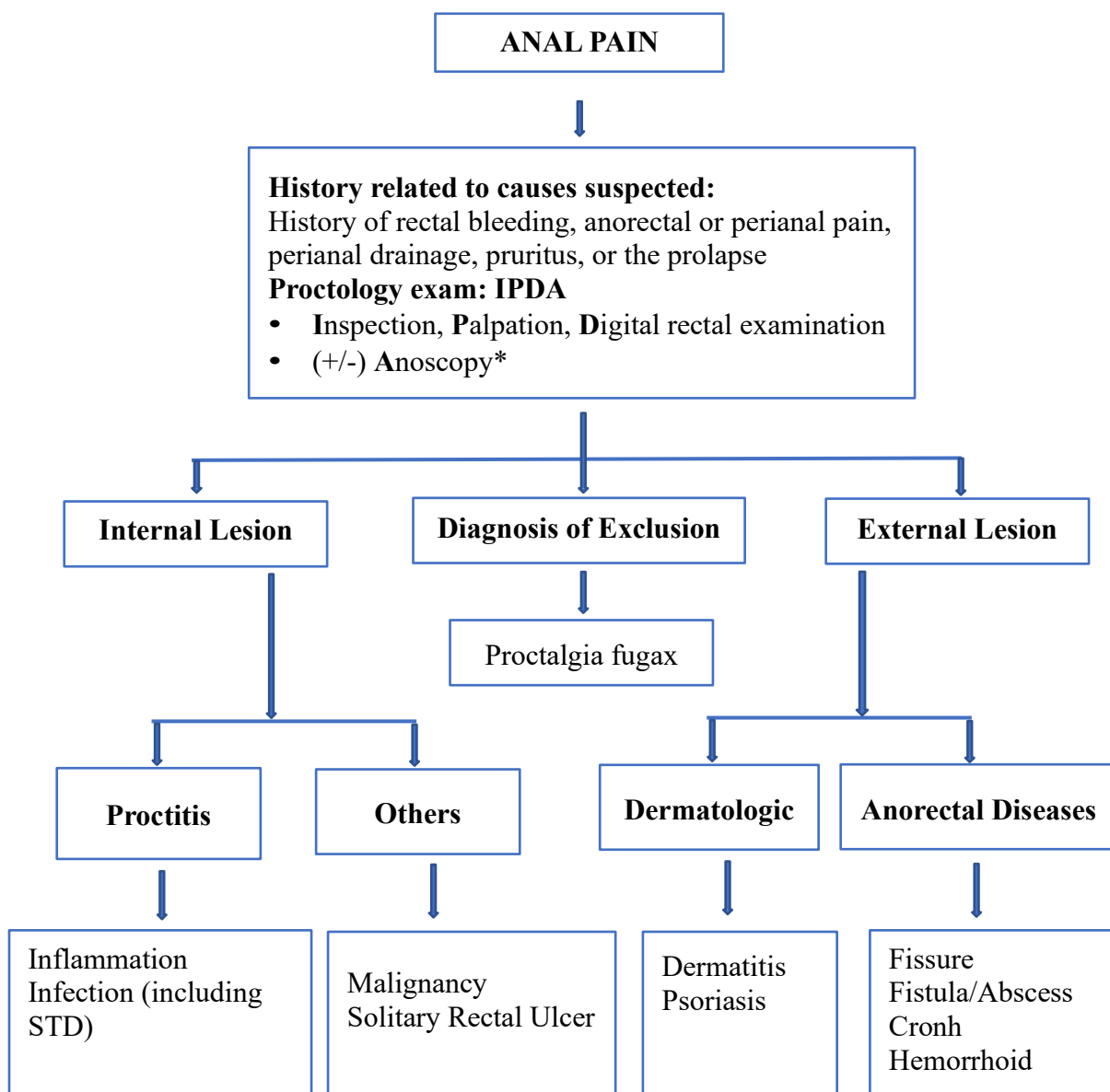
- Initial treatment: in most patients with mild and infrequent symptoms, we suggest reassurance and explanation. The episodes of pain are typically brief, and prevention is not possible.

- For patients who experience severe or frequent symptoms, sitz baths and treatment with topical antispasmodics may be helpful (nitroglycerin 0.2% or topical diltiazem gel 2%).
- If patients with severe or frequent symptoms who fail to respond to topical treatment, we perform additional evaluation with a rectal endoscopic ultrasound or magnetic resonance imaging of the abdomen and pelvis.

6. Treatment adjustment

- Pain: the clinician should first clarify whether the pain is perianal or anal in location, and if perianal. We need to find the exact location (s) of the pain.
- Take a warm bath to ease itching and pain.

7. Diagnostic Algorithm for Anal pain



* = proctoscopy or colonoscopy might be indicated through specific clinical features.; **STDs**: Sexually transmitted infections.

VIII- SPECIAL CASES

1- Anal Pain in Immunocompromised Patients

- Unique Features
 - Higher risk of severe infections and malignancies.
 - Delayed healing and increased susceptibility to complications.
- Examples
 - Perianal abscess or fistula in Crohn's disease.
 - Anorectal infections (e.g., herpes, cytomegalovirus, or fungal infections) in patients with HIV/AIDS.
 - Kaposi's sarcoma or lymphoma presenting as anorectal pain or mass.
- Management
 - Prompt and aggressive treatment of infections.
 - Coordination with infectious disease specialists for systemic therapies.

2- Anal Pain Associated with Systemic Diseases

- Endometriosis
 - Cyclical rectal pain that worsens during menstruation.
 - May be associated with dyspareunia and infertility.
- Inflammatory Bowel Disease (IBD)
 - Crohn's disease with perianal manifestations such as abscesses, fissures, or fistulas.
 - Vasculitis (e.g., Behçet's disease):
 - Painful ulcers in the anal or perianal area.
- Management
 - Treat underlying systemic disease.
 - Symptomatic relief for localized pain.

3. Anal Pain with Neurological Origins

- Pudendal Neuralgia
 - Chronic pain due to entrapment of the pudendal nerve.
 - Pain is positional, worsens with sitting, and may radiate to the genitals.
- Cauda Equina Syndrome
 - Rare but serious condition with anal pain, saddle anesthesia, and incontinence.
 - Requires immediate surgical intervention.
- Management
 - Pudendal neuralgia: Nerve blocks, physical therapy, or decompression surgery.
 - Cauda equina syndrome: Emergent decompression surgery.

IX. EVOLUTION AND PROGNOSIS

Evolution and Prognosis of anal pain depends on the etiology.

X- COMPLICATION

- Anal discomfort.
- Poor quality of life.
- Anemia (tumoral cause)
- Septic shock (anal abscess)

XI- PREVENTION AND RECOMMENDATION

- Hemorrhoids and anal fissure:
 - Adequate water intake and fiber to keep good bowel movement
 - Refrain from Straining or Linger (eg, reading) on the toilet.
 - Regular Physical exercise.
 - Limit their Intake of fatty foods & Alcohol (=> Constipation)
 - Avoid medications causing constipation or diarrhea.
- Anal and perineal hygiene
- Healthy food
- Routine Screening for Colorectal cancer: Regular colonoscopies for individuals aged 45 and older, or earlier for those with a family history or other risk factors.
- Inflammatory Bowel Disease: Optimize medical therapy to avoid complications
- Practice safe sex.
- Seek Regular Medical Follow-Up

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[DIAGNOSIS: Elements of the patient history can provide some guidance:]

HEMORRHOIDS

Asst. Prof. Chea Ong, Dr. Sann Channa

I- DEFINITION

Hemorrhoids (hemorrhoidal diseases) are very common anorectal conditions defined as the symptomatic enlargement and distal displacement of the normal anal cushions (piles).

II- EPIDEMIOLOGY

Hemorrhoidal disease is a common and debilitating clinical entity. The true prevalence of hemorrhoids is uncertain as anorectal discomfort is often attributed to symptomatic hemorrhoids. In a large, cross-sectional survey conducted in the United States, the self-reported prevalence of symptomatic hemorrhoids was 4.4 percent. The prevalence was equal in males and females, peaked between the ages of 45 and 65, and declined thereafter. Development of symptoms prior to 20 years was unusual.

III- ETIOLOGY

Risk Factors:

- ☐ Chronic constipation or diarrhea – Straining during bowel movements increases pressure on rectal veins.
- ☐ Prolonged sitting or standing – Reduces blood circulation and increases pressure in the anal area.
- ☐ Low-fiber diet – Leads to harder stools, making defecation difficult.
- ☐ Pregnancy – Increased abdominal pressure and hormonal changes contribute to hemorrhoid formation.
- ☐ Obesity – Excess weight puts additional pressure on the rectal veins.
- ☐ Aging – Weakening of rectal tissues over time.
- ☐ Heavy lifting – Frequent strain increases the risk of hemorrhoids.
- ☐ Genetic predisposition – Family history may increase susceptibility.

IV- ANATOMY AND PATHOPHYSIOLOGY

Anatomy of hemorrhoids

Their three primary locations (left lateral, right anterior, and right posterior) correspond to the end branches of the middle and superior hemorrhoidal veins.

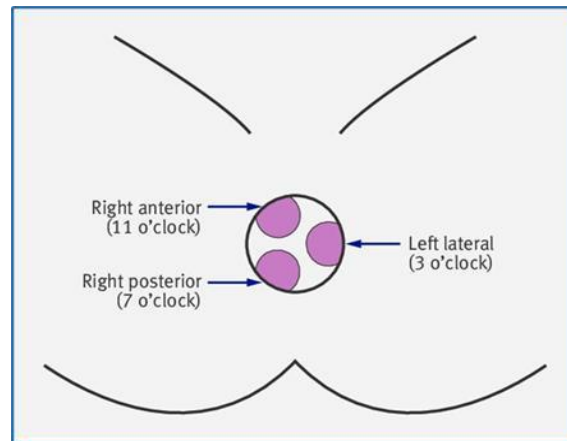


Figure 1: Three Positions of hemorrhoids (patients in supine position with legs in stirrups)

- External hemorrhoids (EH) arise from the inferior hemorrhoidal plexus.
- External and Internal hemorrhoids can be divided by Pectinate line.

Internal / External Haemorrhoids

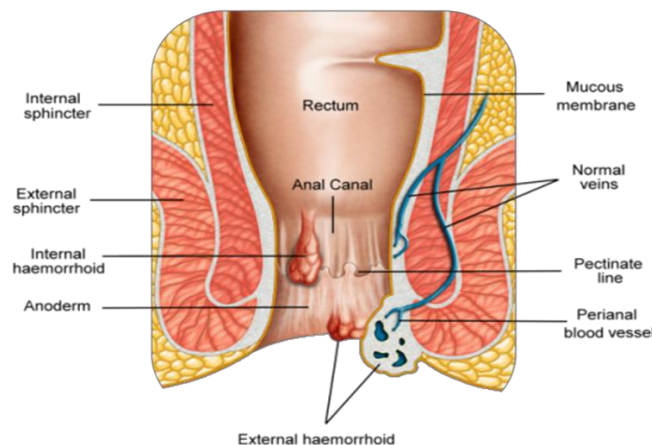


Figure 2: External and internal hemorrhoids

Pathophysiology of hemorrhoids

Four core pathophysiological events are universally recognized:

- first, the sliding process of anal cushions.
- second, the deterioration of the connective tissue of the cushion.
- third, the reduction of venous return from sinusoids to the superior rectal veins (SRV) and middle rectal veins (MRV) during defecation.
- and fourth, the stagnation of blood inside the dilated plexus.

V- DIAGNOSIS

Hemorrhoids can be divided into external, internal and mixed hemorrhoids

- Symptoms and Signs

1. **External hemorrhoids** (or Thrombosis of external) do not have specific grading but can cause severe discomfort, swelling, and thrombosis (blood clot formation).



Figure 3: External hemorrhoid (Thrombosed)

2. **Internal hemorrhoids (IH)** can cause bleeding and or prolapse and progress through different stages.

- Cardinal signs: painless bleeding and intermittent protrusion.

GRADING OF INTERNAL HEMORRHOIDS

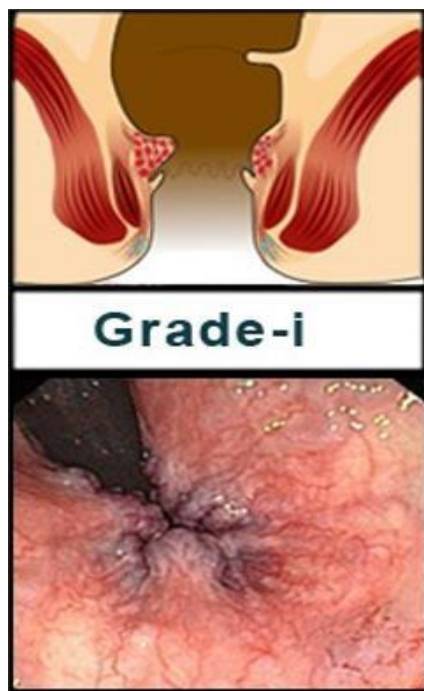


Figure 4 (Grade 1): Hemorrhoid protrudes into the anal canal but does not prolapse outside the anus.

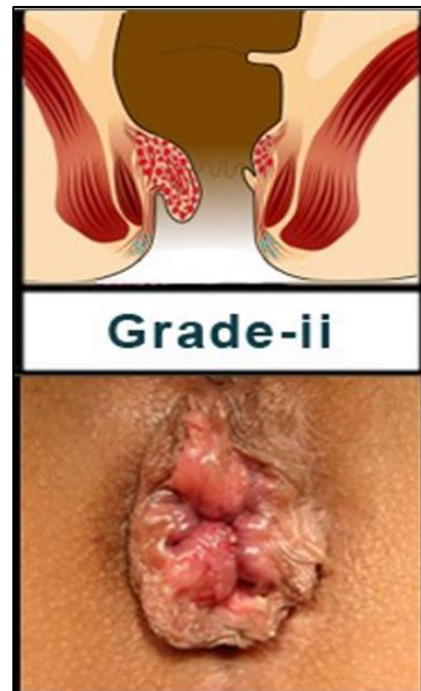


Figure 5 (Grade II): Hemorrhoid protrudes through the anus during straining or evacuation but returns spontaneously.



Figure 6 (Grade III): Hemorrhoid protrudes through the anus during straining or evacuation but needs to be manually returned to position.

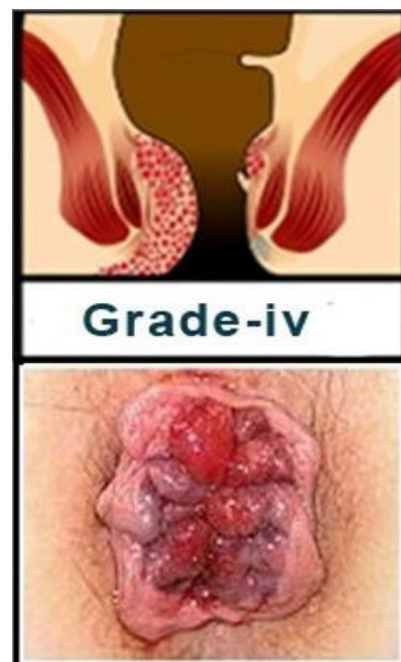


Figure 7 (Grade IV): Hemorrhoid remains prolapsed outside of the anus.

IH Grade I, II, III and IV.

Strangulated IH: hemorrhoids prolapse and strangulate. It can cause extreme pain.

3. Mixed hemorrhoids: Combination of external and internal hemorrhoids.

- **Laboratory evaluation** — Laboratory testing contributes little and is not routinely recommended, but if it is done and anemia or iron deficiency is identified, endoscopic evaluation should be performed.
- **Additional evaluation in selected patients** — Additional evaluation may be needed in selected patients with suspected hemorrhoidal bleeding to exclude other etiologies.
 - **Patients <40 years** – Endoscopic evaluation in patients with any of the following:
 - Clinical manifestations of alternative diagnosis (melena, hematochezia,)
 - Iron deficiency anemia (IDA)
 - Risk factors for colonic disease:
 - Previous gastrointestinal blood loss and previously resected colorectal cancer (CRC) or polyps.
 - Inflammatory bowel disease (IBD).
 - Pelvic radiation therapy.
 - Family history of IBD
 - Family history suggestive of advanced colorectal polyps, cancer, or hereditary CRC syndrome.
 - **Patients ≥ 40 years** – Endoscopic evaluation (colonoscopy) for other causes.

APPROACH TO DIAGNOSING HEMORRHOIDS

The diagnosis of hemorrhoids is primarily clinical, relying on history and physical examination, including anoscopy. Further investigations are warranted in patients with alarm symptoms or risk factors for serious conditions.

- History Taking: Characters of Symptoms and Associated Symptoms.
- Physical Examination (anal region): inspection, palpation and digital rectal exam.
- Anoscopy.

VI- DIFFERENTIAL DIAGNOSIS

Hemorrhoids present with symptoms like rectal bleeding, pain, itching, and prolapse. However, other conditions can mimic hemorrhoids, and a proper evaluation is essential to rule out serious pathologies.

Table 1: Differential Diagnosis for Rectal Pain, Bleeding, or Mass

Diagnosis	History	Physical examination
Abscess	Gradual onset of pain	Tender fluctuant mass
Cancer	Pain, bleeding, changes in bowel movements, weight loss	Ulcerating, indurated lesion
Condyloma	Possible bleeding; anal intercourse	Verrucous lesions
Fissure	Tearing pain with bowel movements	Visible tender fissure
Fistula	Soiling, itching	Visible opening of fistula
IBD	Bloody diarrhea, abdominal pain, family history	Possible fistula; colitis on anoscopy
Polyps	Painless bleeding	Polyps on endoscopy
Diverticular Bleeding	Painless, large-volume rectal bleeding.	Normal examination Confirmed by colonoscopy or CT scan
Rectal Varices (Portal Hypertension)	Painless, massive rectal bleeding, history of liver disease	Signes of cirrhosis Diagnosed via endoscopy
Proctalgia fugax	Painful rectum, no bleeding	Normal examination; diagnosis of exclusion
Proctitis	Painful rectum, bleeding	Tenderness on digital rectal examination
Rectal prolapse	Mass with Valsalva maneuver	Prolapse of rectal mucosa
Skin tags	History of hemorrhoids, no bleeding	Tags covered with normal skin

VII- TREATMENT

1. Objectives

- Symptom Relief (Improve Quality of Life) – Alleviate pain, itching, swelling, and discomfort.
- Reduce Inflammation and Swelling and Promote Healing
- Prevent Complications – Avoid thrombosis, bleeding, prolapse, and infection.
- Prevent Recurrence
- Minimize the need for surgery

2. Critical Signs

- ❑ Severe Rectal Bleeding
- ❑ Strangulated Hemorrhoids
- ❑ Signs of Infection
- ❑ Unexplained weight loss or change in bowel habits (underlying condition, such as Colorectal cancer?)

3. Mortality Risk Factors

- ❑ Massive Rectal Bleeding (rare) → Anemia & Shock
- ❑ Unexplained Weight Loss or Change in Bowel Habits or age > 50 years (underlying condition, such as CRC?)
- ❑ Elderly individuals with chronic bleeding
- ❑ Immunocompromised patients (e.g., diabetes, cancer, HIV)

4. Outpatient Management

Especially for non-surgical hemorrhoids.

5. First-Line Treatment

A. DIETARY AND LIFESTYLE MODIFICATION

- ❑ Consume 20 to 30 g of insoluble fiber per day and drink plenty of water (1.5 to 2 liters per day). Both are essential for producing regular, soft stools, which help reduce straining during defecation. It may take up to six weeks to fully experience the benefits of fiber.
- ❑ Establish a regular toilet routine.
- ❑ Engage in regular physical exercise.
- ❑ Avoid medications that can cause constipation or diarrhea.
- ❑ Limit the intake of fatty foods and alcohol.
- ❑ Avoid eating spicy foods (e.g., red hot chili peppers).

B. MEDICATIONS FOR SYMPTOMATIC RELIEF

- **Topical analgesics and steroids** — Topical ointments containing anesthetics, steroids, emollients, and antiseptics are often self-prescribed by the patients for quick symptomatic relief (*lidocaine-hydrocortisone*, *hydrocortisone*).
- **Venoactive agents** — Phlebotonics are primarily used in Europe and Asia to treat acutely inflamed and bleeding hemorrhoids. Phlebotonics have not been approved by the US Food and Drug Administration for the indication of hemorrhoids.

E.g. Daflon 1000mg

- Acute hemorrhoidal attack: 3 tablets daily (in 3 divided doses) for the first 4 days, then 2 tablets per day (in 2 divided doses) for the next 3 days.
- **Antispasmodic agents** — Several types of agents (nitric oxide donors) can be useful for reducing anal sphincter spasm.

E.g. Nitroglycerin (glyceryl trinitrate) ointment (0.2%, Rectogesic) was effective in controlling bleeding, pain, itch, irritation, and difficulty in defecation.

C. OFFICE-BASED PROCEDURE (OBP)

- Internal Hemorrhoids

- Grade I, II, or III internal hemorrhoids, we recommend rubber band ligation (RBL) rather than another procedure. Compared with sclerotherapy or infrared coagulation, RBL is more effective and requires fewer treatment sessions.



Figure 8: Rubber band ligation (RBL)

D. SURGERY

- External Hemorrhoids (EH)

- **Excision:** Acute pain and thrombosis of the EH within 48-72 hours of onset.
 - Infiltration of a local anesthetic containing epinephrine is followed by elliptical incision and excision of the thrombosed hemorrhoid, its accompanying vein, and overlying skin.
- **Incision:**
Simple incision and clot evacuation is inadequate therapy. So, it is not recommended.
- **For EH after 72 hours** from the start of symptoms, conservative medical therapy is preferable (medical treatment with warm baths).
 - **Warm baths:** bathing in tubs with warm water universally eases painful perianal conditions. Relaxation of the sphincter mechanism and spasm is probably the etiology. Ice can relieve the pain of acute thrombosis.

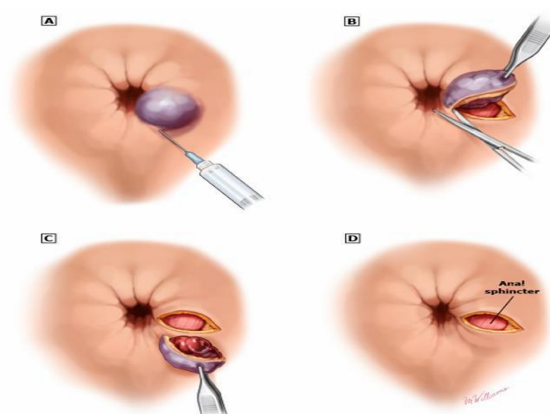


Figure 9: Excision of thrombosed external hemorrhoids

- Internal hemorrhoids (IH)

- IH of any grade with repeated bleeding and severe anemia (uncontrolled by conservative medical treatments).
- IH Grade III with more than one column.
- Prolapsed and incarcerated IH (Grade IV).
- Symptomatic IH (e.g., pain, thrombosis) refractory to conservative measures office-based procedures (OBP), or unable to tolerate OBP because of pain.
- Combined internal and external hemorrhoids.
- Symptomatic IH concomitant with anorectal condition that requires surgery.

INTERNAL HEMORRHOIDECTOMY

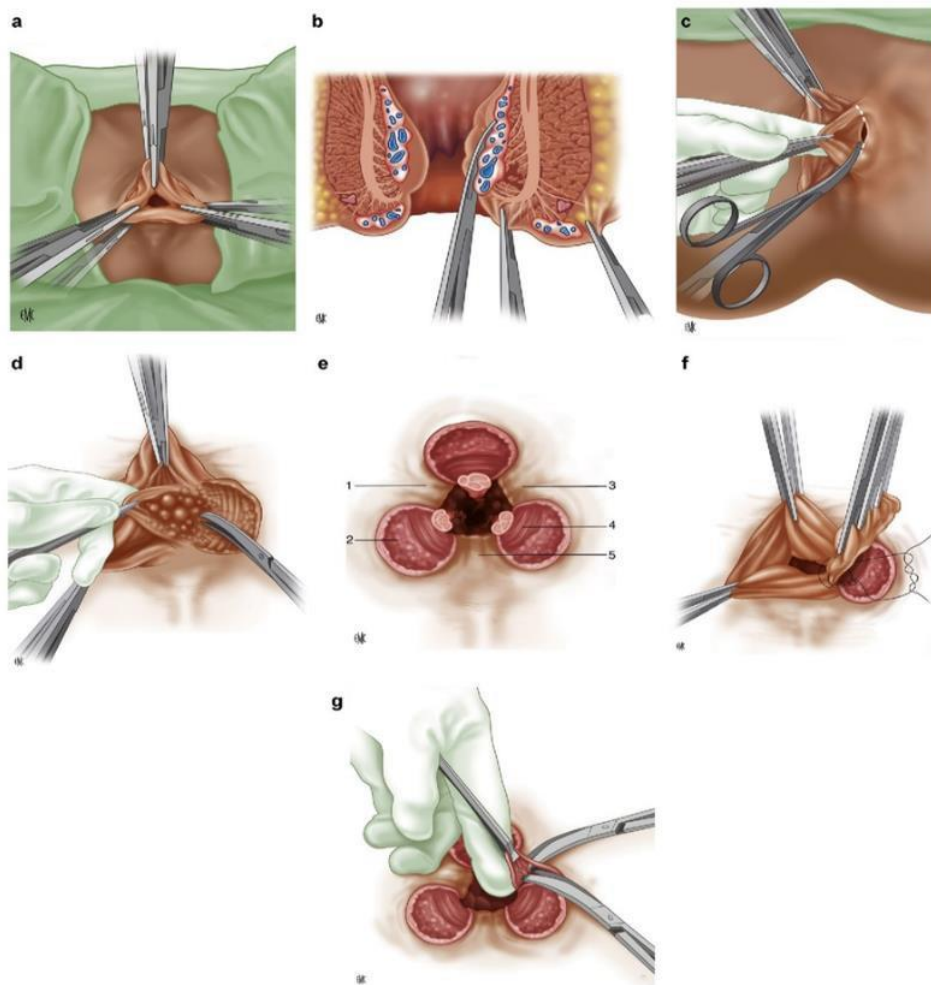


Figure 10: Open hemorrhoidectomy (Milligan-Morgan)

STAPLED HEMORRHOIDOPEXY

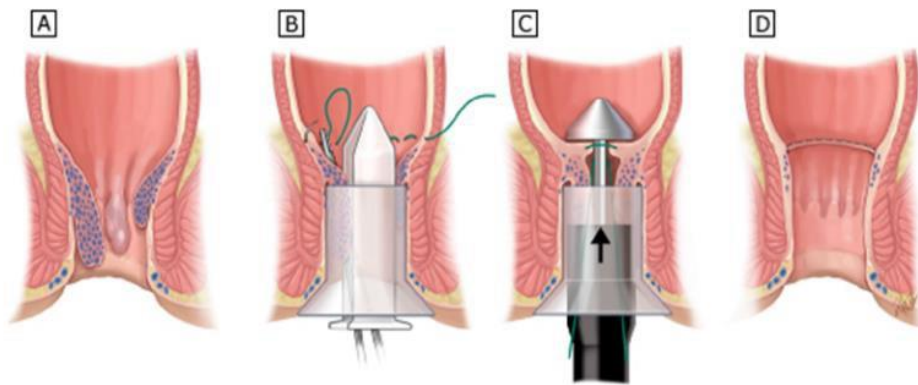


Figure 11: Staped hemorrhoidopexy

DOPPLER-GUIDED TRANSANAL HEMORRHOIDAL DEARTERIALIZATION

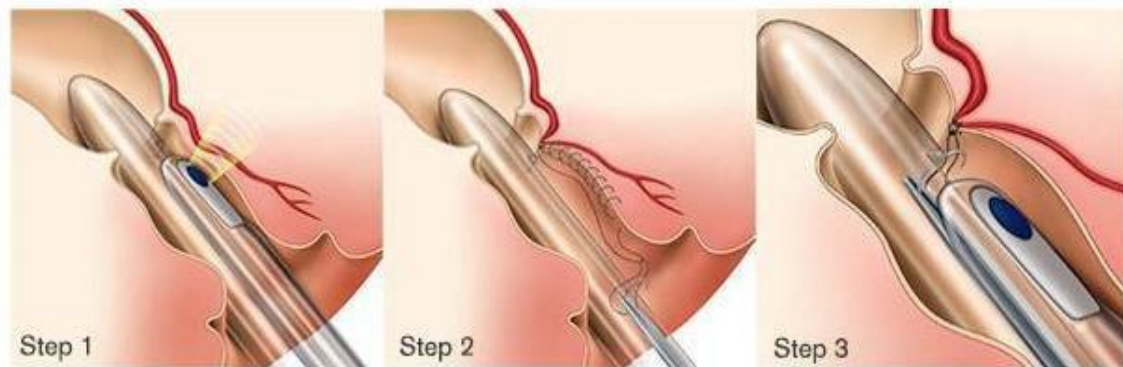


Figure 12: Doppler-guided transanal hemorrhoidal dearterialization

DOPPLER-GUIDED HEMORRHOID ARTERY LIGATION WITH RECTO-ANAL REPAIR

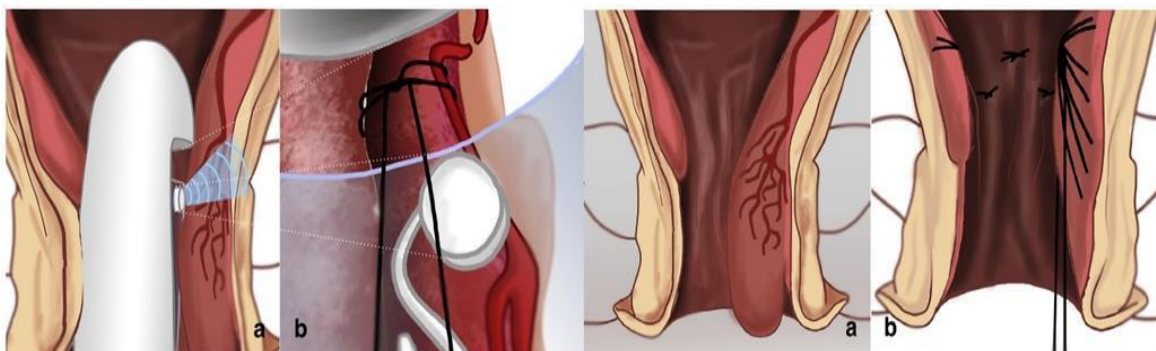


Figure 13: Doppler-guided hemorrhoid artery ligation with recto-anal repair

6. Treatment adjustment

See points A & B in First-Line Treatment

7. Algorithm

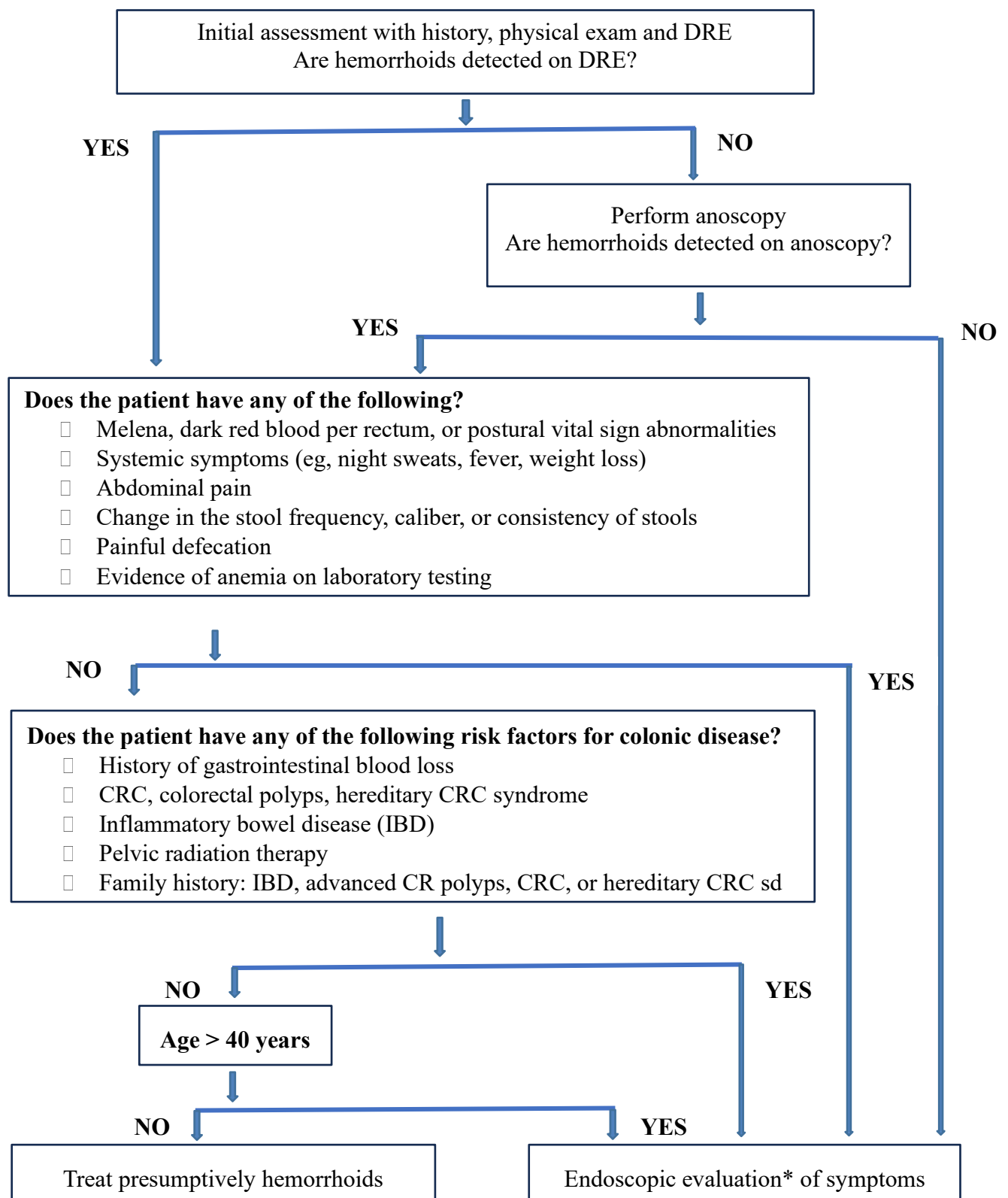


Figure 14: Approach to Patient with suspected hemorrhoidal bleeding

*CRC: Colorectal cancer; DRE: Digital rectal examination; sd: syndrome; * Colonoscopy ± gastroscopy.*

VIII- SPECIAL CASE

Hemorrhoids in pregnancy

First-line treatment

1. Conservative Measures (Non-Pharmacologic)

- Increase Fiber Intake: To prevent constipation and reduce straining during bowel movements
- Adequate Hydration: Drink plenty of water to soften stools and prevent constipation.
- Sitz Baths: Soaking the anal area in warm water for 10–15 minutes several times a day can help reduce pain, itching, and swelling.
- Proper Bowel Habits: Avoid straining, prolonged sitting, or reading while on the toilet.
- Elevate Feet: Use a stool or footrest to elevate the feet while sitting on the toilet to reduce pressure on the hemorrhoidal veins.

2. Topical Treatments: Topical hydrocortisone cream (low strength) can be used for short-term relief of relieve pain and swelling.

3. Stool Softeners: Psyllium or Docusate Sodium

4. Pain Relief

- Acetaminophen (Tylenol): Can be used for pain relief if necessary. Nonsteroidal anti-inflammatory drugs (NSAIDs), such as **Ibuprofen**, should be avoided during pregnancy, especially in the third trimester.

5. Venoactive agents (the same usage in general cases)

Second-line treatment: In case of failure with the first-line treatment => Consult specialist.

IX- EVOLUTION AND PROGNOSIS

□ Mild to Moderate Hemorrhoids

- Good prognosis with conservative treatment (dietary fiber, hydration, sitz baths, and topical medications).
- Symptoms like pain, itching, and bleeding often improve within a few days to weeks.
- Lifestyle modifications (avoiding straining, high-fiber diet) can prevent recurrence.

□ Severe or Chronic Hemorrhoids

- Internal hemorrhoids (Grade III-IV) may require medical intervention, such as rubber band ligation, sclerotherapy, or surgery (hemorrhoidectomy, stapled hemorrhoidopexy).
- If left untreated, they may lead to complications like thrombosis, strangulation, or anemia due to chronic bleeding.

□ Post-Treatment Prognosis

- Minimally invasive procedures have a high success rate and low recurrence risk.
- Surgical treatments offer long-term relief but may involve post-operative pain and complications like infection or anal stenosis.

□ Risk of Recurrence

- Without lifestyle changes, recurrence is common.

- Preventive measures like a high-fiber diet, avoiding prolonged sitting, and maintaining good bowel habits improve long-term outcomes.

X- COMPLICATIONS

- Thrombosed Hemorrhoids (External Hemorrhoids)
- Strangulated internal hemorrhoids
- Prolapse and persistent Discomfort
- Fecal incontinence
- Perianal skin tags
- Rectal ulceration and Fissures
- Anemia (IDA)
- Infection

XI- PREVENTION AND RECOMMENDATION

- Maintain a High-Fiber Diet
- Stay Hydrated
- Avoid Straining During Bowel Movements
- Exercise Regularly
- Maintain a Healthy Weight
- Practice Good Toilet Habits
- Maintain Proper Anal Hygiene
- Manage Chronic Conditions

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CHAPTER V NEPHROLOGY

1. Acute kidney injury
2. Chronic kidney disease
3. Nephrotic syndrome
4. Post streptococcal glomerulonephritis
5. Urinary tract infection
6. Kidney stone

ACUTE KIDNEY INJURY

*Dr. Niv Rathvirak, Dr. Sin Boohavan, Dr. Lim Vadhana, Dr. Thim Pichthida,
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I. DEFINITION AND CLASSIFICATION

- Acute Kidney Injury (AKI) is defined as reduction in kidney function, including decreased GFR and kidney failure.
- In the short term, AKI is associated with an increased length of hospital stay, health care costs, and in-hospital mortality, and its impact extends into the long term, with AKI being associated with increased risks of cardiovascular events, progression to chronic kidney disease (CKD), and long-term mortality.
- Kidney failure is a stage of AKI is defined as a GFR < 15 ml/min per 1.73 m² body surface area, or requirement for RRT.
- The criteria for the diagnosis of AKI and the stage of severity of AKI are based on changes in SCr and urine output. KDIGO was the most commonly used criteria, followed by AKIN, and RIFLE (estimated GFR does not reflect measured GFR in AKI as accurately as in CKD).

Table 1. | Comparison of Recent Consensus AKI Definitions

AKI Stage	Urine Output ^a	KDIGO	AKIN	RIFLE
1	<0.5 mL/kg/h for 6-12 h	Scr to 1.5-1.9 × baseline over 7 d or ≥0.3 mg/dL absolute increase over 48 h	Scr to 1.5-2 × baseline or ≥0.3 mg/dL absolute increase within 48 h	<i>Risk:</i> Scr to ≥1.5 × increase within 7 d, sustained for ≥24 h
2	<0.5 mL/kg/h for ≥12 h	Scr to 2.0-2.9 × baseline	Scr to >2-3 × baseline	<i>Injury:</i> Scr to ≥2 × increase
3	<0.3 mL/kg/h for ≥24 h or anuria for ≥12 h	Scr to ≥3.0 × baseline, or Scr increase to ≥4.0 mg/dL or initiation of RRT	Scr to >3.0 × baseline, or Scr increase to ≥4.0 mg/dL (with increase of 0.5 mg/dL) or initiation of RRT	<i>Failure:</i> Scr to ≥3.0 × increase or Scr increase to ≥4.0 mg/dL (with increase of 0.5 mg/dL) or initiation of RRT <i>Loss:</i> Complete loss of kidney function for >4 wk <i>ESKD:</i> ESKD for >3 mo

II. EPIDEMIOLOGY

- The overall incidence of AKI was 23.2%
- The incidence of community-acquired AKI was 8.3%, while the incidence of hospital-acquired AKI was 20.9%.
- In the critical care setting, every third patient developed AKI (31.7%).
 1. Sepsis (40.7 %)
 2. Hypovolemia (34.1 %)
 3. Drug related (14.4 %)
 4. Cardiogenic shock (13.2 %)
 5. Hepatorenal syndrome (3.2 %)
 6. Obstruction of the urine outflow tract (1.4 %)

III. ETIOLOGY

Table 2. | Causes of Acute Kidney Injury

Prerenal	- Intrarenal vasoconstriction (hemodynamically mediated) ○ Medications: NSAID, * ACE, * ARB, * cyclosporine (Sandimmune), tacrolimus (Prograf) ○ Cardiorenal syndrome* ○ Hepatorenal syndrome ○ Abdominal compartment syndrome ○ Hypercalcemia - Systemic vasodilation (e.g., sepsis, * neurogenic shock) - Volume depletion: ○ Renal loss from diuretic overuse, * ○ osmotic diuresis (e.g., diabetic ketoacidosis*) ○ Extrarenal loss from vomiting, diarrhea, * burns, sweating, blood loss.
	- Glomerular (e.g., postinfectious and other glomerulonephritis) - Interstitial ○ Medications: penicillin analogues, * cephalosporins, * sulfonamides, ciprofloxacin (Cipro), acyclovir (Zovirax), rifampin, phenytoin (Dilantin), interferon, proton pump inhibitors, NSAID. ○ Infections (e.g., direct infection of renal parenchyma or associated with systemic infections) ○ Viruses: Epstein-Barr virus, cytomegalovirus, human immunodeficiency virus ○ Bacteria: Streptococcus species, Legionella species ○ Fungi: candidiasis, histoplasmosis ○ Systemic disease: sarcoidosis, lupus - Tubular ○ Ischemic: prolonged hypotension* ○ Nephrotoxic: exogenous toxins (e.g., radiographic contrast agents, * aminoglycosides, * cisplatin, methotrexate, ethylene glycol, amphotericin B) and endogenous toxins (e.g., hemolysis and rhabdomyolysis [pigment nephropathy], tumor lysis syndrome, myeloma) - Vascular ○ Renal vein thrombosis, malignant hypertension, scleroderma renal crisis, renal atheroembolic disease, * and renal infarction
Post renal	- Extrarenal obstruction: prostate hypertrophy*; neurogenic bladder; retroperitoneal fibrosis; bladder, prostate, or cervical cancer - Intrarenal obstruction: stones, * crystals (acyclovir, indinavir [Crixivan]), clots, tumors

Adapted with permission from Holley JL. Clinical approach to the diagnosis of acute renal failure. In: Greenberg A, Cheung AK, eds. Primer on Kidney Diseases. 5th ed. Philadelphia, Pa.: National Kidney Foundation; 2009:278.

IV. PATHOPHYSIOLOGY

A. Prerenalcauses:

- Decreased renal perfusion associated with intravascular volume depletion or decreased arterial pressure results in a reduced glomerular filtration rate (without damage to the renal parenchyma).
- Several medications can also cause prerenal AKI (table 2).
- In patients with prerenal AKI, kidney function typically returns to baseline after adequate volume status is established, the underlying cause is treated, or the offending drug is discontinued.

B. Intrinsicrenalcauses:

- Intrinsic renal causes are also important sources of AKI and can be categorized by the component of the kidney that is primarily affected (Tubular damage, Glomerular damage, Interstitial damage or Vascular damage)
- Acute tubular necrosis is the most common type of intrinsic AKI in hospitalized patients. The cause is usually ischemic (from prolonged hypotension) or nephrotoxic (from an agent that is toxic to the tubular cells).
- Glomerular causes of AKI are the result of acute inflammation of blood vessels and glomeruli resulting from systemic illness. Renal biopsy is often required to confirm the diagnosis before initiating therapy.
- Acute interstitial nephritis can be secondary to many conditions, but most cases are related to medication use. Eosinophiluria may be found in patients with acute interstitial nephritis, but it is not pathognomonic. A kidney biopsy may be needed to distinguish between allergic interstitial nephritis and other renal causes of AKI.
- Acute events involving renal arteries or veins can also lead to intrinsic acute kidney injury. Renal atheroembolic disease is the most common cause and is suspected with a recent history of arterial catheterization, the presence of a condition requiring anticoagulation, or after vascular surgery.

C. Postrenalcauses:

occurs after acute obstruction of the urinary flow, which increases intra-tubular pressure and thus decreases GFR. In addition, acute urinary tract obstruction can lead to impaired renal blood flow and inflammatory processes that also contribute to diminished GFR.

V. RISK FACTOR

Risk for AKI is increased by exposure to factors that cause AKI or the presence of factors that increase susceptibility to AKI.

Table 3. | Causes of AKI: exposures and susceptibilities for non-specific AKI

Exposures	Susceptibilities
Sepsis	Dehydration or volume depletion
Critical illness	Advanced age
Circulatory shock	Female gender
Burns	Black race
Trauma	Chronic kidney disease
Cardiac surgery	Chronic diseases (heart, lung, liver)
Major non-cardiac surgery	Diabetes mellitus
Nephrotoxic drugs	Cancer
Radiocontrast agents	Anemia
Poisonous plants and animals	
Cardiopulmonary bypass	
Being hospitalized, especially for a serious condition that requires intensive care	

VI. DIAGNOSIS

Clinical presentation:

- Varies with the cause and severity of renal injury, and associated diseases.
- **Patients with mild to moderate AKI** are asymptomatic and are identified on laboratory testing.
- **Patients with severe AKI** : listlessness, confusion, fatigue, anorexia, nausea, vomiting, weight gain, or edema, oliguria (urine < 400 mL per day), anuria (urine < 100 mL per day).
- Other manifestation: **uremic encephalopathy** (manifested by a decline in mental status, asterixis, or other neurologic symptoms), **anemia**, or **bleeding** caused by **uremic platelet dysfunction**.

History taking and Physical examination

Table 4. | History finding and Physical Examination finding in AKI

<i>Type of AKI</i>	<i>History findings</i>	<i>Physical examination findings</i>
Prerenal	Volume loss (e.g., history of vomiting, diarrhea, diuretic overuse, hemorrhage, burns) Thirst and reduced fluid intake Cardiac disease Liver disease	Weight loss, orthostatic hypotension and tachycardia Poor skin turgor Dilated neck veins, S3 heart sound, pulmonary rales, peripheral edema Ascites, caput medusae, spider angiomas

Intrinsic renal

Acute tubular necrosis	History of receiving nephrotoxic medications (including over-the-counter, illicit, and herbal), hypotension, trauma or myalgias suggesting rhabdomyolysis, recent exposure to radiographic contrast agents	Muscle tenderness, compartment syndrome, assessment of volume status
Glomerular	Lupus, systemic sclerosis, rash, arthritis, uveitis, weight loss, fatigue, hepatitis C virus infection, human immunodeficiency virus infection, hematuria, foamy urine, cough, sinusitis, hemoptysis	Periorbital, sacral, and lower-extremity edema; rash; oral/nasal ulcers
Interstitial	Medication use (e.g., antibiotics, proton pump inhibitors), rash, arthralgias, fever, infectious illness	Fever, drug-related rash
Vascular	Nephrotic syndrome, trauma, flank pain, anticoagulation (atheroembolic disease), vessel catheterization or vascular surgery	Livedo reticularis, fundoscopic examination (showing malignant hypertension), abdominal bruits
	Urinary urgency or hesitancy, gross hematuria, polyuria, stones,	Bladder distention, pelvic mass, prostate enlargement
Postrenal	medications, cancer	

Laboratory investigation:

Urinalysis, CBC, and measurement of serum creatinine level and fractional excretion of sodium (FENa).

Table 5. | Role of Fractional Excretion of Sodium and Urea in Acute Kidney Injury Diagnosis

	Fraction Excretion of Sodium (FENa)	Fraction Excretion of Urea (FEUrea)
Calculation	$FENa = 100 \times \frac{(\text{urinary Na} \times \text{serum creatinine})}{(\text{serum Na} \times \text{urinary creatinine})}$	$FEUrea = 100 \times \frac{(\text{urinary urea} \times \text{serum creatinine})}{(\text{serum urea} \times \text{urinary creatinine})}$
Interpretation	- FENa < 1% indicates a prerenal cause of AKI, suggesting that the kidneys are conserving sodium in response to perceived hypoperfusion - FENa > 2% indicates a intrinsic renal damage, suggesting that the kidneys are impariing reabsorb Na due to tubular injury. * The interpretation of FENa can be confounded in patients receiving diuretic therapy. Diuretics induce natriuresis, which can lead to elevated FENa values even in the presence of prerenal causes.	- FEUrea < 35% indicates a prerenal cause of AKI, suggesting that the kidneys are conserving urea in response to perceived hypoperfusion - FENa < 35% indicates a intrinsic renal damage, suggesting that the kidneys are impariing reabsorb Urea due to tubular injury. * The fractional excretion of urea (FEUrea) is a diagnostic tool in the evaluation of acute kidney injury (AKI),in patients on diuretic therapy.

Table 6. | Biochemical parameters in Prerenal, Renal and Post Renal AKI

Parameter	Pre-renal	Intra-renal	Post-renal
Urine sodium (U Na ⁺)	<20 mmol/L	>40 mmol/L	>40 mmol/L
Fractional excretion of sodium (FE Na ⁺)	< 1 %	> 2 %	> 1 %
Fractional excretion of Urea (FE Urea)	< 35 %	> 35 - 50 %	N/A
Urine osmolality	≤ 1,200 mOsm/kg	< 300 mOsm/kg	< 300 mOsm/kg
Urine creatinine: serum creatinine ratio	> 40 : 1	< 20 : 1	< 20 : 1
Urine Specific gravity	> 1.010	< 1.010	Varies

- *Urine Sodium (U Na⁺) and Fractional Excretion of Sodium (FE Na⁺) can be used to evaluate acute kidney injury (AKI) in **absence of diuretic therapy** (diuretics alter the sodium concentration, making the FENa equation unusable) (8)*
- *Fractional excretion of Urea (FE Urea) can still be used in patients **on diuretic therapy**. (9)*

Table 7. | Diagnostic Test Results and Corresponding Diseases in Patients with AKI

Test result	When to order	Associated diseases/conditions
Elevated antineutrophil cytoplasmic antibody, antiglomerular basement membrane antibody	Suspected acute glomerulonephritis pulmonary renal syndromes	Vasculitis, Goodpasture syndrome
Elevated antistreptolysin O titer	Recent infection and clinical picture of acute glomerulonephritis	Poststreptococcal glomerulonephritis
Elevated creatine kinase level, elevated myoglobin level, dipstick positive for blood but negative for red blood cells	Recent trauma, muscle injury	Rhabdomyolysis
Elevated prostate-specific antigen level	Older men with symptoms suggestive of urinary obstruction	Prostate hypertrophy, prostate cancer
Elevated uric acid level	History of rapidly proliferating tumors, recent chemotherapy	Malignancy, tumor lysis syndrome
Eosinophiluria	Fever, rash	Allergic interstitial nephritis
Evidence of hemolysis (schistocytes on peripheral smear, decreased haptoglobin level, elevated indirect bilirubin level, elevated lactate dehydrogenase level)	Fever, anemia, thrombocytopenia neurologic signs	Hemolytic uremic syndrome, thrombotic thrombocytopenic purpura, systemic lupus erythematosus, other autoimmune diseases

Hydronephrosis on renal ultrasonography	Suspected obstruction	Malignancy, prostate hypertrophy, uterine fibroids, nephrolithiasis, ureterolithiasis
Increased anion gap with increased osmolar gap*	Suspected poisoning, unresponsive patient	Ethylene glycol or methanol poisoning
Low complement level	Suspected acute glomerulonephritis	Systemic lupus erythematosus, endocarditis, postinfectious glomerulonephritis
Monoclonal spike on serum protein electrophoresis	Anemia, proteinuria, acute kidney injury in older patients	Multiple myeloma
Positive antinuclear antibody, double-stranded DNA antibody	Proteinuria, skin rash, arthritis	Autoimmune diseases, systemic lupus erythematosus
Positive blood cultures	Intravenous drug use, recent infection, new cardiac murmur	Endocarditis
Positive HIV test HIV = human immunodeficiency virus.	Risk factors for HIV infection	HIV nephropathy

*—Calculations are as follows:
 Anion gap = sodium – (chloride + bicarbonate)
 Calculated serum osmolality = 2(sodium [in mEq per L]) + (blood urea nitrogen [in mg per dL] ÷ 2.8) + (glucose [in mg per dL] ÷ 18)
 Osmolar gap = measured serum osmolality – calculated serum osmolality.

Imaging studies:

Renal ultrasonography for rule out obstruction. Other imaging modalities, such as computed tomography or magnetic resonance imaging, may be required.

Kidney biopsy:

1. Renal biopsy is reserved for patients in whom prerenal and postrenal causes of acute kidney injury have been excluded and the cause of intrinsic renal injury is unclear.
2. Renal biopsy may need to be performed urgently in patients with oliguria who have rapidly worsening acute kidney injury, hematuria, and red blood cell casts.

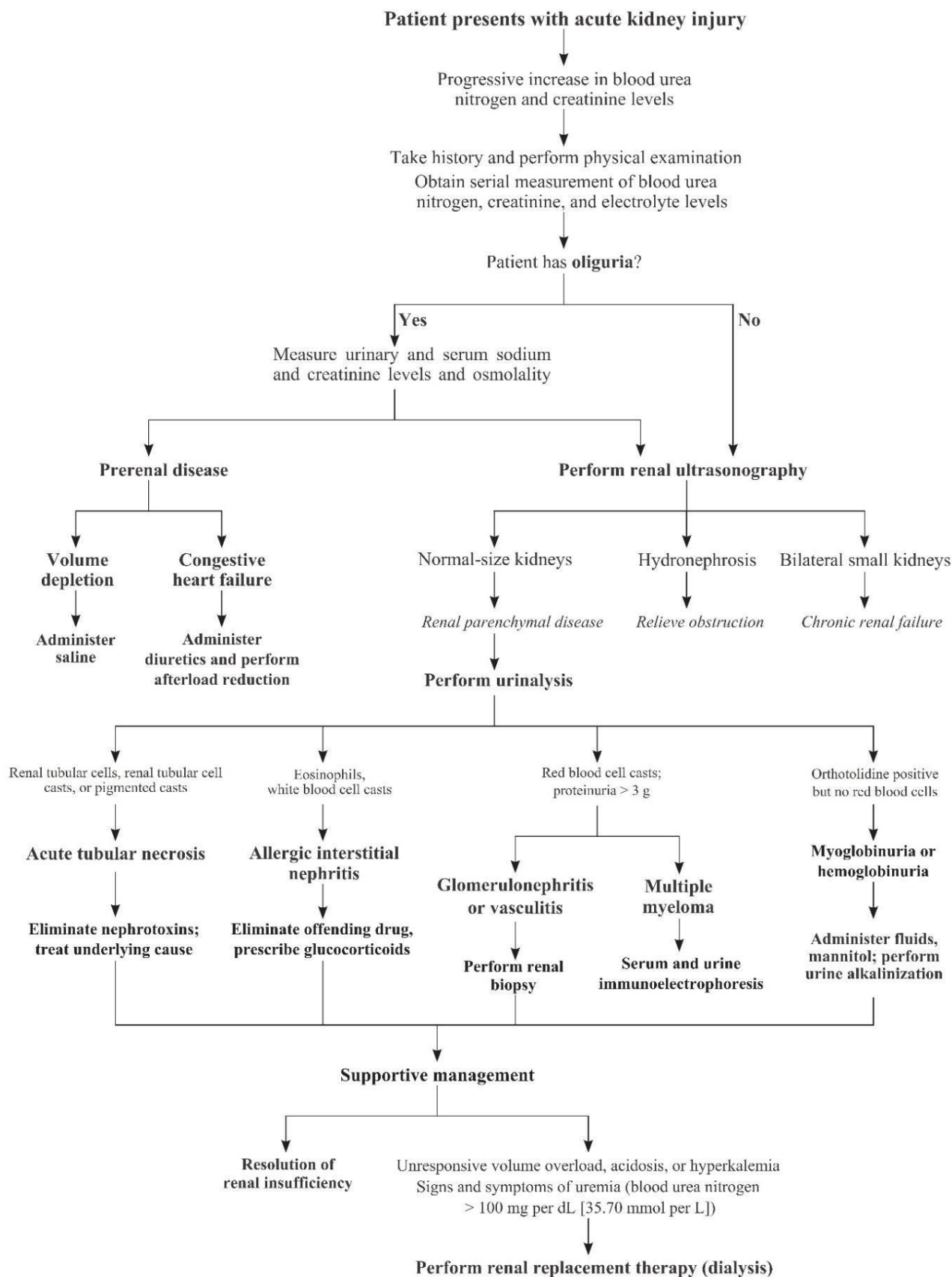


Figure 1. Algorithm for the diagnosis and treatment of acute kidney injury

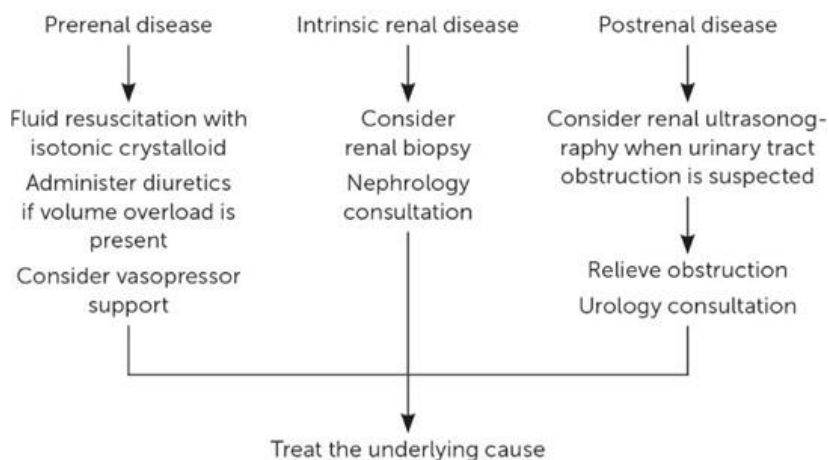
VII. DIFFERENTIAL DIAGNOSIS

- CKD: 3 criteria distinguish AKI from CKD
 1. History taking:
 - No past medical history of kidney disease.
 - Normal baseline of kidney function prior to recent illness.
 - Clinical context exposed to AKI.
 2. Renal ultrasound:
 - Normal kidney size and echo structure
 3. Laboratory investigation:
 - No anemia
 - No hypocalcemia
- But attention:
 - AKI associated with anemia: acute hemolysis, hemorrhagic shock, inflammation, etc.
 - AKI associated with hypocalcemia: rhabdomyolysis, tumor lysis syndrome, etc.
 - AKI with normal kidney size: diabetes, myeloma, amyloses, PKR, etc.

VIII. MANAGEMENT

Management of acute kidney injury is primarily supportive, with the goals of preventing further damage and promoting recovery of renal function.

1. Treatment etiology



2. Treatment complication:

a). Eliminating and avoidance of potential insults

- Medication:
 - STOP nephrotoxic medication
 - Dose adjustment for drugs that are renally cleared.
- Hypotension:
 - Correct volume status +/- inotrope
- Iodinated contrast agents:

- After AKI has already occurred, intravascular iodinated contrast should not be given unless it is required for an emergency or life-saving procedure (the lowest possible dose).

b). Volume assessment and management

- Hypovolemic patient:

- Intravenous fluid therapy should be administered to patients based on clinical history consistent with fluid loss, physical examination and inferior vena cava ultrasound (if available).
- However, fluid therapy should be avoided in patients with pulmonary edema or clear evidence of anuria.
- If hypovolemia is confirmed, we administer 1 to 3 liters of crystalloid with assessment of clinical response (urine output and improvement in GFR), and with persistent evidence of hypovolemia or inability to maintain fluid balance, we continue maintenance isotonic fluids at 75 mL/hour or greater depending upon the ongoing losses.
- Among patients with AKI who have a volume status that is difficult to interpret, such as in older adult patients or patients with reduced effective arterial blood volume, we administer a smaller volume trial (up to 1 liter) of an isotonic fluid. Based upon the response, a decision regarding continuation of fluid therapy may be made.

- Hypervolemic patient

➤ **Role of diuretic:**

- Diuretics may be used to relieve hypervolemia among patients with AKI who are not anuric.
- Loop diuretics are the preferred agents as they provide a greater natriuretic effect than thiazide diuretics.
- In diuretic-naïve patients, we start with 80 mg of intravenous (IV) [furosemide](#), or equivalent, and assess for response.
- If there is no definite augmentation in the urine output within two hours of an IV diuretic dose, then we administer double the initial dose (maximum of 200 mg in a single dose of IV [furosemide](#) or equivalent). Addition of a thiazide diuretic such as [chlorothiazide](#) (500 to 1000 mg IV) is sometimes given in conjunction with furosemide to augment urine output. Lack of response to a 200 mg dose of IV furosemide or equivalent, with or without a thiazide diuretic, may suggest the need for extracorporeal removal of excess volume.
- Among patients who respond to diuretics, we continue to give repeated doses to avoid hypervolemia if kidney function is improving or improvement is thought to be imminent.

➤ **Role of RRT:**

We typically initiate KRT for volume overload in patients who have anuria for more than 24 hours, who fail to respond to diuretics, or whose response to diuretics is insufficient to avoid worsening hypervolemia due to high obligate intake

Table 9. | Indications for Renal Replacement Therapy in Patients with AKI

Condition	Signs or symptoms
Anuria	Negligible urine output for six hours
Hyperkalemia	Potassium level > 6.5 mEq per L (6.5 mmol per L)
Poisoning or intoxication	Ethylene glycol or lithium ingestion
Pronounced azotemia	Urea nitrogen concentrations > 84 mg per dL (30 mmol per L)
Severe metabolic acidosis	pH < 7.2 despite normal or low partial pressure of carbon dioxide in arterial blood
Severe oliguria	Urine output < 200 mL over 12 hours
Uremic complications	Encephalopathy, neuropathy, pericarditis
Volume overload	Pulmonary edema unresponsive to diuretics

c). Managing electrolyte imbalance

- Hyperphosphatemia:

- We restrict dietary phosphorous to <2 g per day in all patients with AKI
- In addition, other patients who should start phosphate binder therapy include:
 - Hyperphosphatemic patients with AKI who have severe hypocalcemia, even if the phosphate concentration is <6 mg/dL or if KRT initiation is imminent
 - Hyperphosphatemic patients with continued release of intracellular phosphate such as rhabdomyolysis and tumor lysis syndrome
- The selection of phosphate binder depends on the level of serum ionized calcium concentration. If the serum ionized calcium concentration is low, calcium-containing phosphate binders such as [calciumacetate](#) or [calcium carbonate](#) may be given to control serum phosphate levels, providing patients can take oral medications.

- Hypocalcemia:

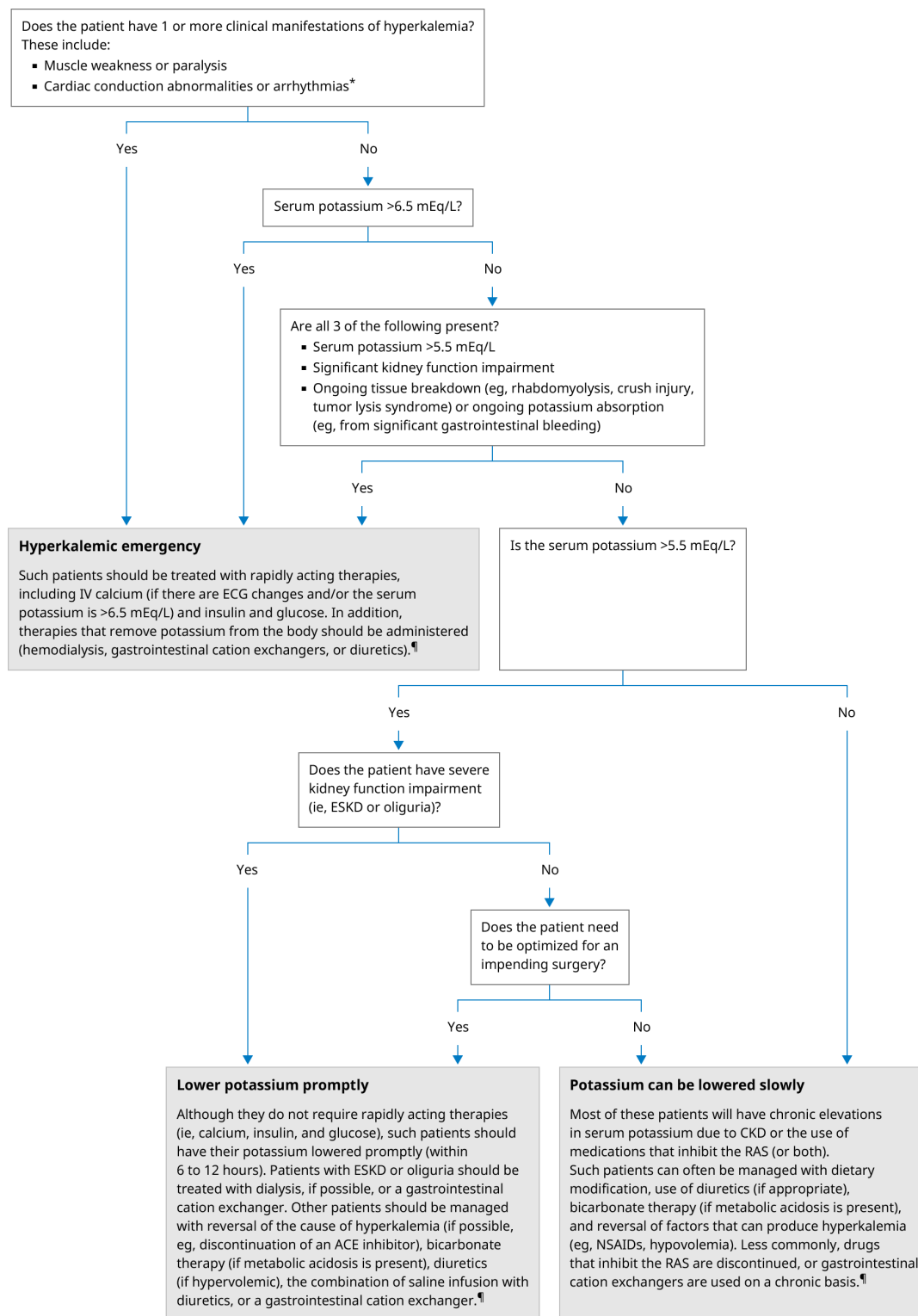
- The treatment of hypocalcemia depends on the acuity, severity, and presence of symptoms.

- Hypomagnesemiaandhypermagnesemia

- Patients with hypomagnesemia and kidney function impairment should be treated cautiously with small doses and frequent monitoring

- Hyperuricemia

- **Hyperkalemia**



d). Managing acid-base disturbance

- Metabolic acidosis:

- Among patients with AKI and severe metabolic acidosis who are not volume overloaded and have no other indication for acute KRT, bicarbonate may be administered instead of RRT; diuretics can be used in nonoliguric patients to prevent hypervolemia and to enhance excretion of acid.
 - The goal serum bicarbonate level is 20 to 22 mEq/L and the goal pH is >7.2.

e). Managing nutrition:

- We suggest achieving a total energy intake of 20–30 kcal/kg/d in patients with any stage of AKI. (2C)
- We suggest to avoid restriction of protein intake with the aim of preventing or delaying initiation of RRT. (2D)
- We suggest administering 0.8–1.0 g/kg/d of protein in noncatabolic AKI patients without need for dialysis (2D), 1.0–1.5 g/kg/d in patients with AKI on RRT (2D), and up to a maximum of 1.7 g/kg/d in patients on continuous renal replacement therapy (CRRT) and in hypercatabolic patients. (2D)

f). Assessing for uremic:

- Nonemergency dialysis may also be indicated for patients with prolonged AKI who develop uremic symptoms.
- Uremic signs and symptoms include anorexia, nausea, vomiting, metallic taste, and altered mental status.
- Physical examination findings such as asterixis and pericardial rub

g). Preventing and managing uremic bleeding:

AKI can cause qualitative platelet dysfunction, which can result in a hemorrhagic diathesis. The major clinical manifestation of this is cutaneous bleeding, but gastrointestinal bleeding also occur.

IX. COMPLICATION AND PROGNOSIS

- Prognosis depend on:
 - Etiology of AKI: septic or hemorrhage shock, convulsion, respiratory failure, severe trauma, acute pancreatitis.etc.
 - Comorbidity: age, cardiopathy, lung disease, diabetes, cancer.etc.
 - Secondary complication of AKI: infection, malnutrition.etc.
- Complications:
 - Hyperkalemia
 - Metabolic acidosis
 - Fluid overload
 - Malnutrition
 - Infection
 - Gastro-intestinal bleeding

X. PREVENTION

In the absence of effective therapeutic interventions on established AKI and due to its significant on morbidity and mortality, we can only rely on AKI prevention and early diagnosis to reduce its incidence and detrimental consequences.

Table 10. | Indications for Renal Replacement Therapy in Patients with AKI

<i>Risk factors</i>	<i>Preventive strategies</i>
Cancer chemotherapy with risk of tumor lysis syndrome ²⁷	Hydration and allopurinol (Zyloprim) administration a few days before chemotherapy initiation in patients at high risk of tumor lysis syndrome to prevent uric acid nephropathy
Exposure to nephrotoxic medications	Avoid nephrotoxic medications if possible Measure and follow drug levels if available Use appropriate dosing, intervals, and duration of therapy
Exposure to radiographic contrast agents ²⁹	Avoid use of intravenous contrast media when risks outweigh benefits If use of contrast media is essential, use iso-osmolar or low-osmolar contrast agent with lowest volume possible Optimize volume status before administration of contrast media; use of isotonic normal saline or sodium bicarbonate may be considered in high-risk patients who are not at risk of volume overload Use of <i>N</i> -acetylcysteine may be considered
Hemodynamic instability	Optimal fluid resuscitation; although there is no consensus, a mean arterial pressure goal of > 65 mm Hg is widely used; isotonic solutions (e.g., normal saline) are preferred over hyperoncotic solutions (e.g., albumin) ¹⁹ Vasopressors are recommended for persistent hypotension (mean arterial pressure < 65 mm Hg) despite fluid resuscitation; choice of vasoactive agent should be tailored to patients' needs ²⁰ Dopamine is not recommended ²¹
Hepatic failure ³⁰	Avoid hypotension and gastrointestinal bleeding Early recognition and treatment of spontaneous bacterial peritonitis; use albumin, 1.5 g per kg at diagnosis and 1 g per kg at 48 hours Early recognition and management of ascites Albumin infusion during large volume paracentesis
Rhabdomyolysis ²⁰	Avoid nephrotoxic medications Maintain adequate hydration Alkalinization of the urine with intravenous sodium bicarbonate in select patients (normal calcium, bicarbonate less than 30 mEq per L [30 mmol per L], and arterial pH less than 7.5)
Undergoing surgery	Adequate volume resuscitation/prevention of hypotension, sepsis, optimizing cardiac function Consider holding renin-angiotensin system antagonists preoperatively ³¹

XI. RECOMMENDATION

- In the absence of effective therapeutic interventions on established AKI and due to its significant on morbidity and mortality, we can only rely on AKI prevention and early diagnosis to reduce its incidence and detrimental consequences.
- High-risk patients for AKI should have a kidney health assessment (KHA) every 12 months, at least 30 days before exposure and two-to-three days after exposure that carries AKI risk in order to define and modify their risk profile
- After an acute exposure to an AKI inciting event, namely nephrotoxic medications, radiocontrast, surgery, or acute disease, medications should be adjusted, further nephrotoxic exposures should be minimized, and AKI occurrence and its complications should be monitored

- At hospital admission, patients should also be screened for AKI risk. In high-risk patients, the early correction of modifiable risk factors should be considered in order to prevent AKI occurrence

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MANAGEMENT OF CHRONIC KIDNEY DISEASE (CKD)

Dr. RITH Sovannara, Prof. LIM Vadhana

I. INTRODUCTION

- Purpose: To provide a standardized approach for the diagnosis, management, and treatment of chronic kidney disease (CKD) in adult patients.
- Scope: This guideline applies to all healthcare professionals involved in the care of patients with CKD.

II. DEFINITION AND STAGING OF CKD

- Definition: CKD is defined as abnormalities of kidney structure or function, present for more than 3 months, with implications for health.
- Staging:
 - Stage 1: GFR ≥ 90 mL/min/1.73 m² with kidney damage
 - Stage 2: GFR 60-89 mL/min/1.73 m² with kidney damage
 - Stage 3a: GFR 45-59 mL/min/1.73 m²
 - Stage 3b: GFR 30-44 mL/min/1.73 m²
 - Stage 4: GFR 15-29 mL/min/1.73 m²
 - Stage 5: GFR < 15 mL/min/1.73 m² or on dialysis

Prognosis of CKD by GFR and albuminuria categories: KDIGO 2012				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				< 30 mg/g < 3 mg/mmol	30–300 mg/g 3–30 mg/mmol	> 300 mg/g > 30 mg/mmol
GFR categories (mL/min/1.73 m ²) Description and range	G1	Normal or high	≥ 90			
	G2	Mildly decreased	60–89			
	G3a	Mildly to moderately decreased	45–59			
	G3b	Moderately to severely decreased	30–44			
	G4	Severely decreased	15–29			
	G5	Kidney failure	< 15			

Green, low risk (if no other markers of kidney disease, no CKD); yellow, moderately increased risk; orange, high risk; red, very high risk.

III PHYSIOPATHOLOGY AND ETIOLOGY

The pathophysiology of CKD involves complex interactions between various factors that contribute to the progressive decline in kidney function. The key pathological processes include:

1. Initiation of Kidney

Damage Primary Causes:

- **Diabetes Mellitus:** The leading cause of CKD, where chronic hyperglycemia leads to glomerular hyperfiltration, increased renal blood flow, and the development of diabetic nephropathy.
- **Hypertension:** High blood pressure causes direct damage to the renal vasculature, leading to ischemic injury and nephrosclerosis.
- **Glomerulonephritis:** Inflammatory diseases of the glomeruli can result in scarring and loss of nephron function.
- **Other Causes:** Polycystic kidney disease, recurrent urinary tract infections, and prolonged use of nephrotoxic drugs.

2. Progression of Kidney Damage

- **Hyperfiltration and Hypertrophy:**

The initial damage to some nephrons leads to compensatory hyperfiltration in the remaining nephrons. This increased glomerular pressure and hyperfiltration, while initially compensatory, eventually cause damage to the glomerular capillaries, leading to glomerulosclerosis.

- **Proteinuria:**

Increased permeability of the glomerular basement membrane allows for the leakage of proteins, particularly albumin, into the urine. Proteinuria is both a marker and a mediator of kidney damage, as proteins in the renal tubules induce inflammation and fibrosis.

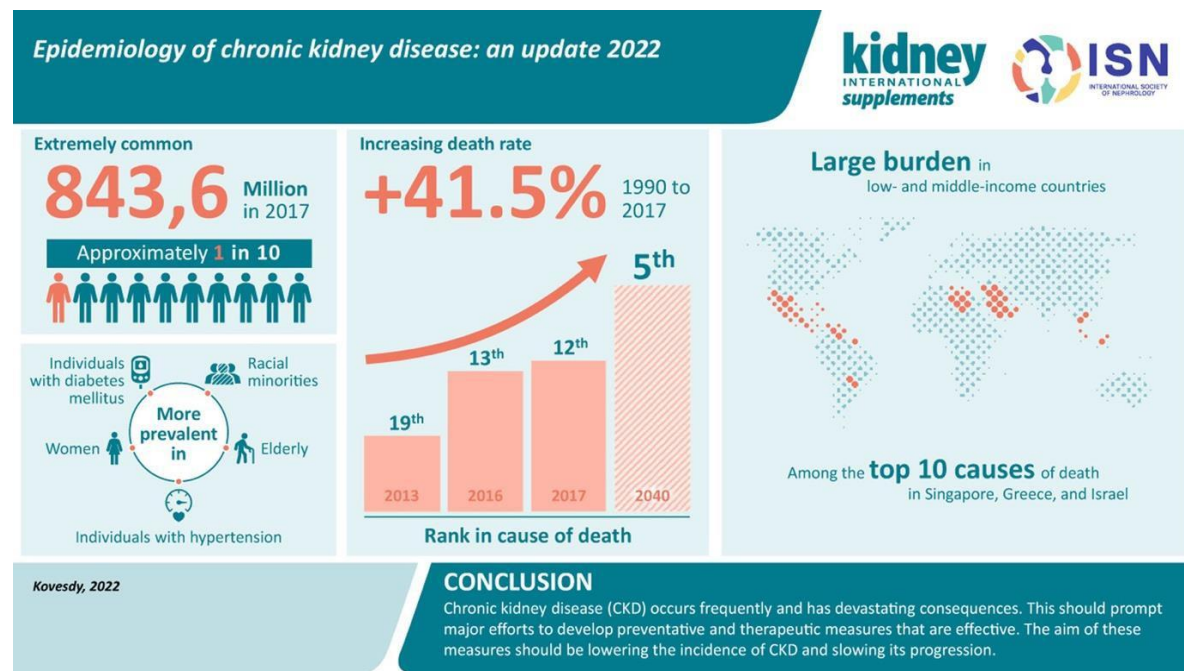
- **Tubulointerstitial Inflammation and Fibrosis:**

Chronic injury leads to inflammation in the renal interstitium, characterized by the infiltration of immune cells, release of pro-inflammatory cytokines, and activation of fibroblasts. Over time, this results in tubulointerstitial fibrosis, which is a major determinant of CKD progression.

- **Vascular Injury:**

Damage to the renal vasculature (e.g., arteriolar hyaline sclerosis) contributes to ischemia and further nephron loss. This vascular injury is exacerbated by hypertension and diabetes, creating a vicious cycle of damage.

IV. PIDEMIOLOGY



V. RISK FACTORS FOR CKD

- Non-Modifiable: Age, family history of CKD, genetic factors.
- Modifiable: Diabetes, hypertension, obesity, smoking, recurrent urinary tract infections, nephrotoxic medications, and cardiovascular disease.

VI. COMPLICATION AND SYSTEMIC COMPLICATIONS OF CKD

• Fluid and Electrolyte Imbalance:

As kidney function declines, the ability to regulate fluid, electrolytes, and acid-base balance is impaired. This can lead to volume overload, hyperkalemia, metabolic acidosis, and other electrolyte disturbances.

• Anemia:

Reduced production of erythropoietin by the failing kidneys leads to anemia, which further contributes to fatigue and cardiovascular complications in CKD patients.

• Mineral and Bone Disorders:

CKD disrupts the balance of calcium, phosphate, and parathyroid hormone (PTH), leading to secondary hyperparathyroidism, renal osteodystrophy, and vascular calcification. This condition is known as CKD-Mineral and Bone Disorder (CKD-MBD).

• Cardiovascular Disease:

CKD is closely associated with cardiovascular disease due to shared risk factors and direct effects of kidney dysfunction on the heart and blood vessels. Hypertension, fluid overload, and vascular calcification contribute to the high cardiovascular mortality in CKD patients.

- Uremic Syndrome:

In advanced CKD (Stages 4-5), the accumulation of toxins normally excreted by the kidneys leads to a clinical syndrome known as uremia. This manifests with symptoms such as fatigue, anorexia, nausea, neurological disturbances, and pericarditis.

End-Stage Renal Disease (ESRD):

As CKD progresses to its final stage (Stage 5), kidney function becomes insufficient to sustain life, necessitating renal replacement therapy (dialysis or kidney transplantation). At this stage, patients exhibit severe complications of uremia, electrolyte imbalances, and cardiovascular instability.

VI. SCREENING AND DIAGNOSIS

- Screening:

- Target high-risk populations (e.g., patients with diabetes, hypertension, cardiovascular disease).
- Utilize eGFR, serum creatinine, and urine albumin-to-creatinine ratio (ACR) as primary screening tools.

- Diagnosis:

- Confirm CKD if $GFR < 60 \text{ mL/min/1.73 m}^2$ and/or $ACR > 30 \text{ mg/g}$ on two separate tests at least 3 months apart (Rule out acute kidney injury (AKI) or reversible causes).

- . Clinical Evaluation

- Medical History: Assess for symptoms, duration of disease, family history, medication use, and comorbid conditions.
- Physical Examination: Evaluate for signs of fluid overload, hypertension, uremia, or other systemic manifestations.

- Laboratory Tests:

- Routine: eGFR, ACR, serum electrolytes, complete blood count (CBC), lipid profile, hemoglobin A1c.
- Additional: Parathyroid hormone (PTH), vitamin D, imaging studies (e.g., renal ultrasound).
- Referral to Nephrology: Consider early referral for advanced CKD (Stage 3b or higher), difficult-to-manage cases, or rapid progression.

VII. MANAGEMENT OF CKD

1- Treat or control the primary causes and General Measures:

- Blood Pressure Control: Target $< 130/80 \text{ mmHg}$, preferably with ACE inhibitors or ARBs.
- Glycemic Control: Target $HbA1c < 7\%$ in diabetics.
- Lifestyle Modifications: Smoking cessation, weight management, dietary sodium restriction ($< 2 \text{ g/day}$).

2- Pharmacologic Therapy:

- Use ACE inhibitors or ARBs for proteinuria.
- Statins for lipid management to reduce cardiovascular risk.
- Diuretics for managing fluid overload.

3- Management of Complications:

- Anemia: Consider erythropoiesis-stimulating agents (ESAs) and iron supplementation.
- Bone Mineral Disease: Phosphate binders, vitamin D analogs, and calcimimetics.
- Metabolic Acidosis: Sodium bicarbonate supplementation if serum bicarbonate < 22 mEq/L.

4- Avoid Nephrotoxic Drugs: NSAIDs, contrast agents, certain antibiotics (e.g., aminoglycosides).

5- Renal Replacement Therapy (RRT) is required when a patient with Chronic Kidney Disease (CKD) reaches a point where their kidneys can no longer maintain adequate function to sustain life. The decision to initiate RRT, which includes dialysis (hemodialysis or peritoneal dialysis) and kidney transplantation, is based on a combination of clinical symptoms, laboratory findings, and the patient's overall health status. Below are the primary indications for initiating RRT in CKD patients:

1. Severe Decline in Glomerular Filtration Rate (GFR)

- $\text{GFR} \leq 15 \text{ mL/min/1.73 m}^2$: RRT is generally considered when the estimated GFR falls below $15 \text{ mL/min/1.73 m}^2$, especially if the patient has symptoms of uremia or other complications of kidney failure. This stage is classified as Stage 5 CKD or End-Stage Renal Disease (ESRD).

2. Uremic Symptoms

- Persistent or Worsening Uremic Symptoms: These include fatigue, anorexia, nausea, vomiting, pruritus, pericarditis, encephalopathy, and a metallic taste in the mouth. These symptoms result from the accumulation of uremic toxins that the kidneys can no longer excrete.
- Uremic Pericarditis: Inflammation of the pericardium due to uremia, often presenting with chest pain, is an absolute indication for immediate initiation of RRT.

3. Fluid Overload

- Refractory Fluid Overload: When conservative management (e.g., diuretics, fluid restriction) fails to control fluid overload, leading to pulmonary edema, hypertension, or congestive heart failure, RRT is indicated to remove excess fluid.

4. Electrolyte Imbalances

- Severe Hyperkalemia (Potassium > 6.5 mEq/L): Persistent or recurrent hyperkalemia that is unresponsive to medical treatment can be life-threatening and necessitates urgent initiation of RRT.

- Severe Acidosis (Metabolic Acidosis with $\text{pH} < 7.2$): When metabolic acidosis is not adequately controlled with bicarbonate therapy, RRT is needed to correct the acid-base imbalance.

5. Severe Azotemia

- Markedly Elevated Blood Urea Nitrogen (BUN) and Serum Creatinine: Although there is no absolute BUN or creatinine level that mandates RRT, very high levels (e.g., $\text{BUN} > 100 \text{ mg/dL}$ or $\text{creatinine} > 10 \text{ mg/dL}$) in symptomatic patients are often used as indicators for starting RRT.

6. Complications of Chronic Kidney Disease

- Refractory Hypertension: Hypertension that cannot be controlled with multiple antihypertensive medications may indicate the need for RRT.
- Chronic Kidney Disease-Mineral and Bone Disorder (CKD-MBD): Severe, treatment-resistant hyperphosphatemia, secondary hyperparathyroidism, or other mineral imbalances may require RRT to manage.
- Anemia of Chronic Kidney Disease: Severe, symptomatic anemia unresponsive to erythropoiesis-stimulating agents (ESAs) and iron therapy may necessitate RRT.

7. Nutritional Deficiencies

- Malnutrition: Severe protein-energy wasting and failure to thrive, despite dietary interventions, can be an indication for starting RRT to improve nutritional status and quality of life.

8. Preparation for Kidney Transplantation

- Pre-Transplantation: Patients who are candidates for kidney transplantation may require RRT while awaiting a donor kidney, particularly if they are in ESRD.

Summary

The decision to initiate Renal Replacement Therapy in CKD is based on a combination of clinical judgment and patient-specific factors. It is typically indicated when there is a significant decline in kidney function ($\text{GFR} \leq 15 \text{ mL/min/1.73 m}^2$), the presence of uremic symptoms, refractory fluid overload, severe electrolyte imbalances, or other serious complications that cannot be managed with conservative therapy. Early discussion with the patient about RRT options, including dialysis modalities and transplantation, is essential for optimal management and patient preparedness.

9. Monitoring and Follow-Up

- Frequency: Adjust based on CKD stage; generally, every 3-6 months for Stage 3, and more frequent (1-3 months) for Stages 4-5.
- Parameters: Monitor eGFR, ACR, blood pressure, electrolytes, hemoglobin, and signs of progression.
- Adjustment of Therapy: Based on the progression of CKD, side effects, and comorbid conditions.

10. Patient Education and Self-Management

- Educate: Patients on the importance of lifestyle changes, medication adherence, and

regular follow-up.

- Dietary Counseling: Recommend consultation with a dietitian for personalized nutritional advice.
- Symptom Monitoring: Teach patients to recognize and report symptoms of worsening kidney function (e.g., swelling, fatigue, decreased urine output).

11. Referral and Coordination of Care

- Indications for Nephrology Referral:
 - Stage 4-5 CKD, persistent proteinuria, rapid progression of CKD, or need for renal replacement therapy (dialysis, transplantation).
- Coordination: Ensure communication between primary care providers, nephrologists, dietitians, and other specialists involved in patient care.

12. End-of-Life Care and Advanced Directives

- Discuss: End-of-life care options, including conservative management and palliative care, especially in elderly patients or those with multiple comorbidities.
- Advanced Directives: Encourage patients to discuss and document their wishes regarding dialysis and other life-sustaining treatments.

13. Documentation and Quality Improvement

- Record Keeping: Maintain accurate and up-to-date patient records, including CKD stage, comorbidities, and management plan.
- Audit and Feedback: Regularly review CKD management practices and outcomes to identify areas for improvement.

14. References and Resources

- Provide a list of key references, guidelines, and patient resources for further reading and patient education.

This guideline can be tailored to fit the specific protocols, patient population, and resources available at your hospital. Regular review and updates of the guideline are recommended to incorporate new evidence and best practices.

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- 3- Kidney International 2022, Epidemiology and Death related CKD.
- 4- Up To Date.

NEPHROTIC SYNDROME

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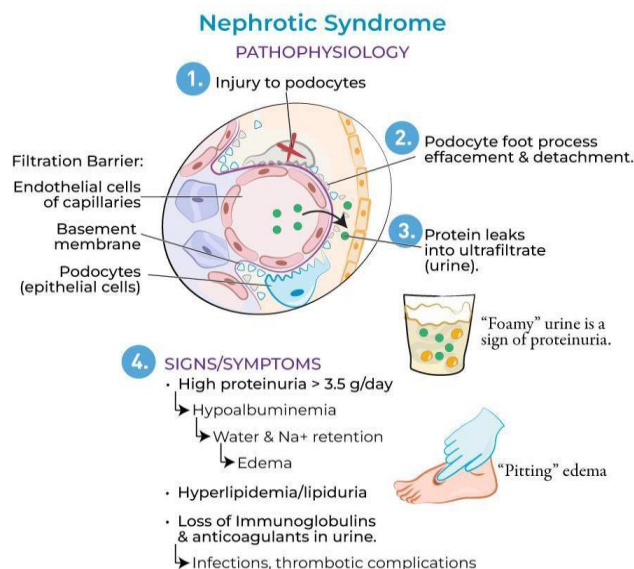
I. DEFINITION

Nephrotic syndrome is defined as:

- 24h proteinuria more than 3.5g
- Serum protein less than 60g/L and serum albumin less than 30g/L
- Edema
- Hyperlipidemia

II. PATHOPHYSIOLOGY

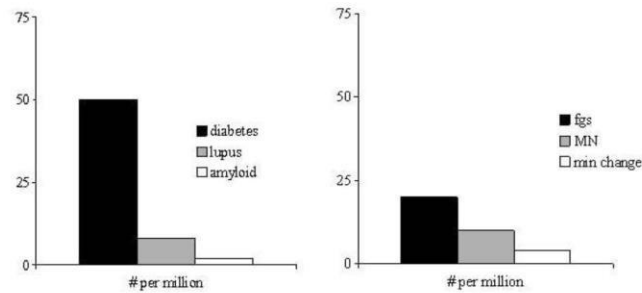
- Non inflammatory damage to the glomerular capillary wall that leads to alteration in the glomerular filtration barrier. This abnormality increases the glomerular permeability to plasma protein into the urine called proteinuria.
- Albumin is the principal urinary protein lost.
- Other plasma proteins lost include: hormones carrying proteins such as vitamin D binding protein, transferrin and clotting inhibitors.
- Edema results from sodium retention and disequilibrium of volume repartition between interstitial and plasmatic sector in extracellular compartment.



III. EPIDEMIOLOGY

Incidence per million populations per year in USA (Medscape 2023):

- Children: 20 cases per million per year
- Adults: diabetic nephropathy is most common at a rate of 50 cases per million per year



Incidence of important causes of nephrotic syndrome, in number per million population. The left panel shows systemic causes, and the right panel lists primary renal diseases that can cause nephrotic syndrome. fgs = focal glomerulosclerosis, MN = membranous nephropathy, min change = minimal-change nephropathy. Data are in part from Swaminathan et al and Bergesio et al.

IV. ETIOLOGY

Nephrotic syndrome may appear as a primary (idiopathic) renal disease or secondary to any of a number of systemic conditions and hereditary disease.

1. Primary causes:

- Minimal change disease (MCD)
- Focal segmental glomerular sclerosis (FSGS)
- Membranous nephropathy (MN)
- Membrano-proliferative glomerulonephritis (MPGN)

2. Secondary causes:

- Diabetes mellitus (most common)
- Systemic lupus erythematosus
- Viral hepatitis B, C, HIV infection
- Amyloidosis
- Leukemia, lymphoma
- Solid malignancy
- Hereditary nephropathies
- Medication or toxic (gold, D-penicillamine, mercury, lithium, NSAID, bevacizumab)

V. DIAGNOSIS

1. Symptoms and signs:

- Can present as asymptomatic proteinuria
- Commonly edema, in area of high intra vascular hydrostatic pressure (feet, ankle), and area in where tissue hydrostatic pressure is lowest (periorbital, scrotal)
- Sometimes ascites and pleural effusion
- Eventually oliguria, hypertension, and rarely dyspnea

2. Laboratory finding:

- Urine dipstick often 3+ or 4+ protein
- 24h proteinuria > 3.5g
- Serum protein < 60g/L, serum albumin < 30g/L
- Hyperlipidemia
- Patients may have elevation in Creatinine and BUN
- Sometimes anemia, hypocalcemia, vitamin D deficiency

3. Special tests:

- Others tests depend on patient's clinical presentation
 - Serum and urine protein electrophoresis to evaluate myeloma or amyloidosis
 - ANA to evaluate systemic lupus erythematosus
 - Hepatitis serology, syphilis, HIV, complements, cryoglobulin and thyroid function test
 - Anti PLA2R (Ab to phospholipase A2 receptor) in primary membranous nephropathy
- Ophthalmoscopic examination to find out diabetic retinopathy
- Imaging: renal ultrasound
- Kidney biopsy: recommend when the etiology of nephrotic range proteinuria is in doubt
 - Help determine the etiology, progression and prognosis
 - Aid in management decision

VI. COMPLICATION

- Hypoalbuminemia and malnutrition
- Hyperlipidemia and acceleration of atherosclerosis
- Thrombosis:
 - . relate to urinary losses of antithrombin III, protein C and S, and factors IX, XI, XII
 - . increased incidence of venous and arterial thrombo-emboli (especially deep vein thrombosis, pulmonary embolism and renal vein thrombosis)
- Vitamin D deficiency, hypocalcemia and osteoporosis
- Infection: due to loss of immunoglobulin and immunosuppressive medication used
- Anemia: due to loss of erythropoietin and transferrin
- Alteration of kidney function

VII. MANAGEMENT

1. Treatment of underlying systemic disease or primary renal disease
- Corticosteroids: typically used for MCD and primary FSGS
 - In severe form: pulse methylprednisolone 500-1000mg per day for 3 days
 - In steroid sensitive form: start high dose Prednisolone 1mg/kg/d for 4-8 weeks, then taper gradually after achieving complete remission (urinary protein <1+ or negative for 3 consecutive days) by decreasing 5mg every 1-2 weeks for total duration at least 24 weeks
 - In steroid dependent form or resistant form (after maximum 16 weeks): need immunosuppressive agents
 - Immunosuppressive agents: indicated in steroid resistant form, or in frequent relapse, or patient develop steroid related side effects
 - Cyclophosphamide: PO 1.5-2.5 mg/kg/d for 3-6 months, OR IV 750mg/m² every 4 weeks for 24 weeks
 - Mycophenolate mofetil: 1000 mg twice daily at least 12 months
 - Cyclosporine: 3-5 mg/kg/d in 2 divided doses with trough level of 100-175 ng/ml for at least 12 months
 - Tacrolimus: 0.05-0.1 mg/kg/d in 2 divided doses with trough level of 5-10 ng/ml for at least 12 months
 - Azathioprine: 2-3 mg/kg/d for at least 12 months

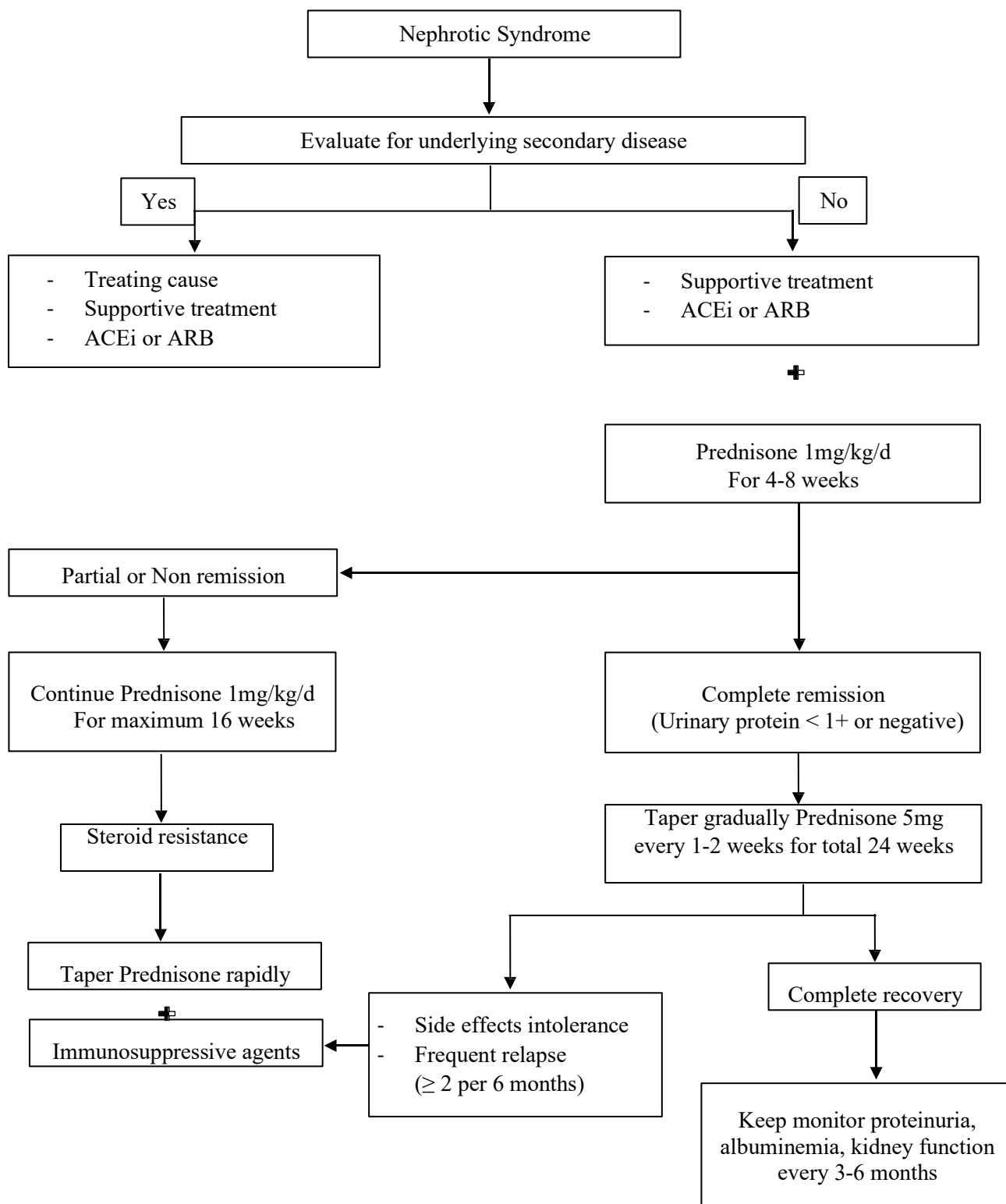
- Rituximab: 1g IV twice within 2 weeks, OR 375mg/m² given 4 times weekly intervals
- Treating underlying disease
 - Control blood pressure
 - Control glycemia in diabetic nephropathy
 - Treat hemopathies and solid malignancy
 - Antimicrobial agents for hepatitis B, C and HIV
- 2. Management of proteinuria
 - ACEi or ARB: first line
- 3. Management of edema
 - Diuretics: loop diuretics (eg. furosemide) are commonly used +/- potassium sparing diuretics
 - Salt and fluid restriction: sodium intake 2g/d, fluid intake depends on severity of edema
- 4. Management of hypoalbuminemia
 - Dietary modification: plant-based protein diet of 1g/kg/d (balance again malnutrition)
 - Albumin infusion: use temporarily to improve oncotic pressure in severe case of hypoalbuminemia with symptomatic edema
- 5. Management of hyperlipidemia
 - Statins
 - Dietary modification: low fat, low cholesterol diet
- 6. Management of thrombosis and prophylaxis
 - If thrombotic event: anticoagulant continue until resolution of nephrotic syndrome
 - Prophylactic anticoagulation may benefit when albuminemia less than 20-25g/L with additional risk factors (eg. immobility, recent surgery ...)

VIII. EDUCATION

- Diet in nephrotic patients should provide adequate calorie and protein (1g/kg/d)
- Low salt diet will limit the fluid retention and edema. Daily sodium intake is about 2g
- Low fat, low cholesterol diet
- No activity restriction. Ongoing activity will reduce the risk of blood clots.
- Eradication of parasitic infection with Albendazole 400mg/d for 3 days before treatment
- Nephrotic syndrome in adults is wax and wane, characterized by relapse and remission.
- There will be illness from the disease and from its treatment. Monitoring is important, not only focus on progression and efficacy, but also on medication toxicity.
- Progression to renal failure will require preparation for dialysis and/or kidney transplantation.

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3. Current Diagnosis and Treatment of Nephrology and Hypertension, 2nd Edition, 2018.



Decision algorithm of management of nephrotic syndrome in adults

POSTSTREPTOCOCCAL GLOMERULONEPHRITIS

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I. DEFINITION

Poststreptococcal glomerulonephritis (PSGN) is characterized by rapid deterioration of kidney functions due to an inflammatory response (type III hypersensitivity reaction) following streptococcal infection. This condition results from specific strains of group A beta-hemolytic streptococci called nephrogenic streptococci. The disease affects the glomeruli and the small blood vessels of the kidneys. PSGN is most frequently present in children 1 to 2 weeks after a sore throat, or 6 weeks after a skin infection (impetigo).

II. EPIDEMIOLOGY

Although PSGN continues to be the most common cause of acute nephritis in children globally, it primarily occurs in developing countries. Of the estimated 470,000 new annual cases of PSGN worldwide, 97 percent occur in regions of the world with poor socioeconomic status, with an annual incidence that ranges from 9.5 to 28.5 per 100,000 individuals.

III. ETIOLOGY

Poststreptococcal glomerulonephritis (PSGN) is an immunologically mediated sequela of pharyngitis or skin infections caused by nephrogenic strains of *Streptococcus pyogenes*.

IV. PATHOPHYSIOLOGY

PSGN appears to be caused by glomerular immune complex disease induced by specific nephrogenic strains of group A beta-hemolytic streptococcus (GAS). The presence of immune complexes leads to the activation of the alternate complement pathway causing infiltration of the leukocytes, and proliferation of the mesangial cells in the glomerulus thus impairing the capillary perfusion and glomerular filtration rate (GFR). Reduction in GFR can lead to renal failure (oliguria or anuria), acid-base imbalance, electrolyte abnormalities, volume overload, edema, and hypertension.

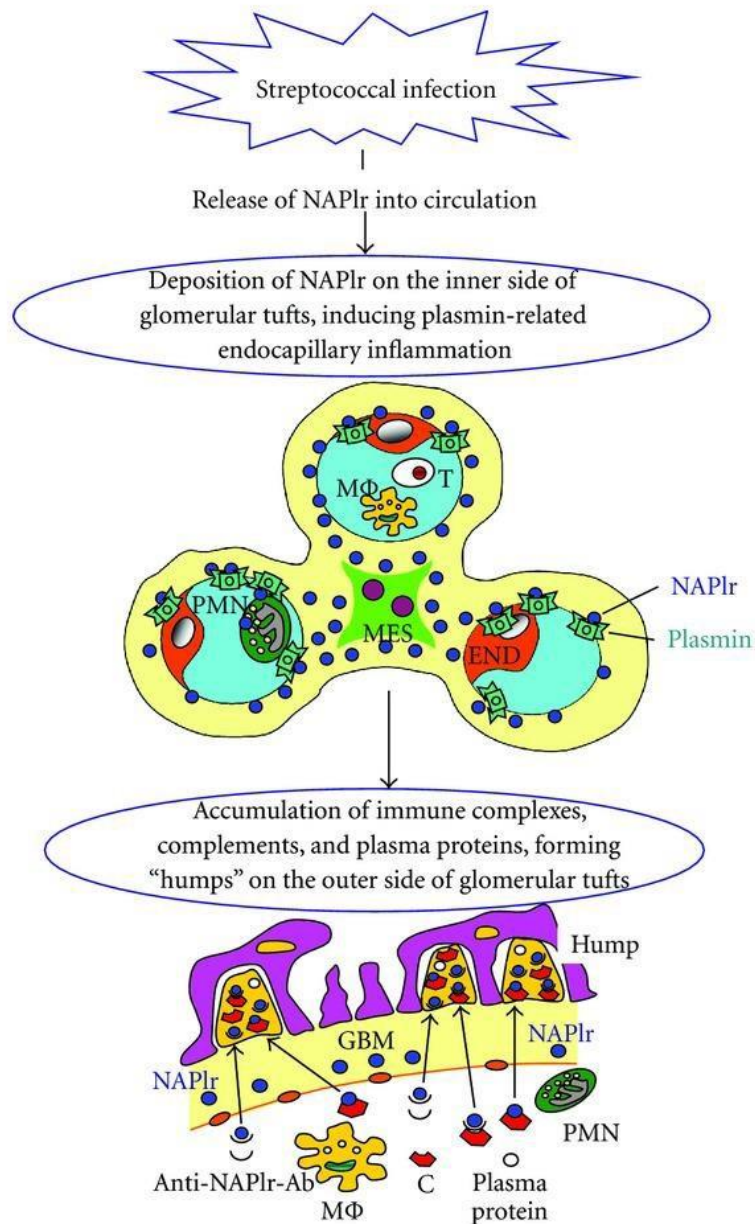


Figure 1. Schematic representation of proposed mechanisms involved in the development of acute poststreptococcal glomerulonephritis (APSGN).

The Role of Nephritis-Associated Plasmin Receptor (NAPlr) in Glomerulonephritis Associated with Streptococcal Infection. Journal of biomedicine & biotechnology. 2012. 417675. 10.1155/2012/417675.

MES: mesangial cell; END: endothelial cell; PMN: polymorphonuclear cell; MΦ: macrophage
T: T lymphocyte; GMB: glomerular basement membrane. C: complement,
Anti-NAPlr-Ab: Anti-NAPlr-antibody, NAPlr: nephritis-associated plasmin receptor

V. DIAGNOSIS

1. Clinical Manifestation

The following symptoms are the most common presenting signs:

- Edema – due to sodium and water retention. In severe cases, fluid overload leads to respiratory distress due to pulmonary edema.
- Gross hematuria – approximately 30 to 50 percent of patients.
- Hypertension – in 50 to 90 percent of patients. Hypertensive encephalopathy is an uncommon but serious complication.
- Decrease urine output

2. Investigation

- Renal function — can be normal or decreased (rise in serum creatinine)
- Urinalysis and urinary protein excretion — hematuria (some of the red cells are typically dysmorphic) with or without red blood cell casts, varying degrees of proteinuria, and often pyuria. Nephrotic range proteinuria (defined as ≥ 1000 mg/m² per day or 40 mg/m² per hour) is uncommon and occurs in approximately 5 percent of cases at presentation.
- Complement — In approximately 90 percent of patients, C3 is significantly depressed in the first two weeks of the disease course. The C3 returns to normal within four to eight weeks after presentation. C4 and C2 levels may be low in some patients, which suggests activation of both classical and alternative pathways.
- Swab culture — only approximately 25 percent of patients will have either a positive throat or skin culture.
- Serology — If only the ASO titer is used to screen for GAS infection, it may be falsely low or negative in patients with skin infections. It remains a useful test in patients with PSGN due to GAS pharyngitis but in some cases, the rise in ASO titer may be blunted in patients with pharyngitis who have received antimicrobial therapy.
- Renal biopsy — is not systemically performed in most patients to confirm the diagnosis of PSGN, since the resolution of PSGN typically begins within one week of presentation. Renal biopsy is performed only when other glomerular pathologies are suspected.

VI. DIFFERENTIAL DIAGNOSIS

A renal biopsy may be needed to differentiate PSGN from these other disorders:

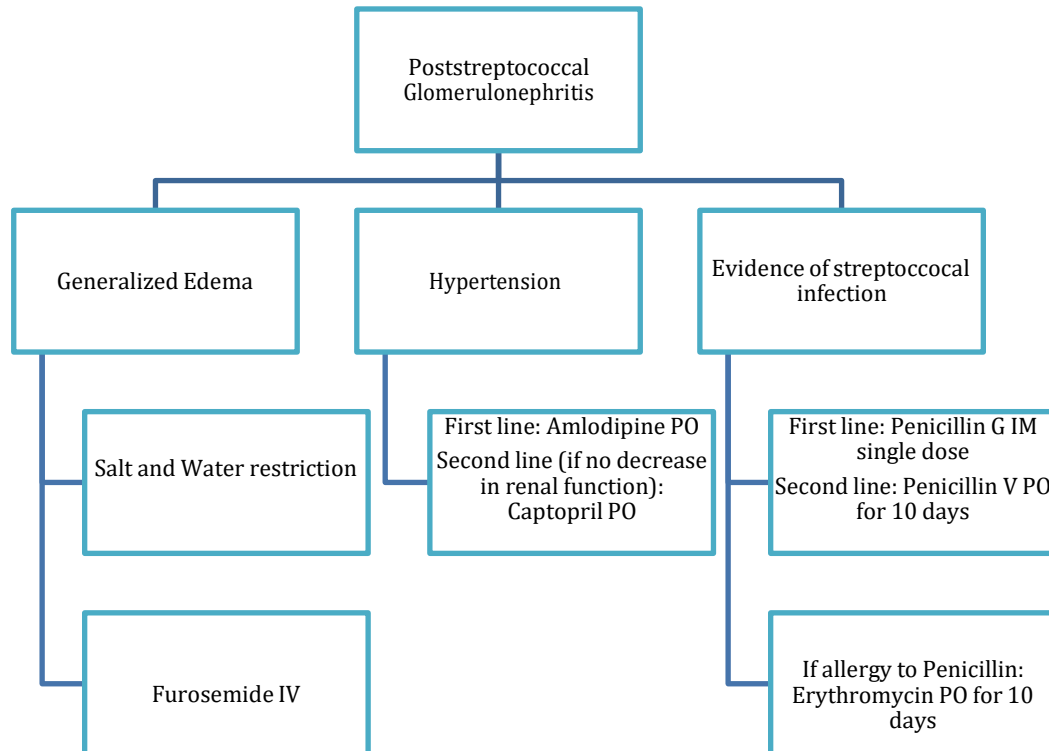
- C3 glomerulopathy
- IgA Nephropathy
- Secondary causes of glomerulonephritis – Lupus nephritis and IgA vasculitis
- Postinfectious GN due to other microbial agents

VII. MANAGEMENT

1. Objective: Treatment of PSGN focuses on managing hypertension and fluid retention. The eradication of the nephrogenic strain can be indicated to prevent the spread of the strain to other people.
2. General Measures
 - Fluid volume restriction: restricted to insensible water loss calculated at 400ml/m² per day, in addition to replacing urine losses
 - Low salt diet: $\leq 1\text{mmol/kg/day}$ for children or $<2\text{g/day}$ for adult
 - Low potassium diet: 0-150mg ($\leq 3.8\text{mmol}$) per serving
 - Daily monitoring of weight
 - Strict input and output chart
 - Close blood pressure monitoring: Hourly initially, then 2-4 hourly if blood pressure is stable
3. Management of complication
 - Decrease fluid overload and sodium retention to prevent hypertensive encephalopathy:
 - Loop diuretic (IV furosemide) is given at an initial dose of 1 mg/kg (maximum 40 mg).
 - Dose can be repeated in an hour and subsequently 4-5hourly to achieve desired diuresis.
 - Treatment of hypertension: Blood pressure can be managed by restricting salt and fluid intake along with diuretics are needed. In cases with uncontrolled blood pressure, the use of calcium channel blockers is recommended. The use of angiotensin-converting enzyme inhibitors (ACEI), angiotensin receptor blockers (ARBs) is recommended in patients with stable GFR and with near-normal potassium levels.
 - Oral Amlodipine:
 - For children 1–5 years of age: Initial dose 0.1 mg/kg once a day, up to a maximum dose 0.6 mg/kg (not to exceed 5 mg) once a day
 - For children ≥ 6 years of age and adult: Initial dose 2.5 mg once a day, up to a maximum dose 10 mg once a day
 - Captopril:
 - For children: Initial dose 0.2–0.5 mg/kg 2 to 4 times a day, or for older children 6.25–12 .5 mg 2 to 4 times a day (maximum daily dose 6 mg/kg)
 - For adolescents and adults: 12.5–25 mg 2 to 4 times a day (maximum daily dose 150 mg)
 - Dialysis — For patients with serious renal impairment, indications for dialysis include:
 - Life-threatening fluid overload (pulmonary edema, heart failure, and hypertension) that is refractory to medical therapy
 - Hyperkalemia (serum or plasma potassium $>6.5\text{ mEq/L}$) unresponsive to medical therapy
 - Severe Acidosis
 - Uremic encephalopathy
4. Antibiotics
 - Indicated if there are clinical features of streptococcal infection (pharyngitis or pyoderma) or serological evidence of streptococcal infection.

- Choice of antibiotics
 - Penicillin G benzathine (preferable) to eradicate the nephritic strain. This will prevent spread of the strain to other people.
Children ≤ 27 kg: IM: 600,000 units as a single dose.

Children >27 kg and adult: IM: 1,200,000 units (1.2 million units) as a single dose.
 - Oral Penicillin V
Children ≤ 27 kg: Oral: 250 mg every 8 to 12 hours for 10 days
Children >27 kg and Adult: Oral: 500 mg every 8 to 12 hours for 10 days
 - Oral Erythromycin (if penicillin allergy) 10mg/kg every 6 hours for 10 days
 - Preventive antibiotic treatment (Penicillin V) may be indicated in household members of index cases in high risk communities.
5. Algorithm



VIII. EVOLUTION AND PROGNOSIS

- Resolution of the clinical manifestations of PSGN is generally quite rapid, assuming concurrent resolution of the infection.
- A diuresis typically begins within one week.
- The serum creatinine returns to the previous baseline by three to four weeks.

- Hematuria usually resolves within three to six months. Proteinuria usually disappears earlier than microscopic hematuria. In severe cases with nephrotic range proteinuria (defined as ≥ 1000 mg/m² per day or 40 mg/m² per hour), this degree of proteinuria may persist for six months or more, long after the hematuria has disappeared.
- Most patients, particularly children, have an excellent outcome.
- Some patients, particularly adults, develop hypertension, recurrent proteinuria (with a relatively normal urine sediment), and renal insufficiency as long as 10 to 40 years after the initial illness.

IX. COMPLICATION

- Acute renal failure leading to acidosis, hyperkalemia, uremia
- Acute pulmonary edema
- Hypertensive encephalopathy
- Chronic kidney disease

X. PREVENTION

Good hand hygiene and respiratory etiquette can reduce the spread of all types of group A strep infection. Treating PSGN patients with antibiotics can stop a nephritogenic strain from circulating in a household. Thus, treating PSG patients can prevent additional infections among close contacts.

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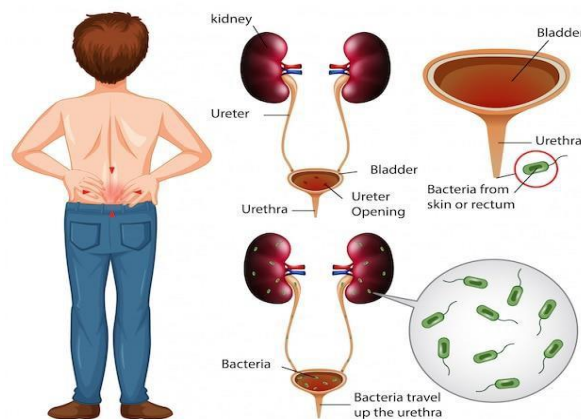
URINARY TRACT INFECTION

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I. DEFINITION

- Urinary tract infection is an infection that affect the urinary tract anywhere from the urethra to the kidney.
- Difference classification exist:
 - **Uncomplicated UTI:** Non pregnancy females, immunocompetent, No known urological abnormalities
 - **Complicated UTI :**
 - Male
 - pregnancy and pre-adolescent females
 - Immunocompromise
 - Metabolic disorder (Poorly controlled diabetes)
 - Presence of indwelling catheter
 - Structural abnormality of urinary tract
 - Other urological abnormalities (Neurological bladder)

Urinary tract infections



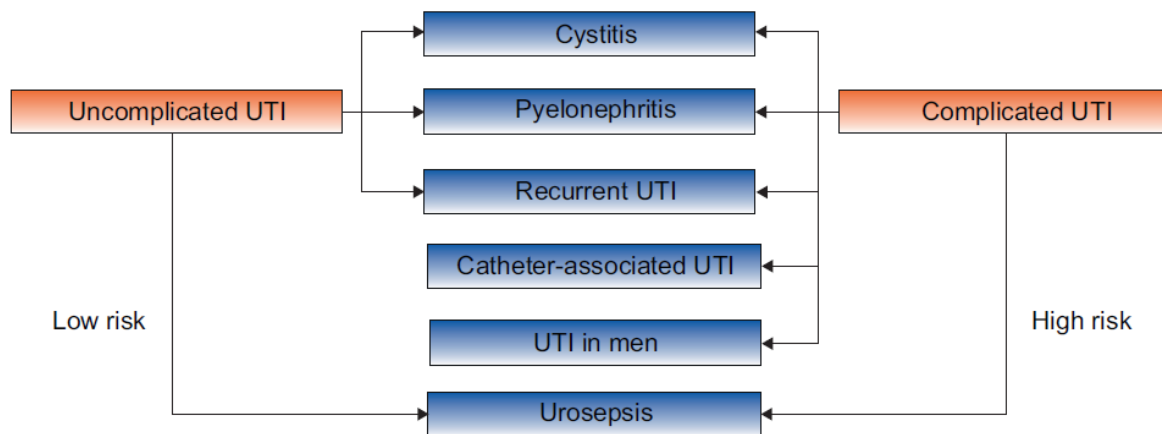


Fig. 1 – Concept of uncomplicated and complicated urinary tract infection (UTI).

II. EPIDEMIOLOGY

- Sex (Women>Men)
- UTIs are most common among sexually active women age 18-29years old
- 50% of women will develop acute cystitis at least once on their lives and a quarter will experience recurrence.

III. ETIOLOGY

- E-coli the most common cause: 70-95%
- Other pathogens: Enterobacterales (Klebsiella spp, Proteus spp), Pseudomonas, Enterococci, and Staphylococci (Methicillin sensitive Staphylococcus aureus: MSSA, Methicillin resistant Staphylococcus aureus: MRSA)

IV. PATHOPHYSIOLOGY

- Bacteria colonizing the urethra or periurethral space migrating into bladder and cause inflammatory respond: **Ascendant pathway**
- Bacteria in the bloodstream migrating to the kidney or bladder: **Hematogenic pathway (Rare)**

V. RISK FACTOR

- Females
- Older age
- Decrease estrogen levels
- Incontinence
- Catheterization
- Procedure
- Kidney stone
- Diabetes mellitus
- Immunocompromised status

VI. DIAGNOSIS

1. History taking

▪ Uncomplicated cystitis:

- Dysuria
- Urinary frequency
- Urinary urgency
- Supra pubic pain
- Absence of systemic symptoms (Fever, chill, rigor)

▪ Pyelonephritis (infection extend to the kidney):

- Fever
- Flank pain
- Rigor
- Costovertebral angle tenderness
- Nausea or vomiting

▪ Acute prostatitis /Chronic prostatitis (in Men)

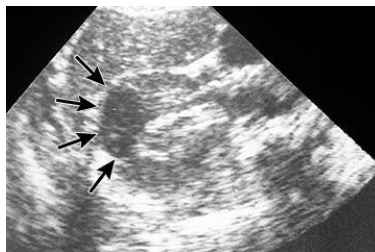
- Fever and chill
- Dysuria
- Irritative urinary symptoms (frequency, urgency, urge incontinent)
- Pelvis or perineal pain
- Acute inflamed prostate → voiding symptom→ urinary retention
- On exam: Prostate firm, edematous and tender

2. Laboratory

- Urinalysis: Urine microscopic (identify the presence of the white blood cell in urine: $>10\text{WBC/uL}$) and Urine dipstick (present of leukocyte esterase and nitrite =Bacteriuria of gram negative).
- Urine culture: Presence of bacteria $>10^5$ CFU/ml.

3. Imaging:

- Abdomen ultrasound
- Contrast enhanced computed tomography scan of the abdomen: to evaluate all of the process to delay respond of therapy or intervention (not clinical respond in 48-72h after treatment). An obstructive cause such as calculus or obstruction, a complication of infection (Ex: Renal abscess...)



Transverse view of a kidney with hypoechoic mass indicative of pyelonephritis or early abscess



Contrast-enhanced CT scan in a patient with acute pyelonephritis showing a large, hypodense region in the right kidney. There is no discrete abscess formation in this setting

***Noted:

- Urine culture should be on midstream voiding.
- For uncomplicated cystitis: Urinalysis and urine culture are not indicated in most case.
- Urine culture should be performed to all cases that suspect pyelonephritis.

VII. DIFFERENTIAL DIAGNOSIS

- **Cystitis symptoms:**
 - Vaginitis
 - Pelvis inflammation syndrome
 - Benign prostatic hyperplasia
- **Pyelonephritis symptoms:**
 - Nephrolithiasis
 - Renal infarct
 - Herpes zoster
 - Intra abdomen cause: Appendicitis, cholecystitis, biliary colic, pancreatitis

VIII. MANAGEMENT

1. Uncomplicated cystitis:

- Non antibiotic treatment: Acetaminophen maybe use for symptoms relief of cystitis.
- Delay antibiotic use with close follow up.
- Antibiotic selection and duration:

Antibiotic	Acute Uncomplicated Cystitis
Nitrofurantoin	• First-line agent • 100mg twice daily for 5 days
Trimethoprim-sulfamethoxazole	• First-line agent • 1 DS tablet twice daily for 3days
Fosfomycin	• First-line agent • 3g as one dose
Oral beta-lactams (eg: amoxicillin-clavulanic acid or cefpodoxime)	• Use if first line cannot use • amoxicillin-clavulanic acid 500/125mg twice daily for 5-7 days • cefpodoxime 100mg twice daily for 5-7 days
Fluoroquinolone (Ciprofloxacin)	• Effective but only alternative oral antimicrobials for acute cystitis

- **Safety Oral antibiotics in Pregnancy:**

Antibiotic	Category
Nitrofurantoin	B
Trimethoprim-sulfamethoxazole	D
Fosfomycin	B
Oral beta-lactams (eg: amoxicillin-clavulanic,..)	B
Fluoroquinolones (eg: ciprofloxacin)	C

****Noted:**

- Nitrofurantoin and Fosfomycin should be avoided if early pyelonephritis is suspect because they have poor drug penetration to renal parenchyma.
- Avoid using fluoroquinolone in uncomplicated cystitis due to highly rates of E-Coli resistance.

2. Complicated Urinary tract infection:

- Consider a 7 days of treatment antibiotics in women below of 65years old without comorbidities.
- Use 14 course of antibiotics to treatment complicated pyelonephritis.
- Hospitalization indication:
 - Septic or critically illness
 - Persistently high fever, pain, inability oral hydration or oral medication
 - When urinary tract obstruction is suspected.
- Empiric microbial therapy (Depend on severity of illness, Risk factor, specific host factor)

Urinary tract obstruction / Critically illness ** Community prevalence multidrug resistant risk is high	• Decompress urinary tract obstruction • Use a broad spectrum for empiric therapy - Anti-pseudomonas Carbapenem: Imipenem 500mg IV every 6 hours or Meropenem 1g IV every 8 hours to cover extended-spectrum beta-lactamase (ESBL) producing organism and pseudomonas. - Plus Vancomycin 15-20mg /kg IV loading dose then continue with maintenance dose 15mg/kg q12h (for maintenance dose please adapt to renal function)
Other hospitalized patients ** No risk of multiple drug resistant	• Ceftriaxone 1g IV dose once daily - Alternatives: · Levofloxacin 750mg IV or orally daily · Ciprofloxacin 400mg IV twice daily / 500mg orally twice daily • If Enterobacter is suspected: Piperacillin-tazobactam

Other hospitalized patients ** risk of multiple drug resistant	• Piperacillin-tazobactam 4.5g IV every 8 hours or • Cefepime 2g IV every 12 hours (not for ESBL) - Anti pseudomonal Carbapenem: Imipenem 500mg IV every 6 hours or Meropenem 1g IV every 8 hours to cover extended-spectrum beta-lactamase (ESBL) producing organism and pseudomonas.
Bacterial prostatitis	• High dose of bacterial antimicrobials, such as broad-spectrum penicillin, a third-generation cephalosporin or fluoroquinolone. • For initial therapy, any of these antimicrobials combined with an aminoglycoside. • After normalization of infection, oral therapy continues for total 2-4 weeks.

IX. COMPLICATION

- Complicated UTI: bacteremia, sepsis, multiple organ dysfunction, shock, acute renal failure.
- Acute pyelonephritis: extend to renal parenchyma→ renal corticomedullary abscess, emphysematous pyelonephritis, papillary necrosis.

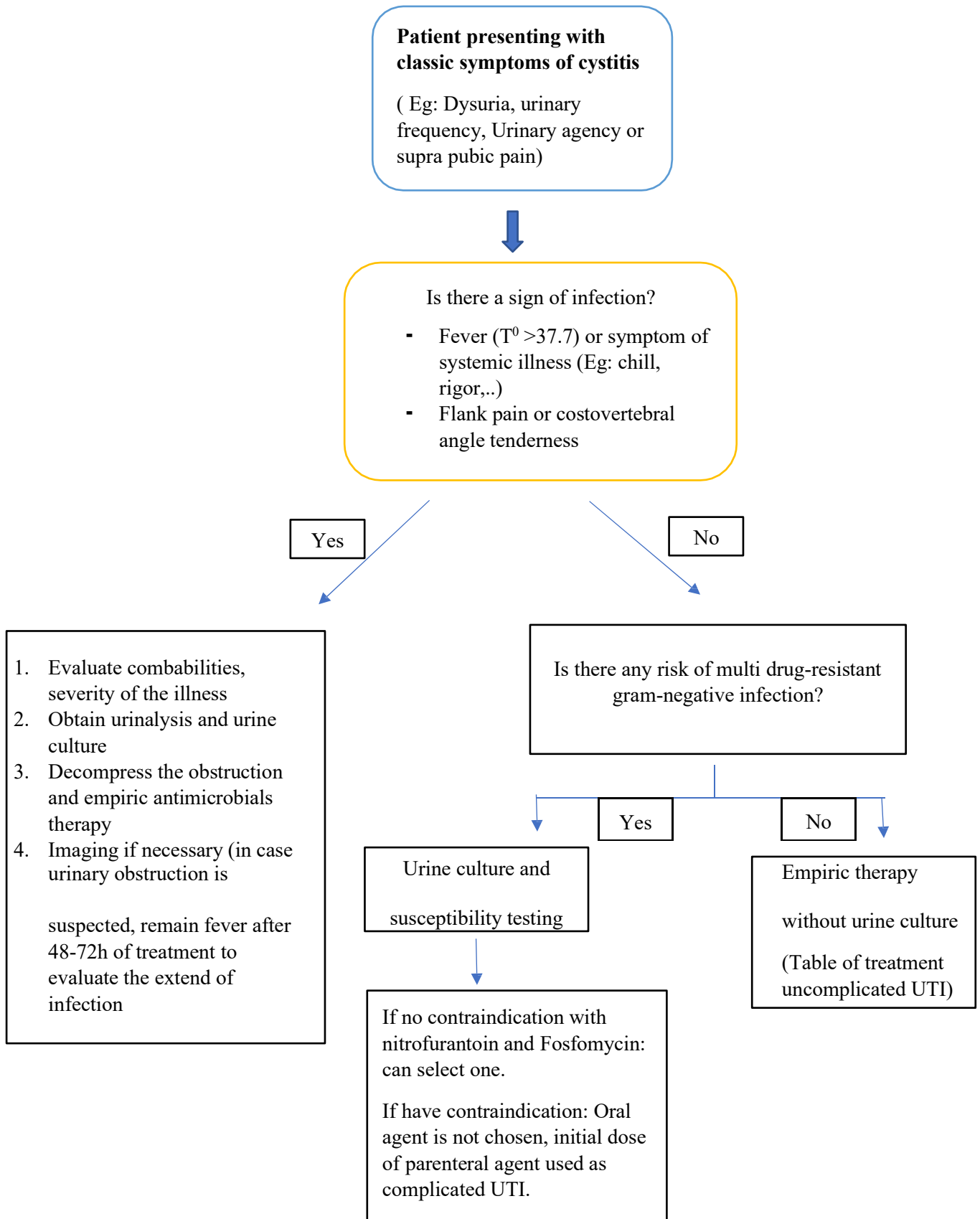
X. PREVENTION AND RECOMMENDATION

1. Patient with resolved symptom no routine laboratory follow-up
2. Patient with recurrence UTIs (>2UTIs in 6months, or 3 episode in 1 year):
 - Encourage menopause to drink water at least 1.5L/day
 - Consider antibiotic prophylaxis
 - Assess for urinary retention for men and postmenopausal women
3. Avoid catheterization

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Algorithm in Urinary tract infection



KIDNEY STONE

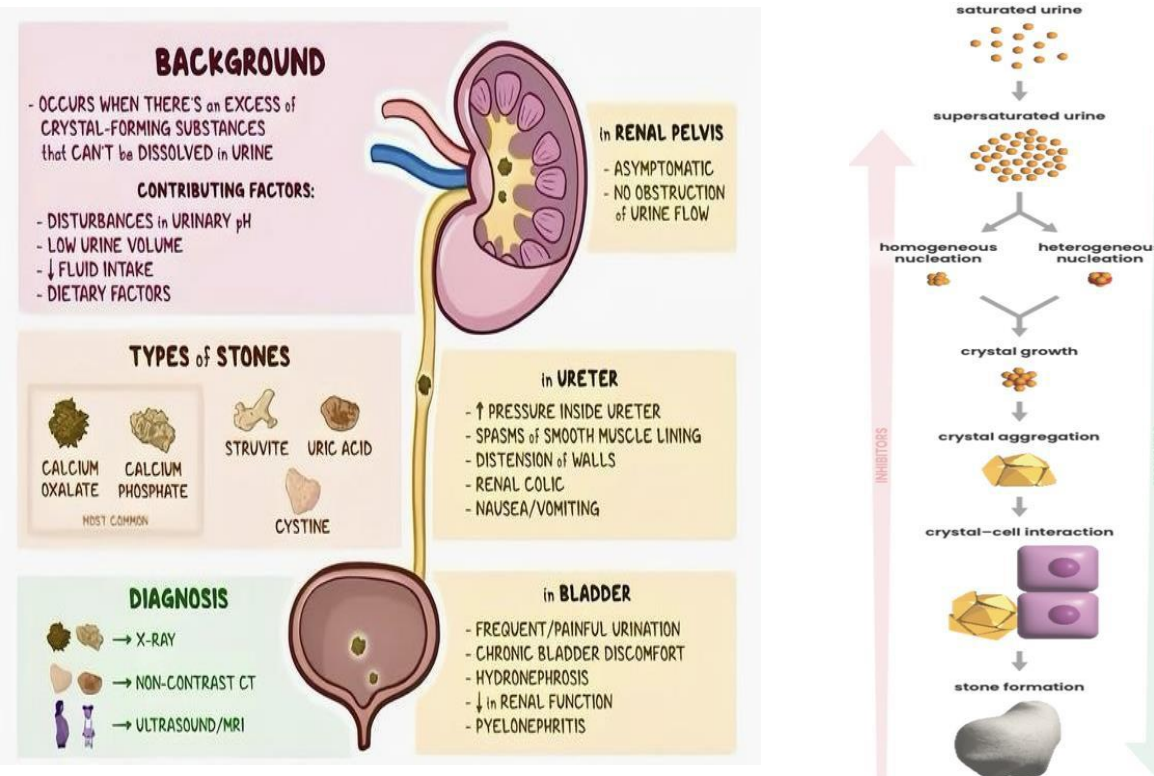
Dr. Hy Chanseila, Dr. Sin Boohavan, Dr. Lim Vadhana, Dr. Niv Rathvirak, Dr. Thim Pichthida, Dr. Lim Sochun, Dr. Rith Sovannara, Dr. Keut Chansophal

I. DEFINITION (what is kidney stone)

Kidney stone (or urolithiasis) are hard mineral deposits that form in the kidneys. Their size can vary, ranging from a few millimeters to several centimeters. The risk of developing kidney stones is three times higher in males than in females.

II. PHYSIOPATHOLOGY (how kidney stone form)

- Insufficient hydration or consuming foods high in mineral salts such as calcium or phosphorus, and large amounts of animal protein can lead to the formation of kidney stones.
- The growth of a kidney stone begins with the formation of small crystals in **saturated** urine, which adhere to the urothelium and initiate the development of the stone.
- Kidney stones are usually asymptomatic until they obstruct the flow of urine. In such cases, the kidney becomes swollen, and symptoms develop, a condition known as renal colic.
- The frequency of kidney stone compositions is as follows: oxalate > phosphate > uric acid > struvite > cystine > medications.



III. DISCOVERY CIRCUMSTANCE

- During a routine health checkup
 - When the patient experiences renal colic
 - Presence of hematuria (microscopic or macroscopic)
 - Urinary tract infection
 - Acute kidney failure (caused by obstruction in both kidneys)
- An abdominal ultrasound is needed for all the symptoms above, and kidney stones are often found during the exam.

IV. IMAGING TEST (which lead to discover kidney stone)

- Abdominal ultrasound to find kidney stone and dilatation of ureter.
- Abdominal X-ray to identify the location of kidney stones.

nature of kidney stone	opacity	morphology
phosphore	+++	homogeneous/smooth
oxalate	+++	smooth or spicular
struvite	+	corraliform
cystine	+	small/irregular
uric acid	-	radiotransparent +++
medicines	-	radiotransparent +++

- A pelvic-abdominal CT scan without contrast injection can replace ultrasound and abdominal X-ray in atypical cases.

V. NATURE OF KIDNEY STONE

- Morphology test: CT urography to analyze the form of the stone and determine its origin.
- Metabolic blood test: ionogram, renal function, uric acid, calcium and phosphor
- Analysis urine sedimentation.

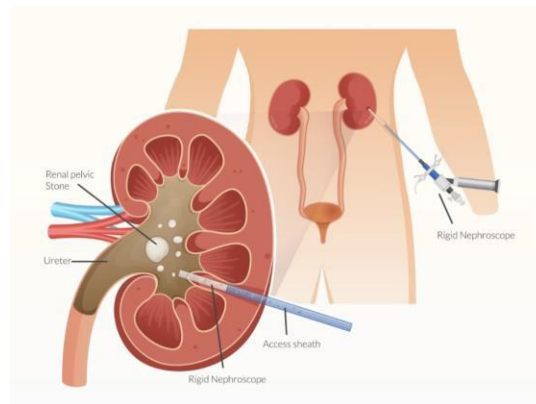
VI. SERVER AND COMPLICATED FORM

- Kidney stone in: pregnancy, renal failure and patient with single kidney.
- Kidney stone with fever: acute pyelonephritis obstructive
- Kidney stone hyperalgie: resistance to NSAID
- Bilateral kidney stone which lead to renal failure (elevation creatinine).

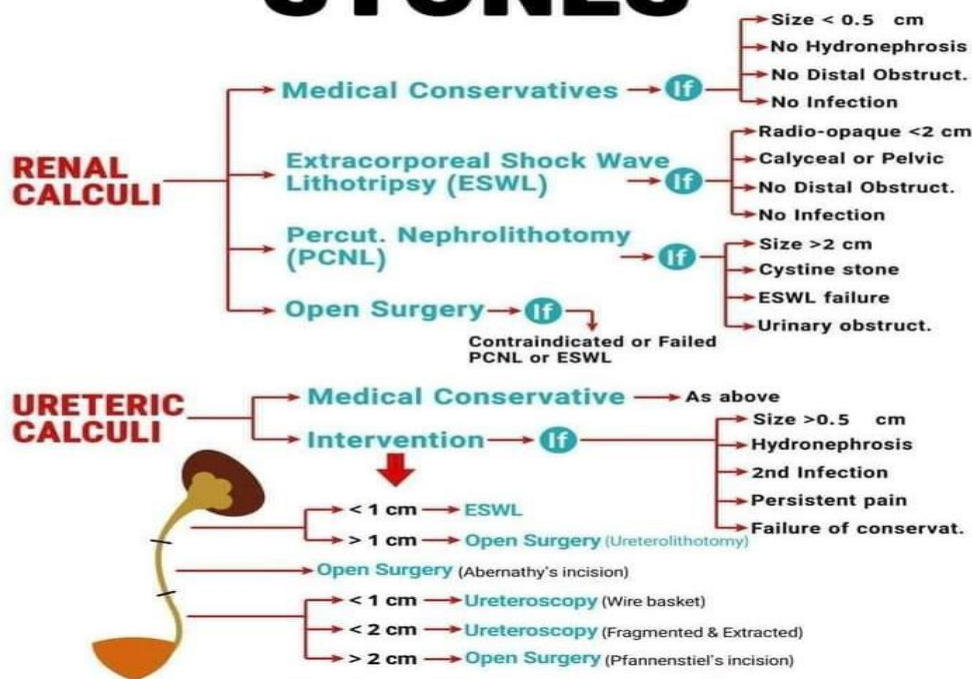
VII. TREATMENT

- Renal Colic:
 - Non-steroidal Anti-inflammatory Drugs (NSAIDs): (Check creatinine and beta-hCG levels before use). Medications such as Ketoprofen or Diclofenac 100mg IV, twice a day for 3 days, followed by 100mg tablets twice a day for 5 days.
 - Painkillers: Paracetamol 1g IV, four times a day for 3 days, or morphine titration subcutaneously if NSAIDs and painkillers fail to relieve the pain.
 - Anti-spasmodic Medications: Phloroglucinol (Spasfon) IV for 3 days, then 2 tablets twice a day for 5 days.

- | | | |
|----------------|--------|---------------------------------------|
| kidney stone | < 2cm | Extracorporeal shock wave lithotripsy |
| | > 2cm | Percutaneous nephrolithotomy |
| ureteral stone | lumbar | Extracorporeal shock wave lithotripsy |
| | iliac | Extracorporeal shock wave lithotripsy |
| | pelvis | urethroscopy |



MANAGEMENT OF URINARY SYSTEM STONES



VIII. REVENTION

- Drink water more than 2 liters per day
- Dietary recommendations based on the origin of kidney stones:
 - lithiase from calcium: limit foods that are high in calcium.
 - lithiase from oxalate: limit the consumption of chocolate or tea.
 - lithiase from uric acid: limit the consumption of meat and use febuxostat if uric acid is more than 70mg/L.
- Treat the underlying diseases, such as hyperuricemia, malformative uropathy or hyperparathyroidism.

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CHAPTER VI RHEUMATOLOGY

1. Rheumatic fever
2. Rheumatoid arthritis
3. Septic arthritis
4. Osteoarthritis
5. Gout
6. Adult onset still disease
7. Rotator cuff tendinopathy
8. Infectious spondylodiscitis
9. Spondylarthritis
10. Complex regional pain syndrome
11. Systemic sclerosis
12. Systemic lupus erythematosus
13. Osteoporosis

RHEUMATIC FEVER

*Dr. KHUN Rithya, Dr. HENG Reaksmeay, Dr. KAO PhengAnn, Dr. UY Putthimundul,
Dr. CHHE Sereyvuth*

I. DEFINITION

Acute rheumatic fever (ARF) is an autoimmune complication of group A streptococcal pharyngitis, Carditis and Arthritis is the most manifestation, ARF occurred about 2 or 3 weeks after Pharyngitis if not treated correctly, resulting over 50% for rheumatic heart diseases.

Carditis and Arthritis are the most common manifestation; Erythema marginatum, subcutaneous nodule, chorea occurs less frequency.

II. EPIDEMIOLOGY

- Acute Rheumatic heart disease (ARF) is worldwide distribution
- The highest reported incidences of ARF are in low-income countries, likely associated with inadequate housing, poor hygiene.
- Children and adolescent age 5 to 15 years old are most affected
 - ARF is rare in children ≤ 5 years old.
 - The highest incidence of ARF is in children aged 10-14 years.
 - The first episode of ARF is rare in adults ≥ 30 years old.

III. ETIOLOGY

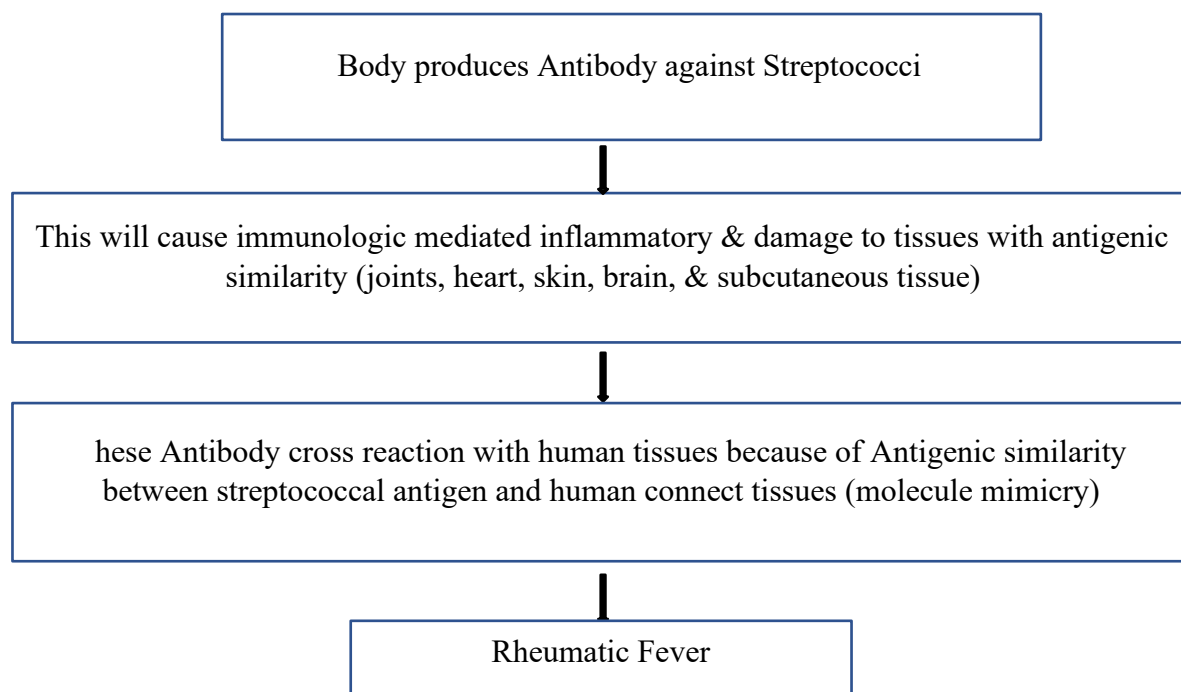
Group A Streptococcus (GAS) or Streptococcus pyogenes: Streptococci are gram-positive cocci in chains and beta-hemolytic, the primary reservoir is human throat and skin, person-to-person transmission occurs primarily via respiratory droplets and direct skin contact.

Risk factors:

- Aged 6 to 15 years old
- Untreated streptococcus throat infection
- Familia predisposition
- Over crowded
- Poverty, poor hygiene
- Lack of access of medical care.

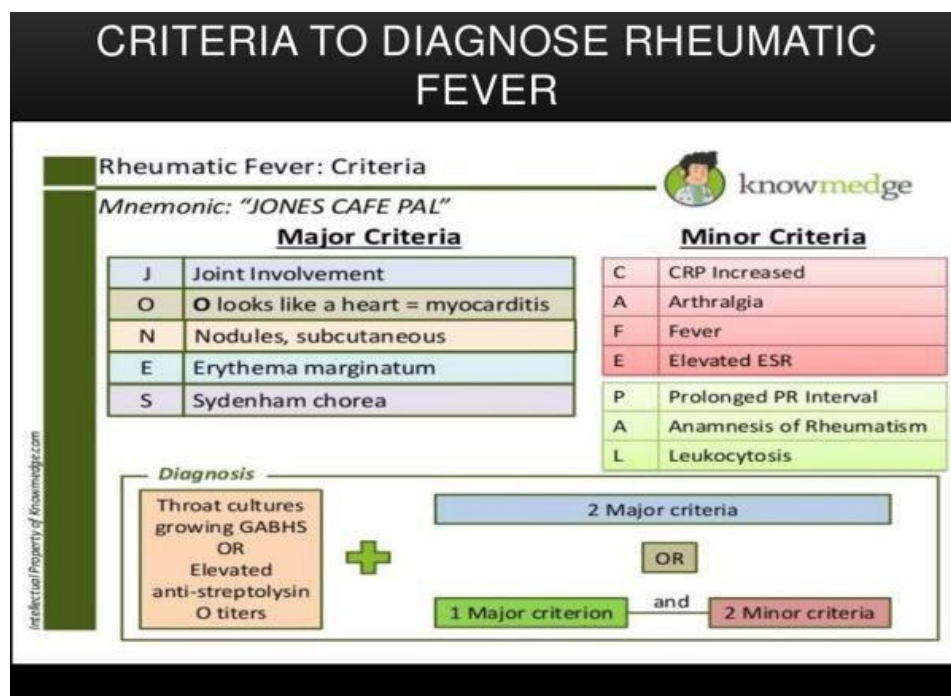
IV. PATHOPHYSIOLOGY

- Acute rheumatic fever (ARF) is a humoral and cellular-mediated autoimmune disorder causing damage to the heart, brain, joints, and/or skin.
- Complex molecular mimicry between group A streptococcal (GAS) antigens and host factors induces autoimmune reaction. leading to damage heart valve, joints, brain and skin.



V. DIAGNOSIS

- Diagnosis of Acute rheumatic fever needs laboratory evidence of prior group A streptococcal infection plus either
 - 2 major criteria
 - Or 1 major plus 2 minor criteria
 - Or 3 minor criteria
- Laboratory evidence of group A strep is:
 - Positive throat culture or
 - Increased or rising Antistreptolysin O titer or anti-deoxyribonuclease B (anti-DNase B).



VI. DIFFERENTIAL DIAGNOSIS

- Reactive arthritis
- Septic arthritis
- Juvenile idiopathic arthritis
- Systemic lupus erythematosus
- Mixed connective tissue disease
- Viral infection

VII. MANAGEMENT

1. Objectives

- Eradication of group A beta hemolytic streptococcus (GAS)
- Symptomatic relief of acute disease manifestation (e.g joints pain, fever)
- Manage of rheumatic heart disease (RHD, e.g Carditis, heart failure if present)
- Manage of Chorea if present
- Prophylaxis against future GAS infection to prevent progression of cardiac disease
- Provision of education for patient and patient care's giver

2. Critical signs

- The frequency attacked of GAS is the permanent of heart valve damaged, and it can do undetectable until the patient present with debilitating heart failure.

3. Mortality and Risk factors

- Pericarditis: sharp, cutting pain, relieve on leaning forward, Pericardial friction rub.
- Myocarditis: is lethal as the myocardium become loose, flabby and doesn't contract well
- Mitral or Aortic regurgitation-endocardium involved
- Congestive heart failure, Arrhythmia mostly atrial fibrillation
- Infective endocarditis, embolic event.

4. Outpatient management

- The major goals of therapy are eradication of GAS infection, and supportive care, directed at organ-specific manifestations.
- Patient work-up: CBC, ESR, CRP. Kidney function, transaminase, EKG, chest X ray, Heart ultrasound (if available), throat culture, Anti - streptolysin O titer (ASLO), Anti-deoxyribonuclease B (anti-DNase B).

5. First line treatment

- Patients with newly diagnosed ARF are started on antibiotic therapy to eradicate group A *Streptococcus* (GAS) carriage
- GAS pharyngitis
- Penicillin V: 500 mg orally two to three times daily for 10 days
- Amoxicillin: 500 mg orally twice daily for 10 days

- Cephalexin: 500 mg orally twice daily for 10 days
 - Cefuroxin: 250 mg orally twice daily for 10 days
 - Penicillin G benzathine 1.2 million units IM as single dose.
 - If Penicillin allergic: Azithromycin 12mg/kg/day 1 time daily for 5 days (maximum 500mg/day)
 - GAS pyodermic: treatment the same as GAS pharyngitis
 - Household contact: should be screen if positive need full course of Antibiotic even if asymptomatic
 - Arthritis management: non steroid anti-inflammatory (NSAID) Ibuprofen, Diclofenac until the symptom resolved; alternative low dose steroid.
 - Carditis: management: prednisolone 1 to 2 mg /kg for 4 to 6week then taper
 - Sydenham chorea management: usually mild and self-limited, if sever need to neurologist
6. Treatment adjustment: patient who have had an ARF need the secondary prevention for developing and progression of RHD (rheumatic heart disease)
- Preferred regiment: Penicillin G benzathine 1.2 million units IM every month or penicillin V oral 500mg BID; if Penicillin allergic consider Azithromycin
 - Rheumatic fever carditis with and persistent Valvular heart disease: 10 years since last episode or until 40 ys old, sometime lifelong prophylaxis.
 - Rheumatic fever with carditis and no valvular heart disease: 10 years since last episode or until 21 years old (sometime longer)
 - Rheumatic fever without carditis: 5 years or until 21 years old (sometime longer)
7. Algorithm management

Duration of prophylaxis for Rheumatic fever	
- Rheumatic fever without Carditis	- 5 years or until 21 years old
- Rheumatic fever with Carditis but no Valvular heart disease	- 10 years or until 40 years old
- Rheumatic fever with Carditis and has Valvular heart disease	- At least 10 years since last episode or last until 40 years - Sometime lifelong prophylaxis

VIII. SPECIAL CASES

Patient with severe valvular heart disease who develop severe symptomatic valvular heart disease (ie, mitral stenosis, aortic stenosis, aortic insufficiency), New York Heart Association class III or IV heart failure, or ventricular dysfunction (ie, ejection fraction <50 percent), we recommend prophylaxis with an oral antibiotic rather than IM Benzathine penicillin.

IX. EVOLUTION AND PROGNOSIS

Patient with primary acute rheumatic fever:

- Mortality is low, Death was reported 0.6%
- Most manifestation of ARF are self- limited
- Carditis may lead to irreversible cardiac damage and exacerbated by recurrent ARF attacks.

X. COMPLICATION

Rheumatic heart disease

- Patients with previous acute rheumatic fever (ARF) are at elevated risk of recurrence which may worsen RHD.
- Patients may progress to RHD even if carditis was not present during primary episode of ARF.

XI. PREVENTION AND RECOMMENDATIONS

- Hand hygiene: hand hygiene is very important for preventing the spread to others
- Post exposure prophylaxis: Testing and treatment of asymptomatic persons who have been exposed to a patient with group A Streptococcus (GAS) pharyngitis are not routinely recommended.
- Patient with a history of acute rheumatic fever: Patients with a history of acute rheumatic fever are at high risk for recurrent rheumatic fever and the development of chronic valvular heart disease with any subsequent GAS infection. We educate these patients on the risk of recurrence and its complications and recommend long-term antibiotic prophylaxis.
- Vaccination: not yet available

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RHEUMATOID ARTHRITIS

KHUN Rithya, HENG Reaksmeay, KAO PhengAnn, UY Putthimundul, CHHE Sereyvuth

I. DEFINITION

Rheumatoid Arthritis (RA) is a symmetric, inflammatory, peripheral polyarthritis of unknown etiology. leading to joint destruction through the erosion of cartilage and bone. Untreated or uncontrolled, it will lead to loss of physical function of daily task, and may have other health risks including higher rates of cardiovascular disease, osteoporosis, and certain types of cancer (eg, lymphoma).

II. EPIDEMIOLOGY

Rheumatoid arthritis (RA) is estimated to affect approximately 0.24 to 1 percent of the population and to be twice as common in women compared with men. The annual incidence of RA is estimated to be approximately 40 per 100,000 persons.

III. ETIOLOGY: IS AUTOIMMUNE DISEASE OF UNKNOWN

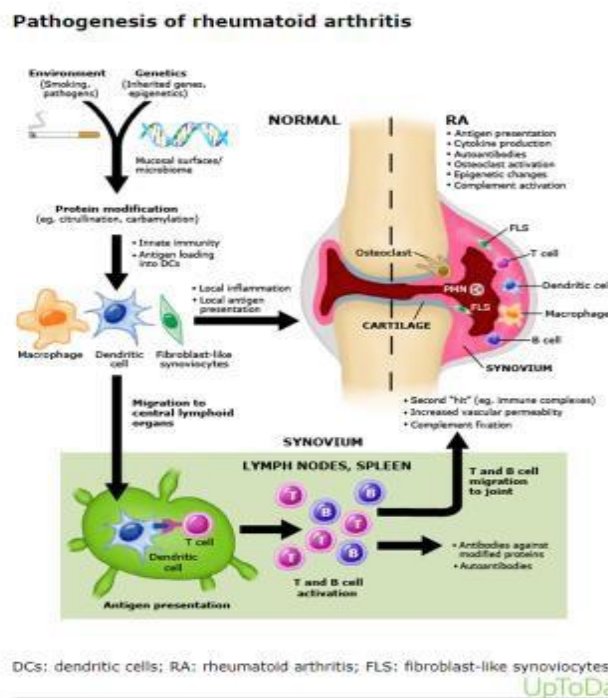
Family history and Genetic risk factors: Human leukocyte antigen (HLA)-DRB1 genes represent the strongest genetic risk factor for RA, and having a first-degree relative with RA increases the risk of RA approximately threefold.

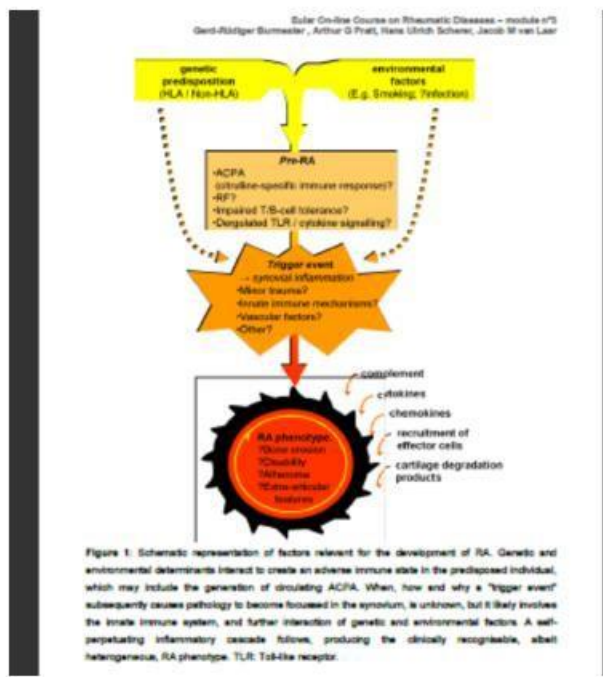
Lifestyles risk factors: Cigarette smoking, poor diet quality, obesity, and physical inactivity are lifestyle factors Infections

- Periodontal infection
- Viral infection: Hepatitis B and C, Rubella, Chikungunya, EBV can trigger RA

IV. PATHOPHYSIOLOGY

1. Multiple genetic, environmental, immunologic, and other factors contribute to the pathogenesis of rheumatoid arthritis





V. DIAGNOSIS

1. Is based on family history, history taken and physical exam : constitutional symptoms, morning stiffness more then one hour , elevated ESR or CRP, and a small-joint arthritis affecting the hands and feet. Joint erosions or deformity appear late in the disease process.
2. EULAR/ ACR 2010 criteria, a score >6/10 is fulfill diagnosis of RA

Classification criteria for RA (Score base algorithm: add score of categories A-D A score of >6/10 is needed for a definite classification of a	
A. Joint involvement	
1 large joint	0
2-10 large joints	1
1-3 small joints (with or without involvement of large joints	2
4- 10 small joints (with or without involvement of large joints	3
>10 joints (at least one small joint)	5
B. SEROLOGY (at least one test result is needed for classification)	
Negative RF AND negative ACPA	0
Low positive RF or ACPA	2
Hight positive RF or high positive ACPA	3
C. ACUTE PHASE REACTANTS (at least one test	
Normal CRP and normal ESR	0
Abnormal CRP or abnormal ESR	1
D. Duration of symptoms	
< 6 weeks	0
> 6 weeks	1

3. ACR 1987: Does the patient has 4 of 7 of the following symptoms

- Morning stiffness > one hour
- Involvement > 3 joints
- Arthritis of hand joints (wrists, MCP, PIP)
- Symmetric arthritis
- Rheumatoid nodule
- Rheumatic factor positive
- Radiography evidence of erosion

VI. DIFFERENTIAL DIAGNOSIS

Viral arthritis: Chickungungia, Hepatitis

B & C Septic Arthritis

Others systematic diseases: SLE, Sjorgen's disease, Mixed connect tissue disease Psoriasis arthritis

Osteoarthritis, Fibromyalgia, Carpal tunnel syndrome

VII. MANAGEMENT

1. Objective:

- RA is a serious disease requiring early appropriate management
- Assessment of disease activity, risk factors, and comorbidities is important in treatment selection
- Initiating the DMARDs soon after the diagnosis has been made
- Monitoring for clinical response and adverse events is critical, with the goal of remission or low disease activity (Treat to target)

2. Critical sign

- Female gender
- Duration symptoms more than 3 months
- Hight tender and swollen joints (functional limitation)
- Hand involvement
- Cigarette smoking
- Hight ESR, CRP
- Rheumatic factors or Anti-CCP positive
- Join Erosions
- Or DAS > 5.1 (DAS < 3.2 low disease activity, DAS 3.2 to 5.1 moderate disease activity, DAS 5.1 sever disease activity)

3. Mortality risk factors is related to:

- Infection:
 - Hepatitis B : would be Collaboratory with hepatologist and non hepatotoxic drug is prefer Sulfassalazine/ chloroquine
 - Hepatitis C : would be Collaboratory with hepatologist and non hepatotoxic drug is prefer Sulfassalazine/ chloroquine
 - Tuberculosis: need to treat correctly before starting treatment with DMARDs
 - Serious infections : all the DMARDs should be temporarily hold until resolution of infection and completion of antimicrobial therapy
- Malignancys: nonmelanoma skin cancer, melanoma skin cancer, solid tumors
- Lungs diseases : Comorbid interstitial lung disease (RA-ILD) and chronic obstructive pulmonary disease (COPD) are common in patients with RA, need to avoid to used Methotrexate (MTX)
- Cardiovascular diseases: Comorbid cardiovascular disease can occur in patients with RA and may also be a complication of therapy also, need to start statin , LDL should less then 100mg/dl

- Diabetes : steroid need to be caution in RA and Diabetes. The DM should be good controlled
 - The comorbidities like serious infection, RA-ILD, CVDs etc..
 - The complication of RA : Cervical subluxation
4. Outpatient management
- Enough painkiller either NSAIDs or consider Prednisolone if renal impairment
 - Strongyloidiasis prevention with Albendazole 400mg bid for five days
 - Provide Calcium/vit D ; if patient on Prednisolone > 7.5mg per day more than 3 month consider Alendronate 70 mg every week.
 - Advice birth spacing both man , woman before start DMARDs
 - Stop smoking , management hyperlipemia, or DM to decrease CVDs risk
 - Well cooked food, psycho-social intervention,
 - Patient education regarding RA and its management
 - Rest , exercise, physical and occupational therapy
 - Pretreatment evaluation :
 - General testing for all patients : CBC, ESR, CRP, Transaminase, kidney function , LDL, triglyceride , HbA1C , C x R if suspected TB or lung disease ,
 - Screening : Hep B (HBs Ag , HBs Ab, anti HBc) , HCV, Bacteria infection if suspected and need to treat appropriate before start DMARDs,
 - Ophthalmology screening for Hydroxychloroquine/ chloroquine used if possible
 - DMARDs : Drugs classified as disease-modifying antirheumatic drugs (DMARDs), which all have the potential to reduce or prevent joint damage and to preserve joint integrity and function, include:
 - Non biologic (traditional or conventional) DMARDs: Methotrexate (MTX), Leflunomide (LEF), Sulfasalazine (SSZ), Chloroquine or Hydroxychloroquine (HCQ).
 - Targeted synthetic DMARDs: JAK inhibitor (Janus kinase inhibitor): Tofacitinib ,Baricitinib.
 - Modality treatment
 - Monotherapy : one DMARDs with or without Prednisolone
 - Combination therapy : two or three with or without Prednisolone:
 - MTX or LEF or SSZ + HCQ
 - MTX + SSZ + HCQ
 - LEF + SSZ + HCQ
 - MTX or LEF or SSZ + Tofacitinib or Baricitinib
 - Low dose prednisolone : dose less than 7.5 mg /d ; Prolong steroid used : is used more than three month.
5. First line treatment
- For mild to moderate disease activity used monotherapy :
 - Methotrexate (2.5mg) 4 tablets PO every week + folic acid 5mg PO every week + Prednisolone 5mg to 10mg PO qd (can decrease and stop when no more joints pain , usually 3 to 6 months) ; MTX can increased each month as tolerance till no more joints pain (15mg , 20 mg , up to 25 mg)
 - For whom can not used MTX :
 - Sulfasalazine (500mg) 1 tablet PO BiD (SSZ can increased each month till 3g /day) + Prednisolone 5mg to 10 mg PO qd
 - Leflunomide (20mg)1 tablet qd + Prednisolone 5mg to 10 mg.
 - Repeat transaminase , CBC, creatinine every month for three month if stable will do every 3to6 month interval.

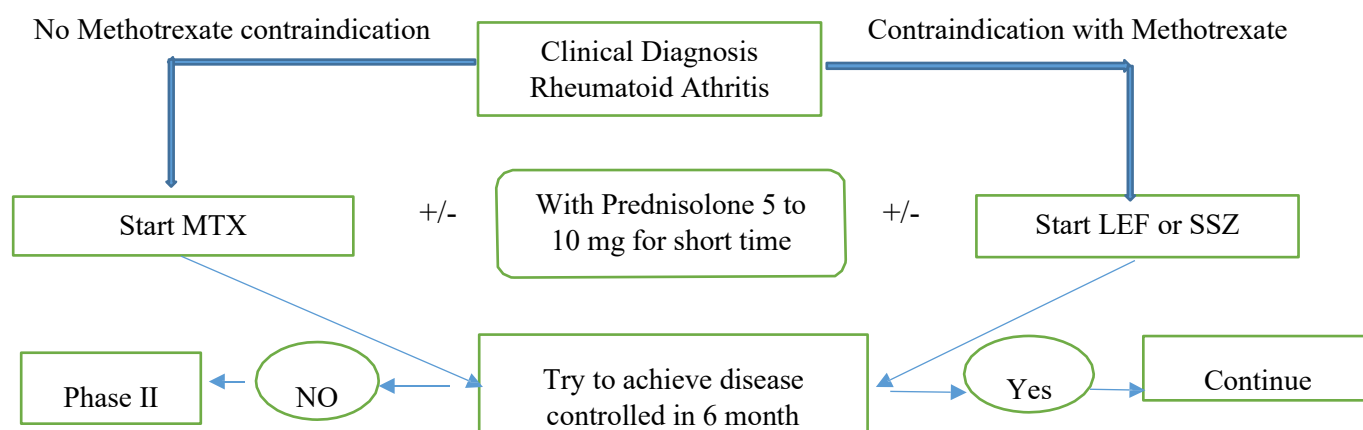
- If the disease still not get controlled after optimal therapy in 6 month ; go to subsequence treatment.

6. Treatment adjustment

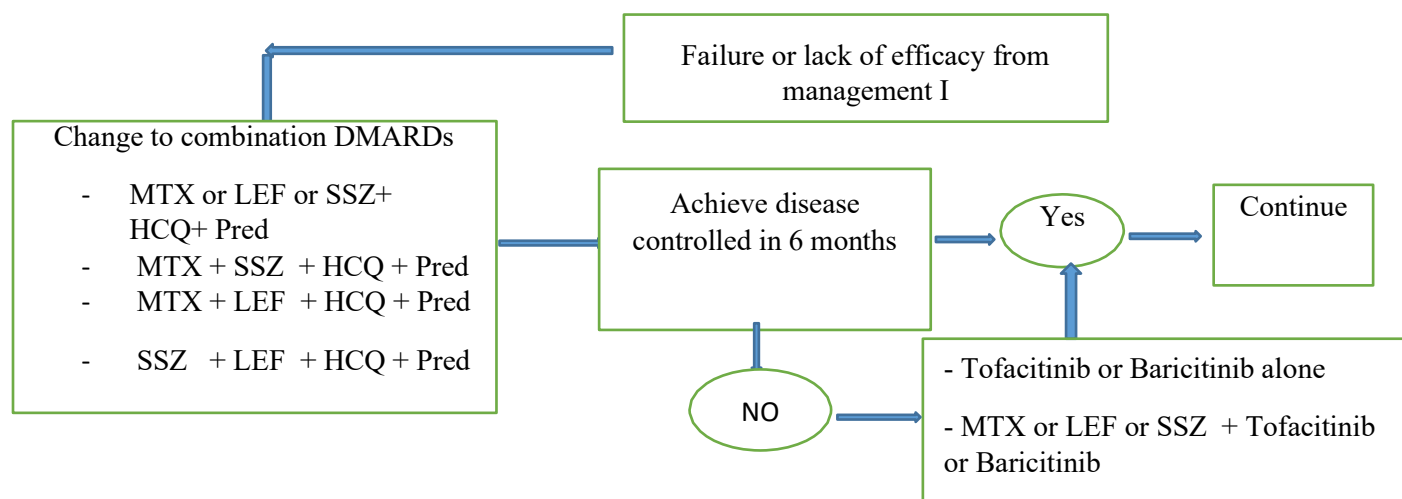
- For moderate to severe disease or whom failed with first line consider Bitherapy or tripletherapy
 - MTX or LEF or SSZ + HCQ + prednisolone 5 to 10 mg (bitherapy)
 - MTX + SSZ + Chloroquine + prednisolone 5 to 10 mg
 - MTX + LEF + Chloroquine + Prednisolone 5 to 10 mg (need to controlled transaminase more frequency in first 6 month)
 - MTX or LEF or SSZ + Tofacitinib 5mg BID or Baricitinib 2mg QD.
- When the disease get remission more then one year, we can taper the DMARDs rather than stop all with close monitoring of flair.

7. Treatment algorithm

- First line management



- Treatment adjustment
Prognostic unfavorable or risk factors present (Anti CCP + or RF + or high disease activity)



Abbreviation : RA : Rheumatoid arthritis , RA-ILD : rheumatoid arthritis interstitial lung disease , MTX : Methotrexate , LEF : Leflunomide , SSZ : Sulfasalazine , HCQ : Hydroxychloroquine , Pred : Prednisolone , EULAR : European league against rheumatology , ACR : American committee Rheumatology .

VIII. SPECIAL CASES: (NEED TO REFER TO SPECIALIST)

RA with Lungs disease RA with malignancies RA with Renal diseases

Pregnancy : pregnancy make the diseases become improve or in remission

- If planning for pregnancy : need to stop MTX 3 months ; for LEF need to stop one year ahead.

IX. EVOLUTION AND PROGNOSTIC

Early Diagnosis and appropriate management with DMARDs making the disease in remission has decrease joint surgery and early mortality

Factors make adverse prognostic :

- Functional limitation
- Extraarticular disease
- Rheumatic factor or anticyclic citrullinated peptide (CCP) antibody positive
- Bony erosion
- Cigarette smoking,
- Has others medical problem

X. COMPLICATIONS

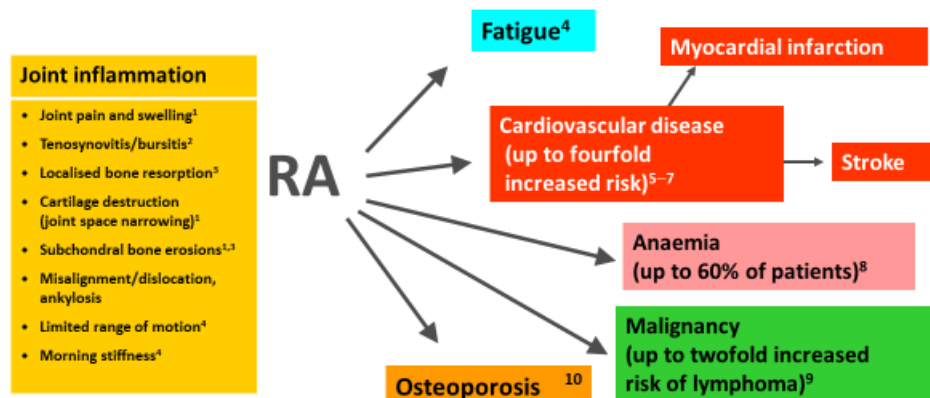
Joints destruction and disabling

Osteoporosis can be from the disease and steroids Cardiovascular disease : ischemic

heart disease , stroke Anemia

Malignancy

RA is a chronic systemic autoimmune inflammatory arthritis associated with extra-articular manifestations



¹Smolen J, et al. *Nat Rev Drug Disc* 2003; **2**:473–488. ²Grassi W, et al. *Eur J Radiol* 1998; **27** (Suppl. 1):S18–24.

³Firestein G. *Nature* 2003; **423**:356–361. ⁴Smolen J, et al. *Lancet* 2007; **370**:1861–1874.

⁵Turesson C, et al. *Ann Rheum Dis* 2004; **63**:952–955. ⁶del Rincón I, et al. *Arthritis Rheum* 2001; **44**:2737–2745.

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XI. INFORMATION FOR PATIENT AND PREVENTION

1. Education and counselling :

- Rheumatoid arthritis (RA) is a chronic inflammatory condition. It primarily affects the joints, but it can affect many different parts of the body. The cause of RA is unknown but likely, early recognition and appropriate treatment make the good prognosis .

2. Exercise :
 - Inactivity can lead to a loss of joint motion, contractions, and a loss of muscle strength , regular exercised can help to prevent this effect ,If your joint symptoms make it difficult for you to move or be active, a physical therapist can help.
3. Nutrition and Diet :
 - There are no Diet or herbal known to cure RA, the addition of fish oils rich in omega-3 fatty acids and taking vitamin D3 have modestly improved arthritis pain and joint swelling.
 - If you are obese or overweight need to lose weight.
 - Stop smoking and alcohol

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SEPTIC ARTHRITIS

Rheumatology Team: KAO Pheng Ann, UY Putthimundul, HENG Reaksmeay, CHHE Sereyvuth, KHUN Rithya Infectiology team: CHHE Sereyvuth, KOY Sophany, YOUNG Yuda, MENG Sreyhouch, CHEA Sereywata, VANN Pisey, BORY Sotharith

I. DEFINITION

Septic arthritis is an infection of a joint caused by bacteria, viruses, or fungi, leading to inflammation, pain, and potential joint destruction if untreated. It is a medical emergency requiring prompt diagnosis and treatment.

II. EPIDEMIOLOGY

Septic arthritis occurs worldwide, with an annual incidence of 2-10 cases per 100,000 people. Risk is higher in individuals with underlying joint disease, immunosuppression, or prosthetic joints. The condition is more common in children and older adults.

III. ETIOLOGY

- Bacterial pathogens:
 - Gram-positive: *Staphylococcus aureus* (most common), *Streptococcus* species.
 - Gram-negative: *Neisseria gonorrhoeae*, *Escherichia coli*, melioidosis.
 - Tuberculosis.
- Fungal causes: *Candida* species (in immunocompromised).
- Viral causes: Hepatitis B, C, and HIV (less common).

IV. PATHOPHYSIOLOGY

- Pathogens invade the joint via:
 - Hematogenous spread (most common).
 - Direct inoculation (e.g., trauma, surgery).
 - Contiguous spread from adjacent infected tissues.
- The infection triggers an inflammatory response, releasing cytokines and proteolytic enzymes that damage cartilage and synovial tissues, potentially leading to joint destruction.

V. DIAGNOSIS

1. Clinical evaluation:

- Acute onset of joint pain, swelling, warmth, and restricted motion.
- Fever and systemic symptoms may be present.

2. Laboratory tests:

- Elevated inflammatory markers (e.g., CRP, ESR).
- Synovial fluid analysis: Elevated white blood cells ($>50,000/\text{mm}^3$) with neutrophil predominance.
- Gram stain and culture of synovial fluid.

3. Imaging:

- X-ray: May show joint effusion or joint destruction.
- MRI: Detects early joint inflammation and abscesses.
- Ultrasound: Useful for detecting effusions and guiding aspiration.

VI. DIFFERENTIAL DIAGNOSIS

- Gout or pseudogout.
- Rheumatoid arthritis flare.
- Reactive arthritis.
- Lyme disease.
- Osteoarthritis.

- Hemarthrosis.

V. MANAGEMENT

1. Objectives

- Eradicate infection and prevent joint damage.
- Restore joint function and mobility.
- Address underlying risk factors.

2. Critical sign

- Severe joint pain with rapid onset.
- Fever, tachycardia, and hypotension (sepsis).
- Signs of systemic involvement (e.g., multi-organ dysfunction).

3. Mortality risk factor

- Delayed diagnosis and treatment.
- Older age or comorbidities (e.g., diabetes, renal failure).
- Presence of prosthetic joints.
- Polymicrobial or resistant infections.

4. Outpatient management

- Reserved for mild cases in stable patients with confirmed low-virulence pathogens.
- Oral antibiotics with close monitoring of symptoms and lab markers.

A. Eligibility Criteria

- Hemodynamically stable patient with no signs of systemic sepsis.
- Confirmed low-virulence or well-characterized pathogen based on culture results.
- Adequate initial improvement on inpatient intravenous (IV) therapy.
- No prosthetic joint or immunocompromising conditions.
- Reliable patient or caregiver compliance with outpatient therapy and follow-up.

B. Therapeutic Strategy

a. Transition to Oral Antibiotics:

- Criteria for switch: Clinical improvement, afebrile for 48-72 hours, decreasing inflammatory markers (e.g., CRP, ESR).
- Common oral antibiotics:
 - + **For Gram-positive cocci (e.g., *Staphylococcus aureus*):**
 - Linezolid 600 mg PO every 12 hours.
 - Clindamycin 300-450 mg PO every 6 hours.
 - Cephalexin 500 mg PO every 6 hours (if MRSA not suspected).
 - + **For Gram-negative rods:**
 - Ciprofloxacin 500-750 mg PO every 12 hours.
 - Levofloxacin 500-750 mg PO once daily.
 - Amoxicillin-clavulanate 875/125 mg PO every 12 hours (polymicrobial coverage).

+ Total duration: Typically, 2-6 weeks depending on severity and pathogen.

C. Monitoring and Follow-up:

- Schedule frequent outpatient visits (e.g., weekly) during the therapy course.
- Regular blood tests to monitor inflammatory markers and renal/hepatic function.
- Repeat imaging (e.g., ultrasound or MRI) as needed for persistent symptoms.

5. First-line treatment

A-Empiric Antibiotic Therapy:

- should begin immediately after obtaining joint fluid samples for culture and sensitivity testing.

a. Suspected Gram-positive pathogens (most common, e.g., *Staphylococcus aureus*, including MRSA):

- Vancomycin (15-20 mg/kg IV every 8-12 hours).
- Alternative for penicillin-allergic patients: Linezolid.

b. Suspected Gram-negative pathogens (e.g., *Escherichia coli*, *Pseudomonas*):

- Ceftriaxone (2 g IV daily) or cefotaxime (2 g IV every 8 hours).

c. Suspected *Neisseria gonorrhoeae*:

- Ceftriaxone (1 g IV daily) is highly effective.

d. Suspected Polymicrobial infections (e.g., post-trauma or surgery):

- Piperacillin-tazobactam (4.5 g IV every 6-8 hours) or Meropenem for broad-spectrum coverage, including anaerobes.

B. Joint Drainage:

- Arthrocentesis: Repeated aspiration in cases of large effusions or for synovial decompression.
- Surgical drainage in complicated case.

C. Supportive Care:

- Analgesia: Non-steroidal anti-inflammatory drugs (NSAIDs) or acetaminophen.
- Joint immobilization initially to reduce pain and prevent further damage.
- Early mobilization with physical therapy once the acute inflammation subsides.

D. Duration of Treatment

- Total antibiotic duration is typically 2-4 weeks:
 - IV therapy for 1-2 weeks, followed by oral antibiotics based on clinical response.
 - Severe or complicated cases (e.g., prosthetic joint infections): May require 4-6 weeks or longer.

6. Treatment adjustment

A. Adjusting Antibiotics

- Narrow the spectrum of antibiotics once culture and sensitivity results are available to minimize resistance and adverse effects.
- If symptoms persist after 48-72 hours, reassess the treatment plan:
 - Repeat joint aspiration or imaging to detect abscess or joint effusion.
 - Broaden antibiotic coverage to include resistant or atypical pathogens.
- Once clinical improvement is observed, transition to oral antibiotics for outpatient management.

B. Monitoring Therapeutic Response

- Clinical improvement: Pain reduction, joint mobility restoration.
- Laboratory markers: Decreasing levels of CRP and ESR.
- Synovial fluid analysis: Repeat aspiration if clinical improvement stalls.

C. Modifying Duration

- Adjust the duration of therapy based on infection severity (Complicated cases 4-6 weeks or longer) and joint type such as weight bearing joint (knee, Hip).

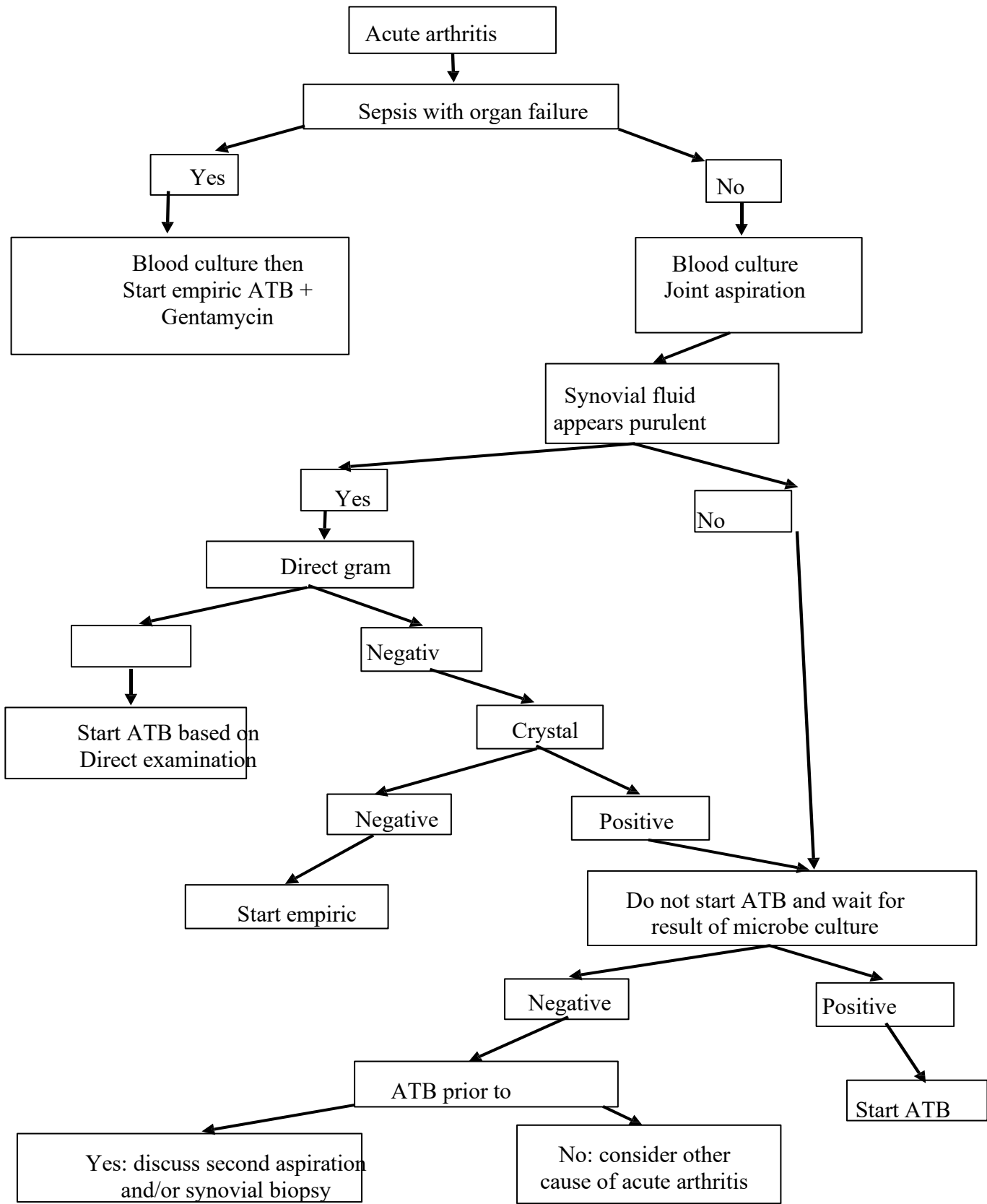
D. Surgical Interventions

- If antibiotics fail to resolve the infection, surgical options like arthroscopic lavage, open debridement, or prosthetic joint removal may be necessary.
- Consider long-term suppressive antibiotics if surgical options are limited (e.g., in patients unfit for surgery).

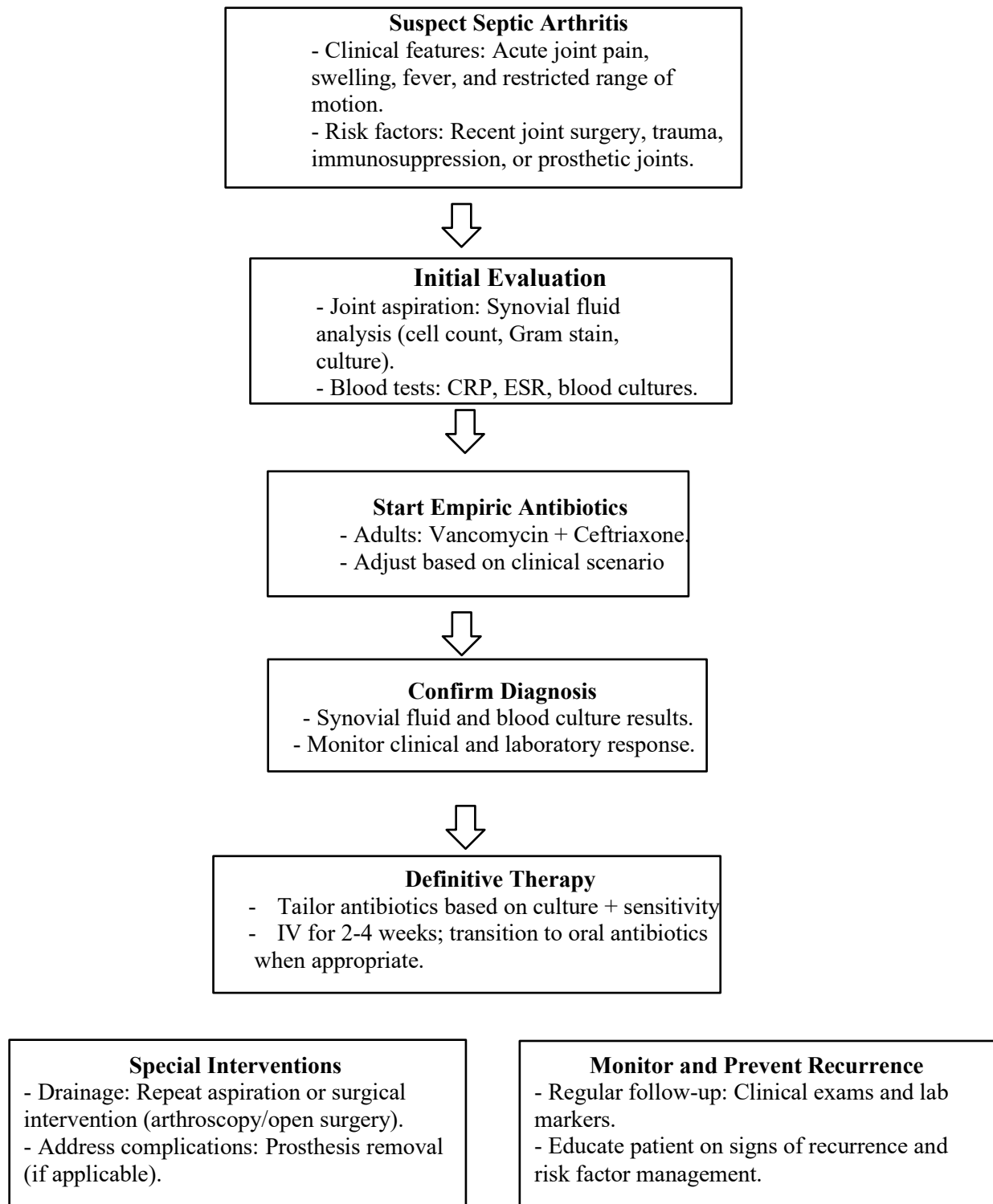
5. Patient Factors

- Renal or hepatic dysfunction: Modify antibiotic doses to avoid toxicity (e.g., vancomycin requires renal dosing).
- Immunocompromised patients: May need longer therapy and broader coverage.

7. Algorithm (1)



Algorithm (2)



VI. PECIAL CASE

1. Prosthetic Joint Infections

- Biofilm formation on prosthetic surfaces makes these infections difficult to treat.
- Surgical intervention: Two-stage revision surgery is standard, involving prosthesis removal, debridement, and antibiotic spacer placement.
- Prolonged antibiotic therapy: Often 6-12 weeks, guided by pathogen sensitivity.

○ Immunocompromised Patients

- Higher risk of atypical pathogens (e.g., fungal infections, Mycobacterium tuberculosis).
- Empiric therapy should include coverage for fungal and resistant bacterial pathogens.
- Longer duration of antibiotic therapy is often necessary.
- Close monitoring for systemic involvement and opportunistic infections.

2. Gonococcal Septic Arthritis

- Population: Sexually active young adults.
- Often associated with migratory polyarthritis, tenosynovitis, and dermatitis.
- Synovial cultures may be negative; diagnosis relies on blood or mucosal swab cultures.

3. Septic Arthritis in Rheumatoid Arthritis

- Differentiating septic arthritis from an RA flare can be difficult.
- Stop or adjust immunosuppressive therapy temporarily.
- Treat with broad-spectrum antibiotics until pathogen-specific results are available.

4. Atypical Pathogens

- Tuberculous arthritis: Associated with chronic, indolent presentations. Requires antitubercular therapy for 6-12 months.
- Fungal arthritis: Occurs in immunosuppressed patients or after joint trauma. Treat with antifungals like amphotericin B or fluconazole.

5. Polyarticular Septic Arthritis

- More common in older adults and immunocompromised patients.
- Often involves Staphylococcus aureus or streptococcal species.
- Management with broad-spectrum antibiotics and multiple joint aspirations or surgeries may be required.

6. Post-Surgical or Trauma-Related Septic Arthritis

- Early surgical debridement.
- Antibiotics based on hospital-acquired pathogen profiles (e.g., MRSA, multidrug-resistant Gram-negative bacteria).

7. Viral or Reactive Arthritis Mimicking Septic Arthritis

- Common viruses: Parvovirus B19, hepatitis B/C, HIV.
- Supportive care for viral arthritis. Avoid unnecessary antibiotics unless bacterial infection is confirmed.

VI. VOLUTION AND PROGNOSIS

- Favorable outcomes with early diagnosis and treatment.
- Delayed treatment can lead to joint destruction, chronic pain, and disability.
- Prognosis depends on patient comorbidities and infection severity.

VII. COMPLICATION

- Joint destruction and loss of function.
- Chronic osteomyelitis.
- Sepsis and multi-organ failure.
- Recurrence or reinfection.
- Growth disturbances in children (if affecting growth plates).

VIII. REVENTION AND RECOMMENDATION

- Prompt treatment: Seek medical care for joint pain and swelling.
- Infection prevention: Proper wound care and aseptic techniques during joint procedures.
- Risk factor management: Control diabetes, avoid immunosuppressive overuse, and address substance abuse.
- Vaccination: Protect against pathogens like *Neisseria meningitidis* and *Haemophilus influenzae*.
- Regular follow-up: Monitor for relapse or complications.

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OSTEOARTHRITIS

UY Putthimundul, HENG Reaksmeay, KAO Pheng Ann, CHHE Sereyvuth, KHUN Rithya

I. DEFINITION

Osteoarthritis (OA) is a joint disease characterized by progressive degradation of cartilage associated with marginal osteophytes, changes in the subchondral bone and minimal transient inflammation of the synovial membrane.

II. EPIDEMIOLOGY

OA is the most common joint disorder. Globally, the incidence is 492 cases per 100,000. There are 528 million people lived with OA in 2019 according to WHO. With the aging population and increasing rates of obesity and injury, the prevalence is expected to continue to rise. The most frequent joints affected is knee, followed by hip and hand. There is female predominance. About 75% of patient are older than 55 years.

III. ETIOLOGY

- OA is a degenerative disorder. No cause is recognized in primary OA.
- However, OA could be secondary to
 - bone defect such as Paget bone disease, aseptic osteonecrosis,
 - joint malformation or traumatism (dislocation)
 - other inflammatory joint disease like gout, pseudogout, septic arthritis or inflammatory rheumatism.
- Risk factors of OA
 - General factors: age, female, hormonal status, overweight, genetic factor.
 - Local factors: joint trauma, instability or dislocation, non-repaired ACL injury, joint surgery history.

IV. PATHOGENESIS

OA is a disease of the entire synovial joint. A synovial joint is mainly composed of four tissues: cartilage, the subchondral bone, synovial membrane, and capsule.

During osteoarthritis, under the effect of mechanical, inflammatory, metabolic stresses, all of these tissues will undergo modifications including:

- degradation of cartilage by proteolytic enzymes capable of destroying collagen and proteoglycan fibers;
- inflammation of the synovial membrane with the recruitment of immune cells under the effect of cartilaginous debris;
- remodeling of the subchondral bone leading to geodes, subchondral sclerosis and the development of bony outgrowths called osteophytes.

OA is characterized by excess mechanical stress on the joints, but also an increase in the production of inflammatory mediators. This low-grade inflammation will induce the increased production of proteolytic enzymes such as metalloproteinases, capable of digesting the extracellular matrix of the cartilage.

V. DIAGNOSIS

Diagnosis of OA is made on clinical sign and imaging.

OA can be diagnosed without imaging in the presence of typical symptoms and signs in the at-

risk group.

5.1- Clinical manifestation

- Non-inflammatory pain
 - No nocturnal pain
 - Morning stiffness less than 30 minutes
 - Improved by rest
 - Worsening with usage
- Joint line tenderness
- Limitation of movement
- Bony swelling
- Joint deformity

5.2- OA flare

There is also episode of flare: worsening of the pain at a certain moment and can be accompanied by the effusion.

5.3- Joint distribution

OA has a predilection for the knees, hips, finger interphalangeal joints, first carpo-metacarpal joints, first MTP joints, and apophyseal (facet) joints of the lower cervical and lower lumbar spine. Other joints may be affected but less common.

The signs and symptoms are variable depending on the joint affected

5.3.1- Knee joint

In the knee joint, there are three compartments

- Patellofemoral
- Medial femorotibial
- Lateral femorotibial

Knee OA can affect one or all compartments. In knee pain, it is essential to rule out hip pain.

5.3.2- Hip joint

In hip OA, coxo-femoral joint is affected. It manifests by pain in the groins while walking. Early physical examination finds limitation of internal rotation.

5.3.3- Hand

There are two type of hand OA: thumb OA and OA of distal and proximal interphalangeal joint. DIP joint OA is often accompanied by Heberden and Bouchard nodes.

5.3.4- Generalized OA

It is characterized by slow accumulation of multiple joint involvement over several years. The American College of Rheumatology and the European Alliance of Associations for Rheumatology suggests that generalized OA is present if there is OA at either the spinal or hand joints, respectively, and in at least two other joint regions.

5.4- Laboratory and imaging 5.4.1-

Laboratory investigation

In OA, ESR and CRP, or another inflammatory marker are normal. They are useless in making diagnosis but can be useful in excluding other diagnoses, particularly inflammatory rheumatism diseases.

Rheumatoid factor and anti-cyclic citrunillated peptide (anti-CCP) antibodies may be checked if rheumatoid arthritis cannot be excluded with clinical signs.

In case of effusion, arthrocentesis is important. Synovial fluid is usually noninflammatory with less than 2000 white blood cell/mm³ and polynuclear neutrophil is less than 50%.

5.4.2- Imaging

X-ray of affected joint is first line imaging study. Ideally, it is done bilaterally.

Characteristic radiographic features of OA are

- Marginal osteophytes
- Localized joint space narrowing
- Subchondral sclerosis
- Cysts

Nevertheless, there is no correlation between clinical and radiographic signs. Radiographic OA is a common incidental asymptomatic finding in older people.

Ultrasonography can detect synovial inflammation, effusion and osteophytosis.

There is no role of IRM in diagnosis of OA but it can be used to find alternative diagnosis like meniscal tear.

VI. Differential diagnosis

The differential diagnosis for OA depends largely on the location of the affected site as well as the presence of absence of additional systemic symptoms.

- Rheumatoid arthritis
- Psoriatic arthritis
- Gout or pseudogout
- Hemochromatosis
- Infectious arthritis

VII. ANAGEMENT

1- Objectives

- To relieve symptoms and improve joint function
- To slow down the anatomical progression of the disease.

2- Critical signs

- Repeated OA flare with effusion.
- Pain and joint stiffness lead to daily functional impairment.

3- Mortality risk factor

OA-associated mortality is largely driven by cardiovascular death. It may be explained by

- Reduction of levels of physical activity
- Adverse effects of medications used to treat symptomatic OA, particularly nonsteroidal anti-inflammatory drugs (NSAIDs).

4- Outpatient management

- Weight loss
- Maintaining physical activity
- Joint-friendly measures: avoid overuse of the joints

5- First line treatment

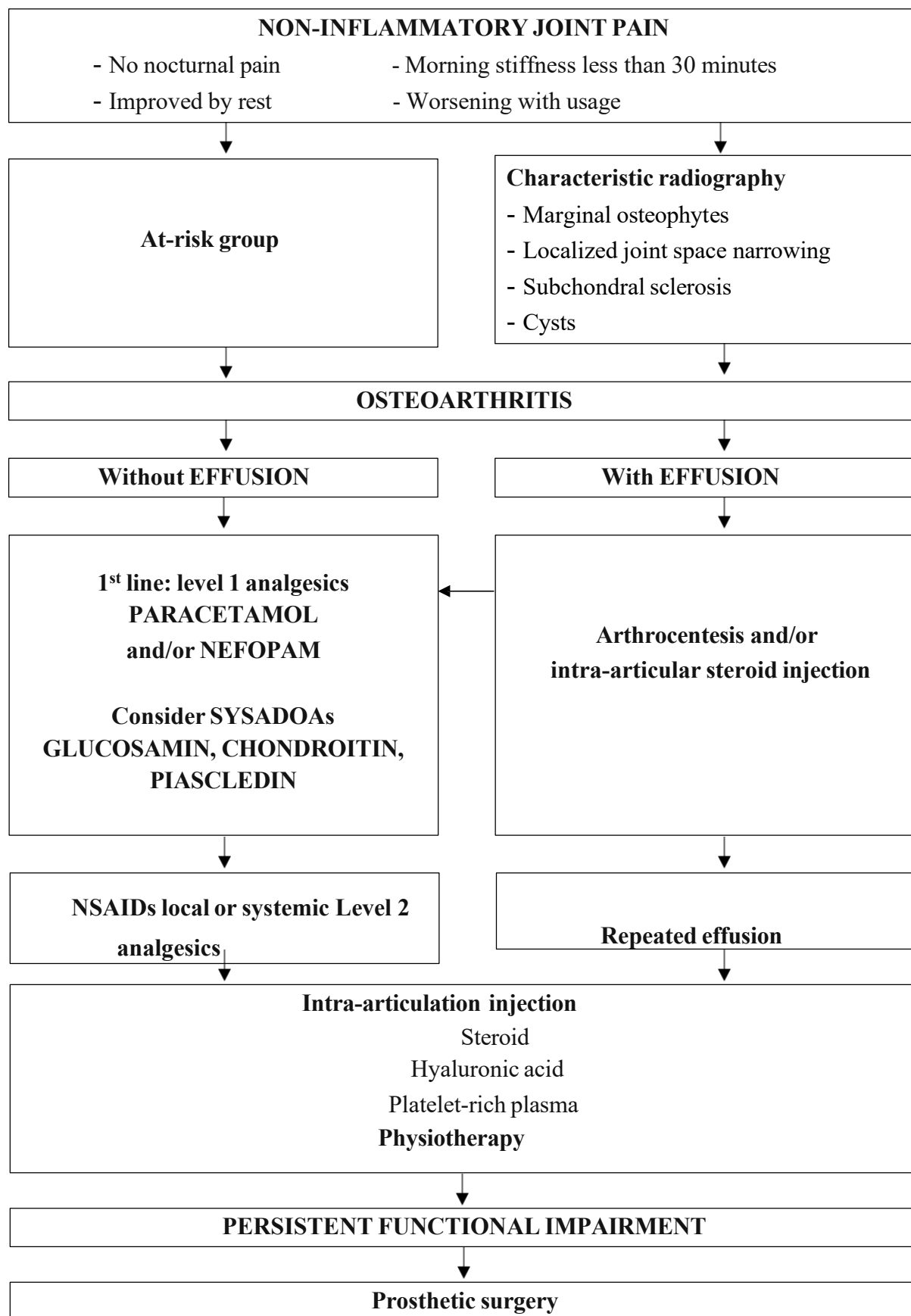
- Treat only symptomatic OA.
- No benefit in treating radiographic OA without clinical manifestation.

- Pain management
 - Level 1 analgesics, PARACETAMOL and/or NEFOPAM
 - Level 2 or 3 analgesics should be avoided
 - Local or systemic NSAIDs occasionally in case of resistance to analgesic or during more painful or flare periods.
 - Small period steroid per os can be used especially in case of OA flare with effusion. Intra-articular steroid injection is an alternative choice.
 - Other intra-articular injection such as hyaluronic acid or platelet-rich plasma can be considered.
- Chondroprotective agent or symptomatic slow-acting anti-OA drugs (SYSADAOs), GLUCOSAMIN, CHONDROITIN, PIASCLEDIN, are conditionally prescribed because there is only small benefit proven. The analgesic effect acts after 1 month and lasts for 1 to 3 months.
- Physiotherapy
- Joint replacement

6- Treatment adjustment

OA is elder disease. Past medical histories need to be considered, especially hypertension, diabetes mellitus, chronic kidney disease, atherosclerosis and heart failure. The usage of NSAIDs is cautiously. The least harmful NSAIDs like NAPROXEN is recommended.

7- Algorithm



VIII. SPECIAL CASE

8.1- Rapidly progressive osteoarthritis

- Typically occurs in older females over 70 years old
- Subacute onset of pain, typically progressing from mild to severe over a period of several months.
- Inflammatory markers are normal.
- Joint replacement surgery is often required within months to a few years of symptom onset.

8.2- Osteoarthritis with calcium pyrophosphate deposition

- Simultaneous presence of CPPD with OA modifies the symptoms
- Longer early morning stiffness and more signs of synovitis.
- Marked pain, swelling, and tenderness lasting from a few days to one to two weeks; intermittent; or low-grade and persistent.

IX- EVOLUTION AND PROGNOSIS

- Painful periods alternating with periods of calm.
- No role of control radiographs except in case of suspicion of rapidly progressive OA.
- The presence of recurrent or chronic joint effusion is a poor prognostic factor for the progression.

X- COMPLICATION

- Risk of cardiovascular events favored by decline of physical activities and side effects of NSAIDs.
- Daily functioning impairment

XI- PATIENT EDUCATION AND PREVENTION

- Weight loss
- Physical activities

XII- REFERENCES

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GOUT

HENG Raksmeay, UY Putthimundul, KAO Phengann, CHHE Sereyvuth, KHUN Rithya

I. DEFINITION

Gout (monosodium urate crystal deposition disease) is a chronic arthritis caused by too much uric acid build-up in your body, which is reflected in the blood by hyperuricemia, with serum or plasma urate concentrations exceeding 6mg/dl (360micromol/l).

II. EPIDEMIOLOGY

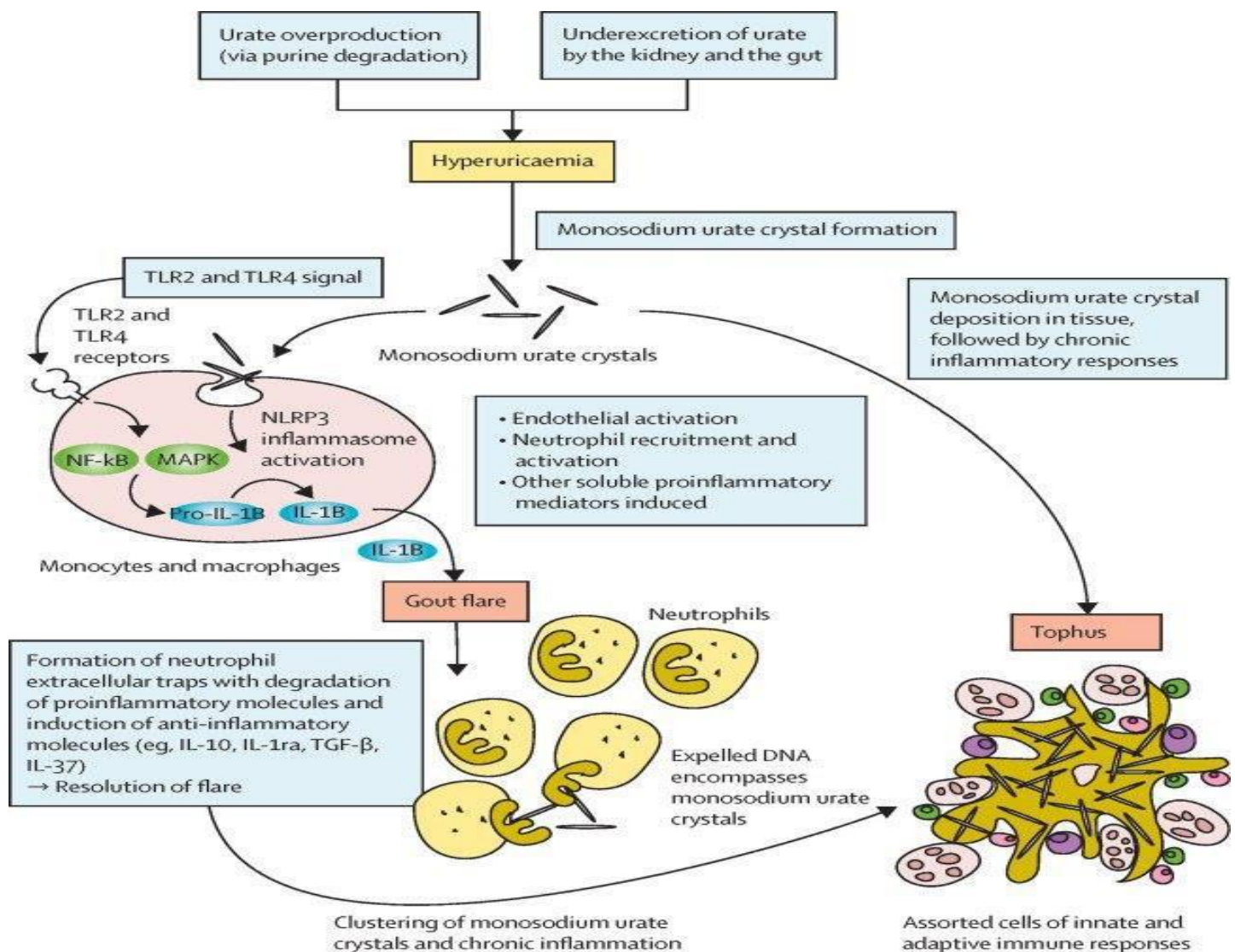
The worldwide burden of gout has grown in recent decades, although reports suggest a plateauing of incidence in some geographic regions.

In general, the prevalence of gout is 1 to 4%. The prevalence of gout in men (3 to 6%) is 2 to 6-fold higher than in women (1 to 2%). In Women, it is only observed after menopause.

III. ETIOLOGY

The etiology of gout is usually multifactorial, including genetic risk factors, medical comorbidities (cardiovascular, metabolic, renal disease) and dietary factors.

IV. PHYSIOPATHOLOGY



V. DIAGNOSIS

Symptom:

Typical gout flare:

- Severe pain redness, warmth, swelling and disability maximal severity of the flare is usually reached within 12 to 24 hours. Complete resolution of the earliest flares almost always occurs within a few days to several weeks.
- Onset more frequent overnight and in early morning (between midnight and 8am)
- Lower extremity involvement (at least 80% of initial flares involve a single joint, first metatarsophalangeal joint, known as podagra).

Polyarticular gout flares < 20%

History taking:

- Anterior gout flare
- Factors provoking gout flares: trauma, surgery, dehydration, alcohol consumption ...

Physical examination:

- Signs of inflammation beyond the joint (this feature may give the impression of dactylitis or cellulitis or may actually be due arthritis in several contiguous joints or tenosynovitis)
- Tophaceous gout



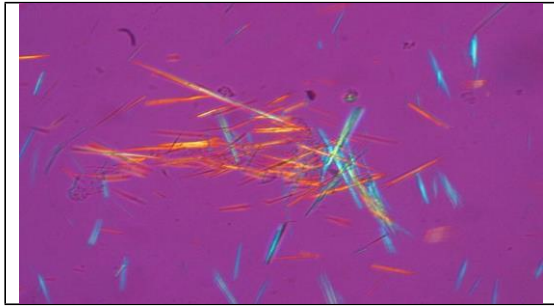
1. Investigation

1.1 Blood tests

- Urate level may be high, normal or low
- Neutrophilic leukocytosis and/or elevation of the erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP) are common in gout flares

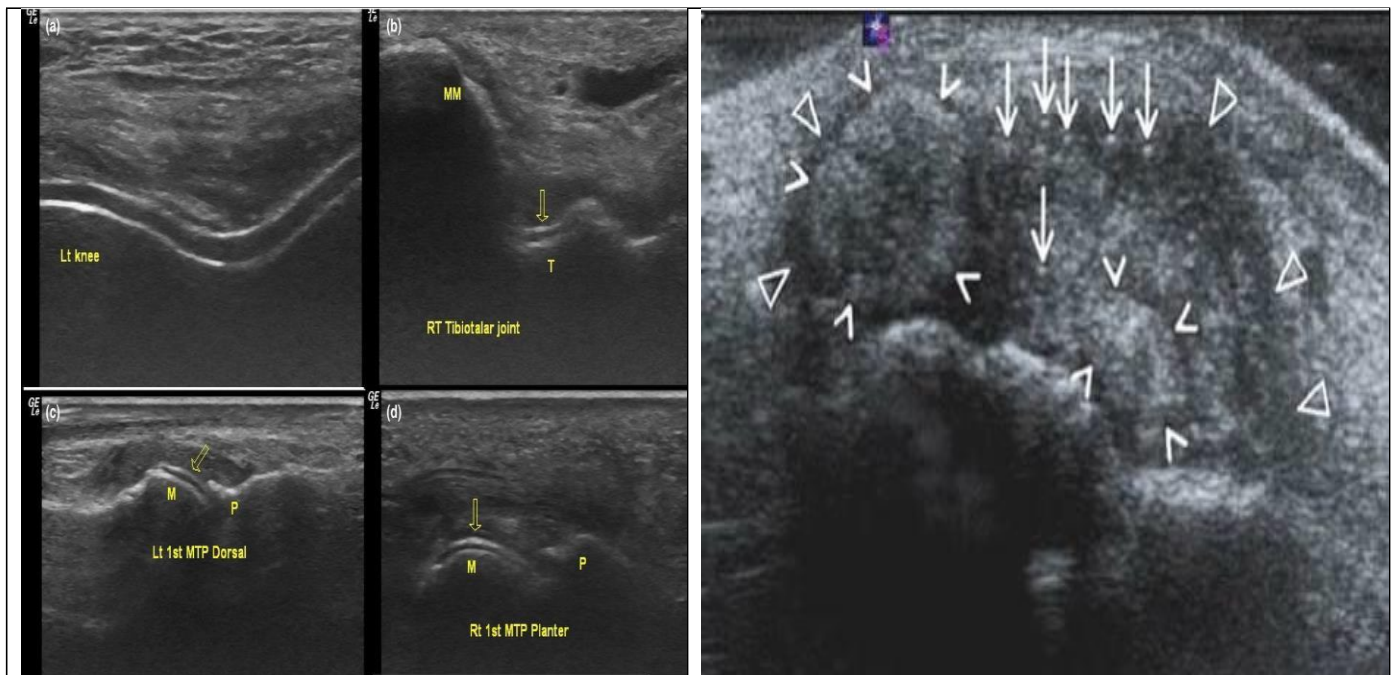
1.2 Synovial fluid analysis

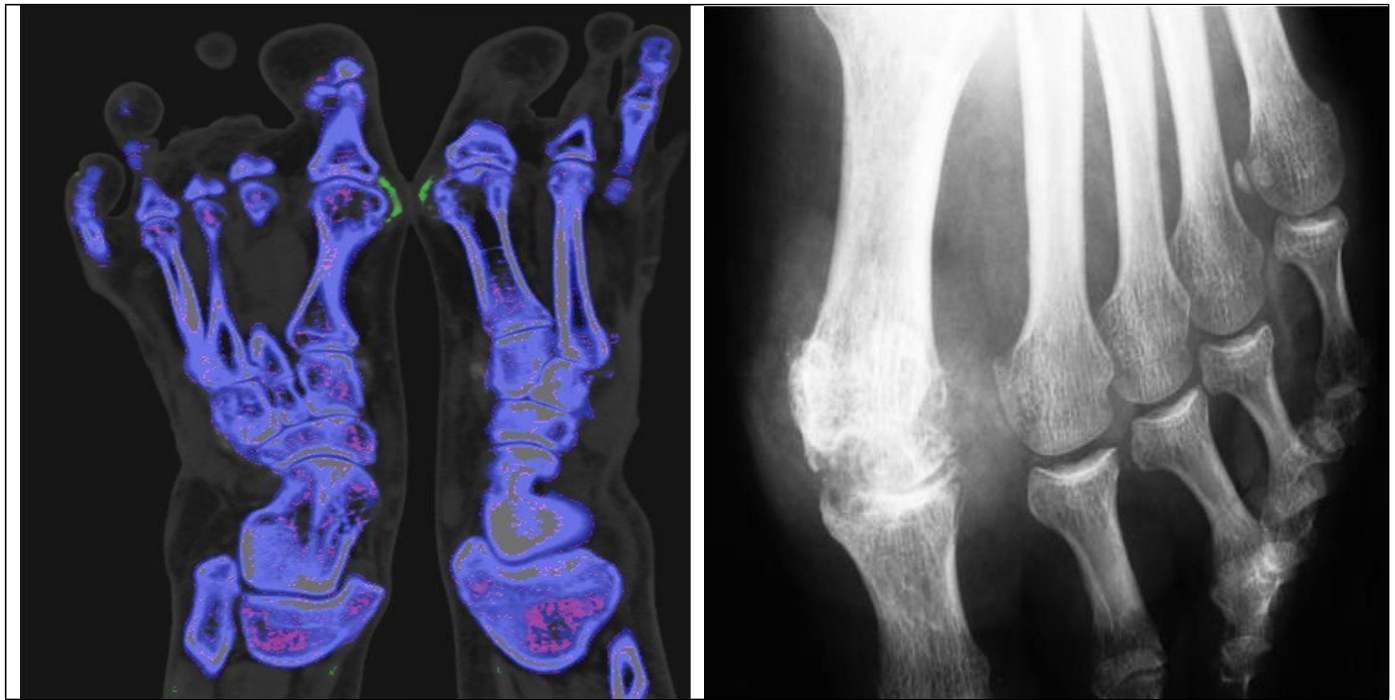
- The synovial fluid is inflammatory with white blood cell counts in joint fluid that usually range between 10.000 to 100.000 with neutrophil predominance.
- The presence of mono sodium urate (MSU) crystals in the synovial fluid obtained from joints or bursas visualized by direct examination of a sample of fluid using compensated polarized light microscopy



1.3 Imaging

- **Ultrasonography**: hyperechoic linear density (double contour sign overlying the surface of joint cartilage) or tophaceous appearing deposits in joints or tendon which are represented by an ovoid stippled signal (hyperechoic cloudy area surrounded by a hypoechoic border).
- **Dual energy computed tomography (DECT)**: identify urate deposits in articular and periarticular location
- **Radiography and magnetic resonance imaging**: sub cortical bone cysts apparent on plan radiography or MRI examination can be suggestive of gouty tophi or erosion.





2015 ACR-EULAR Gout Classification Criteria

Criterion	Characteristics	Points
Pattern of involvement	Ankle/ foot	1
	MTP-1	2
Symptoms:		
- Erythema of joint	1/3 symptoms	1
- Affected joint very sensitive to touch	2/3 symptoms	2
- Movement very limited	All 3 symptoms	3
Criteria for typical episode:		
- Pain develops within 24 hours	1 typical episode	1
- Resolves after ≤ 14 days	Multiple typical episodes	2
- Complaint-free intervals		
Evidence of tophus	Present	4
Laboratory tests:		
- Serum uric acid (preferably without treatment, 4 weeks before/after an episode or highest value during episode)	< 4mg/dl	-4
	6 to 8mg/dl	2
	>8 to < 10mg/dl	3
	≥ 10 mg/dl	4
- Join fluid analysis following puncture (if performed)	No urate crystals	-2
Imaging:		
- Urate deposits on ultrasound or DECT	Present	4
- Erosion characteristic of gout on conventional X-ray	Present	4

DECT: Dual-energy computed tomography; MTP-1: Metatarsophalangeal joint 1

+ If swelling score is 8 or more out of a maximum of 23 points, arthritis can be classified as gout with a sensitivity of 92% and a specificity of 89%. Evidence of uric acid crystals on puncture is a sufficient criterion.

VI. DIFFERENTIAL DIAGNOSIS

Gout flare:

- Septic arthritis
- Calcium pyrophosphate crystal deposition disease (pseudo gout)
- Cellulitis
- Basic calcium phosphate crystal disease (apatite

crystals) Tophaceous gout:

- Rheumatoid arthritis
- Dactylitis

VII. MANAGEMENT

1. Objectives

- Prevention recurrence of gout flares
- Reduce uremia $\leq 60\text{mg/l}$ ($360\mu\text{mol/L}$) or 50mg/L in cases of tophaceous gout
- Treat comorbidities and eradicate accessible risk factors

2. Critical sign

Bacterial superinfection

3. Mortality risk factor: infection and disabled.

4. Outpatient management

- Individualized lifestyle advice
- Pain killer

5. First -line treatment

Non pharmacologic treatment:

- Education about the disease
- Individualized lifestyle advice
- Screening for comorbidities and current medication

Pharmacologic treatment:

The management of gout flares

- Colchicine: 1mg followed 1hour later by 0.5mg than 0.5mg twice or three daily for patients with normal renal function. Colchicine 0.5-1mg per day, or less if renal failure. Colchicine-induced neuromyopathy in patients treated with strong CYP3A4 or P-glycoprotein inhibitors (cyclosporine, clarithromycin, ketonazole, ritonavir).
- NSAIDs (Classic or coxibs + PPI if appropriate): NSAIDs should not be used in patients with moderate or more severe renal impairment or significant heart failure.
- Prednisolone: 30-35mg per day for 3 to 5 days.
- Intraarticular injections of corticosteroid

→ The 2020 American College of Rheumatology (ACR) guidelines support the use of colchicine prophylaxis 0.5-1mg per day for at least three to six months.

The Urate-Lowering drug

- Allopurinol: we usually initiate allopurinol treatment at 100mg daily in patients with weight - adjusted creatinine clearance > 60ml/minute and titrate the dose every two to four weeks to the minimum dose required to achieve and maintain the goal range of serum urate. An initial dose of 100mg or less of allopurinol is associated with reduced risks of developing severe cutaneous adverse reactions and gout flares that are otherwise common during initiation of urate-lowering therapy.
- Febuxostat: received approval from the FDA for treatment of hyperuricemia in gout patients at daily doses of 40 and 80mg. The American manufacturers recommended that the starting dose is 40mg once daily. Titration to a daily dose of 80mg is suggested for patients whose serum urate level does not fall to <6mg/dL after two weeks of treatment on the lower dose. In Europe, higher doses have also received regulatory approval up to 120mg/day.

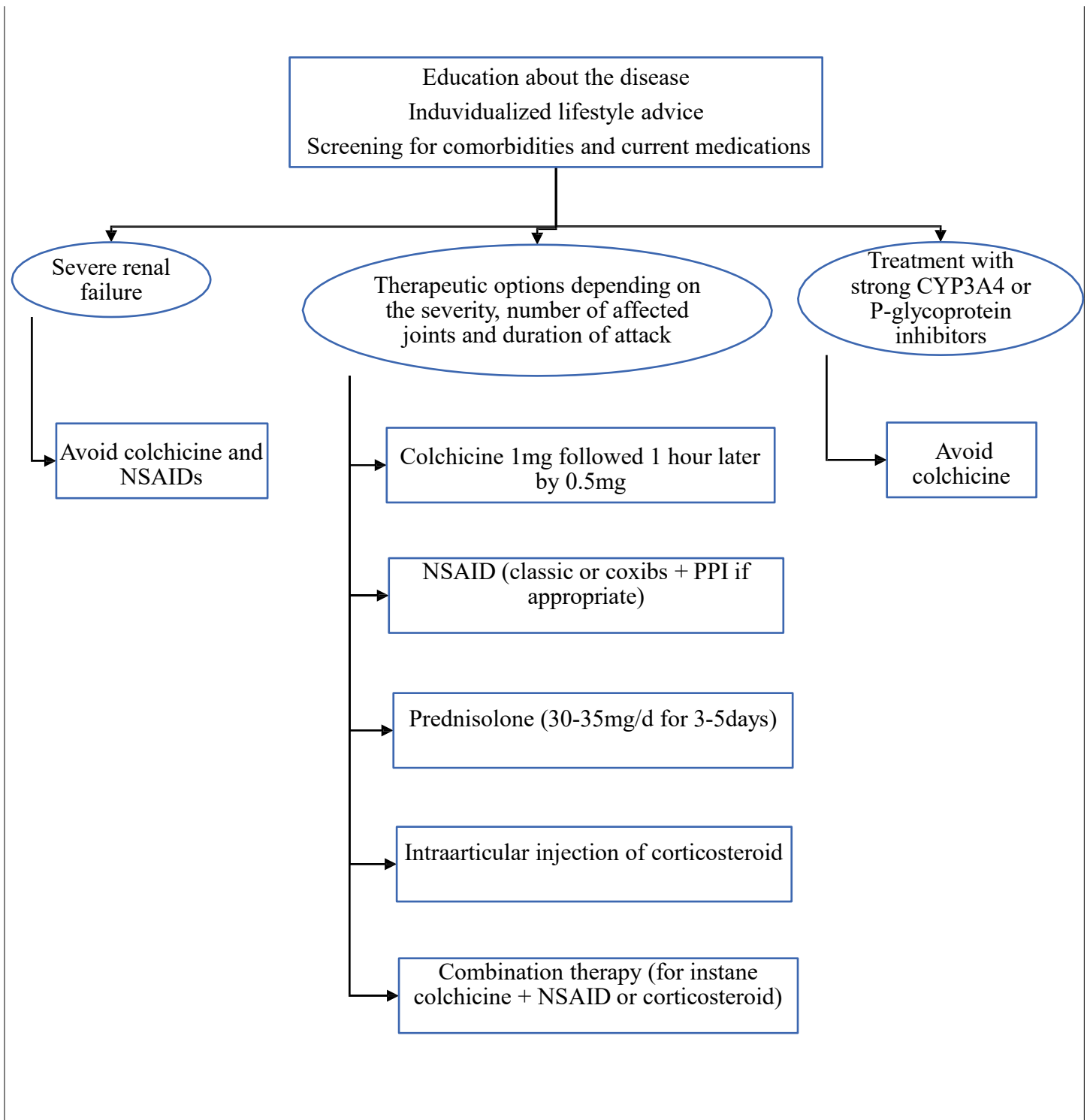
6. Treatment adjustment

- Individualized lifestyle advice
- Screening for risk factors and treating comorbidities (cardiovascular, renal, metabolic syndrome)

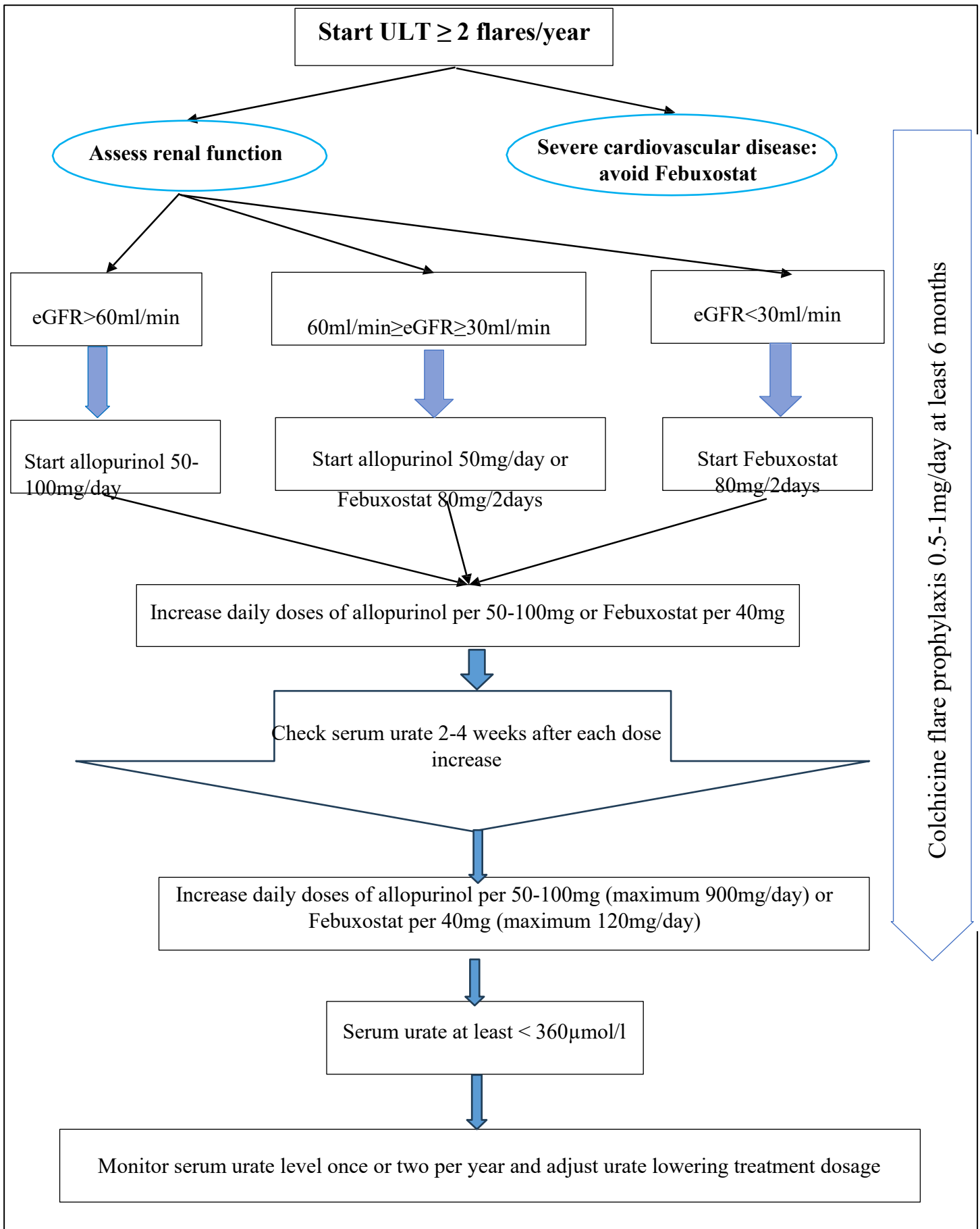
7. Algorithm

2020 recommendation from the French Society of Rheumatology for the management of gout flares

Treat as early as possible



The Urate-Lowering drug



VIII. SPECIAL CASES

- Asymptomatic hyperuricemia: isolated hyperuricemia is not an indication for pharmacological treatment but for dietary and medication adjustment measures.
- Acute hyperuricemia: treatment and prophylaxis of acute hyperuricemia are necessary to prevent acute renal failure in patient suffering from hematological malignancies with high tumor mass and at risk of rapid tumor reduction or lysis upon treatment initiation of chemotherapy.
- Indications for surgery: surgery for gout is appropriate for patients with complications of tophaceous disease, including infection, nerve compression, joint deformity and intractable pain. Some patients also desire surgery for cosmetic reasons.

IX. EVOLUTION AND PROGNOSIS

- Recurrent flares
- Chronic gouty arthritis
- Tophaceous gout

X. COMPLICATIONS

- Urolithiasis
- Osteoarthritis
- Poor kidney function

XI. PREVENTION AND RECOMMENDATION

- A progressive weight loss in cases of obesity or overweight
- Reduction of alcoholic beverages
- Ban on spirits and beer (very rich in purines even when alcohol-free) and stopping sugary sodas and reducing fruit juices rich in fructose.
- Moderation of meat and seafood consumption
- Gout risk factors and comorbidities (cardiovascular, renal, metabolic syndrome) should be screened for and treated systematically.

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ADULT ONSET STILL DISEASE

Dr. CHHE Sereyvuth, Dr. KAO Pheng An, Dr. HENG Raksmei, Dr. UY Puthimundu, Dr. KHUN Rithya

I. DEFINITION

A rare, systemic inflammatory disorder characterized by high spiking fevers, a salmon-pink rash, arthritis, and elevated levels of inflammatory markers. It is classified as an autoinflammatory disease, meaning that it is caused by dysregulation of the immune system, leading to inflammation without an infectious cause.

II. EPIDEMIOLOGY

- Incidence: estimated incidence of 0.16 to 0.4 cases per 100,000 people per year globally.
- Prevalence: 1 to 34 cases per million people.
- Age of Onset: Majority of cases occurring between the ages of 16 and 35 years. A second peak in incidence is sometimes seen in adults aged 40–50 years.
- Sex Ratio: female-to-male ratios ranging from 1.5:1 to 3:1.
- Geographic Variation: does not seem to be restricted to any specific ethnic or geographic group.
- Regional Differences: slightly higher incidences in Asian populations, particularly in Japan and Korea.

III. ETIOLOGY

A. Genetic Factors:

- Associations with specific HLA alleles (e.g., HLA-DRB1, HLA-B17, HLA-B18).

B. Immune Dysregulation:

- Overproduction of pro-inflammatory cytokines (e.g., IL-1, IL-6, IL-18).
- Dysregulation of the innate immune system.

C. Environmental Factors:

- Viral infections: Epstein-Barr virus (EBV), Cytomegalovirus (CMV), Parvovirus B19, Hepatitis viruses (HBV, HCV)
- Bacterial infections: *Yersinia enterocolitica*
- Stress (physical or emotional) may contribute to disease onset.

D. Hormonal Factors:

- Potential influence of hormones, given the higher incidence in women.

IV. PATHOPHYSIOLOGY

1. Cytokine Dysregulation:

- Overproduction of pro-inflammatory cytokines: IL-1, IL-6, IL-18.

2. Immune System Activation:

- Dysregulation of the innate immune system, leading to persistent inflammation, Hyperactivation of macrophages and monocytes.

3. Autoinflammatory Mechanism:

- Characterized by recurrent episodes of systemic inflammation without an infectious trigger.

4. Genetic Predisposition:

- Certain HLA allele associations (e.g., HLA-DRB1, HLA-B17, HLA-B18) may increase susceptibility to the disease.

5. Systemic Inflammatory Response:

- Activation of the acute phase response leads to symptoms such as: Fever, Rash, Arthritis

6. Vascular Changes:

- Inflammatory cytokines can lead to endothelial dysfunction and increased vascular permeability, contributing to swelling and erythema.

7. Bone and Joint Involvement:

- Inflammation can lead to joint destruction and damage over time if not effectively managed, resulting in chronic arthritis.

8. Hemodynamic Changes:

- Patients may experience hypotension and other circulatory changes during flares due to the systemic inflammatory response.

V. DIAGNOSIS

1. Clinical Presentation

- Fever: Spiking fever $\geq 39^{\circ}\text{C}$, typically occurring in the evening and returning to normal between spikes.
- Evanescent Salmon-Pink Rash: A fleeting maculopapular rash, often appearing during febrile episodes, usually on the trunk and extremities.
- Arthritis or Arthralgia: Pain and swelling in multiple joints, typically involving the wrists, knees, and ankles.
- Sore Throat: Common, especially during febrile episodes.
- Lymphadenopathy and Hepatosplenomegaly: Enlarged lymph nodes and/or spleen.
- Systemic Symptoms: Fatigue, weight loss, pleuritis, pericarditis.

2. Laboratory Findings

- Elevated Inflammatory Markers: C-reactive protein (CRP), Erythrocyte sedimentation rate (ESR).
- Ferritin: Significantly elevated levels (often $>1000\text{ ng/mL}$).
- Leukocytosis: Elevated white blood cell count (usually $>10,000/\text{mm}^3$), often with neutrophilia.
- Liver Enzymes: Mild elevation of aspartate aminotransferase (AST) and alanine aminotransferase (ALT).
- Anemia: Normocytic, normochromic anemia may be present.
- Negative Autoimmune Tests: Rheumatoid factor (RF) and antinuclear antibodies (ANA) are usually negative.

3. Imaging Studies

- X-rays or MRI of joints: May show synovitis, effusion, or erosive changes.
- Ultrasound or CT scans: To assess lymphadenopathy, splenomegaly, or pleural/pericardial effusion.

effusion.

4. Diagnostic criteria

Yamaguchi Criteria (requires ≥ 5 criteria, with ≥ 2 being major criteria)
Major Criteria: - Fever $\geq 39^{\circ}\text{C}$ lasting ≥ 1 week. - Arthralgia or arthritis lasting ≥ 2 weeks. - Evanescent, non-pruritic, salmon-pink maculopapular rash. - Leukocytosis ($\geq 10,000/\text{mm}^3$) with neutrophils $\geq 80\%$. Minor Criteria: - Sore throat. - Lymphadenopathy or splenomegaly. - Abnormal liver function tests. - Negative RF and ANA.
Exclusion Criteria: - Infection, malignancy, and other autoimmune diseases must be excluded.

VI. DIFFERENTIAL DIAGNOSIS

1. Infectious Diseases:
 - Viral Infections: Epstein-Barr virus (EBV) infection, Cytomegalovirus (CMV) infection, HIV/AIDS., Hepatitis B or C.
 - Bacterial Infections: Endocarditis, Tuberculosis.
 - Parasitic Infections: Malaria, Leishmaniasis.
2. Rheumatic and Autoimmune Diseases:
 - Systemic Lupus Erythematosus (SLE), Rheumatoid Arthritis (RA), Systemic Vasculitis
3. Malignancies:
 - Lymphoma, Leukemia, Other Solid Tumors.
4. Other Autoinflammatory Disorders:
 - Hemophagocytic Lymphohistiocytosis (HLH), Macrophage Activation Syndrome (MAS).
5. Granulomatous Diseases:
 - Sarcoidosis.
6. Other Causes of Fever of Unknown Origin (FUO):
 - Drug-induced Fever, Post-infectious Syndromes.
7. Miscellaneous Conditions:
 - Kikuchi-Fujimoto Disease, Adult Rheumatic Fever.

VII. MANAGEMENT

1. Objectives

- Control Inflammation.
- Symptom Relief.
- Prevent and Manage Complications such as Prevent Joint Damage, Avoid Organ Involvement, prevent Long-term Disability.
- Achieve Disease Remission: Induce and Maintain Remission, adjust therapy to maintain remission and minimize side effects.
- Minimize Side Effects of Treatment.
- Improve Quality of Life.
- Monitor Disease Progression: Regularly assess disease activity using clinical evaluations and laboratory markers (e.g., ferritin, CRP, ESR) and monitor for signs of disease flare-ups and organ involvement.

2. Critical sign:

- Fever, salmon-pink rash, arthritis, sore throat, and elevated ferritin are critical for diagnosing AOSD.
- These signs, along with elevated inflammatory markers, form the basis for clinical diagnosis and help differentiate AOSD from other febrile and systemic diseases.

3. Mortality risk factor:

- Macrophage Activation Syndrome (MAS).
- Cardiac involvement (e.g., pericarditis, cardiac tamponade, myocarditis).
- Pulmonary complications (e.g., pleural effusion, ARDS).
- Acute liver failure.
- Acute kidney injury (AKI).
- Disseminated intravascular coagulation (DIC).

- Severe infections and sepsis.
- Chronic disease with multi-organ involvement.
- Treatment-resistant disease.
- Advanced age and comorbidities.

4. Outpatient management:

A. Pharmacologic Treatment

- Non-steroidal Anti-inflammatory Drugs (NSAIDs), Corticosteroids.
- Disease-modifying Antirheumatic Drugs (DMARDs) such as methotrexate, Hydroxychloroquine or azathioprine.
- Bisphosphonates.

B. Monitoring and Follow-up

- Regular Assessment: disease activity, treatment efficacy, and adjust medications as needed.
- Monitor laboratory markers like CRP, ESR, and ferritin to gauge inflammation levels.
- Bone Density Monitoring: are recommended to assess for osteoporosis.

C. Preventing and Managing Complications

- Osteoporosis Prevention: bisphosphonates, calcium, and vitamin D.
- Infection Prevention: monitored for signs of infection, and vaccinations.

D. Referral to Specialists

- Rheumatology: For ongoing disease management, particularly in complex or refractory cases.
- Pulmonology/Cardiology: For patients with severe organ involvement (e.g., pleuritis, pericarditis).

5. First-line treatment

A. Non-steroidal Anti-inflammatory Drugs (NSAIDs)

- Use: For mild to moderate symptoms, especially to manage fever, joint pain, and inflammation.
- Examples:
 - Ibuprofen: 400–800 mg 3–4 times a day.
 - Naproxen: 250–500 mg twice daily.

B. Corticosteroids (Prednisone)

- Use: Particularly effective in controlling fever, rash, and arthritis, as well as systemic involvement (e.g., pleuritis, pericarditis).
- Prednisone: Initial dose: 0.5–1 mg/kg/day (usually 40–60 mg daily).

C. Disease-modifying Antirheumatic Drugs (DMARDs)

a. Methotrexate:

- Often introduced early, especially if there is significant joint involvement or if patients require long-term corticosteroid therapy to reduce steroid dependence.
- Initial dose: 10–15 mg once weekly, increased gradually to 20–25 mg/week as needed.

b. Azathioprine:

- Use: Azathioprine is an immunosuppressant often used as a steroid-sparing agent in AOSD to control systemic inflammation and prevent reliance on long-term corticosteroid therapy.
- Initial dose: 1–2.5 mg/kg/day orally. Typical maintenance dose: 100–150 mg/day, depending on the patient's weight and response.
- Monitoring: Regular blood tests are required to monitor for potential side effects like bone marrow suppression and liver toxicity.

c. Hydroxychloroquine:

- Use: Hydroxychloroquine is often used for its anti-inflammatory and immunomodulatory properties, particularly in managing joint symptoms and skin manifestations in AOSD.
- Typical dose: 200–400 mg/day orally, taken in one or two divided doses.
- Monitoring: Regular eye exams (every 6–12 months) are recommended to monitor for potential retinal toxicity, especially with long-term use.

D. Antiosteoporosis

a. Alendronate

- Dose: 70 mg orally once weekly.
- Use: Alendronate is commonly used to increase bone density and prevent fractures in patients at risk of osteoporosis, including those with AOSD on long-term corticosteroid therapy.

b. Zoledronic Acid (Intravenous)

- Dose: 5 mg IV infusion once yearly.
- Use: Zoledronic acid is often used for patients who cannot tolerate oral bisphosphonates or for those with severe osteoporosis.

6. Treatment adjustment

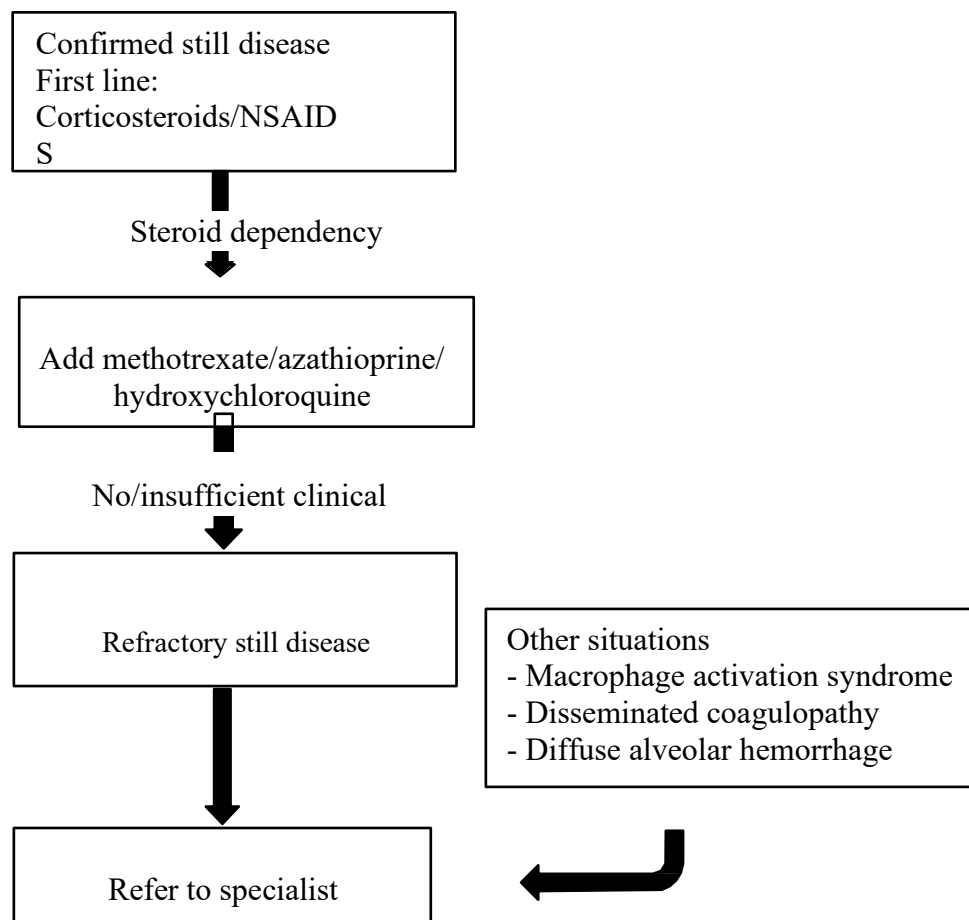
A. Tapering or Adjusting Corticosteroids

- Tapering Dosage: If the disease is well-controlled, corticosteroids (e.g., prednisone) should be gradually tapered to minimize side effects.
- Re-escalation During Flares: Increase the corticosteroid dose temporarily if there is a disease flare, then taper as symptoms subside.
- Switch to Steroid-sparing Agents: If long-term corticosteroids are required, transition to immunosuppressants or biologics to reduce steroid dependency.

B. Adding or Adjusting DMARDs

- Methotrexate: Commonly used if there is significant joint involvement or need for steroid-sparing effect. Increase the dose gradually to maintain control.
- Azathioprine or Hydroxychloroquine: Consider these alternatives or add them if methotrexate is insufficient or poorly tolerated.

7. Algorithm



VIII. SPECIAL CASE

A. Macrophage Activation Syndrome (MAS)

- Description: MAS is a life-threatening complication of AOSD characterized by excessive immune activation, leading to multi-organ failure.
- Signs and Symptoms: High fever, cytopenia (low blood cell counts), liver dysfunction, extremely high ferritin levels, coagulopathy, and neurological symptoms.
- Management: Requires immediate intervention with high-dose corticosteroids, immunosuppressive drugs. Close monitoring in a hospital setting is crucial.

B. AOSD with Severe Cardiac Involvement

- Description: Cardiac manifestations such as pericarditis, myocarditis, or cardiac tamponade can occur in AOSD and may be life-threatening.
- Signs and Symptoms: Chest pain, dyspnea, muffled heart sounds, hypotension, jugular venous distention (JVD), and arrhythmias.
- Management: Requires high-dose corticosteroids and, in severe cases, biologics to control inflammation. Pericardiocentesis may be needed if there is cardiac tamponade. Cardiology consultation is essential.

C. AOSD in Pregnancy

- Description: Managing AOSD during pregnancy is challenging due to the limited treatment options that are safe for the fetus.

Management: Low-dose corticosteroids are preferred. NSAIDs and methotrexate are generally avoided due to fetal risks.

IX. EVOLUTION AND PROGNOSIS

1. Disease Evolution

- AOSD generally has three clinical patterns, and each has implications for long-term management and prognosis:

A. Self-limited (Monocyclic) Course:

- Characterized by a single episode that eventually resolves, usually within one year.
- After treatment, patients achieve remission without long-term complications.
- Prognosis: Good, with minimal risk of recurrence or chronic disability.

B. Intermittent (Polycyclic) Course:

- Recurrent episodes or flare-ups interspersed with periods of remission.
- Patients may experience months or years of symptom-free intervals, followed by sudden flare-ups.
- Prognosis: Generally good, but frequent recurrences may require long-term treatment, especially during flares.

Chronic (Persistent) Course:

- Continuous or prolonged symptoms, often associated with chronic arthritis.
- Higher risk of complications, including joint damage and systemic involvement.
- Prognosis: More challenging, with a higher risk of long-term disability, especially if joint involvement leads to deformities or functional limitations.

- Prognosis
- Remission and Quality of Life: Many patients with AOSD achieve remission with appropriate treatment, especially those with self-limited or intermittent disease courses.
- Chronic Disability: In patients with a chronic, persistent course, the risk of long-term disability is higher due to joint destruction and systemic complications.
- Mortality Risk: While AOSD is generally not fatal, complications like MAS, cardiac tamponade, severe infections, and multi-organ failure can increase mortality risk in severe cases.

X. COMPLICATION

- Macrophage Activation Syndrome (MAS).
- Chronic Arthritis and Joint Damage.
- Cardiac Complications: Pericarditis and Myocarditis, Valvular Disease.
- Pulmonary Complications: Pleuritis and Pleural Effusion, Acute Respiratory Distress Syndrome.
- Liver Involvement and Hepatitis.
- Renal Complications: Acute Kidney Injury (AKI), chronic kidney disease (CKD).
- Disseminated Intravascular Coagulation (DIC).
- Infections.
- Osteoporosis.
- Anemia and other blood disorders.

XI. PREVENTION AND RECOMMENDATION

- Prevention and recommendations for managing Adult-onset Still's disease (AOSD) aim to reduce disease flares, minimize complications, and enhance the patient's quality of life.
- Since AOSD has no known prevention, focus is placed on early intervention, managing symptoms, and monitoring for complications.

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ROTATOR CUFF TENDINOPATHY

HENG Raksmeay, UY Putthimundul, KAO Phengann, CHHE Sereyvuth, KHUN Rithya

I. DEFINITION

Rotator cuff tendinopathy is an inflammation of the tendons that make up the rotator cuff. This condition is often seen along with shoulder impingement and can present acutely following an injury or as a result of chronic, repetitive overuse activities.

II. EPIDEMIOLOGY

The incidence of shoulder complaints is approximately 11.2 cases per 1000 patients per year. In working populations, the incidence of shoulder-related symptoms may be as high as 14 to 18 percent.

The prevalence of rotator cuff disease, symptomatic or asymptomatic, that is identified by surgery or imaging increases with age from 9.7 percent under age 20 to 62 percent over age 80.

III. ETIOLOGY

According to Lewis et al the definitive cause of rotator cuff tendinopathy remains uncertain but proposed mechanisms include intrinsic, extrinsic or combined factors.

IV. PHYSIOPATHOLOGY

- Intrinsic factors are specific to the individual and cannot be modified: age (aging tendons develop microtear, calcification and fibrovascular proliferation), genetics, anatomy (morphology of the acromion for the shoulder, for example)
- Extrinsic factors are related to the environment: work, sport (swimming, tennis, throwing, golf, weightlifting, volleyball and gymnastics), etc.

V. DIAGNOSIS

5.1 Clinical presentation

Symptom:

- Shoulder pain with overhead activity
- Painful daily activities may include putting on a shirt or brushing hair
- Localize the pain to the lateral deltoid and often describe pain at night, especially when lying on the affected shoulder

Physical examination:

Inspection:

- Muscle atrophy: with long-standing rotator cuff tendinopathy, atrophy of supraspinatus and infraspinatus muscles may be present.
- Scapula appearance and motion: dyskinesia scapula



Muscle atrophy



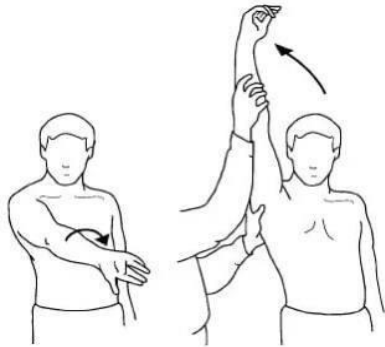
Dyskinesia scapula

Palpation: look for pain points. Shoulder pain may be referred from the neck. It is important to examine the neck and perform a screening neurologic examination to rule out cervical spine pathology.

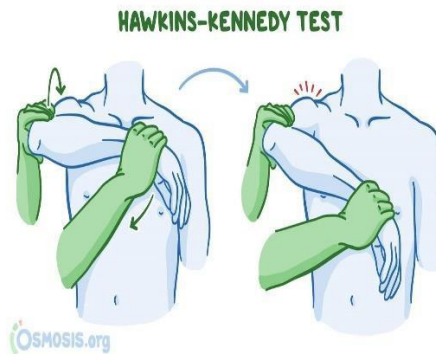
Range of motion: pain with greater than 90 degrees of shoulder abduction or with internal rotation suggests rotator cuff tendinopathy.

1. Test for shoulder impingement:

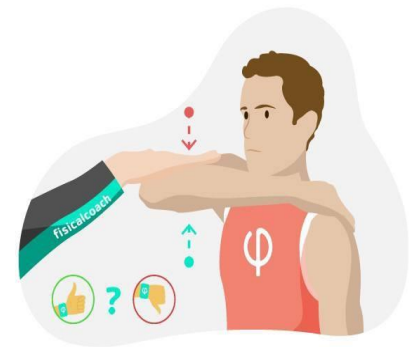
- Neer test
- Hawkin's test
- Yocum's test



Neer test



Hawkin's test



Yocum's test

2. Tendo-muscle test:

- **Supraspinatus tendon:** Jobe test
- **Infraspinatus tendon :** External rotation test
- **Subscapularis tendon:** Push off or Gerber's test
- **Biceps tendon:** Palm up test



Jobe test



External rotation test



Push off or Gerber's test



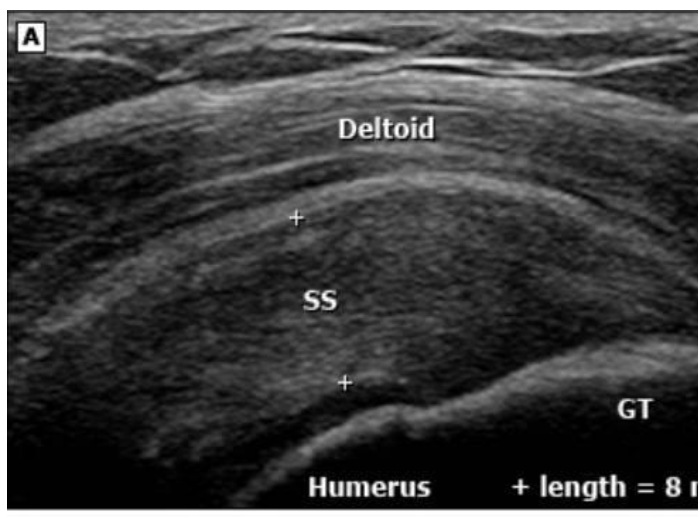
Palm up test

5.2. Investigation

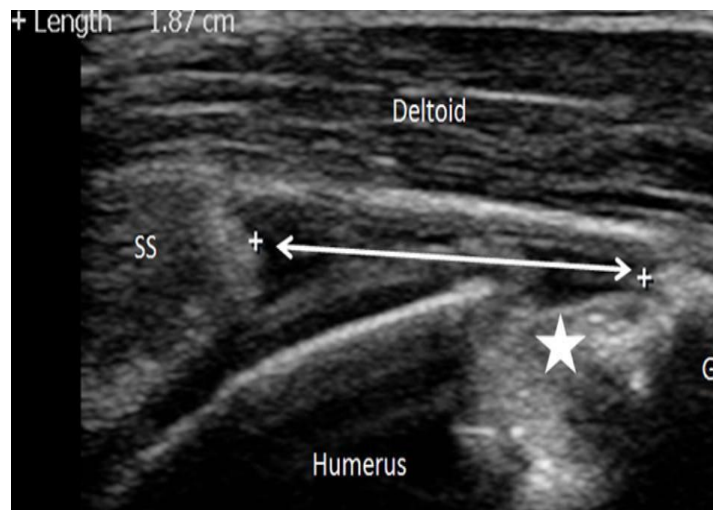
5.2.1 Radiography: better showing the subacromial space often involved in rotator cuff pathologies. Proximal humeral migration and decreases in the acromiohumeral interval to $< 7\text{mm}$.

5.2.2 Musculoskeletal ultrasound: to be the gold standard of the initial evaluation of tendon disorders, including rotator cuff disease. Pathologic findings: the supraspinatus tendon is involved most often.

- Tendon hypo echogenicity or thickening with or without internal hypo or hyperchoic foci
- Tendon retraction
- Calcification
- Fibrillar disruptions
- Fatty infiltration
- Impingement during dynamic examination



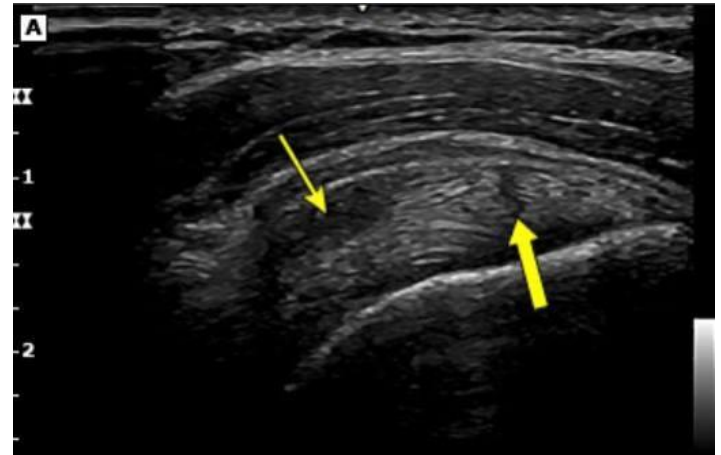
Thickening of the supraspinatus tendon



Tendon retraction



Calcification



Fibrilla disruption

5.2.3 Magnetic resonance imaging

MRI is used to rule out rotator cuff tear when conservative therapy fails, to assess for tear suspected on clinical grounds, and to aid in diagnosis when shoulder pathology is unclear. Injuries to the labrum, glenohumeral joint, and other adjacent structures can be identified by MRI.

VI. DIFFERENTIAL DIAGNOSIS

- Cervical radiculopathy
- Acromioclavicular osteoarthritis
- Subacromial bursitis
- Rotator cuff tear
- Glenoid labrum tear
- Adhesive capsulitis

VII. MANAGEMENT

1. Objectives

- To relieve symptoms and improve joint function
- To prevent rotator cuff tear

2. Critical sign

- Pain and joint stiffness

3. Mortality risk factor:

- Reduction of levels of physical activity
- Adverse effects of medication used to treat symptomatic rotator cuff tendinopathy, particularly nonsteroidal anti-inflammatory drugs (NSAIDs)

4. Outpatient management

- Ice
- Pain killer

5. First -line treatment

Non pharmacologic treatment:

- Cryotherapy
- Relative rest
- Physical therapy

Pharmacologic treatment:

- NSAIDs: for acute injuries, we give a short course (ie, 7 to 10 days)
- Glucocorticoids injection
- Topical nitroglycerin (glyceryl trinitrate)

6. Treatment adjustment: surgical intervention should be individually tailored based on the patient's symptoms, improvement with nonoperative modalities, and overall goals of the patient. If pain continues after several months of these treatments and steroid injections have provided only temporary relief, then surgery becomes an option.

VIII. SPECIAL CASES

- Incidental rotator cuff pathology revealed via shoulder MRIs in asymptomatic patient population: there is no reliable evidence that surgical intervention prevents tear propagation or the development of clinical symptoms.
- Surgical repair is usually recommended for people with a complete rotator cuff tear, especially if the person is young and/or active. Surgery is usually recommended soon after the injury, if possible, to prevent the tendon and muscle from shrinking.

IX. EVOLUTION AND PROGNOSIS

- Persistent pain/recurrent symptoms
- Risk of tear propagation, lack of healing, fatty infiltration, atrophy and retraction

X. COMPLICATION

- Frozen shoulder (adhesive capsulitis)
- Rotator cuff tear

XI. PREVENTION AND RECOMMENDATION

- Avoid lifting heavy objects above shoulder height
- Avoid lifting heavy objects with an outstretched arm
- Avoid repetitive overhead activities. Find way to modify the task and ensure you take frequent breaks
- If you have pain, avoid lying on that shoulder while you sleep
- Ensure that the load is close to your body to reduce the strain on the shoulder
- Perform regular strengthening exercises for the rotator cuff to ensure the muscles are strong enough to cope with the demands of your work or sport.

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INFECTIOUS SPONDYLODISCITIS

KAO Pheng Ann, UY Putthimundul, HENG Reaksmeay, CHHE Sereyvuth,
KHUN Rithya

I. DEFINITION

Infectious spondylodiscitis is the infection of an intervertebral disc and adjacent vertebral bodies.

II. EPIDEMIOLOGY

Presence 20 % of joint and bone infection. Risk factors include immunocompromised such as alcoholism, diabetes, neoplasia, renal or hepatic failure hemodialysis, intravenous drug addiction, chronic inflammatory rheumatism, HIV infection, immunosuppressive therapy. Males present 70% of cases after 50 years (mean age 60 years).

III. ETIOLOGY

- Staphylococcus (S. aureus or S. epidermidis) about 50 % of cases
- Gram-negative bacillus, representing about 15% (Escherichia coli, Pseudomonas, Proteus, Klebsiella, Enterobacter, Salmonella)
- Streptococcus, responsible for about 15% of bacterial spondylodiscitis, readily associated with endocarditis.
- Tuberculosis, whose spinal infection is also called "Pott's disease"

IV. PATHOPHYSIOLOGY

There are two main ways of infection:

1. **Hematogenous route:** this is the most frequent mode of contamination, from a distance infectious focus.
2. **Direct inoculation:** it follows a surgical or percutaneous procedure on the spine.

V. DIAGNOSIS

CLINIC

- Vertebral pain: acute inflammatory vertebral pain, tenacious, without triggering factor and poorly calmed by rest.
- Stiffness
- Fever
- Paraplegia in case of spinal cord compression.

PARACLINIC

- BIOLOGIC MARKER

- Elevation of white blood cell
- Elevation of CRP
- Blood culture can identify the infectious microbe.

- IMAGERY:

1. SPINE X-RAY:

- X-ray can be normal in the first 4 weeks
- After, radiographic sign could be:
 - a pinch of the disc;
 - a blurred aspect of the vertebral plateaus, mirrored;
 - erosions of the vertebral plateaus that develop on either side of the intervertebral disc even later.



2. MRI:

- This is the gold standard for the early diagnosis of spondylodiscitis. However, it may be normal in the first 10 days of the infection. It makes it possible to examine the entire spine. MRI signs are very characteristic:
- inflammatory signal (hypersignal in STIR or T1 sequences after injection of gadolinium, hypo signal in T1) of the disc and adjacent vertebral plateaus;
- peri vertebral abscess or infectious epidiorites



- DISCOVERTEBRAL PUNCTION BIOPSY:

- Disco vertebral punction biopsy should be done under guide control of CT scan
- Can identify the pathogenic microbe.

VI. DIFFERENTIAL DIASGNOSIS

Clinic: another cause of inflammatory pain of the spine:

- Axial manifestation of spondyloarthropathy
- Spine tumor
- Modic discopathy

VII. MANAGEMENT

OBJECTIVE: Our management is aim to:

- Kill microbe agent from infected site
- Control the pain
- Prevent neurological complication

CRICAL SIGN:

- Severe sepsis when Quick SOFA score ≥ 2 .
- Neurological complication

Parameter	Value
Systolic blood pressure	< 100 mmHg
Respiratory rate	> 22/min
Level of consciousness	GCS < 15

MORTALITY RISK FACTOR

- Immunocompromised such as alcoholism, diabetes, neoplasia, renal or hepatic failure hemodialysis, intravenous drug addiction, chronic inflammatory rheumatism, HIV infection, immunosuppressive therapy.

OUTPATIENT MANAGEMENT

- Pain killer

FIRST LINE TREATMENT

Antibiotic:

- **In case of presence of gravity sign or septic choc**, antibiotic therapy must be started urgently after systematic infectious sampling (blood cultures, urine culture). In this case, Cefazoline 80 to 100 mg/kg/day is suggested for suspect staphylococcus aureus MSSA or Ceftriaxone 2g/d if suspicion of enterobacteria.

- **Without gravity sign:** Antibiotic will be started after causal microbe is identified:

Bacteria	Initial Antibiotic	PO antibiotic
Methicillin sensible <i>staphylococcus aureus</i>	Penicillin M (cloxacillin or oxacillin) or C1G (cefazoline)	Rifampicin + fluoroquinolone
Methicillin resistance <i>staphylococcus aureus</i>	Vancomycin or teicoplanin	Rifampicin + fluoroquinolone
<i>Streptococcus spp</i> <i>Enterococcus spp</i>	Amoxicillin (+ gentamycin if enterococcus)	Amoxicillin Clindamycin
GNB (except <i>P. aeruginosa</i>)	C3G (ceftriaxone or cefotaxime)	Fluoroquinolone
<i>Pseudomona s aeruginosa</i>	Piperacillin-Tazobactam or ceftazidime + amikacin	N/A

- **Total duration of antibiotic: 6 weeks**

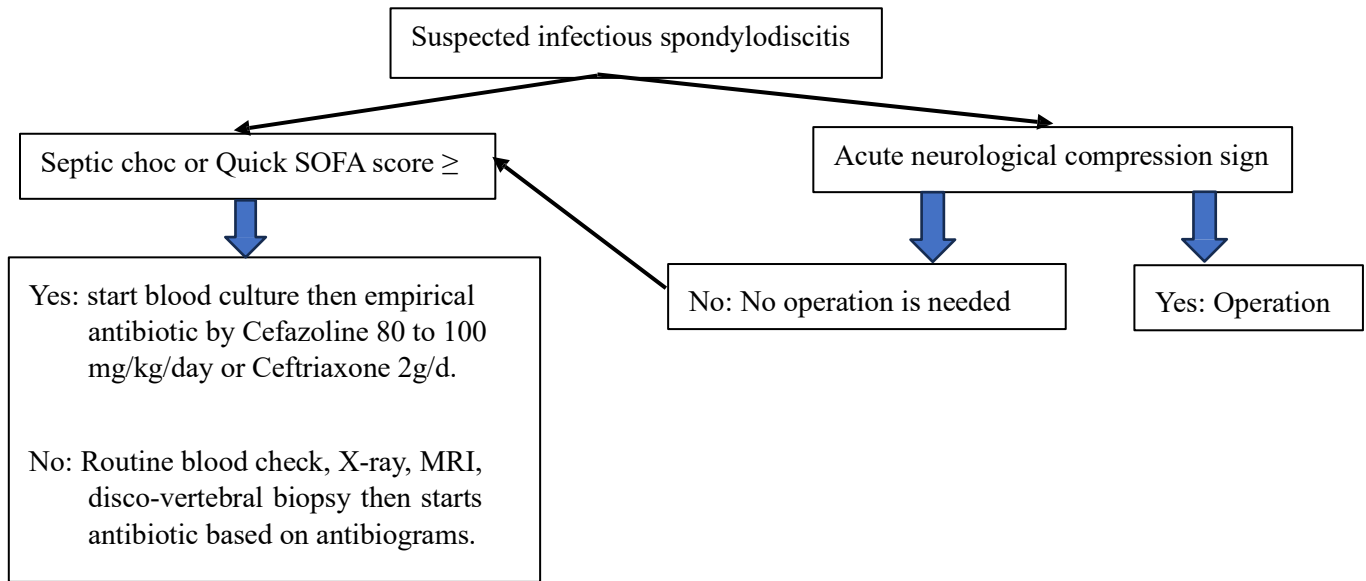
TREATMENT ADJUSTMENT

- After microbe identifying, antibiotic should must be started.
- Some antibiotic should be started by regarding of renal, liver function.
- Interaction of drug some be considered.
- **OTHER MANGAMENT**
 - 1. Immobilization:**
 - Immobilization of patients has two objectives:
 - The first is the prevention of neurological complications in of cases potentially unstable lesions.
 - The second is to get an analgia allowing a reverticalization of patients as early as possible to prevent complications of decubitus (Bedsore, thromboembolic diseases, malnutrition, constipation, orthostatic hypotension...)
 - Duration:
 - 3 months for cervical-dorsal spine involvement
 - 1 to 3 weeks for lumbar spine involvement
 - 2. Rehabilitation**
 - Rehabilitation is fundamental to allow a recovery of autonomy as quickly as possible.
 - As soon as sufficient analgia is made and a corset is put on, standing and sitting should be performed to reduce the risk of complications of decubitus.
 - A high protein diet is recommended to prevent amyotrophy and skin complications as much as possible.

3. Operation:

- Surgery is generally not necessary unless there is severe acute neurological compression or severe and disabling spinal deformity at a distance.
- Surgical drainage if presence of local abscess.

Algorithm



VIII. SPECIAL CASES

If mycobacterium tuberculosis is identified, antituberculosis should be started. Total duration of management due to Cambodian national guideline for osteoarticular tuberculosis is 1 year (two months of RHZE et 10 months of RH).

IX. EVOLUTION AND PRONOSIS:

- Good evolution and prognosis if microbes are identified and sensible to antibiotic et early management.
- Bad prognosis if cervical and dorsal spine involvement: risk of spinal cord compression.

X. COMPLICATION

- Septic choc
- Endocarditis
- Neurological compression

XI. PREVENTION AND RECOMMENDATION

- Hygiene and avoid risk factor such as alcohol
- Regular follow up with doctor if taking immunosuppressor or anticancer agent.
- Consult with doctor immediately if having fever and back pain during treatment of immunosuppressor or anticancer agent.

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SPONDYLOARTHRITIS

*KAO Pheng Ann, UY Putthimundul, HENG Reaksmey, CHHE Sereyvuth,
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I. DEFINITION

Spondylarthritis refers to chronic inflammatory arthritis that the most important features are:

- Chronic low back pain, positivity with a blood test for HLA B27, and radiographic or MRI change in sacroiliac joint in patients with axial SpA.
- Oligo-arthritis, heel enthesitis, and dactylitis in patients with peripheral SpA.

II. EPIDEMIOLOGY

- 0.35% of French population
- Sex-ratio males/females: 1.5
- Age: 20 to 40 years old

III. ETIOLOGY

The cause is still unknown but many predisposing factors involve in this pathophysiology mechanism:

- Genetic factor: HLA B27 positivity
- Immune factor: Activation of TH17 by IL23 stimuli the production of proinflammatory cytokine IL17
- Environmental factor: Cigarette smoking, Infection (Chlamydia trachomatis, Gonococcus).

IV. PATHOPHYSIOLOGY

All these factors provoke the inflammation of the specific site:

- Enthesitis: sacroiliac joint, pubic joint, talon, intervertebral disk, manubriosternal, sternoclavicular or acromioclavicular joint, etc.
- Synovitis: inflammation of the synovial membrane.

V. DIAGNOSIS

• CLINIC

- Pelvic and rachial syndrome: Chronic inflammatory
 - Low back pain
 - Cervical and dorsal spine pain
 - Pygalgia (buttock pain)
- Peripheral joint syndrome ⑨ Oligo-arthritis of the large joint
- Enthesitis syndrome ⑨
 - all entheses can be touched, the most frequent in plantar fasciitis (talalgia)
 - Sausage finger
 - Chest wall pain
- Extra-articular syndrome:
 - Anterior uveitis
 - Psoriasis
 - IBD

- **Clinical classification of spondyloarthritis (SpA) :**

- Axial SpA : non radiographic axSpA and radiographic axSpA
- Peripheral SpA : Articular SpA and enthesitis SpA
 - ⇒ with or without extra-articular manifestation :
 - IBD
 - Anterior uveitis
 - Psoriasis

- **INVESTIGATION:**

- Blood test:

- CRP and ESR: can be elevated in active disease
- HLA B27 can be positive

- X-ray of pelvis and spine:

- Can be normal in the early stage of disease
- In late stage: we can see the inflammation of sacroiliac joint, syndesmosis and bamboo sign of spine



- MRI of pelvis:

- Hypo-signal in T1 sequence and Hypersignal in T2 sequence



- **Diagnosis criteria:**

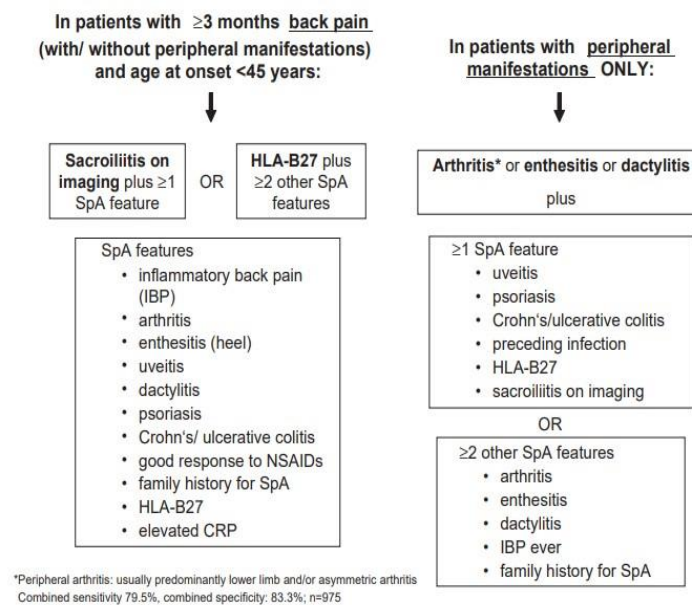


Figure 2 Combined use of the Assessment of SpondyloArthritis international Society (ASAS) criteria for axial spondyloarthritis (SpA) and the ASAS criteria for peripheral SpA in the entire SpA population. In patients with predominantly axial involvement (back pain) with or without peripheral manifestations, the ASAS criteria for axial SpA⁶ are applied. In patients with peripheral manifestations only, the ASAS criteria for peripheral SpA are applied. In the entire ASAS population of 975 patients sensitivity and specificity of the combined use of the two sets of criteria were 79.5% and 83.3%, respectively.

VI. DIFFERENTIAL DIAGNOSIS

Other causes of back pain:

- Sciatica pain
- Infectious spondylodiscitis
- Vertebral fracture
- Etc.

VII. MANAGEMENT

- **OBJECTIVES**

- To inhibit activity of disease by evaluating BADAI and BASFI score
- BASDAI ≤ 2 or BASFI ≤ 4

Instruction: Please indicate your level of ability with each of the following activities during the past week.

- How would you describe the overall level of fatigue/tiredness you have experienced?
None (0) Very Severe (10)
- How would you describe the overall level of AS neck, back or hip pain you have had?
None (0) Very Severe (10)
- How would you describe the overall level of pain/swelling in joints other than neck, back or hips you have had?
None (0) Very Severe (10)
- How would you describe the overall level of discomfort you have had from any areas tender to touch or pressure?
None (0) Very Severe (10)
- How would you describe the overall level of discomfort you have had from the time you wake up?
None (0) Very Severe (10)
- How long does your morning stiffness last from the time you wake up? hours

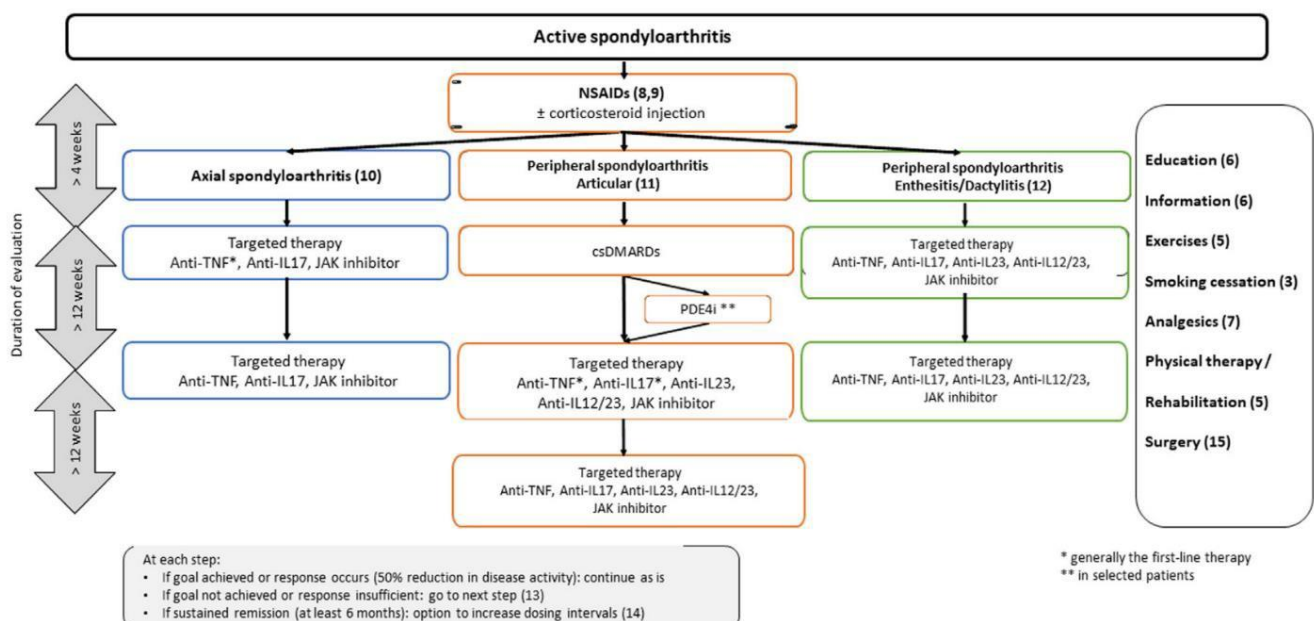
BASDAI score

Please evaluate on a scale from 0 = easy to 10 = impossible how hard you find the following situations to be:

- Putting on your socks or tights without help or aids (e.g. sock aids)?
- Bending forward from the waist to pick up a pen from the floor without an aid?
- Reaching up to a high shelf without help or aids (e.g. helping hand)?
- Getting up out of an armless dining room chair without using your hands or any other help?
- Getting up off the floor without any help from lying on your back?
- Standing unsupported for 10 minutes without discomfort?
- Climbing 12-15 steps without using a handrail or walking aid (one foot on each step)?
- Looking over your shoulder without turning your body?
- Doing physically demanding activities (e.g. physiotherapy exercises, gardening or sports)?
- Doing a full day activities whether it be at home or work?

BASFI score

- **CRITICAL SIGN**
 - Hip involvement
 - Extra-articular manifestations: uveitis, IBD.
- **MORTALITY RISK FACTOR**
 - Neurological compression due to vertebral fracture
 - Heart and lung involvement: restrictive lung disease et heart block
 - Kidney involvement: renal amyloidosis
- **OUTPATIENT MANAGEMENT:**
 - Pain killer
- **FIRST-LINE TREATMENT**
 - NSAIDs, local steroid injection and physiotherapy are recommended as first line therapy for all form of spondyloarthritis.
 - If still high disease activity (no response to NSAIDs), pass to second line treatment
- **ADJUSTMENT TRETMENT (SECOND LINE TREATMENT)**
 - For axial spondylarthritis: Anti-TNF, Anti-IL17, JAK inhibitors
 - For peripheral articular spondyloarthritis:
 - ⇒ Conventional DMARDs or PDE inhibitors: Methotrexate, Sulfasalazine, leflunomide, Apremilast
 - ⇒ If no efficacy: Anti-TNF, Anti-IL17, JAK inhibitors, Anti-12/23, Anti-IL23
 - For peripheral enthesitis spondyloarthritis: Anti-TNF, Anti-IL17, JAK inhibitors, Anti-12/23, Anti-IL23.
- **ALGORITHM**



VIII.SPECIAL CASES

- **Psoriasis arthritis:**
 - First line treatment: NSIAIDs, oral corticosteroid and local steroid injection.
 - Second line treatment:
 - ⇒ Conventional DMARDs: Methotrexate, Leflunomide, Sulfasalazine.
 - ⇒ No efficacy:
 - Axial form: Anti-TNF, Anti-IL17, JAK inhibitors
 - Peripheral form: Anti-TNF, Anti-IL17, JAK inhibitors, Anti-12/23, Anti-IL23

IX. EVOLUTION

- Chronic paroxysmal evolution
- Abortive evolution

X. COMPLICATION

- Vertebral fracture
- Kidney amyloidosis
- Restrictive lung disease
- Cardiovascular complication: Heart block, coronary artery disease
- Dural sac dilatation
- Drug induce infection.

XI. PREVENTION AND RECOMMANDATION

- Stop smoking
- Preservation to inhibit STD (chlamydia and gonococcus infection).

XII. REFERENCE

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COMPLEX REGIONAL PAIN SYNDROME

Dr. CHHE Sereyvuth, Dr. KAO Pheng An, Dr. HENG Raksmei, Dr. UY Puthimundul Dr. KHUN Rithya

I. DEFINITION

Complex regional pain syndrome (CRPS) is a neuropathic pain disorder characterized by ongoing, disproportionate pain relative to the injury and is accompanied by sensory, motor, and autonomic abnormalities. It is commonly associated with trauma, fractures, or surgeries, although some cases are spontaneous.

Type of CRPS:

- Type I (Reflex Sympathetic Dystrophy): Occurs without nerve injury.
- Type II (Causalgia): Occurs after a known nerve injury.

II. EPIDEMIOLOGY

- Incidence varies geographically, with more frequent cases reported in females.
- Common triggers include fractures, especially in the extremities.

III. ETIOLOGY

The etiology of Complex Regional Pain Syndrome (CRPS) is multifactorial and not fully understood. Here are the main factors contributing to the onset of CRPS:

1. Trauma or Injury:

- Fractures, Surgery, Sprains, Contusions, Crush Injuries or prolonged Immobilization.

2. Spontaneous Cases:

- Spontaneous onset without a clear injury or noxious event has also been documented.

3. Psychological Factors:

- Stress or psychological distress

4. Genetic Predisposition:

- Genetic factors may contribute to CRPS susceptibility.

5. Surgical Risk Factors:

- like carpal tunnel release or Dupuytren contracture surgery, patients undergoing closed reduction of fractures also show increased risk.

6. Miscellaneous Causes:

- Can be triggered by seemingly innocuous events such as intravenous line placement.

IV. PATHOPHYSIOLOGY

- Inflammatory changes: Elevated pro-inflammatory cytokines.
- Immunological factors: Autoantibodies have been identified.
- Peripheral sensitization: Increased sensitivity of the peripheral nervous system.
- Central sensitization: Involves neuroplasticity and changes in spinal and cortical structures.
- Autonomic dysfunction: Manifests as vasomotor and sudomotor abnormalities.

V. DIAGNOSIS

1. Clinical manifestation

- Sensory symptoms (allodynia, hyperalgesia), motor dysfunction (weakness, tremor, dystonia), and autonomic symptoms (temperature, color changes).
- CRPS can severely affect the quality of life, leading to depression and anxiety.

2. Diagnostic criteria

- Based on the **Budapest Criteria**
- Diagnosis is largely clinical, with tests like bone scans used to exclude other conditions.

Budapest criteria

All of the following criteria must be met:

- Continuing pain that is disproportionate to the inciting event
- 1 sign in 2 or more of the categories below

No other diagnosis can better explain the signs and symptoms

Category	Sign or symptoms
Sensory	Allodynia (pain to light touch or temperature sensation and/or deep somatic pressure and/or joint movement) and/or hyperalgesia (to pinprick)
Vasomotor	Temperature asymmetry and/or skin color changes and/or skin color asymmetry
Sudomotor/Edema	Edema and/or sweating changes and/or sweating asymmetry
Motor /Trophic	Decreased range of motion and/or motor dysfunction (weakness, tremor, dystonia) and/or trophic changes (hair, nail, skin)

VI. DIFFERENTIAL DIAGNOSIS

- Peripheral Neuropathy
- Cellulitis
- Erythromelalgia
- Deep Vein Thrombosis (DVT)
- Vasculitis
- Raynaud's Phenomenon
- Lymphedema
- Fibromyalgia
- Rheumatoid Arthritis
- Post-Traumatic Stress Disorder (PTSD)
- Multiple Sclerosis (MS)
- Mononeuritis Multiplex
- Guillain-Barré Syndrome
- Ischemic Limb Pain (Arterial Insufficiency)
- Phlebothrombosis

VII. MANAGEMENT

1. Objectives

- Pain Reduction
- Functional Restoration
- Prevention of Disability
- Psychosocial Support
- Improvement of Quality of Life

2. Critical sign: N/A

3. Mortality risk factor: N/A

4. Outpatient management:

It requires a multidisciplinary approach that focuses on:

- Pain relief: NSAIDs, Opioids, neuropathic pain agent, Antidepressant, Bisphosphonates, corticosteroids
- Physical therapy
- Functional recovery, and psychological support.

5. First-line treatment

A. Pharmacotherapy

a. Neuropathic Pain Medications

1. Gabapentin:

- Initial dose: 300 mg daily, titrated to 300 mg three times daily over a few days.
 - Maintenance dose: 900–1800 mg/day in divided doses (up to 3600 mg/day in some cases).
1. Pregabalin:
- Initial dose: 75 mg twice daily.
 - Maintenance dose: 150–300 mg/day in divided doses (up to 600 mg/day if necessary).
2. Amitriptyline (TCA):
- Initial dose: 10–25 mg daily at bedtime.
 - Maintenance dose: Titrate up to 25–75 mg/day, depending on response and tolerance.
3. Duloxetine (SNRI):
- Initial dose: 30 mg once daily.
 - Maintenance dose: Increase to 60 mg once daily, with a maximum dose of 120 mg/day.

b. Non-Steroidal Anti-Inflammatory Drugs (NSAIDs)

1. Ibuprofen:

- Dose: 200–400 mg every 4–6 hours as needed (maximum 3200 mg/day).

2. Naproxen:

- Dose: 250–500 mg twice daily (maximum 1000 mg/day).

c. Corticosteroids

1. Prednisone (oral):

- Initial dose: 30–60 mg/day for 1–2 weeks.
- Tapering dose: Gradually taper the dose by 5–10 mg/week depending on symptom resolution.

d. Bisphosphonates

1. Alendronate:

- Dose: 70 mg once weekly.

2. Zoledronic Acid

(IV):

- Dose: 5 mg IV infusion once per year for osteoporosis or osteopenia.

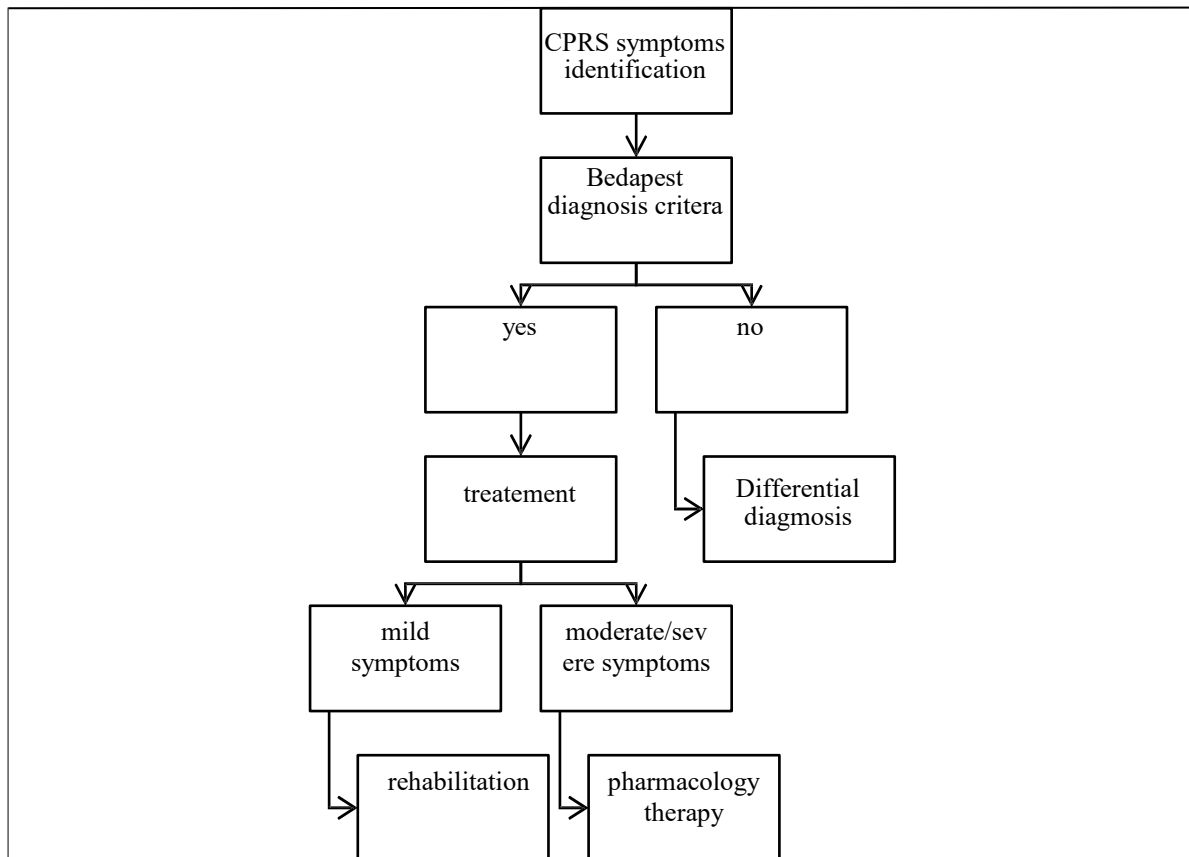
B. Psychological Interventions

- Used to address the psychological impact of CRPS, such as anxiety and depression, and to teach coping strategies for chronic pain.

6. Treatment adjustment

- Neuropathic Pain Medications: If first-line treatments like gabapentin or pregabalin are ineffective, consider increasing doses or switching to alternative medications like tricyclic antidepressants (e.g., amitriptyline) or serotonin-norepinephrine reuptake inhibitors (e.g., duloxetine).
- Bisphosphonates: If there is bone involvement, treatment with bisphosphonates may be extended or adjusted to reduce pain and improve bone density.
- Corticosteroids: For persistent inflammatory symptoms, a short course of low-dose corticosteroids may be reintroduced or tapered depending on the response.
- Opioids: Can be introduced for severe, refractory pain, but should be used with caution and for the shortest duration possible due to the risk of dependency.

7. Algorithm



VI. PECIAL CASES

1. CRPS After Stroke

Presentation

- CRPS can develop after hemiplegic strokes, commonly affecting the paralyzed limb.
- Often presents with severe pain, swelling, and joint stiffness.

Management

- Early mobilization and physical therapy post-stroke are critical.
- Corticosteroids and bisphosphonates may be used to manage inflammation and bone pain.
- Treat comorbid conditions such as spasticity or contractures.

Prognosis: Can be challenging to treat due to the underlying neurological deficits.

2. CRPS in Post-Surgical Patients:

Presentation

- May develop several weeks post-surgery, with intense pain, swelling, and stiffness in the operated limb.

Management

- Early rehabilitation and pain control to prevent the development of chronic symptoms.
- Corticosteroids are often used early in post-surgical CRPS, with a multimodal approach including physical the repay

.Prognosis: Variable, with early treatment improving outcomes.

CRPS in Pregnancy Presentation

- CRPS during pregnancy is rare but may be triggered by hormonal changes or trauma.
- Presents with typical symptoms of pain, swelling, and autonomic changes in the extremities.

Management

- Medication use is limited during pregnancy, so the focus is on physical therapy and pain management using non-pharmacologic methods.

- Opioids and corticosteroids are used cautiously.

Prognosis: Management can be challenging due to pregnancy-related restrictions on medications, but most cases stabilize postpartum.

VI. VOLUTION AND PROGNOSIS

A. Evolution of CRPS

- CRPS generally follows three stages, though these stages are not rigid, and progression can vary:

1. Acute Stage (0–3 months)

- Symptoms: Intense, burning pain, localized swelling, warmth, and skin color changes (redness or mottling).
- Vasomotor changes: The affected limb may feel warmer than the other.
- Trophic changes: Skin may be shiny, hair growth may increase or decrease, and nails may become brittle.
- Early Intervention: If treated promptly, patients may experience significant pain relief and prevent chronic disability.

2. Dystrophic Stage (3–12 months)

- Symptoms: Increased pain intensity, skin becomes cool and bluish, and swelling may persist or worsen.
- Loss of function: Decreased range of motion and weakness in the affected limb due to disuse and pain.
- Worsening of trophic changes: Skin, hair, and nail changes become more prominent.
- Progression: Some patients may show signs of improvement, while others might progress to a chronic state.

3. Atrophic Stage (Chronic, >12 months)

- Symptoms: Pain may decrease, but joint stiffness, muscle atrophy, and contractures may set in.
- Chronic trophic changes: Severe skin thinning, reduced sweating, and irreversible joint deformities may occur.
- Cold CRPS: The affected limb may feel cold and stiff with irreversible tissue damage.
- Risk of Disability: Functional impairment is common, and long-term disability may occur without aggressive management.

B. Prognosis of CRPS

- The prognosis of CRPS can range from full recovery to chronic pain and disability.
- Early intervention, multidisciplinary treatment, and addressing both physical and psychological factors are crucial in improving outcomes and preventing long-term complications.

VII. COMPLICATION

- Chronic Pain
- Functional Impairment and Disability
- Trophic Changes: Skin, Hair, and Nail Changes, Ulcers and Skin Breakdown
- Bone Demineralization (Osteopenia/Osteoporosis)
- Psychological and Emotional Impact: Depression and Anxiety, Social Isolation, Suicidal Ideation
- Spread of CRPS
- Autonomic Dysregulation: Sympathetic Nervous System Dysfunction, Cardiovascular Complications
- Sleep Disturbances
- Gastrointestinal and Urological Dysfunction
- Systemic Complications: Immune System Dysfunction (increased susceptibility to infections), Endocrine Dysregulation (adrenal insufficiency).
- Cognitive Impairment
- Increased Risk of Cardiovascular (effects of immobility, autonomic dysregulation, and psychological stress).
- DVT and Pulmonary Embolism
- Complicated Surgical Recovery

VIII. REVENTION AND RECOMMENDATION

- Multidisciplinary Approach: Treatment should involve a team of healthcare providers, including pain specialists, physical and occupational therapists, psychologists, and primary care physicians.
- Early Intervention: early and aggressive treatment, including physical therapy, pain management, and psychological support.

- Pain Education: pain management strategies, graded activity, and the importance of early movement.
- Customized Treatment Plans: Tailor treatment to each patient, considering their medical history, psychological state, and specific needs.

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SYSTEMIC SCLEROSIS

Dr. CHHE Sereyvuth, Dr. KAO Pheng An, Dr. HENG Raksmeay, Dr. UY Puthimundul Dr. KHUN Rithya

I. DEFINITION

Systemic sclerosis (Ssc), also known as, Scleroderma is a connective tissue disease characterized by fibrosis of the skin and internal organs, pronounced alterations in the microvasculature and frequent cellular and humoral immunity abnormalities.

There are 3 mains types of Systemic

- sclerosis: ○ Limited cutaneous
- systemic sclerosis ○ Diffuse
- cutaneous systemic sclerosis ○
- Systemic sclerosis sine
- scleroderma.

II. EPIDEMIOLOGY

- It can occur at any age, but it is most common in individuals aged 30-50 years.
- It is approximately four times more common in women than in men.
- Global prevalence varies significantly, with the highest rates in North America, Europe, and South America.

III. ETIOLOGY

The exact cause is unknown, but researchers believe it is a complex interaction of several factor.

IV. ATHOPHYSIOLOGY

1. Immune System Dysfunction:

- An overactive immune system attacking healthy tissue may be the main cause.

2. Genetic Predisposition:

- Certain genes can increase an individual's susceptibility to scleroderma. Having a family history of the disease is a risk factor.

3. Environmental Triggers:

- Exposure to certain environmental factors, such as silica dust, organic solvents, and some medications

4. Hormones:

- Women are diagnosed with the disease significantly more than men, suggesting hormonal influences.

5. Other factors:

- Viral infections, gut microbiome composition, and exposure to certain chemicals.

V. DIAGNOSIS

1. Clinical manifestation

Musculoskeletal	- Contractures or difficulty opening finger, mouth - Loss of distal bone (sclerodactyly) - Tendon tightness, audible rubs
Gastrointestinal	- Dysphagia - Malabsorption
Pulmonary	- Interstitial lung disease - Pulmonary hypertension
Renal	- Malignant hypertension - Acute renal failure
Scleroderma and clinical presentation	
Raynaud phenomenon	- May precede other manifestation for month or year - Digital ulcer
Skin	- Tight indurated skin, especially of the face and hand - Facial telangiectasias - Hyperpigmentation - Calcinosis

2. Diagnostic criteria

Item	Sun-items	Weight
Skin thickening of the fingers of both hands extending proximal to the metacarpophalangeal joints		9
Skin thickening of the fingers	Puffy finger	2
	Whole finger, Distal to MCP	4
Finger tips lesions	Digital tip ulcers	2
	Pitting scars	3
Telangiectasia		2
Abnormal nailfold capillaries		2
Pulmonary arterial hypertension and/or interstitial lung disease		2
Raynaud's phenomenon		3
Scleroderma related antibodies	Any of anti-centromere, anti-topoisomerase9anti-Scl 70)	3
Patient having a total score of 9 or more are classified as having definite systemic sclerosis.		

VI. DIFFERENTIAL DIAGNOSIS

- Lichen sclerosis
- Eosinophilic fasciitis
- Dermatomyositis
- Systemic lupus erythematosus
- Fibromyalgia
- Vasculitis
- Polymyositis
- Amyloidosis

VII. MANAGEMENT

1. Objectives

- To limit or stop the progression of the disease
- To reduce its sequelae
- To improve the quality of life of patients.

2. Critical sign

- Emergency HTA in renal crisis
- Acute decompensation of right heart failure secondary to HTAP
- Acute decompensation of pulmonary fibrosis
- Gastro intestinal bleeding
- decompensate of cardiomyopathy
- Gastro intestinal tract dysmotility

3. Mortality risk factor

- Predictors of early mortality include: male sex, late onset, diffuse disease, pulmonary arterial hypertension, and renal crisis.
- Patients with diffuse skin disease tend to have a more aggressive clinical course and eventually develop visceral complications

4. Outpatient management

A. Regular Monitoring and Assessment

- Routine Check-ups: to monitor disease progression, organ involvement, and treatment efficacy.
- Laboratory Tests: Periodic blood tests to check for markers of inflammation, renal function, liver function, and other relevant parameters.
- Imaging: Echocardiograms, pulmonary function tests, and high-resolution CT scans of the lungs to monitor for complications like pulmonary arterial hypertension and interstitial lung disease.

B. Lifestyle Modifications

- Smoking Cessation, regular exercise, balance diet with adequate nutrition

C. Specialist Referrals

- Refer to specialists as new symptoms arise or existing symptoms worsen.

5. First-line treatment

A. Raynaud's Phenomenon

- Calcium Channel Blockers: Nifedipine or amlodipine are often the first line to improve blood flow and reduce the frequency and severity of attacks.
- Lifestyle Modifications: Keeping warm, avoiding cold exposure, and smoking cessation.

B. Skin Involvement

- Moisturizers and Emollients: Regular application to maintain skin hydration.
- Topical Treatments: For localized skin thickening.

C. Gastrointestinal Symptoms

- Proton Pump Inhibitors (PPIs): For managing gastroesophageal reflux disease (GERD) symptoms.
- Prokinetic Agents: For motility issues.

D. Interstitial Lung Disease (ILD)

- Immunosuppressive Therapy: Mycophenolate mofetil or cyclophosphamide may be initiated to manage lung involvement.

E. Pulmonary Arterial Hypertension (PAH)

- Phosphodiesterase-5 Inhibitors: Sildenafil or tadalafil.

F. Immunosuppressive Therapy

- Methotrexate or Mycophenolate Mofetil: Often used as first-line immunosuppressants for overall disease control.

G. General Measures

- Patient Education: Educating the patient about the disease, its course, and self-management strategies.
- Regular Monitoring: Regular follow-ups to monitor for disease progression and treatment response.

6. Treatment adjustment

A. Raynaud's Phenomenon:

- If calcium channel blockers are insufficient, consider adding or switching to phosphodiesterase-5 inhibitors (sildenafil).

B. Skin Involvement:

- For progressive skin thickening, immunosuppressants like methotrexate, mycophenolate mofetil, or even intravenous cyclophosphamide may be used.

C. Gastrointestinal Symptoms:

- Increase the dose or switch to more potent PPIs for GERD. Add prokinetic agents like metoclopramide or domperidone for severe motility issues.

D. Interstitial Lung Disease (ILD):

- If ILD progresses, consider switching from mycophenolate mofetil to cyclophosphamide.

E. Pulmonary Arterial Hypertension (PAH):

- Better refer the patient to specialist center.

F. Renal Crisis:

- Initiate ACE inhibitors (captopril) immediately if there are signs of scleroderma renal crisis. Adjust the dose based on blood pressure and renal function.
- Refer the patient to specialist center.

G. Joint and Muscle Pain:

- NSAIDs or low-dose corticosteroids (<10mg) may be used, but avoid using high dose of steroid (risk of scleroderma renal crisis)

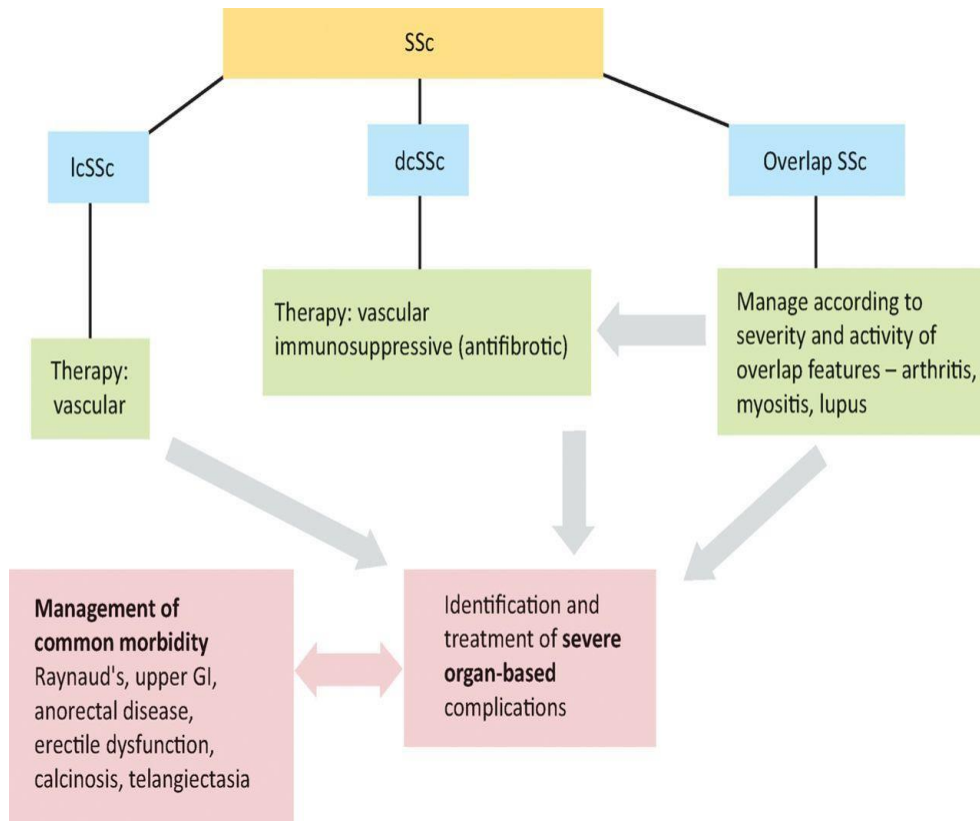
H. Fatigue and General Well-being:

- Ensure optimal control of all symptoms, and consider medications like low-dose prednisone if systemic inflammation is significant.

I. Mental Health:

- Address anxiety, depression, or other psychological impacts through counseling, support groups, or psychiatric medications if necessary.

7. Algorithm



VIII. SPECIAL CASES

For below special case, should refer to patient to specialist center.

A. Scleroderma Renal Crisis (SRC)

- Presentation: Acute onset of severe hypertension, rapidly progressive renal failure, and microangiopathic hemolytic anemia.

B. Pulmonary Arterial Hypertension (PAH)

- Presentation: Progressive dyspnea, fatigue, chest pain, and syncope.

C. Interstitial Lung Disease (ILD)

- Presentation: Progressive dyspnea, cough, and reduced exercise tolerance.

D. Overlap Syndromes

- Presentation: Features of SSc along with symptoms of other autoimmune diseases (e.g., lupus, polymyositis).

E. Pregnancy in Systemic Sclerosis

- Preconception Counseling: To assess disease stability and organ involvement.
- Multidisciplinary Care: Close monitoring by a team including rheumatologists,

obstetricians,
and other specialists.

- Monitoring: Frequent assessments for maternal and fetal well-being.

F. Juvenile Systemic Sclerosis

- Presentation: often with rapid progression and significant impact on growth and development.
- Multidisciplinary Approach: Including pediatric rheumatologists, dermatologists, and other specialists.

IX. EVOLUTION AND PROGNOSIS

- Overall, 10-year survival is 92% for limited systemic sclerosis and is 65% for diffuse systemic sclerosis.
- Predictors of early mortality include male sex, late onset, diffuse disease, pulmonary arterial hypertension, and renal crisis.
- Patients with diffuse skin disease tend to have a more aggressive clinical course and eventually develop visceral complications (usually within the first 3 to 5 years), which, if severe, can lead to death.
- Patients with limited systemic sclerosis may have disease that is nonprogressive for long periods; visceral changes eventually develop, but the course is often remarkably benign.

VII. COMPLICATION

- Skin: Ulcers, Calcinosis, Contractures
- Musculoskeletal: Joint pain and stiffness, Muscle weakness, Raynaud's phenomenon
- Pulmonary: Interstitial lung disease (ILD), Pulmonary hypertension
- Cardiovascular: Pericardial effusion, Myocardial fibrosis, Arrhythmias
- Gastrointestinal: Gastroesophageal reflux disease (GERD), Scleroderma-associated small intestinal bacterial overgrowth (SIBO), Gastroparesis
- Renal: Scleroderma renal crisis, chronic kidney disease
- Other potential complications: Fatigue, Nutritional deficiencies, Mental health problems

VIII. REVENTION AND RECOMMENDATION

1. General Recommendations and prevention

- Avoiding silica dust: use appropriate protective gear like masks and gloves.
- Minimizing exposure to organic solvents: avoid using organic solvents or wear protective gear when handling them.
- Maintain a healthy lifestyle: Eat a balanced diet, Exercise regularly, get enough sleep and manage stress effectively. Quit smoking
- avoid cold temperatures and stress. Wear warm clothing and gloves in cold weather.
- Avoid sun exposure and Use sunscreen and skin Moisturizer.
- Regular checkups with your doctor are follow the prescription

2. Recommendations and prevention for Specific Complications

- Skin ulcers: Keep the ulcers clean and dry. Avoid putting pressure on the ulcers.
- Joint pain and stiffness: Exercise regularly to keep your joints flexible. Apply heat or ice to painful joints.
- Interstitial lung disease (ILD): Avoid smoking and exposure to irritants. Get vaccinated

against the flu and pneumonia.

- Gastroesophageal reflux disease (GERD): Avoid eating large meals before bedtime. Elevate your head while sleeping. Avoid foods that trigger your symptoms.
- Scleroderma renal crisis: Seek immediate medical attention if you experience symptoms such as high blood pressure, shortness of breath, and decreased urination.

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SYSTEMIC LUPUS ERYTHEMATOUS

KHUN Rithya, HENG Reaksmey, KAO PhengAnn, UY Putthimundul, CHHE Sereyvuth

I. DEFINITION

Systemic lupus erythematosus (SLE) is a chronic autoimmune disease of unknown cause that can affect any organ of the body. Immunologic abnormalities, especially the production of a number of antinuclear antibodies (ANA), are a prominent feature of the disease. Patients present with variable clinical features ranging from mild joint and skin involvement to life-threatening kidney, hematologic, or central nervous system involvement; that's make it difficult to diagnosis.

II. EPIDEMIOLOGY

1. Geography and race:

- The prevalence is high among Asian, African American, African Caribbean. The estimated incidence is 1 to 25 per 100,000 in North America, South America, Europe, and Asia.

2. Sex:

- in adults, in women childbearing years the ratio female to male is from 7:1 to 15:1.
- In children the ratio female to male is 3:1
- In older peoples especially, menopause women the ratio approximately is 8:1

3. Age of onset:

- 65% of SLE mostly happen at age 16 and 55 years old, 20% before 16 and 15% after 55 ages

4. Factors affected disease outcome:

- SLE has poorer renal prognosis in Asian and African then white peoples.
- Men with lupus tend to have higher frequencies of renal disease, skin manifestations, cytopenia, serositis, neurologic involvement, thrombosis, cardiovascular disease, and difficult to treat.
- SLE in children tends to be more severe than in adults, with a high incidence of malar rashes, nephritis, pericarditis, hepatosplenomegaly, and hematologic abnormalities.

III. ETIOLOGY

The etiology of SLE remains unknown and is clearly multifactorial. Many observations suggest a role for genetic, hormonal, immunologic, and environmental factors.

IV. PATHOPHYSIOLOGY

1. Although the exact etiology of systemic lupus erythematosus (SLE) remains obscure, it is clear that many of the clinical manifestations of SLE are mediated directly or indirectly by antibody formation and the creation of immune complexes (IC).
2. Kidney disease: Initiation of kidney disease in SLE is likely due to the deposition or formation of immune complexes in the mesangium, subendothelial, or subepithelial spaces, with subsequent activation of complement.
3. Cells surface antibody: SLE make antibody to a number of surface antigen and may cause organs damage.
4. Environment factor: sunlight is the obvious that exacerbated SLE and crystalline silica, viral infection like EBV
5. Hormonal factor: SLE has more women childbearing and tend to decrease after menapauses.

V. DIAGNOSIS

1. Clinical:

Table clinical manifestation in series of 1000 European patients

SLE manifestation	Prevalence (%)
Arthritis	8
Malar rash	5
Fewer	5
Photosensitive	4
Nephropathy	3
Serositis	3
Raynaud phenomenon	3
Neurologic involvement	2
Oral ulcer	2
Thrombocytopenia	2
Sicca syndrome	1
Livedo reticularis	1
Thrombosis	1
Lymphadenopathy	1
Discoid lesion	1
Myositis	9
Hemolytic anemia	8
Lung involvement	7
Subacute cutaneous lesion	6
Chorea	2

2. Paraclinical

A. Laboratory studies

- **Standard laboratory studies:** Complete blood count (CBC) with differential, Serum creatinine Urinalysis with microscopy
- **Other laboratory studies:** Erythrocyte sedimentation rate (ESR) or C-reactive protein (CRP)
- Complement levels, Liver function tests, Creatine kinase assay, Spot protein/spot creatinine ratio
- **Auto antibodies tests:** ANA, Anti ENA

B. Imaging study

- Standard imaging study: Chest X-Ray, abdominal ultrasound, heart ultrasound
- Other imaging study: chest CT scan, MRI is depended on clinical manifestation

The 2019 European League Against Rheumatism/American College of Rheumatology classification

criteria for SLE

Entry criterion

Antinuclear antibodies (ANA) at a titer of $\geq 1:80$ on Hep-2 cells or equivalent positive test

↓

If absent, do not classify as SLE.

If present, apply additive criteria

Additive criteria

Do not count a criterion if there is more likely explanation than SLE. Occurrence of a criterion on at least one occasion is sufficient.

SLE classification requires at least one clinical criterion and ≥ 10 points.			
Criteria need not occur simultaneously. Within each domain, only the highest weighted criterion is counted toward the total score.			
Clinical domains and criteria	Weight	Immunology domains and criteria	Weight
Constitutional		Antiphospholipid antibodies	
Fever	2	Anti-cardiolipin antibodies OR	
Hematologic		Anti- β 2GP1 antibodies OR	
Leukopenia	3	Lupus anticoagulant	2
Thrombocytopenia	4	Complement proteins	
Autoimmune hemolysis	4	Low C3 OR low C4	3
Neuropsychiatric		Low C3 AND low C4	4
Delirium	2	SLE-specific antibodies	
Psychosis	3	Anti-dsDNA antibody OR	
Seizure	5	Anti-Smith antibody	6
Mucocutaneous			
Non-scarring alopecia	2		
Oral ulcers	2		
Subacute cutaneous OR			
discoid lupus	4		
Acute cutaneous lupus	6		
Serosal			
Pleural or pericardial effusion	5		
Acute pericarditis	6		
Musculoskeletal			
Joint involvement	6		
Renal			
Proteinuria $>0.5\text{g}/24\text{h}$	4		
Renal biopsy Class II or V			
lupus nephritis	8		
Renal biopsy Class III or IV			
lupus nephritis	10		
Total Score:			

Classify as Systemic Lupus Erythematosus with a score of 10 or more if entry criterion fulfilled.

The 1997 modified classification criteria for SLE

Malar rash	Fixed erythema, flat or raised, over the malar eminences, tending to spare the nasolabial folds 2.
Discoid rash	Erythematous raised patches with adherent keratotic scaling and 3 follicular plugging; atrophic scarring may occur in older lesions
Photosensitivity	Skin rash as a result of unusual reaction to sunlight, by patient 4 history or physician observation
Oral ulcer	Oral or nasopharyngeal ulceration, usually painless, observed by a 5 physician
Arthritis	Non-erosive arthritis involving 2 or more peripheral joints, 6 characterized by tenderness, swelling or effusion
Serositis	A-Pleuritis: Convincing history of pleuritic pain or rub heard by a physician or evidence of pleural effusion B-Pericarditis: documented by ECG or rub or evidence of pericardial effusion
Renal disorder	A- Persistence proteinuria Proteinuria: greater than 0.5 grams per day or greater than +++ if quantification not performed. B- Cellular cast Casts: may be red cell, hemoglobin, granular, tubular or mixed
Neurologic disorder	Seizure or psychosis
Hematologic disorder	A- Hemolytic anemia - With reticulocytotic B- Leucopenia - Less than 4000/mm ³ total on 2 or more occasions C- Lymphopenia - Less than 1500/mm ³ total on 2 or more occasions D- Thrombocytopenia - Less than 100,000/mm ³ in the absence of offending drugs
Immunologic disorder	A- Anti DNA Antibody to native DNA in abnormal titer B- Anti Smith Presence of antibody to Sm nuclear antigen C- Positive finding of Antiphospholipid antibody 1. Abnormal serum level of IgG or IgM anticardiolipin antibodies 2. A positive test result for lupus anticoagulant using a standard method 3. A false positive serologic test for syphilis, known to be positive for at least 6 months and confirmed by <i>Treponema pallidum</i> 7 immobilization or fluorescent treponema antibody absorption test
Anti-nuclear antibody	Abnormal titer of antinuclear antibody by immunofluorescence or an equivalent assay at any point in time, and in the absence of drugs known to be associated with 'drug induced lupus' syndrome

This classification is based on 11 criteria. For the purposes of identifying patients in clinical studies, a person must have SLE if any four or more of the 11 criteria are present, serially or simultaneously, during any interval of observation.



Malar rash



Subacute cutaneous lupus

VI. DIFFERENTIAL DIAGNOSIS

- Rheumatoid Arthritis
- Mixed connective tissue diseases
- Systemic sclerosis
- Vasculitis
- Malignancy
- Infections
- Adult still's disease
- Multiple sclerosis

VII. MANAGEMENT

1. Objectives

- **Induction of remission:** aimed at rapidly controlling disease activity for prolonged periods
 - **Maintenance therapy:** aimed at continuing remission and preventing flares,
 - **Treatment of comorbidities:** aimed at reducing the side effects of drugs employed to control activity,
- controlling other associated conditions (hypertension, diabetes) and minimizing damage.

2. Critical sign

- **Mild disease:** has arthritis and skin involvement
- **Moderate disease:** has serositis (pleuritis, pericarditis), Hematologic, Vasculitis
- **Sever disease:**
 - (1) SLE affected kidney (abnormal renal function, decreased urine output, edema, HTN)
 - (2) SLE affected Nervous system (seizure, loss of consciousness)

3. Mortality risk factor

- Patients with SLE have mortality rates ranging from two to five times higher than that of the general population.
- The major causes of death in the first few years of illness are active disease (eg, CNS and kidney) or infection due to immunosuppression,

- While causes of late death include complications of SLE (eg, end stage kidney disease), treatment complications, and cardiovascular disease. Patients are also at risk for significant morbidity due to both active disease and the side effects of drugs such as glucocorticoids and cytotoxic agents.

4. Outpatient management

- Laboratory evaluation: CBC, ESR, CRP, Creatinine, Urea, electrolytes, urine dipstick or spot urine protein and creatinine, C3, C4 compliment if possible
- Screen for Tuberculosis and others infection, Hepatis B and C, before started the SLE management
- Ideal target values for classic risk factors and indications for additional interventions in SLE patients

Risk factors	Ideal target values
Blood pressure LDL cholesterol Diabetes mellitus Smoking Obesity	<130mmHg systolic and < 80mmHg diastolic < 2,6mmol/l Fasting blood sugar 7mmol/l Stop smoking Body mass index < 25 kg/m2
Additional measurement	indications
Aspirin ACE inhibitor	Known as vascular disease
	Prevalent cardiovascular disease including heart failure. Left ventricular hypertrophy or diabetes. Preferred second drug for hypertension

5. First line treatment

- All patients with SLE should be on chloroquine 250mg qd or hydroxychloroquine 200mg to 400mg qd
- **Mild disease (skin and joints):** low dose prednisolone (0.1 to 0.2 mg/kg/day) plus Chloroquine/ Hydroxychloroquine
- **Moderate disease (serositis, hematologic, vasculitis):** low (0.1 to 0.2 mg/kg/day) /moderate (0.3 to 0.5 mg/kg/day) dose prednisolone/ plus Chloroquine/ Hydroxychloroquine/ immunosuppressive (AZA/ MMF/ CYC/MYX/LEF
- **Severe disease:** Lupus Nephritis and Lupus Neuropsychiatric involvement: the treatment has two phase, induction phase and maintenance phase, Oral CYC is administered at doses ranging from 1 to 2 mg/kg/day for 6-9 months or intravenous CYC at dose 60-120mg/m² every month for 6 months or MMF at 2000mg/day for 6month with Prednisolone 0.5-1 mg/kg/day for 6-8 weeks, followed by tapering to a maintenance dose of 5-10 mg/day (36,41). Either AZA, MMF could be considered for maintenance therapy.

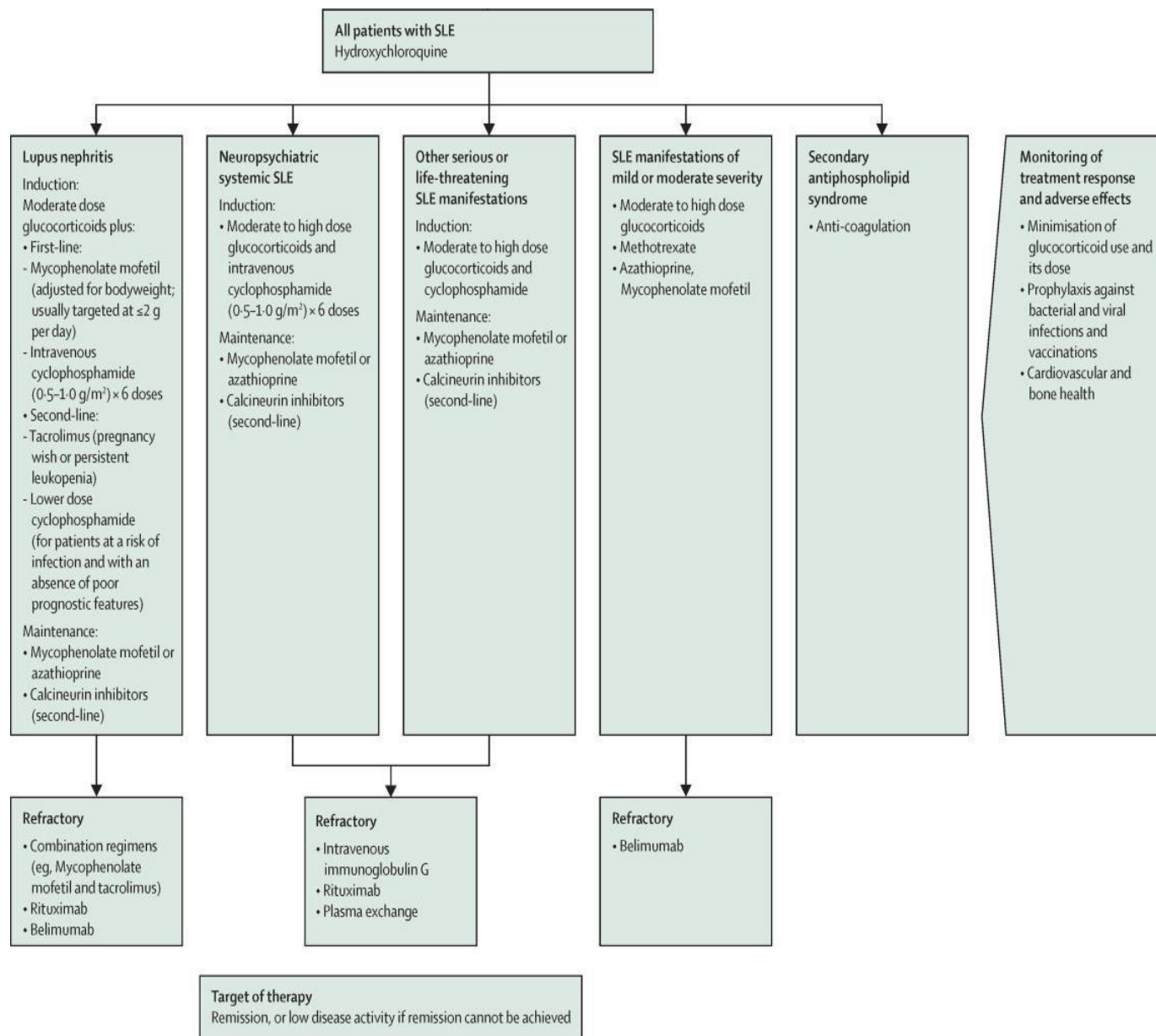
6. Treatment adjustment: Therapy of active SLE is mainly guided by the type and severity of each single manifestation.

1. **Treatment for serositis:** medium dose of Prednisolone plus AZA has been effective with pleuritis, pericarditis.
2. **Treatment for Hematologic:** medium dose of Prednisolone and Cyclosporin A for Thrombocytopenia, Azathioprine for Hemolytic Anemia, (alternative MMF for refractory case)

3. **Treatment for articular manifestation:** low dose Prednisolone with Chloroquine or Methotrexate or Leflunomide (AZA for refractory case).

7. Algorithm

APLAR Consensus Statement on managing SLE



Abbreviation: APLAR: Asia Pacific league against rheumatologic AZA: Azathioprine, APS: Anti phospholipid syndrome, CYC: Cyclophosphamide, CNIs: Calcineurin inhibitor, GC: Glucocorticoid, HCQ: Hydroxychloroquine, IVIG: intravenous Immunoglobulin G LEF: Leflunomide, LN: Lupus Nephritis, NPSLE: Neuropsychiatric Systemic Lupus Erythematosus MMF: Mycophenolate mofetil, SLE: Systemic lupus erythematosus, TAC: Tacrolimus

VIII. SPECIAL CASES

- SLE with serious infections, need to hold immunosuppressive, double dose of current Prednisolone and start appropriate Antibiotic
- SLE with pregnancy stop immunosuppressive, others drugs affected pregnancy, continue Chloroquine and Prednisolone, suggested to see specialist.

IX. EVOLUTION AND PROGNOSIS

- SLE can run a varied clinical course, ranging from a relatively benign illness to a rapidly progressive disease with fulminant organ failure and death
- Clinical remission after appropriate therapy is uncommon, and, when it is achieved, it is often not sustained.
- Prognostic factors associated with increased mortality in SLE include: affected kidney, hypertension, male sex, young age/old age at presentation, low socioeconomic status, being a black person, affected many organs.

X. COMPLICATIONS

1. From SLE itself:

- End stage renal failure
- Heart failure
- Hematologic (Anemia, leucopenia, bleeding)
- Neurologic

2. Infections

3. Thrombosis

4. Malignancy

XI. PREVENTION AND RECOMMENDATION

- 1. Photoprotection:** exposure to sun light may exacerbate or induce systemic manifestations of systemic lupus erythematosus (SLE), Sunscreens that block both UV-A and UV-B and have a sun protection factor (SPF) ≥ 55 are suggested. Medication that's cause photosensitive should be avoid.
- 2. Diet and nutrition**
 - Good hygiene and well-cooked food
 - Need a good balance diet, if not has to take daily multivitamin
 - Vitamin D levels should be monitored periodically and patients with low vitamin D levels should be treated with supplemental vitamin D.
 - In patients with hypertension and/or nephritis, dietary measures such as salt restriction may be required.
- 3. Exercise**
- 4. Smoking cessation**
- 5. Vaccination**
- 6. contraceptive**

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OSTEOPOROSIS

UY Putthimundul, HENG Reaksmey, KAO PhengAnn, CHHE Sereyvuth, KHUN Rithya

I. DEFINITION

Osteoporosis (OP) is a skeletal disorder characterized by compromised bone strength, predisposing a person to an increased risk of fracture.

Osteoporotic fractures (OFx) are occurring from a fall from a standing height or less, without major trauma. OFx are classified as non-severe and severe or major. A severe OFx is defined as a fracture of proximal humerus, distal and proximal femur, proximal tibia, vertebrae, pelvis and 3 ribs adjacent. The fracture other from the severe is considered as non-severe OFx.

II. EPIDEMIOLOGY

Postmenopausal osteoporosis is a common disease, affecting one in three women after menopause.

Globally, it is estimated that there will be 4.5 million hip OFx in 2050.

III. ETIOLOGY

OP is categorized as primary and secondary. Primary OP includes postmenopausal OP and senile OP, while secondary OP is characterized as having an etiologic mechanism

- Prolonged glucocorticoid use
- Endocrine disorder such as Cushing syndrome, Primary hyperparathyroidism, hyperthyroidism, hypogonadism, diabetes, hyperprolactinemia
- Anorexia nervosa
- Alcohol use
- Smoking
- Chronic kidney disease
- Malabsorption
- Rheumatic disease: rheumatoid arthritis, spondyloarthropathies
- Mastocytosis
- Prolonged bed rest and immobilization.

The most important risk factors for osteoporosis are advanced age, white or Asian race, low body mass index, and family incidence of the disease. Other lifestyle risks include low calcium intake, premature ovarian failure, smoking, alcohol use, and low level of physical activity.

IV. PATHOGENESIS

Osteoporosis is a result of imbalance between bone formation and resorption. There is a period of bone formation with acquisition of peak bone mass between the ages of 20 and 30. This peak remains in a plateau until around 40 years old. Then bone loss occurs before the menopausal period, significant during the menopausal period, particularly in the 10 years following menopause, linked to estrogen deficiency. Over the course of their lives, women lose 35 to 50% of their bone mass.

V. DIAGNOSIS

Diagnosis of OP can be made in the presence of one of the following:

- Fragility fracture, particularly at the spine, hip, wrist, humerus, rib and pelvis.
- T-score less than or equal to -2.5 based upon bone mineral density (BMD) measurement by dual-energy x-ray absorptiometry (DXA).

5.1- Clinical manifestation

OP is a silent disease until OFx happens. However, painless vertebral fractures can manifest as height loss or marked kyphosis.

5.2- Laboratory and imaging

There is no role of laboratory investigation to confirm the diagnosis of OP, but to eliminate other common secondary cause and to initiate the treatment. Minimal test should include calcemia, phosphatemia, albuminemia, and 25-OH vitamin D.

Radiography may show previous fracture and generalized bone transparency but there is no specificity.

5.3- Bone mineral density measurement

The gold standard measurement is DXA. It should be measured at the spine and hip.

T-score > -1 SD	Normal
-1 SD > T-score > -2.5 SD	Osteopenia
T-score ≤ -2.5 SD	Osteoporosis

5.4- Fracture risk assessment tool – FRAX score

FRAX score allows to estimate the 10-year probability of hip fracture and major OFx for untreated patients between ages 40 and 90 years. The elements for calculation include easily obtainable clinical risk factors for fracture and femoral neck BMD by DXA, if available. Clinical risk factors include weight, height, previous fracture, parent fractured hip, current smoking, glucocorticoids, rheumatoid arthritis, secondary osteoporosis, and alcohol consumption.

VI. DIFFERENTIAL DIAGNOSIS

- Osteomalacia
- Primary hyperparathyroidism
- Malignant disease: bone metastasis, multiple myeloma or other hematological disorder
- Renal osteodystrophy

VII. ANAGEMENT

1- Objective

- To prevent fracture if no fracture
- To prevent next fracture in previous fracture patient
- To increase BMD and T-score to more than -2.5 SD

2- Critical signs

In spite of T-score, OP is considered as severe in presence of severe fracture: proximal humerus, distal and proximal femur, proximal tibia, vertebrae, pelvis and 3 ribs adjacent.

3- Mortality risk factor

- Hip fracture
- Vertebral fracture

4- Outpatient management

- Adequate calcium and vitamin D intake (1g of calcium and 400UI of vitamin D/day)

- Exercise
- Smoking cessation
- Avoidance of alcohol consumption
- Fall prevention

5- First line treatment

Indication to treatment depends on risk factors, history of fracture and T-score.

	Severe fracture	Non-severe fracture	No fracture
T-score > -1	Specialist advice	No treatment	No treatment
-2 < T-score ≤ -1	Treatment	Specialist advice	No treatment
-3 < T-score ≤ -2	Treatment	Treatment	Specialist advice
T-score ≤ -3	Treatment	Treatment	Treatment

The first-line treatment is bisphosphonate if not contraindicated.

In absence of fracture, oral bisphosphonate, ALENDRONATE or RISEDRONATE should be prescribed. It can be given daily, weekly or monthly according to patient preference. Oral bisphosphonate must be taken first thing in the morning on an empty stomach with a big glass of water. Do not take it with other beverages. Patient must remain upright sitting or standing and no lying down for 30–60 minutes after taking the medication. Do not take any additional medications, beverages, or food for 30–60 minutes after taking the medication.

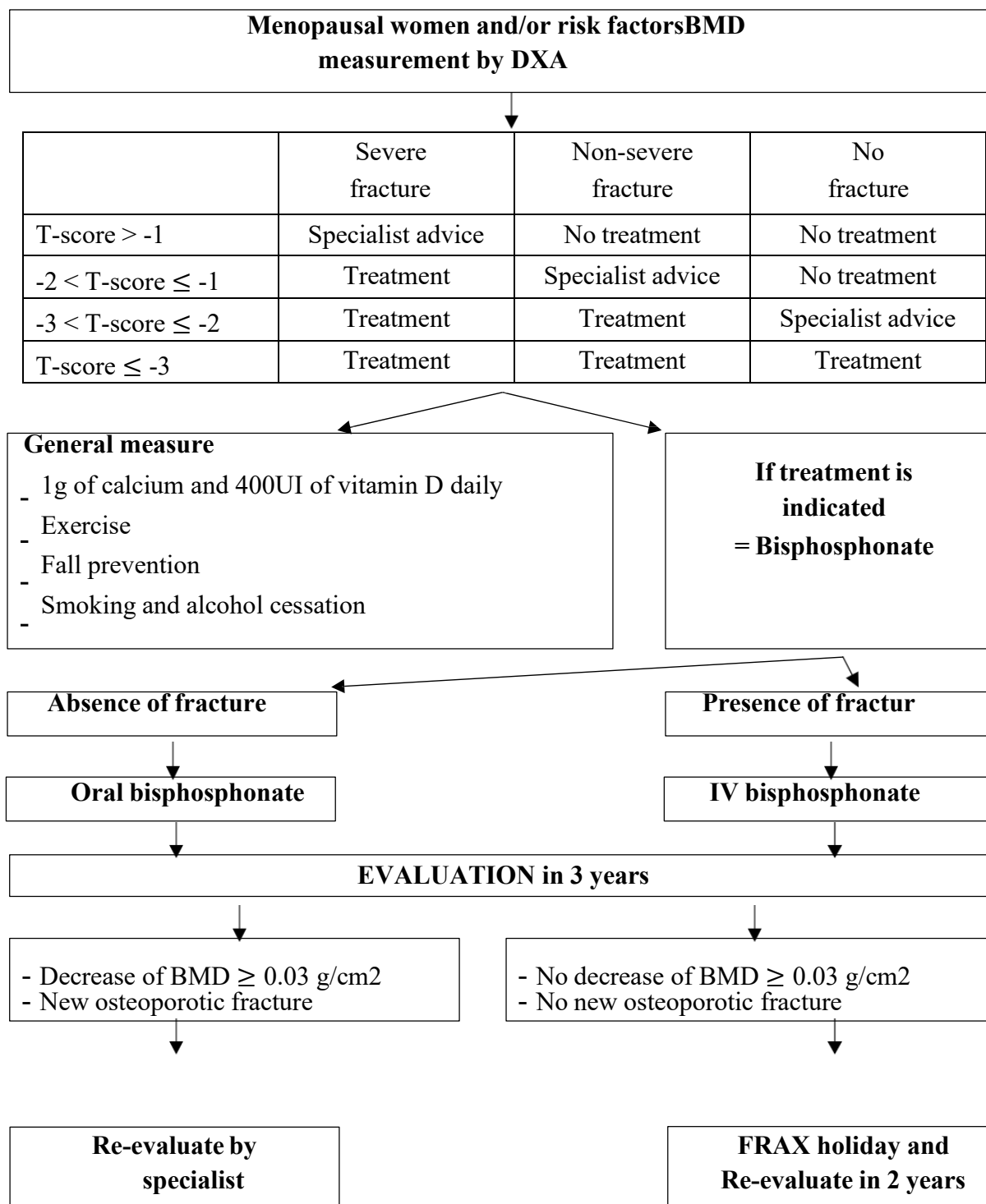
In presence of fracture, intravenous bisphosphonate, ZOLEDRONIC ACID is prioritized. The posology is 5mg annually.

6- Treatment adjustment

If there is new fracture after 1 year of treatment, a re-evaluation is needed. If patient is on oral treatment, we can switch to intravenous treatment or DENOSUMAB. If patient is already on intravenous treatment, DENOSUMAB is indicated. There must be at least 1-year treatment by bisphosphonate after DENOSUMAB.

Re-evaluation is 3 years after treatment. If there is no decrease in BMD more than or equal to 0.03g/cm² or no new fracture, patient can take a FRAX holiday for 2 years. Otherwise, they need to be evaluated by specialist.

7- Algorithm



VII. SPECIAL CASE

In patient who also suffers from hot flush, hormone replacement therapy might be used. If there is an OFx in stage 4 of CKD patient, renal osteodystrophy needs to be eliminated. Indication to OP specific treatment is discussed case by case. If needed, RISEDRONATE might be given 35mg every other week but for not more than 3 years. However, DENOSUMAB is preferred.

VIII. EVOLUTION AND PROGNOSIS

Patient, who is candidate to treatment but refuses, probably suffers the fracture.

Hip and vertebral fractures are catastrophic. Less than half of survived hip fracture patients will walk unaided again, while a significant proportion will never regain their former degree of mobility. Vertebral fractures adversely affectsufferers physically and mentally. These lead to a significant negative repercussion on routine activities of daily living.

Mortality during the 5 years after a hip or vertebral fracture is approximately 20%higher than would be expected, with most premature deaths occurring within thefirst 6 months after hip fracture.

The risk of recurrence of fracture is greatest in the year following the fracture. After a first fracture, the risk of recurrence is multiplied by 2, and this risk is multiplied by 5.3 in the following year.

IX. COMPLICATION

- Osteoporotic fracture
- Handicap and mortality
- Side effect of bisphosphonate and denosumab: osteonecrosis of the jaw, atypicalfemoral fracture

X. PATIENT EDUCATION AND PREVENTION

- Optimize calcium and vitamin D intake
- Daily Exercise
- Smoking and alcohol cessation
- Fall prevention

XI. REFERENCES

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